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PE EXAM SPECIFICATIONS

PE



CIVIL

STRUCTURAL

EXAM PREP

**2025
2026**

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- **6 FULL-LENGTH** PRACTICE EXAMS



E-LEARNING PLATFORM INCLUDED

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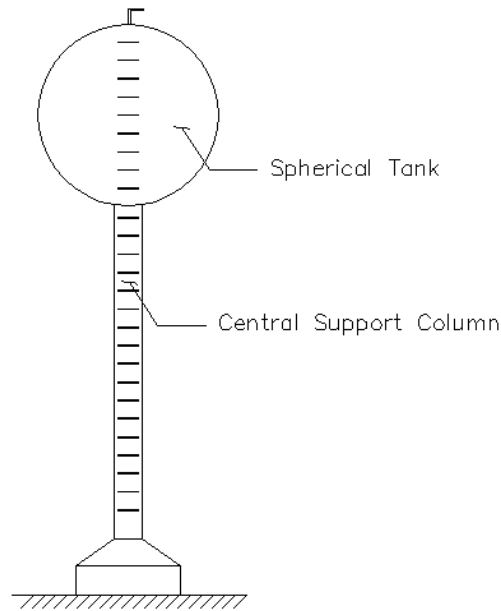
1. Analysis of Structures–Loads and Load Applications

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1.1 Dead Load

[Click here for solutions 1.1](#)

1. An elevated reinforced concrete spherical water tank can store 2000 gallons of water. The walls of the tank are 6 inches thick. What is the total unfactored dead load acting on the supporting column?

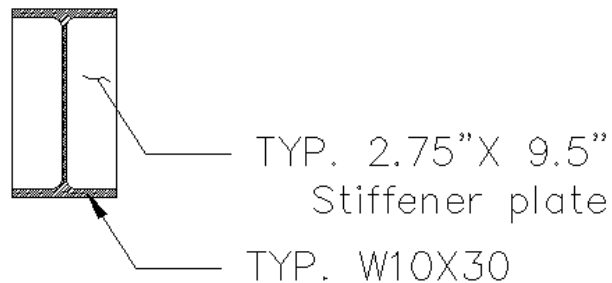


- (A) 17 kips
 - (B) 24 kips
 - (C) 30 kips
 - (D) 34 kips
-
2. Which of the following is not true about dead loads on the roof of a building?
 - (A) Dead loads consist of the weight of materials of construction incorporated in the building roof including insulation layers,
 - (B) Installed solar panels and their support systems including framing and ballast are all considered dead loads,
 - (C) Weight of soil layers in landscaped roofs should be considered as dead loads, and the moisture content should be taken as the average of dry and fully saturated conditions.
 - (D) Dead loads include fixed service equipment such as cranes and material handling system

3. A floor system is constructed using a 2-inch steel deck and a 3-inch lightweight concrete topping. W8X18 Secondary Beams are spaced at 6 feet center to center. What is the factored dead load (to be used in combination with live loads) acting on each secondary beam?
Weight of steel deck = 5 psf

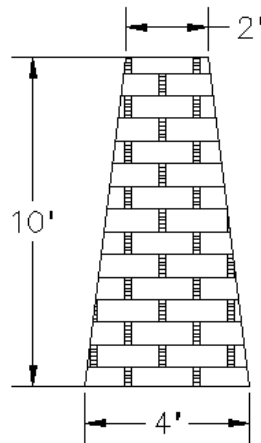
(A) 192 lb/ft
(B) 220.5 lb/ft
(C) 264.6 lb/ft
(D) 202.5 lb/ft

4. A W10X30 beam is 20 ft long and has stiffener plates welded on both sides throughout the span, spaced at 4ft center to center. The end bearing stiffeners are 0.5 inches thick on both ends and the inside web stiffeners are 0.25 inches thick. What is the total weight of the beam including the plates?



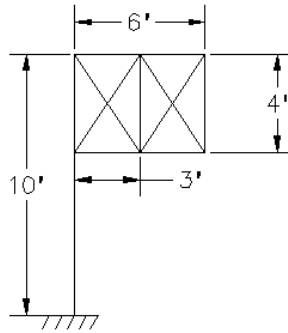
(A) 600 lb
(B) 630 lb
(C) 648 lb
(D) 670 lb

5. A trapezoidal stone masonry abutment is constructed to support a culvert. The abutment is 12 ft in length and the cross section of the abutment is as shown below. What is most nearly the self-weight of the abutment?



- (A) 35 kips
(B) 61 kips
(C) 76 kips
(D) 92 kips
6. A residential building is undergoing underpinning and retrofitting of the foundation. The total dead load on the East wall is calculated as 2020 kN. The wall must be jacked up but the ratings on the hydraulic ram are only available in US (short) tons only. Select the correct rating of the jack that can be used for this job to satisfy the minimum requirement.
- (A) 100 tons
(B) 200 tons
(C) 225 tons
(D) 250 tons

7. A cantilevered traffic light pole is designed as shown below. The vertical and horizontal members are fabricated with HSS 6X6X1/2 and the diagonal members are fabricated with HSS 2X2X1/4. The whole assembly is to be erected at the job site. What is the capacity of the hoisting mechanism required to lift the framing assembly? Assume additional rigging weight of 10% of the framing assembly.

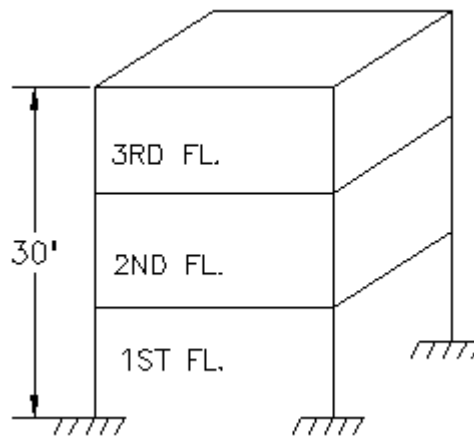


- (A) 1165 lb
(B) 1282 lb
(C) 1382 lb
(D) 1400 lb
8. A timber beam at the roof of a building supports a parapet wall along its length. The wall is 12 ft long and 4 ft tall. The wall is constructed with hard brick masonry and is 9 inches thick. What is the factored dead load to be used in combination with live loads, due to the masonry's weight acting on the beam?
- (A) 390 lb/ft
(B) 468 lb/ft
(C) 561 lb/ft
(D) 673 lb/ft

1.2 Live Load

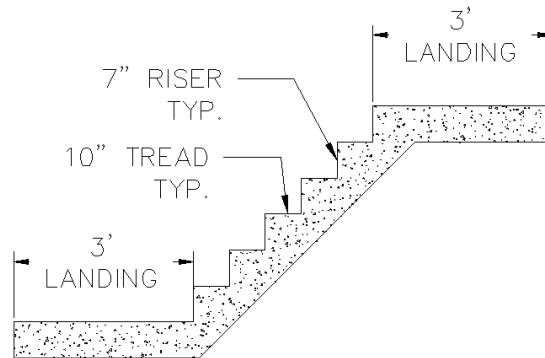
[Click here for solutions 1.2](#)

9. A 3-story office building is to be constructed. The 1st floor will be used as an open parking lot and the 2nd and 3rd floors will be rented to two separate businesses. The businesses are not permanent, and the partition wall layouts will be changed to meet the requirements of the tenants. The typical floor height is 10ft and the typical floor area is 30ftX30ft. What is the total estimated partition wall load acting on the building?



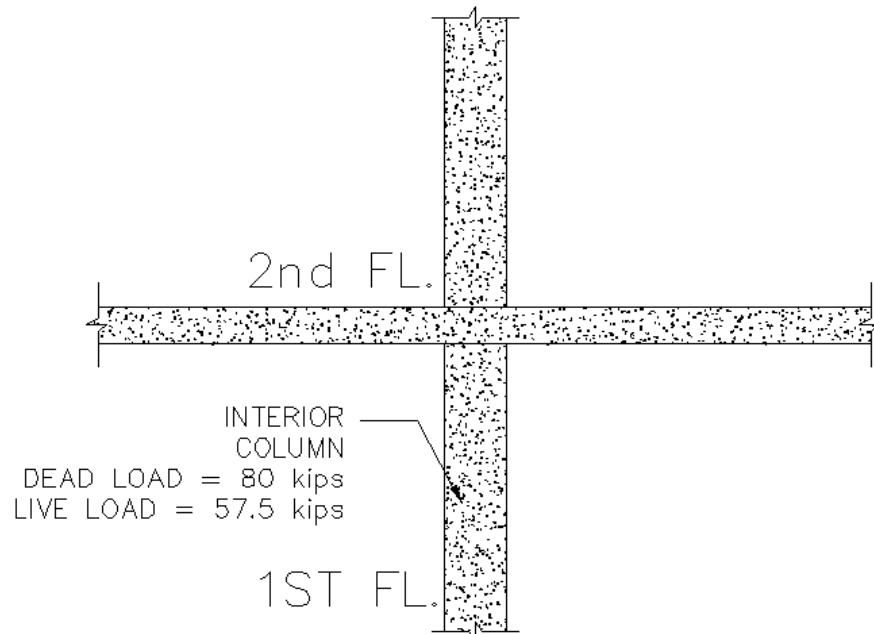
- (A) 13 kips
(B) 20 kips
(C) 27 kips
(D) 35 kips
10. A single story 50ftX100ft storage warehouse is built in Arizona. The warehouse will be heavily loaded with electronic goods. What is the total factored live load acting on the floor slab of the warehouse?
- (A) 390 kips
(B) 625 kips
(C) 1250 kips
(D) 2000 kips

11. The flight of stairs shown below belongs to a school building. The width of the stairs and landing is 4 ft. Calculate the total live load following the provisions of International Building Code.



- (A) 1626 lb
(B) 2601 lb
(C) 4067 lb
(D) 2000 lb
12. A two-story gymnasium is to be built in New York. The ground floor slab is on grade and the first floor will be supported by steel beams. The tributary area for an interior beam of length 40 ft is 1250ft². What is the minimum live load acting on the interior beam? Apply reduction factors as applicable.
- (A) 6.25 kips/ft
(B) 3.125 kips/ft
(C) 2.5 kips/ft
(D) 1.5 kips/ft

13. A three-story retail store has a heavily loaded interior column on the first floor. The design is to be optimized, and the engineers are looking to reduce the live load demands on the column. The tributary area of the column is 500 ft². Calculate the reduced live load demand acting on the column.

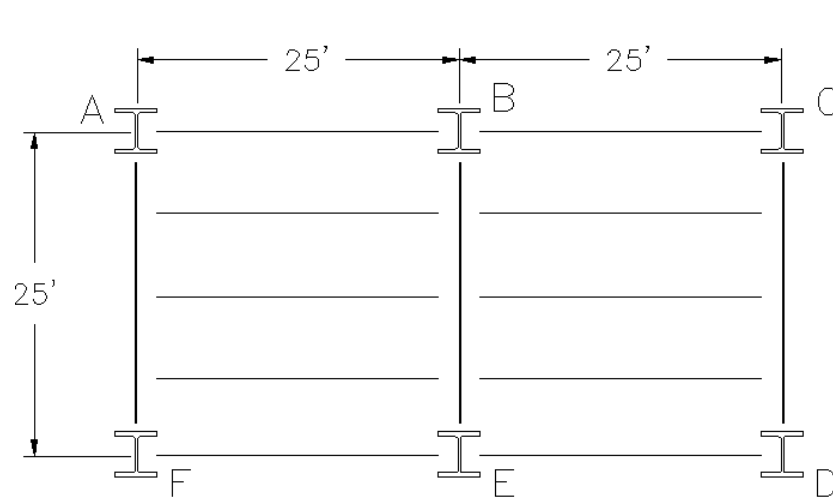


- (A) 20 kips
(B) 25 kips
(C) 34 kips
(D) 40 kips
14. According to IBC 2018, which of the following factors does not affect the reduction of design live loads?
- (A) The expected intensity of loading
(B) Tributary Area of the member
(C) The expected service life of the structure
(D) The utility of the structure

15. A school classroom has an area of 12ftX12ft. Furniture and equipment are estimated to weigh a total 2000 pounds. The class can seat only 10 students and 2 teachers for any class. What is the minimum total live load to be considered to design the classroom? Assume the maximum weight of each student and teacher to be 200 pounds.

(A) 4400 lb
(B) 5760 lb
(C) 7040 lb
(D) 14400 lb

16. The partial plan below shows the roof of a single family home. The roof is flat and not intended for occupancy. Calculate the applicable reduction factors and total roof live load acting on column A

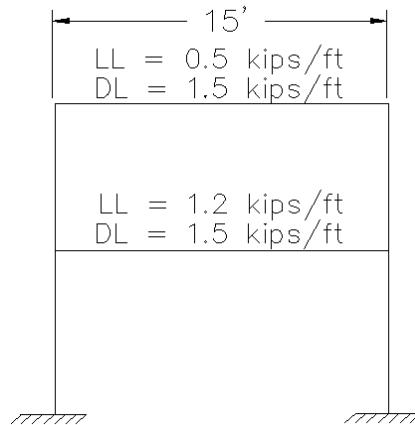


(A) No roof live load for roof without occupancy
(B) 1562 lb
(C) 3125 lb
(D) 6248 lb

17. The unfactored roof live load acting on a beam is 200 lb/ft . The roof rises 6 inches for every foot of run. The tributary area of the beam is 300ft². What is the factored reduced live load acting on the beam to be used for strength design?

(A) 240 lb
(B) 260 lb/ft
(C) 320 lb/ft
(D) 400 lb/ft

18. The frame shown below belongs to the bay of a building. The columns shown are exterior columns. Tributary area for each column is 112.5 ft^2 . What is the total live load acting on the base of a ground floor column? Apply all reduction factors as applicable.



- (A) 6 kips
(B) 12.75 kips
(C) 12.2 kips
(D) 4.4 kips
19. For a sloped roof, what is the basis for applying slope adjustment factors to reduce the live loads per the provisions of ASCE7-16 and IBC 2018?
- (A) Structural capacity of roofing members increases with inclination
(B) Seismic load governs for sloped roof and live loads can be neglected
(C) Sloped roofs have reduced accessibility and are less likely to be occupied
(D) Snow load accumulation is lesser in sloped roofs
20. ASCE 7-16 Table 4.3-1 provides minimum distributed live loads and minimum concentrated live loads to be used for design for structures. Which of the following is not true about concentrated loads?
- (A) Concentrated live loads are assumed to act uniformly over an area of 0.58 m^2
(B) ASCE 7-16 specified concentrated live loads are summed with minimum uniformly distributed live loads to create the most severe effect in the structural member
(C) Live load reduction factors cannot be applied to concentrated loads.
(D) Concentrated loads are usually critical over uniformly distributed loads during slab design.

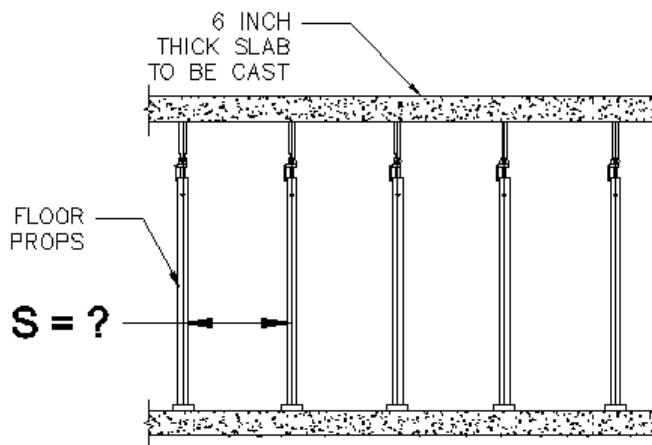
1.3 Construction loads

[Click here for solutions 1.3](#)

21. For a deep foundation construction job, H piles are used in support of excavation systems. The section to be used is HP 10X57. The yield stress of the material is 50 ksi. What is the allowable driving load for the top driven pile?



- (A) 500 kips
(B) 750 kips
(C) 1000 kips
(D) 1500 kips
22. A 6-inch thick fresh concrete slab is to be cast on a new floor. The construction load is estimated as 50 psf including equipment and weight of formwork. The newly cast floor is to be shored with floor props that have allowable carrying capacity 2 kips. The props are to be laid out following a square layout, with equal spacing in both directions. What is the maximum permitted spacing? Unit weight of concrete = 150 pcf

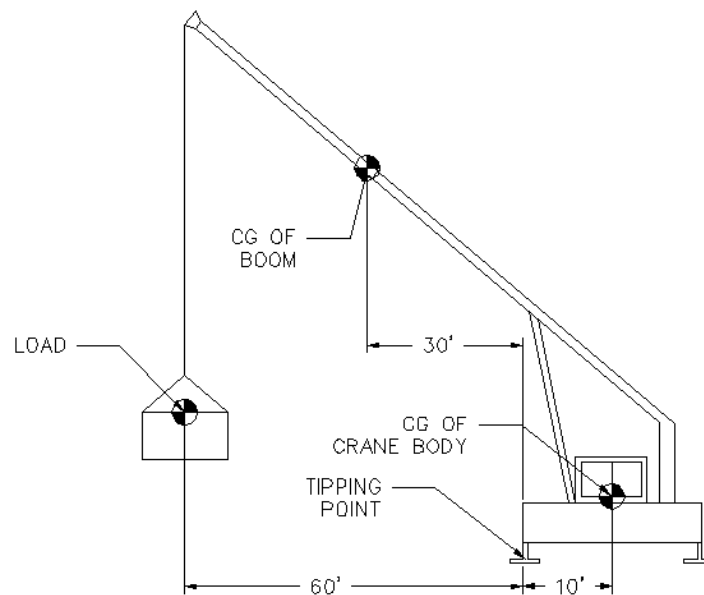


- (A) 2ft
(B) 3ft
(C) 4ft
(D) 5ft

23. Which of the following is not true about construction loads?

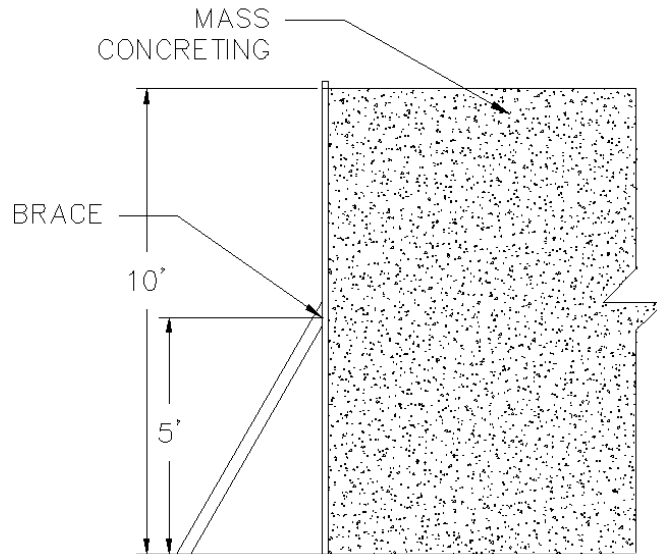
- (A) Construction loads can be higher than permanent live loads acting on the building
- (B) When considering construction loads, wind loads can be reduced, adjusting it based on the duration of construction.
- (C) The load factors for permanent live loads and construction loads are different in ASD methodology.
- (D) Concentrated loads are both static and dynamic.

24. A Crane is set at a job site as shown below. The weight of the crane body is 150 kip and the weight of the boom is 15 kip. The crane must lift a load and transport it to a radius of 60 feet from the pick point. What is the maximum tipping load that is allowable to be lifted by the crane. Assume factor of safety 1.5 against overturning.



- (A) 5 kips
- (B) 9 kips
- (C) 12 kips
- (D) 15 kips

25. In a mass concreting job, fresh concrete is poured to cast an embankment. The first pour is 10 ft high. The formwork boards are braced with a 4X4 and are prevented from sliding at the base. The boards are braced at mid height as shown in the sketch below. What is the lateral force exerted by the freshly poured concrete at the point of bracing? Unit weight of fresh concrete = 150 pcf

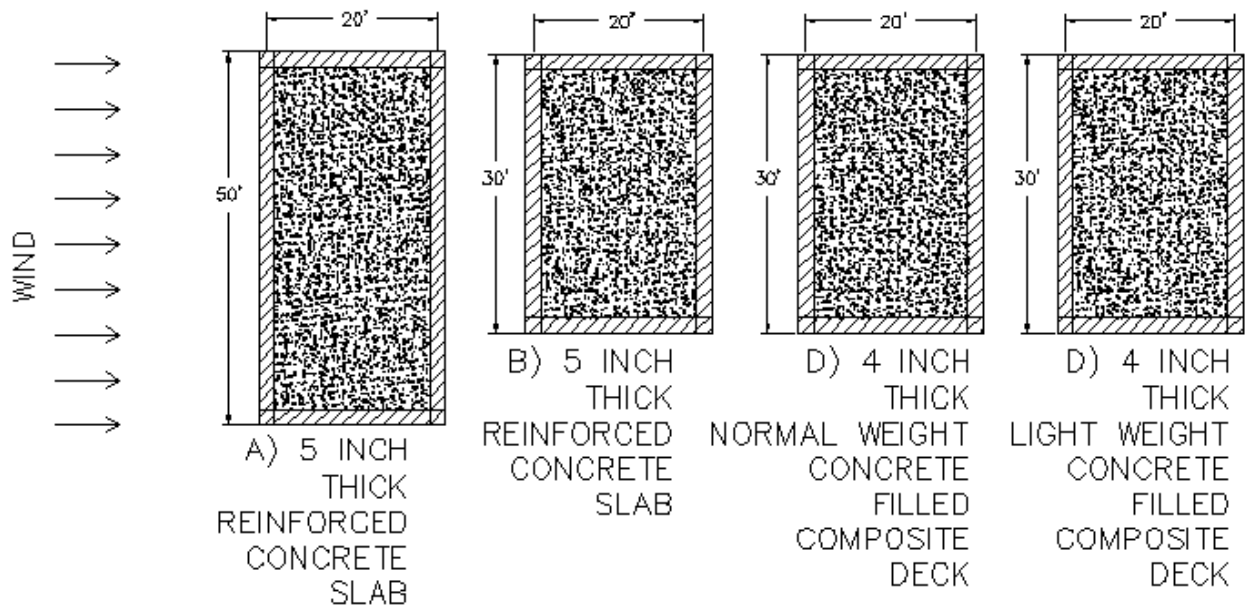


- (A) 5 kips/ft
- (B) 7 kips/ft
- (C) 10 kips/ft
- (D) 12 kips/ft

1.4 Wind loads

[Click here for solutions 1.4](#)

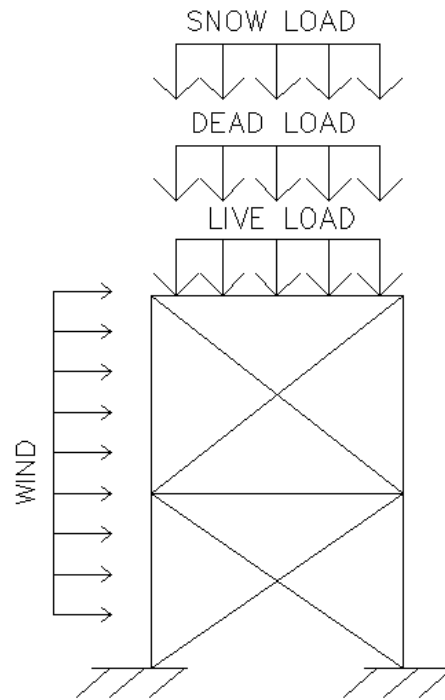
26. The plans below show Roof slabs for four different buildings. Which of the following cannot be idealized as a rigid diaphragm for analysis under wind loads per ASCE 7 -16?



- (A) A
(B) B
(C) C
(D) D
27. An 80 ft tall commercial building is constructed in a river valley located 2000 ft above sea level. The basic wind speed in the area is 90 mph. The building is classified as Risk category III and lies in the Exposure B category terrain. The building can be considered enclosed. Topographic effects in the area can be neglected. Calculate the velocity pressure acting on the MWFRS at the top of the building. The structure can be considered rigid.

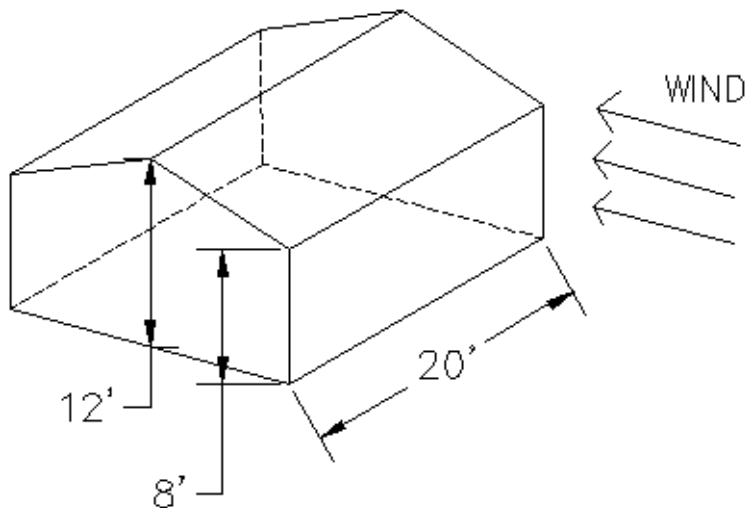
- (A) 11 lb/ft²
(B) 12.81 lb/ft²
(C) 15.24 lb/ft²
(D) 22.3 lb/ft²

28. Structural stability is to be evaluated for the framed structure below. The structure is located in a mountainous region with significant snow and wind load. To evaluate the stability of the structure against foundation uplift, which combination must be used?



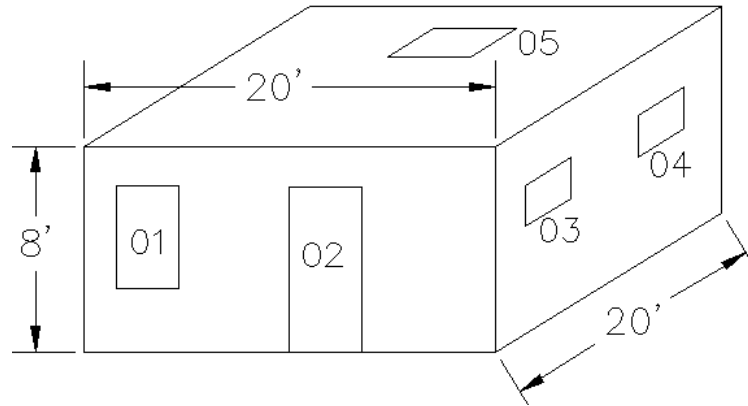
- (A) Dead + Live + Snow + Wind
 (B) Wind + Snow
 (C) Dead + Wind
 (D) Live + Wind
29. An HVAC unit is to be installed at the roof of a 5 story building. The unit weighs 200 pounds. The unit is 6 ft tall. The length and width of the unit is 5ft each. Center of gravity of the unit can be assumed to be at the geometric center of the unit. Calculate the lateral wind load acting on the unit to be considered for anchorage design. The properties of the building are as follows:
- Width of building = 40 ft Roof height = 50 ft
 $V = 95$ mph,
 Topographic factor $K_{zt} = 1$,
 Ground elevation factor $K_e = 1$
- (A) 0.15 kips
 (B) 0.25 kips
 (C) 0.5 kips
 (D) 0.9 kips

30. A regular shaped enclosed low rise building qualifies for wind analysis using envelope procedure. The calculations yielded a horizontal wind design pressure of 7 psf in the windward direction. What is the total wind load in the projected vertical plane of the windward side as shown below? Follow minimum design requirements per ASCE 7-16.



- (A) 1680 lb
(B) 3200 lb
(C) 3840 lb
(D) 4800 lb
31. A 10 story hospital is to be built in New York. The infrastructure is classified as an essential structure to the county. The site is flat and topographic effects can be neglected. There are numerous closely spaced commercial houses and up to 2500 ft from the site upto which the area can be classified as suburban. Beyond that, there is open terrain with scattered obstructions. What is the wind exposure category to be used in the design of the building?
- (A) Exposure A
(B) Exposure B
(C) Exposure C
(D) Exposure D

32. A small shed shown below is to be constructed near a coastal area. The building has openings in two walls and roof as shown below. The other two walls are solid and has no openings



The sizes of openings are as follows

01 = 2ftX3ft 02 = 2ftX6ft

03 = 2ftX2ft 04 = 2ftX2ft

05 = 4ftX4ft

The front wall with openings 01 and 02 receives positive wind pressure.

Based on the information above, the shed can be categorized as?

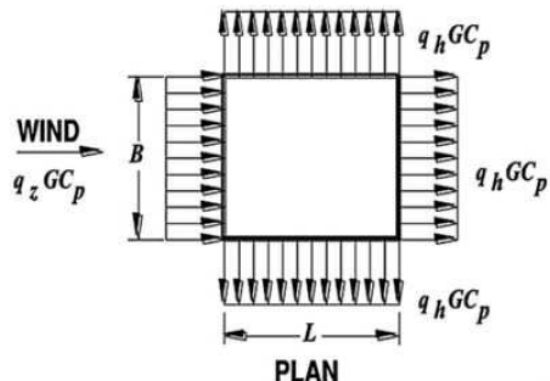
- (A) Enclosed
- (B) Open
- (C) Partially enclosed
- (D) Partially open

33. Using the directional procedure, compute the design wind pressure in the windward walls of a rigid enclosed 1 story building located in Exposure C terrain. The basic wind speed of the area is 100 mph. Neglect any ground elevation and topographic effects.

$L = 30\text{ft}$

$B = 30\text{ft}$

Height of wall $z = h = 10\text{ft}$



- (A) 9.25 psf, 15.91 psf
- (B) 15.91 psf, -9.25 psf
- (C) 9.25psf, -15.91 psf
- (D) -9.25psf, -15.91 psf

34. A curtain wall is to be replaced in an Exposure C zone. The basic wind speed of the area specified by ASCE 7-16 chapter 26 is 100 mph. The specifications of the material of the curtain wall only list allowable stress values and the strength of the wall against wind loads cannot be checked in accordance with LRFD methodology. What is the basic wind speed to be used in ASD design of the wall?

- (A) 100 mph
- (B) 77.5 mph
- (C) 85 mph
- (D) 90 mph

35. Using the directional procedure, calculate the design wind pressure (wind normal to the ridge) in the windward side of the roof of a rigid (assuming internal pressure coefficients for an open structure) 1 story building located in Exposure C terrain. The basic wind speed of the area is 100 mph.

Slope of the roof = 10 degrees

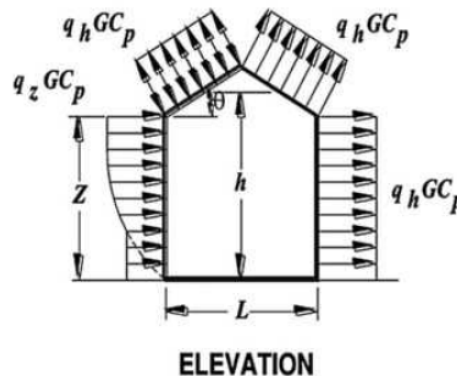
$L = 20\text{ft}$

$B = 50\text{ft}$

Topographic factor $K_{zt} = 1$

Elevation factor $K_e = 1$

Height of roof $h = 10\text{ft}$

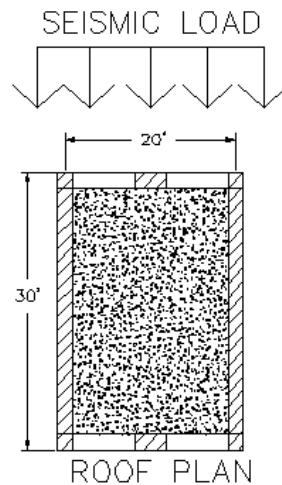


- (A) 14.15 psf, 2.83 psf
- (B) -14.15 psf, -2.83 psf
- (C) -14.15 psf, 2.83 psf
- (D) 14.15psf, -2.83 psf

1.5 Seismic loads

[Click here for solutions 1.5](#)

36. The plan below shows the roof plan of a one story building. The roof is a concrete slab, supported by masonry walls on longer sides and columns on shorter sides as shown by the hatches. During a seismic event, the lateral load acting at the roof slab level is 200 lb/ft and is along the direction as shown below. What is most nearly the loads shared by each wall?

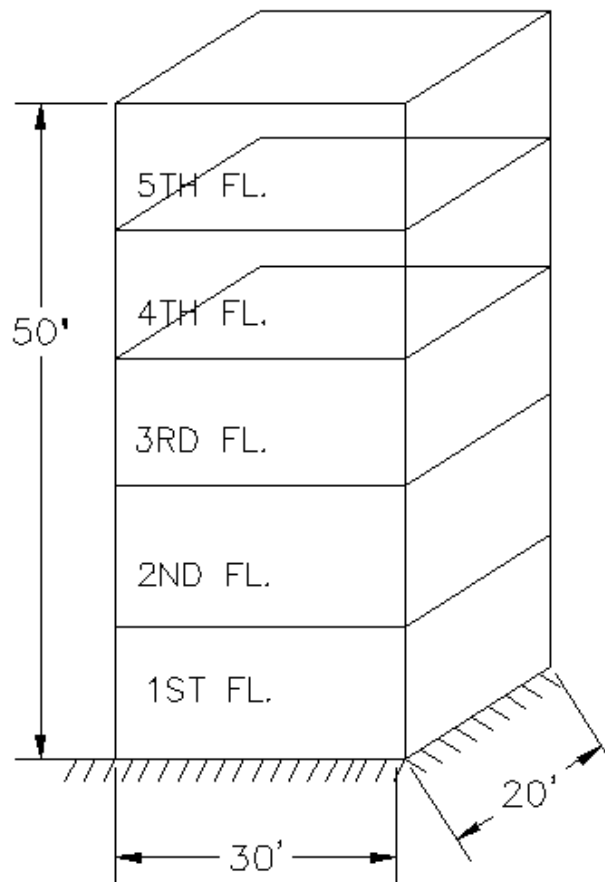


- (A) 1 kips
(B) 2 kips
(C) 3 kips
(D) 4 kips
37. A 3 story building is analyzed for seismic loads. The effective seismic weight of the structure is calculated as 800 kip. However, the design needs to be optimized. Which of the following can be removed from the calculation of seismic weight so that the seismic demands on the structural members can be reduced?
- (A) Weight of partition walls for the second floor office
(B) Weight of landscaping and permanent HVAC unit on the roof
(C) 20% of uniform design snow load
(D) 25% of floor live load for 'slab on grade' ground floor storage

38. A 10 story building is identified as a critical infrastructure and is assigned risk category IV. The building is located in an active seismic zone with $S_s = 1g$ and $S_1 = 0.4g$. The bearing stratum is explored to consist of stiff soil. Determine seismic design category of the structure following procedures of ASCE 7-16.

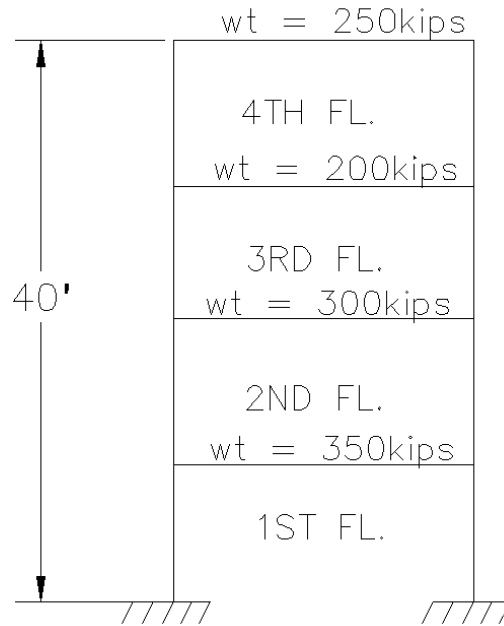
- (A) Seismic Design Category F
- (B) Seismic Design Category A
- (C) Seismic Design Category C
- (D) Seismic Design Category D

39. The dimensions of a 5 story building are as shown in the sketch below. The building is made of reinforced concrete moment resisting frames in which the frames resist all lateral and gravity loads. What is the estimated fundamental time period of the building?



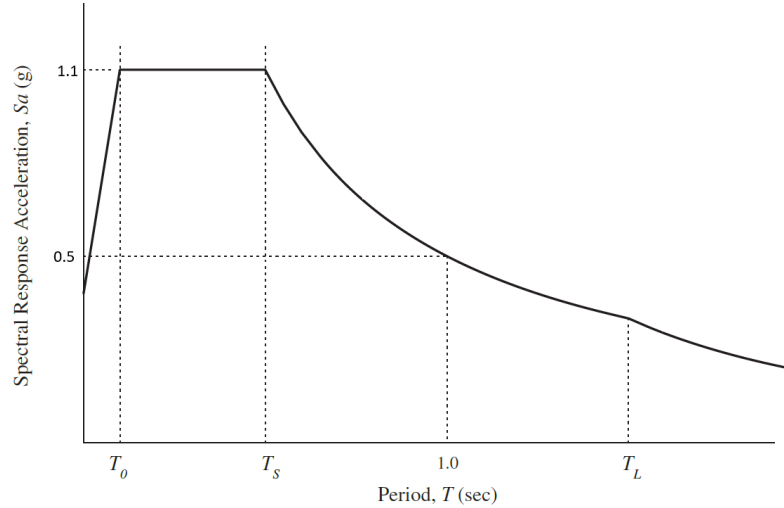
- (A) 0.45 seconds
- (B) 0.5 seconds
- (C) 0.55 seconds
- (D) 0.6 seconds

40. The design base shear of a 4 story building is calculated as 500 kips. The average story height is 10ft. The fundamental time period of the building is 0.4s. The effective seismic weights of each floor is given below. Calculate the seismic lateral force acting on the 3rd floor of the building.



- (A) 118 kips
(B) 125 kips
(C) 200 kips
(D) 250 kips
41. Which of the following is not true about the response modification coefficient “R” in ASCE 7-16 Table 12.2-1
- (A) Forces transferred from the upper system to the lower system shall be increased by multiplying by the ratio of the higher response modification coefficient to the lower response modification coefficient,
(B) Seismic Loads scaled down using the response modification factor is to be used for LRFD load combinations,
(C) Higher value of “R” indicates that the system is weaker and has lower resistance to lateral loads,
(D) The value of the response modification coefficient, R, used for design in the direction under consideration shall not be greater than the least value of R for any of the systems used in that direction.

42. For a site, the horizontal ground motion design response spectrum generated per the procedures of ASCE 7-16 section 11.4.6 is given in the sketch below. The structural design also requires incorporation of vertical ground motions. What is the design vertical response spectral acceleration to be used in the seismic analysis of a building with a vertical period of vibration 0.1 second located at site class D?

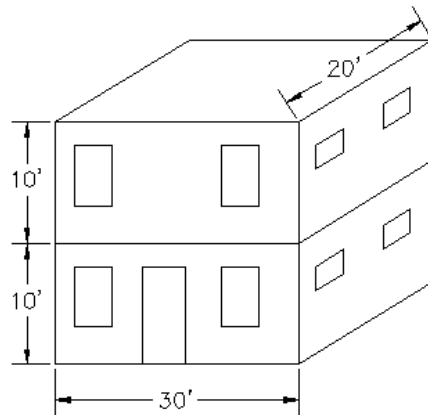


- (A) 1.1g
(B) 1.26g
(C) 1.65g
(D) 2g
43. A 2 story ordinary reinforced concrete moment frame building has an effective seismic weight of 700 kips. The Maximum considered Earthquake short period spectral acceleration is 0.5g. The fundamental period of the structure is 0.2 seconds and the long-period transition period is 4 seconds. Calculate the seismic base shear acting in the structure following equivalent lateral force procedure. Assume risk category II.
- (A) 50 kips
(B) 67 kips
(C) 78 kips
(D) 116 kips

44. Calculate the seismic demand on a cantilevered unbraced parapet wall located at the roof of a building. The importance factor of the component can be taken as 1. The weight of the wall is 50 kips and the design short period spectral acceleration is 0.5g.

- (A) 25 kips
- (B) 30 kips
- (C) 40 kips
- (D) 50 kips

45. A 2 story ordinary plain masonry wall building is identified as seismic design category B. The mapped short period spectral acceleration $S_s = 0.5g$. The site is rocky and the total seismic weight of the building is 600 kips. Calculate the seismic base shear using simplified lateral force analysis procedure per ASCE 7-16.

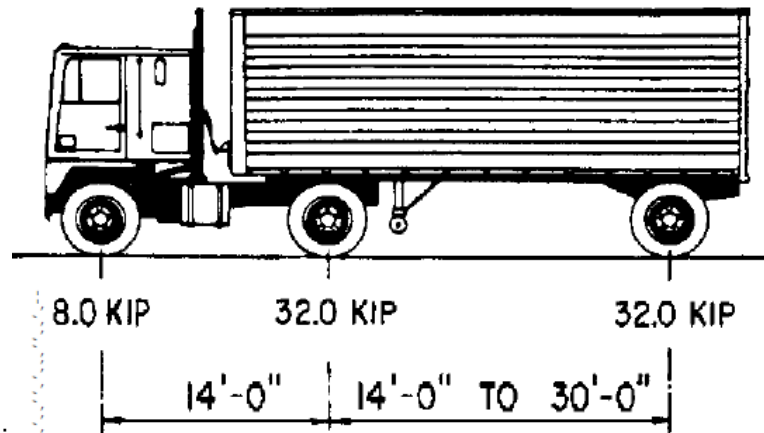


- (A) 100 kips
- (B) 150 kips
- (C) 250 kips
- (D) 300 kips

1.6 Moving loads

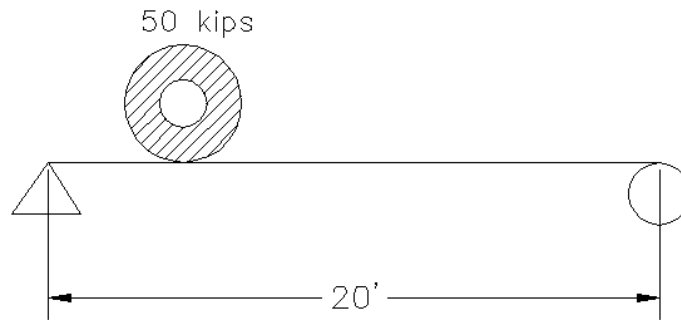
[Click here for solutions 1.6](#)

46. The figure below shows a standard design Truck from AASHTO LRFD bridge design specifications. What is the pressure exerted by tires on each side of the rear axle?

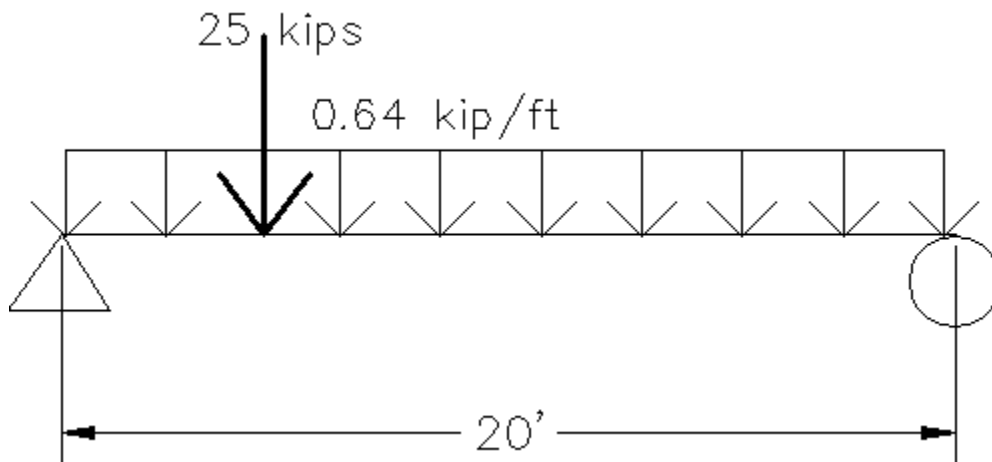


- (A) 40 psi
(B) 60 psi
(C) 80 psi
(D) 160 psi
47. According to AASHTO LRFD specifications, to calculate the total design vehicular live load, which one of the following must be considered?
- (A) Design lane load and Design truck load,
(B) Tandem axle of 25 kip each spaced 4.0 ft apart,
(C) Design truck load plus design tandem axle load plus design lane load,
(D) Design truck or Design tandem plus 0.64 klf of uniformly distributed lane load

48. A 50 kip wheel is moving along the span of a beam as shown below. What is the maximum bending moment acting in the beam due to the wheel load ?

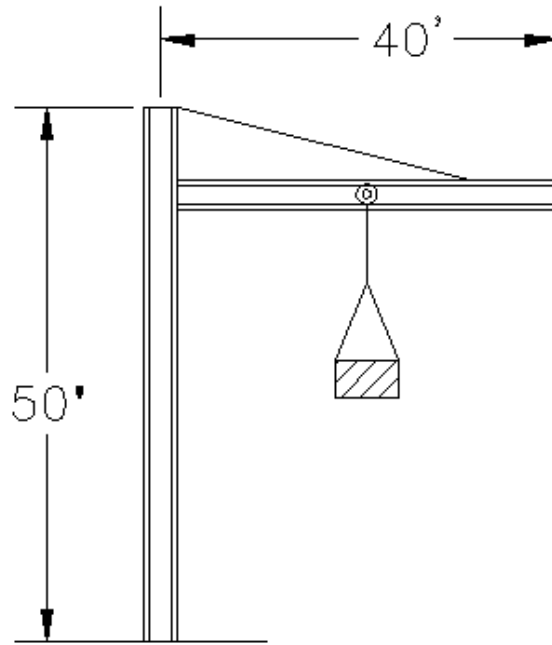


- (A) 100 kip*ft
(B) 250 kip*ft
(C) 500 kip*ft
(D) Cannot be determined
49. A single span simply supported bridge is subjected to a uniform lane load of 0.64 kip/ft and a moving concentrated wheel load of 25 kips. What is the maximum reaction acting on the support of the bridge?



- (A) 18.9 kips
(B) 25 kips
(C) 31.4 kips
(D) 37.8 kips

50. The figure below shows a mini crane that can pick and land a maximum weight of 1ton at a maximum radius of 40ft. The flexural capacity of the girder and column is at 100% utilization at that load and radius. If the hoisting motor, cables and all other components have sufficient residual capacity, what is the estimated maximum weight that the crane can pick and land safely at a minimum operating radius of 5 ft?

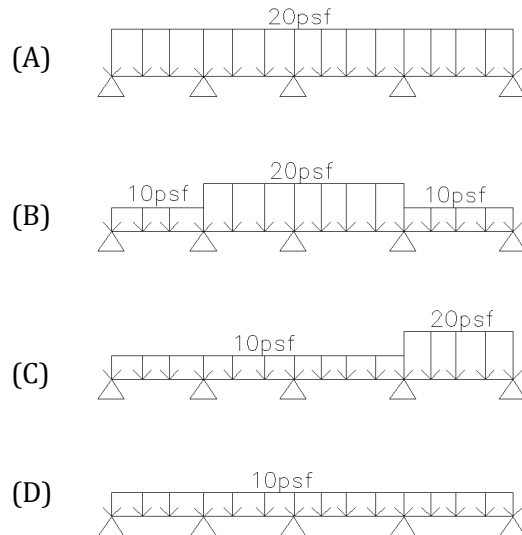


- (A) 2 kips
(B) 8 kips
(C) 12 kips
(D) 16 kips
51. Two 25 kips concentrated loads spaced 6 ft apart are moving along the span of a simply supported beam. Calculate the maximum moment acting on the beam. The span of the beam is 25 ft.
- (A) 200 kip*ft
(B) 221 kip*ft
(C) 242 kip*ft
(D) 276 kip*ft

1.7 Snow, rain, and ice

[Click here for solutions 1.7](#)

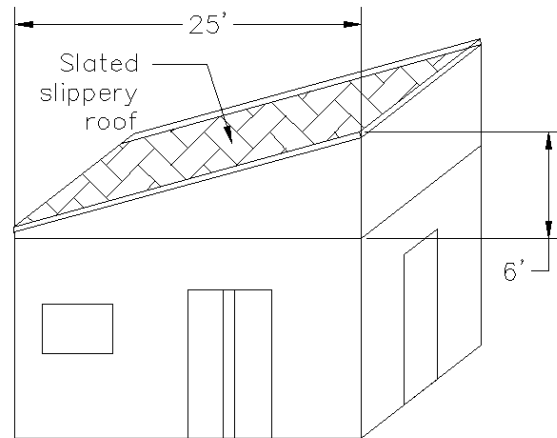
52. In addition to uniform balanced snow load, ASCE 7-16 requires that designs incorporate partial loadings and unbalanced snow loads. For a roofing system with 4 spans, the full uniform snow load is 20psf. Which of the following loading conditions is not recommended per ASCE 7-16 provisions?



53. Calculate the design snow loads acting on the roof of a building located in Anchorage Alaska. The building is classified as Risk Category III, The roof is unheated and partially exposed. The site is in an urban area with numerous closely spaced obstructions.

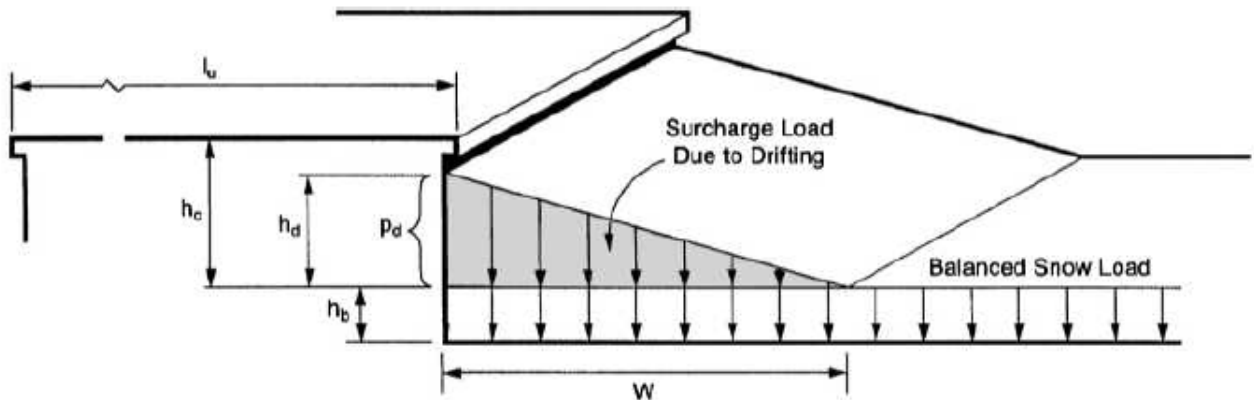
- (A) 42.6 psf
(B) 46.2 psf
(C) 50 psf
(D) 55psf

54. Calculate the design snow loads acting on the roof of a building located in Arlington, Washington shown below. The ground snow load is 17 psf. The building is classified as Risk Category II and the surface roughness category is C. The building is well heated and the roof is exposed on all sides.



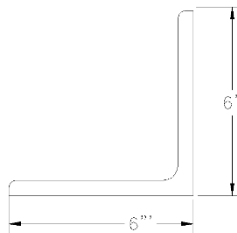
- (A) 9.63 psf
(B) 17 psf
(C) 20 psf
(D) 25psf
55. Which of the following is incorrect about rain on snow surcharge load per ASCE 7-16?
- (A) Rain on surcharge load is applied in addition to uniform roof snow load for balanced loading condition,
(B) For an area with ground snow load 18 psf, rain surcharge may apply,
(C) Rain surcharge for a roof with slope 20 degrees and horizontal distance from eave to ridge 20ft is 5 psf,
(D) For a roof with slope 0.5 degrees and horizontal distance from eave to ridge is 30ft, if the uniform roof snow load is 16 psf, the total load including rain surcharge is 21psf

56. Calculate the maximum intensity of leeward drift surcharge ' p_d ' for the lower roof shown below. The ground snow load for the area is 30 psf. Vertical separation between the two roofs is 15ft and the length of the upper roof is 50ft. The snow importance factor of the building is 1.



- (A) 30 psf
(B) 36.75 psf
(C) 40.5 psf
(D) 44.75psf
57. During an extreme rainfall event, for a roof in Miami, the hydraulic head for the design flow of the secondary drainage system is 2 inches. When the primary drainage system is blocked, the volume of ponding on the 30ftX30ft roof area is 5,000 gallons . What is the design rain load on the roof?
- (A) 13 psf
(B) 29psf
(C) 37psf
(D) 57psf
58. Calculate the ice load due to weight of ice formed on the members of a mounting frame at the roof of a 10 story building that is 100ft tall. The frame is constructed using L6X6X1/2 members. The building is classified as risk category III. The nominal ice thickness formed by freezing rain at a height of 33 ft in the area is 1.5 inches. Topographic effects can be neglected for the area.

- (A) 24.6 lb/ft
(B) 28.3 lb/ft
(C) 30.1 lb/ft
(D) 32.7 lb/ft



59. Which of the following is incorrect about ice sensitive structures and their design per ASCE 7-16?

- (A) In areas with freezing rain and snow, lattice structures, guyed masts, overhead lines, light suspension and cable-stayed bridges, aerial cable systems are sensitive to ice loads,
- (B) While analyzing a structure for ice loads, the load due to weight of ice and the load due to wind on ice formed areas are considered separately to avoid conservatism in design,
- (C) For temperature-sensitive structures, the load shall include the effect of temperature change from everyday conditions to the design temperature for ice and wind on ice,
- (D) Dynamic load effects due to atmospheric icing are not covered in ASCE 7-16.

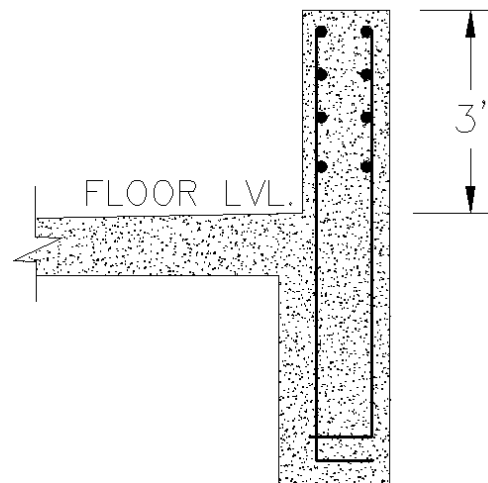
1.8 Impact loads

[Click here for solutions 1.8](#)

60. On a simply supported two lane bridge, the reaction due to design truck load is calculated as 40 kips. The design lane load reaction is calculated as 65 kips. What is the estimated braking force per AASHTO LRFD?

- (A) 2.5 kips
- (B) 5.25 kips
- (C) 10 kips
- (D) 26.25 kips

61. Given detail below is the section of a barrier in a 2nd floor parking lot. The barrier system is designed per the provisions of ASCE 7-16 for impact by passenger vehicles. What is the maximum factored design moment acting at the slab level?

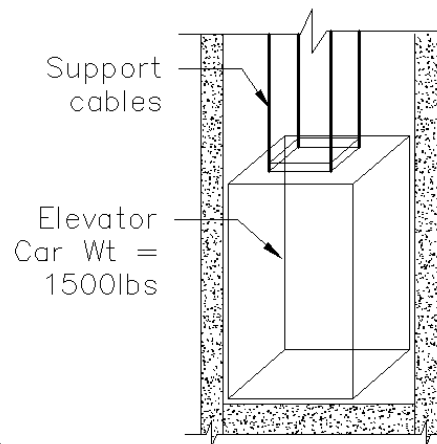


- (A) 9 kip*ft
- (B) 13.5 kip*ft
- (C) 14.4 kip*ft
- (D) 21.6 kip*ft

62. The maximum moment acting on a single lane bridge girder due to truck load is 500 kip*ft. The maximum moment due to design lane load is 600 kip*ft. What is the total unfactored moment including any impact factor per AASHTO LRFD bridge design specifications?

- (A) 1100 kip*ft
- (B) 1265 kip*ft
- (C) 1385 kip*ft
- (D) 1518 kip*ft

63. An elevator car weighs 1500 lbs and is rated for a total passenger weight of 2000 lbs. The elevator is supported by hoisting cables on four corners of the car as shown in the sketch below. What is the minimum safe working load limit of each cable supporting the car? The impact factor to be used for design is 100% .



- (A) 1 kip
- (B) 2 kips
- (C) 3 kips
- (D) 4 kips

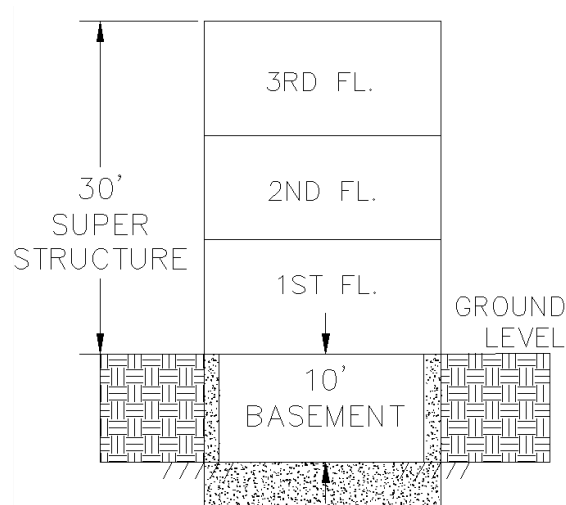
64. A structural designer wants to account for impact loads in a semi-commercial building. Which one of the following provisions is not true per ASCE 7-16?

- (A) Loads due to unusual vibrations should be accounted for while designing machine foundations. These load are not provided in ASCE 7-16 and should be provided by the equipment manufacturer,
- (B) The operating weights of motor driven light machinery should be increased by 20%,
- (C) The vertical impact due to wheel loads of pendant operated bridge crane is taken as 10% of the wheel load,
- (D) The equipment/moving load factored with appropriate impact factors need not have the additional 1.6 live load factor for LRFD design.

1.9 Earth pressure and surcharge loads

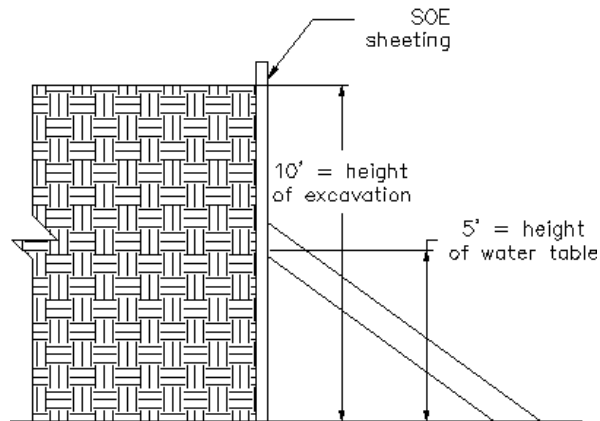
[Click here for solutions 1.9](#)

65. A three story building shown below has a 10 ft deep basement parking level. 12 inch thick reinforced concrete foundation walls extend from the mat to the 1st floor slab level. The soil retained by the foundation wall is well graded gravel-sand mixes. Calculate the total lateral load exerted by the soil on the East foundation wall. The total length of the East wall is 40 feet. Water table can be assumed below the foundation mat.



- (A) 60 kips
(B) 90 kips
(C) 120 kips
(D) 150 kips
66. When designing support of excavation systems in a site with varying soil deposits, which of the following statements is correct to account for expansive pressure of soil?
- (A) Additional pressure due to swelling should be considered for all types of soil due to varying porosity,
(B) Clayey soils are more susceptible to swelling due to varying moisture content during the course of excavation,
(C) Sandy soils are more susceptible to swelling due to varying moisture content during the course of excavation,
(D) Lateral soil pressure can be assumed at optimal moisture content if expansive soils are present at site.

67. What is the total horizontal force acting per unit length of the SOE sheeting as shown in the figure below? The height of excavation is 10ft and the water table is at 5ft. The active earth pressure coefficient is 0.33. The dry density of the soil is 110 pcf and the submerged density is 65 pcf.



- (A) 1.1 kips/ft
- (B) 1.3 kips/ft
- (C) 1.9 kips/ft
- (D) 2.4 kips/ft

68. What is the total lateral stress acting on the retaining wall below with uniform surcharge 50psf?

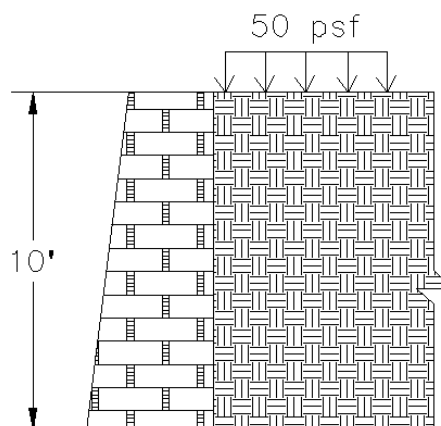
Density, $\gamma = 115$ pcf

Angle of internal friction, $\phi=30$ degree

Cohesion $c=0$

Depth of water table is below the wall

Neglect Wall friction



- (A) 1150.5 lb/ft
- (B) 2081.25 lb/ft
- (C) 2120.25 lb/ft
- (D) 3150.4 lb/ft

69. A reinforced concrete retaining wall has 12ft backfill on one side and 3ft backfill on the other. What is the factor of safety against sliding?

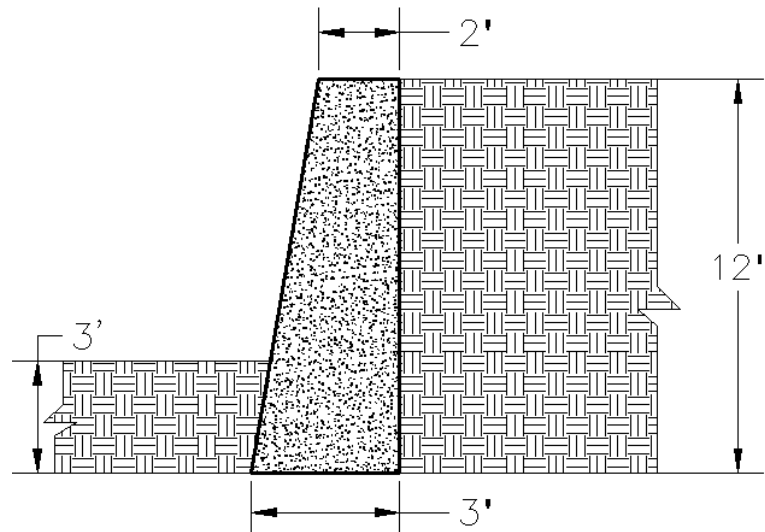
Density of soil, $\gamma = 115$ pcf

Density of concrete, $\gamma_c = 150$ pcf

Active earth pressure coefficient
= 0.33

Passive earth pressure
coefficient = 3

Coefficient of friction between
concrete and
soil = 0.4



- (A) 0.9
(B) 1
(C) 1.2
(D) 1.5

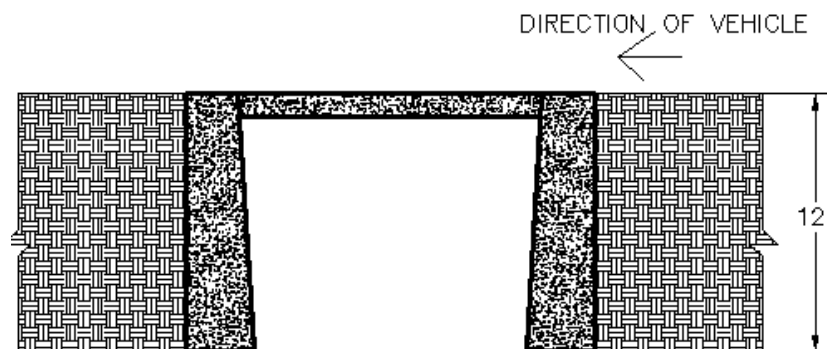
70. The height of an abutment wall supporting a bridge deck is 12ft. The abutment has backfill on one side as shown below. What is the additional lateral load due to vehicular load on top of the embankment? Calculate live load surcharge per AASHTO LRFD provisions.

Active earth pressure coefficient = 0.33

Earth pressure coefficient at rest = 0.5

Passive earth pressure coefficient = 3

Density of soil, $\gamma = 110$ pcf



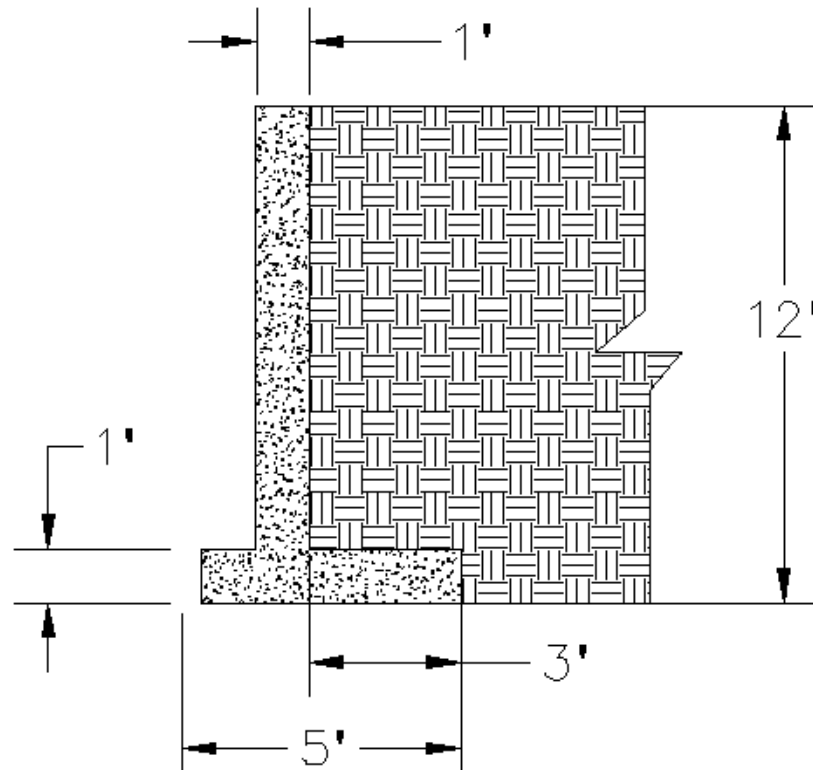
- (A) 1288 lb/ft
(B) 1848 lb/ft
(C) 2048 lb/ft
(D) 2288 lb/ft

71. A reinforced concrete retaining wall shown below has backfill on one side of the wall.
Calculate the factor of safety against overturning.

Density of soil, $\gamma = 115$ pcf

Density of concrete, $\gamma_c = 150$ pcf

Active earth pressure coefficient = 0.33



- (A) 1.2
(B) 1.4
(C) 1.6
(D) 1.8
72. A long line load of intensity 150 lb/ft acts on top of an embankment, parallel to a retaining wall. The line load acts at a distance 5ft from the face of the wall. The total depth of the wall is 20ft. What is the intensity of horizontal pressure on the wall acting 4ft below the top of the embankment due to this surcharge load per AASHTO LRFD specifications?

- (A) 10.13 lb/ft²
(B) 11.38 lb/ft²
(C) 12.88 lb/ft²
(D) 14.69 lb/ft²

73. Which of the following is not correct about lateral earth pressure?

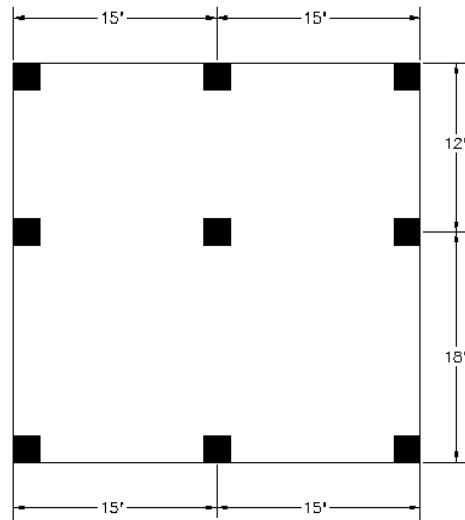
- (A) Lateral earth pressure coefficient is the ratio of horizontal stress to vertical stress exerted by the soil,
- (B) The strain required to achieve passive pressure is significantly larger than the strain required to reach active pressure,
- (C) Active earth pressure cannot be negative regardless of properties of soil,
- (D) The lateral pressure exerted by soil retained by bridge abutments connected to the superstructure deck above is calculated by using the coefficient of earth pressure at rest.

1.10 Tributary areas and load paths

[Click here for solutions 1.10](#)

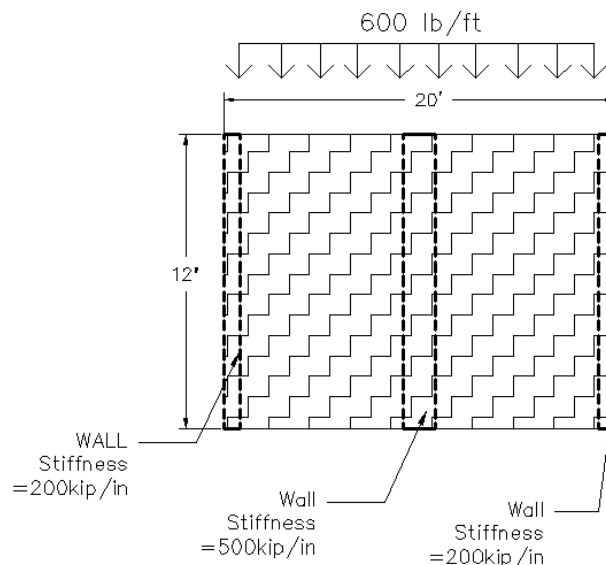
74. The plan below shows a typical framing plan of a three story building. The dead load on each floor is 60psf and the superimposed dead load is 30 psf. The live load on each floor is 50 psf. What is the total unfactored gravity load acting on the interior column at foundation level?

- (A) 55 kips
- (B) 82.5 kips
- (C) 94.5 kips
- (D) 115 kips

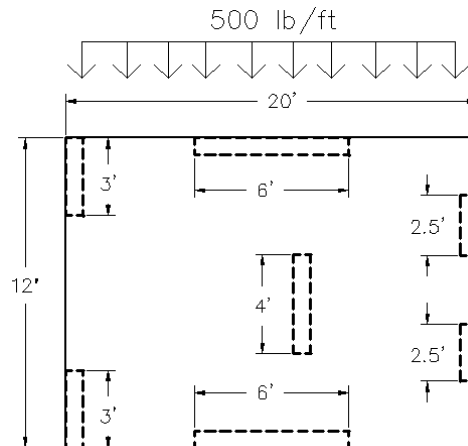


75. The plan shown below is a light gauge steel deck floor and is supported by three walls underneath. The stiffness of the walls supporting the floor is indicated in the figure. A lateral load of 600 lb/ft is applied on the diaphragm. What is the shear load per unit length acting on the central wall?

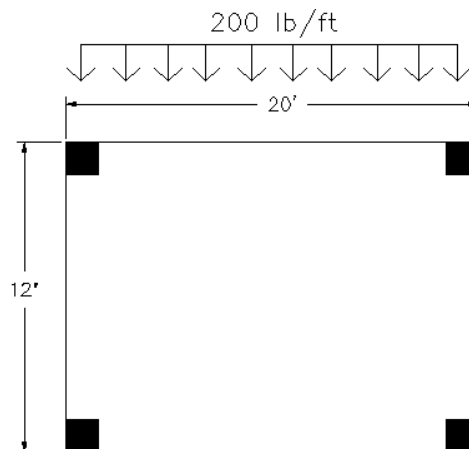
- (A) 500 lb/ft
- (B) 600 lb/ft
- (C) 666.66 lb/ft
- (D) 700 lb/ft



76. The plan shown below is a reinforced concrete floor and is supported by shear walls underneath. The in plane stiffness of the walls supporting the floor is directly proportional to the length of the wall. Out of plane stiffness can be neglected. The center of mass and center of eccentricity coincides with each other and any accidental eccentricity can be neglected. What is the shear load acting on the center wall?

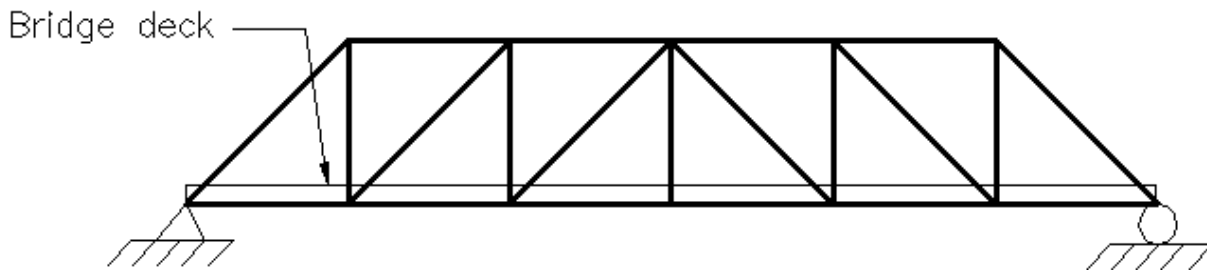


- (A) 2000 lbs
 (B) 2666.67 lbs
 (C) 3000 lbs
 (D) 3333.33 lbs
77. The plan shown below is a reinforced concrete floor and is supported by columns underneath. A lateral load of 200 lb/ft acts on the floor. What is the chord force in the diaphragm?



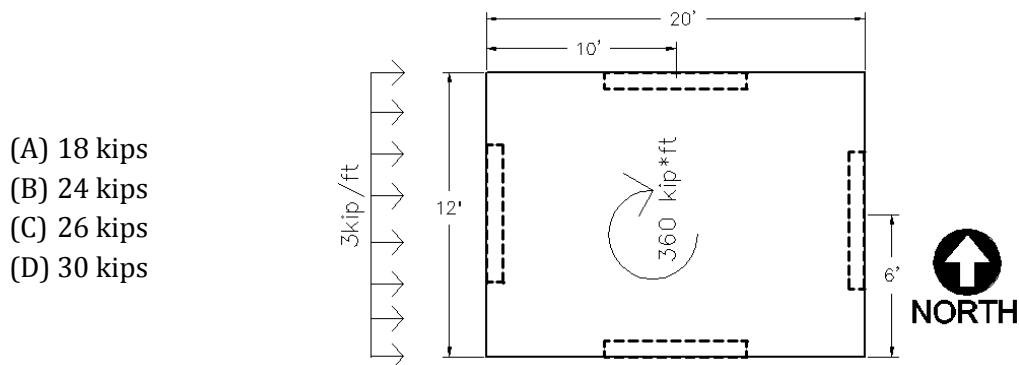
- (A) 200 lbs
 (B) 533.33 lbs
 (C) 833.33 lbs
 (D) 1000 lbs

78. The elevation of a single span simply supported truss bridge is shown below. When a car passes through the deck, what is the correct distribution of forces in the members of the truss? Assume lateral loads are not acting.



- (A) Diagonals in Tension, vertical members in compression, upper chord in compression and bottom chord in tension,
- (B) Diagonals in Compression, vertical members in tension, upper chord in compression and bottom chord in tension,
- (C) Diagonals in compression, vertical members in tension, upper chord in tension and bottom chord in compression,
- (D) Diagonals in Tension, vertical members in compression, upper chord in tension and bottom chord in compression.

79. The rigid diaphragm shown below has a lateral load and a torsional moment acting at the center. The center of rigidity is at the geometric center of the diaphragm. The in-plane stiffness of each wall is equal. What is the maximum shear load on an individual wall?



- (A) 18 kips
- (B) 24 kips
- (C) 26 kips
- (D) 30 kips

1.11 Load combinations

[Click here for solutions 1.11](#)

80. The maximum unfactored axial loads on a column supporting a roof per ASCE 7-16 are as follows:

- a. Dead Load = 25 kips
- b. Load from an equipment = 5 kips
- c. Roof live load = 10 kips
- d. Snow load = 8 kips
- e. Wind load = 15 kips

What is the governing factored load for strength design?

- (A) 41.5 kips
- (B) 50 kips
- (C) 55 kips
- (D) 62 kips

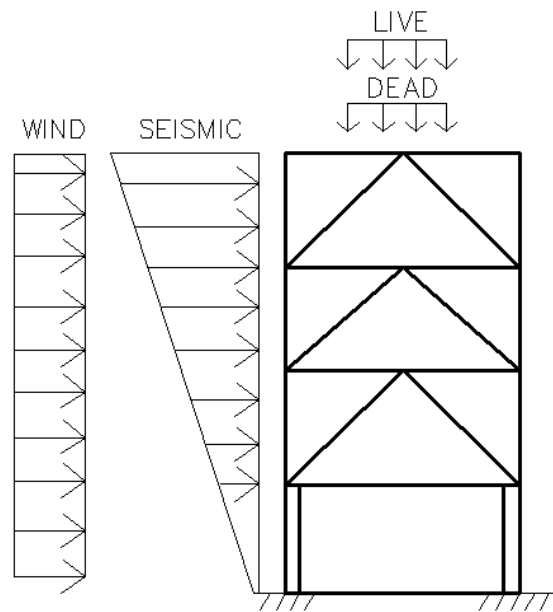
81. The maximum bending moments on a roof beam due to different load cases per ASCE 7-16 are as follows:

- a. Loads from slab and parapet wall = 40 kip*ft
- b. Load from a water tank = 75 kip*ft
- c. Roof live load = 5 kip*ft
- d. Rain load = 10 kip*ft
- e. Snow load = 8 kip*ft
- f. Vertical seismic load = 5 kip*ft
- g. Horizontal seismic load = 25 kip*ft

What is the governing moment for allowable stress design?

- (A) 112 kip*ft
- (B) 125 kip*ft
- (C) 136 kip*ft
- (D) 137 kip*ft

82. For the building shown below, which of the following load combinations must be considered to calculate the governing factor of safety for stability against overturning?



- (A) Dead + Live + Wind
- (B) Dead + Seismic + Wind
- (C) Dead + Seismic or Wind
- (D) Dead + Live + Seismic + Wind

83. Which of the following is not true about load combinations and associated limit states in AASHTO LRFD bridge design specifications?

- (A) For less important bridges, the load factors recommended by chapter 3 can be scaled down.
- (B) For design using extreme event combinations, inelastic behavior which may restrict operation is not permitted
- (C) Typically, 'Strength IV' combination governs over 'Strength I' if the dead load effects are at least two times the live load effects,
- (D) The design forces for jacking in service shall not be less than 1.3 times the permanent load reaction at the bearing, adjacent to the point of jacking

84. The load effects on the main girder of a single lane bridge are as follows:

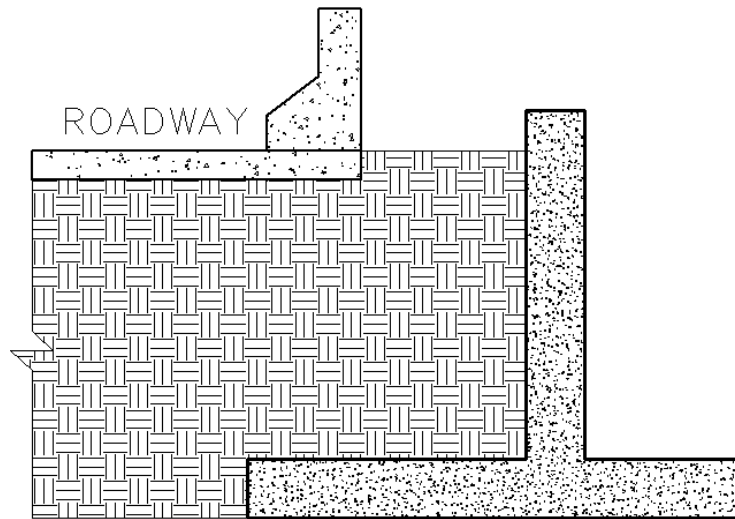
- a. Service dead load moment = 800 kip*ft
- b. Service live load moment due to design truck = 250 kip*ft
- c. Service live load moment due to lane load = 300 kip*ft

Force effects due to temperature, creep, settlement, water load and friction can be neglected,

What is the maximum factored moment acting on the girder per Strength I combination?

- (A) 2150 kip*ft
- (B) 2328 kip*ft
- (C) 2785 kip*ft
- (D) 2966 kip*ft

85. Partial section of an elevated roadway made on an embankment is shown below. The moment due to the back fill's unfactored active earth pressure on the wall is 10 kip*ft/ft. The moment due to live load surcharge from a vehicle travelling parallel to the wall is 5.5 kip*ft/ft. Moment due to wind load on the exposed stem of the wall above the backfill is 1.5 kip*ft/ft



What is the maximum factored moment acting on the wall per Strength V combination?

- (A) 20 kip*ft/ft
- (B) 22 kip*ft/ft
- (C) 24 kip*ft/ft
- (D) 26 kip*ft/ft

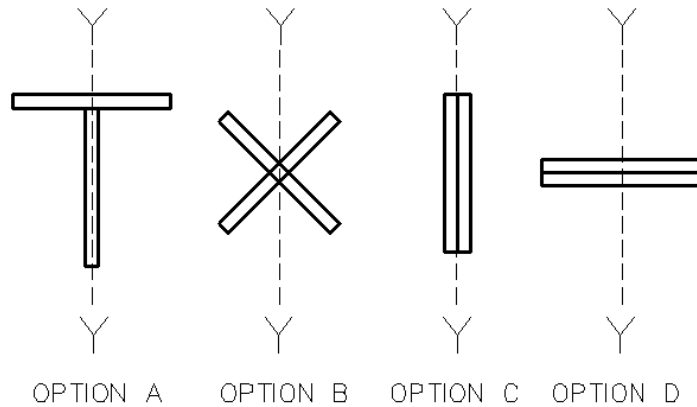
2. Analysis of Structures–Forces and Load Effects

Sub- Headings	Number of Problems
Diagrams	86-103
Axial	104-123
Shear	124-140
Flexure	141-158
Combined stresses	159-174
Deflection	175-193
Special topics	194-207
Total number of Problems	122

2.1 Diagrams

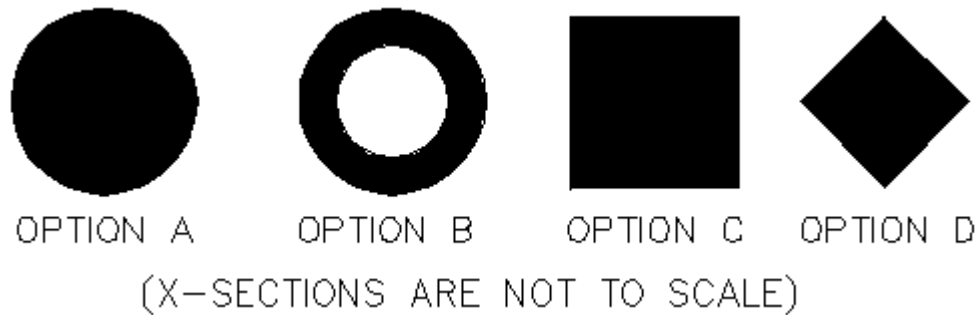
[Click here for solutions 2.1](#)

86. Two 1 inch by 12 in plates are used to form a cross section. The four shapes formed are shown as follows



Which shape will have the highest flexural stiffness about Y-Y axis?

- (A) Option A
 - (B) Option B
 - (C) Option C
 - (D) Option D
87. The cross-sectional areas of each of the shapes shown below are equal.



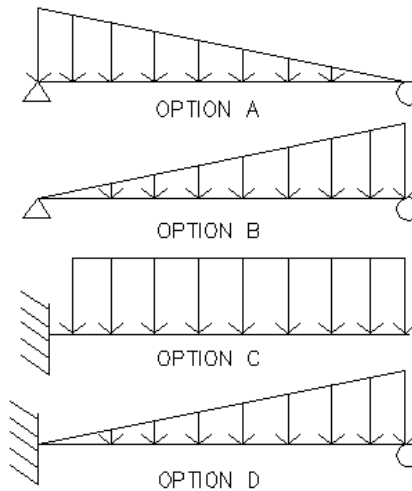
Which shape is best suited for torsional applications?

- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

88. For a single span beam, the equation for bending moment diagram is given as

$$M = 13 + 27x - \frac{2x^3}{3} \text{ where "x" is the distance from the left support.}$$

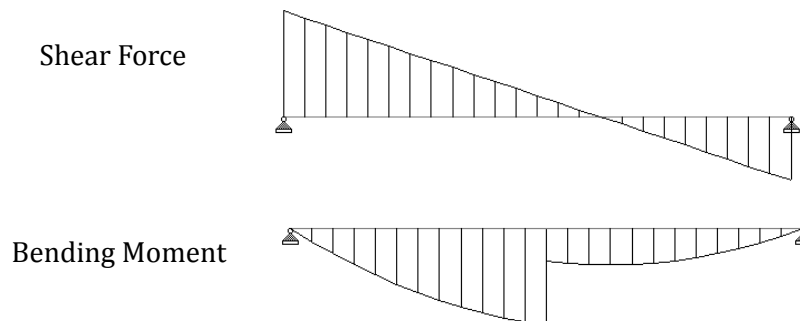
Which of the following diagrams describes the beam and the loading?



- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

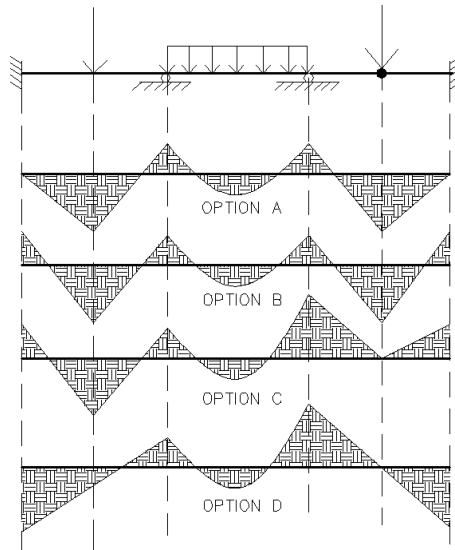
89. For a simply supported beam, the shear force diagram and the bending moment diagram are shown below. Which of the following best describes the loading condition in the beam?

Moment diagram is drawn on the tension side of the beam.



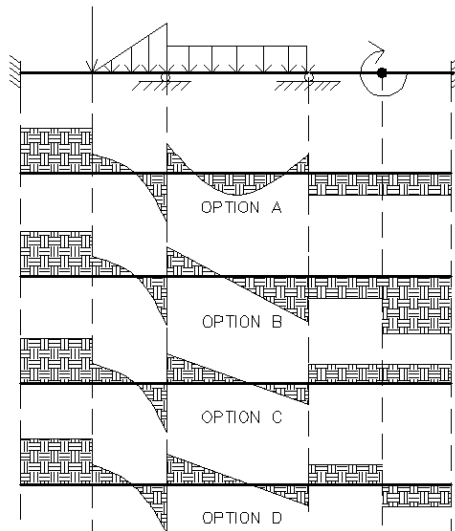
- (A) Uniformly distributed load with a concentrated load at the midspan
- (B) Uniformly distributed load with anti-clockwise moment at the midspan
- (C) Uniformly distributed load with clockwise moment at the midspan
- (D) Uniformly varying load with a couple at the midspan

90. For the given three span beam with a hinge at the center of the third span, what is the correct bending moment diagram?



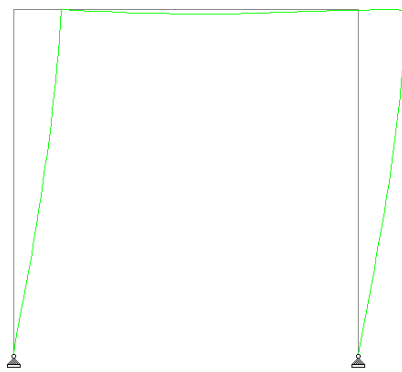
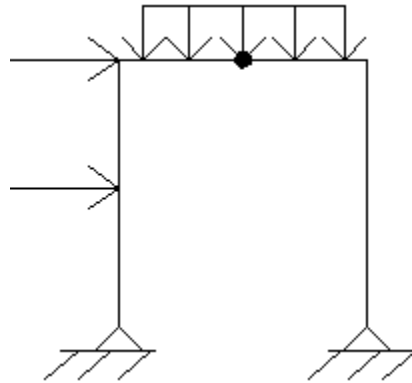
- (A) Option A
(B) Option B
(C) Option C
(D) Option D

91. For the given three span beam with a hinge at the center of the third span, what is the correct shear force diagram?

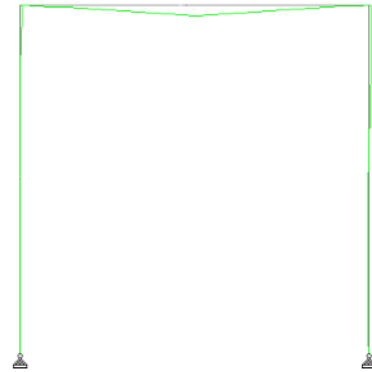


- (A) Option A
(B) Option B
(C) Option C
(D) Option D

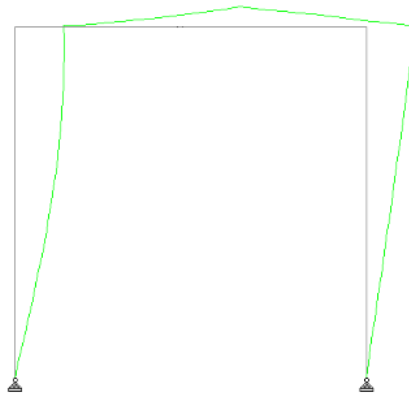
92. For the given frame with a hinge at the center span of the beam, what is the true deflected shape?



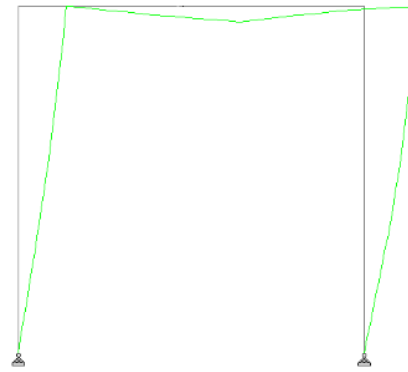
Option A



Option B



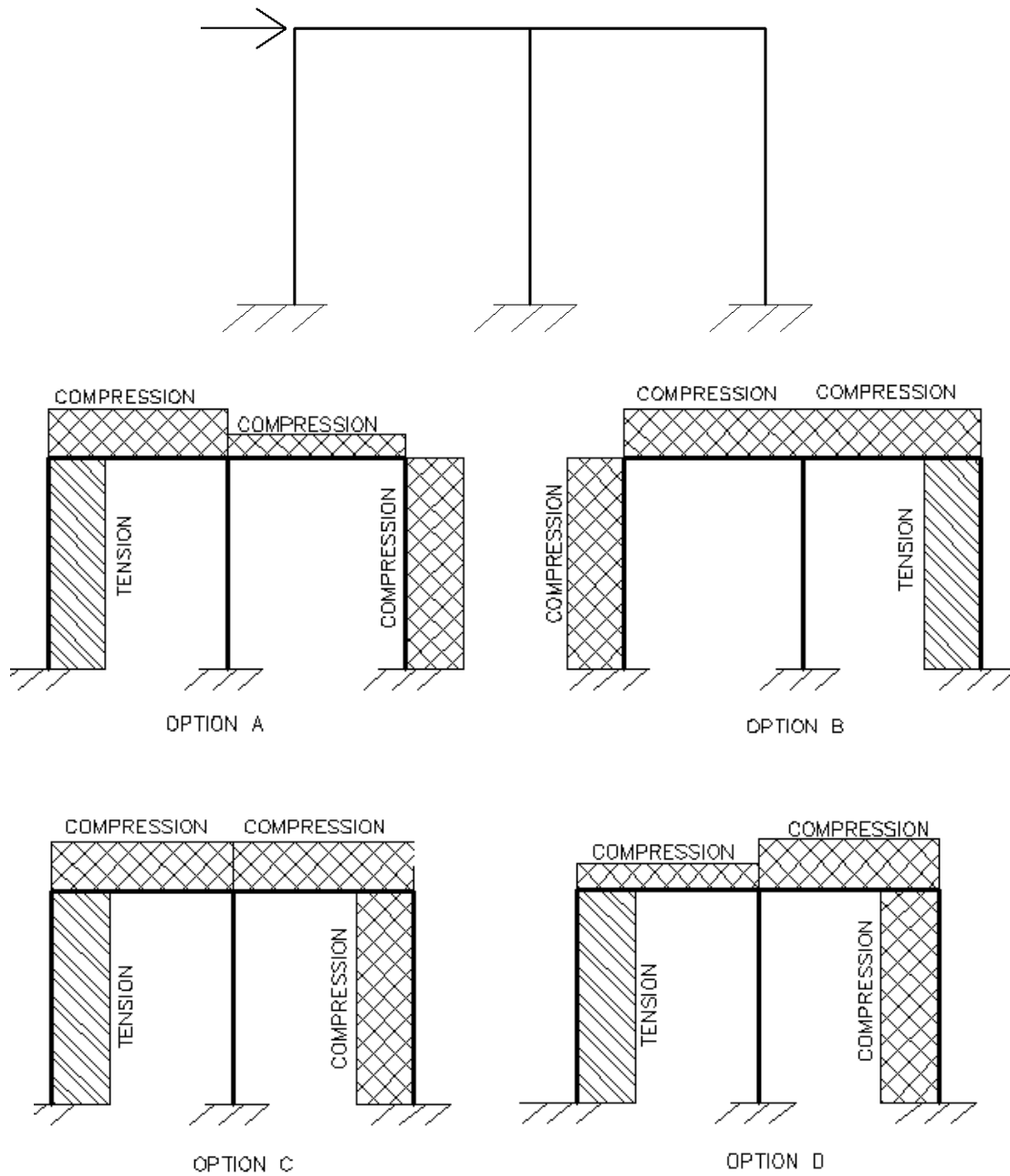
Option C



Option D

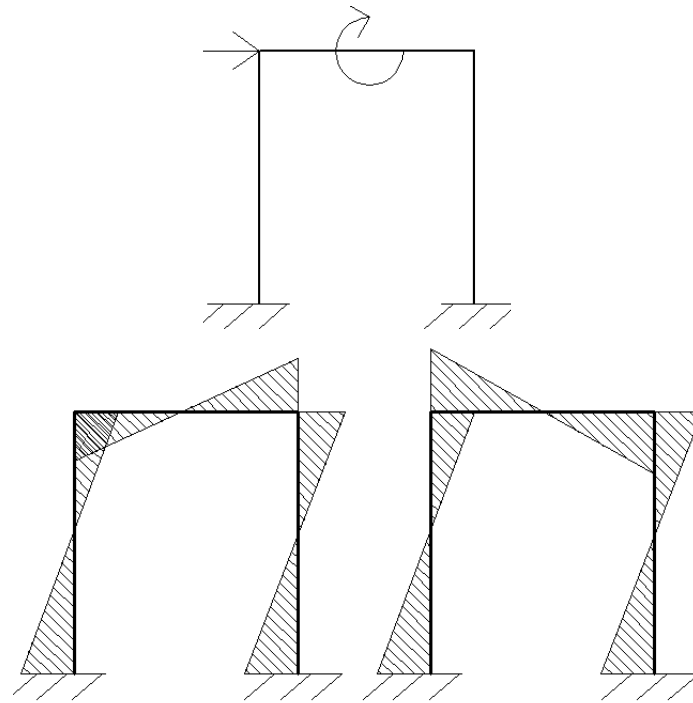
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

93. The given multi bay frame has a lateral load acting on the top. Select the correct axial force diagram.



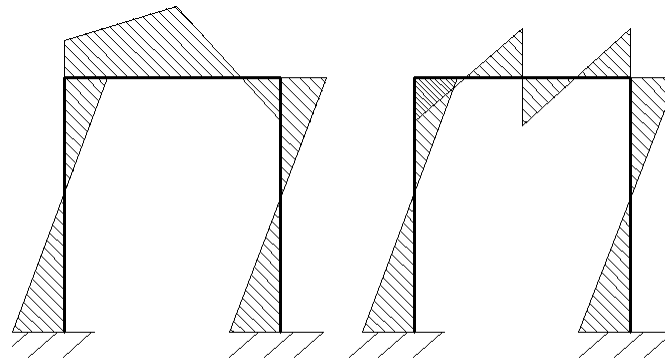
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

94. The frame below has a lateral load acting on the top and an external clockwise moment acting on the center of the beam. Select the correct bending moment diagram.



OPTION A

OPTION B

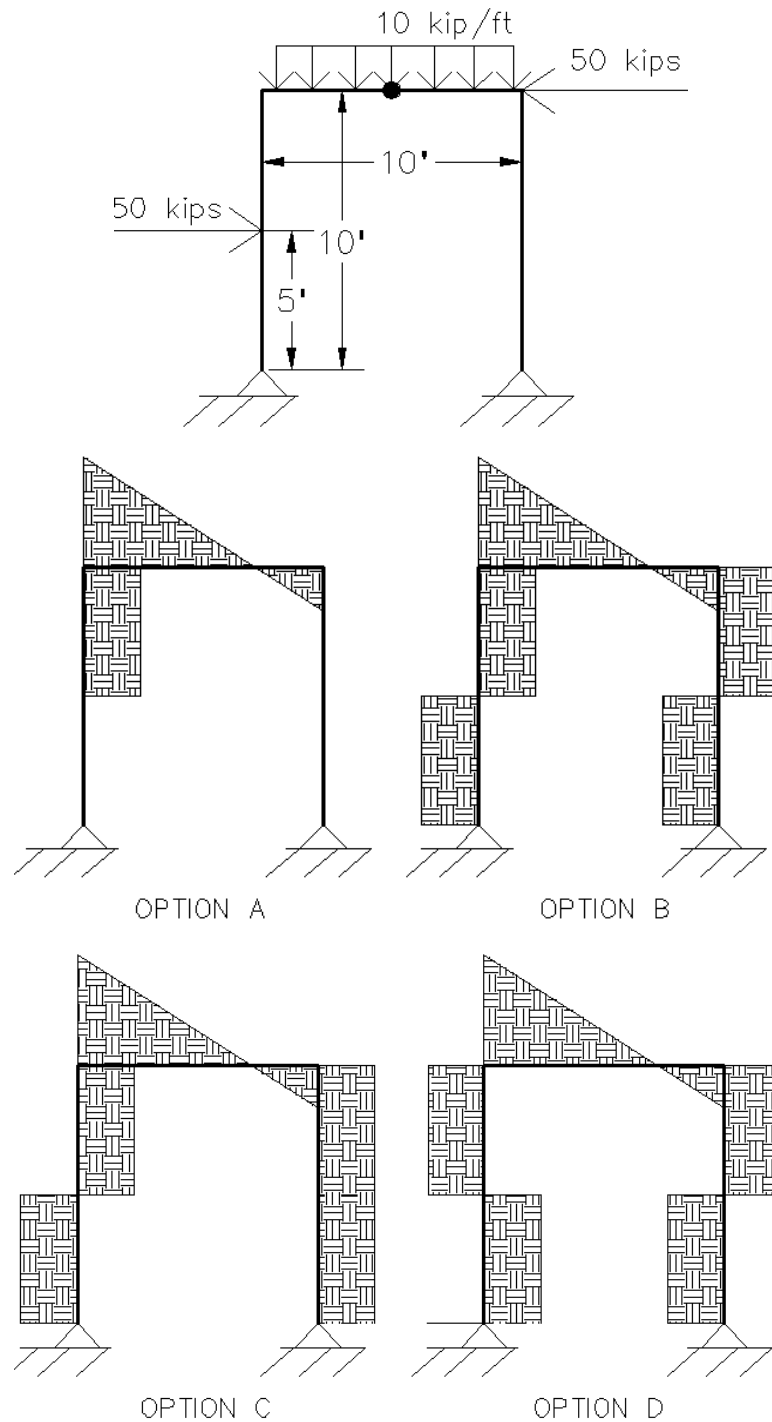


OPTION C

OPTION D

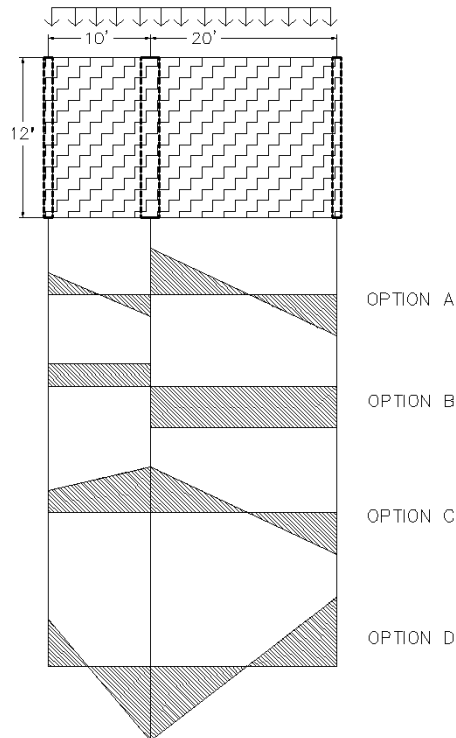
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

95. The frame shown below has a hinge at the center of the beam. Select the correct shear force diagram.



- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

96. A flexible diaphragm is supported by three walls. The center wall is twice as stiff as the outer walls. Select the correct shear force diagram for the diaphragm.



- (A) Option A
(B) Option B
(C) Option C
(D) Option D

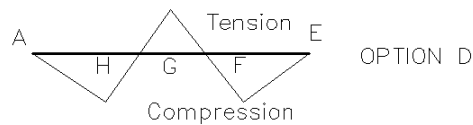
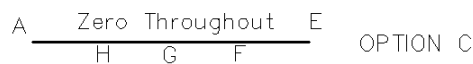
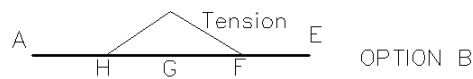
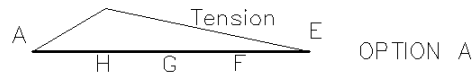
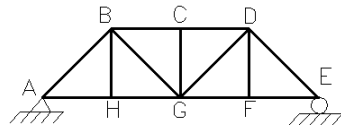
97. Which of the following is not correct about shear force and bending moment diagrams?

- (A) Shear force between two unloaded points in a beam is constant,
- (B) The slope of the shear force diagram at any point gives the intensity of moment at that point,
- (C) The area between the shear force diagram between two points gives the change in moment between those points,
- (D) Maximum bending moment occurs at the point where shear force is zero.

98. Which of the following is not true about influence line diagrams?

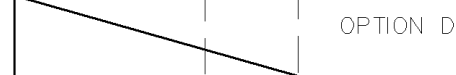
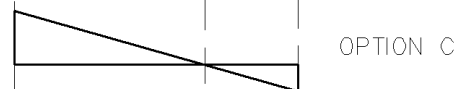
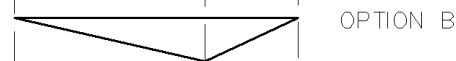
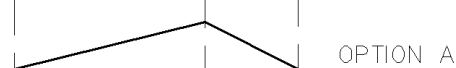
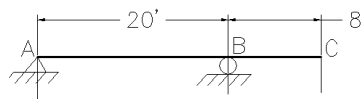
- (A) Influence line diagrams for reactions and internal forces for statically determinate structures are linear,
- (B) The shape of influence line diagram depends upon the system of units in the structure i.e. SI units or US customary units,
- (C) In case of a simply supported beam with overhangs, the influence lines for reactions and internal forces for section between supports should be constructed as for beam without overhangs and extended along the overhangs,
- (D) Mueller – Breslau’s Principle can be used to obtain influence line diagrams for determinate and indeterminate structures.

99. For the truss shown below, select the correct influence line diagram for member CG when a point load passes through A to E



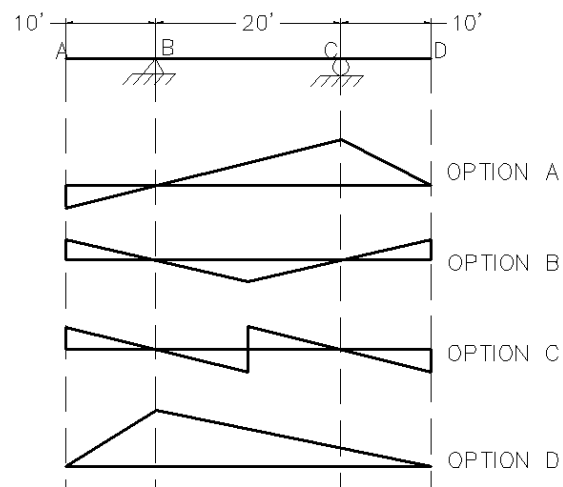
- (A) Option A
(B) Option B
(C) Option C
(D) Option D

100. For the overhanging beam shown below, select the correct influence line diagram for maximum reaction at A, when a point load passes from point A to point C

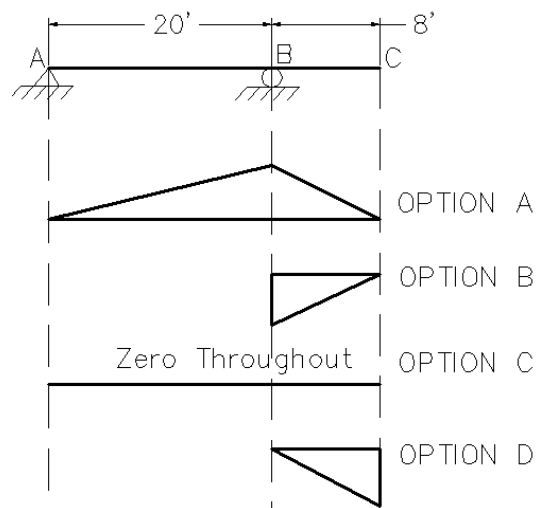


- (A) Option A
(B) Option B
(C) Option C
(D) Option D

- 101.** For the beam shown below, select the correct influence line diagram for maximum shear at midspan, when a concentrated load passes from point A to point D.

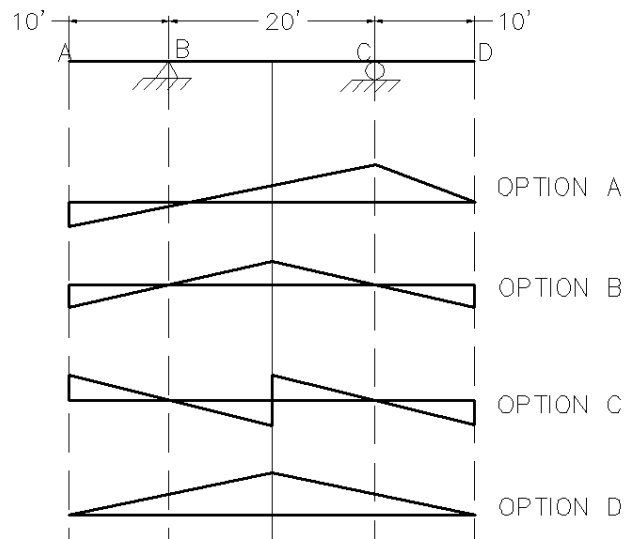


- (A) Option A
(B) Option B
(C) Option C
(D) Option D
- 102.** For the overhanging beam shown below, select the correct influence line diagram for maximum moment at B, when a concentrated load passes from point A to point C



- (A) Option A
(B) Option B
(C) Option C
(D) Option D

103. For the beam shown below, select the correct influence line diagram for maximum moment at midspan, when a concentrated load passes from point A to point D



- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

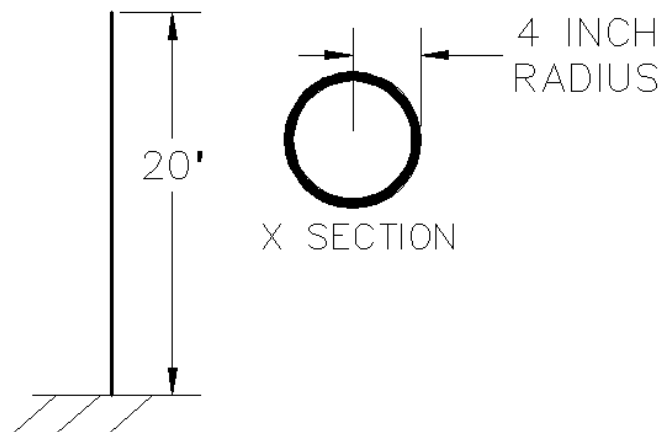
2.2 Axial

[Click here for solutions 2.2](#)

104. Calculate the buckling load for a column that is 8 ft tall. Modulus of elasticity of the material is 29000 ksi . The cross section of the column is square with dimensions 4 inches by 4 inches. The column is free on one end and is pin supported (free to rotate) at the other end.

- (A) 200 kips
- (B) 300 kips
- (C) 400 kips
- (D) Cannot be determined

105. Calculate the critical buckling load for the column shown below. The column is 20 ft tall. Modulus of elasticity of the material is 29000 ksi . The cross section of the column is a circular tube with radius 4 inches and wall thickness 1 inch. The column is free on one end and fixed at the other end



- (A) 150.8 kips
- (B) 170.7 kips
- (C) 300.2 kips
- (D) 682.9 kips

- 106.** A 30 ft long member is subjected to tensile stresses and temperature strains. The material is linearly elastic with $E = 29000$ ksi and is subjected to a tensile load of 200 kips. The length of the material was measured at 32°F . What is the length of the member under the present loading condition at a temperature of 100°F ? Coefficient of thermal expansion can be taken as $0.000007 / ^\circ\text{F}$. The Cross section Area of the member is 20 in^2 .

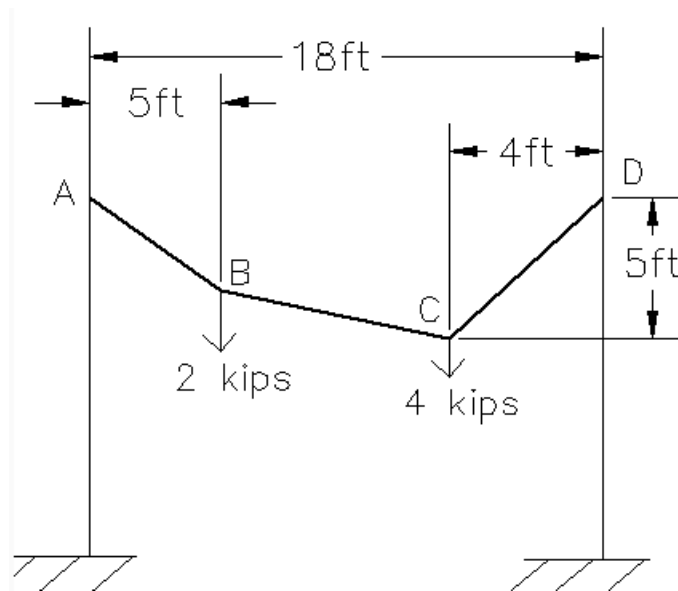
- (A) 0.124 in
- (B) 0.171 in
- (C) 0.295 in
- (D) 0.51 in

- 107.** A 30 ft long beam with cross sectional area 15 in^2 is shown below. The beam is fixed on one end and has a 0.1 inch clearance to the support at the other end to allow for thermal expansion. The beam was installed when the ambient temperature was 30°F . The material is linearly elastic with $E = 29000$ ksi and not subjected to any load. The 30ft length of the beam was measured at 30°F . What is the axial load in the member during summer at a temperature of 100°F ? Coefficient of thermal expansion can be taken as $0.000007 / ^\circ\text{F}$.



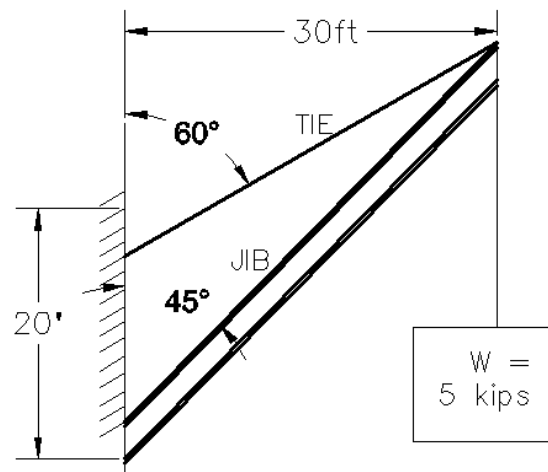
- (A) 0 kips
- (B) 92.3 kips
- (C) 158.3 kips
- (D) 213.15 kips

- 108.** A cable profile is shown below. The cables are supporting concentrated loads at points B and C. The sag at point C is 5ft. What is the total tension in the cable at section AB? The weight of the cable can be neglected.

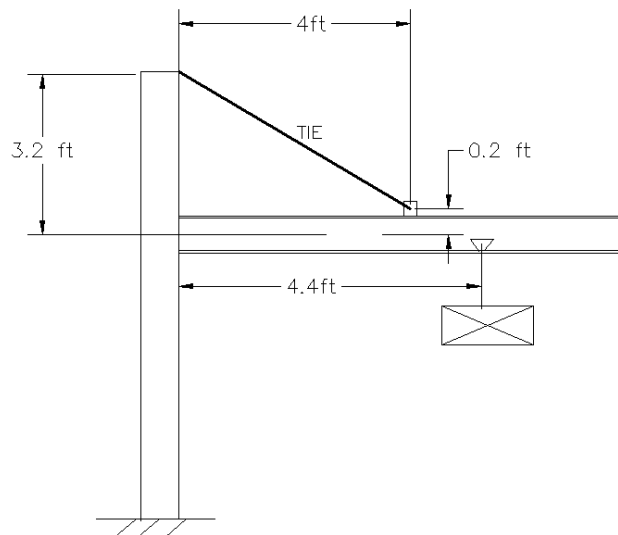


- (A) 2.5 kips
(B) 3.75 kips
(C) 4.5 kips
(D) 5 kips
- 109.** A cable is suspended between two points that are 50 ft high. The only load that is acting in the cable is its own weight. What is the shape of the cable?
- (A) Parabolic Curve
(B) Linear
(C) Catenary curve
(D) Circular arc
- 110.** A cable is suspended between two points and is subjected to a uniformly distributed load of 200 lb/ft. The cable profile is parabolic. Which point in the cable is subjected to the lowest tension?
- (A) Support points
(B) Mid span at lowest sag point
(C) Quarter spans
(D) Cannot be determined

- 111.** The configuration of a maintenance unit crane is as shown below. The pick weight is 5 kips. Both the tie and the jib are rigidly connected to a building wall. What is the axial force acting on the tie?

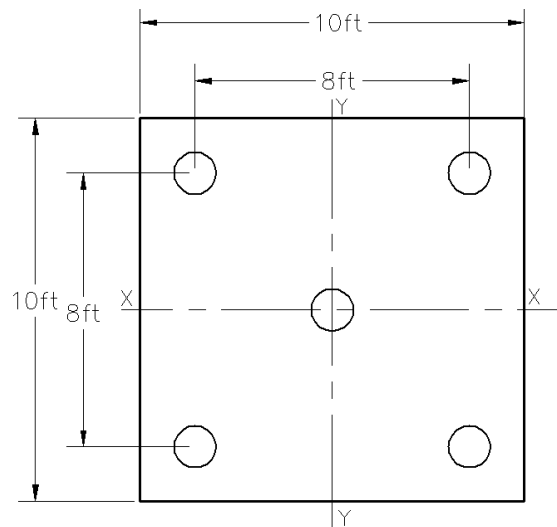


- (A) 8.66 kips
 (B) 11.23 kips
 (C) 13.66 kips
 (D) 16.73 kips
- 112.** For the configuration below, the weight of the pick is 500 lbs. The tie is offset from the center of the beam by 0.2 ft. What is the maximum tension in the tie? Self weight of the jib can be neglected.

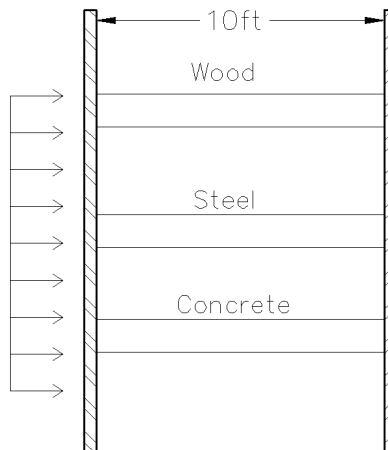


- (A) 820 lbs
 (B) 860 lbs
 (C) 900 lbs
 (D) 920 lbs

- 113.** The plan below shows a rigid concrete pile cap with 5 piles underneath. The maximum axial load acting on the pile cap is 2000 kips and the maximum moments are $M_{xx} = 500$ kip*ft and $M_{yy} = 300$ kip*ft. What is the maximum axial load acting on an individual pile?

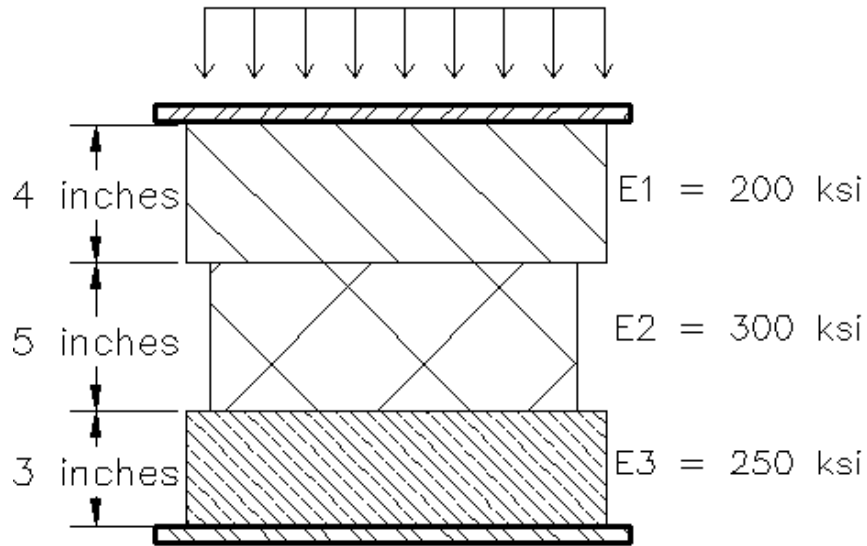


- (A) 400 kips
(B) 450 kips
(C) 500 kips
(D) 600 kips
- 114.** Three horizontal braces are arranged as shown below. The modulus of elasticity of wood is 1500 ksi, modulus of elasticity of concrete is 3500 ksi and modulus of elasticity of steel is 29000 ksi. What is the total horizontal axial stiffness of the system? Cross section area of each brace is 15 in^2 .



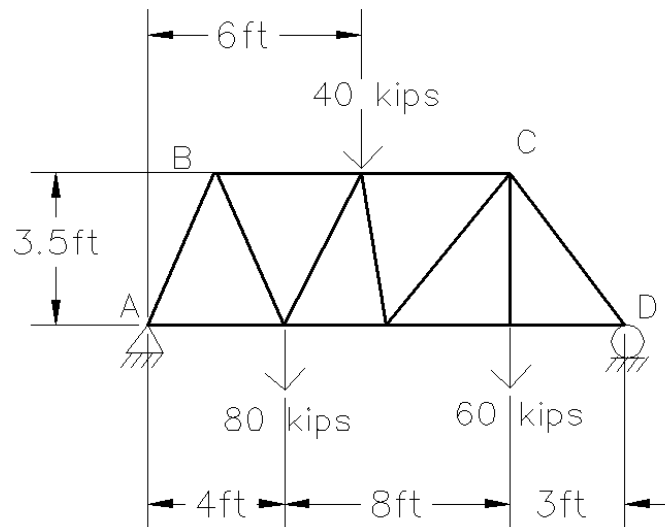
- (A) 3050 kip/in
(B) 3450 kip/in
(C) 4000.75 kip/in
(D) 4250 kip/in

- 115.** A bearing system is developed using three materials as shown below. The cross sectional areas of the top and bottom layers are 20 in^2 and the middle layer is 16 in^2 . What is the total axial stiffness of the bearing system?

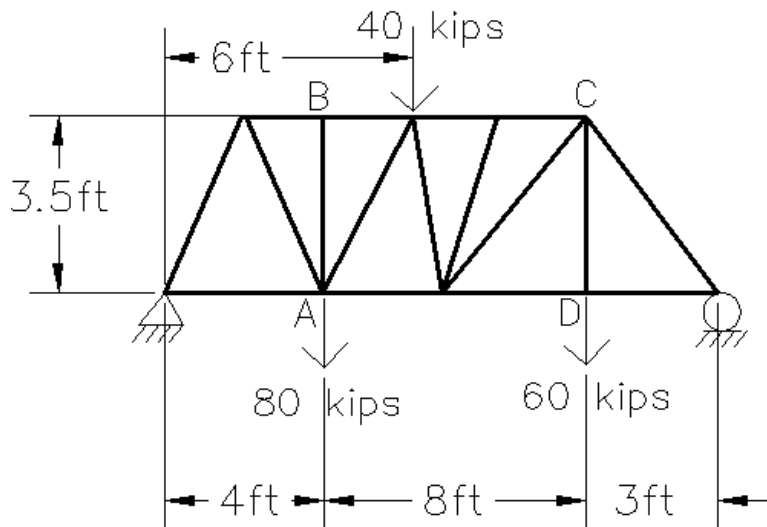


- (A) 378.5 kip/in
(B) 588.5 kip/in
(C) 1000.75 kip/in
(D) 3626 kip/in
- 116.** For axial load analysis, what is the primary assumption made in Euler's buckling theory to calculate failure load in slender columns?
- (A) The self weight is considered for heavier columns,
(B) The material of the column undergoes non linear behavior,
(C) The column experiences considerable axial and shear deformations,
(D) The column is subjected to purely axial load directly along the column axis.

117. For the simply supported truss shown below, calculate the force acting in member CD.

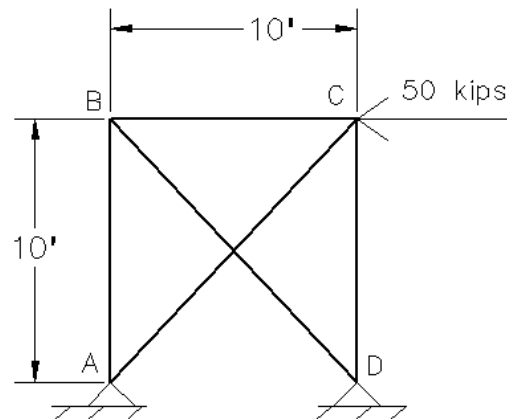


- (A) 112 kips (compression)
 (B) 112 kips (tension)
 (C) 131 kips (compression)
 (D) 131 kips (tension)
118. For the simply supported truss shown below, calculate the force acting in members AB and CD.

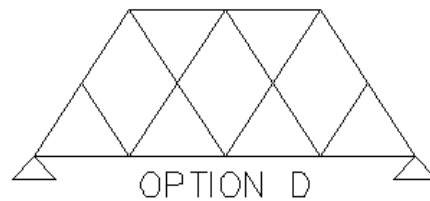
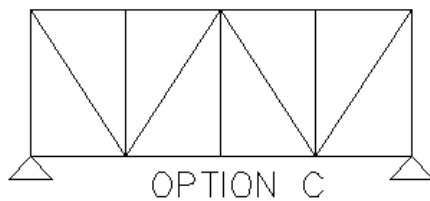
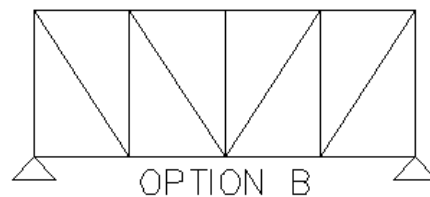
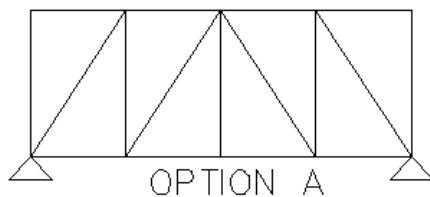


- (A) AB = 0 CD = 0
 (B) AB = 80 kips CD = 60 kips
 (C) AB = 0 kips CD = 60 kips
 (D) AB = 80 kips CD = 0 kips

- 119.** The pin connected frame shown below has a wind load of 50 kips acting on the top. The frame is braced with two threaded tie rods AC and BD forming an X bracing. Calculate the maximum axial load on the braces.

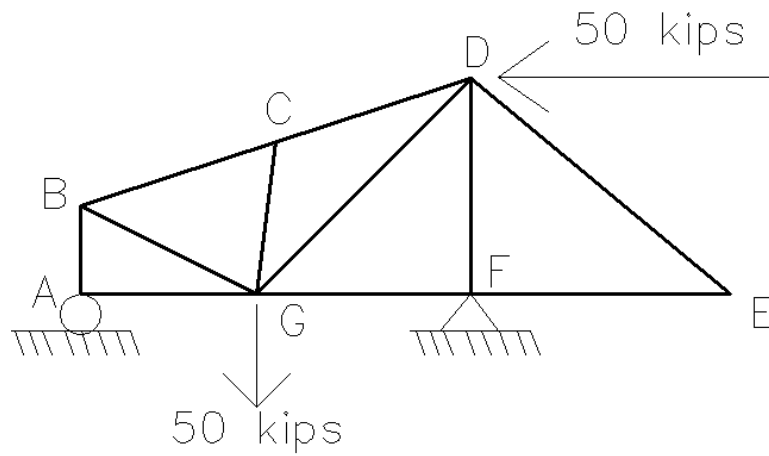


- (A) 35.35 kips
 (B) 50 kips
 (C) 70.7 kips
 (D) 89 kips
- 120.** Which of the following truss models have only compressive forces in all diagonal members ?

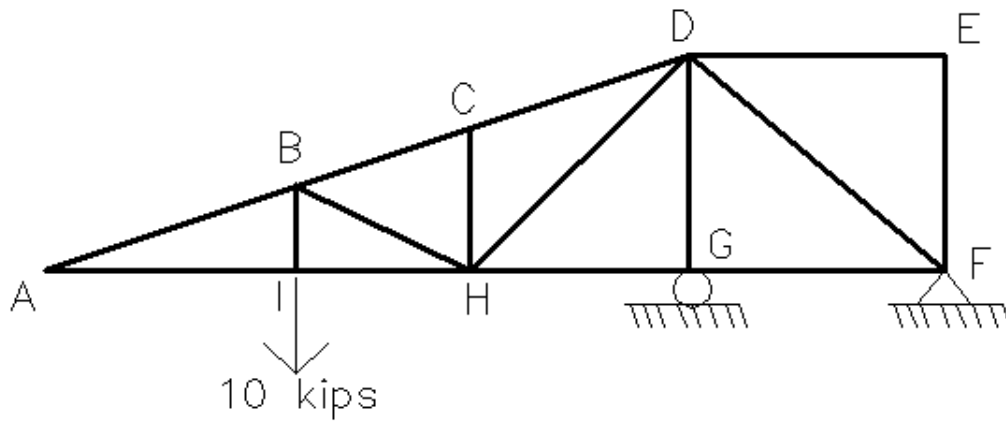


- (A) Option A
 (B) Option B
 (C) Option C
 (D) Option D

- 121.** Find the number of zero force members in the truss shown below. Vertical load is acting on point G and horizontal load is acting on point D.

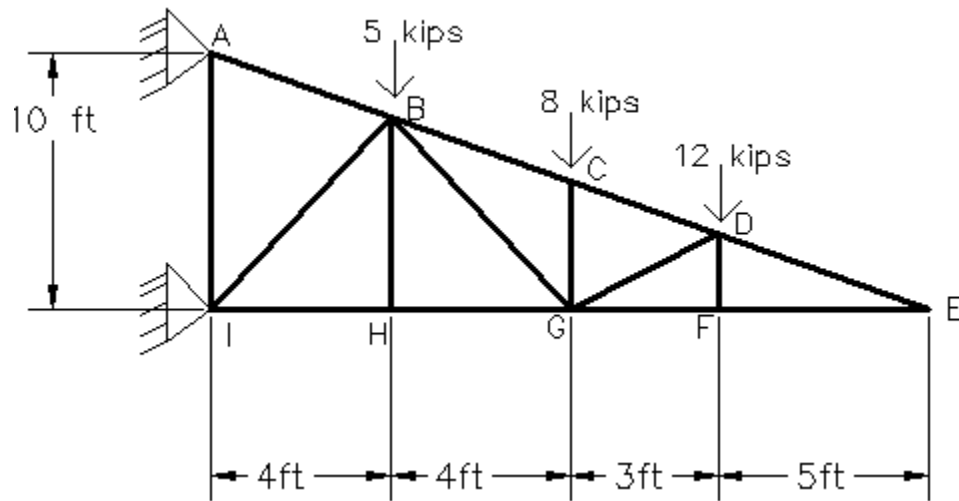


- (A) 2
(B) 3
(C) 4
(D) 5
- 122.** Find the number of zero force members in the truss shown below. Horizontal load is acting on point D.



- (A) 3
(B) 4
(C) 5
(D) 6

123. For the truss shown below, calculate the forces in members BH and DF.

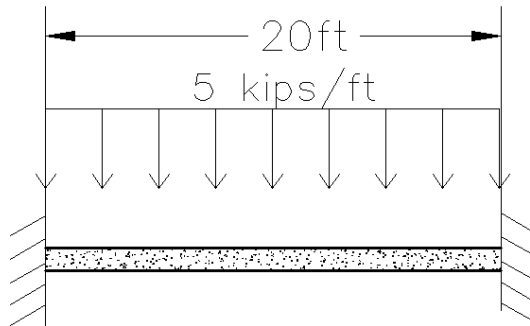


- | | |
|-----------------|--------------|
| (A) BH = 0 | DF = 12 kips |
| (B) BH = 0 | DF = 0 |
| (C) BH = 5 kips | DF = 12 kips |
| (D) BH = 5 kips | DF = 0 |

2.3 Shear

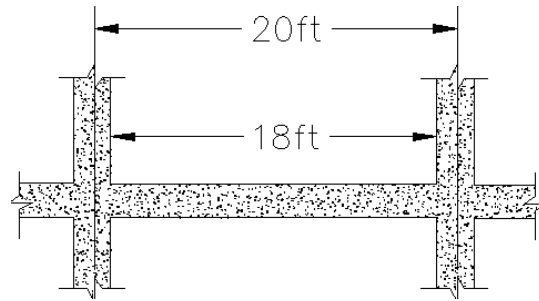
[Click here for solutions 2.3](#)

- 124.** A Reinforced Concrete beam spans between two columns. The span of the beam is 20ft. The only load acting on the beam is a uniform distributed load of 5 kips/ft (factored load) throughout the span. The effective depth 'd' of the beam is 18 inches. What is the critical factored shear force to be considered for design of the beam?

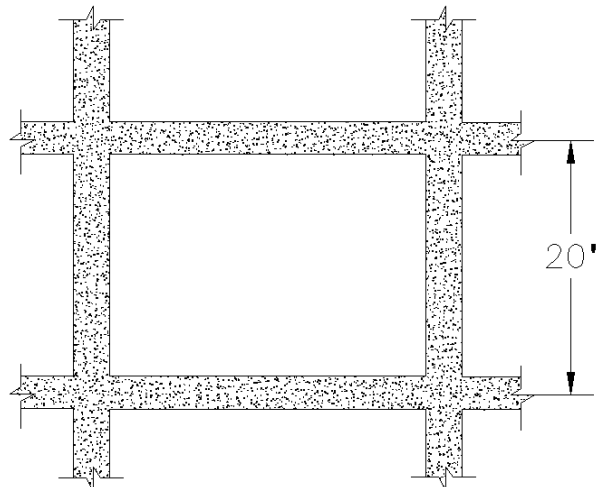


- (A) 40 kips
(B) 42.5 kips
(C) 45 kips
(D) 50 kips
- 125.** Which of the following is not true about shear failure in beams and columns?
- (A) Short beams and columns are more susceptible to shear failure,
(B) In short beams, the shear stress is more concentrated making it more susceptible to fail in shear before it develops significant flexural cracks,
(C) Shear failure may occur in over reinforced sections with high amount of steel,
(D) The critical value of one way shear in a beam is always at a distance 'd' i.e. effective depth of beam from the face of the support.

- 126.** Partial elevation of an interior reinforced concrete beam is shown below. The frame is designed and analyzed as a special moment resisting frame. During sidesway, the maximum probable flexural strengths at the left and right ends are 500 kip*ft and 600 kip*ft. The unfactored uniform dead load acting on the beam is 2 kip/ft. The unfactored live load acting on the beam is 1 kip/ft. What is the factored design shear to be considered for the beams?



- (A) 36 kips
(B) 60 kips
(C) 92 kips
(D) 105 kips
- 127.** The frame below shows partial elevation of beams and columns in an intermediate moment resisting, reinforced concrete system. The center to center vertical distance between the beams i.e. center to center height of columns is 20 ft. The depth of both top and bottom beams is 24 inches. The nominal flexural strength at the top and bottom of the columns is 450 kip*ft and 600 kip*ft. The unfactored shears acting on the columns are as follows:
Dead load shear = 5 kips
Live load shear = 10 kips
Earthquake load shear = 20 kips
What is the factored design shear to be considered for the column?



- (A) 45 kips
(B) 60 kips
(C) 76 kips
(D) 89 kips

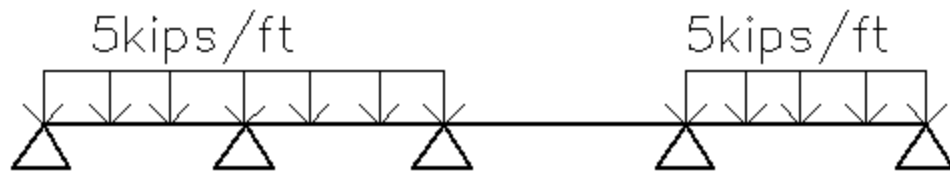
128. The equation of bending moment in at a certain span of a beam system is $M = x^3 - 5x^2 - 3x$, where x is in ft and M is in kip*ft. What is the value of the shear force in that span when $x = 3$ ft?

- (A) -6 kips
- (B) 6 kips
- (C) 27 kips
- (D) -27 kips

129. A beam with a rectangular cross section is subjected to uniformly varying load and multiple point loads along the span. Which of the following best describes the distribution of shear stress in the section?

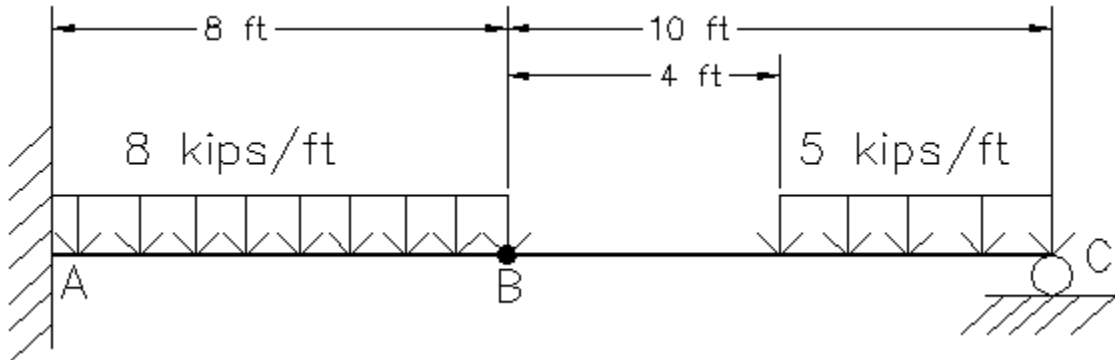
- (A) For a symmetrical rectangular section, the shear stress is uniform along the height of the section,
- (B) Shear stresses are maximum at the top and bottom surfaces and minimum at the neutral axis,
- (C) Shear stresses are maximum at the neutral axis and zero at the top and bottom of the section,
- (D) Shear stress varies linearly from top to bottom of the section.

130. A continuous beam system is shown below. Each of the four spans are equal in length and = 10ft. What is the shear force at a distance 13 ft from the exterior left support?

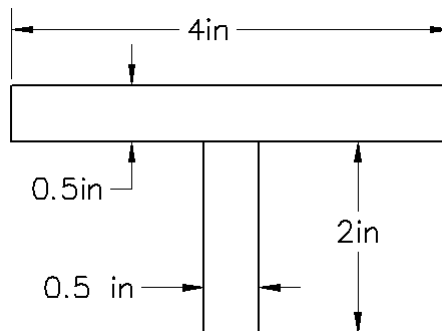


- (A) 10.35 kips
- (B) 15.15 kips
- (C) 20.3 kips
- (D) 30.15 kips

131. For the beam system shown below with a hinge at point B, what is the maximum shear force at hinge location B? Follow general beam sign convention.

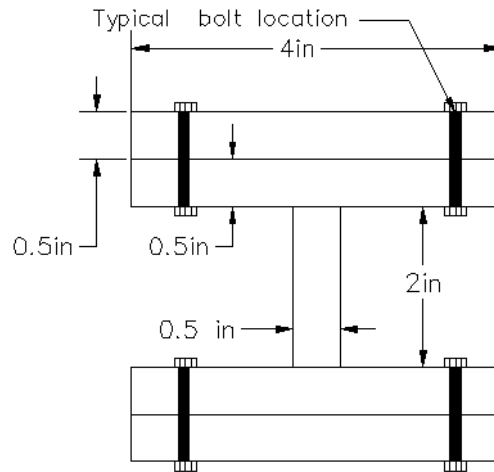


- (A) 0 kips
 (B) -9 kips
 (C) +9 kips
 (D) -29 kips
132. The maximum shear force acting on the T beam section shown below is 2.5 kips. What is the shear stress at the junction of the flange and web?

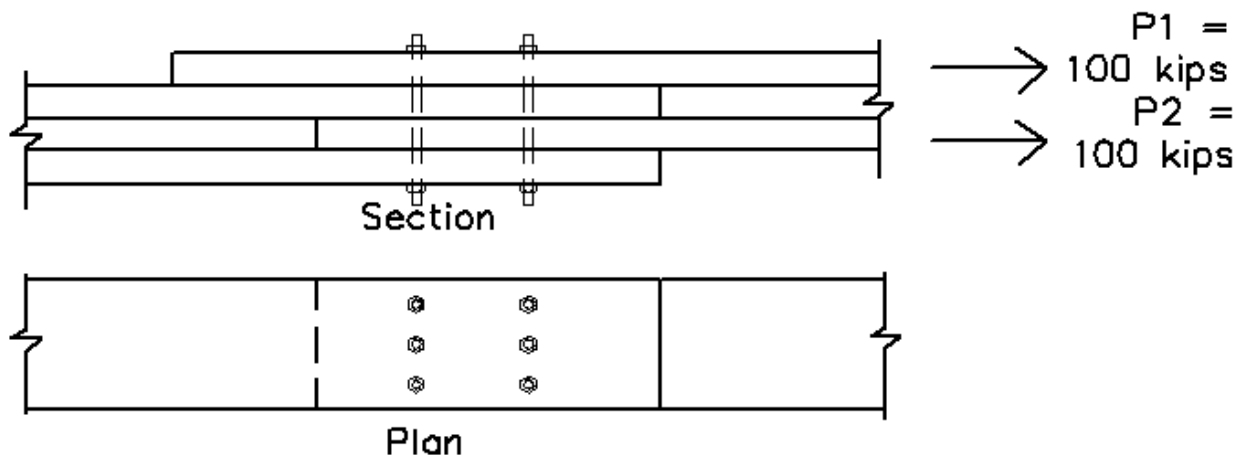


- (A) 1313 psi
 (B) 1978 psi
 (C) 2618 psi
 (D) 2943 psi

133. A built up girder system is shown below. The maximum shear force acting in the section is 15 kips. The top and bottom flange plates are connected to the web by 2 bolts on either side of the web as shown below. What is the maximum shear load on each bolt if the bolts are spaced at 12 inches center to center along the span?

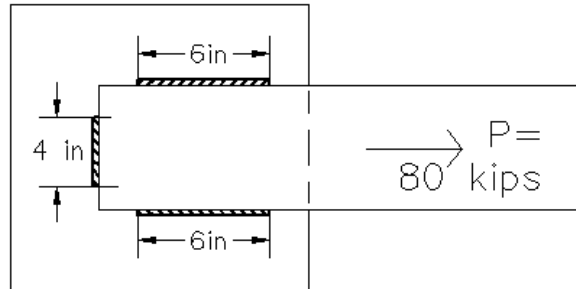


- (A) 12.2 kips
(B) 16.5 kips
(C) 21.6 kips
(D) 24.8 kips
134. The figure below shows the section and plan of a 4 plate connection system. The top plate is subjected to a load of 100 kips and the 3rd plate is also subjected to a load of 100 kips. There are a total of 6 bolts as seen in the plan. What is the maximum nominal shear stress acting on the bolts? The bolts are 1 inch in diameter.



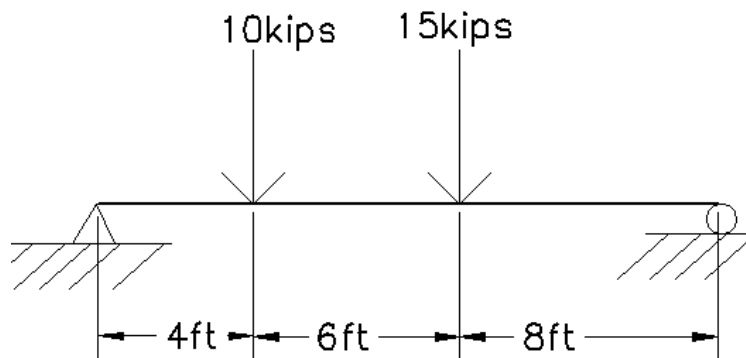
- (A) 21.23 ksi
(B) 42.46 ksi
(C) 63.69 ksi
(D) 84.92 ksi

- 135.** Two plates are welded as shown in the plan below. A load of 80 kips acts in the smaller plate concentric to the centroid of the weld group. What is the shear acting on the welds?

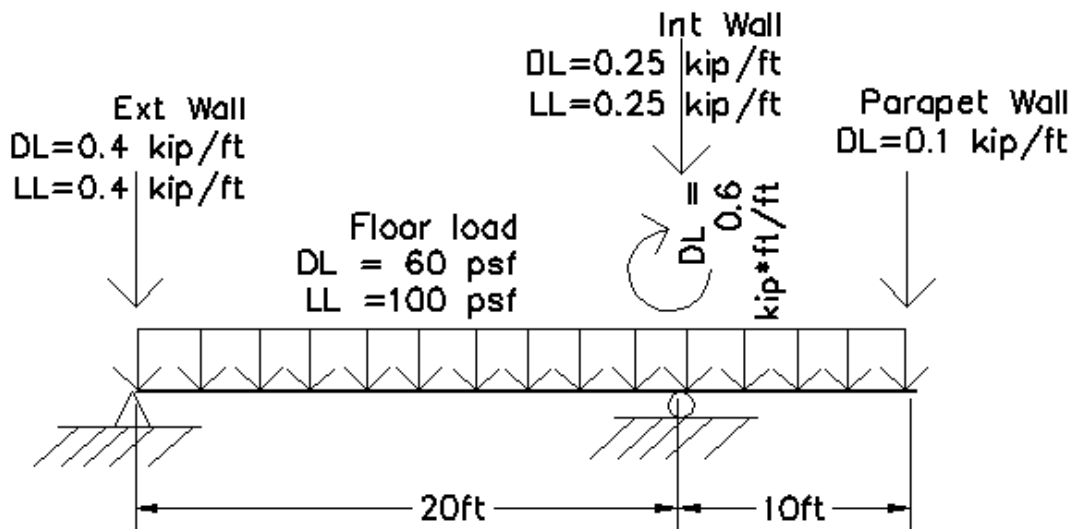


- (A) 3 kip/in
 (B) 4 kip/in
 (C) 4.5 kip/in
 (D) 5 kip /in
- 136.** In a continuous span beam system , point load is applied directly to the shear center of the beam section. What is true about the load effect and internal forces?
- (A) Beam will only be subjected to pure shear,
 (B) Beam will be subjected to shear and torsion,
 (C) Beam will be subjected to bending moments and shear without torsion,
 (D) Net shear displacement will be zero.

137. A simply supported beam subjected to two point loads is shown below. What is the magnitude of equivalent uniformly distributed load that produces the same amount of maximum shear in the beam?

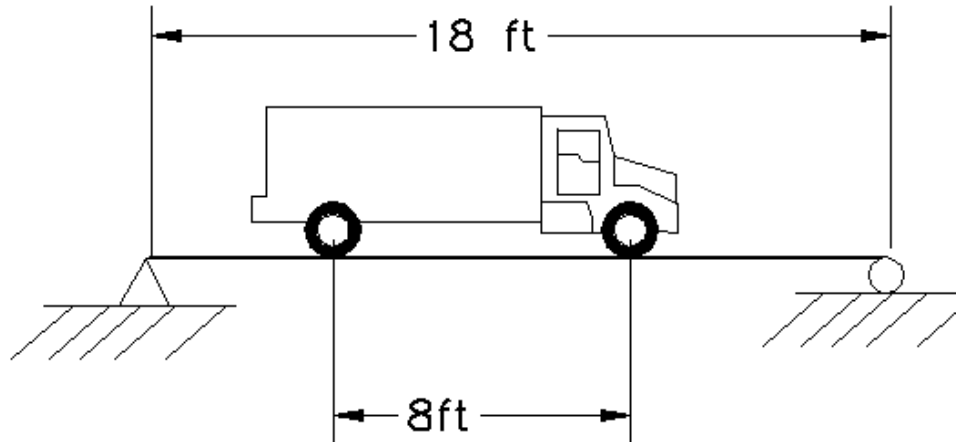


- (A) 1506 lbs/ft
 (B) 1604 lbs/ft
 (C) 2500 lbs/ft
 (D) 2666 lbs/ft
138. Loads acting on a simply supported floor beam is shown below. The floor beams are spaced 8 ft c/c. What is the maximum service load shear acting on the beam ?



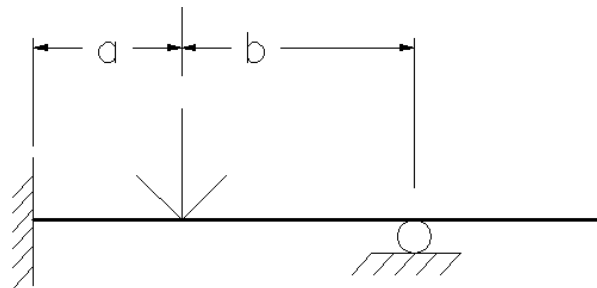
- (A) 2.08 kips
 (B) 8.96 kips
 (C) 15 kips
 (D) 16.64 kips

- 139.** A truck travels along a simply supported beam. The distance between the front and rear axle is 8ft. The span of the beam is 18 ft. If the front axle weighs 3 kips and the rear axle weighs 8 kips, what is the maximum possible shear in the beam without considering dynamic load effects?



- (A) 3 kips
(B) 8 kips
(C) 9.67 kips
(D) 11 kips
- 140.** A propped cantilever beam with a concentrated load is shown in the figure below. To reduce the maximum shear at fixed support, what design intervention can be used?

- I) Reduce span b
II) Reduce the magnitude of load
III) Increase the depth of beam section

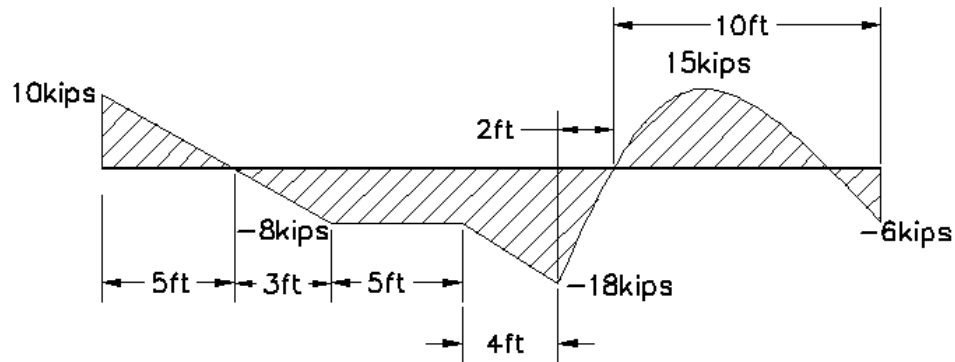


- (A) I, II and III
(B) I and II
(C) II only
(D) II and III

2.4 Flexure

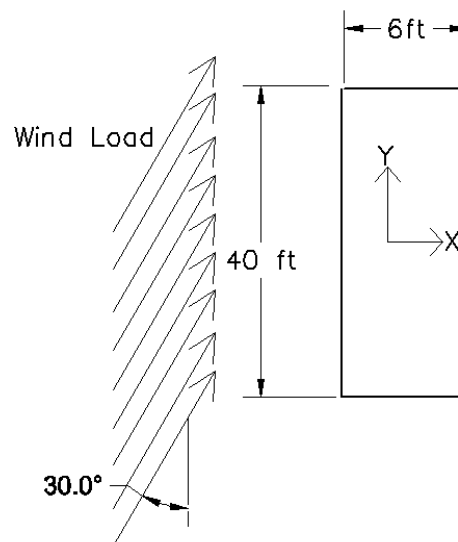
[Click here for solutions 2.4](#)

- 141.** The shear force diagram for a simply supported beam is given below. What is the value of the bending moment at a section 11 ft from the left support?

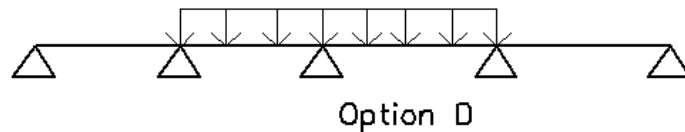
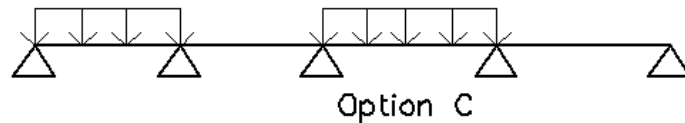
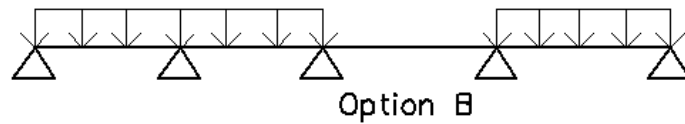
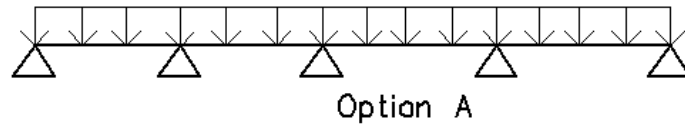


- (A) 9 kip*ft
 (B) -9 kip*ft
 (C) -11 kip*ft
 (D) 13 kip*ft
- 142.** The plan of a 40 ft by 6 ft pile cap is shown below. The pile cap is 3 feet tall. It is supported by a pile that is 10 feet tall. What is the moment at the base of the pile about Y axis due to wind load of 50 psf acting at 30 degree inclination to the Y axis?

- (A) 21.7 kip*ft
 (B) 32.8 kip*ft
 (C) 33.5 kip*ft
 (D) 43.3 kip*ft

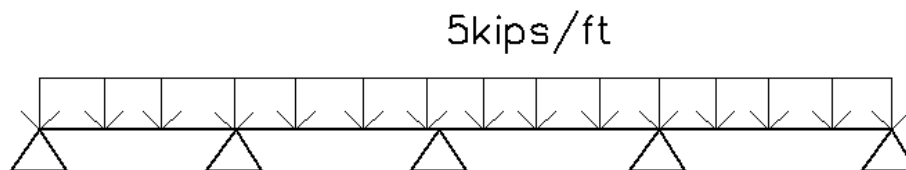


- 143.** Four span continuous beam is subjected to uniformly distributed loadings in different spans as shown below. Which loading configuration produces maximum negative moment in an internal support?



- (A) Option A
(B) Option B
(C) Option C
(D) Option D

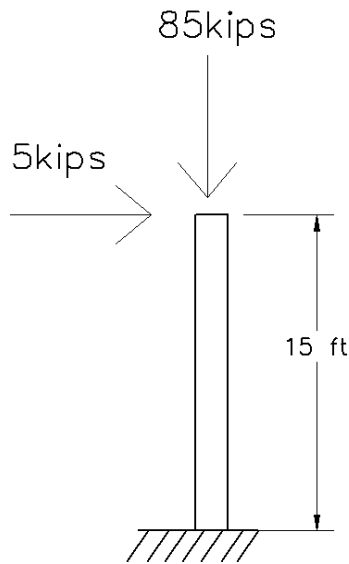
- 144.** The continuous beam system shown below has a uniform distributed load of 5 kips/ft throughout every span. Total length is 40 ft and all four spans are 10 ft in length. W 10 X45 section is used. What is the maximum elastic flexural stress acting on the beam about the strong axis?



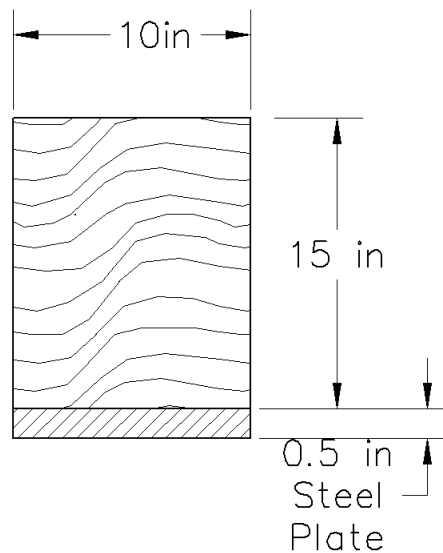
- (A) 7 ksi
(B) 9 ksi
(C) 11.5 ksi
(D) 13 ksi

- 145.** A cantilevered column is subjected to 5 kip lateral load at the top and an 85 kips concentric axial load. The lateral stiffness of the column is 2.5 kips/in. What is the total moment acting at the base of the column?

- (A) 60 kip*ft
(B) 75 kip*ft
(C) 90 kip*ft
(D) 100 kip*ft

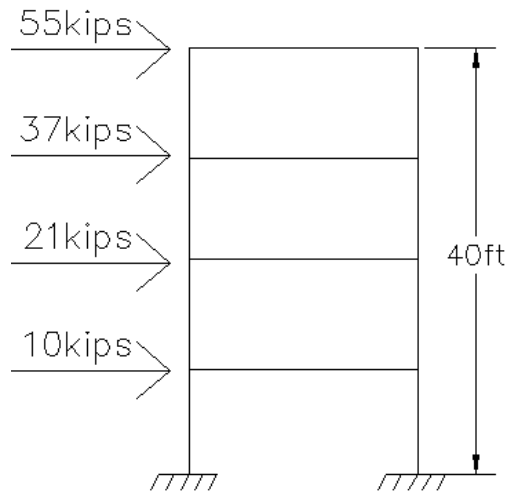


- 146.** A composite beam constructed using wood, reinforced with 0.5 inch thick steel plate at the bottom as shown below is subjected to a bending moment of 150 kip*ft. What is the flexural stress acting on the steel? Modulus of elasticity of wood = 2500 ksi and modulus of elasticity of steel = 30000 ksi.

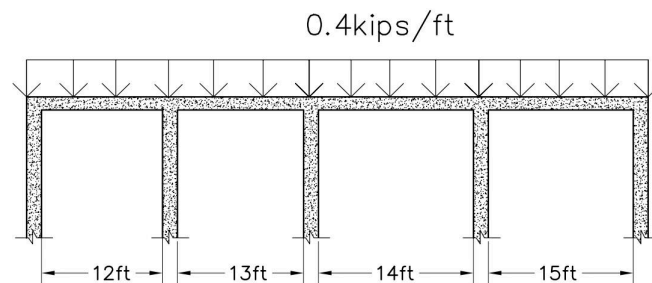


- (A) 23.2 ksi
(B) 25.3 ksi
(C) 30.6 ksi
(D) 33.8 ksi

- 147.** The figure below shows lateral seismic load distribution on each floor level. What is the total overturning moment at the base of the building? Assume all story heights are equal.

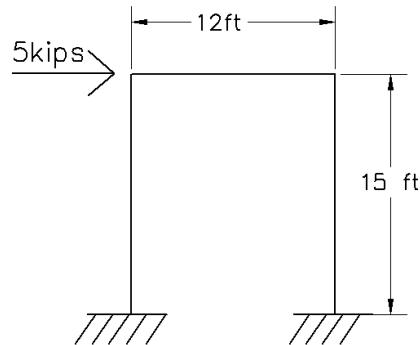


- (A) 2365 kip*ft
 (B) 2870 kip*ft
 (C) 3292 kip*ft
 (D) 3830 kip*ft
- 148.** The figure below shows a monolithically constructed reinforced concrete beam and column frame. The factored load acting on the beam is 0.4 kips/ft. What is the approximate maximum positive and negative moment acting on the concrete beams? Assume all conditions for ACI's simplified analysis are satisfied.

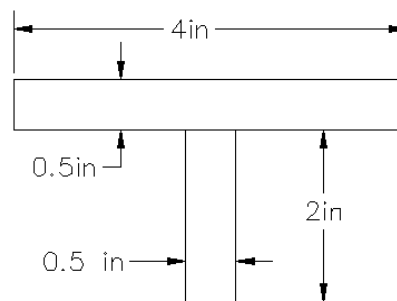


- (A) +3.4 kip*ft , -6.8 kip*ft
 (B) + 4.7 kip*ft , -6.6 kip*ft
 (C) + 6.4 kip*ft , -8.4 kip*ft
 (D) + 8.2 kip*ft , -11.8 kip*ft

- 149.** What is the maximum bending moment acting on the beam of the frame loaded as shown below? The point of inflection can be assumed at $0.55H$ from the base where H is the height of the column.



- (A) 15 kip*ft
 (B) 17 kip*ft
 (C) 20 kip*ft
 (D) 37.5 kip*ft
- 150.** Which of the following is not true about point of contraflexure and point of inflection?
- (A) Point of inflection is the point where the bending moment changes from convex to concave or vice versa,
 (B) Point of contraflexure is a point of zero bending moment,
 (C) The shape of elastic curve changes direction at the point of inflection,
 (D) Shear force is either minimum or maximum at the point of inflection.
- 151.** For the T section shown below, what is the location of the plastic neutral axis from the base of the section?

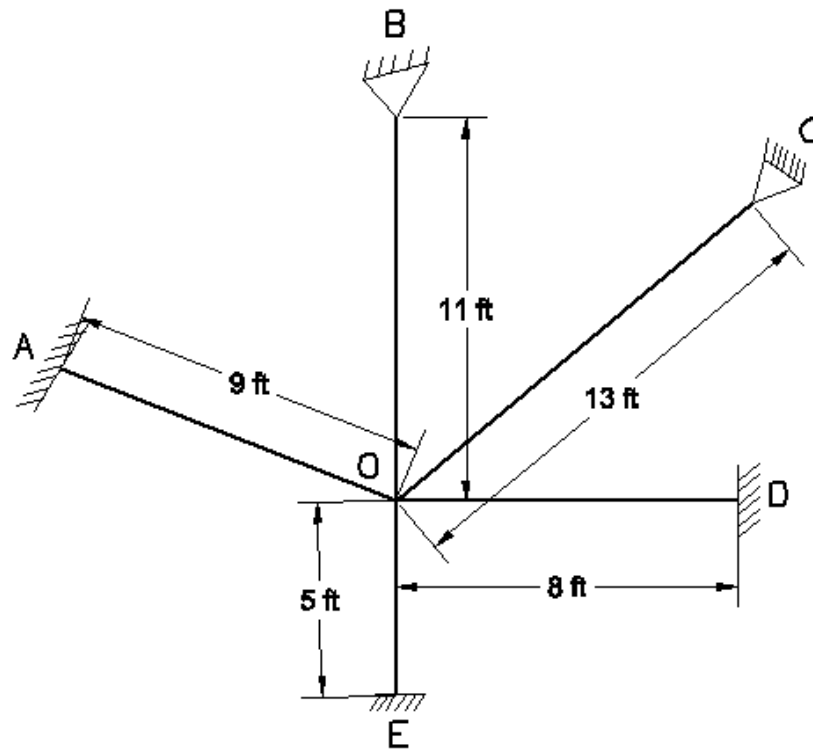


- (A) 1.5 in
 (B) 1.833 in
 (C) 2 in
 (D) 2.125 in

152. Which of the following is not true about plastic section modulus ?

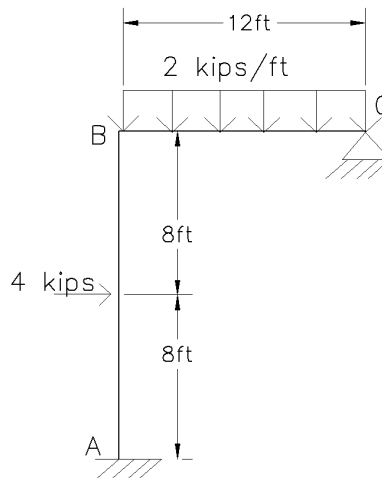
- (A) The shape factor, i.e. the ratio of plastic to elastic section modulus of I sections is lower than rectangular sections,
- (B) Structures designed using plastic section modulus is generally lighter than structures designed using elastic section modulus,
- (C) When calculating plastic section modulus, the neutral axis divides the areas of compression block and tension block equally,
- (D) Plastic section modulus is always independent of the material property.

153. For the framing configuration below, a correction moment of 50 kip*ft is applied at joint O. What proportion of correction moment is distributed to member OB? The moment of inertia and modulus of elasticity of all the sections are uniform.

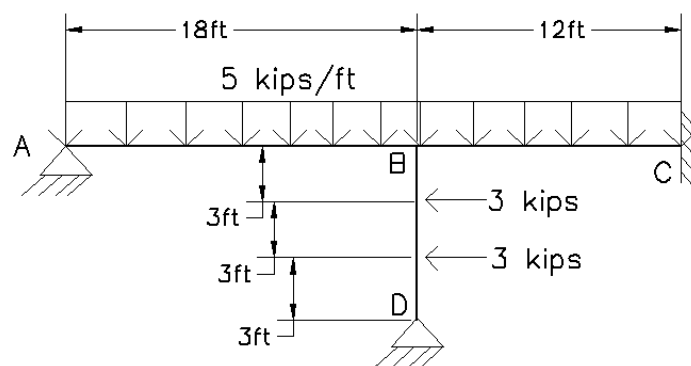


- (A) 5.23 kip*ft
- (B) 6.07 kip*ft
- (C) 9.18 kip*ft
- (D) 15.34 kip*ft

- 154.** For the framing configuration below, what is the moment reaction under column AB? The modulus of elasticity is constant. Moment of inertia of beam BC = 800 in^4 and column AB = 550 in^4 .

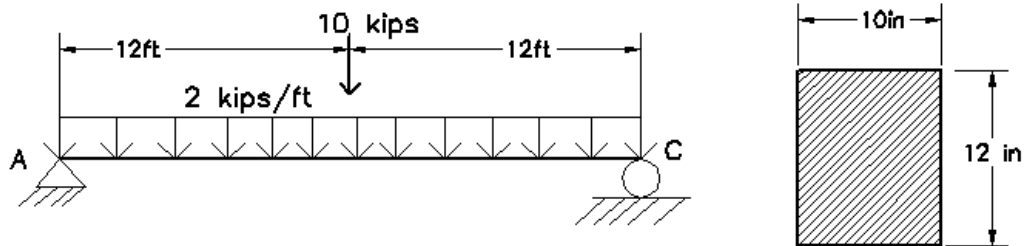


- (A) 1.2 kip*ft
 (B) 2.3 kip*ft
 (C) 3.5 kip*ft
 (D) 5.8 kip*ft
- 155.** For the framing configuration below, what is the reaction moment at point C? The material for the whole framing system is homogeneous. Moment of inertia of beam ABC = 700 in^4 and Moment of inertia for column DB = 1100 in^4 .

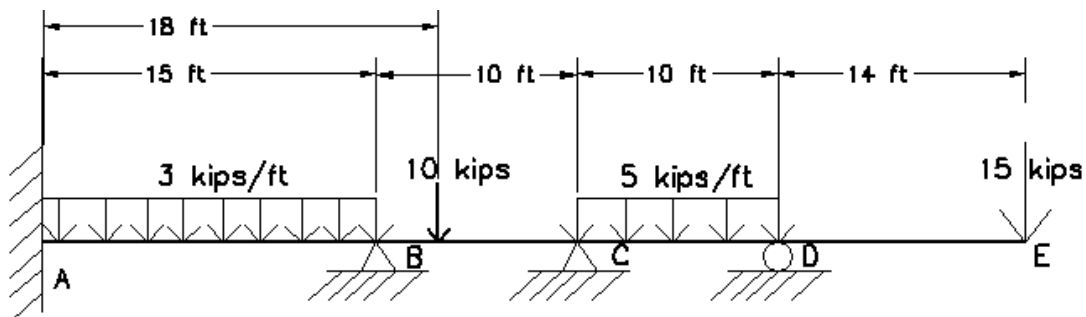


- (A) 23.2 kip*ft
 (B) 31.43 kip*ft
 (C) 38.24 kip*ft
 (D) 55.8 kip*ft

156. For the beam shown below, what is the maximum flexural stress acting on the section?
The dimensions of the rectangular cross section is 10in X 12 in.

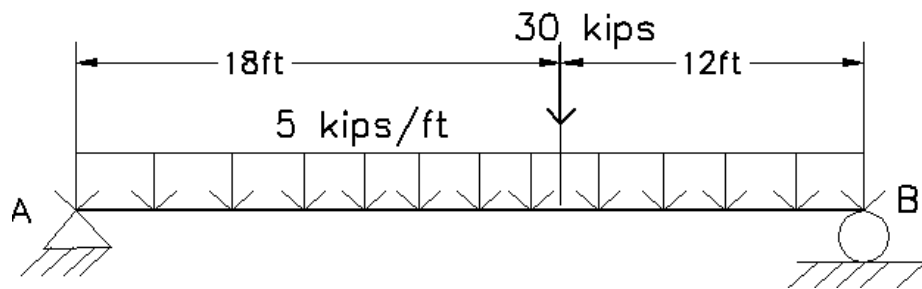


- (A) 9.5 ksi
(B) 10.2 ksi
(C) 11.95 ksi
(D) 13.1 ksi
157. For the continuous beam shown below, using the moment distribution method, what is the unbalanced portion of fixed end moments for joint C?



- (A) 23.2 kip*ft
(B) 31.43 kip*ft
(C) 35.37 kip*ft
(D) 38.22 kip*ft

158. For the beam shown below, what is the location of maximum bending moment from support A?



- (A) 17.4 ft
- (B) 18 ft
- (C) 18.6 ft
- (D) 19.2 ft

2.5 Combined stresses

[Click here for solutions 2.5](#)

159. The state of stresses in the wall of a vessel is given as follows:

- a. Compressive stress (along XX) = 10 ksi
- b. Tensile stress (along YY) = 25 ksi
- c. Shear stress = 15 ksi

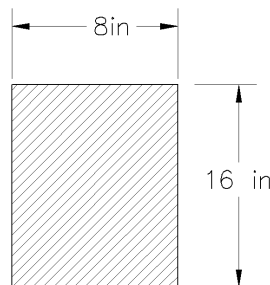
Calculate the principal stresses acting on the body.

- (A) 30.5 ksi compressive, 15.5 ksi tensile
- (B) 15.5 ksi compressive, 30.5 ksi tensile
- (C) 42.2 ksi compressive, 13.8 ksi tensile
- (D) 13.8 ksi compressive, 42.2 ksi tensile

160. A rectangular beam is subjected to a compressive load of 150 kips. In addition to the compressive force it is subjected to biaxial bending. The dimensions of the beam is 8 in X 16 in. What is the maximum normal stress acting on the beam section?

Moment about stronger axis = 100 kip*ft

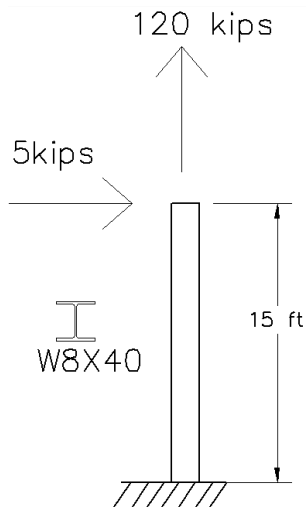
Moment about weaker axis = 50 kip*ft



- (A) 1.2 ksi
- (B) 5.6 ksi
- (C) 8.2 ksi
- (D) 12.1 ksi

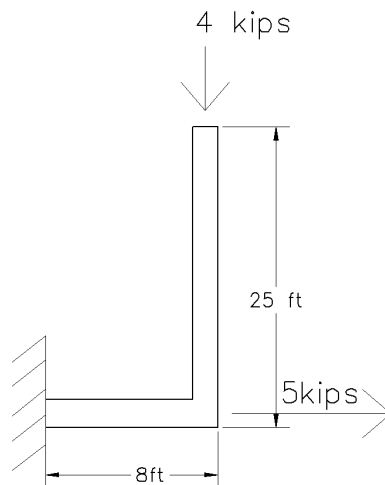
- 161.** A W8X40 cantilevered column is subjected to a tensile load of 120 kips. A lateral load of 5 kip acts at the top causing bending about the stronger axis. The column is 15 ft tall. What is the maximum compressive stress acting at the fibres of the column section?

- (A) 0 ksi
(B) 5.5 ksi
(C) 8 ksi
(D) 15 ksi



- 162.** A circular pipe bend is shown below. A 4 kip load acts at the top and another force of 5 kip is exerted on the bend as shown below. The pipe's outer diameter is 7 inches and inner diameter is 6in. What is the maximum normal stress acting in the pipe section?

- (A) 12.32 ksi
(B) 17.56 ksi
(C) 21.23 ksi
(D) 25.26 ksi



163. Which of the following biaxial states of stress represents a body in pure shear?

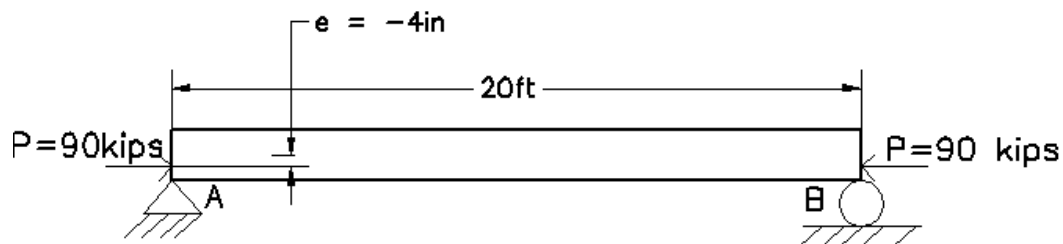
(A)
$$\begin{bmatrix} \sigma_1 & \tau_{xy} \\ \tau_{yx} & \sigma_2 \end{bmatrix}$$

(B)
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

(C)
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & -\sigma_1 \end{bmatrix}$$

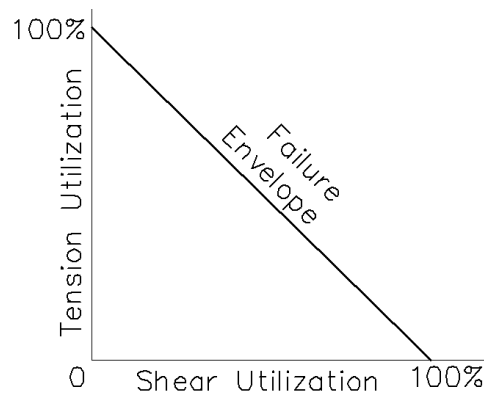
(D) None of the above

164. A beam is subjected to a prestressing force of 90 kips applied 4 inches below the centroid of the section. A uniformly distributed load of 3 kip/ft acts along the entire span of the beam. The area of the beam section is 240 in² and the section modulus is 800 in³. What is the maximum normal stress at the top fibre of the beam due to combined Uniform load and prestress ?

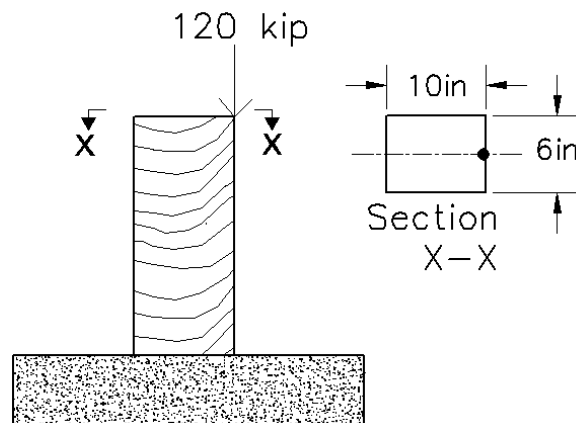


- (A) 1425 psi compressive
- (B) 1425 psi tensile
- (C) 2175 psi tensile
- (D) 2175 psi compressive

165. For a section subjected to combined tension and shear, the failure envelope is given by the linear graph below. What is not true about the failure due to interaction of combined stresses?

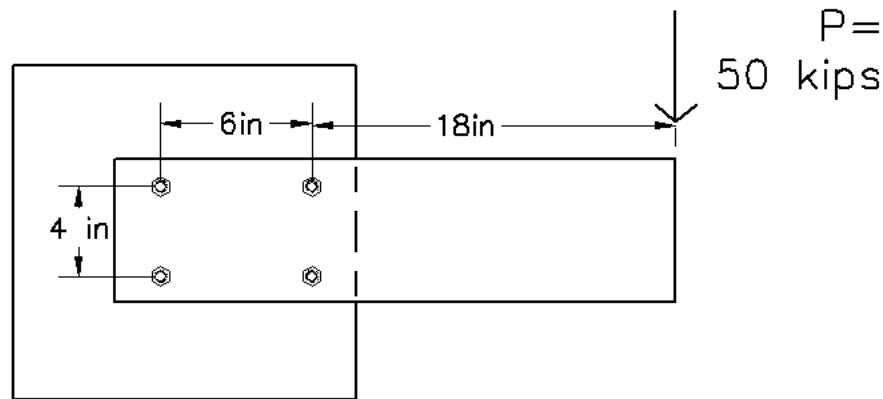


- (A) The section can have residual tension capacity and residual shear capacity but still fail in interaction,
 (B) The section can have no tension load but still fail when interaction is considered,
 (C) The section can have no shear load but still fail when interaction is considered,
 (D) If the section is utilized 60% for tension and 50% for shear, then it passes the design.
166. A wooden post rests on a concrete base. A 120 kip load acts on the top of the post at the center of the right edge of the post, as shown in section X-X. What is the maximum compressive stress exerted by the post on the concrete base?

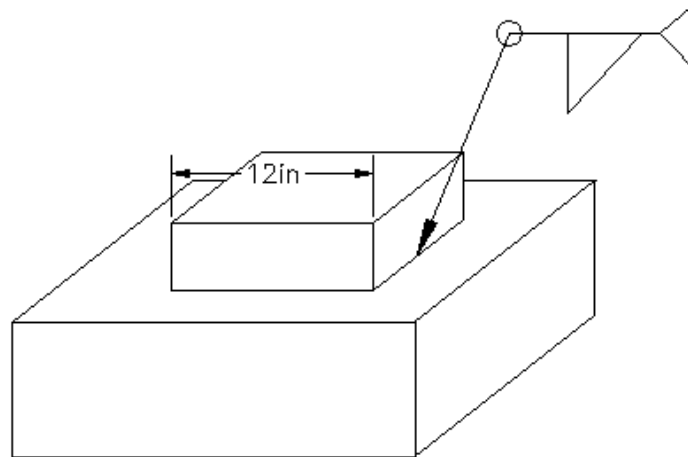


- (A) 2 ksi
 (B) 4 ksi
 (C) 8 ksi
 (D) 10 ksi

167. The figure below shows a bolted connection that is supporting a 50 kip load. The load is eccentric to the connection geometry creating additional torsion in the connection. What is the maximum shear load on an individual bolt?

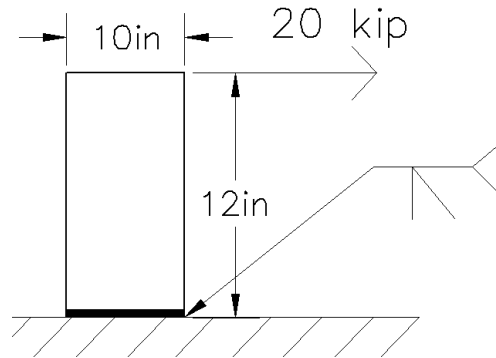


- (A) 12.5 kips
(B) 40.4 kips
(C) 73.1 kips
(D) 83.5 kips
168. A 12inX12 in steel box is welded on top of a base with all round fillet welds as shown below. The net tension load acting on the box is 100 kips and the net horizontal shear acting on the box is 150 kips. What is the resultant load acting on the welds due to combined shear and tension?



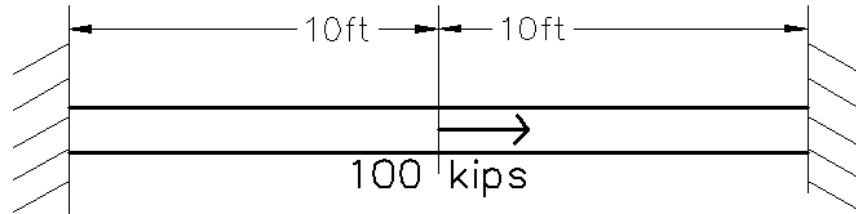
- (A) 2.08 kip/in
(B) 3.125 kip/in
(C) 3.75 kips/in
(D) 5.2 kips/in

- 169.** A 10 in X 0.5 in X 12 in plate is rigidly connected using full penetration weld as shown in the figure below. The plate is subjected to a lateral load of 20 kip at the top. What is the resultant stress in the weld?



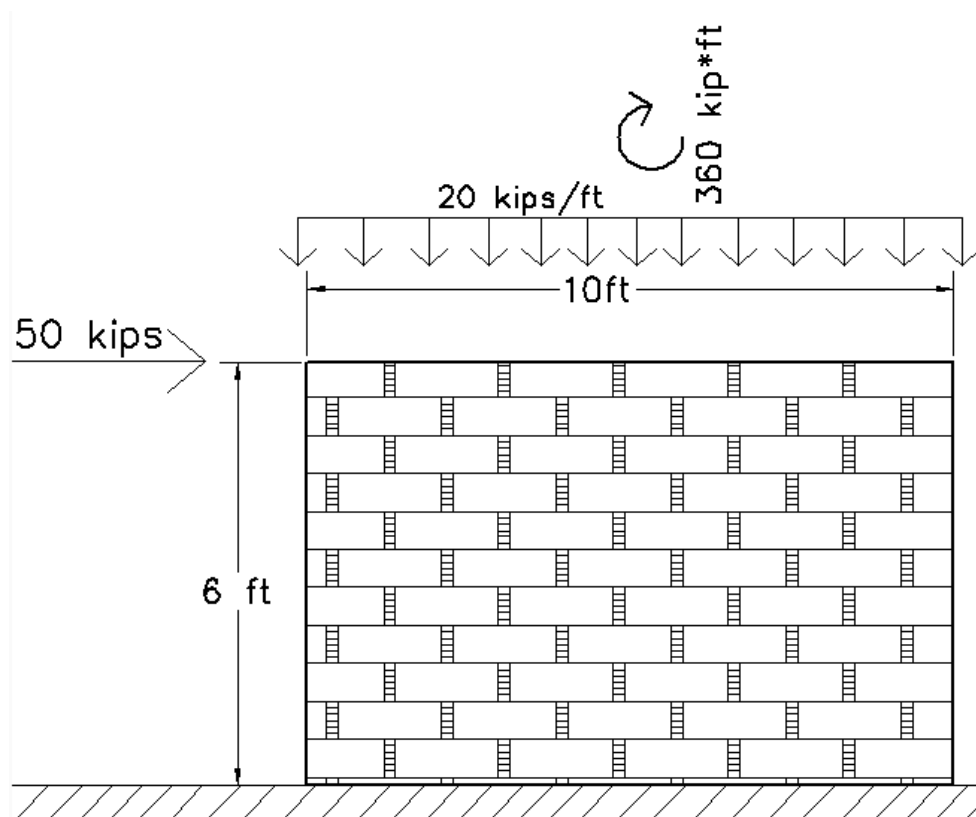
- (A) 2 kip/in
(B) 5.63 kip/in
(C) 7.81 kips/in
(D) 14.54 kips/in
- 170.** A solid circular shaft is subjected to a torsional moment of 10 ft-kips. The shaft is also subjected to a bending moment of 15 kip*ft. What is the equivalent twisting moment that when acting alone, produces the same maximum shearing stress as the combined effect of both moments acting currently?
- (A) 15 kip*ft
(B) 18 kip*ft
(C) 25 kip*ft
(D) 28 kip*ft

- 171.** A fixed-fixed beam with area 11 in^2 is subjected to an axial load at the center of the span. The beam was installed at 32°F with no room for expansion and the ambient temperature at service is 102°F . What is the total combined compressive stress acting in the beam section? Modulus of Elasticity = 29000 ksi Coefficient of thermal expansion = $0.000007 / ^\circ\text{F}$.



- (A) 4.54 ksi
 (B) 9.09 ksi
 (C) 18.76 ksi
 (D) 20.34 ksi
- 172.** When a threaded rod is tightened with a wrench, which of the following stresses are acting in the assembly?
- I) Bending of rod due to wrench torque,
 - II) Bearing stresses under locking nuts and washers,
 - III) Tension in rod due to preload.
- (A) I, II and III,
 (B) II only,
 (C) I and II,
 (D) II and III

173. A masonry wall is subjected to distributed load, a moment and a lateral load at the top as shown in the figure below. What is the maximum tensile stress acting on the masonry wall? The thickness of the wall is 6 inches.



- (A) 0.27 ksi
 (B) 0.55 ksi
 (C) 0.75 ksi
 (D) 0.88 ksi
174. For a complete state of stress in a three dimensional body , how many independent stress components are there?
- (A) Two normal components and one shear component,
 (B) Three normal components and three shear components,
 (C) Three normal components and six shear components,
 (D) Two normal components and four shear components.

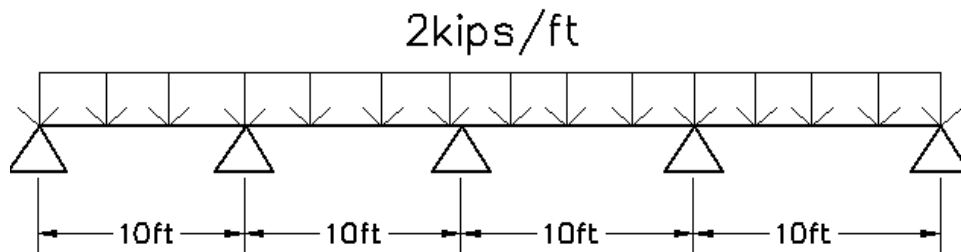
2.6 Deflection

[Click here for solutions 2.6](#)

175. A wooden block is fixed rigidly at the base. A tangential load is applied to the upper surface of the block. The deflection caused at the top of the block is due to

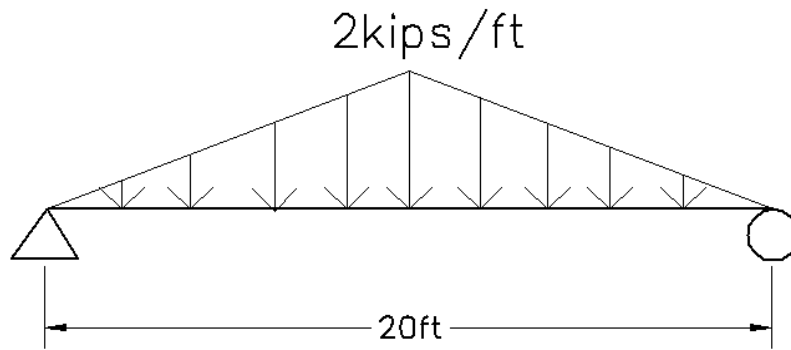
- (A) Shear and Bending moment,
- (B) Bending moment only,
- (C) Shear and torsional moment,
- (D) Shear only,

176. A four span W10X45 steel beam has a uniformly distributed load of 2 kips/ft as shown below. What is the maximum deflection in the beam? The modulus of elasticity of steel $E = 29000$ ksi. The bending moment acts about the weaker axis of the beam.

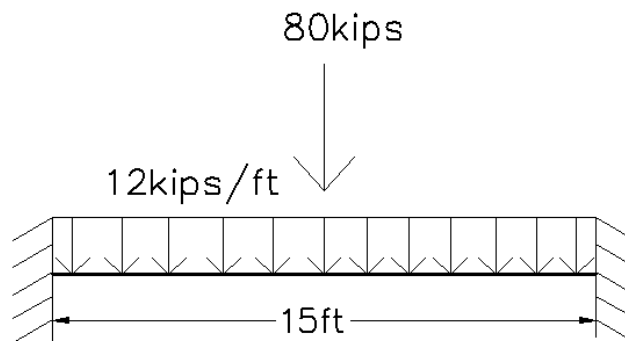


- (A) 0.1 in
- (B) 0.15 in
- (C) 0.2 in
- (D) 0.25 in

177. A simply supported beam is subjected to a uniformly varying load as shown below. The moment of inertia of the beam section is 60 in^4 and the modulus of elasticity is 29000 ksi . What is the intensity of an equivalent uniformly distributed load that causes the same deflection in the beam?

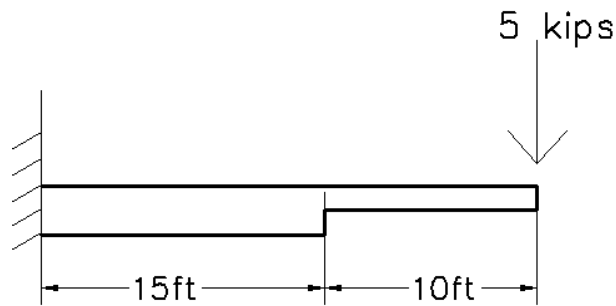


- (A) 0.7 kip/ft
(B) 1 kip/ft
(C) 1.3 kip/ft
(D) 1.5 kip/ft
178. A fixed-fixed beam is subjected to a uniformly distributed load and a point load as shown below. The moment of inertia of the beam section is 84 in^4 and the modulus of elasticity is 29000 ksi . What is the maximum deflection in the beam due to the combined loading?



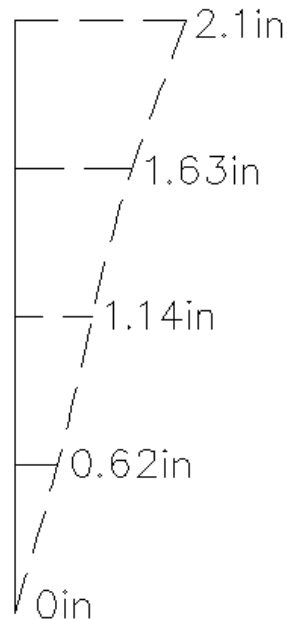
- (A) 1 in
(B) 1.12 in
(C) 1.52 in
(D) 2.12 in

- 179.** A cantilevered beam is loaded at the tip as shown below. The heavier first 15 ft length of the beam has a moment of inertia 1000 in^4 and the lighter 10ft length has a moment of inertia of 500 in^4 . What is the deflection at the tip of the beam? $E = 29000 \text{ ksi}$



- (A) 1 in
(B) 1.35 in
(C) 1.65 in
(D) 2 in
- 180.** A timber joist has a span 12 ft and is not supporting any ceiling. The joist is subjected to combined dead and live loading. What is the maximum allowable deflection for the joist?
- (A) 0.4 in
(B) 0.8 in
(C) 1 in
(D) 1.2 in
- 181.** A lintel supports unreinforced masonry above it. The span of the lintel is 8ft. What is the maximum allowable deflection in the lintel under combined service dead and live loads?
- (A) 0.16 in
(B) 0.26 in
(C) 0.53 in
(D) 1 in

- 182.** A four story steel special moment resisting frame building is identified as an essential infrastructure and a seismic design category C. The elastic deflections of the floors are as follows. What is the maximum design story drift per ASCE 7-16?



- (A) 2.1 in
(B) 2.27 in
(C) 2.5 in
(D) 2.78 in
- 183.** The average story height of a three story steel frame building is 11ft. It is identified as Seismic Risk category III and seismic design category D. The structure has extreme torsional irregularity. What is the allowable story drift per ASCE 7-16?

- (A) 2 in
(B) 2.64 in
(C) 3 in
(D) 3.2 in

- 184.** A simply supported truss is shown below. A 30 kips load acts at joint B. The forces resulting from the load on each member is shown on each member (tension +ve). What is the maximum horizontal deflection on joint B due to the load? Area = 10 in² and E = 29000 ksi for all members.

Lengths of members

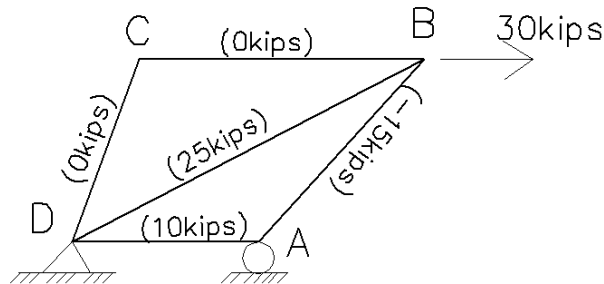
AB = 120 in

BC = 150 in

CD = 100 in

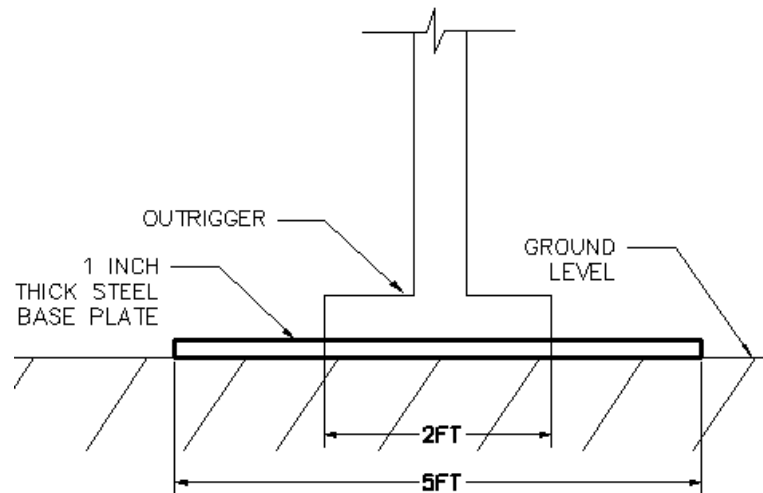
AD = 100 in

BD = 180 in



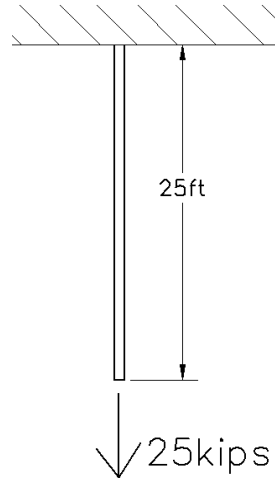
- (A) 0.011 in
(B) 0.017 in
(C) 0.023 in
(D) 0.027 in

- 185.** An outrigger from a mobile crane is laid on a 5ftX5ft X1in steel base plate. The foot of the outrigger is 2ftX2ft. The maximum concentric axial thrust from the outrigger is 50 kips. What is the maximum deflection of the tip of the steel base plate? E = 29000 ksi



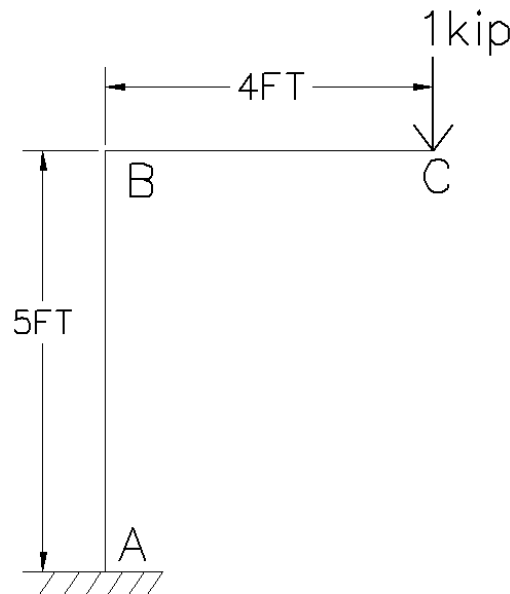
- (A) 0.015 in
(B) 0.03 in
(C) 0.05 in
(D) 0.075 in

- 186.** A hanger rod of 1.85 inch diameter is measured to be 25ft long when the ambient temperature was 100°F. During winter, a load of 25 kips is applied to the rod and the ambient temperature is 0°F. What is the deflection at the tip of the rod?
 Modulus of Elasticity = 29000 ksi Coefficient of thermal expansion = $0.000007 / ^\circ\text{F}$.



- (A) 0.11 inch downwards
- (B) 0.21 inch downwards
- (C) 0.11 inch upwards
- (D) 0.21 inch downwards

- 187.** Calculate the maximum downward deflection at point C due to a load of 1 kip acting as shown in the drawing below.
 Moment of inertia of column AB = $2I$
 Moment of inertia of beam BC = I .
 Modulus of elasticity = E .



- (A) $184/3EI$
- (B) $284/3EI$
- (C) $328/4EI$
- (D) $584/5EI$

188. A simply supported beam is subjected to uniformly distributed load . The beam has a rectangular cross section. If the intensity of loading is halved and the depth of beam is doubled, what is the ratio of maximum deflections of the new beam and the original beam?

- (A) One third
- (B) One quarter
- (C) One eighth
- (D) One sixteenth

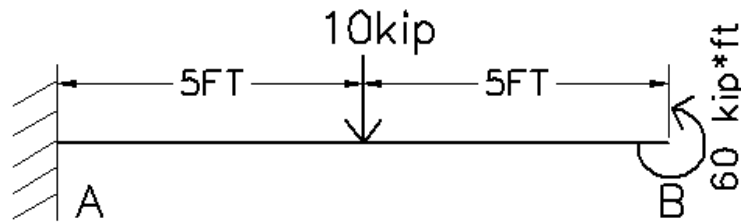
189. Two beams with identical cross sections and lengths are manufactured with two materials. One of them is manufactured with gray cast iron and another is manufactured with mild steel. Both of them are fixed in two ends identically and identical uniform loading is applied. Which of the following is true about their deflections?

- (A) Gray cast iron beam will deflect less than mild steel beam,
- (B) Mild steel beam will deflect less than gray cast iron beam,
- (C) For identical cross section, loading and statically determinate conditions, the deflections will be equal for both materials.
- (D) Deflections will depend upon the allowable yield stresses of both materials.

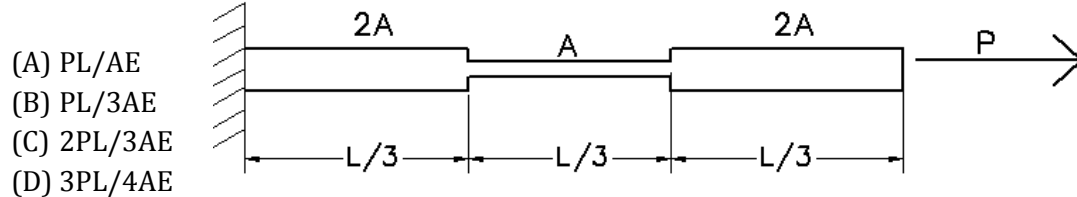
190. A fixed-fixed beam with a rectangular cross section is designed to support a concentrated load at the center of the span. Due to construction extensions, the beam has to be doubled in length. The new beam will be supporting the same concentrated load at mid span as the original beam. To restrict the deflections to the original configuration, what changes should be made to the new beam?

- (A) Double the depth,
- (B) Double the breadth,
- (C) Double the width and breadth,
- (D) Increase cross section area two times.

191. What is the deflection at the free end of the cantilever beam due to the loads shown below? $I = 50 \text{ in}^4$ $E = 29000 \text{ ksi}$

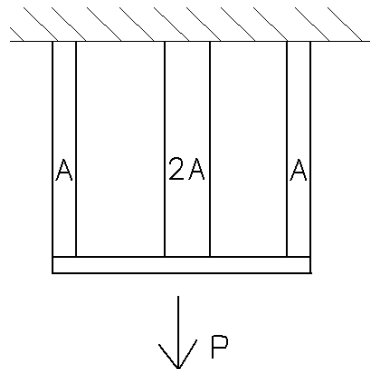


- (A) 1.2 in
(B) 2.33 in
(C) 3.1 in
(D) 4.8 in
192. A beam is subjected to axial load as shown. The cross section areas of the beam vary at different spans as shown below. What is the total deflection of the beam? Modulus of Elasticity is uniform.



- (A) PL/AE
(B) $PL/3AE$
(C) $2PL/3AE$
(D) $3PL/4AE$
193. What is the deflection at the free end of the system? The center bar has two times the area of the edge bar. All three bars are equal in length.

$A = 10 \text{ in}^2$
 $E = 29000 \text{ ksi}$
 $L = 15 \text{ ft}$
 $P = 200 \text{ kips}$



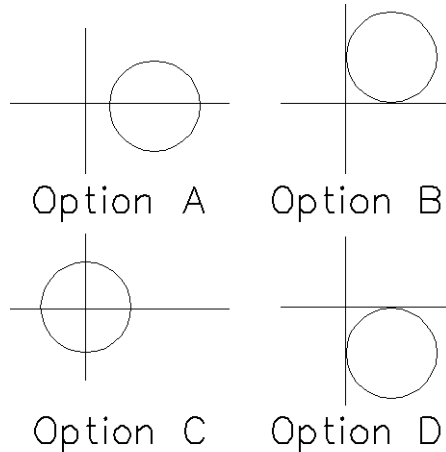
- (A) 0.02 in
(B) 0.03 in
(C) 0.04 in
(D) 0.05 in

2.7 Special topics

[Click here for solutions 2.7](#)

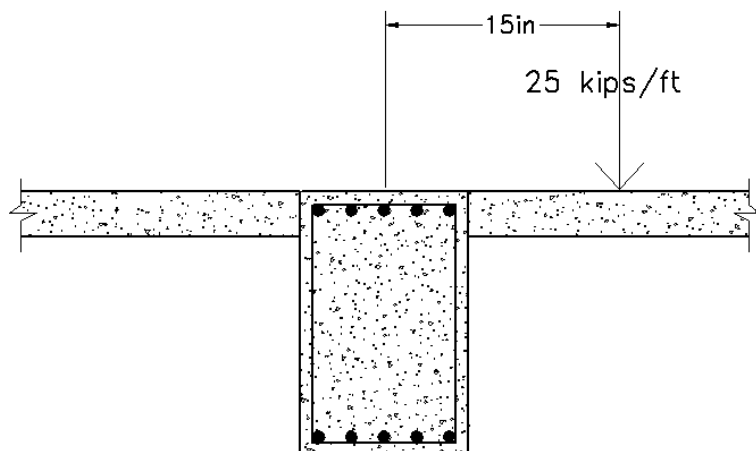
194. What is the shape of Mohr's circle for pure torsion?

- (A) Option A,
- (B) Option B,
- (C) Option C,
- (D) Option D



195. A reinforced concrete beam has a span of 20 ft. The beam is subjected to a live load of 25 kips/ft at a distance 15 in from the center of the beam as shown below. The beam is restrained against torsion only at one end. What is the torsional moment acting on the beam?

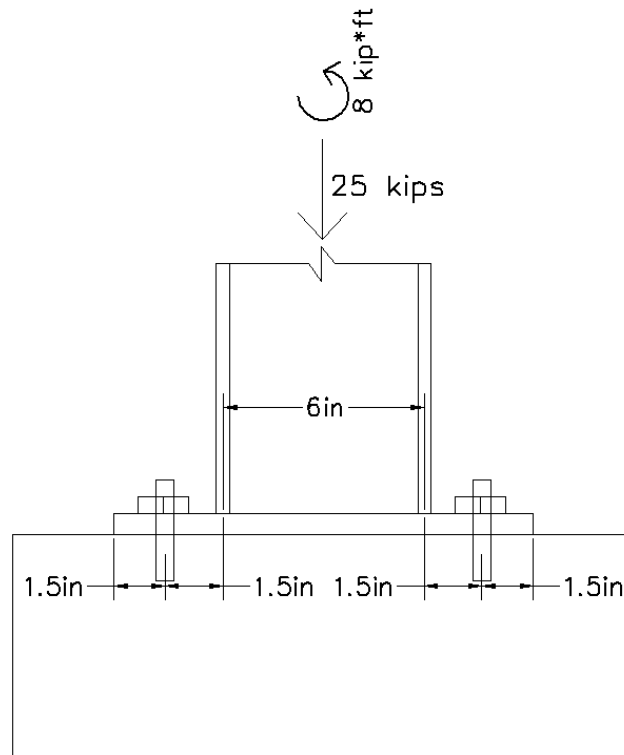
- (A) 125 kip*ft
- (B) 250 kip*ft
- (C) 312.5 kip*ft
- (D) 625 kip*ft



196. A circular shaft of diameter 8 inches and length 12 ft is fixed at both ends against deflections and rotations about all directions. It is subjected to a torsional moment of 500 kip*ft at the center. What is the angle of twist in the shaft? Modulus of elasticity, $E = 29000$ ksi and Poisson's ratio is 0.3.

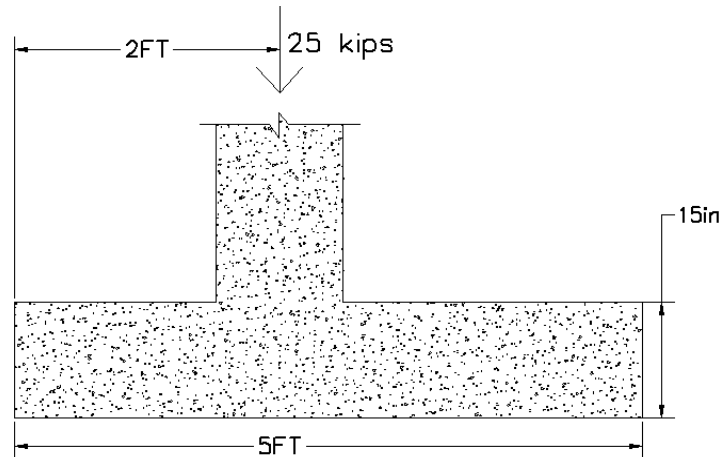
- (A) 0.048 rad
- (B) 0.096 rad
- (C) 0.13 rad
- (D) 0.192 rad

197. The figure below shows a column base plate. The square base plate is 12in X 12 in. Assuming the centroid of the compressive force in the base plate is at the flange of the column, what is the maximum bearing pressure exerted by the loads shown below?



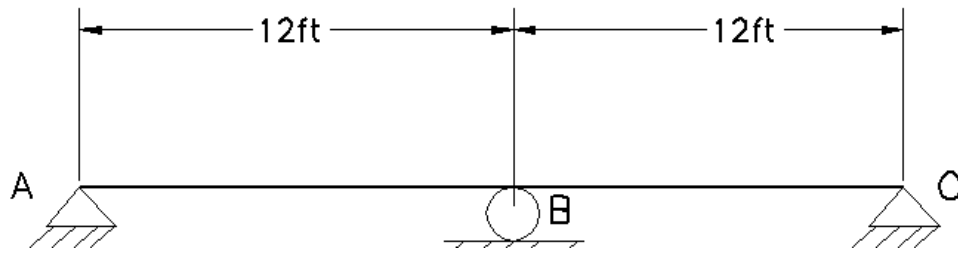
- (A) 0.25 ksi
- (B) 0.375 ksi
- (C) 0.5 ksi
- (D) 0.75 ksi

- 198.** The figure below shows a 5 ft X 5ft square footing. The column load including self weight of the column acting on the footing is 25 kips. What is the maximum bearing pressure? The unit weight of concrete = 0.15 kcf.



- (A) 1 ksf
(B) 1.6 ksf
(C) 1.8 ksf
(D) 2.8 ksf
- 199.** Which of the following is not true about steel column buckling?
- (A) Inelastic buckling strength is less than elastic buckling strength,
(B) Euler's buckling equation predicts buckling load which when released, causes the member to return to its undeformed form,
(C) Buckling deflection is sudden and catastrophic,
(D) Elastic buckling strength of a material is always less than the yield strength of the material.
- 200.** Which of the following structural members are most susceptible to lateral torsional buckling?
- (A) Unbraced beams subjected to significant axial loads,
(B) Unbraced columns subjected to significant axial loads,
(C) Unbraced beams subjected to significant bending moments,
(D) Roof truss members subjected to significant wind loading.

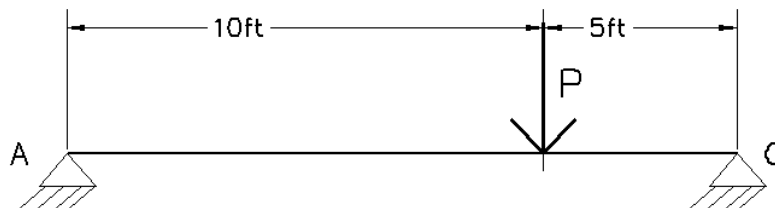
201. A continuous beam shown below has equal span lengths of 12 ft. What is the Moment developed at support B due to a settlement of 2 inches at support A? Modulus of elasticity = 29000 ksi, Moment of inertia = 1200 in^4



- (A) 380.8 kip*ft
 (B) 419.5 kip*ft
 (C) 450.7 kip*ft
 (D) 510.2 kip*ft
202. A building is to be analyzed for progressive collapse after sudden failure of ground floor column. Which of the following analysis is not required for alternate load path analysis?

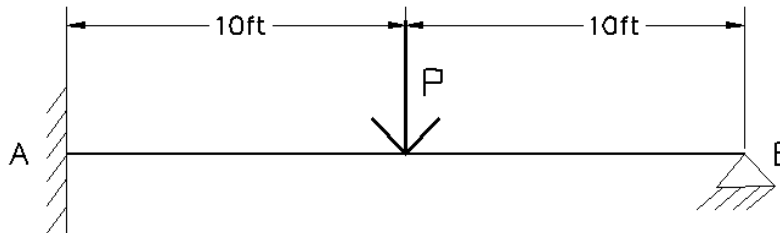
- (A) Static analysis with factored service loads,
 (B) Nonlinear analysis for large deformations,
 (C) Dynamic analysis simulating sudden removal of column,
 (D) Foundation uplift and bearing under failed column.

203. A beam with a span of 15 ft is shown below. The plastic moment capacity of the beam section is M_p . What is the value of the collapse load P ?



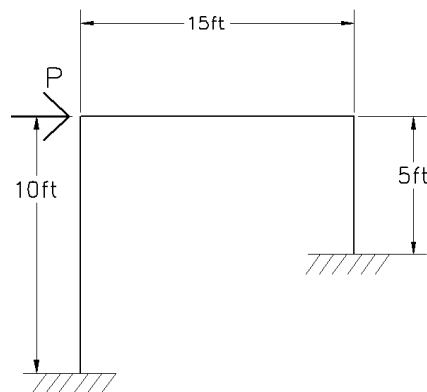
- (A) $3M_p/10$
 (B) $4M_p/9$
 (C) $6M_p/13$
 (D) $7M_p/19$

- 204.** A propped cantilever beam is shown below. The plastic moment capacity of the beam section is 100 kip*ft. What is the value of the collapse load P when the beam is analyzed via plastic analysis?



- (A) 28 kip
- (B) 30 kip
- (C) 32 kip
- (D) 34 kip

- 205.** For the frame shown below, the plastic moment capacity of the section is 150 kip*ft. What is the value of lateral collapse load P ?

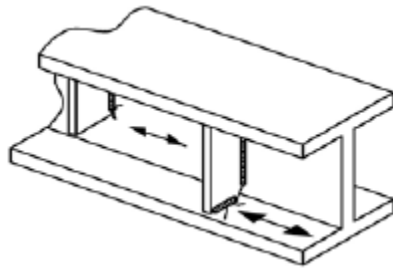


- (A) 60 kip
- (B) 70 kip
- (C) 80 kip
- (D) 90 kip

206. A steel connection is subjected to a maximum stress range of 10 ksi in its service life. Through experimental testing, the constant amplitude fatigue limit for the detail is determined as 12 ksi. What is the estimated fatigue life of the connection?

- (A) 0,
- (B) 10000,
- (C) Infinite,
- (D) Cannot be determined without the modulus of elasticity.

207. A gusset plate is welded to the web of a bridge girder as shown below, near an interior support. The span of the girder is 60 ft and is a part of a continuous girder system. Calculate the minimum fatigue resistance of the welded plate. The single lane average daily truck traffic is 2000 vehicles per day.



- (A) 3.77 ksi
- (B) 4.21 ksi
- (C) 5.38 ksi
- (D) 7.63 ksi

3. Temporary Structures and Other Topics

Sub- Headings	Number of Problems
Special inspections	208-217
Submittals	218-222
Formwork, falsework, scaffolding, shoring and reshoring, bracing, and anchorage	223-235
Impact of construction on adjacent facilities	236-239
Safety	240-244
Total number of Problems	37

3.1 Special inspections

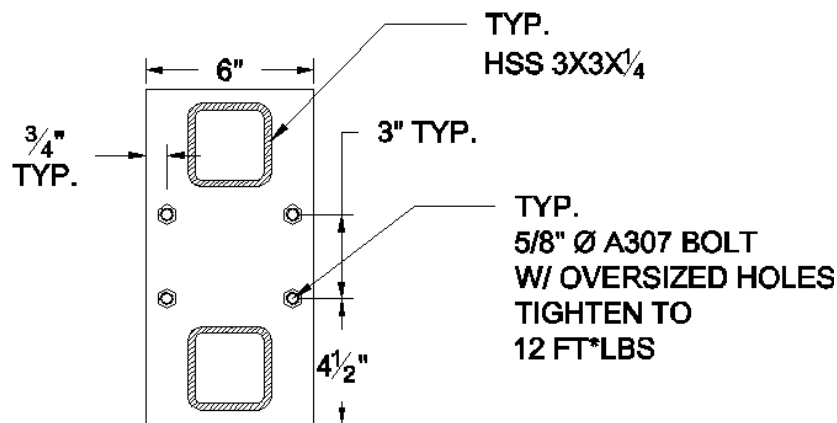
[Click here for solutions 3.1](#)

- 208.** A real estate agency hires a class B contractor to construct a 12 story building. Which of the following parties can qualify as a special inspector?
- (A) Department of building official,
 - (B) Engineer of record of the building to be constructed,
 - (C) Registered professional employed by the contractor to supervise construction,
 - (D) An employee from the subcontractor's team who has necessary experience and complete information on material specifications to be used in the project.
- 209.** Which of the following is not true about procedures related to special inspections?
- (A) The owner, engineer of record, contractor and building officials have to approve the competency of the party serving as the special inspector,
 - (B) For a welding job, in addition to non destructive evaluation reports, the qualifications of the fabricator also needs to be included as a part of inspection documentation,
 - (C) Structural observations or special inspections are not mandatory for every project, regardless of the cost of the project,
 - (D) The special inspector is not responsible for the contractor's means and methods.
- 210.** Which of the following constructions does not mandate special inspections?
- (A) An isolated concrete footing for a 4 story building that is supported on rocky soil,
 - (B) An over reinforced continuous footing that is constructed with 3 ksi concrete and is analyzed to have sufficient structural strength with 25% lesser rebar,
 - (C) An under reinforced footing that is constructed with 3 ksi concrete and is analyzed to have sufficient structural strength with 25% lesser compressive strength of concrete,
 - (D) A 5 inch thick slab on grade that serves as a foundation to the diesel generator.

211. The statement of special inspections that is to be submitted to the building officials is prepared by

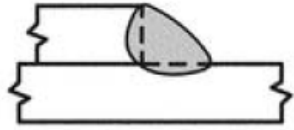
- (A) A firm hired by the owner to serve as the engineer of record of the project,
- (B) The contractor's team coordinating all construction permits ,
- (C) The special inspection agency hired for the project who is independent of the contractor,
- (D) The owner's technical personnel who oversee overall construction scheduling and official permits.

212. The figure below shows a 4 bolt connection detail. A special inspector needs to verify the bolting compliance on site. What is the maximum diameter of holes that he can approve?

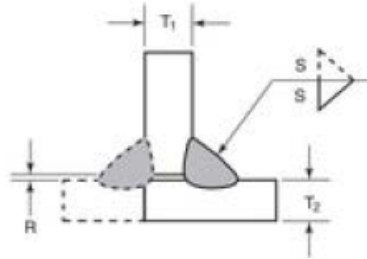


- (A) 5/8 inch
- (B) 11/16 inch,
- (C) 13/16 inch.
- (D) 15/16 inch

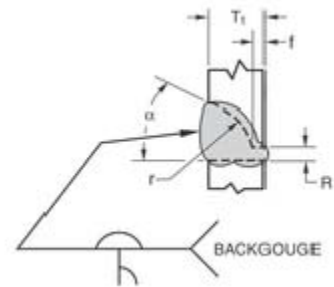
213. Which of the following welds require special inspections for joint preparation prior to welding?



Option I



Option II



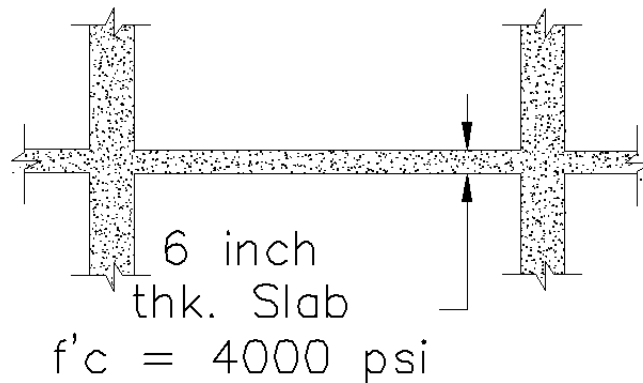
Option III

- (A) Option I
- (B) Option II
- (C) Option III
- (D) Option I and II

214. Which of the following wooden truss elements are subjected to the requirements of special inspection per IBC chapter 17

- (A) A wooden Pratt truss with overall height of 3 feet,
- (B) Site built King truss with a span 65 feet and metal plate connections,
- (C) Site built Queen truss with a span 50 feet and metal plate connections,
- (D) Prefabricated truss with span of 80 ft, the fabricator approved to work according to IBC clause 1704.2.5.1

- 215.** For a concrete construction job site, which of the following can be inspected intermittently and does not require constant monitoring by the special inspector?
- (A) Verifying proper spacing, cover and size of rebars,
 - (B) Adhesive anchors installation to fasten electrical panel unit on a shear wall cast two weeks ago,
 - (C) Verifying groove welds used for rebar lapping,
 - (D) Application of prestressing forces on a multi span concrete girder.
- 216.** For a 6 inch thick concrete slab shown below, #5 rebars are spaced at 3 inch c/c at the negative moment zones. What is the maximum aggregate size that can be approved by the special inspector for the pour?



- (A) 1.25 inch
 - (B) 1.5 inch
 - (C) 1.75 inch
 - (D) 2 inch
- 217.** Which of the following does not typically require special inspection per IBC chapter 17?
- (A) Installation of prestressing tendons,,
 - (B) Soil bearing capacity for shallow foundations ,
 - (C) Thickness of fireproofing applications,
 - (D) Operations of a concrete batching plant.

3.2 Submittals

[Click here for solutions 3.2](#)

- 218.** Which document governs the content, format, overall procedures and approvals of construction submittals?
- (A) International Building Code ,
 - (B) OSHA Technical manual,
 - (C) ACI 318-14,
 - (D) Contract documents and specifications.
- 219.** What is the contractor's role during the submittal process?
- (A) To create and approve all submittal documents,
 - (B) To compile and review all submittal documents before forwarding it to the building engineer of record,
 - (C) To revise and document design assumptions based on discrepancies on site and capabilities of subcontractor's shop,
 - (D) To deliver Department of Building approved design drawings to the engineer of record.
- 220.** Which of the following is typically not a part of the general contractor's submittal package for a commercial building project?
- (A) Vendor provided flooring product data sheet to be used in the building,
 - (B) Concrete cylinder break reports provided by the testing lab,
 - (C) Structural calculations for construction loads,
 - (D) Mill test reports.

221. A shop drawing package submitted to the building engineer of record was returned to the contractor with remarks as follows:

WE ARE SENDING:	SUBMITTED FOR:	ACTION TAKEN:
☒ Shop Drawings	☒ Approval	☐ Approved as Submitted
☐ Letter	☐ Your Use	☒ Approved as Noted
☐ Prints	☐ As Requested	☐ Resubmit
☐ Change Order	☐ Review and Comment	☐ Submit
☐ Plans	☐ Other:	☐ Returned
☐ Samples	SENT VIA:	☐ Returned for Corrections
☐ Specifications	☒ Attached	☐ Due Date:
☐ Other:	☐ Separate Cover Via:	☐ Other:

What does the action taken here typically mean?

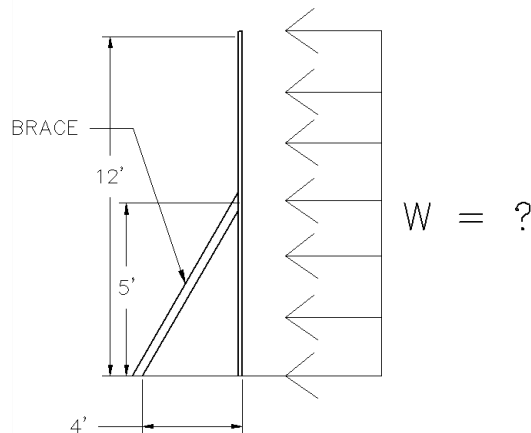
- (A) Revisions are required before proceeding,
 - (B) The contractor can proceed with field installation and the minor changes are acceptable,
 - (C) Submittal package needs to be resubmitted,
 - (D) Architect's approval is required before proceeding any further.
222. Which of the following is not the purpose of construction submittals?
- (A) To verify overall compliance with design intent and specifications,
 - (B) To facilitate record keeping, communication and coordination between different parties,
 - (C) To serve as a means of quality control and risk reduction during project implementation,
 - (D) To assist the owners with budgeting and financial planning during subsequent phases of the project.

3.3 Formwork, falsework, scaffolding, shoring and reshoring, bracing, and anchorage

[Click here for solutions 3.3](#)

- 223.** A 12' tall wall is braced at a height of 5 feet as shown in the figure below. The wall is subjected to a uniformly distributed lateral load W lbs/ft. The bottom of the wall can be considered pin jointed. The allowable axial strength of the brace is 20 kips. What is the maximum allowable lateral load ' W ' on the brace?

- (A) 550 lbs/ft
- (B) 750 lbs/ft
- (C) 850 lbs/ft
- (D) 950 lbs/ft



- 224.** A concrete deadman block is used to anchor a cable that provides lateral stability to a falsework system. The maximum force acting on the cables is 5 kips and the horizontal inclination of the cable is 30 degrees. Due to site constraints, the maximum height of the anchor block is limited to 6ft. What is the minimum area of the block that can safely stabilize the system? The cable is attached at the center of the top surface of the block.

Factor of safety = 1.5,

Coefficient of friction between concrete and soil = 0.3,

Density of concrete = 150 lbs/ft³

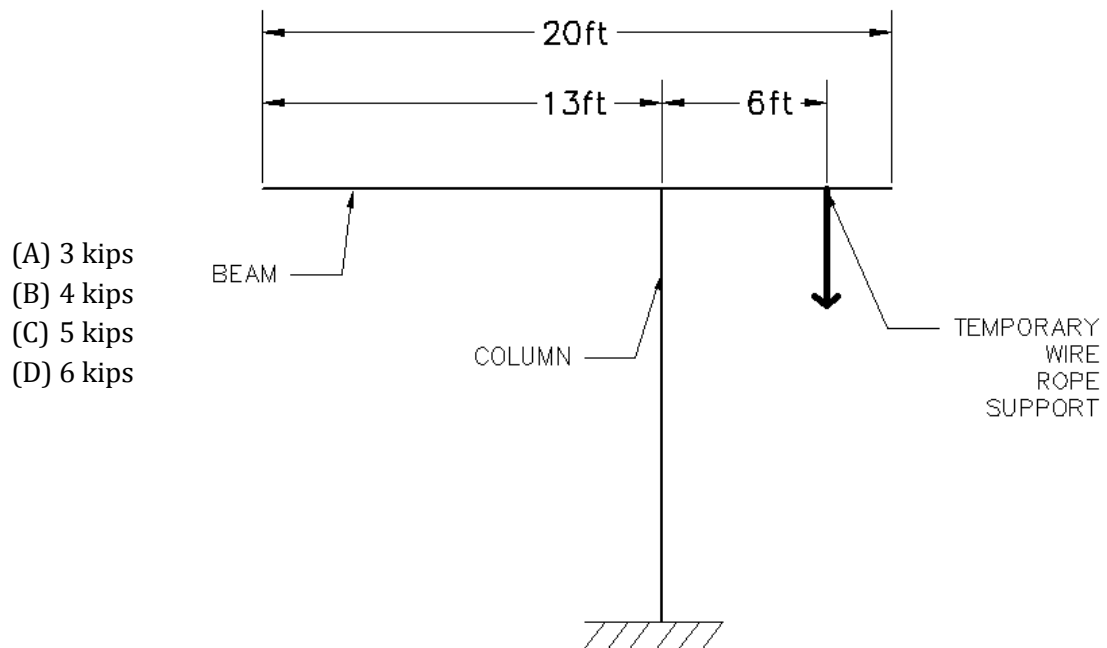
- (A) 5 ft X 5 ft
- (B) 6 ft X 6 ft
- (C) 7 ft X 7 ft
- (D) 8 ft X 8 ft

- 225.** Guy ropes are used to stabilize a bridge falsework system. The maximum pretension in the guy rope is 2.5 kips. The guy rope is inclined at 45 degrees to the horizontal and is anchored to a 4 ksi concrete slab. The table below shows ultimate forces that the anchors can withstand in lbs. A factor of safety of 2 can be used to convert the ultimate forces to allowable forces. What is the minimum diameter of the anchor that can be used?

Anchor Dia. (in)	Required Torque (ft-lb)	Embedment depth (in)	$F'_c = 2000$ psi		$F'_c = 4000$ psi	
			Tension	Shear	Tension	Shear
$\frac{1}{4}$	8	$2\frac{1}{8}$	2,260	1,680	3,300	1,680
$\frac{3}{8}$	25	4	4,800	4,000	5,940	4,140
$\frac{1}{2}$	55	6	5,340	7,240	9,640	7,240
$\frac{5}{8}$	90	$7\frac{1}{2}$	7,060	9,600	15,020	11,900

- (A) 1/4 in
(B) 3/8 in
(C) 1/2 in
(D) 5/8 in

- 226.** During a steel erection job, a 20 ft long beam is set on top of a column as shown below. The beam is temporarily supported by wire rope behind the column. If the weight of the beam is 200 lbs/ft, what is the maximum reaction on the column?



- (A) 3 kips
(B) 4 kips
(C) 5 kips
(D) 6 kips

227. For a concrete construction project, what is the purpose of reshoring?
- (A) To support the weight of concrete beams and slabs during concrete placement,
 - (B) To support the beams and slabs after stripping off formwork and redistribute loads safely to the base of the structure,
 - (C) To control lateral displacements in the falsework system,
 - (D) To help the concrete members gain compressive strength faster.

228. What is the most important factor that governs the right time to remove the formwork under beams and slabs?
- (A) The number of days that the member has been cured and the compound used to cure,
 - (B) The ambient temperature during curing and construction,
 - (C) The compressive strength that the concrete has gained before removal,
 - (D) The depth of beams and slabs that are cast.

229. The table shows allowable loads for shoring props. A 550 prop is to be used to shore a floor that is subjected to a 100 psf dead load, 75 psf construction load and 15 psf load due to weight of all formwork and falsework. If the floor height is 15ft, what is the maximum spacing of the props that can be used to shore the floor?

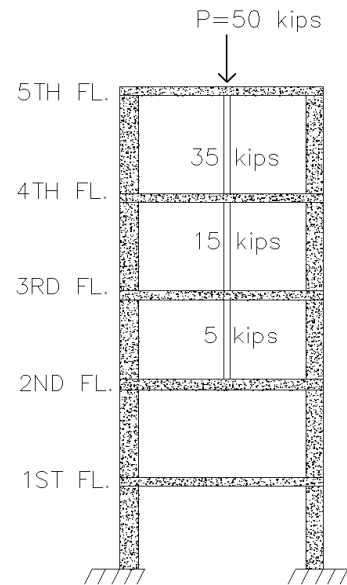
- (A) 4 ft X 4 ft,
- (B) 4.5 ft X 4.5 ft
- (C) 5 ft X 5 ft
- (D) 5.5 ft X 5.5 ft

Permitted prop loads [kN]

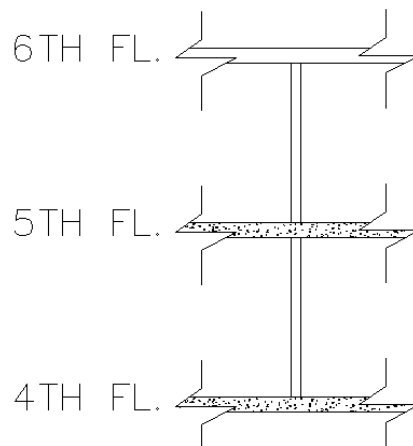
Prop height [m]	Eurex 20				
	250	300	350	400	550
5.5					
5.4					
5.3					
5.2					
5.1					
5.0					
4.9					
4.8					20.0
4.7					
4.6					
4.5					
4.4					
4.3					
4.2					
4.1					
4.0					20.6
3.9					21.7
3.8				20.0	22.9
3.7					24.1
3.6					25.5
3.5					26.9
3.4				20.8	28.5
3.3				22.0	30.3
3.2				23.4	32.2
3.1			20.0	25.0	34.3
3.0				26.7	35.0

- 230.** Three floors are shored to support a concentrated load of 50 kips acting on the fifth floor. The axial loads acting on the 5th floor prop is 35 kips, on the 4th floor prop is 15 kips and on the 3rd floor prop is 5 kips. What is the maximum load distributed to an individual floor slab?

- (A) 15 kips
- (B) 20 kips
- (C) 35 kips
- (D) 50 kips



- 231.** The 6th floor of a concrete building is about to be cast. The shoring posts on each floor are laid at a spacing 5 ft X 5 ft. The thickness of all floor slabs are 6 inches. The weight of formwork and falsework can be assumed to be 15 psf. Density of concrete is 150 pcf. The construction load on the 5th floor is 50 psf and on the 6th floor is 75 psf. Assuming that all loads from the 6th floor and 5th floor are carried by formwork and shoring posts, what will be the maximum load acting on the shoring posts under 5th floor?



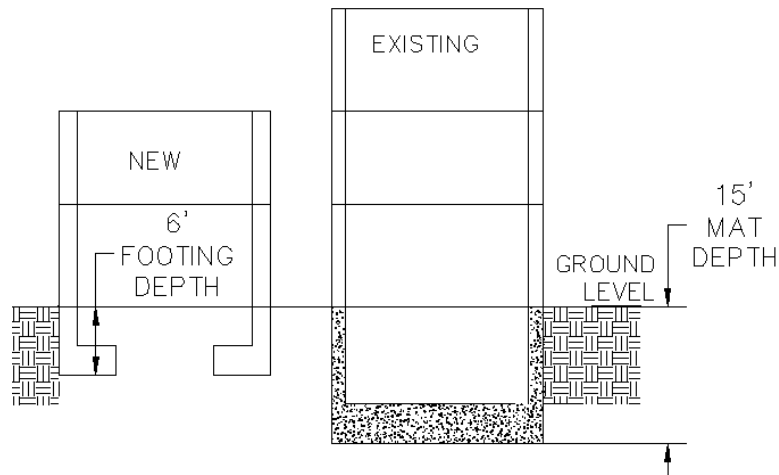
- (A) 4.7 kips
- (B) 5.8 kips
- (C) 6.3 kips
- (D) 7.6 kips

- 232.** For a concrete wall formwork, the plywood sheathing is supported by vertical posts and horizontal wales. The wales are spaced vertically at 24 inches center to center and the vertical posts are spaced 30 inches center to center horizontally. A bundle of six tie wire strands are used to tie the wales and posts together. Every alternate vertical post is tied to the wales. If the rate of concrete placement is 3 inches per hour and the lateral pressure exerted by the concrete on the forms is 400 psf, what is the maximum load acting on the tie wire?
- (A) 2 kips
 - (B) 4 kips
 - (C) 6 kips
 - (D) 8 kips
- 233.** A 10 ft X 5 ft scaffold platform is intended to support a load of 20 psf. The platform is supported by simply supported beams on the two longer sides. Calculate the design moment on the scaffold beam per OSHA regulations. The self weight of the beam is 40 plf.
- (A) 1 kip*ft
 - (B) 2 kip*ft
 - (C) 3 kip*ft
 - (D) 4 kip*ft
- 234.** A multi level guard rail is installed on a single point adjustable suspension scaffold. What is the design moment acting on the midrail per OSHA regulations, if it is welded between two posts spaced 5 ft apart.
- (A) 50 lbs*ft
 - (B) 75 lbs*ft
 - (C) 100 lbs*ft
 - (D) 150 lbs*ft
- 235.** For a 30 feet high, two story wooden scaffold supporting a uniform load of 50 psf, what is the maximum longitudinal and transverse spacing of a 4 in X 4 in wooden pole?
- (A) 6 ft X 5 ft
 - (B) 8 ft X 5 ft
 - (C) 10 ft X 5 ft
 - (D) 12 ft X 5ft

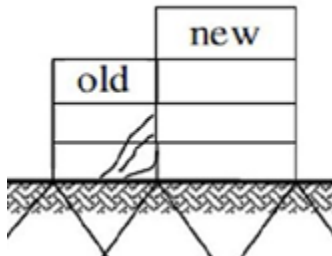
3.4 Impact of construction on adjacent facilities

[Click here for solutions 3.4](#)

- 236.** A two story building is to be constructed next to an existing building with a basement. The depth of foundation of the new building is 6 feet and the depth of basement mat of the existing building is 15 feet below grade. Assuming that the footing load due to the new structure follows a 45 degree influence line, what is the minimum distance between the two buildings so that the existing building is away from the influence of new building loads?



- (A) 6 ft
(B) 8 ft
(C) 9 ft
(D) 15 ft
- 237.** A new building is to be constructed next to an old building causing stress overlaps under their foundations. Which of the following is not a remedial measure to counter settlements in the old building due to the new construction?



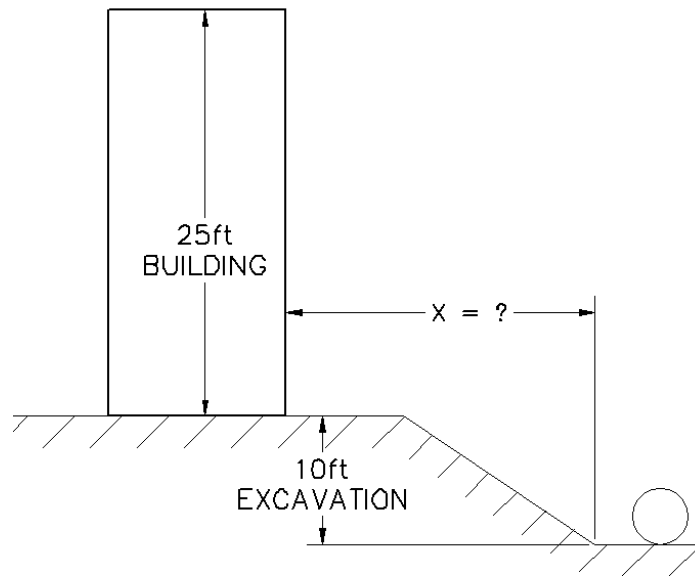
- (A) Improve the strength parameters of the soil underneath,
(B) Using pile foundations for the new building,
(C) Placing the foundation of the new building at a depth above the foundation of the old building,
(D) Designing lateral load bearing systems of the new building to be placed at the center of the new building. .

- 238.** Two adjacent buildings are to be constructed without setbacks from the property line. One of the buildings is 5 story tall and has a maximum inelastic response displacement of 7 inches per ASCE 7-16. Another building is 6 story tall and has a maximum inelastic response displacement of 9 inches. What is the required separation between the two buildings to prevent seismic pounding?
- (A) 9 in
 - (B) 12 in
 - (C) 16 in
 - (D) 20 in
- 239.** Per IBC 2018, which of the following is not true about protecting adjoining existing facilities during construction activities ?
- (A) Owners of adjoining buildings should be notified via a written notice before any excavation work.
 - (B) Proper provisions for water runoff during construction or demolition should be made,
 - (C) Owners of adjacent buildings should be notified so that all tenants will vacate before a deep foundation excavation job in an adjacent lot with appropriate financial compensation.
 - (D) Protections should be provided for adjacent footings, roofs, chimneys, skylights and adjacent walls.

3.5 Safety

[Click here for solutions 3.5](#)

- 240.** Utility lines are to be laid next to a 25 feet tall building at a depth of 10 feet below grade. An undisturbed perimeter of at least half the height of the building has to be maintained around the building. What is the minimum horizontal distance “X” from the face of the building to the bottom of the excavated slope following OSHA’s excavation safety regulations? The soil around the building is Type C per OSHA classifications.



- (A) 18 ft
(B) 22.5 ft
(C) 27.5 ft
(D) 30.75 ft
- 241.** A foreman at a demolition site is exposed to the following noise levels during an 8 hour working shift:
- 8 a.m. to 12 p.m. = 90 dBA
 - 12 p.m. to 2 p.m. = 100 dBA
 - 2 p.m. to 4 p.m. = 95 dBA
- What is the noise dose per OSHA regulations?
- (A) 80%
(B) 90%
(C) 100%
(D) 200%

242. A wire rope bundle is used in a powered platform mechanism and the maximum static tension on the bundle including the platform's self weight is 10 kips. Per OSHA regulations, what is the minimum required rated strength of the wire rope bundle

- (A) 10 kips
- (B) 40 kips
- (C) 60 kips
- (D) 100 kips

243. Which of the following is not true about fall protection per OSHA regulations?

- (A) Anchorages used for attachment of personal fall arrest equipment needs to be capable of withstanding at least 5 kips.
- (B) Personal fall arrest systems when stopping a fall shall bring an employee to a complete stop and limit maximum deceleration distance an employee travels to 6 feet,
- (C) Each employee on a walking surface with an unprotected edge which is 6 feet or more above a lower level shall be protected from falling,
- (D) Each employee on walking/working surfaces shall be protected from falling through holes more than 6 feet above lower levels,

244. In a pre-engineered building, a 2nd story bolted rigid moment connection joint has 8 bolts on each side of the web. During erection, what is the minimum number of bolts on each side of the web that needs to be installed and tightened before the hoisting equipment can be safely released?

- (A) 2
- (B) 4
- (C) 6
- (D) 8

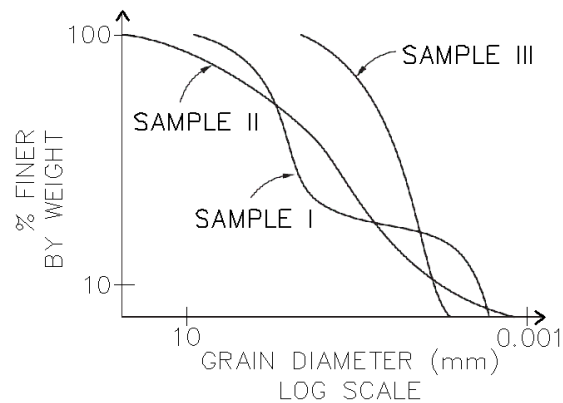
4. Design and Details of Structures–Materials and Material Properties

Sub- Headings	Number of Problems
Soil classification and soil properties	245-259
Concrete	260-275
Steel	276-290
Timber	291-298
Masonry	299-307
Material test methods and specification conformance	308-315
Total number of Problems	71

4.1 Soil classification and soil properties

[Click here for solutions 4.1](#)

245. The graph below shows particle size distribution for three soil samples I, II and III. Which of the following denotes the correct grading of the samples?



- (A) I - Well graded II- Gap graded III- Uniform
(B) I - Gap graded II- Well graded III- Uniform
(C) I - Uniform II- Well graded III- Gap graded
(D) I - Gap graded II- Uniform III- Well Graded
246. The particle size distribution of an inorganic soil sample is as follows:

Liquid Limit = 51
Plastic Limit = 19

Determine the USCS classification of the soil sample.

- (A) GC
(B) SP
(C) CH
(D) MH

Soil Size (mm)	Fraction retained on sieve
<0.002	0.3
0.002-0.005	0.12
0.005-0.05	0.11
0.05-0.075	0.16
0.075-1.18	0.31
>1.18	0

247. The particle size distribution of an inorganic soil sample is as follows:

Liquid Limit = 55

Plastic Limit = 17

Determine the AASHTO classification of the soil sample.

- (A) A-2-7 (14)
- (B) A-2-7 (22)
- (C) A-7-6(14)
- (D) A-7-6(22)

Soil Size (mm)	Fraction retained on sieve
<0.005	0.42
0.005-0.05	0.11
0.05-0.075	0.12
0.075-2.0	0.35
>2.0	0

248. Classify the following inorganic soil sample according to USCS.

Percentage passing through 0.075 mm sieve = 11%

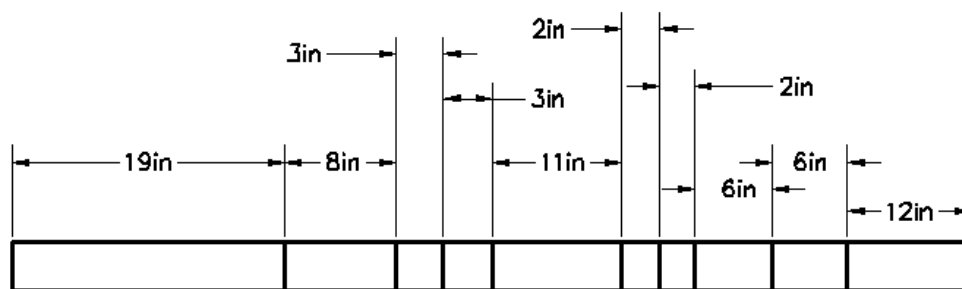
Percentage passing through 4.75 mm sieve = 70%

Liquid Limit = 46 Plastic Limit = 36 $D_{10} = 0.12$ $D_{60} = 0.8$

Determine the USCS classification of the soil sample.

- (A) SM
- (B) SC
- (C) SW-SM
- (D) SW-SC

249. A 6ft core drilling was carried out at a site with bedrock and the core recovery is:



What is the quality of rock?

- (A) Fair
- (B) Good
- (C) Very Good
- (D) Excellent

- 250.** The figure below shows the subsurface profile of a site. The groundwater is at EL. 58 ft. What is the effective stress at elevation 45 ft?

GRAVEL, 120 pcf	EL. 75ft
SILT A, 130 pcf	EL. 60ft
SILT B, 128 pcf	EL. 55ft
SAND, 125 pcf	EL. 40ft
ROCK, 140 pcf	EL. 20ft
	EL. 0ft

- (A) 2920 psf
 (B) 3110 psf
 (C) 3230 psf
 (D) 3400 psf
- 251.** Standard penetration test was conducted at a site. At the required depth , the effective overburden pressure was 4 ksf and the energy corrected N_{60} value was 20 blows. What is the estimated in situ dry unit weight of the soil sample?
 Data obtained from lab tests are:
 $e_{max} = 0.4$ $e_{min} = 0.2$ specific gravity of soil sample , $G_s = 2.7$
- (A) 115.2 pcf
 (B) 120.8 pcf
 (C) 125.4 pcf
 (D) 131.6 pcf

252. A soil sample has a density of 1900 kg/m^3 . The water content ratio is 19% and the specific gravity of the sample is 2.7. What is the degree of saturation of the sample?

- (A) 55%
- (B) 65%
- (C) 75%
- (D) 85%

253. An excavation trench is to be filled using soil from a borrow pit nearby. The in situ density of the soil in the borrow pit is 1600 kg / m^3 and the in situ water content is 10 %. For the fill area, the required dry density of the soil must be 1800 kg/m^3 and the water content must be 12 %. What is the volume of soil to be excavated from the borrow pit area if the volume of the trench to be filled is $20,000 \text{ m}^3$?

- (A) 22500 m^3
- (B) 24750 m^3
- (C) 26250 m^3
- (D) 28000 m^3

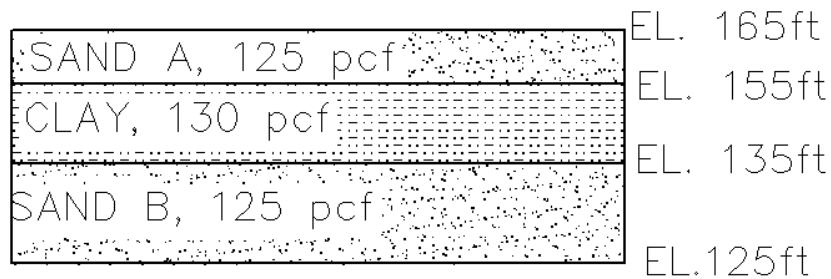
254. Standard proctor test is conducted on a soil sample and the results were as follows:

Mass of sample filling standard mold(lbf)	Water content
4.11	8
4.22	9
4.35	10
4.31	11
4.28	12

The sample tested at site had a total unit weight of 120 pcf and dry unit weight of 112.7 pcf . What is the estimated relative compaction?

- (A) 81%
- (B) 85%
- (C) 91%
- (D) 95%

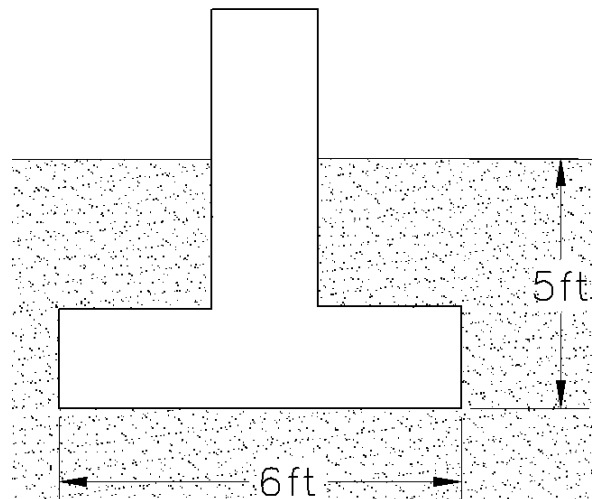
255. Clay layer is bound between two sand layers as shown below:



The coefficient of consolidation of the clay is $0.29 \text{ ft}^2/\text{day}$. Determine the time in days for 60% average degree of consolidation.

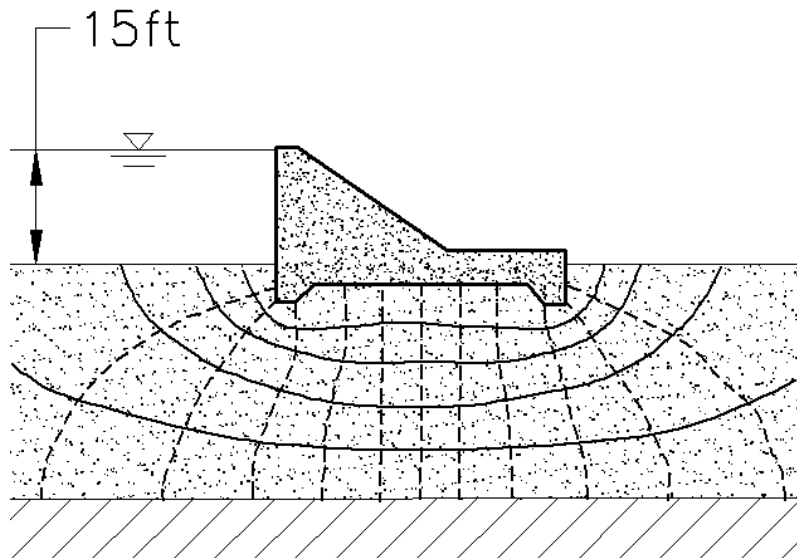
- (A) 59 days
 - (B) 99 days
 - (C) 180 days
 - (D) 395 days
256. Which of the following is not true about liquefaction of soil?
- (A) Foundations built in submerged sandy soils are susceptible to liquefaction.
 - (B) Clayey soils have higher cohesion and density and are immune to liquefaction.
 - (C) During a strong earthquake, the pore water pressure increases and the soil loses its ability to support loads and liquefies.
 - (D) Liquefied soil causes higher lateral earth pressures on retaining walls.

257. Calculate the net allowable bearing capacity for the square footing shown below. The factor of safety against bearing is 3. The groundwater table is observed at a depth of 25 feet. The density of soil is 115 pcf, the angle of internal friction is 20° and the cohesion = 350 psf.

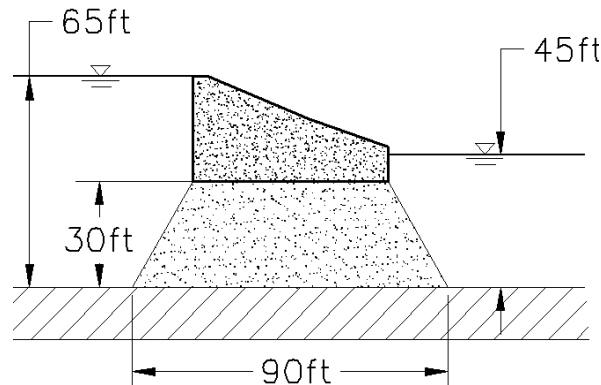


- (A) 3.57 ksf
- (B) 4.33 ksf
- (C) 5.22 ksf.
- (D) 6.18 ksf

258. The figure below shows a flow net for seepage under an embankment. Under the flow net, there is an impervious layer. The permeability of the soil is 0.0001 ft/min. What is the estimated seepage per day?



- (A) 0.9 cubic feet/ day/feet
 (B) 1.1 cubic feet/ day/feet
 (C) 1.3 cubic feet/ day/feet
 (D) 1.5 cubic feet/ day/feet
259. An embankment is made with impervious material on top of permeable fill. The permeable fill is 30 ft tall, 90 ft in length and has a permeability of 0.004 ft/min. The headwater and tailwater elevations are shown below. What is the discharge through the permeable layer? The width of the embankment is 300 ft.



- (A) 2 ft³ /min
 (B) 4 ft³ /min
 (C) 6 ft³ /min
 (D) 8 ft³ /min

4.2 Concrete

[Click here for solutions 4.2](#)

260. A pile cap is about to be cast in a coastal area and it is exposed occasionally to tides. The design strength of concrete is 4000 psi. Which of the following mixes should be chosen to cast the pier?

- (A) Type I cement, sand, aggregates, water, chemical accelerator,
- (B) Type I cement, sand, aggregates, water, super plasticizer,
- (C) Type II cement, sand, aggregates, water, chemical accelerator,
- (D) Type II cement, sand, aggregates, water, super plasticizer,

261. Three standard 4 inX 8 in cylinders were sampled from a slab pour job. The samples were cured and tested after 28 days with the following results:

Sample number	Axial compressive load during failure
1	50 kips
2	45 kips
3	57 kips

What is the 28 day compressive strength of concrete that is to be used for compliance verification?

- (A) 3581 psi
- (B) 3979 psi
- (C) 4032 psi
- (D) 4536 psi

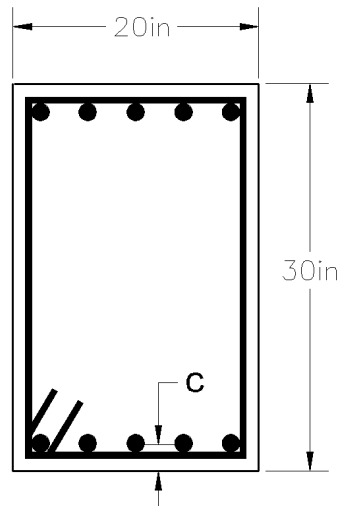
- 262.** The test results for standard 4 inX 8 in cylinders from a job site for the first six days of pour, cured for 28 days yielded the following results:

Day number	Compressive strength
1	3.12 ksi
2	3.4 ksi
3	4 ksi
4	4.1 ksi
5	2.98 ksi
6	3.9 ksi

If the design strength of concrete mix is 3500 psi, which of the following is true?

- (A) Mix is not compliant because tests from Day 1, Day 2 and Day 5 is below 3500 psi,
 - (B) Mix passes because overall average of the tests is above 3500 psi,
 - (C) Mix passes because the average of any three consecutive tests is above 3500 psi,
 - (D) Mix is not compliant because the test result from day 5 is below 3000 psi.
- 263.** A concrete mix is designed for a roof slab that will be subjected to snow and ice accumulation and frequent rain. The nominal aggregate size is 3/4 inch and the compressive strength of concrete is 4500 psi. What is the recommended water - cement ratio and total air content in the concrete?
- (A) Target air content 6 % and minimum w/c ratio 0.45
 - (B) Target air content 5 % and minimum w/c ratio 0.55
 - (C) Target air content 6 % and maximum w/c ratio 0.45
 - (D) Target air content 5 % and maximum w/c ratio 0.55
- 264.** What is the modulus of rupture of a 4ksi lightweight plain concrete when it is subjected to bending?
- (A) 0.35 ksi
 - (B) 0.4 ksi
 - (C) 0.45 ksi
 - (D) 0.5 ksi

- 265.** The section of a precast reinforced concrete beam that is manufactured under controlled conditions is shown below. The beam is a part of an open parking lot in a rainy area. What is the minimum distance “c” shown in the figure below? The main bars are number 8 bars and the stirrups are number 4 bars.



- (A) 2 in
(B) 2.5 in
(C) 3 in
(D) 3.5 in
- 266.** 250 ft³ of concrete mix is to be produced. The mix proportion by weight is 1: 2 :3. Six gallons of water is to be used per bag of cement. How many bags of cement will be required for the job if the target air content in the mix is 5%?

Density of cement = 195 lbf/ft³

Density of fine aggregate = 160 lbf/ft³

Density of coarse aggregate = 164 lbf/ft³

Weight of cement bag = 94 lbs

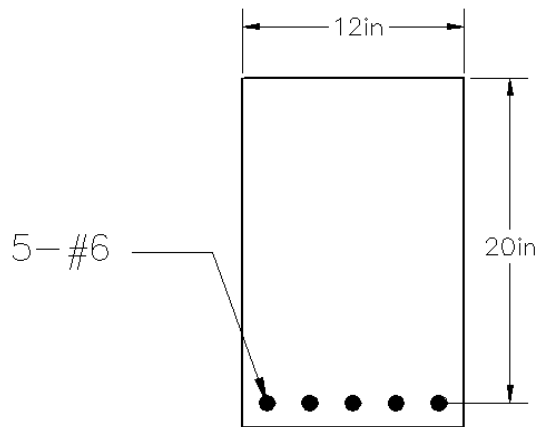
- (A) 40 bags
(B) 50 bags
(C) 60 bags
(D) 70 bags

267. A 12in X 18 in simply supported non prestressed singly reinforced concrete beam section is to be designed with an effective cover of 2 inches. The strength of concrete is 3500 psi and the rebar yield strength is 60 ksi. What is the minimum number of #4 main flexural reinforcement bars that should be used following ACI recommendations?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

268. For the section shown below, what is the strength reduction factor for flexure per ACI 318-14? Grade of steel is 60 ksi and concrete strength is 4000 psi.

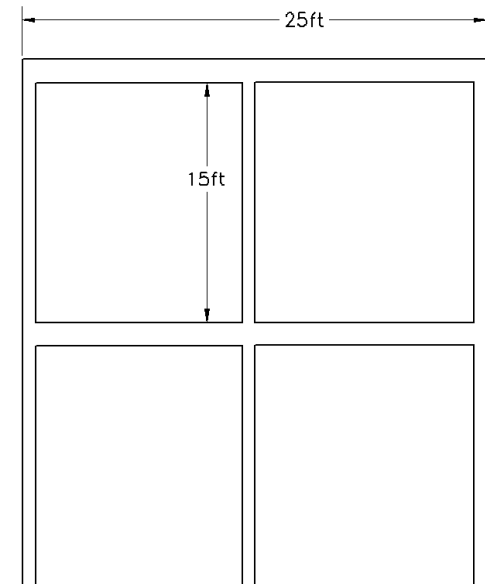
- (A) 0.65
- (B) 0.75
- (C) 0.85
- (D) 0.9



269. What is the cracked moment of inertia of a singly reinforced rectangular section with depth 24 in and width 12 in?. The beam is reinforced with 4#6 bars at the bottom and the effective depth is 22 in. The modulus of elasticity of concrete is 3600 ksi and the modulus of elasticity of rebar is 29000 ksi.

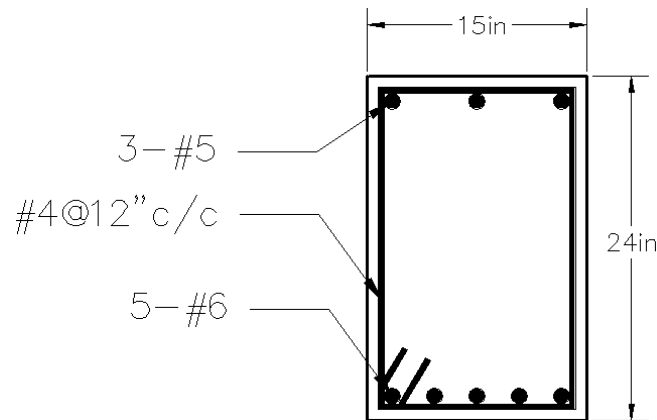
- (A) 924 in⁴
- (B) 1875 in⁴
- (C) 4494 in⁴
- (D) 4804 in⁴

- 270.** What is the slenderness ratio of the center-top column shown below per ACI 318-14? The span of each beam is 12.5 ft and height of each column is 15ft. The dimensions of all beams are 12 in X 24 in and the dimensions of all the columns are 14 in X 14 in. The same grade concrete is used for all floors and members.



- (A) 22
(B) 31
(C) 39
(D) 47
- 271.** What is the development length of #8 compression reinforcement with yield strength 60 ksi for 6000 psi normal weight concrete?
- (A) 16 in
(B) 18 in
(C) 20 in
(D) 30 in
- 272.** The design strength of concrete during initial prestress is 5000 psi and the 28 day design strength is 6500 psi. What is the allowable compressive stress at the midspan of the uncracked section due to prestressing plus a service live load that causes significant time dependent deflections per ACI?
- (A) 2250 psi
(B) 2925 psi
(C) 3000 psi
(D) 3900 psi

273. The section of a simply supported RCC beam is shown below. If the immediate deflection of the beam is 1 inch, what is the estimated long term deflection after 8 years? Effective cover can be assumed as 2 inches.

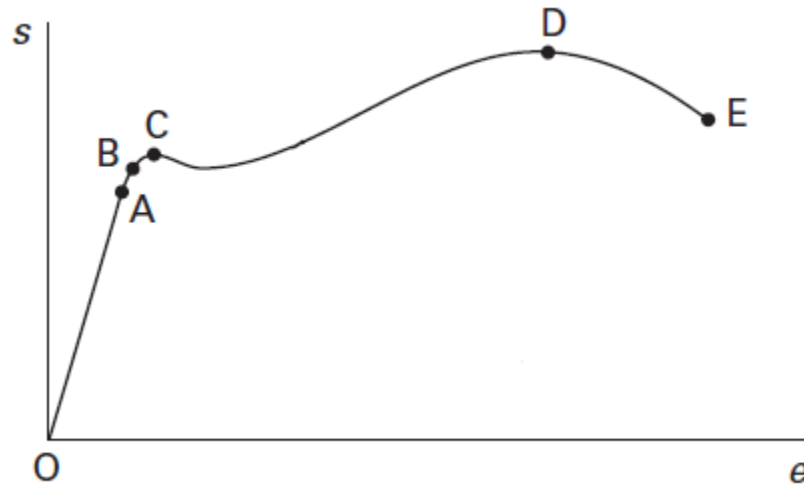


- (A) 1.25 in
(B) 1.5 in
(C) 1.75 in
(D) 2 in
274. What is the modular ratio of a generic non prestressed rebar and a concrete of strength 8000 psi and density 155 lb/ft³?
- (A) 5.1
(B) 6.2
(C) 7.3
(D) 8.4
275. The gross area of a prestressed girder is 500 in². The profile of tendons are concentric with the gross area of the girder throughout the span and the jacking force is 350 kips. The concrete is normal weight concrete and the strength of concrete during initial prestressing is 5000 psi. What is the estimated loss in prestressing because of shortening of concrete?
- (A) 3.75 ksi
(B) 4.46 ksi
(C) 5.45 ksi
(D) 6.51 ksi

4.3 Steel

[Click here for solutions 4.3](#)

276. A typical stress strain curve for mild steel is shown below. Which of the following is the correct descriptions of the points shown in the curve



- (A) A - Yield Point, C - Proportionality limit, D - ultimate strength, E - Fracture point
(B) A - Proportionality limit, B - Elastic limit, C - Yield point, E - Fracture point
(C) A - Elastic limit, C - Lower Yield point, D - Higher yield point, E - Ultimate strength
(D) A - Proportionality limit, C - Elastic limit, D - Yield point, E - Fracture point
277. Which of the following is not true about strain hardening of steel?
- (A) Strain hardening occurs when material is strained beyond the yield point.
(B) Strain hardening increases the hardness of the material but reduces the ductility of the and yield strength of the material.
(C) Strain hardening is used for steel used in Pipelines, storage tanks and pressure vessels.
(D) Certain plastics like steel also exhibit strain hardening capabilities.

278. Which of the following is true about effects of temperature on different properties of steel?

- (A) Increase in temperature increases the yield and ultimate strength of steel.
- (B) Increase of temperature reduces the ductility and hardness of steel.
- (C) Temperature has no effect on the modulus of commonly used structural steel.
- (D) At lower temperatures, the stiffness of steel members increases.

279. The yield strength of the threaded rod shown below is 120 ksi. The minimum ultimate strength is 150 ksi. What is the minimum allowable tensile strength and the minimum allowable shear strength for bearing type connections using the rod if the material conforms to AISC specifications?

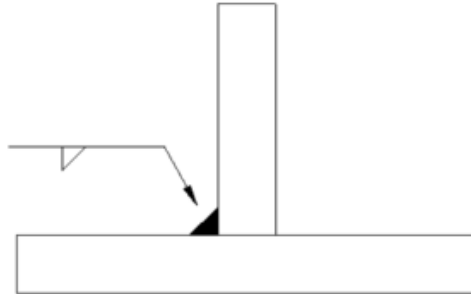


- (A) 56 ksi and 33 ksi
- (B) 84 ksi and 50.625 ksi
- (C) 90 ksi and 54 ksi
- (D) 112.5 ksi and 67.5 ksi

280. An ASTM A307 bolt is subjected to a shear stress of 10 ksi. What is the reduced allowable tensile stress in the bolt material when subjected to shear tension interaction?

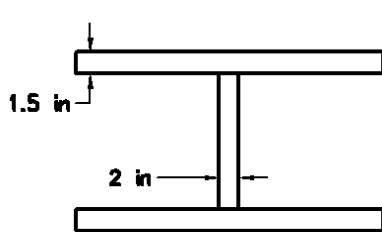
- (A) 8 ksi
- (B) 12 ksi
- (C) 25 ksi
- (D) 35 ksi

- 281.** The strength of filler metal for the weld below is 70 ksi. The weld size is $\frac{3}{8}$ inch. What is the allowable strength of the weld per unit length when it is loaded transversely to its longitudinal axis?

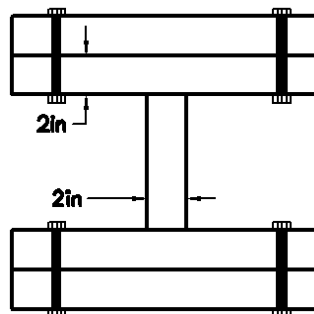


- (A) 5.57 kip/in
- (B) 8.35 kip/in
- (C) 11.14 kip/in
- (D) 16.7 kip/in

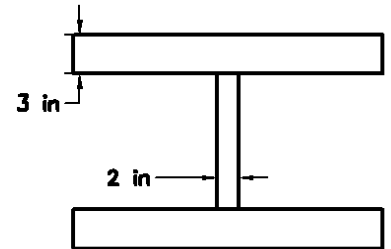
- 282.** Which of the following steel sections require an impact test per AISC specifications?



Section I



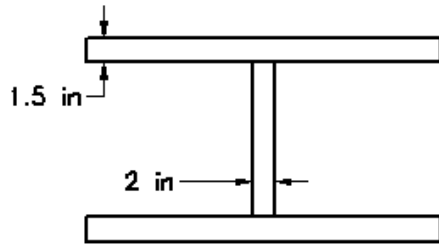
Section II



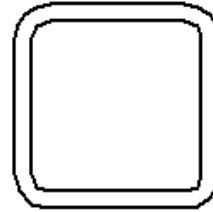
Section III

- (A) I, II and III
- (B) I and II only
- (C) I only
- (D) III only

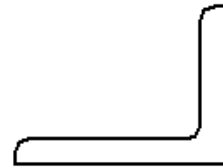
283. Match the preferred material specification with the appropriate steel section per AISC specifications.



(I) Plate Girder



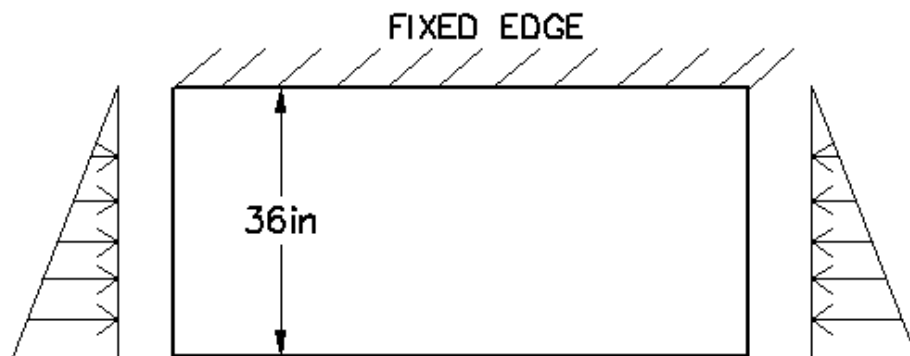
(II) HSS 8X8



(III) L8X6

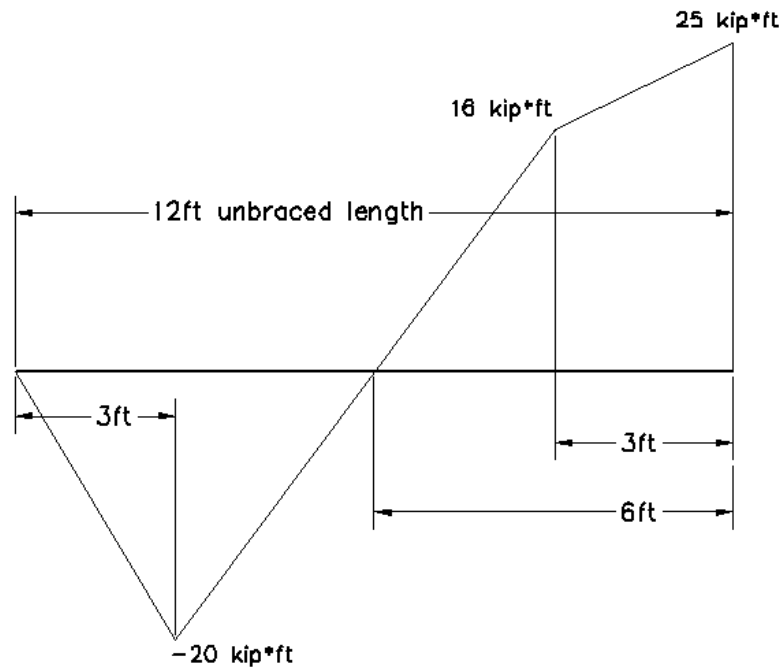
- | | | |
|--------------|-----------|------------|
| (A) I - A992 | II - A572 | III - A709 |
| (B) I - A572 | II - A500 | III - A36 |
| (C) I - A992 | II - A572 | III - A618 |
| (D) I - A500 | II - A572 | III - A36 |

284. A plate is rigidly welded at the top along the length as shown below and loaded along the thickness with a uniformly varying load. The thickness of the plate is 1 in and $E = 29000000$ psi. What is the nominal elastic buckling stress? The yield strength of the plate is 50 ksi.



- (A) 32.5 ksi
 (B) 42 ksi
 (C) 48.5 ksi
 (D) 50 ksi

285. For a doubly symmetric member, the moment diagram between two braced points is shown below. What is the lateral torsional buckling modification factor C_b ?

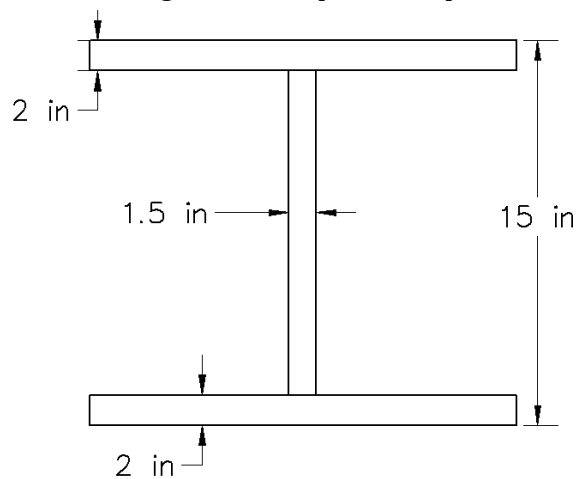


- (A) 1
(B) 1.67
(C) 1.83
(D) 2
286. An I beam has flange thickness of 0.8 in and flange width of 7 in. Both flanges have two bolt holes, one on each side of the web. The diameter of the holes, adjusted per AISC 360-16 B 4.3b, is 0.875 in. What is the nominal moment capacity of the section?
The elastic section modulus is 200 in^3 and the plastic section modulus is 240 in^3 . The yield strength of the material is 50 ksi and the ultimate strength is 65 ksi.
- (A) 6500 kip*in
(B) 7820 kip*in
(C) 9750 kip*in
(D) 12000 kip*in

287. Which of the following is not true about the properties of cold formed steel ?

- (A) Cold formed steel exhibits anisotropic material behavior due to the forming process,
- (B) Cold formed steel sections develop longitudinal residual stresses due to differential plastic deformation during forming,
- (C) Cold forming process induces plastic deformation, particularly at the corners, leading to strain hardening which leads to increased yield strength in these regions,
- (D) Because of the difficulty of cold working instead of being hot rolled, the dimensional tolerances for cold formed steel are higher than that for hot rolled sections.

288. A built up plate girder is made using A572 Gr 50 steel plates. The webs of the girder are stiffened with transverse stiffeners with 40 inches clear spacing between the stiffeners. What is the web plate shear buckling coefficient per AISC specifications?



- (A) 5
- (B) 5.34
- (C) 5.53
- (D) 5.7

- 289.** A Wide flange section has 1 inch cover plates welded to the bottom flange. The moment of inertia of the composite section is 2238 in^4 . The location of the centroid from the top flange is 9.8 in. What is the maximum flexural stress acting on the cover plate due to a moment of 350 kip*ft?



- (A) 11.6 ksi
(B) 18.4 ksi
(C) 22.3 ksi
(D) 27.5 ksi
- 290.** A threaded rod is pretensioned to 102 kips during installation. The slip critical connection is in double shear and the surface can be classified as class A. The bolt holes are standard size. What is the allowable shear strength of the connection? There is only one shim plate between the connected parts.

- (A) 20 kips
(B) 22 kips
(C) 40 kips
(D) 44 kips

4.4 Timber

[Click here for solutions 4.4](#)

- 291.** Which of the following properties of structural timber is most significantly impacted due to the presence of knots?
- (A) Maximum density of the member,
 - (B) Allowable compression parallel to the grain,
 - (C) Allowable tension perpendicular to the grain,
 - (D) Modulus of elasticity and stiffness
- 292.** Which of the following is true about the influence of specific gravity of wood on its mechanical properties?
- (A) Specific gravity of wood only affects thermal and moisture related properties of wood,
 - (B) Specific gravity of wood is directly proportional to the modulus of elasticity and inversely proportional to the modulus of rigidity,
 - (C) Specific gravity has negligible effects on mechanical properties except during dynamic loading conditions,
 - (D) Wood with higher specific gravity tends to swell and shrink more.
- 293.** A 4X10 Aspen No. 1 is to be used as a post. What is the ASD adjusted value of compression parallel to the grain for live load conditions? Assume 90°F temperature and wet service conditions. Take $C_i = 1$, $C_p = 1$.
- (A) 480 psi
 - (B) 540 psi
 - (C) 600 psi
 - (D) 700 psi

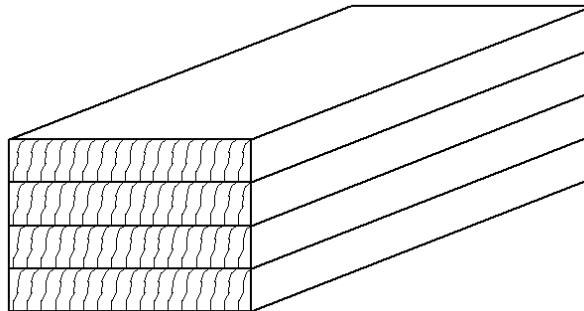
294. A 4X8 Alaska Yellow cedar No. 2 is to be used as a flexural member. What is the ASD adjusted value of bending for wind load conditions? Assume 110°F temperature and moisture content of 19 %. Take $C_i = 1$, $C_L = 0.9$.

- (A) 800 psi
- (B) 935 psi
- (C) 1127 psi
- (D) 1258 psi

295. A 4X6 Southern Pine No. 2 is to be used as a tension member. What is the ASD adjusted value of tension for construction loads? Assume 95°F temperature and moisture content of 20 %. Take $C_i = 0.9$.

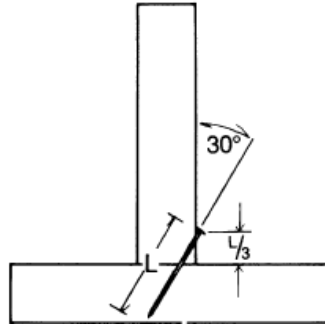
- (A) 600 psi
- (B) 675 psi
- (C) 750 psi
- (D) 900 psi

296. Determine the ASD adjusted modulus of elasticity for stability calculations of a 5 DF L1 with 4 laminations. The member is primarily subjected to axial loads. Service temperature = 120°F and service condition is wet.



- (A) 794 ksi
- (B) 832 ksi
- (C) 1060 ksi
- (D) 1128 ksi

- 297.** A toe nail connection is made using 3.5inch long 16D common wire nails between two Douglas Fir Larch members as shown below. The dressed size of the side member is 1.5X1.5. Determine the ASD adjusted lateral load capacity for live loads. Assume dry service conditions during fabrication and service and temperature 100°F. Take end grain factor = 1.



- (A) 91 lbs
(B) 117 lbs
(C) 141 lbs
(D) 155 lbs
- 298.** A simply supported wooden floor beam is subjected to a uniformly distributed live load of 50plf. The center to center distance between the end bearing plates is 12 ft. What is the minimum required moment of inertia of a Select structural Douglas Fir Larch dimensional lumber to satisfy IBC deflection limits? Assume $C_i = 1$ and dry conditions, service temperature less than 100°F.
- (A) 31 in⁴
(B) 35 in⁴
(C) 41 in⁴
(D) 50 in⁴

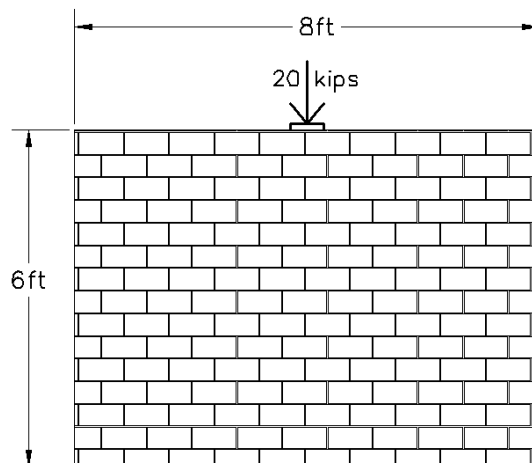
4.5 Masonry

[Click here for solutions 4.5](#)

299. The specified compressive strength of a concrete masonry unit is 1500 psi. The masonry unit is constructed with Type M mortar using ordinary portland cement. The unit is unreinforced, fully grouted. What is the allowable flexural tensile stress normal to the bond ?

- (A) 33 psi
- (B) 40 psi
- (C) 53 psi
- (D) 65 psi

300. What is the axial stress acting in the wall at mid height due to a 20 kips load acting on a 5 in X 5 in bearing plate on the top? The masonry units are standard hollow 8X8X16 CMU with face shell thickness of 1.25 in and web shell thickness of 1 in. Mortar is not provided at the webs of the units.

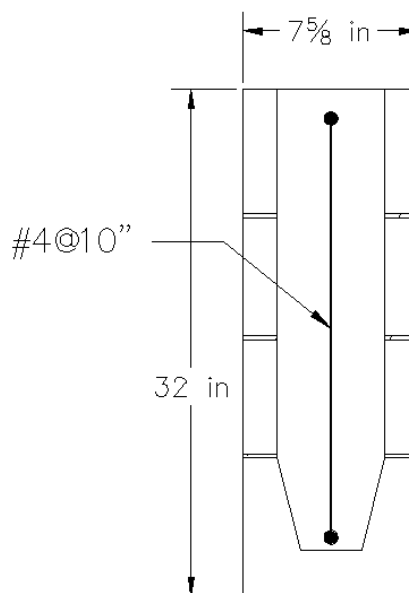


- (A) 133 psi
- (B) 157 psi
- (C) 195 psi
- (D) 250 psi

301. Which of the following is not true about the property of mortar used in masonry construction?

- (A) During construction, the temperature of the mortar mix should always be maintained less than 120°F and above 40°F.
- (B) The strength of the mortar should always be greater than the strength of the individual masonry units,
- (C) Type M mortars have high strength and is used for heavy duty applications,
- (D) Type N and Type O mortars have high workability.

302. The section of a fully grouted masonry beam is shown below. Grade 60 #4 bars @ 10 inches center to center are used as shear reinforcement. Assuming that the shear force is entirely resisted by the steel reinforcement, what is the maximum allowable shear force?



- (A) 8 kips
- (B) 10 kips
- (C) 12 kips
- (D) 14 kips

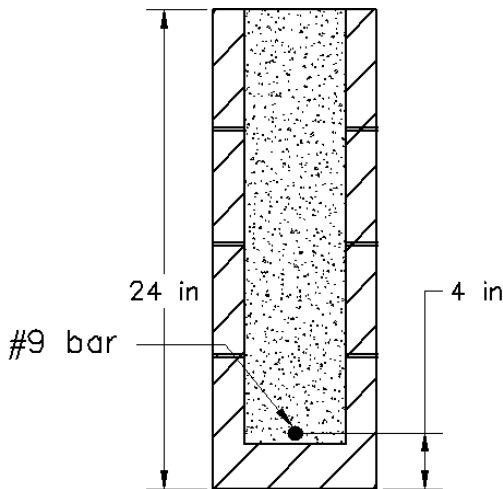
303. An unreinforced, fully grouted masonry wall spans 12 ft vertically from the slab on grade to the roof level. The load bearing wall is laterally supported by cross walls at 10 ft center to center. What is the minimum thickness of the wall to be considered for design?

- (A) 4 in
- (B) 6 in
- (C) 8 in
- (D) 10 in

304. What is the development length of a No. 6 bar used as tensile reinforcement in a masonry wall with specified strength 2000 psi ? The clear spacing between the bar splices is 3.5 inches and the minimum cover is 3 inches. All rebars used are Grade 60.

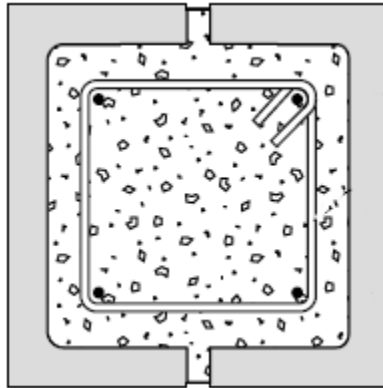
- (A) 12 in
- (B) 36 in
- (C) 43 in
- (D) 48 in

305. The section below shows a fully grouted concrete masonry beam. The compressive strength of masonry is 2000 psi. The section is flexurally reinforced with one # 9 bar (Gr. 60). Determine the tensile stress on the rebar due to a moment of 25 kip*ft acting on the beam. Width of the section = 7.625 in



- (A) 11 ksi
- (B) 13 ksi
- (C) 17 ksi
- (D) 21 ksi

- 306.** What is the minimum longitudinal reinforcement permitted in the masonry column shown below? The outer dimensions of the column are 9.625 in X 9.625 in.

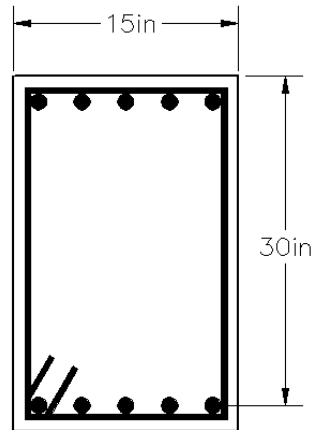


- (A) Two #3
 - (B) Two #4
 - (C) Three #3
 - (D) Four #3
- 307.** What is the fire resistance rating of a concrete masonry wall constructed of calcareous gravel that has an equivalent thickness of 5 inches?
- (A) 1.5 hour
 - (B) 2.25 hour
 - (C) 2.75 hour
 - (D) 3 hour

4.6 Material test methods and specification conformance

[Click here for solutions 4.6.](#)

- 308.** A reinforced concrete beam is designed and approved to be constructed as shown in the section below. What is the minimum tolerable dimension of the effective depth 'd' while placing the rebars at the construction site per ACI 318-14?

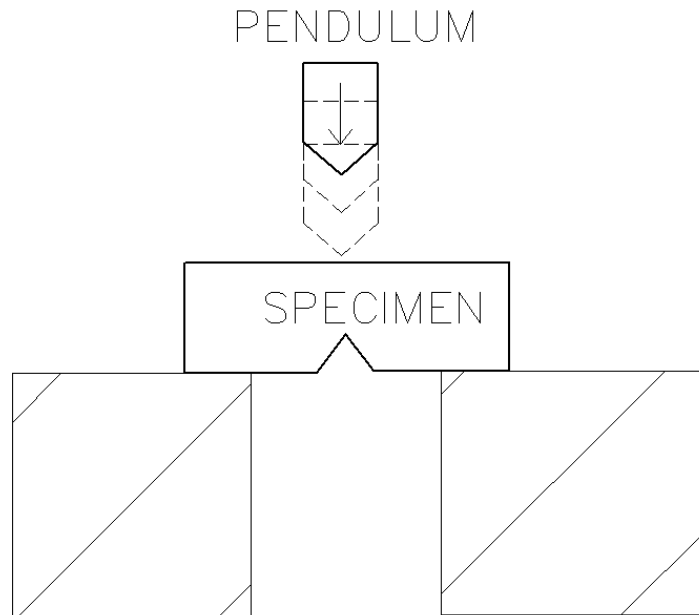


- (A) 29 in
(B) 29.375 in
(C) 29.5 in
(D) 29.75 in
- 309.** Which of the following is not true about concrete compressive strength tests methods and sampling?
- (A) A strength test shall be the average of strengths of at least two 6X12 or 4X8 cylinders.
(B) Samples of the concrete shall be made at the point of delivery of concrete.
(C) Concrete cylindrical specimens should be tested per ASTM C39
(D) Sampling of concrete cylinders should be done per ASTM C172

- 310.** Two rebars shown below are identified as ASTM A615 and ASTM A706. Which of the following is true about the rebars?



- (A) Rebar with marking C denotes carbon steel Rebar
 - (B) Rebar with marking S denotes ASTM A706 low-alloy steel rebar.
 - (C) ASTM A615 rebars are more suitable for seismic reinforcement than A706 rebars because they are stronger than A706 rebars.
 - (D) ASTM A615 rebars are more affordable and less ductile than A706 rebars.
- 311.** A V- notched steel specimen is centered on simple supports and is struck with a falling pendulum. The kinetic energy of the falling pendulum to cause failure of the specimen is recorded at standard temperature. Which of the following tests is represented by this procedure?



- (A) Toughness test
- (B) Hardness test
- (C) Fatigue test
- (D) Fracture test

312. During a standard penetration test, the potential energy of the hammer before the drop was measured as 467 J and the energy transferred to the drill rod was 375 J. If the number of blows measured during the test was 8, what is the correct N_{60} value of blows?

- (A) 6
- (B) 9
- (C) 10
- (D) 11

313. After a standard penetration test, which of the following properties of the soil can be correlated with the corrected number of blows 'N' ?

- I) Unconfined compressive strength
- II) Relative density
- III) Angle of internal Friction

- (A) I and II
- (B) II only
- (C) I, II and III
- (D) I only

314. A concrete masonry assembly is constructed using individual units of strength 2000 psi tested per ASTM C90 and using Type N mortar. Thickness of mortar beds is $\frac{1}{2}$ inch. What is the specified compressive strength 'f'm' of the masonry assembly?

- (A) 1500 psi
- (B) 1750 psi
- (C) 2000 psi
- (D) 2650 psi

315. For the bolt shown below, which of the following material tests are required to comply with ASTM specs?



- I) Tensile strength test
- II) Proof load test
- III) Coating test

- (A) I and II
- (B) II only
- (C) I, II and II
- (D) I only

5. Design and Details of Structures–Component Design and Detailing

Sub- Headings	Number of Problems
Horizontal members	316-355
Vertical members	356-390
Systems	391-415
Connections	416-467
Shallow foundations	468-487
Deep foundations	488-494
Retaining walls	495-501
Total number of Problems	186

5.1 Horizontal members

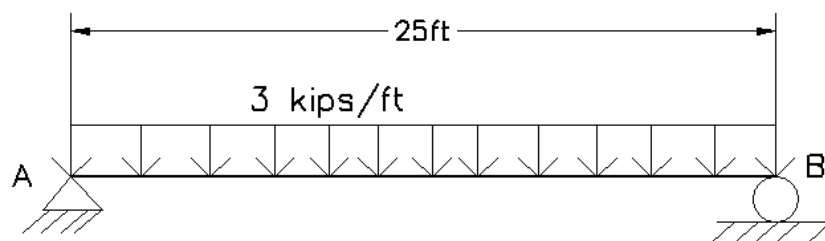
[Click here for solutions 5.1](#)

- 316.** A simply supported steel beam has a span of 25 ft and is used as a floor member. The beam is subjected to a concentrated service live load of 80 kips at the midspan and bends about the major axis. What is the lightest A992 section that will satisfy the IBC 2018 serviceability requirements?

$E = 29000 \text{ ksi}$

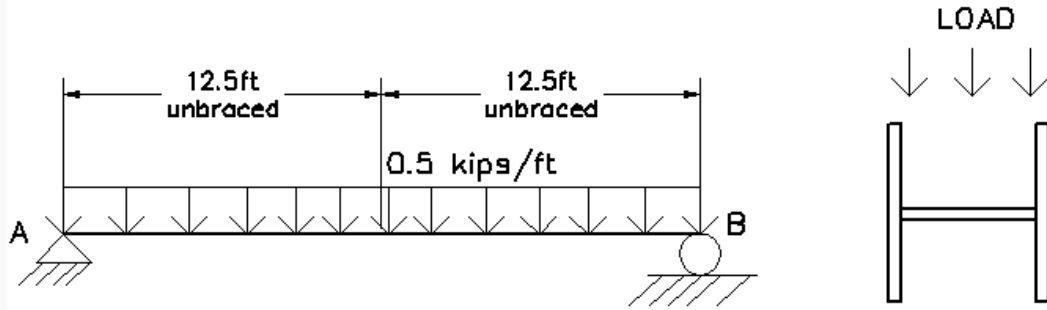


- (A) W18X86
(B) W18X97
(C) W18X106
(D) W18X119
- 317.** A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed live load of 3 kip/ft and bends about the major axis. What is the lightest Gr. 50 section that will satisfy the flexural strength requirements?

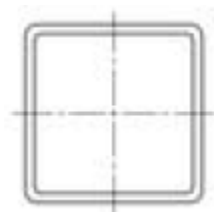


- (A) W18X40
(B) W18X50
(C) W18X55
(D) W18X60

- 318.** A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed service live load of 0.5 kip/ft and bends about the minor axis. What is the lightest Gr. 50 section that will satisfy the flexural strength requirements?

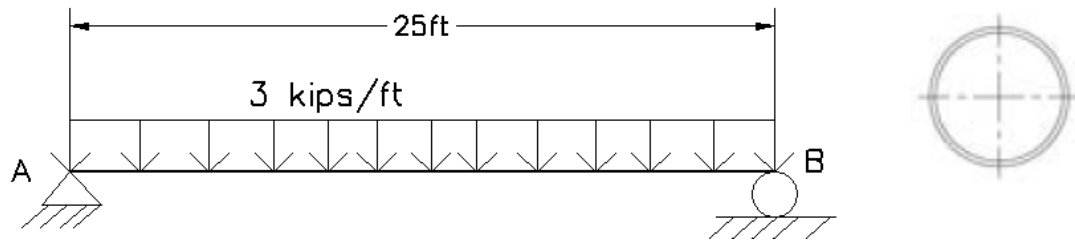


- (A) W18X46
 (B) W18X50
 (C) W18X55
 (D) W18X60
- 319.** A simply supported HSS steel beam has a span of 30 ft and is used as a floor member. The beam is subjected to a service live load of 5 kip/ft and a service dead load of 1 kip/ft and bends about the major axis. What is the lightest section that will satisfy the IBC 2018 serviceability requirements?
 $E = 29000$ ksi

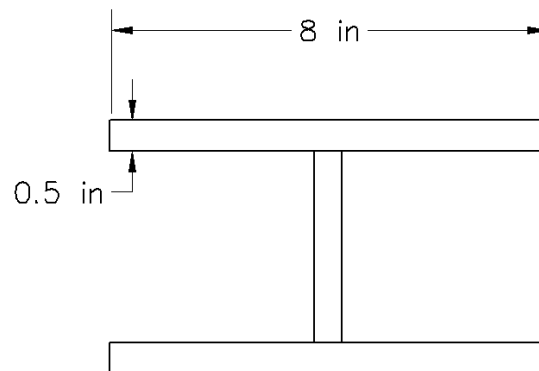


- (A) HSS 20 X 20 X 1/2
 (B) HSS 20 X 20 X 5/8
 (C) HSS 20 X 20 X 3/4
 (D) HSS 22 X 22 X 3/4

- 320.** A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed live load of 3 kip/ft. What is the lightest ASTM A500 Gr. C, 46 round HSS section that will satisfy the flexural strength requirements?

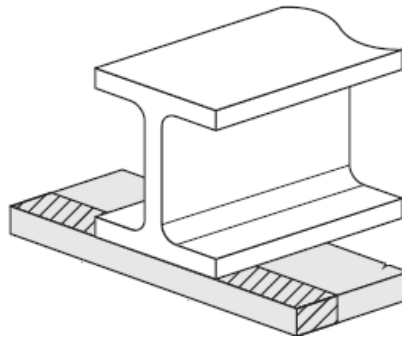


- (A) HSS 16.000 X 0.375
 (B) HSS 16.000 X 0.438
 (C) HSS 18.000 X 0.375
 (D) HSS 18.000 X 0.5
- 321.** A symmetric plate girder is constructed using A 572 Gr. 50 plates. The section of the girder is shown below. Based on the flange dimensions, which of the following is true about the compactness of the flange if the girder is subjected to axial loads and moments?



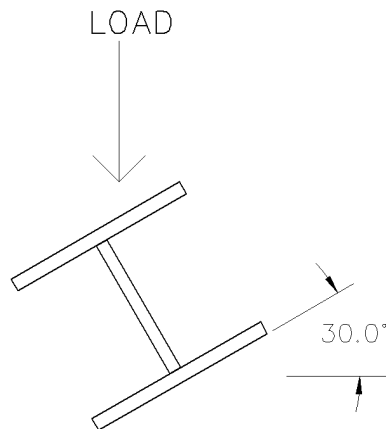
- (A) Compact
 (B) Non Compact
 (C) Slender
 (D) Cannot be determined with given information

- 322.** A simply supported Gr. 50 W30X90 beam is subjected to a uniformly distributed factored load of 10 kip/ft throughout the span of the beam. The span of the beam is 25 ft. What is the minimum length of the bearing plate 'lb' along the span of the beam?



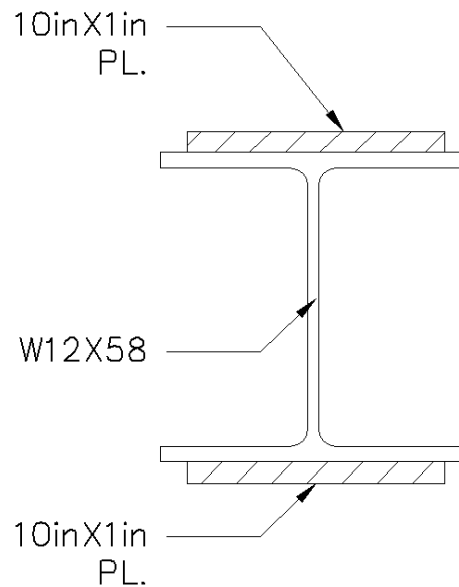
- (A) 2.5 in
- (B) 3.5 in
- (C) 4.5 in
- (D) 5.5 in

- 323.** A simply supported roof purlin has a span of 12 ft. The uniformly distributed load acting on the purlin is 1 kip/ft. If the allowable flexural strengths for bending in strong axis and weak axis are 30 kip*ft and 10 kip*ft respectively, what is the total flexural utilization of the section?



- (A) 50%
- (B) 75%
- (C) 90%
- (D) 150%

- 324.** 10 inch X 1 inch cover plates are welded effectively to the top and bottom flanges of a W12X58 beam section as shown below. What is the nominal plastic moment of resistance of the section about the major axis? $F_y = 50$ ksi for both plates and rolled section.



- (A) 850 kip*ft
 (B) 910 kip*ft
 (C) 980 kip*ft
 (D) 1055 kip*ft
- 325.** Determine the major axis nominal shear strength of a Gr 50 Channel section C15X40 beam. No transverse stiffeners are used. $E = 29000$ ksi



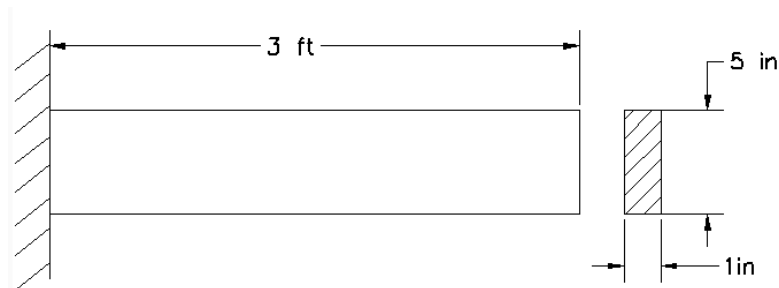
- (A) 168 kips
 (B) 192 kips
 (C) 234 kips
 (D) 263 kips

- 326.** Standard wide flange sections are used as horizontal members in a steel construction project to resist transverse loads. Transverse stiffeners are not used in the S beams. The analysis results show that maximum service level shear acting in the beam is 46 kips. What is the most economical section that can be used in the project? All beams have sufficient axial and flexural strengths.



- (A) S6X17.25
- (B) S8X18.4
- (C) S8X23
- (D) S10X25.4

- 327.** A cantilevered horizontal bar of cross section 5in X 1in is cantilevered with $L = 3\text{ft}$ as shown below. The yield strength of the material is 50 ksi and $E = 29000\text{ ksi}$. What is the allowable flexural strength of the bar? Assume $C_b = 1$.



- (A) 12.1 kip*ft
- (B) 13.2 kip*ft
- (C) 14.9 kip*ft
- (D) 15.5 kip*ft

- 328.** What is the maximum allowable concentrated load that can be applied to a Gr 50 ksi W10X30 section without the use of web stiffener plates. Assume load is applied sufficiently away from supports and length of concentrated load along the span = 1 in.

Section properties:

bf = 5.81 in
tf = 0.51 in
d = 10.5 in
tw = 0.3in
k = 0.81 in

- (A) 48 kips
- (B) 50 kips
- (C) 70 kips
- (D) 75 kips

- 329.** A built up section is to be used as a horizontal beam. The moment of inertia of the section about the stronger axis is 786 in^4 and about the weaker axis is 311.7 in^4 . The cross sectional area is 68 in^2 . What is the maximum unbraced length permitted in the beam section to prevent lateral torsional buckling?

Material properties:

Yield Strength = 50 ksi
Modulus of Elasticity = 29000 ksi

- (A) 5 ft
- (B) 7.5 ft
- (C) 9 ft
- (D) 11.5 ft

- 330.** A steel HSS gantry beam is subjected to the following loads:

Axial compression = 20 kips
Strong axis moment = 220 kip*in
Shear = 4 kips
Torsion = 45 kip*in

If the allowable moment strength = 500 kip*in, allowable shear strength = 20 kips and the allowable torsional strength is 200 kip*in, what is the minimum required axial strength of the beam?

- (A) 50 kips
- (B) 70 kips
- (C) 100 kips
- (D) 130 kips

331. What is the allowable bending moment for the reinforced masonry beam section below?

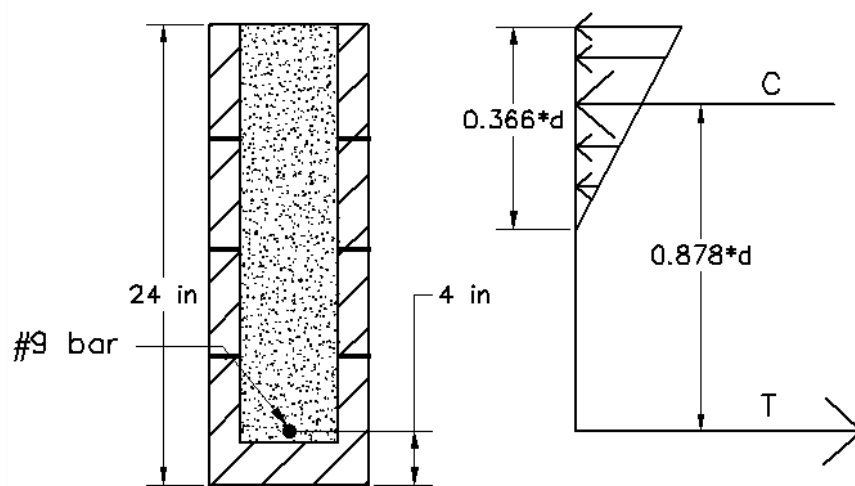
The section is flexurally reinforced with one no. 9 bar .

Section properties:

Rebar $f_y = 60$ ksi

Masonry $f'_m = 2000$ psi

Width of the section = 7.625 in.



- (A) 20 kip*ft
- (B) 27.5 kip*ft
- (C) 32.25 kip*ft
- (D) 36.75 kip*ft

332. A fully grouted reinforced masonry beam is subjected to a shear force of 13 kips. The compressive strength of masonry is 1500 psi. What is the minimum dimension of the masonry beam that satisfies the shear force requirement? Assume $M/Vdv > 1$.

- (A) $b = 7.625$ in , $dv = 16$ in
- (B) $b = 7.625$ in , $dv = 24$ in
- (C) $b = 7.625$ in , $dv = 32$ in
- (D) $b = 7.625$ in , $dv = 40$ in

333. A fully grouted reinforced masonry beam is subjected to a shear force of 13 kips. If a single legged #4 bar is used as shear reinforcement, what is the shear stress carried by the shear reinforcement?

Section properties:

$$f'_m = 1500 \text{ psi.}$$

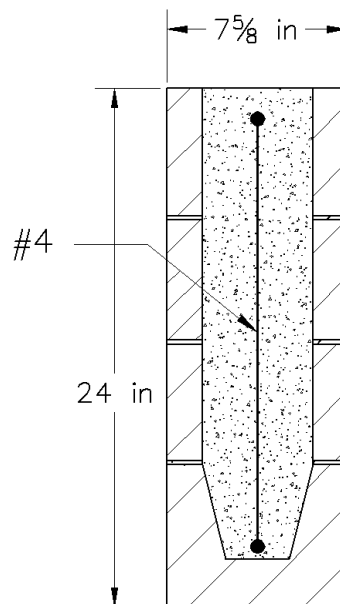
$$M/Vdv > 1$$

$$b = 7.625 \text{ in}$$

$$dv = 24 \text{ in}$$

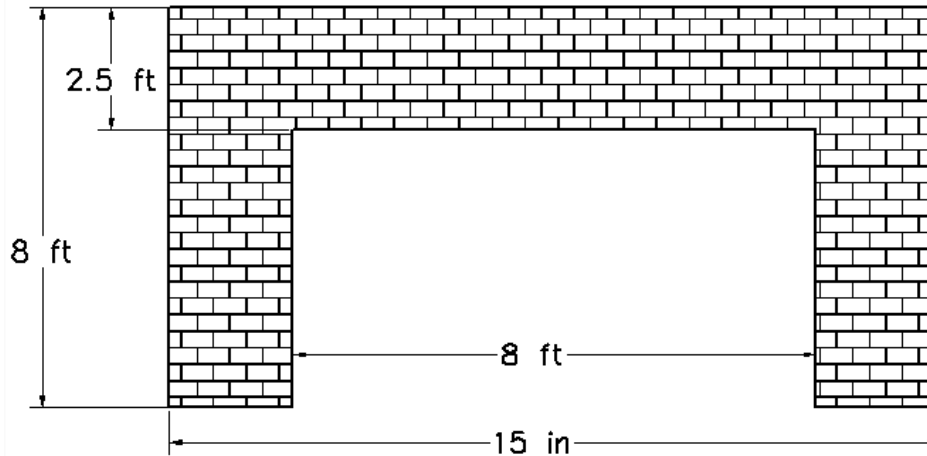
$$d = 20 \text{ in}$$

- (A) 19.2 psi
(B) 21 psi
(C) 22.5 psi
(D) 27.5 psi
334. For the masonry beam section shown below, shear stress resisted by masonry, $F_{vm} = 43.6 \text{ psi}$ and shear stress F_{vs} resisted by shear reinforcement = 27.5 psi . If a single legged #4 bar is used as shear reinforcement, what is the maximum spacing of the #4 bar permissible? Effective depth for tension bar, $d = 20 \text{ in}$.



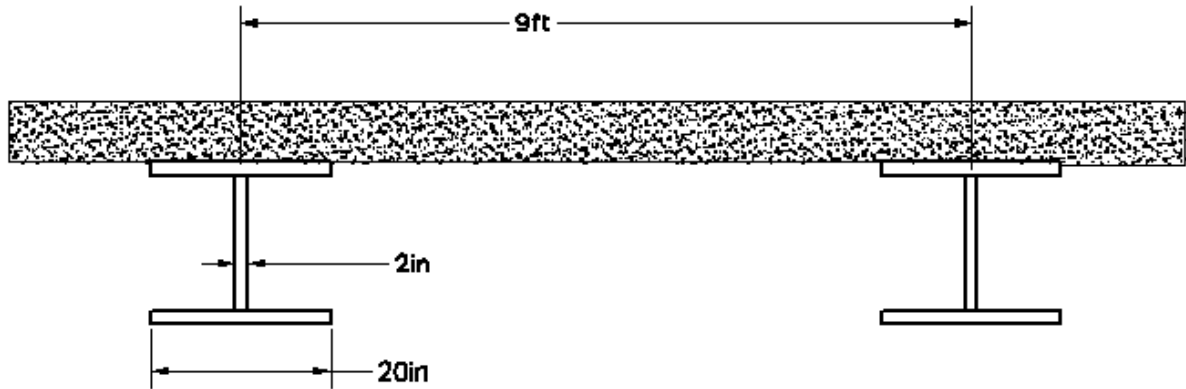
- (A) 6in
(B) 9 in
(C) 15 in
(D) 24 in

335. What is the effective span of the masonry beam shown below? Beam is not built integrally with the piers.

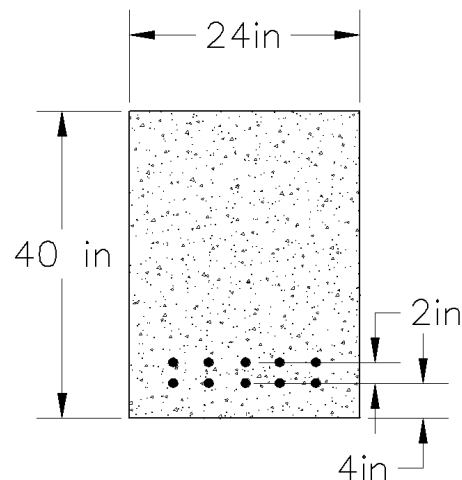


- (A) 8 ft
(B) 10.5 ft
(C) 11.5 ft
(D) 15 ft
336. Which of the following masonry beams can be considered a deep beam?
- I) A simply supported masonry beam with effective span 10 ft and depth 65 in
 - II) A continuous span masonry beam with span length 15 ft and depth 65 in
 - III) A continuous span masonry beam with section width 7.625 in and depth 72 in
- (A) I only
(B) I and II
(C) III only
(D) I, II and III

337. For the bridge section shown below, what is the effective span length 'Leff' of the concrete deck to be used for empirical design per AASHTO LRFD specification?



- (A) 7 ft
 (B) 7.667 ft
 (C) 8.083 ft
 (D) 9 ft
338. A prestressed girder section as shown below only has prestressing strands and no mild steel tensile reinforcement. The compressive stress at the top fiber of the beam due to self weight is 0.75 ksi. The strands are 1/2 in diameter low relaxation strands. Area of each strand is 0.153 in². After initial losses, the stress at release per strand is 160 ksi. What is the total top fiber stress at release at the midspan?



- (A) 0 ksi
 (B) 0.35 ksi tensile
 (C) 0.43 ksi compressive
 (D) 0.43 ksi tensile

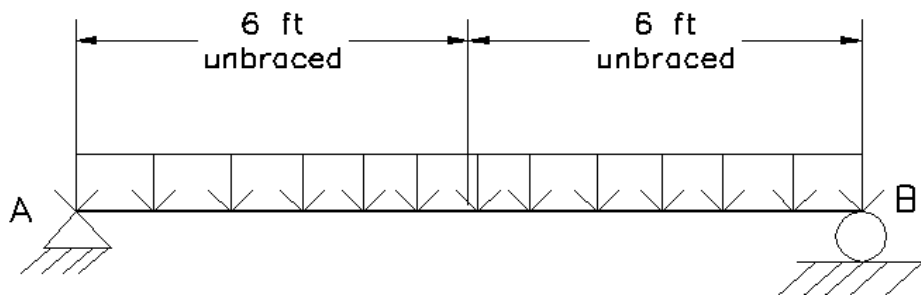
339. A 15 ft long simply supported 24F-V1 SP/SP timber beam has thickness of 4 inches and depth of 16 inches. If the beam is subjected to a uniformly distributed load of 100 lb/ft, what is the volume factor 'Cv' of the beam?

- (A) 0.8
- (B) 1
- (C) 1.01
- (D) 1.21

340. What is the minimum required section size of a DFL select structural grade lumber, used as a joist that is subjected to a live load moment of 980 lb*ft? Assume normal moisture and temperature conditions. C_i and $C_L = 1$. (Sizes shown are final dressed dimensions)

- (A) 2X3
- (B) 2X4
- (C) 2X5
- (D) 2X6

341. A simply supported 4X8 (nominal dimensions) southern pine section is subjected to a uniformly distributed load. The beam is braced laterally at midpoint. What is the beam slenderness ratio?



- (A) 7
- (B) 8
- (C) 9
- (D) 10

- 342.** A DFL select structural grade, simply supported beam has a span of 20 ft and is laterally unbraced. Determine the critical buckling design value F_{bE} for the beam if the beam slenderness $R_b = 8.3$. Assume wet conditions and normal temperature.

$C_i = 1$ and buckling stiffness factor, $C_T = 1$

- (A) 8.1 ksi
- (B) 9.7 ksi
- (C) 10.8
- (D) 12.5 ksi

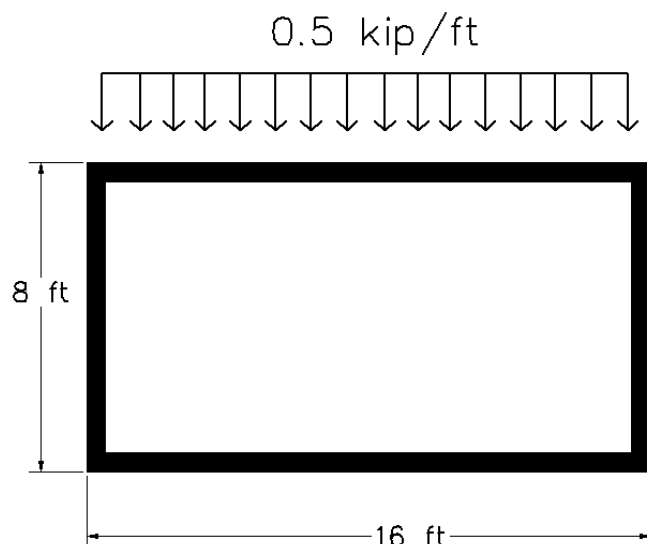
- 343.** An Eastern white pine select structural grade, simply supported rafter spans 15 ft. The rafter is spaced 22 inches center to center. The rafter is subjected to a uniform snow load of 50 plf and is loaded along the thickness. What is the minimum required section modulus of the lumber? Assume dry condition and normal temperature.

$C_L = C_i = C_f = 1$

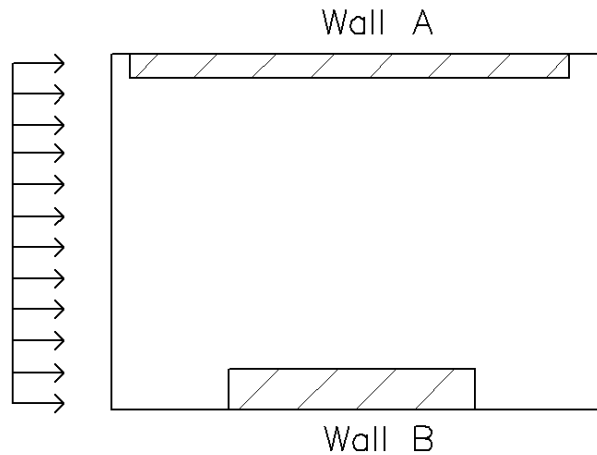
- (A) 10.2 in³
- (B) 12.3 in³
- (C) 14.2 in³
- (D) 15.7 in³

- 344.** The figure below shows a plan of a single story wood frame with Southern pine framing and blocked Structural I, 3/8 inch thick WSP diaphragm. The seismic loading is 0.5 kip/ft. What is the allowable strength of the diaphragm if 8d common nails are used at the diaphragm boundaries with a spacing of 4 inches. Assume nominal width of nailed face at panel edges and boundaries = 2 inches and dry conditions.

- (A) 360 plf
- (B) 720 plf
- (C) 1060 plf
- (D) 1200 plf

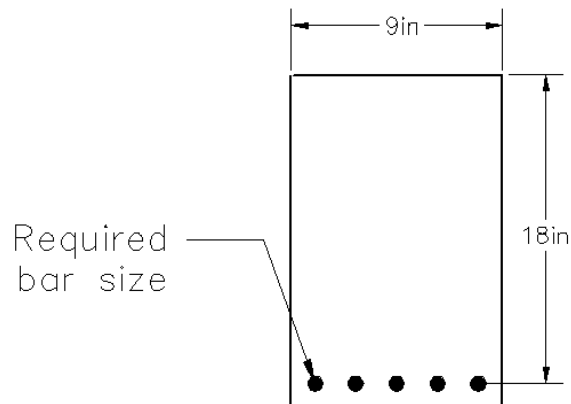


- 345.** The plan of a concrete floor diaphragm is shown below. The diaphragm is supported by two walls and subjected to seismic loads. The story drift of Wall A is analyzed as 0.45 inch and the story drift of wall B is analyzed as 0.6 inch. The in plane deflection of the diaphragm itself is 1.1 inches. Which of the following best describes the diaphragm?



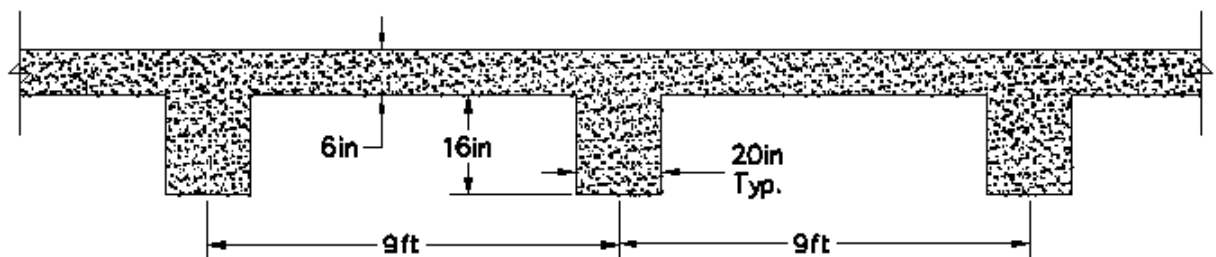
- (A) Rigid
(B) Rigid and torsionally irregular
(C) Flexible and torsionally irregular
(D) Flexible
- 346.** A four span continuous one way slab is constructed using Gr 60 rebar and lightweight concrete with $f'_c = 4000$ psi and unit weight 110 pcf. The interior spans are 21 ft in length and exterior spans are 17 ft in length. What is the minimum required thickness of the slab?
- (A) 6 in
(B) 8.5 in
(C) 9 in
(D) 10 in

- 347.** Concrete beam section shown below has to resist factored moment of 100 kip*ft. The compressive strength of the concrete is $f'_c = 4$ ksi and Gr. 60 rebars are to be used as flexural reinforcements. Determine the smallest bar size that satisfies the design requirement if five bars are to be used? Assume section satisfies minimum and maximum rebar requirements for all bar sizes and section is tension controlled.



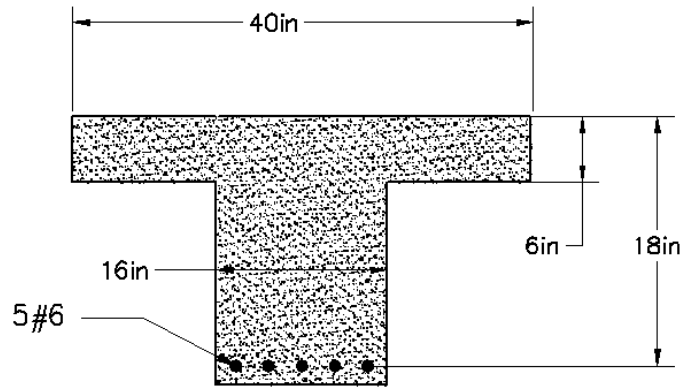
- (A) #4
- (B) #5
- (C) #6
- (D) #7

- 348.** What is the effective T- beam flange width for the center beam shown below ? The span of the RCC beams are 30ft and the clear span is 27ft.

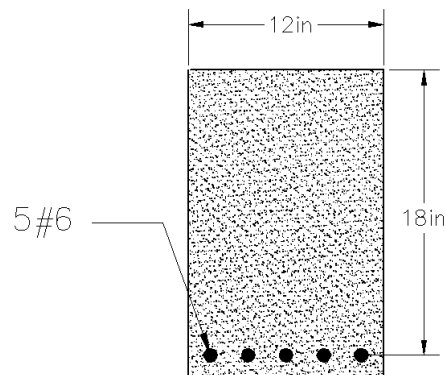


- (A) 48 in
- (B) 96 in
- (C) 101 in
- (D) 108 in

- 349.** What is the nominal moment capacity of the reinforced concrete T beam section shown below? The compressive strength of concrete is 4000 psi and the rebars are Grade 60.



- (A) 150 kip*ft
 (B) 170 kip*ft
 (C) 190 kip*ft
 (D) 230 kip*ft
- 350.** The factored shear to be resisted by the reinforced concrete section shown below is 51 kips. If two legged #3 stirrups are used, what is the maximum spacing of the stirrups required to satisfy the design requirements? Normal weight concrete with compressive strength $f'_c = 4$ ksi, all rebars are Grade 60.

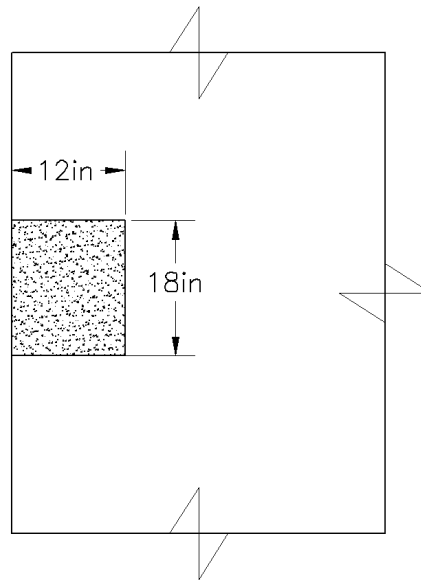


- (A) 6 in
 (B) 9 in
 (C) 12 in
 (D) 10 in

351. The factored shear to be resisted by a RCC beam section is 70 kips. The dimensions of the beam is 12inX24 in and the compressive strength of concrete is $f'_c = 4000$ psi. What is the maximum permissible spacing of stirrups per the provisions of ACI 318-14 for the beam section? Effective cover for tension bars = 2.5in. Concrete is normal weight.

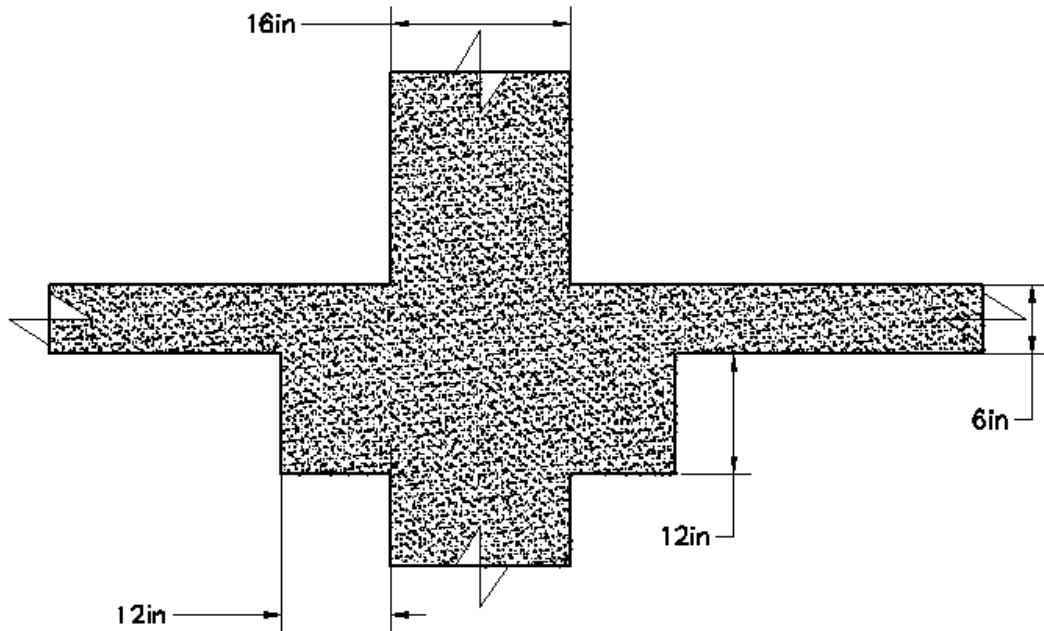
- (A) 6 in
- (B) 9 in
- (C) 12 in
- (D) 24 in

352. The plan below shows a column at the edge of a floor slab. The thickness of the slab is 8in and the effective cover is 1.5in. What is the design punching shear capacity of the slab at the location around the edge column? The compressive strength of the concrete slab is 6 ksi.



- (A) 59 kips
- (B) 72 kips
- (C) 83 kips
- (D) 103 kips

353. A two way flat slab is strengthened around an interior column using drop panels as shown below. Both the column and the panel are square in dimension. If the nominal two way shear strength of the concrete is 250 psi, what is the design two way shear strength of the slab at the column location? Assume effective cover of tension rebars in the slab is 1.25 in



- (A) 98 kips
(B) 115 kips
(C) 160 kips
(D) 189 kips
354. For a one way slab with thickness 5 inches and effective depth 4 inches, what is the maximum permitted spacing of flexural reinforcement (Gr 60 deformed bars)?
- (A) 9 in
(B) 12 in
(C) 15 in
(D) 18 in

- 355.** The maximum factored moment acting on a 4 inch thick one way slab is 0.7 kip*ft/ft. The compressive strength of concrete is 4 ksi and rebars used are grade 60. What is the area of reinforcement required to satisfy the design requirements per ACI 318-14? Assume effective cover for tension bars is 1in.
- (A) 0.064 in²/ft
 - (B) 0.086 in²/ft
 - (C) 0.098 in² /ft
 - (D) 0.12 in²/ft

5.2 Vertical members

[Click here for solutions 5.2](#)

- 356.** A fixed-fixed HSS 8X8X1/2 column is subjected to concentric axial compression. The height of the column is 19.25 ft. What is the maximum allowable axial compression that can be resisted by the column? $F_y = 70$ ksi and $E = 29000$ ksi



- (A) 200 kips
(B) 300 kips
(C) 440 kips
(D) 710 kips
- 357.** A pinned-fixed HSS column is subjected to a concentric unfactored axial compression of 15 kips. The height of the column is 13.5 ft and is unbraced in both directions. What is the lightest section that can withstand the applied load?



- (A) HSS 3 X 3 X 3/16
(B) HSS 3 X 3 X 1/4
(C) HSS 3 X 3 X 5/16
(D) HSS 3 X 3 X 3/8

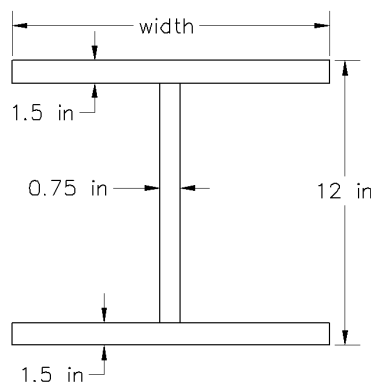
358. A cantilevered W8X40 column is subjected to compression load. The effective length in stronger - X direction, $KL_x = 28\text{ft}$. The effective length in weaker - Y direction, $KL_y = 14\text{ ft}$. The material of the column is A992 Gr 50. What is the axial compressive strength of the column to be used in strength design?

- (A) 100 kips
- (B) 150 kips
- (C) 250 kips
- (D) 321 kips

359. An unbraced W8X40 column is subjected to an axial load of 100 kips and the effective length in both directions is 12 ft. The column is also subjected to a strong axis flexure of 25 kip*ft and a weak axis flexure of 10 kip*ft. The compression flange is continuously braced against lateral torsional buckling. What is the total utilization of the column section?
 $F_y = 50\text{ ksi}$

- (A) 60.3%
- (B) 73.9%
- (C) 82.8%
- (D) 90.1%

360. For the built up column shown below, subjected to axial compression only, what is the maximum flange width permissible, so that the flanges are non slender?
 $F_y = 50\text{ ksi}$, $E = 29000\text{ ksi}$

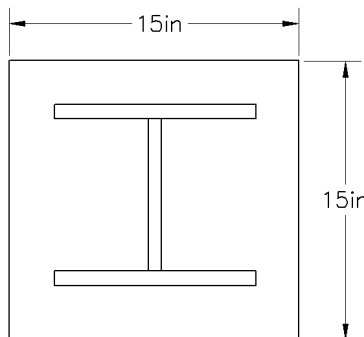


- (A) 20 in
- (B) 28 in
- (C) 35 in
- (D) 40 in

361. A 16 in X 18 in steel column base plate supporting a HSS 9X8X0.5 member is centered on a concrete pedestal. The dimensions of the pedestal are 2 ft X 2 ft. The strength of the concrete is 4 ksi and the steel plate is Gr 50. What is the maximum area of the concrete bearing frustum on which the bearing load is dispersed?

- (A) 288 in²
- (B) 528 in²
- (C) 576 in²
- (D) 1152 in²

362. A W section is centered on a 15 in X 15 in base plate as shown below. The overall depth of the section, $d = 11$ in and flange width, $bf = 12$ in. What is the minimum required thickness of the base plate to support a factored axial load of 650 kips? $F_y = 50$ ksi



- (A) 1 in
- (B) 1.125 in
- (C) 1.375 in
- (D) 1.625 in

363. A W section column is centered on a 15 in X 15 in base plate. The plate is centered on a 18 in X 18 in concrete pedestal. The maximum concentric axial compression acting on the plate is 1000 kips. What is the minimum strength of concrete pedestal required to bear the load safely using AISC specifications?

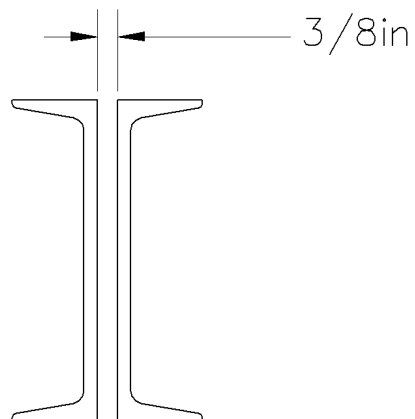
- (A) 4.5 ksi
- (B) 7.5 ksi
- (C) 8.5 ksi
- (D) 10.5 ksi

364. A vertical steel post consists of an angle section loaded in compression, away from its centroid. The designer wants to neglect the effect of eccentricity of the load and evaluate the member as axially loaded only. Which of the following conditions should be met so that the designer's assumption is valid.

- I) Slenderness should be less than 200,
- II) Members should be loaded on the same leg on both ends,
- III) The length of legs of the angle should be equal.

- (A) III
- (B) I and II
- (C) I and III
- (D) I, II and III

365. A built up column section is made with two channels C 12X30 spaced at $\frac{3}{8}$ in back to back as shown below. The effective length in both X and Y direction is 10 ft. What is the nominal critical buckling stress for the column? $F_y = 36$ ksi, $E = 29000$ ksi



- (A) 15 ksi
- (B) 20 ksi
- (C) 25 ksi
- (D) 30 ksi

- 366.** A W21X111 Grade 60 column section spans 15 ft vertically between two floors. The column is unbraced in both directions. What is the design compressive strength for the column to be used in LRFD design?

$E = 29000 \text{ ksi}$

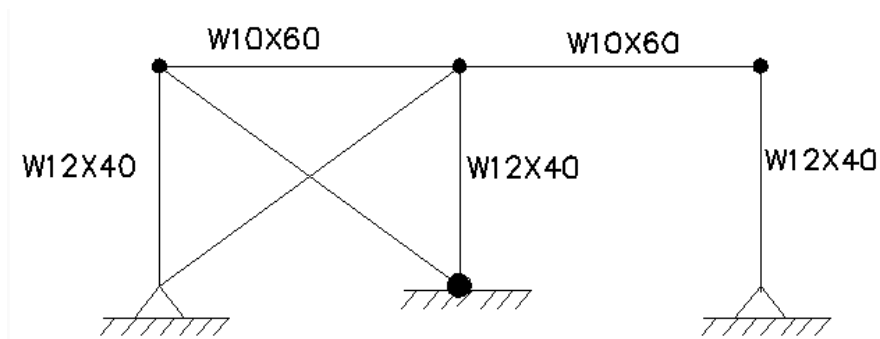
Effective length factor = 1

- (A) 1000 kips
- (B) 1050 kips
- (C) 1100 kips
- (D) 1250 kips

- 367.** The frame shown below belongs to an interior bay of a warehouse. The height of the center column is 15 ft and each beam is 18 ft in length. What is the effective length factor 'K' of the center column?

$E = 29000 \text{ ksi}$

$F_y = 50 \text{ ksi}$

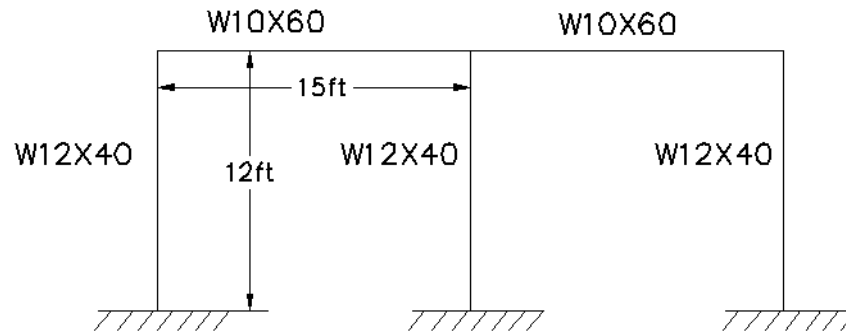


- (A) 0.5
- (B) 0.7
- (C) 1
- (D) 1.2

368. For the moment resisting frame below, what is the effective length factor for the leftmost column?

$E = 29000$ ksi

$F_y = 50$ ksi

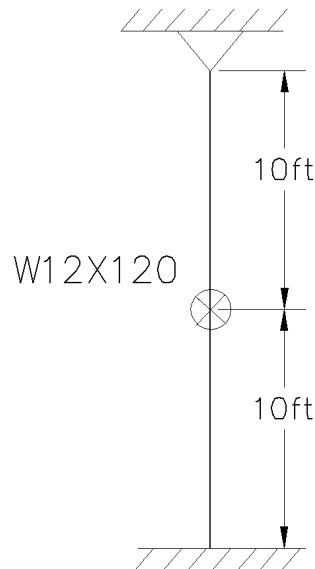


- (A) 1
- (B) 1.2
- (C) 1.3
- (D) 2

369. The W12X120 column shown below is braced at mid height to prevent buckling against the weaker axis only. What is the governing slenderness ratio KL/r of the column?

$E = 29000$ ksi

$F_y = 50$ ksi



- (A) 34
- (B) 38
- (C) 44
- (D) 48

370. The effective length of a W14X82 Grade 50 column is 15ft in both directions. What is the maximum allowable axial compression it can carry, if it is also subjected to a strong axis flexure of 80 kip*ft and weak axis flexure of 40 kip*ft. Lateral torsional buckling is prevented by bracing the compression flanges only.

$E = 29000 \text{ ksi}$

- (A) 125 kips
- (B) 201 kips
- (C) 233 kips
- (D) 489 kips

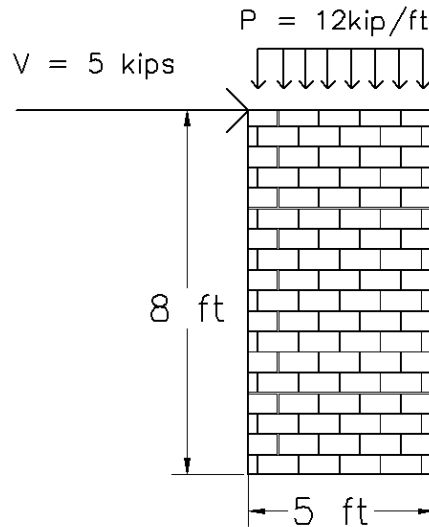
371. For a masonry column, which of the following is true per requirements set by TMS 402/602-16?

- (A) For a 16in X 16 in masonry column, at least four vertical rebars have to be used and the total area of reinforcement should be atleast 1 in^2 .
- (B) For a cantilevered column with least radius of gyration 4 in, unbraced height of 17ft is not permitted.
- (C) Minimum side dimensions of the column should be at least 16in.
- (D) 28 in X 48 in columns can be tied with #4 bars spaced 28 inches O.C. vertically.

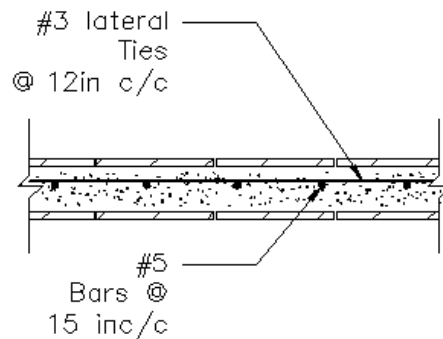
372. What is the allowable compressive force on a 23.625in X 23.625in masonry column section with specified compressive strength $f'_m = 2000 \text{ psi}$? The section has eight #8 Gr. 60 vertical rebars and is fully grouted. The effective height of the column is 20 ft. Neglect accidental eccentricity.

- (A) 325 kips
- (B) 385 kips
- (C) 457 kips
- (D) 650 kips

373. A fully grouted masonry wall is subjected to a 12 kip/ft distributed load at the top and a shear of 5 kips acting at the top. The thickness of the wall is 7.625in and the total area of reinforcement on the both ends is 4 in². The effective cover for the rebars are 4 in. What is the maximum tensile stress acting on the section?



- (A) 0
(B) 0.131 ksi
(C) 0.105 ksi
(D) 0.15 ksi
374. A masonry wall section is shown below. The thickness of the section is 7.625in and the slenderness ratio $H/r = 100$, The vertical reinforcement is Gr. 60 #5 bars spaced at 15 in c/c. If the compressive strength of the masonry is 2000 psi, what is the allowable axial load in the wall section?



- (A) 22 kips/ft
(B) 38 kips/ft
(C) 45 kips/ft
(D) 65 kips/ft

375. A single story building uses Southern Pine wood framed $\frac{3}{8}$ in thick, blocked Structural -I WSP shear wall. 8d nails with spacing 2 inches are used at the panel edges. What is the allowable seismic shear capacity of the shear wall? Assume dry conditions.

- (A) 610 plf
- (B) 1220 plf
- (C) 1330 plf
- (D) 1500 plf

376. Select Structural DFL 2X6 studs are braced effectively about the weaker axis with gypsum boards on both sides as shown below with adequate fastener arrangement. What is the maximum allowable axial compression per stud?

Section properties

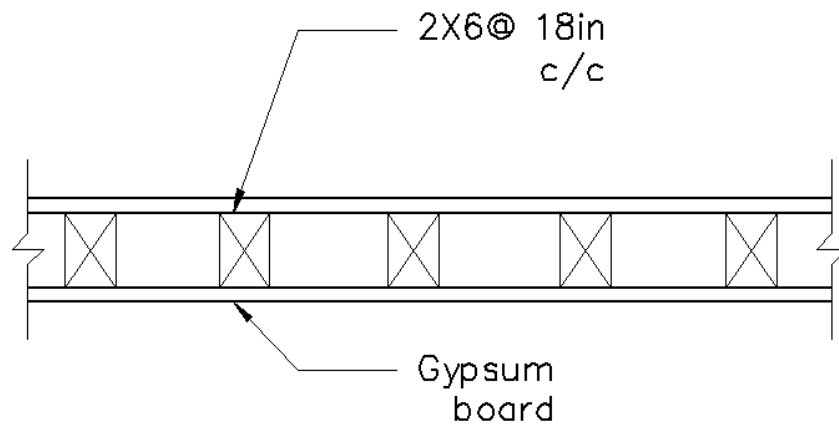
$$C_d = 1$$

$$C_f = 1.15$$

$$C_m = C_t = C_i = 1$$

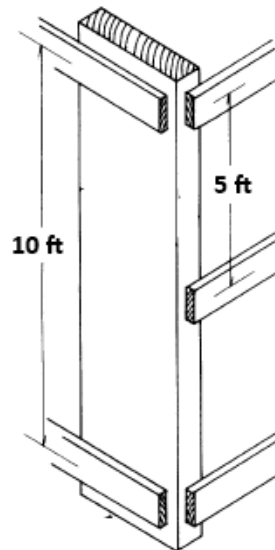
$$C_p = 0.33 \text{ about strong axis}$$

$$C_p = 1 \text{ about weak axis}$$



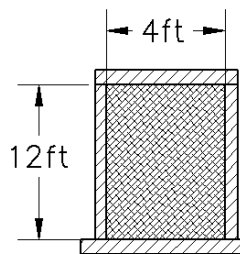
- (A) 2 kips
- (B) 5 kips
- (C) 8 kips
- (D) 16 kips

- 377.** A 2X8 DFL 'No. 1 and better', vertical post is braced effectively in both stronger and weaker directions with pin connected bracing planks as shown below. Determine the critical buckling stress F_{cE} using allowable stress design for the section. Assume dry and normal temperature conditions.
 $C_i = 1$, $C_T = 1$.

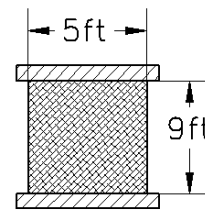


- (A) 339 psi
 (B) 775 psi
 (C) 1257 psi
 (D) 1980 psi

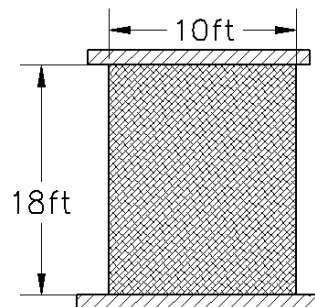
- 378.** Which of the following WSP shear walls does not comply with dimensional requirements of AWC- SDPWS provisions for wood framed shear wall?



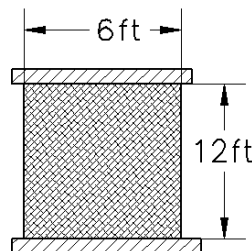
WALL A



WALL B



WALL C



WALL D

- (A) Wall A
 (B) Wall B
 (C) Wall C
 (D) Wall D

379. What is the maximum number of #9 longitudinal bars that can be used for a 10 inch X 10 inch RCC column section per ACI 318-14?

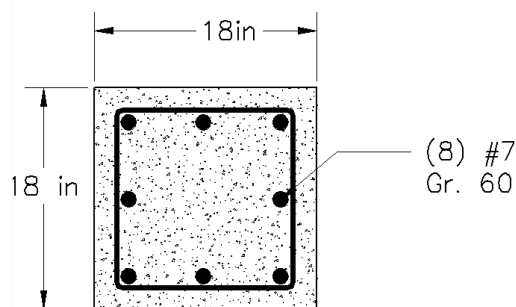
- (A) 4
- (B) 6
- (C) 8
- (D) 12

380. What is the nominal axial load capacity of a 16 in X 16in RCC column section with $f'_c = 6000$ psi? The column has eight # 8 longitudinal reinforcement bars effectively tied per requirements of ACI 318-14. All rebars are Gr 60.

- (A) 859 kips
- (B) 1322 kips
- (C) 1652 kips
- (D) 1943 kips

381. The column section shown below is subjected to a factored moment of 100 kip*ft. What is the design axial load capacity of the column?

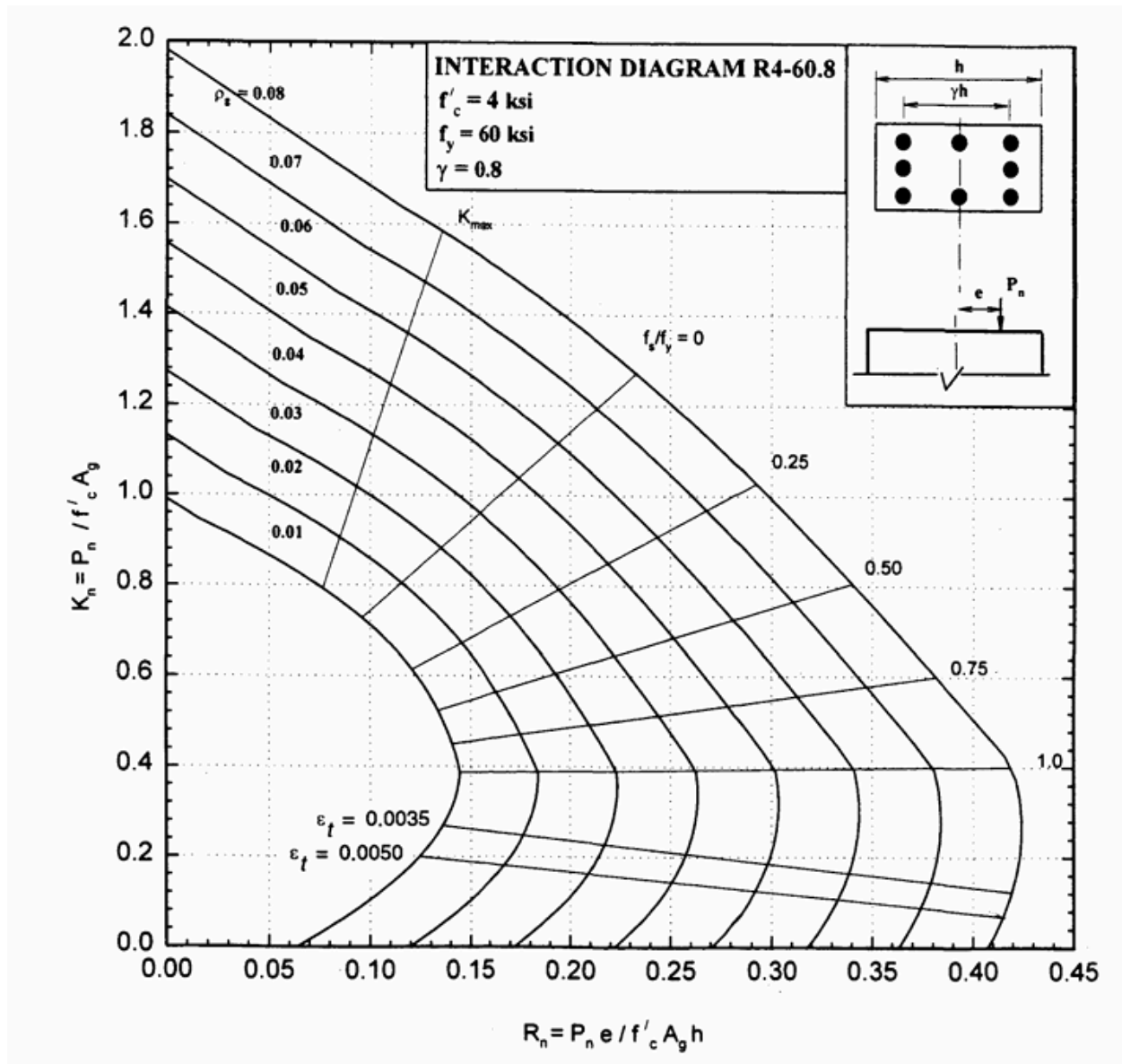
$f'_c = 5$ ksi, $f_y = 60$ ksi , Area of #7 bar = 0.6 in^2 , $\Phi = 0.65$



- (A) 855 kips
- (B) 1125 kips
- (C) 1315 kips
- (D) 1630 kips

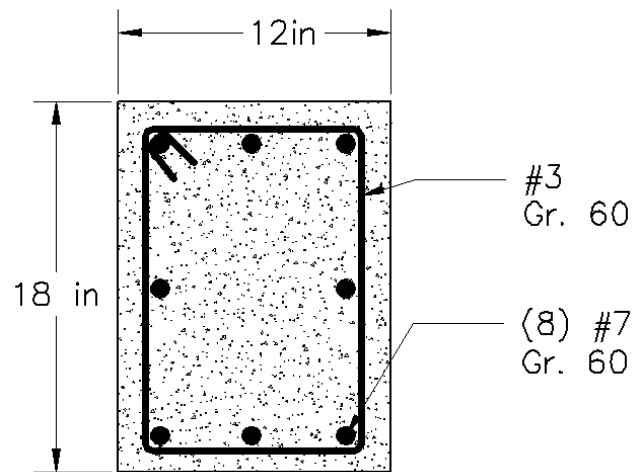
382. A 16 inch X 16 inch square column is subjected to a factored axial load of 798 kips and a factored moment of 66.5 kip*ft. The interaction diagram for the section is shown below. What is the required area of steel for the section?

$$f'_c = 4 \text{ ksi} \quad f_y = 60 \text{ ksi} \quad \Phi = 0.65$$

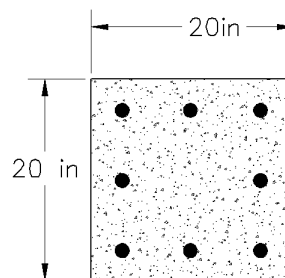


- (A) 3 in²
- (B) 5.5 in²
- (C) 8.6 in²
- (D) 10.5 in²

- 383.** For the column section shown below, #3 ties are used as shear reinforcements. What is the maximum permitted tie spacing per ACI 318-14?

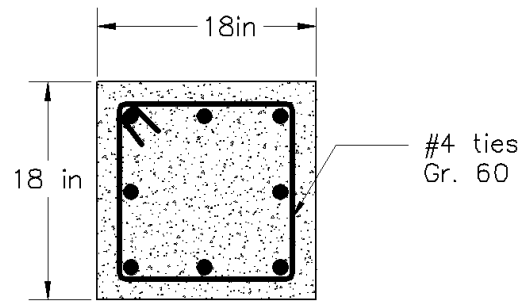


- (A) 6 in
 (B) 12 in
 (C) 14 in
 (D) 18 in
- 384.** What is the design shear strength of the column section shown below? The column is subjected to a factored axial load of 150 kips.
 $f'_c = 4000$ psi $f_y = 60$ ksi

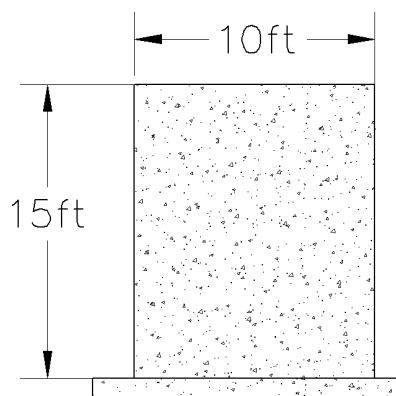


- (A) 25 kips
 (B) 38.3 kips
 (C) 45.1 kips
 (D) 52.5 kips

- 385.** The column shown below is subjected to a factored shear of 75 kips. What is the required tie spacing to satisfy the strength requirement. Assume that the entire shear is taken by the reinforcement only. Effective cover for longitudinal bars = 3 inches.



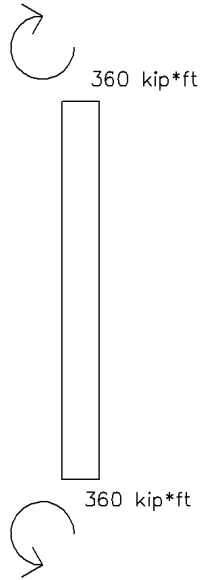
- (A) 3.5 in
 (B) 4.75 in
 (C) 6 in
 (D) 9 in
- 386.** What is the axial load capacity of the cantilevered concrete wall as shown below? The thickness of the wall is 12 in. Assume the resultant of loads is located within the middle third of the wall. $f'_c = 5000$ psi. Wall reinforcement complies to minimum requirements by ACI 318-14.



- (A) 250 kips
 (B) 310 kips
 (C) 480 kips
 (D) 520 kips

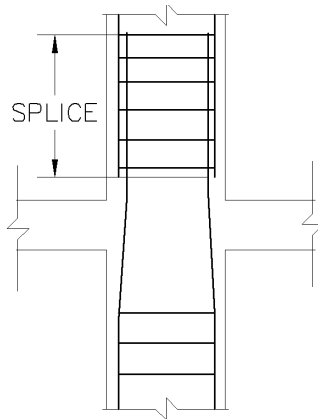
- 387.** A slender column is subjected to 360 kip*ft factored first order moments as shown below. The column is a part of a non-sway frame and is also subjected to a factored axial compression of 350 kips. The critical buckling load for the column P_c is 1400 kips. What is the moment magnification factor for the slender column per ACI 318-14 provisions?

- (A) 1
- (B) 1.2
- (C) 1.38
- (D) 1.5



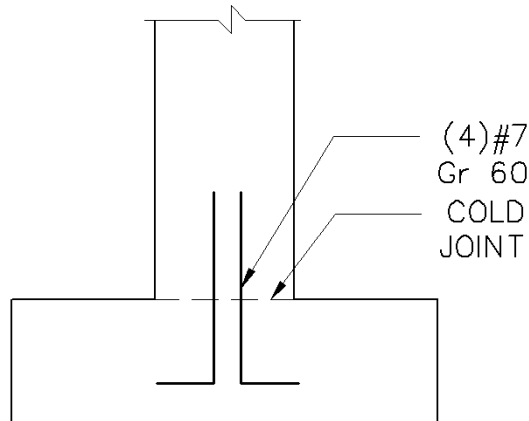
- 388.** What is the required splice length of #8 longitudinal column rebars shown below? The rebars are grade 60 and concrete $f'_c = 4$ ksi. #3 ties are spaced at 5 inches c/c in the splice location. The dimensions of the column is 24 in X 24 in and the factored load in the column is compressive.

- (A) 12 in
- (B) 25 in
- (C) 30 in
- (D) 48 in



- 389.** A 20in X 20in column is cast over an existing footing as shown below. Four #7 dowels connecting the column and the footing are Gr. 60 rebars. The compressive strength of the footing's concrete is 3 ksi and the compressive strength of the column's concrete is 4 ksi. What is the design capacity of the column base in shear friction?
 $\Phi = 0.75$, Area of #7 bar = 0.6 in^2

- (A) 65 kips
- (B) 86 kips
- (C) 108 kips
- (D) 140 kips

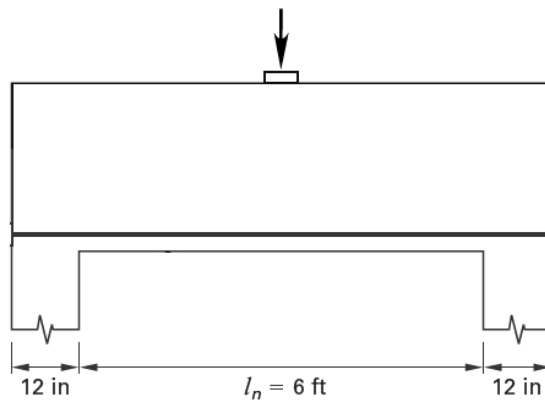


- 390.** What is the minimum transverse Gr. 60 rebar requirement for a 8 in thick RCC wall per ACI 318-14? The factored shear acting on the wall is 50 kips and the design shear strength of the section's concrete only 'V_c' is 110 kips.
- (A) #3 at 18 inches O.C. both faces
 - (B) #4 at 18 inches O.C. both faces
 - (C) #5 at 18 inches O.C. both faces
 - (D) #6 at 18 inches O.C. both faces

5.3 Systems

[Click here for solutions 5.3](#)

- 391.** A deep beam has a clear span of 6 ft and bears on two 12 in thick concrete columns as shown below. The beam is loaded at the center with a concentrated load of 50 kips. Idealizing the deep beam as a simple truss with concrete struts and bottom rebar ties, what is the approximate tension in the beam's bottom reinforcement? Effective depth of beam = 21 inches.



- (A) 25 kips
(B) 50 kips
(C) 75 kips
(D) 100 kips
- 392.** A building frame consists of a combined dual wall and moment frame system to resist lateral loads. What is the primary purpose of the moment frame during a seismic event?
- (A) Transfer lateral loads directly to foundation,
(B) Protect the shear walls from uncertainty in seismic loads,
(C) Resist all the vertical loads,
(D) Enhance lateral stiffness and resistance while the stiffer wall takes the majority of the lateral load.

393. A building is to be designed in a high seismic region. Which of the following systems is most likely to provide the highest level of redundancy and performance for resisting lateral seismic loads?

- (A) Specially reinforced cantilevered columns,
- (B) Flat concrete slabs with drop panels,
- (C) Reinforced masonry shear walls ,
- (D) Combined moment frame and shear wall.

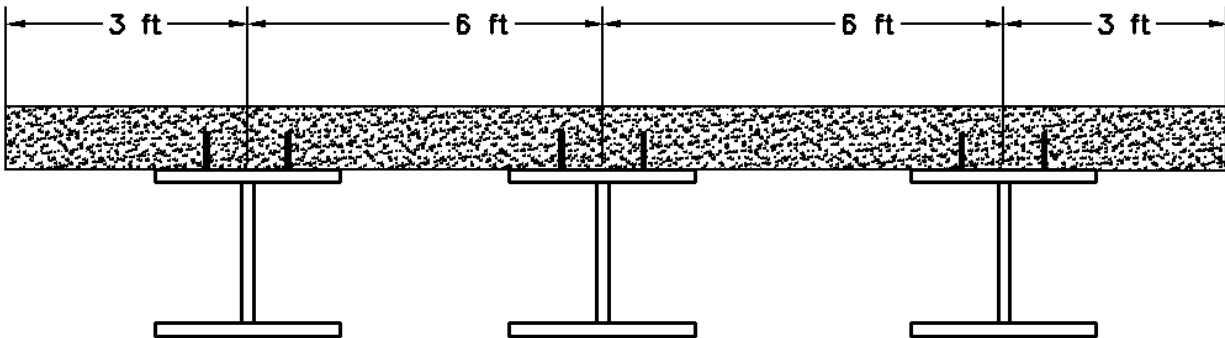
394. In a high wind zone, a five story building is designed with a braced frame system. Which of the following components primarily provides lateral stability in this system?

- (A) Floor diaphragms
- (B) Vertical bracing
- (C) Moment connections
- (D) Foundation anchors

395. A thirty story building with a steel braced frame system is subjected to wind-induced vibrations. The building's natural frequency is close to the vortex shedding frequency of the wind. Which of the following is the most effective strategy to prevent structural damage due to large amplitude vibrations caused by resonance?

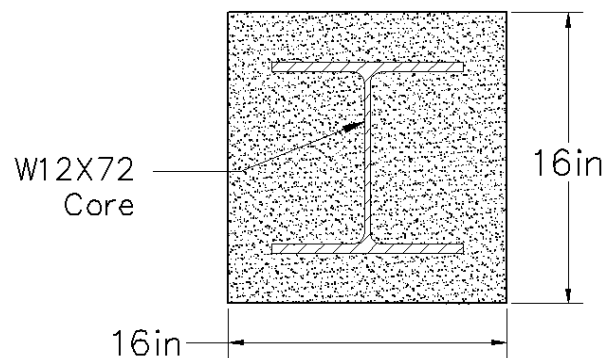
- (A) Increase the overall mass of the structure to counter wind loads and increase stability,
- (B) Increase the height of the building that leads to altered natural frequency of the building,
- (C) Install a tuned mass damper,
- (D) Reduce the stiffness of the braces to alter the natural frequency of the building.

396. The section of a composite deck system is shown below. What is the total effective width of the concrete slab to be considered for the composite beam on the edge? The center to center distance between supports of the beam along the span is 16ft.



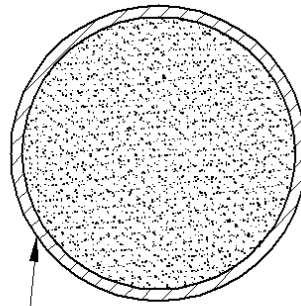
- (A) 2ft
- (B) 4ft
- (C) 6ft
- (D) 8ft

397. The critical elastic buckling load of a composite 16in X 16in concrete column with A992 Gr. 50 W12X72 steel core is 850 kips. What is the allowable compressive load on the composite section? $f'_c = 4000$ psi



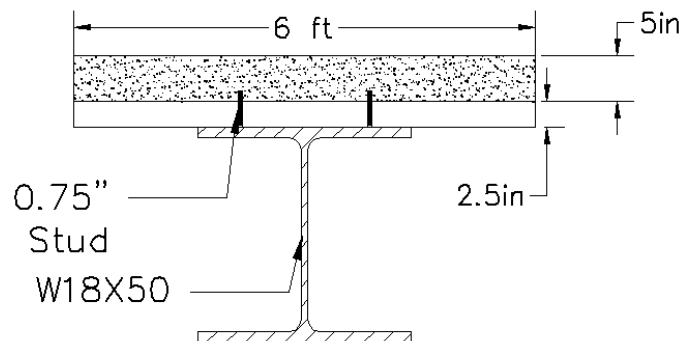
- (A) 372 kips
- (B) 425 kips
- (C) 600 kips
- (D) 758 kips

- 398.** Gr 46 HSS 14.000X0.5 is filled with concrete ($f'_c = 4\text{ksi}$). What is the allowable compressive strength of the composite column section? Elastic critical buckling load of the section is 600 kips. Modulus of elasticity of steel, $E_s = 29000\text{ ksi}$



HSS14.000X0.5

- (A) 263 kips
(B) 318 kips
(C) 450 kips
(D) 526 kips
- 399.** A simply supported composite beam with span 28 ft has ribs of deck perpendicular to the beam span as shown below. The effective width of the slab is 6ft and two 0.75 inch diameter headed studs are used per rib. For full composite action, what is the total number of studs required? The beam is subjected to uniformly distributed load. Nominal shear strength of stud, $Q_n = 15\text{ kips}$.
W18X50 Gr. 50
Slab $f'_c = 4\text{ ksi}$

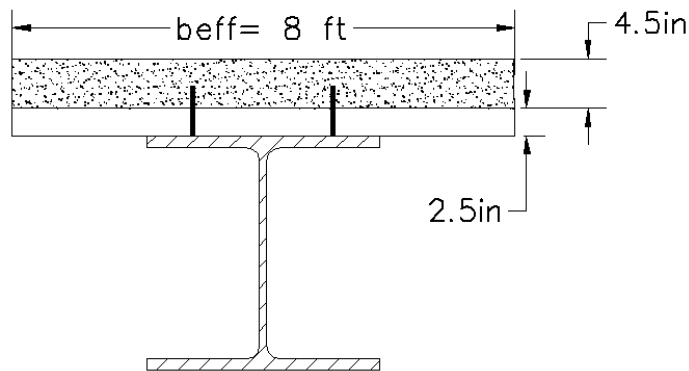


- (A) 36
(B) 49
(C) 98
(D) 116

400. A composite steel deck with A992 Gr. 50 steel beams has the steel decking ribs perpendicular to the span of the beam. The concrete used is normal weight 4 ksi. Single 0.5 inch diameter stud per rib is used at the center of the beam. The distance from the edge of the stud shank to the deck web is 2.5 inches. What is the nominal shear strength of the stud if the tensile strength of stud is 65 ksi?

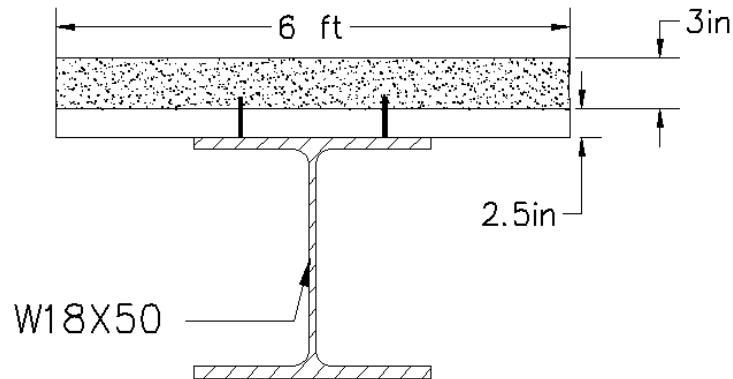
- (A) 6.51 kips
- (B) 7.66 kips
- (C) 9.57 kips
- (D) 11.9 kips

401. A composite steel beam with W18X50 steel section has the steel decking ribs perpendicular to the span of the beam as shown below. The modular ratio of steel to concrete is 8. What is the location of the elastic neutral axis (strong axis) of the composite member from the top of the composite section?



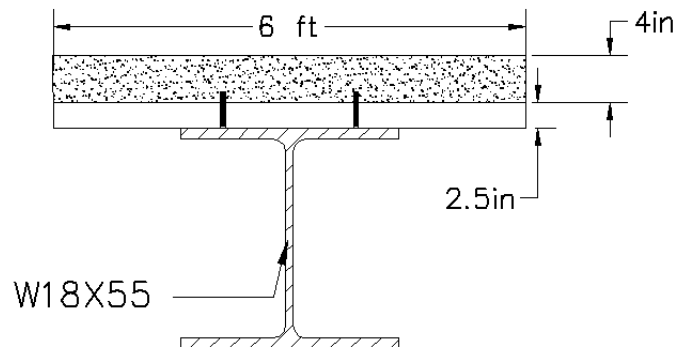
- (A) 5.19 in
- (B) 6.23 in
- (C) 11.32 in
- (D) 16.11 in

- 402.** A simply supported composite steel beam subjected to uniformly distributed load has two studs per rib as shown below. The total number of studs in the span is 64 and the nominal shear strength of each stud is 16.28 kips. What is the allowable moment capacity of the composite section? Strength of concrete slab $f'_c = 4250$ psi



- (A) 319 kip*ft
- (B) 446 kip*ft
- (C) 513 kip*ft
- (D) 544 kip*ft

- 403.** A simply supported composite steel beam subjected to uniformly distributed load has two studs per rib as shown below. The total number of studs in the span is 40 and the nominal shear strength of each stud is 16.8 kips. Full composite action is not developed due to the limited number of studs. What is the moment of inertia of the effective composite section developed? Strength of concrete slab $f'_c = 5550$ psi

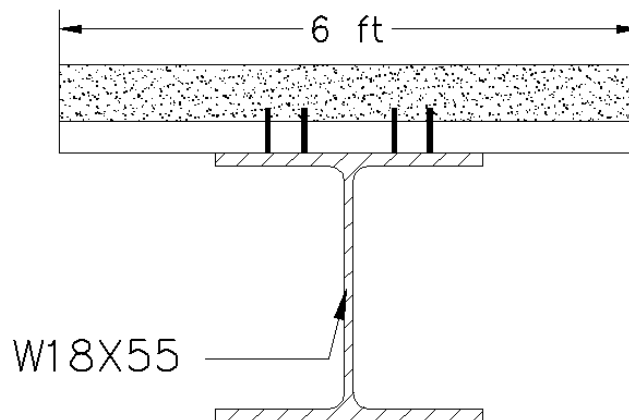


- (A) 1319 in⁴
- (B) 1631 in⁴
- (C) 1970 in⁴
- (D) 2980 in⁴

404. A 1 inch diameter headed steel stud is embedded in a composite concrete slab with strength $f'_c = 6000$ psi and modulus of elasticity, $E = 4400$ ksi. There are two studs per deck rib located at strong positions. Assuming that the failure of steel does not control, what is the nominal strength of the headed anchor?

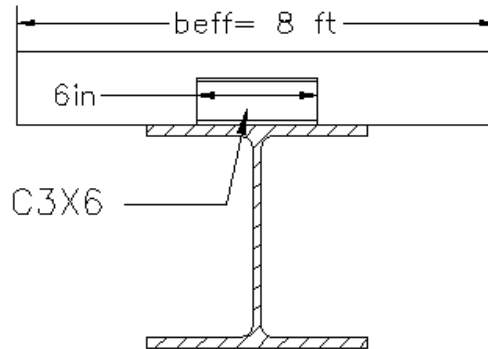
- (A) 51.2 kips
- (B) 55.9 kips
- (C) 63.8 kips
- (D) 72.3 kips

405. A composite steel beam has four studs in a row as shown below, distributed throughout the span. The beam is subjected to a dead load moment of 150 kip*ft and live load moment of 290 kip*ft. The distance between the top of the steel beam and the concrete compression flange force is 3.5 in. Which of the following is most nearly the total nominal shear carried by the studs due to the applied moments? The capacity of the steel beam and concrete slab is sufficient.



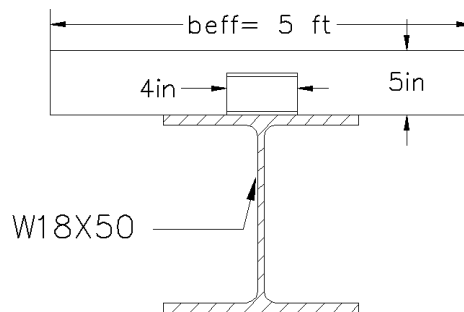
- (A) 335 kips
- (B) 450 kips
- (C) 638 kips
- (D) 750 kips

- 406.** A simply supported composite steel beam subjected to uniformly distributed load has Channel C3X6 welded to the top of the steel beam and embedded in a concrete slab with effective width 8 ft. The compressive strength of the concrete is 4 ksi and the modulus of elasticity is 3600 ksi. If the nominal shear to be transferred by the embedded channels is 380 kips, how many channels are required over the full 15 ft span of the beam?



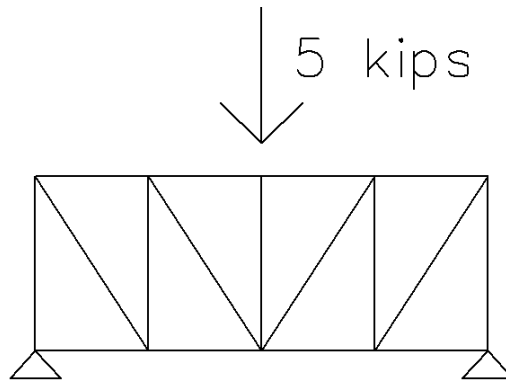
- (A) 2
- (B) 4
- (C) 6
- (D) 8

- 407.** A composite steel beam has four inch long channels, welded throughout the span to transfer horizontal shear. The beam is subjected to a factored dead load moment of 240 kip*ft and a factored live load moment of 400 kip*ft. The distance between the top of the steel beam and the concrete compression flange force is 4 in. Which of the following is most nearly the total nominal shear carried by the channels due to the applied moments? The capacity of the steel beam and concrete slab is sufficient.



- (A) 325 kips
- (B) 430 kips
- (C) 515 kips
- (D) 760 kips

- 408.** A truss shown below is subjected to a concentrated factored load of 5 kips at the center span. The span of the truss is 20 ft and the height of the truss is 5 ft. What is the maximum design tension force at the bottom chord of the truss?



- (A) 4 kips
- (B) 5 kips
- (C) 6 kips
- (D) 7 kips

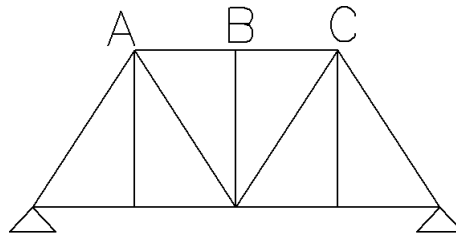
- 409.** The truss shown below is subjected to external loads. After analysis, the internal forces in member BC for two service loading conditions are as follows:

Case A) Dead load = 5 kip tension

Live load = 15 kip tension

Case B) Dead load = 5 kip tension

Live load = 18 kip compression

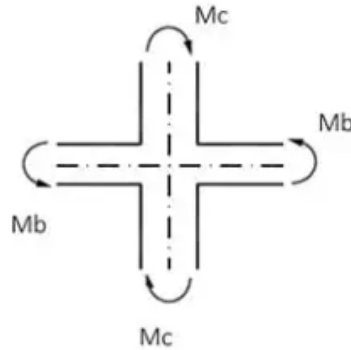


What is the most economical double angle section that satisfies the design requirement?
Span of the truss = 20ft and unbraced length of member BC = 8ft.

Gr. 36 Steel

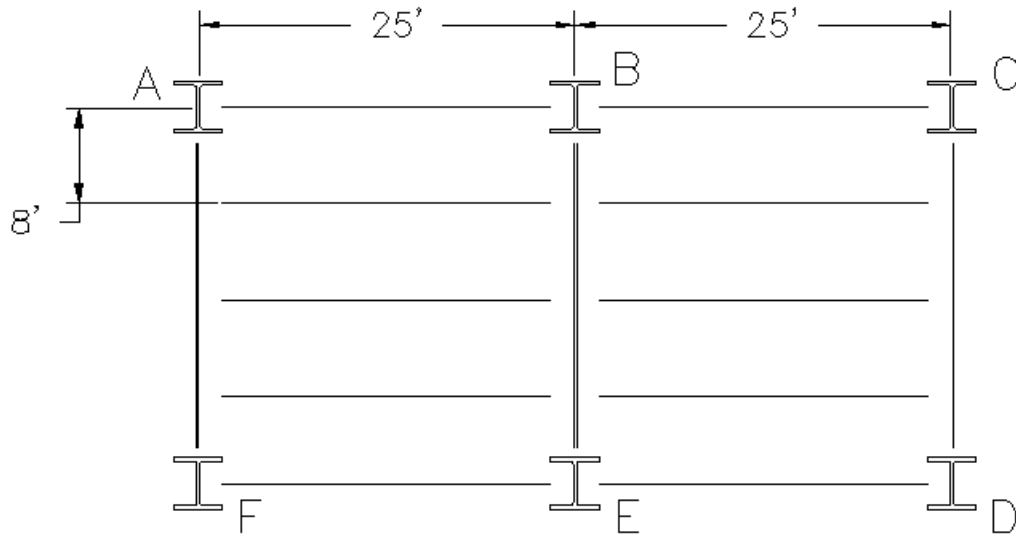
- (A) 2 L 2X2X1/8
- (B) 2 L 2X2X1/4
- (C) 2 L 2X2X5/16
- (D) 2 L 2X2X3/8

- 410.** A reinforced concrete special moment frame is to be designed for a building. For a critical joint, the sum nominal flexural strengths of columns framing into it is 1200 kip*ft. What is the maximum permissible sum of nominal flexural strength of beams framing into the joint so that strong -column-weak-beam criteria is satisfied ?



- (A) 600 kip*ft
(B) 800 kip*ft
(C) 1000 kip*ft
(D) 1200 kip*ft
- 411.** A special moment resisting frame joint has a 30 inX 30 in column and 20 in wide beams framing into it. According to ACI 318-14, which of the following is true about the transverse reinforcement in the joint?
- (A) Transverse reinforcement of the joint must match the transverse reinforcement of the columns above and below the joint,
(B) The joint nominal shear strength is equal to $20\lambda\sqrt{f'_c}$ times joint area
(C) No transverse moment is required if the column is fully in compression.
(D) Spacing should not exceed 8 inches.
- 412.** A reinforced concrete beam in a special moment frame has a width of 12 in. and a total depth of 18 in. The beam is designed per ACI 318-14. The negative moment reinforcement at the face of the column consists of four #8 bars . What is the minimum required positive moment reinforcement at the same location to satisfy ACI 318-14 requirements?
- (A) Six #4 bars.
(B) Four #5 bars.
(C) Four #6 bars.
(D) Four #8 bars.

- 413.** Partial framing plan of a building is shown below. The spandrel beam AB supports a 13 ft tall, 8 in thick wall with unit weight 80 lbf/ft^2 . The floor dead load and live loads are 75 lbf/ft^2 and 100 lbf/ft^2 . Assuming simple connections and full lateral support of compression flange, what is the most economical section that satisfies strength requirements for beam AB? Material used is Gr. 50.



- (A) W12X30
 (B) W12X40
 (C) W12X45
 (D) W12X53
- 414.** A composite steel deck with 5 inch thick concrete slab and Gr. 50-W18X71 beams, designed for full composite action is unshored before pouring. The factored dead load moment including weight of slab and beam is $75 \text{ kip}\cdot\text{ft}$ and factored live load moment is $120 \text{ kip}\cdot\text{ft}$. What is the factored bending stress at the bottom fibre of the steel beam immediately after pouring?

Section modulus of transformed section measured to the bottom of the section = 185 in^3

- (A) 4.86 ksi
 (B) 7.1 ksi.
 (C) 10.3 ksi
 (D) 13.4 ksi

415. A wall system has masonry in the superstructure and plain concrete foundation wall in the substructure. The 11in thick masonry wall is supported by the 8ft tall concrete foundation wall. The foundation wall is supported at the top and bottom adequately. The foundation wall is also retaining 6ft high backfill composed of well graded clean gravel mixes on the exterior side. The interior side has no backfill. The height of the wall is 8ft. What is the minimum thickness of concrete foundation wall per prescriptive design requirements of IBC 2018?

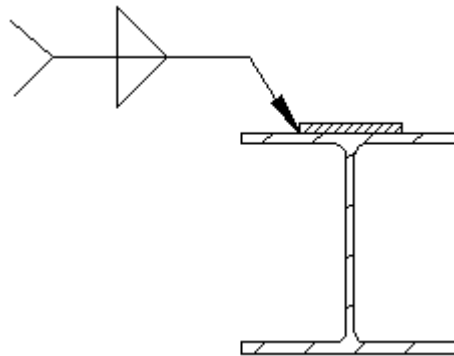
- (A) 7.5 in
- (B) 9.5 in
- (C) 11 in
- (D) 12 in

5.4 Connections

[Click here for solutions 5.4](#)

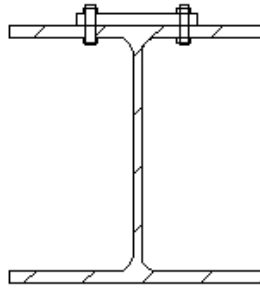
- 416.** Which of the following is not true about restrained and unrestrained steel connections between a beam and a column?
- (A) Fully restrained connections are generally more simpler to analyze than partially restrained connections,
 - (B) Partially restrained connections allow unrestrained relative rotation between the connected members and transmit moments partially,
 - (C) Fully restrained connections allow negligible relative rotation between the connected members and transmit significant moments,
 - (D) Fully restrained connections have high stiffness and partially restrained connections have intermediate level of stiffness

- 417.** A 0.5 inch thick plate is to be welded on a W section flange as shown below. What is the minimum and maximum size of welds allowable? The flange thickness is greater than the plate thickness.



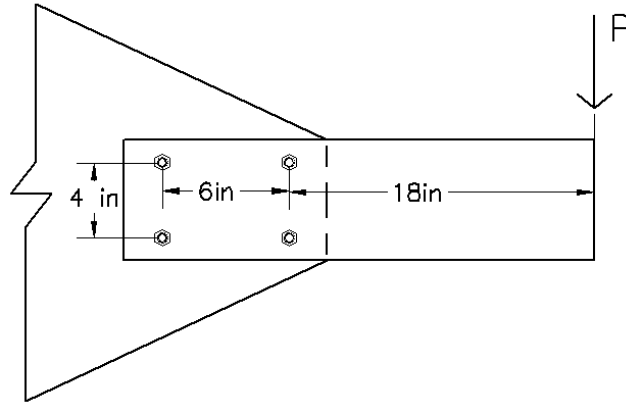
- (A) Min $\frac{1}{8}$ in, Max $\frac{1}{2}$ in
- (B) Min $\frac{5}{16}$ in, Max $\frac{1}{2}$ in
- (C) Min $\frac{3}{16}$ in, Max $\frac{7}{16}$ in
- (D) Min $\frac{1}{4}$ in, Max $\frac{7}{16}$ in

- 418.** A $\frac{3}{4}$ inch thick plate is to be bolted on top of a W section flange as shown below using standard size holes. The diameter of the bolt is $\frac{1}{2}$ inches. What is the minimum clear distance from the edge of the plate to the edge of the hole?

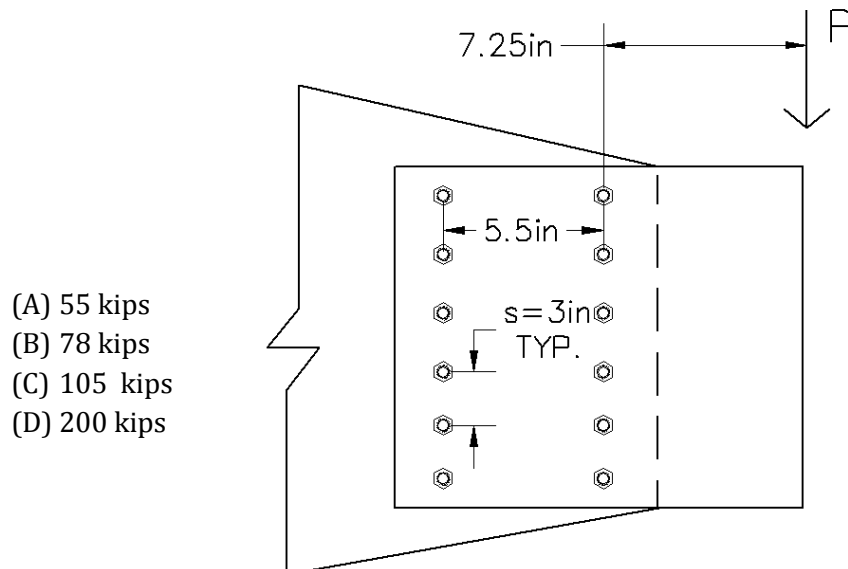


- (A) $\frac{15}{32}$ in
(B) $\frac{9}{16}$ in
(C) $\frac{5}{8}$ in
(D) $\frac{3}{4}$ in
- 419.** An angle brace is connected to a gusset plate using a combination of high strength bolts and fillet welds. The design permits sharing of loads between bolts and welds. The maximum tension load acting on the brace is 100 kips. What is the minimum load that the bolts alone must be able to withstand?
- (A) 33 kips
(B) 50 kips
(C) 66 kips
(D) 75 kips
- 420.** A 0.5 inch thick plate is connected to the face of a column using $\frac{1}{4}$ inch thick fillet welds. The total length of the end loaded fillet welds is 70 inches. What is the effective length of the fillet weld?
- (A) 37 in
(B) 42 in
(C) 45 in
(D) 58 in

- 421.** The figure below shows a bolted connection that is supporting an eccentric load. The governing failure of the connection is bolt shear strength. If the allowable shear strength of the bolt is 22 kips, what is the maximum load “P” that the connection can support?

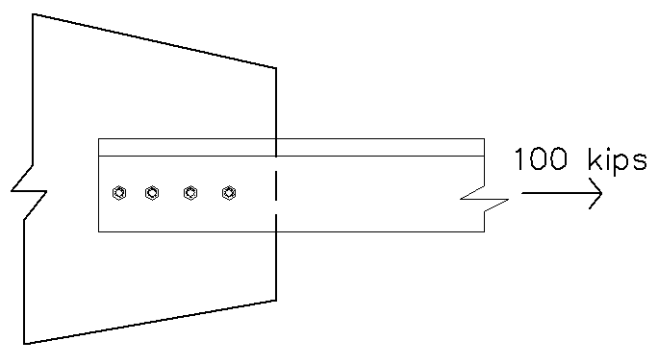


- (A) 13.2 kips
 (B) 22 kips
 (C) 36.7 kips
 (D) 88 kips
- 422.** Using the instantaneous center of rotation method, what is the maximum allowable load “P” that the bolted connection shown below can withstand? The governing failure of the connection is bolt shear strength and allowable shear strength of each bolt is 20 kips.

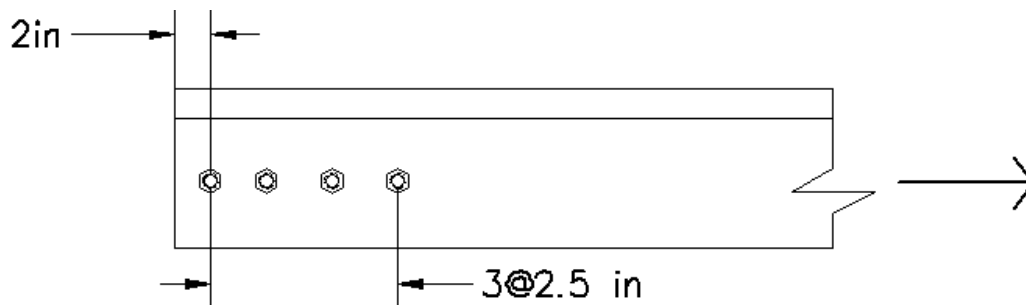


- (A) 55 kips
 (B) 78 kips
 (C) 105 kips
 (D) 200 kips

- 423.** An angle section is connected to a plate with four bolts as shown below. Assuming that the failure of the connection is gross yielding of the section, what is the most economical angle section that satisfies the design requirement? Yield strength of angle is 36 ksi. Loads shown are service loads.

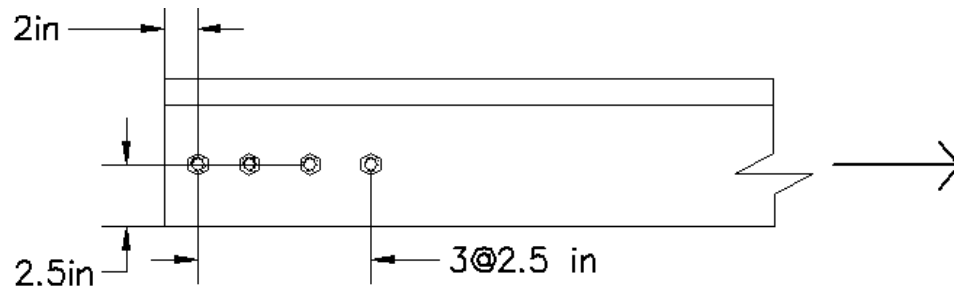


- (A) L 5 X 3 X $\frac{1}{2}$
 (B) L 5 X $3\frac{1}{2}$ X $\frac{1}{2}$
 (C) L 5 X 5 X $\frac{1}{2}$
 (D) L 5 X 5 X $\frac{3}{4}$
- 424.** An angle section 5X3X $\frac{1}{2}$ used as a tension member, is connected to a plate with four 0.75 inch diameter bolts as shown below. What is the governing allowable bolt bearing and tearout strength of an individual bolt if the holes are standard sizes? Yield strength of angle is 36 ksi and tensile strength is 58 ksi. Deformation at bolt hole at service load is a design consideration.

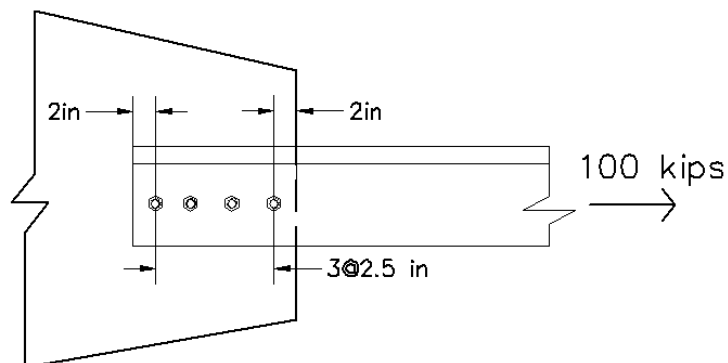


- (A) 20 kips
 (B) 22 kips
 (C) 26 kips
 (D) 28 kips

425. An angle section 5X3X1/2 is used as a tension member, connected to a plate with four 0.75 inch diameter bolts as shown below. What is the allowable block shear strength of the angle? Yield strength of angle is 36 ksi and tensile strength is 58 ksi.

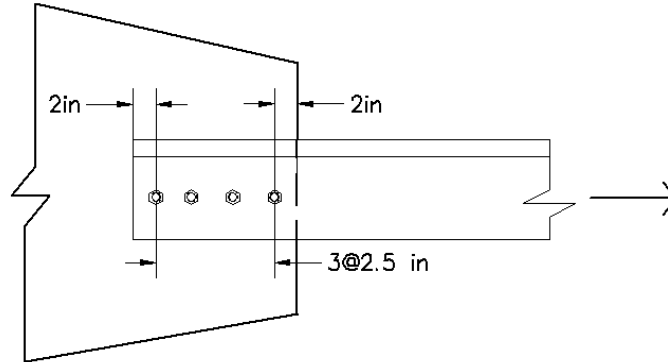


- (A) 68 kips
(B) 81 kips
(C) 85 kips
(D) 135 kips
426. Determine the required minimum thickness of a gusset plate that connects an angle with four bolts as shown below. Assume tensile yielding of the plate governs. Yield strength of plate is 36 ksi and tensile strength is 58 ksi. The loads shown are service loads.

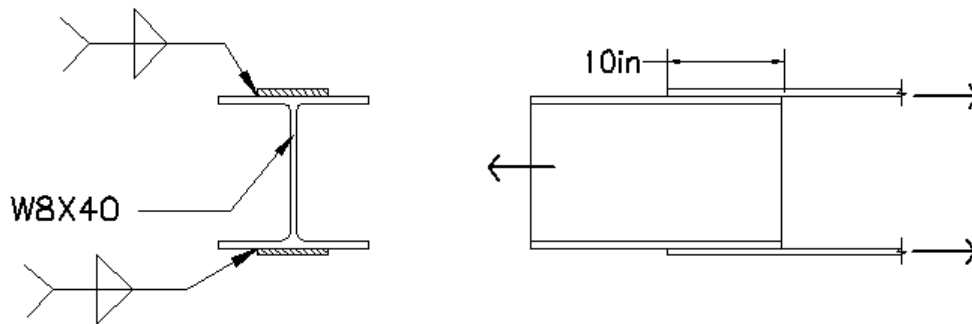


- (A) 1/2 in
(B) 5/8 in
(C) 3/4 in
(D) 1 in

- 427.** For the L 5X3X1/2 connection below, what is the net effective area “Ae” for tensile rupture of the angle as defined by AISC specifications? The bolts used are 0.75 inch diameter. The longer leg of the angle is connected to the plate.
Nominal Hole diameter = 13/16in

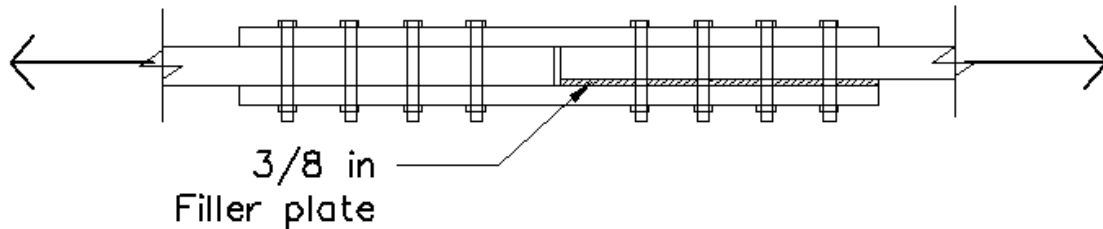


- (A) 2.72 in²
(B) 2.98 in²
(C) 3.32 in²
(D) 3.37 in²
- 428.** Two plates are welded to the top and bottom of a W8X40 and tension load is applied as shown below. What is the shear lag factor “U” to calculate the net effective area of the section per AISC specifications?

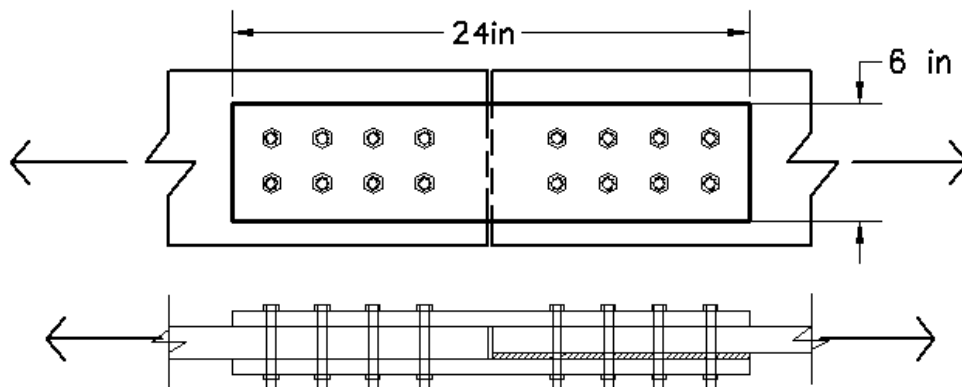


- (A) 0.8
(B) 0.87
(C) 0.93
(D) 1

- 429.** A tension splice is constructed for two plates with different thickness as shown below. To make up for the gap, a $\frac{3}{8}$ in thick filler plate is used. The fasteners used are $\frac{5}{8}$ inch diameter A307 grade bolts. What is the allowable shear strength of an individual bolt to be used for the connection?

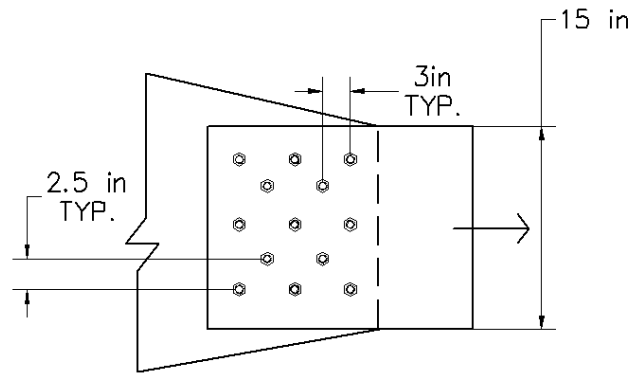


- (A) 4.14 kips
 (B) 7.87 kips
 (C) 8.29 kips
 (D) 12.5 kips
- 430.** The plan and elevation of a tension splice is shown below. The tension load in the connection is 100 kips (factored). What is the required minimum thickness of the splice plate? $F_y = 36$ ksi and $F_u = 58$ ksi for the plate. Nominal hole dimension = $\frac{13}{16}$ in.

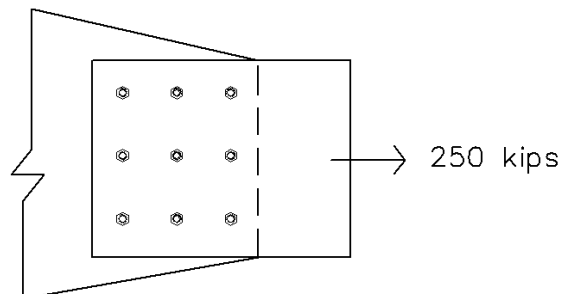


- (A) $\frac{1}{4}$ in
 (B) $\frac{3}{8}$ in
 (C) $\frac{1}{2}$ in
 (D) $\frac{5}{8}$ in

- 431.** Two plates are connected to each other as shown below. The thickness of the thinner top plate is 0.5 in. The nominal size of bolt holes is $\frac{13}{16}$ in. What is the nominal tensile rupture strength of the top plate? $F_u = 58$ ksi

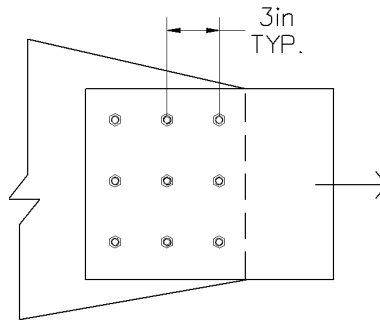


- (A) 269 kips
 (B) 315 kips
 (C) 359 kips
 (D) 390 kips
- 432.** Two plates are connected as shown in the plan below and are subjected to a service load of 250 kips. The thickness of the top plate is $\frac{5}{8}$ in. The bolts are $\frac{3}{4}$ inch in diameter and the holes are standard size. What is the bearing stress on the bolts holes?



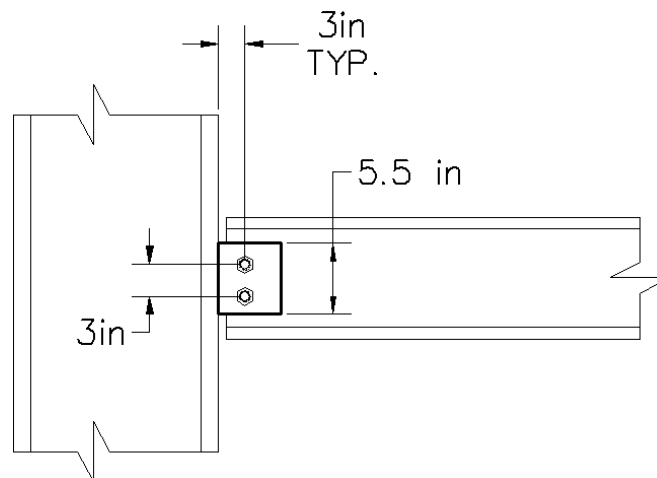
- (A) 26 ksi
 (B) 32 ksi
 (C) 47 ksi
 (D) 55 ksi

- 433.** Two plates are connected as shown below using nine $\frac{7}{8}$ inch diameter A325 bolts with standard holes. Threads are excluded from the shear plane. The thickness of the top plate is $\frac{1}{4}$ in. What is the allowable strength of the connection? Assume plates have sufficient strength for yielding, rupture and block shear and sufficient edge distances. Tensile strength of the plate is 58 ksi.



- (A) 137 kips
- (B) 183 kips
- (C) 210 kips
- (D) 255 kips

- 434.** What is the allowable vertical reaction that can be transferred by the single plate shear connection shown below? The $\frac{3}{8}$ inch thick shear plate is sufficiently welded to the column and has sufficient edge distances per AISC requirements. The bolts used are $\frac{3}{4}$ inch diameter A325 and threads are not included in the shear plane. Plate $F_y = 36$ ksi.



- (A) 21.1 kips
- (B) 25.3 kips
- (C) 27 kips
- (D) 29 kips

435. Which of the following statements best describes the function of designing a bolted connection to be slip critical instead of bearing connection?

- (A) Slip critical connections with high pretension and friction allow for increased shear resistance, compared to bearing type connections with same bolts,
- (B) Slip critical connections are used to prevent slip under service loads for stiffness and fatigue requirements.
- (C) Slip critical connections allows for larger bolt hole tolerances during fabrication,
- (D) Slip critical connections are designed to eliminate the need for bolt bearing and tearout checks.

436. For a beam to column moment connection, which of the following factors most significantly affects the rotational stiffness of the connection?

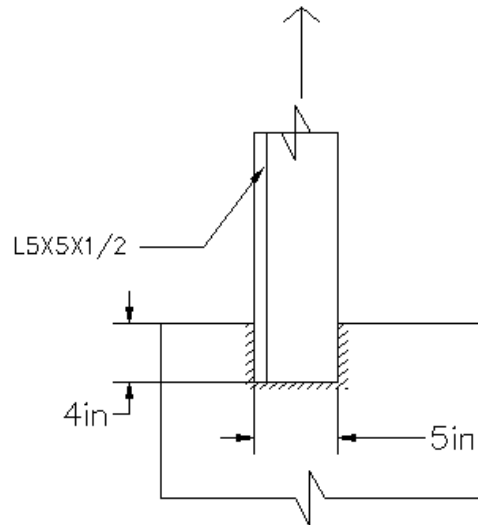
- (A) The grade of steel used for beams and columns,
- (B) The length of the fillet welds along the beam web,
- (C) The diameter of bolts used in the flange connection plates,
- (D) The thickness and size of flange plates or angles,

437. Which of the following design considerations is most critical to ensure adequate ductility and energy dissipation in steel connections that are subjected to earthquake loading?

- (A) High strength bolts should be used to maximize all connection capacities during cyclic loadings,
- (B) The weld sizes in the connections should be increased to reduce stress concentrations during dynamic earthquake loads,
- (C) The connection should be designed such that connected parts yield before the fasteners or welds fails,
- (D) The number of bolts should be minimized to simplify the seismic load path,

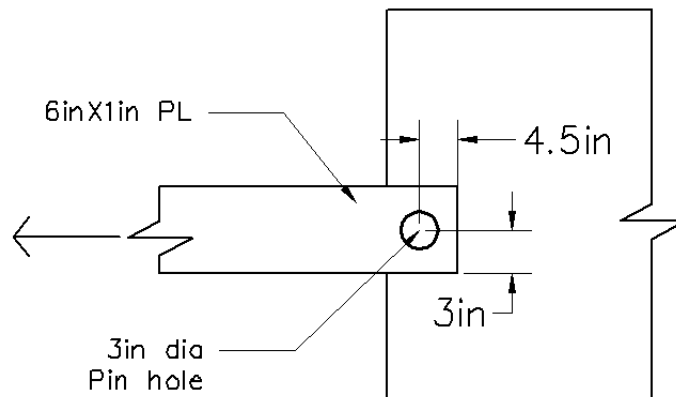
- 438.** An angle section is connected to a plate by welds on three sides and subjected to a tensile loading in the direction shown below. The welds and plate do not govern failure of the connection. What is the design strength of the angle section? $F_y = 36$ ksi and $F_u = 58$ ksi for angle and plates.

- (A) 134.4 kips
(B) 155.2 kips
(C) 160.1 kips
(D) 175.5 kips



- 439.** A 6 in. X 1 in. pin plate is connected to a structural member as shown below. The diameter of the pin is 2.875 in. and the hole diameter is 3 in. What is the design strength of the pin plate in tension? Assume pin failure in shear and bearing do not control. $F_y = 36$ ksi and $F_u = 58$ ksi for plates.

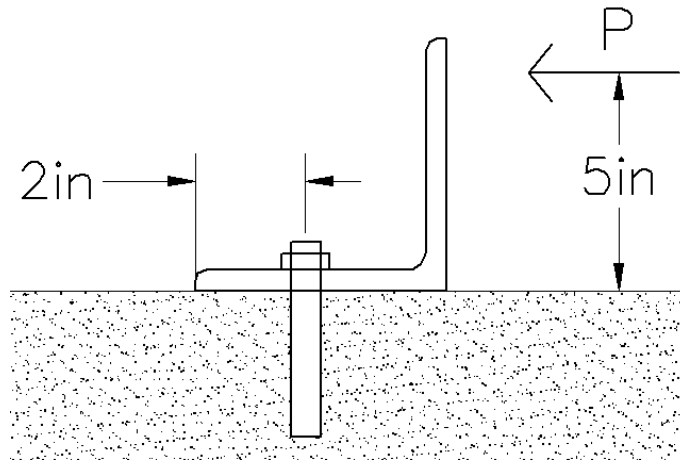
- (A) 130.5 kips
(B) 194.4 kips
(C) 231.6 kips
(D) 258.2 kips



- 440.** A pin connected hanger plate is connected to a support beam. The hanger plate is subjected to tension due to pick loads. The diameter of the pin is 3 inches and the thickness of the plate is 1.5 in. What is the allowable bearing strength of the connection?
 $F_y = 36$ ksi and $F_u = 58$ ksi for plates.

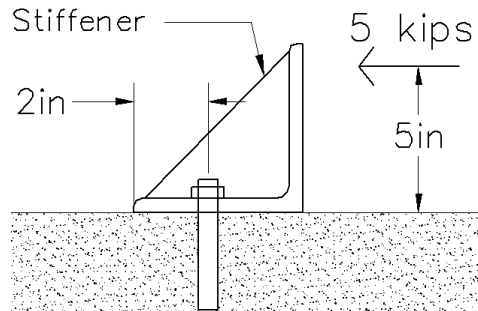
(A) 120.5 kips
(B) 124.4 kips
(C) 145.8 kips
(D) 194.4 kips

- 441.** An angle section 6X6X1/2 is connected to a concrete slab with a bolt as shown below. The length of the angle section is 4 inches. What is the maximum unfactored load P that can be applied to the angle section as shown below? Assume that the bolts and concrete do not govern the failure and the yield strength of the angle section is 36 ksi.

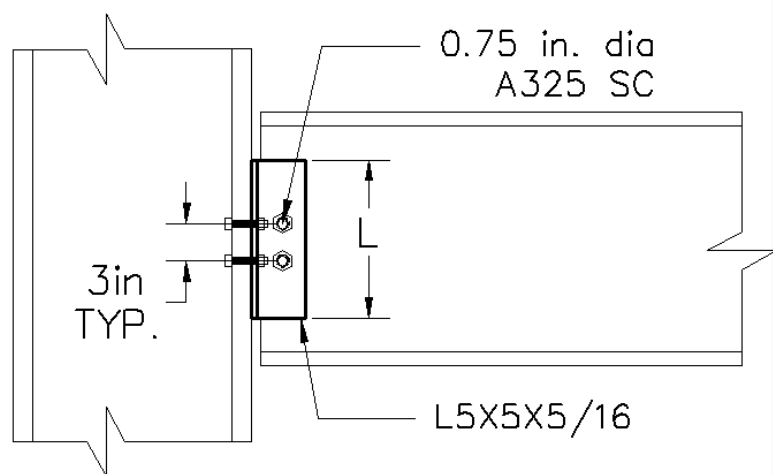


(A) 0.75 kips
(B) 1.2 kips
(C) 1.5 kips
(D) 1.75 kips

- 442.** An angle section is connected to a concrete slab with a bolt as shown below. The length of the angle section is 4 inches. Stiffener plates are welded to the angle section at the ends to make the connection rigid. What is the minimum tension acting in the bolts?



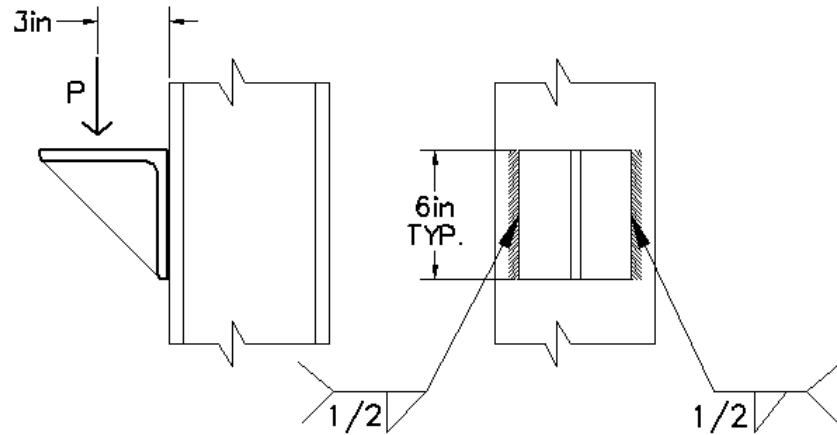
- (A) 5 kips
(B) 8.5 kips
(C) 10.2 kips
(D) 12.5 kips
- 443.** A double angle beam to column shear connection is made using 5X5X5/16 angles on both sides of the beam web as shown below (number of bolts shown for illustration of connection type only). The spacing of the bolts will be kept at 3 inches consistent and Length of the angle 'L' varies per number of bolts used such that the top and bottom bolt edge distances are at least 1.25 in. The horizontal edge distance for the bolts will be maintained at 1.375 in. minimum. The connection has to transfer a service dead load reaction of 20 kips and a service live load reaction of 50 kips. Assuming failure of connection is governed by the angle and bolts, what is the number of slip critical (class A) A325 , 3/4 inch diameter bolts required for the beam web? Beams and columns are A992. Yield strength of angle = 36 ksi. Tensile strength = 58 ksi. All holes are standard size.



- (A) 2
(B) 4
(C) 6
(D) 8

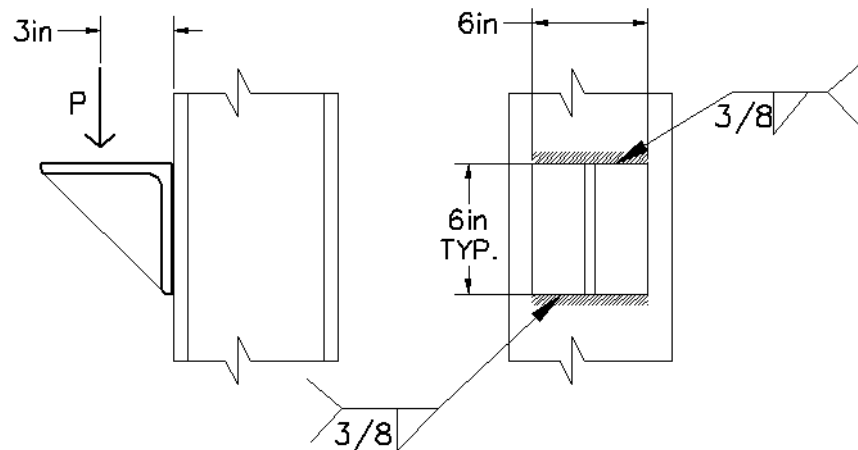
444. A bracket is welded on the face of the column as shown below. Using elastic method, what is the maximum allowable load that 'P' that the bracket can carry, assuming that the welds govern the failure of connection.

$F_{exx} = 70 \text{ ksi}$



- (A) 17.5 kips
- (B) 28.2 kips
- (C) 32.2 kips
- (D) 41.2 kips

445. A bracket is welded on the face of the column as shown below. Using elastic method, what is the maximum allowable load that 'P' that the bracket can carry, assuming that the welds govern the failure of connection. $F_{exx} = 70 \text{ ksi}$

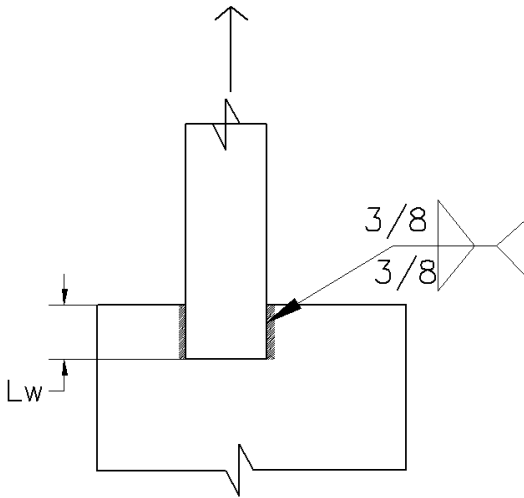


- (A) 29.5 kips
- (B) 38.7 kips
- (C) 61.3 kips
- (D) 70.9 kips

- 446.** A plate is connected to a gusset by welding on two sides as shown below. The service tension load acting concentrically on the welds are 10 kips dead load and 25 kips live load. Assuming that the base metal failure does not govern, what is the required length of weld "L_w" ?

F_{exx} = 70 ksi

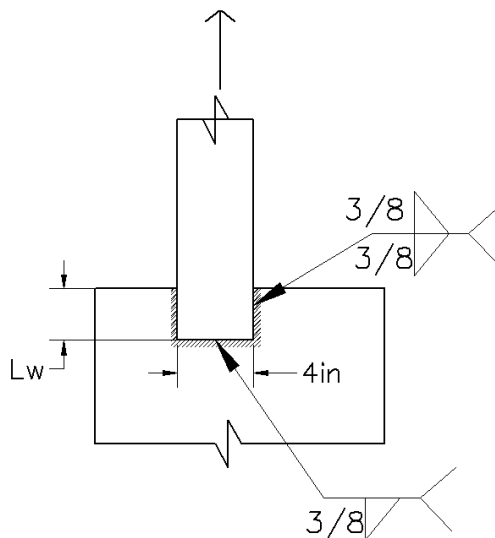
- (A) 2.75 in
(B) 3.25 in
(C) 3.5 in
(D) 4 in



- 447.** A plate is connected to a gusset by welding on two sides as shown below. The service tension load acting concentrically on the welds are 20 kips dead load and 30 kips live load. Assuming that the base metal failure does not govern, what is the required length of weld "L_w" ?

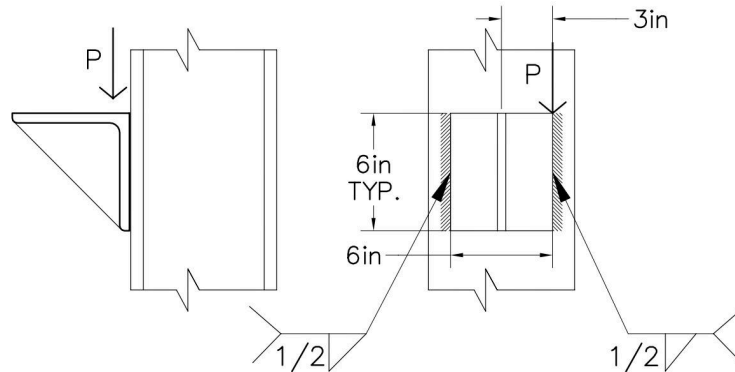
F_{exx} = 70 ksi

- (A) 1.5 in
(B) 2 in
(C) 2.25 in
(D) 3 in



- 448.** A bracket is welded on the face of the column as shown below. Using instantaneous center of rotation method, what is the maximum allowable load that 'P' that the bracket can carry, assuming that the welds govern the failure of connection.

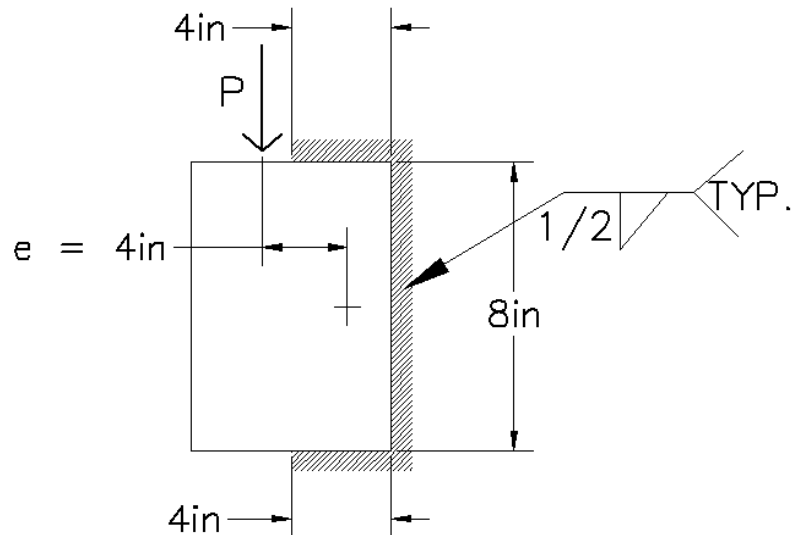
$F_{exx} = 70 \text{ ksi}$



- (A) 41.2 kips
- (B) 48.2 kips
- (C) 62.4 kips
- (D) 71.3 kips

- 449.** A plate has welds on three sides as shown below. Using instantaneous center of rotation method, what is the maximum allowable load that 'P' that the plate can carry, assuming that the welds govern the failure of connection.

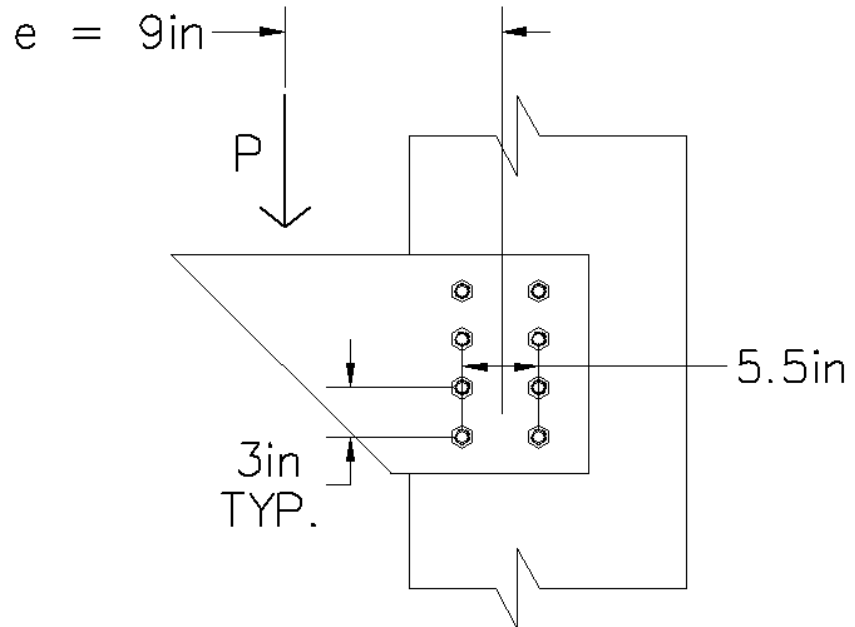
$F_{exx} = 70 \text{ ksi}$



- (A) 51.3 kips
- (B) 66.4 kips
- (C) 78.4 kips
- (D) 91.2 kips

- 450.** A plate is bolted to a column with eight A490 Slip critical (Class A surface) bolts. The total load acting in the plate 'P' = 20 kips, unfactored. What is the minimum diameter of the bolts required for standard size holes assuming bolt failure governs. Use instantaneous center of rotation method.

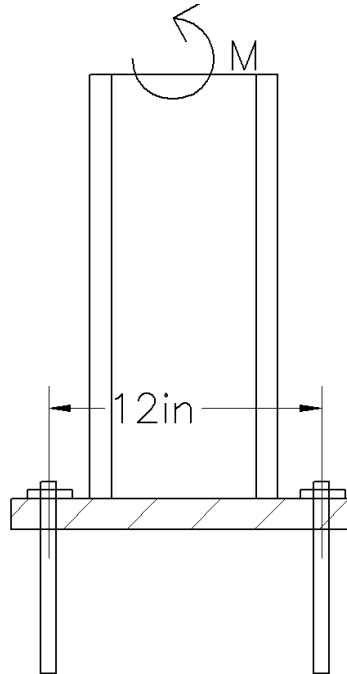
- (A) 1/2 in
(B) 5/8 in
(C) 3/4 in
(D) 1 in



- 451.** In a bearing type connection, a 1 in. diameter ASTM A325 bolt is subjected to both shear and tension loads (single shear with threads not included in shear plane). The shear load in the bolt is 5 kips. What is the allowable tensile strength of the bolt when both shear and tension are acting together on the bolt?

- (A) 24.7 kips
(B) 26.3 kips
(C) 29.1 kips
(D) 35.3 kips

- 452.** A column is connected to its base with four 1 in. diameter ASTM A325 bolt. The four bolts are laid in a square pattern with spacing in each direction = 12 inches. What is the allowable moment capacity of the connection assuming the failure is governed by the bolts?

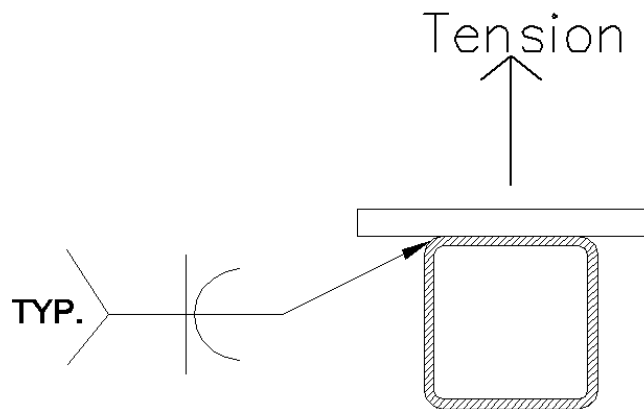


- (A) 35.3 kip*ft
(B) 46.2 kip*ft
(C) 70.6 kip*ft
(D) 83.1 kip*ft
- 453.** A welded connection uses a double-sided fillet weld to connect a plate to a column flange. The weld is designed to resist a shear load, but nondestructive testing reveals a 1-inch-long lack of fusion in one of the welds. Which statement is not true about the impact of this imperfection on the connection's performance?
- (A) The lack of fusion reduces the effective throat, decreasing the weld's shear capacity proportionally,
(B) The lack of fusion affects the weld's fatigue life and has negligible effect on its static strength,
(C) The lack of fusion increases the likelihood of brittle fracture under loadings,
(D) The imperfection acts as a stress concentration point, increasing the likelihood of crack propagation.

454. In a slip critical connection in single shear, a $\frac{3}{4}$ in. diameter ASTM A325 bolt with standard hole is subjected to both shear and tension loads. The bolts are torqued per AISC requirements for slip critical connections. The tension load in the bolt is 5 kips. What is the allowable shear strength of the bolt due to shear tension interaction? Assume only one filler between the connected parts and class A surface.

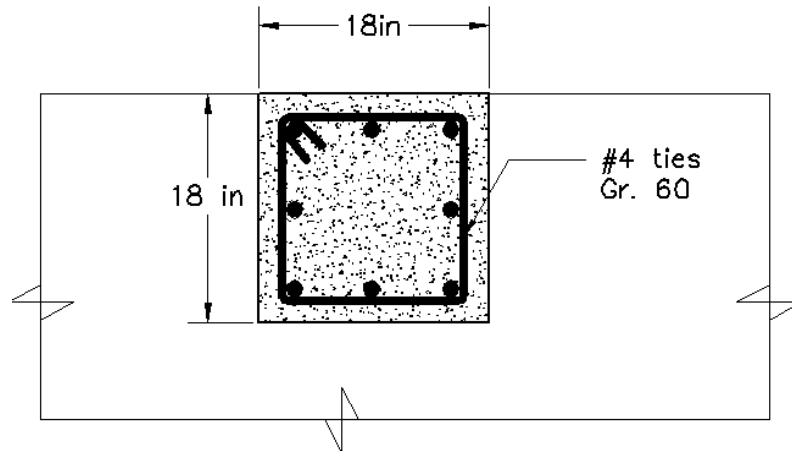
- (A) 2.85 kips
- (B) 3.61 kips
- (C) 4.83 kips
- (D) 6.33 kips

455. HSS 8X8X $\frac{1}{4}$ section is welded to a top plate with flare bevel welds flush to the outer wall of the HSS as shown below. The length of the plate along the HSS span is 6 inches. What is the allowable concentric tension load on the plate connection? $F_{exx} = 70$ ksi (SMAW). Assume HSS walls and plate do not govern failure.



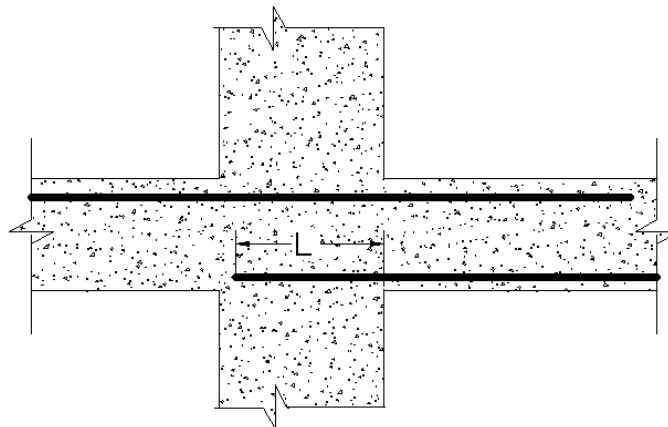
- (A) 19.5 kips
- (B) 39 kips
- (C) 48 kips
- (D) 55.3 kips

- 456.** Plan of an exterior column with slab on three sides is shown below. The grade of concrete of both column and slab is 4 ksi. The thickness of the slab is 8 in and the structure is not in a critical seismic zone. What is the minimum number of Gr. 60 #4 ties required inside the slab- column joint ?



- (A) 1
(B) 2
(C) 3
(D) 4

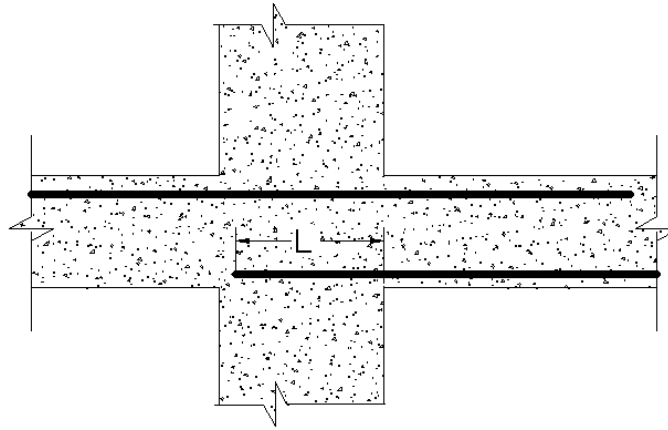
- 457.** The figure below shows a continuous beam column joint. The bottom bars are #8 Gr. 60 deformed bars and the effective depth of the beam is 21 inches. Compressive strength of concrete is 4 ksi. There is no positive moment at the face of the column. The point of inflection is 2.5 ft away from the right face of the column.. What is the minimum required extension 'L' of bottom bars from the face of the column? The beam is not a part of the primary lateral force resisting system.



- (A) 0 in
(B) 6 in
(C) 12 in
(D) 21 in

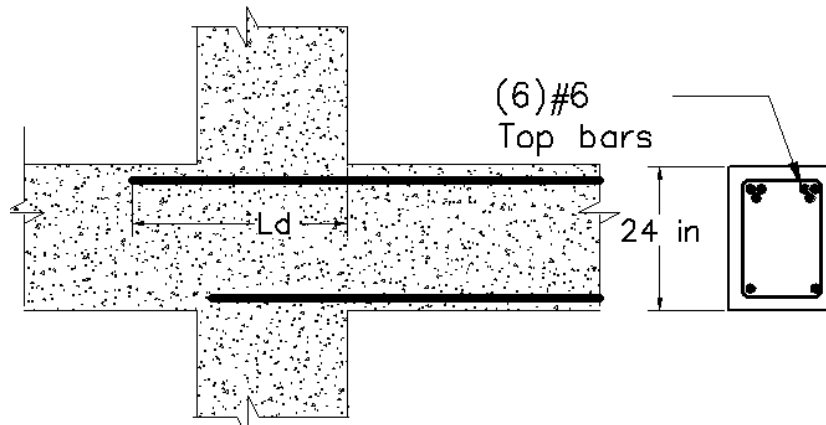
- 458.** The beam on the right of the column shown below is a part of the primary lateral force resisting system. The bottom bars are #8 Gr. 60 deformed bars and the effective depth of the beam is 21 inches. Compressive strength of concrete is 4 ksi. There is no positive moment at the face of the column. The point of inflection is 2.5 ft away from the right face of the column. The clear cover is 1.5 in and the longitudinal bar clear spacing is 3 in. What is the minimum required extension 'L' of bottom bars from the face of the column? $\Psi_e = 1$, $\lambda = 1$

- (A) 0 in
(B) 6 in
(C) 21 in
(D) 48 in



- 459.** The beam column connection shown below has bundled #6 Gr. 60 deformed bars (three rebars in one bundle) at the top. What is the development length 'Ld' of the top bars in tension from the face of the column? The clear cover is 2 in and the clear spacing between the bundles is 7 in.
 $f'_c = 4000$ psi, $\Psi_e = 1$, $\lambda = 1$

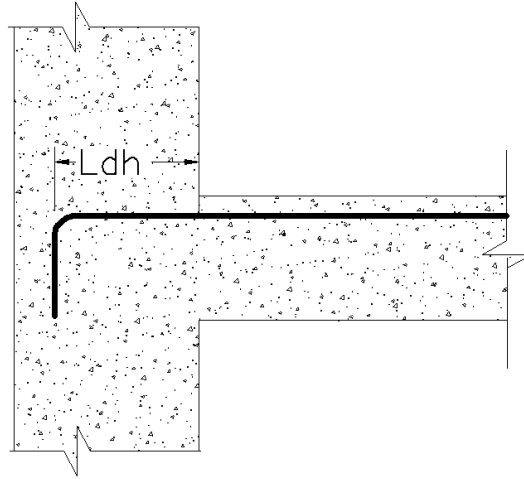
- (A) 6 in
(B) 21 in
(C) 37 in
(D) 45 in



- 460.** The figure below shows a beam column connection for an exterior column. The Gr. 60 #6 bar is hooked into a column that continues to the floors below. What is the minimum required hook development length 'Ldh' as shown below?

$f'_c = 4000$ psi, $\Psi_e = 1$, $\lambda = 1$

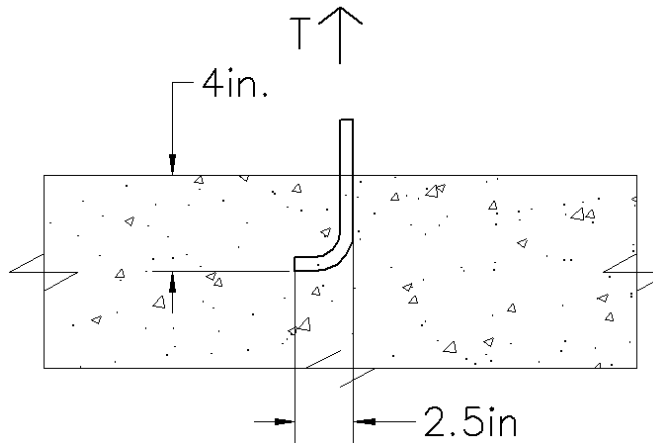
- (A) 6 in
- (B) 10 in
- (C) 12 in
- (D) 18 in



- 461.** J bolt is anchored to concrete as shown below. The diameter of the bolt is 0.5 in. The edge distance from the bolt center is 12 inches on all sides. Concrete is uncracked under service loads. What is the nominal concrete breakout strength under tension load?

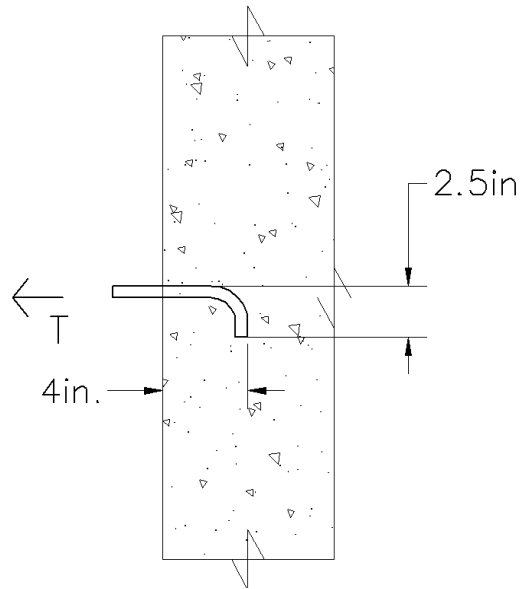
$f'_c = 4000$ psi, $\lambda_a = 1$

- (A) 6.7 kips
- (B) 9.9 kips
- (C) 12.4 kips
- (D) 15.7 kips



- 462.** J bolt is anchored to concrete as shown below. The diameter of the bolt is 0.5 in. Concrete is cracked under service loads. What is the design pullout strength under tension load?

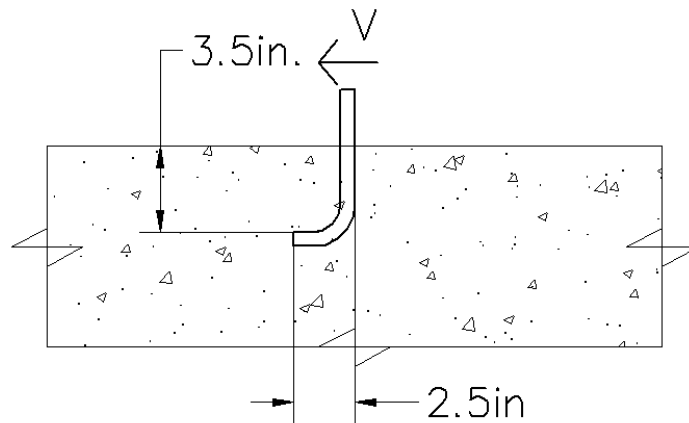
$f'_c = 4000 \text{ psi}$, $\lambda = 1$



- (A) 2.5 kips
- (B) 3.6 kips
- (C) 4.5 kips
- (D) 5 kips

- 463.** J bolt is anchored to concrete as shown below. The diameter of the bolt is 0.5 in. The edge distance from the bolt center is 12 inches on all sides. Concrete is cracked under service loads and supplementary reinforcement is not used. What is the design pryout strength under shear load?

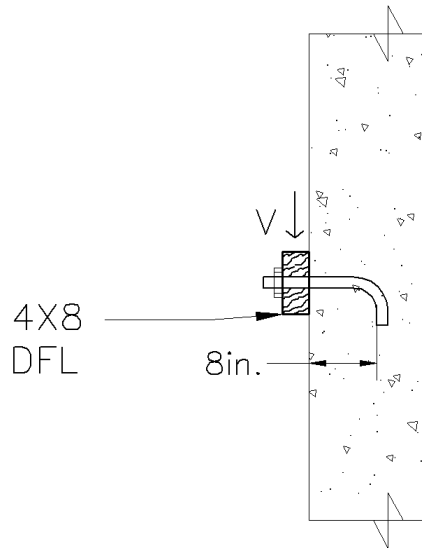
$f'_c = 3000 \text{ psi}$, $\lambda = 1$



- (A) 6.5 kips
- (B) 9.75 kips
- (C) 12 kips
- (D) 15 kips

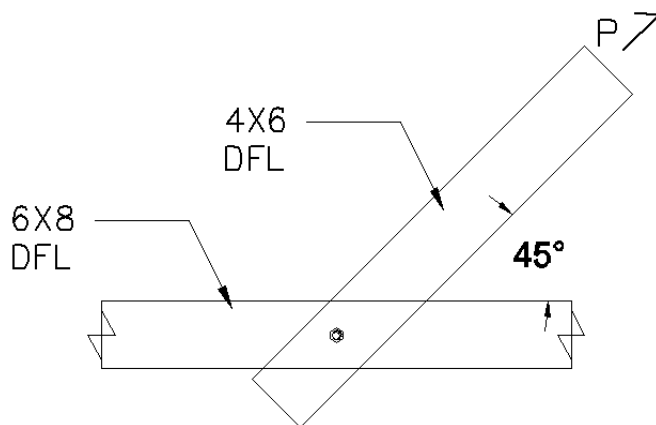
- 464.** A 1 inch diameter J bolt is used to anchor a 4X8 Douglas Fir-Larch ledger to a concrete shear wall as shown below. The bolt has a bending yield strength of 45000 psi and the concrete strength, bearing strength does not govern failure of the connection. What is the allowable vertical dead load on the ledger if the bolts are spaced at 4 ft c/c? Assume all adjustment factors except load factor are 1.

- (A) 165 plf
- (B) 255 plf
- (C) 300 plf
- (D) 455 plf



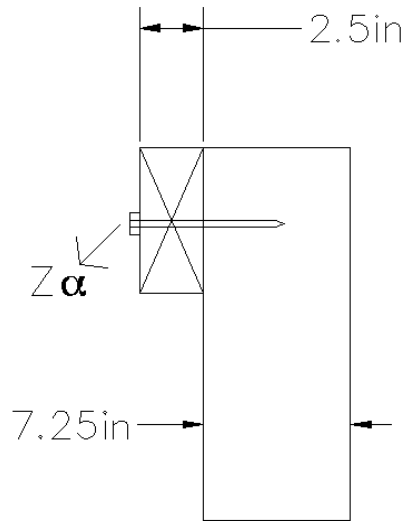
- 465.** A 6X8 DFL member is connected to a 4X6 DFL member with a single 1 inch diameter bolt. The connection is loaded at an angle 45° as shown below. What is the maximum allowable load 'P' in the connection due to the loading angle? Material of the bolt and bearing doesn't govern failure. All other adjustment factors are equal to 1.

- (A) 1390 lbs
- (B) 1522 lbs
- (C) 1826 lbs
- (D) 2260 lbs



- 466.** A standard 7 inch long $\frac{3}{8}$ inch diameter hex lag screw with minimum thread length is used to fasten two Douglas Fir-Larch members as shown below. The fastener is subjected to a loading at an angle 45° . What is the allowable load in the fastener ? Assume all other adjustment factors are equal to 1.

- (A) 140 lbs
(B) 250 lbs
(C) 305 lbs
(D) 325 lbs



- 467.** A 12 gauge steel plate is fastened to a 4X8 DFL member using 16d common nails. If the total lateral load to be transferred in the connection is 2800 lbs, what is the minimum number of nails required to secure the connection? Assume all adjustment factors are equal to 1.

- (A) 10
(B) 15
(C) 20
(D) 25

5.5 Shallow foundations

[Click here for solutions 5.5](#)

- 468.** A shallow strip footing is located 5 ft below the ground level. The footing is 3ft wide. The density of soil is 115 pcf. The angle of internal friction of the soil is 25° and the cohesion = 300 psf. Calculate the ultimate bearing capacity for the footing. The factor of safety against bearing is 3.
- (A) 3.25 ksf
 - (B) 4.73 ksf
 - (C) 8.28 ksf.
 - (D) 14.24 ksf
- 469.** A shallow strip footing is bearing on a soil layer with no cohesion. The footing is 2.5 ft wide and located at a depth of 5ft from the ground level. The density of soil is 110 pcf. The water table is measured at a depth 2.5 ft from the ground level. What is the ultimate bearing capacity of the foundation? The angle of internal friction of the soil is 30° .
- (A) 9130 psf
 - (B) 12350 psf
 - (C) 14700 psf
 - (D) 15200 psf
- 470.** Which of the following statements is not true about controlling differential settlements and its effects in a building with shallow foundations?
- (A) Planting trees and adequate vegetation around the perimeter of the property help stabilize and regulate soil moisture and control differential settlement,
 - (B) Proper surface and subsurface drainage systems in the property prevent water accumulation and minimize water related differential settlement.
 - (C) Expansion joints play a crucial role in controlling the effects of differential settlement,
 - (D) Soil compaction and stabilization reduces the risk of differential settlement.

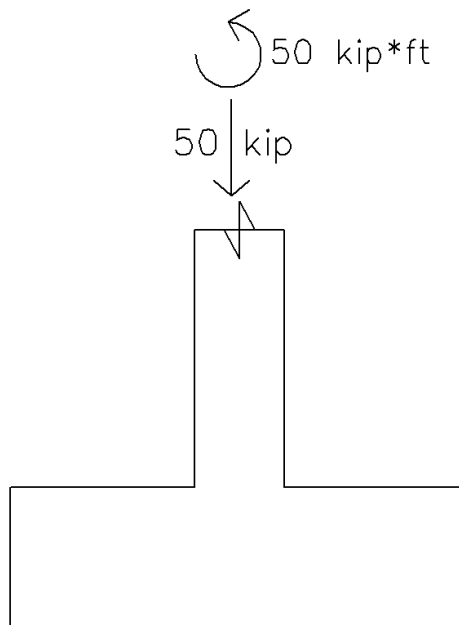
- 471.** The top of a 25ftX50ft concrete mat foundation is 5 ft below the ground level. The dead load acting on the mat due to superstructure is 1500 kips and the uplift due to wind load is 1800 kips. What is the minimum required depth of the mat if a factor of safety of 1.5 is desired against the uplift?

Unit weight of concrete = 145 pcf

Unit weight of soil = 110 pcf

- (A) 2 ft
- (B) 2.75 ft
- (C) 3 ft
- (D) 4 ft

- 472.** A 5ft X 5ft square RCC footing bears on dense sandy soil. The concentric axial load acting on the footing is 50 kips including the footing weight. The superstructure column also transfers a moment of 50 kip*ft on the footing. What is the maximum bearing pressure acting under the footing?

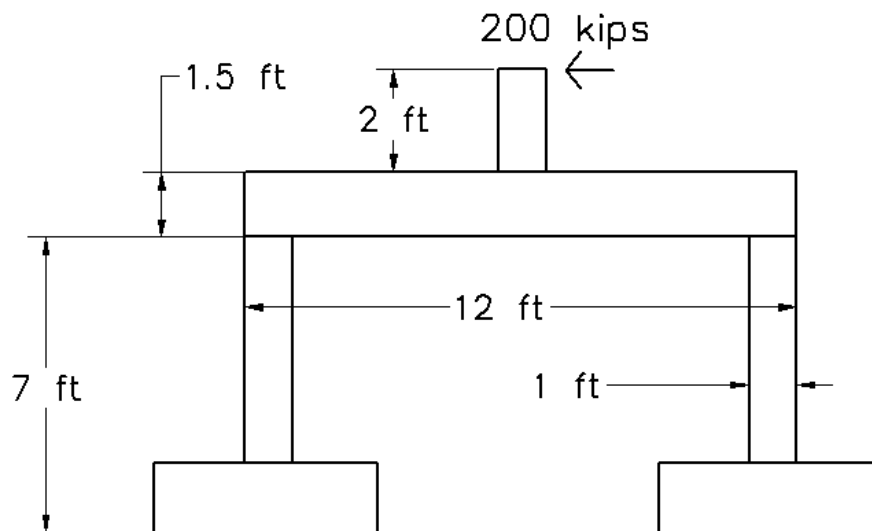


- (A) 2.3 ksf
- (B) 3.55 ksf
- (C) 4.44 ksf
- (D) 5.12 ksf

473. A 6ft X 4ft shallow footing is proposed at 2 ft below grade in a region with a frost depth of 3.5 ft. The soil is sandy with an allowable bearing pressure of 2.5 ksf. What is the most appropriate measure to ensure frost protection?

- (A) The footing depth can be kept at 2ft and soil should be insulated from thermal effects,
- (B) The footing depth has to be increased to 3.5 ft or deeper.
- (C) The footing size has to be reduced to limit frost heaving,
- (D) The foundation design has to be revised to pile foundations with sufficient capacity.

474. The figure below shows two columns supported by 4ftX4ft square footings. The vertical modulus of subgrade reaction of the bearing strata is 150 lb/in³. What is most nearly the settlement under the footing due to the lateral load of 200 kips. Self weight of the structure can be neglected and the stiffness of the structure can be assumed to be very high.



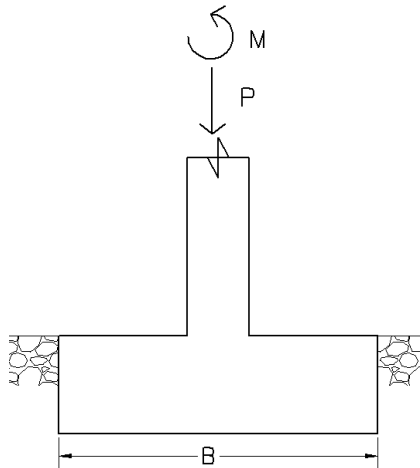
- (A) 0.24 in
- (B) 0.38 in
- (C) 0.55 in
- (D) 0.67 in

- 475.** A square concrete footing is subjected axial loads and moments (including footing weight) from the column as follows:

Dead load = 100 kips Dead load moment = 50 kip*ft

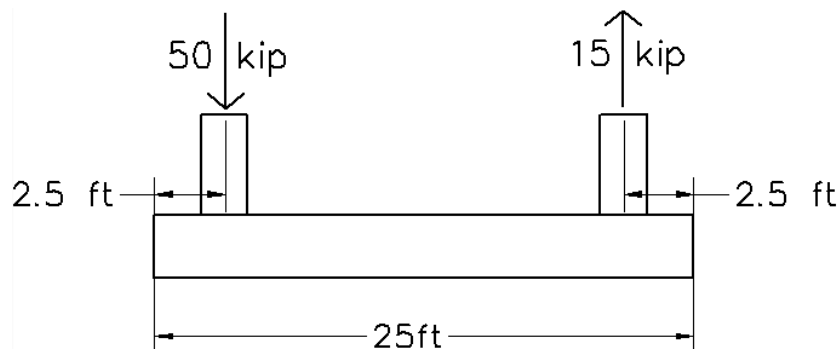
Live load = 150 kips, Live load moment = 80 kip*ft

What should the minimum dimension 'B' of the footing be such that the entire footing area is effective in bearing the loads?



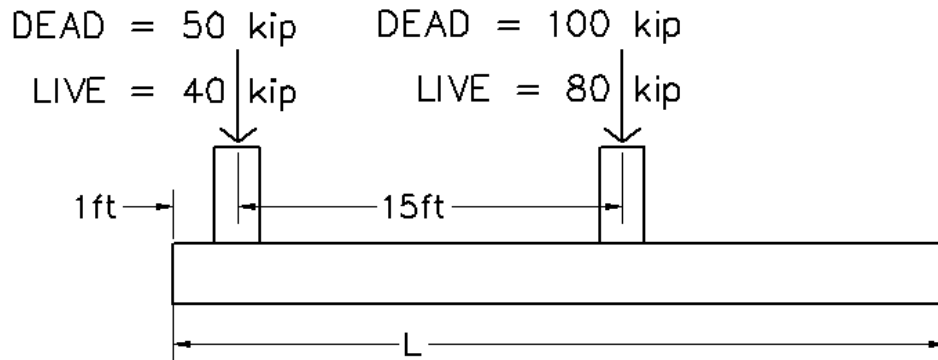
- (A) 2.75 ft
- (B) 3 ft
- (C) 3.25 ft
- (D) 4 ft

- 476.** A combined footing supports two columns as shown below. The self weight of the footing is 1.8 klf and the width of the footing is 6ft. What is the maximum bearing pressure under the footing?

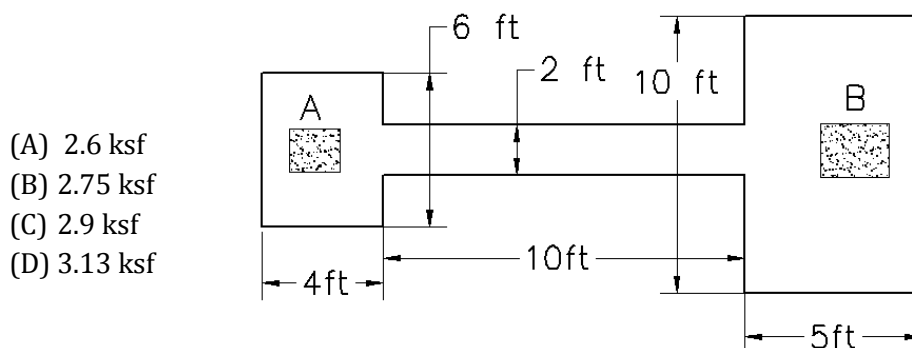


- (A) 0.53 ksf
- (B) 1.55 ksf
- (C) 1.74 ksf
- (D) 2.03 ksf

477. A combined footing supports two columns with axial loads as shown below. What should the length of footing 'L' be such that there is uniform bearing pressure under the combined footing.

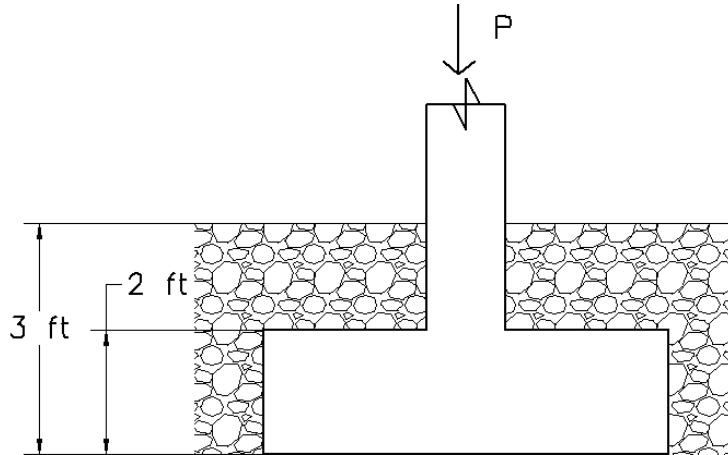


- (A) 18 ft
(B) 22 ft
(C) 25 ft
(D) 32 ft
478. Which theoretical assumption is most commonly used when designing shallow foundations for soils that can deform under stress?
- (A) The soil behaves as a rigid body with infinite stiffness.
(B) The soil can be modeled as a series of independent linear springs.
(C) The soil has no shear strength and relies solely on cohesion.
(D) The footing is infinitely flexible and does not distribute loads.
479. The plan of a strap footing is shown below. The total compression load in column A is 58 kips and the total compression load in column B is 121 kips. The depth of the footings are 2.5 ft and the depth of strap beam is 2 ft. Bearing under strap beam is prevented by using a styrofoam layer underneath. What is most nearly the maximum bearing pressure under footing B? Unit weight of concrete = 150 pcf.



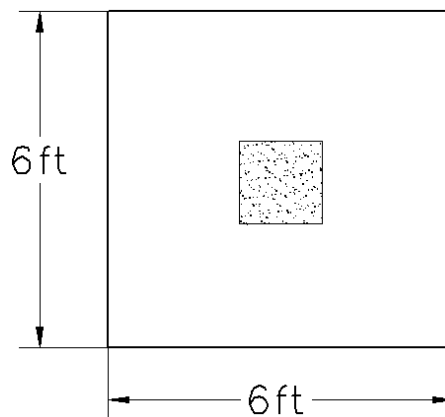
- (A) 2.6 ksf
(B) 2.75 ksf
(C) 2.9 ksf
(D) 3.13 ksf

- 480.** A square footing is located at a site where the bearing layers are classified as sandy gravel and gravel (GW and GP). What is the required width of the footing if a concentric compression of 40 kips acts in the column following IBC provisions?
Unit weight of soil = 115 pcf. Unit weight of concrete = 150 pcf.



- (A) 3.5 ft
- (B) 4 ft
- (C) 4.5 ft
- (D) 5 ft

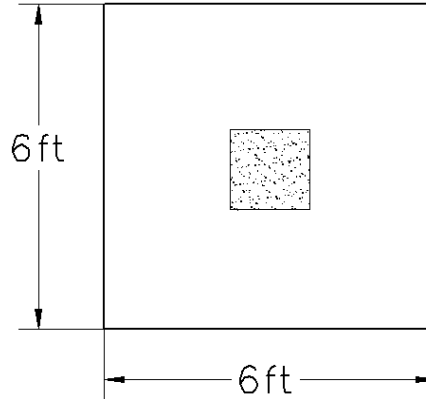
- 481.** The plan of a square RCC footing is shown below. The dimensions of the column at the center of the footing is 18 inX 18in. The factored concentric axial load acting in the column is 50 kips. What is the required design flexural strength of the footing?



- (A) 12 kip*ft
- (B) 18 kip*ft
- (C) 21 kip*ft
- (D) 25 kip*ft

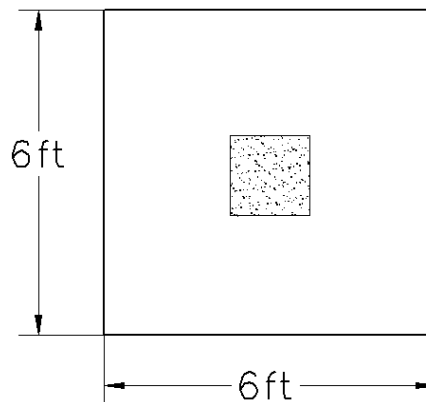
- 482.** The plan of a square RCC footing is shown below. The critical factored shear acting on the footing is 150 kips. The effective cover of the bottom flexural bars is 3 in and there are no shear reinforcements. What is the required depth of footing for one way shear strength?
 $f'_c = 4 \text{ ksi}$ $\lambda = 1$

- (A) 18 in
 (B) 22 in
 (C) 25 in
 (D) 28 in



- 483.** The plan of a square RCC footing is shown below. The dimensions of the column at the center of the footing is 18 in X 18 in. The effective depth of the footing is 21 in and there are no shear reinforcements. What is the design two way shear strength of the footing?
 $f'_c = 4 \text{ ksi}$ $\lambda = 1$

- (A) 418 kips
 (B) 505 kips
 (C) 622 kips
 (D) 828 kips



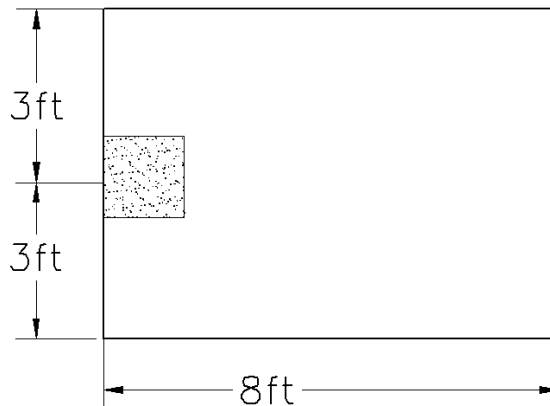
- 484.** A 7ftX7ft square RCC footing is bearing on dense sand and has a depth of 18 inches. Per ACI 318-14 which of the following rebar layout is the most economical option that satisfies the minimum flexural rebar requirement for Gr 60 rebar?

$$f'_c = 4 \text{ ksi} \quad \lambda = 1$$

- (A) #4 @ 12 in. c/c
- (B) #5 @ 12 in. c/c
- (C) #6 @ 12 in. c/c
- (D) #7 @ 12 in. c/c

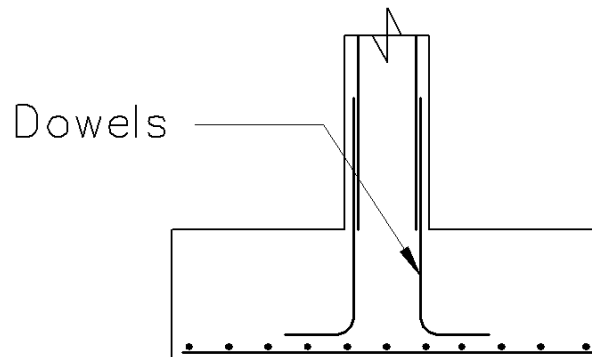
- 485.** The plan of a RCC 6ftX8ft footing is shown below. The footing has an eccentric column at the edge. The dimensions of the column is 18inX18in. What is the design bearing capacity of the column on the footing if the effective depth of the footing is 21 in?

$$f'_c = 4 \text{ ksi} \quad \lambda = 1$$



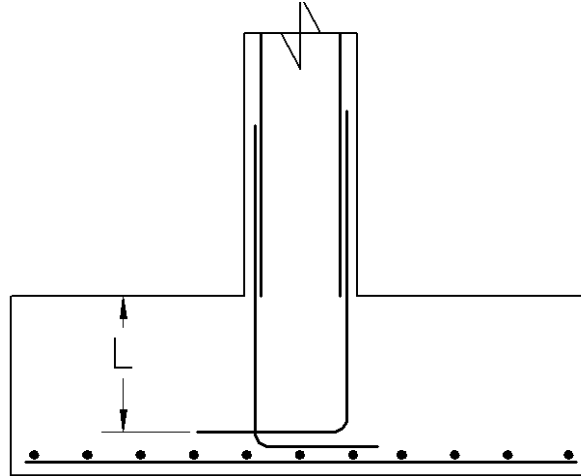
- (A) 612 kips
- (B) 716 kips
- (C) 941 kips
- (D) 1100 kips

- 486.** A 7ftX7ft RCC footing supports a 24 inX 24 in column at the center. The factored axial compression transferred by the column is 1500 kips and the design bearing capacity is 2500 kips. What is the minimum dowel reinforcement required between the column and the footing? $f'_c = 4 \text{ ksi} \quad \lambda = 1$



- (A) Four #6
- (B) Four #7
- (C) Four #8
- (D) Four #9

- 487.** A footing supports a 20 inX 20 in column as shown below. The column transfers moments and axial load into the footing. What is the minimum embedment of the #8 dowel reinforcement 'L' into the footing as shown below? Assume Grade 60 deformed bars.
 $f'_c = 4 \text{ ksi}$ $\lambda = 1$



- (A) 12 in
- (B) 15 in
- (C) 18 in
- (D) 20 in

5.6 Deep foundations

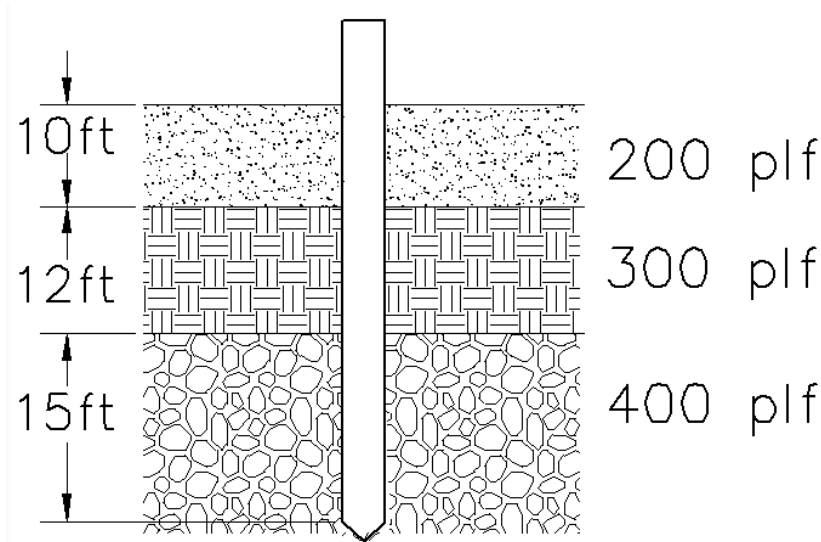
[Click here for solutions 5.6.](#)

- 488.** Which of the following is true about end bearing piles and friction piles?
- (A) End bearing piles are shorter than friction piles,
 - (B) The bearing capacity of end bearing piles depends upon pile material whereas the bearing capacity of skin friction piles are only affected by the properties of surrounding soil,
 - (C) The phenomenon of negative skin friction can affect both end bearing piles and friction piles,
 - (D) End bearing piles are used in cohesionless soil whereas friction piles are used in cohesive soil.
- 489.** During the design of a pile group, when is it necessary to consider group effects ?
- (A) When the number of piles in a group is greater than four,
 - (B) When the diameter of piles is greater than one third of the center to center spacing of the piles,
 - (C) When the piles are end bearing,
 - (D) When piles groups are driven in cohesive soil with significant scour depth.
- 490.** What is negative skin friction in piles and when does it occur in piles?
- (A) It is the downward drag on the piles due to consolidation of surrounding soil,
 - (B) It is the additional uplift force in the piles acting in the negative direction of applied load.
 - (C) It is the reduction in the frictional bearing capacity of the piles due to pile driving methods,
 - (D) It is the skin friction acting in the negative direction during extraction of piles.

- 491.** A concrete caisson is driven 20 ft inside the ground. The thrust acting on the caisson is 75 kips and the uplift acting is 45 kips. The thrust and uplift act in the caisson separately. The allowable end bearing is 12 ksf and the allowable skin friction is 300 psf. What is the minimum required diameter of the caisson ?

(A) 2 ft
(B) 2.5 ft
(C) 2.75 ft
(D) 3 ft

- 492.** A friction pile is driven through different layers of soil strata in a river bed as shown below. The allowable skin friction of each layer per unit length of pile is given next to the layer. The hydraulic specialist determined that the scour depth due to water currents in the river bed is equal to 8 ft. What is the allowable axial load capacity of the piles?

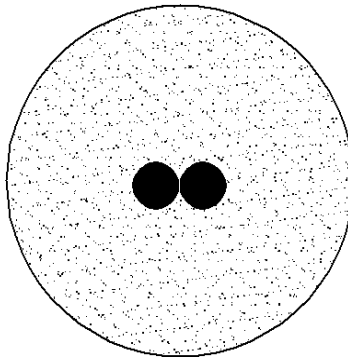


(A) 10 kips
(B) 11.5 kips
(C) 12 kips
(D) 15 kips

493. A pile is driven using a single acting air hammer weighing 700 lbs. The freefall height of the hammer is 12 ft. Every three blows from the hammer drives the pile 1 inch into the ground. What is most nearly the allowable load for the pile?

- (A) 15 ton
- (B) 19 ton
- (C) 23 ton
- (D) 30 ton

494. The figure below shows the cross section of a reinforced concrete micropile that is cast in a rock socket. The diameter of the micropile is 12 in. The compressive strength of grout is $f'_c = 6000$ psi. The two center rebars used are #18 Grade 75. What is the allowable compression in the micropile? Area of #18 bar = 4 in^2

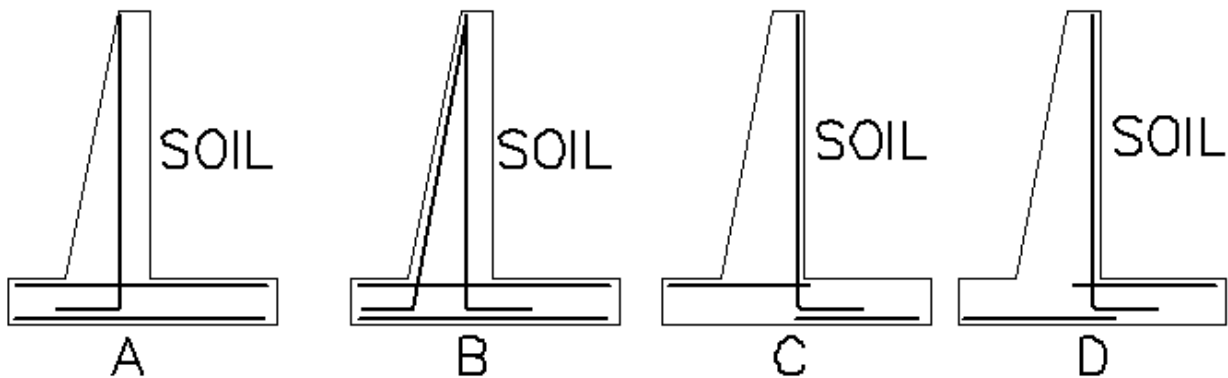


- (A) 328 kips
- (B) 368 kips
- (C) 402 kips
- (D) 448 kips

5.7 Retaining walls

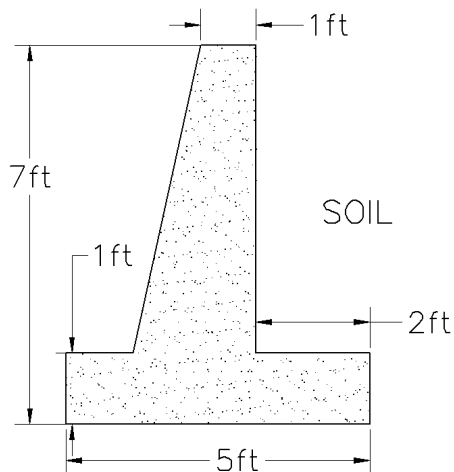
[Click here for solutions 5.7](#)

495. The cross section of a cantilevered retaining wall that is retaining soil on the vertical face of the wall is shown below. What is the correct reinforcement layout for the wall ?



- (A) Option A
(B) Option B
(C) Option C
(D) Option D

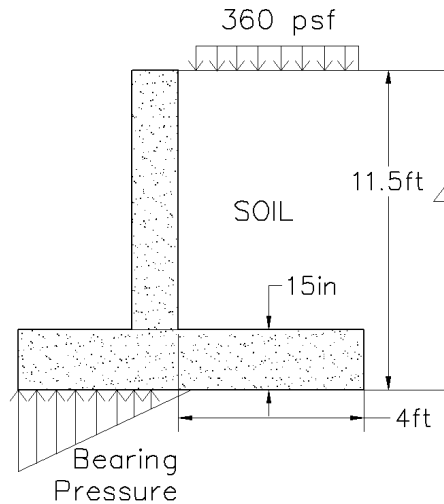
496. A cantilevered wall is retaining soil on one side only as shown below. The height of the backfill is 7ft. What is the maximum factored flexural demand on the stem of the wall?
Density of soil = 110 pcf Active earth pressure coefficient, $K_a = 0.33$



- (A) 1100 lb*ft/ft
(B) 1306 lb*ft/ft
(C) 1893 lb*ft/ft
(D) 2091 lb*ft/ft

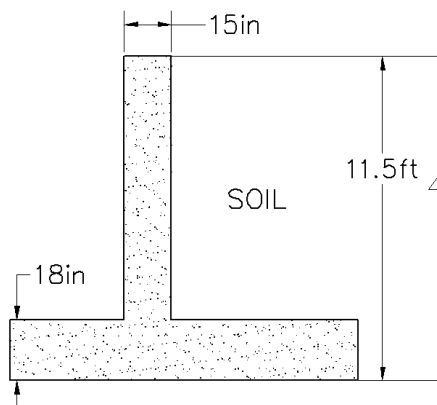
- 497.** A reinforced concrete cantilevered wall is retaining 11.5 ft tall backfill. A surcharge load of 360 psf acts on top of the backfill. The bearing pressure at the base of the wall due to combined eccentric loading is shown below. What is the maximum factored flexural demand on the heel? Consider equal load factor for soil and surcharge.
 Density of soil = 120 pcf Unit weight of concrete = 150 pcf

- (A) 11520 lb*ft/ft
 (B) 13226 lb*ft/ft
 (C) 18457 lb*ft/ft
 (D) 22152 lb*ft/ft



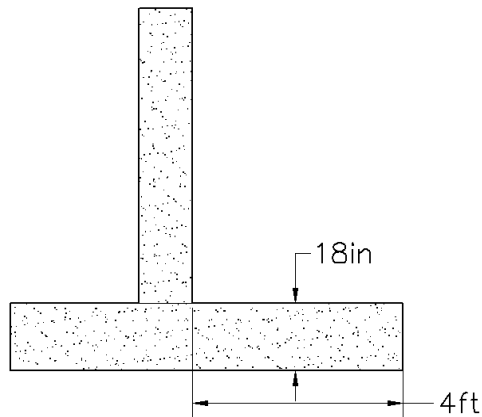
- 498.** A reinforced concrete cantilevered wall is retaining 11.5 ft tall backfill. What is the distance of the critical location for one way shear demand calculation of the stem, from the top of the wall? Effective cover of tension bars at the stem = 3 in. Density of soil = 120 pcf, Unit weight of concrete = 150 pcf, Active earth pressure coefficient = 0.33

- (A) 5.75 ft
 (B) 9 ft
 (C) 10 ft
 (D) 11.5 ft



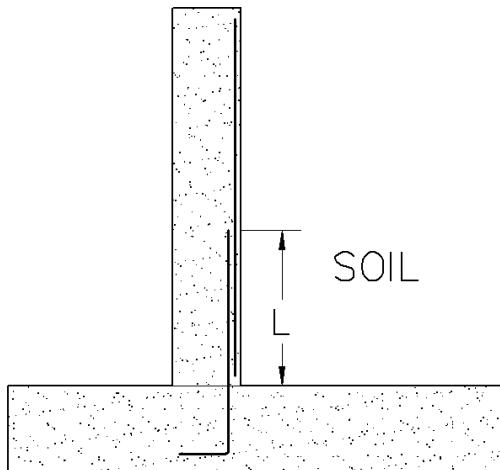
- 499.** The heel of the retaining wall is subjected to 13583 lb*ft/ft factored flexural demand. What is the required area of tension reinforcement for the heel? Effective cover of the tension bars = 3.5in. $f'_c = 4.5$ ksi , $F_y = 60$ ksi

- (A) 0.201 in²/ft
 (B) 0.583 in²/ft
 (C) 0.671 in²/ft
 (D) 0.852 in²/ft

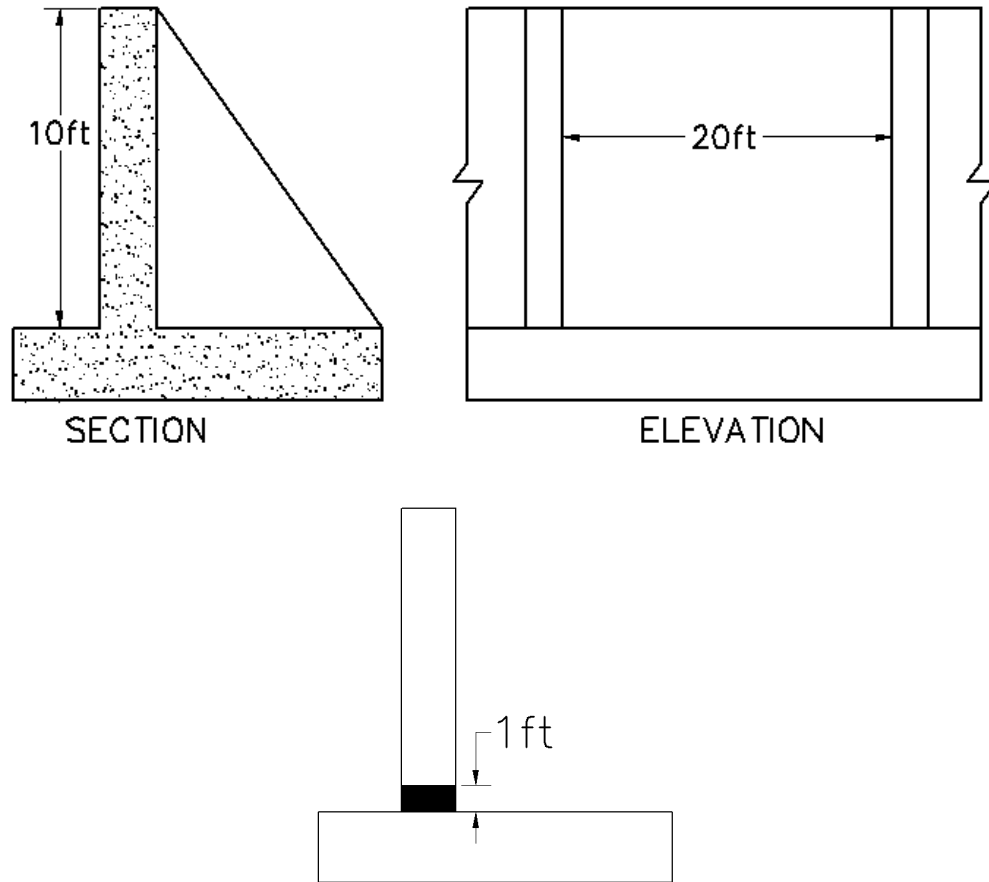


- 500.** A cantilever retaining wall retains soil on one side of the wall as shown below. The reinforcement used is Gr. 60 uncoated deformed bars. What is the required splice length 'L' of the #5 bars spaced 12 inches c/c along the run of the wall. Clear cover of the bars is 2in. The compressive strength of concrete $f'_c = 4000$ psi. $\lambda = 1$

- (A) 18 in
 (B) 24 in
 (C) 32 in
 (D) 38 in



- 501.** The interior segment of a 100 ft long counterfort RCC retaining wall system is shown below. The wall is retaining backfill on the counterfort side. The height of the backfill is equal to the height of the wall. What is most nearly the maximum factored moment to be considered for the design of horizontal flexural reinforcement for the bottom 1 ft strip of the stem? Density of soil = 120 pcf, Active earth pressure coefficient, $K_a = 0.33$



- (A) 25 kip*ft
- (B) 30 kip*ft
- (C) 35 kip*ft
- (D) 40 kip*ft

SOLUTIONS



1. Analysis of Structures–Loads and Load Applications

1.1 Dead Load

[Click here for Problems 1.1](#)

1. Problem

Correct Answer: (D) 34 kip

Explanation:

Calculate the weight of water

Weight of 1 gallon = 8.34 lbf {Ref NCEES handbook page 2}

Weight of 2000 gallons = $2000 \times 8.34 = 16680$ lbf

Calculate weight of tank

Internal volume of tank = 2000 gallons

Solve for internal radius 'r' of tank

$$\frac{4}{3}\pi r^3 = 2000 * 0.134 \text{ cubic feet} \text{ {Ref NCEES handbook page 3 for conversion factors}}$$

$$r = 4 \text{ ft}$$

Calculate volume of 6 inches thick spherical concrete shell

$$\frac{4}{3}\pi(4 \text{ ft} + 6 \text{ inches})^3 - \frac{4}{3}\pi(4 \text{ ft})^3 = 113.62 \text{ cubic feet}$$

{Ref ACI 318-14 handbook page 34 for weight of normal weight concrete}

Weight of tank = $113.62 \text{ cubic feet} * 150 \text{ lb/ft}^3 = 17043$ lbf

Total dead load = Weight of water + Weight of tank

$$= 16680 \text{ lbf} + 17043 \text{ lbf} = 33723 \text{ lbf} \approx \mathbf{34 \text{ kips}}$$

2. Problem

Correct Answer: (C)

Explanation:

Per ASCE 7-16, section 3.1.4 'The weight of all landscaping and hardscaping materials shall be considered as dead load. The weight shall be computed considering both fully saturated soil and drainage layer materials and fully dry soil and drainage layer materials to determine the most severe load effects on the structure.'

Hence, option C that says the average moisture content of soil should be taken, is against the code and is not true.

3. Problem

Correct Answer: (C) 264.6 lb/ft

Explanation:

Weight of steel deck = 5 psf

Density of lightweight concrete = 115 lb/ft³ {Ref ACI 318-14 page 33}

Calculate the weight of slab per unit area

Slab weight = 3 inches* 115 lb/ft³ = 28.75 psf

Calculate Floor load acting on steel beam spaced at 6ft c/c

{Ref AISC steel manual 15th Ed. page 1-29 Table 1-1 for self weight of W8X18}

Floor Load = deck weight + slab weight = (5psf + 28.75 psf)* 6ft = 202.5 lb/ft

Total load acting on beam = self-weight of W8X18 beam + Floor Load

= 18 lb/ft + 202.5 lb/ft

= 220.5 lb/ft

Total Factored Dead Load (used in live load combination)= 220.5 lb/ft *1.2 = **264.6 lb/ft**

4. Problem

Correct Answer: (B) 630 lb

Explanation:

Length of beam = 20ft

Weight of W10X30 = 30 lb/ft {Ref AISC manual table 1-1}

Beam weight = 30lb/ft* 20ft = 600 lb

Density of steel = 490 lb/ft³

Calculate the weight of stiffener plates

Plate 1 = 490 lb/ft³ * 2.75in * 9.5in * 0.25in = 1.85 lb

Plate 2 = 490 lb/ft³ * 2.75in * 9.5in * 0.5in = 3.7 lb

Spacing of stiffeners = 4ft c/c

Total number of stiffened points = 20ft/4ft+1 = 6. These plates are on both sides of the web.

Thus, there are 2X2 end bearing plates and 4X2 inside web stiffeners.

Total weight = Weight of beam + weight of end plates + weight of inside plates
= 600lb + 2*2*3.7lb+4*2*1.85lb = 629.6 lbs $\approx$ **630 lb**

5. Problem

Correct Answer: (B) 61 kip

Explanation: {Ref NCEES manual page 24 for area of common shapes}

Area of trapezoidal cross section = 0.5 * height * (sum of opposite parallel sides)
= 0.5*10ft* (2ft +4ft)
= 30ft²

Length of abutment = 12ft

Volume of abutment = Area*Length
= 12ft*30ft²
=360 ft³

Density of stone masonry = 0.17 kcf

{Ref AASHTO LRFD Table 3.5.1-1 Unit weights}

Weight of abutment = density *volume
= 0.17 kcf*360 ft³
=61.2 kip
 $\approx$ **61 kip**

6. Problem

Correct Answer: (D) 250 tons

Explanation: {Ref NCEES manual page 3 for conversion factors}

Convert kN to lbf

$$1\text{ N} = 0.225\text{ lbf}$$

$$1\text{ kN} = 1000 * 0.225\text{ lbf}$$

$$1\text{ kN} = 225\text{ lbf}$$

$$2020\text{ kN} = 2020 * 225\text{ lbf} = 454500\text{ lbf}$$

Convert lbf to (short) ton

$$2000\text{ lbf} = 1\text{ ton}$$

$$454500\text{ lbf} = (1/2000) * 454500\text{ tons}$$

$$= 227.25\text{ tons}$$

Thus a jack with capacity greater than 227.25 tons is required. Use **250 tons** jack from the options.

7. Problem

Correct Answer: (B) 1282 lb

Explanation: {Ref AISC Steel manual page 1-97 and page 1-99 Table 1-12 for weight of HSS sections}

$$\text{Weight of HSS } 6\text{X}6\text{X}1/2 = 35.24\text{ lb/ft}$$

$$\text{Weight of HSS } 2\text{X}2\text{X}1/4 = 5.41\text{ lb/ft}$$

$$\text{Total length of vertical and horizontal HSS } 6\text{X}6\text{X}1/2 \text{ members} = (10+6+6+4+4)\text{ ft} = 30\text{ft}$$

$$\text{Length of individual diagonal members} = \sqrt{3^2 + 4^2}\text{ ft} = 5\text{ft}$$

$$\text{Total length of diagonal members} = (5+5+5+5)\text{ft} = 20\text{ft}$$

$$\text{Total weight of assembly} = \text{Weight of } 6\text{X}6\text{X}1/2 \text{ members} + \text{Weight of } 2\text{X}2\text{X}1/4 \text{ members}$$

$$= 30\text{ft} * 35.24\text{lb/ft} + 20\text{ft} * 5.41\text{ lb/ft}$$

$$= 1165.4\text{ lb}$$

$$\text{Capacity of hoisting mechanism required} = \text{Total weight} + 10\% \text{ Total weight for rigging}$$

$$= 1.1 * 1165.4\text{ lb}$$

$$= 1281.91\text{ lb} \approx \mathbf{1282\text{ lb}}$$

8. Problem

Correct Answer: (B) 468 lb

Explanation:

{Ref ASCE 7-16 page 430 Table C3.1-2 for unit weight of brick masonry}

Unit Weight of hard brick masonry = 130 lb/ft^3

Area of masonry wall = $9 \text{ inches} \times 4 \text{ ft} = 3 \text{ ft}^2$

Uniformly distributed dead load due to wall = $130 \text{ lb/ft}^3 \times 3 \text{ ft}^2$

= 390 lb/ft

Factored dead load to be used in combination with live load = $1.2 \times 390 = 468 \text{ lb/ft}$

1.2 Live Load

[Click here for Problems 1.2](#)

9. Problem

Correct Answer: (C) 27 kips

Explanation:

{Ref ASCE 7-16 page 13, section 4.3.2 for provisions for partition}

Minimum partition wall load to be considered = 15 psf

Since the 1st floor is an open parking lot, the partition wall requirement does not apply.

Total floor area to be considered for partition walls = 30ftX30ft*2 floors

= 30ftx30ftx2

=1800 ft²

Total load = 15 psf*1800 ft²

=27000 lb

= **27 kips**

10. Problem

Correct Answer: (D) 2000 kips

Explanation:

{Ref ASCE 7-16 page 15, Table 4.3-1 for provisions for warehouse live loads}

Uniformly distributed warehouse load to be considered for heavy storage= 250 psf

Total area to be considered for storage = 50ftX100ft

=5000 ft²

Total load = 250 psf*5000 ft²

=1,250,000 lb

=1250 kips

Factored Live load = 1.6*1250 kips = **2000 kips**

11. Problem

Correct Answer: (C) 4067 kips

Explanation:

{Ref IBC 2018 page 368, Table 1607.1 for provisions for staircase live loads}

Uniformly distributed staircase load to be considered for school buildings=100 psf

Calculate total area of horizontal projection of stairs and landing.

No of treads = 5

Depth of treads = 10 inches

Width of stairs and landings = 4ft

Area of horizontal projection of treads = $5 \times 10 \text{ inches} \times 4 \text{ ft} = 16.67 \text{ ft}^2$

Area of two landings = $2 \times 3 \text{ ft} \times 4 \text{ ft} = 24 \text{ ft}^2$

Total area = $24 \text{ ft}^2 + 16.67 \text{ ft}^2 = 40.67 \text{ ft}^2$

Total load = $100 \text{ psf} \times 40.67 \text{ ft}^2$

=4067 lb

12. Problem

Correct Answer: (B) 3.125 kips/ft

Explanation:

{Ref ASCE 7-16 page 15, Table 4.3-1 for provisions for gymnasium live loads}

Uniformly distributed load to be considered for gymnasiums $L_0=100 \text{ psf}$

Tributary Area, $A_T=1250 \text{ ft}^2$

Per ASCE 7-16 page 17, section 4.7.5, for assembly usage, live loads cannot be reduced.

Hence, total live load acting on the beam = $A_T \times L_0$

$=1250 \text{ ft}^2 \times 100 \text{ psf}$

$=125000 \text{ lb}$

$=125 \text{ kips}$

Uniformly distributed load

$=125 \text{ kip/Length of beam}$

$=125 \text{ kip}/40 \text{ ft}$

= 3.125 kips/ft

13. Problem

Correct Answer: (C) **34 kips**

Explanation:

{Ref ASCE 7-16 page 16 and 17 for live load reduction provisions}

$K_{LL} = 4$ Live load element factor From Table 4.7-1

$L_o = 57.5$ kips

$A_T = 500$ ft²

Calculate reduced live load using equation 4.7-1

$$L = L_o \left(0.25 + \frac{15}{\sqrt{K_{LL} * A_T}} \right)$$

$$L = 57.5 \text{ kips} * \left(0.25 + \frac{15}{\sqrt{4 * 500}} \right)$$

$$L = 57.5 \text{ kips} * 0.585 \text{ (Check reduction should not be less than } 0.4 L_o, \text{ OK)}$$

$$L = 33.63 \text{ kips} \approx \mathbf{34 \text{ kips}}$$

14. Problem

Correct Answer: (C) The expected service life of the structure

Explanation:

Per IBC 2018 section 1607.11,

Live load reduction is not permitted for heavy loads exceeding 100 psf,

Live load reduction factor depends on the tributary area in equation 16-23,

Live load reduction is not permitted in structures designed for storage.

There are no clauses discussing the service life of structures and corresponding live load reduction.

Hence option C is the correct answer as it does not affect the reduction.

15. Problem

Correct Answer: (B)5760 lbs

Explanation:

Per IBC 2018 table 1607.1, page 368, minimum live load to be considered for a school classroom is 40psf.

Calculate the total expected load from given information and check against this minimum requirement.

Total expected live load due to students and furniture

$$= (10+2)*200 \text{ lb} + 2000 \text{ lb}$$

$$= 4400 \text{ lb}$$

Calculate minimum design load per IBC.

$$\text{Area} = 12\text{ft} * 12\text{ft} = 144 \text{ ft}^2$$

$$\text{Load} = 40\text{psf} * 144\text{ft}^2$$

$$= 5760 \text{ lbs}$$

Minimum requirement governs over expected load, hence correct answer is **5760 lbs**

16. Problem

Correct Answer: (C)3125 lbs

Explanation:

Per ASCE 7-16 page 15, Table 4.3-1 minimum live load on roof without occupancy is 20psf.

Per ASCE 7-16 , page 17, section 4.8.2

Check applicable reduction factors,

Tributary area for column A

$$A_T = (25\text{ft} * 25\text{ft}) / 4 = 156.25 \text{ ft}^2 \text{ (less than } 200\text{ft}^2\text{)}$$

For flat roof, pitch factor $F = 0$

Thus reduction factors R_1 and R_2 are both 1, hence the roof live load cannot be reduced.

Live load to be considered = 20 psf

$$\text{Total live load acting on column A} = 20\text{psf} * 156.25 \text{ ft}^2 = \mathbf{3125 \text{ lbs}}$$

17. Problem

Correct Answer: (B)260 lb/ft

Explanation:

Tributary area = 300 ft²

Per ASCE 7-16 , page 17, section 4.8.2, the reduction factors for roof live load are

$$R_1 = 1.2 - 0.001 \cdot A_T \text{ for } 200 \text{ ft}^2 < A_T < 600 \text{ ft}^2$$

$$= 1.2 - 0.001 \cdot 300$$

$$= 0.9$$

Calculate pitch of roof

Pitch = 6 inches/1 feet

Thus, F = 6

$$R_2 = 1.2 - 0.05 \cdot F \text{ for } 4 < F < 12$$

$$= 1.2 - 0.05 \cdot 6$$

$$= 0.9$$

Calculate total reduced load $L_R = 200 \text{ lb/ft} \cdot 0.9 \cdot 0.9$

$$= 162 \text{ lb/ft}$$

Factored roof live load for strength design = $1.6 \cdot 162 \text{ lb/ft} = 259.2 \text{ lb/ft} \approx \mathbf{260 \text{ lb/ft}}$

18. Problem

Correct Answer: (C)12.2 kips

Explanation:

Total live load on ground floor column

$$= (0.5 \text{ kips/ft} + 1.2 \text{ kips/ft}) \cdot 15 \text{ ft} / 2$$

$$= 12.75 \text{ kips}$$

Tributary area = 112.5 ft²

$K_{LL} = 4$ Live load element factor for exterior column From Table 4.7-1

$$K_{LL} \cdot A_T = 4 \cdot 112.5 = 450 \text{ ft}^2$$

$$L = L_o \left(0.25 + \frac{15}{\sqrt{K_{LL} \cdot A_T}} \right)$$

$$L = 12.75 \text{ kips} \left(0.25 + \frac{15}{\sqrt{4 \cdot 112.5}} \right) = \mathbf{12.2 \text{ kips}}$$

19. Problem

Correct Answer: (C) Sloped roofs have reduced accessibility and are less likely to be occupied

Explanation:

Strength of roofing members depend on materials and section properties, not inclination, Snow loads and seismic loads have nothing to do with reduction factors of live loads.

Hence the correct answer is because of decreased access in roofs with higher slopes, loads are permitted to be reduced.

20. Problem

Correct Answer: (B) ASCE 7-16 specified concentrated live loads are summed with minimum uniformly distributed live loads to create the most severe effect in the structural member

Explanation:

Per ASCE 7-16 page 13 section 4.4, concentrated loads are assumed to act uniformly over an area of $762\text{mm} \times 762\text{mm} = 0.58 \text{ m}^2$.

ASCE 7-16 does not provide provisions for reductions of concentrated live loads.

Concentrated loads govern punching and flexural failure of slabs and are usually more critical in slab design than uniform loads.

All other options are correct except that minimum concentrated loads and minimum uniform loads given in the code are summed. They are not summed. They are checked separately to check which condition governs the design.

1.3 Construction loads

[Click here for Problems 1.3](#)

21. Problem

Correct Answer: (B) 750 kips

Explanation:

Cross section area of HP 10X57 = 16.7 in^2 {Ref AISC Steel manual page 1-36}

Yield stress of material = 50 ksi

Allowable pile driving stress in H piles = $0.9 f_y$ {Ref NCEES PE handbook page 62}

= $0.9 \times 50 \text{ ksi} = 45 \text{ ksi}$

Allowable pile driving load = Allowable driving stress * cross section area

= $45 \text{ ksi} \times 16.7 \text{ in}^2$

= 751.5 kips

≈ **750 kips**

22. Problem

Correct Answer: (C) 4ft

Explanation:

Construction load = 50 psf

Density of slab = 150 pcf

Thickness of slab = 6 inch = 0.5 ft

Dead load = $150 \text{ pcf} \times 0.5 \text{ ft} = 75 \text{ psf}$

Total load during concrete placement = construction load + self-weight of slab

= $75 \text{ psf} + 50 \text{ psf}$

= 125 psf

Allowable carrying capacity of floor props = 2 kips = 2000 lb

Area that can be safely supported = prop capacity / loading intensity

= $2000 \text{ lb} / (125 \text{ lb} / \text{ft}^2)$

= 16 ft^2

Maximum permitted spacing in each direction = $\sqrt{16} \text{ ft}$

S = 4ft

23. Problem

Correct Answer: (C) The load factors for permanent live loads and construction loads are different in ASD methodology.

Explanation:

The statement is false because the load factor is 1 in ASD methodology for both permanent live loads and construction loads.

24. Problem

Correct Answer: (B) 9 kips

Explanation:

Sum of moments about the tipping point must be zero.

Resisting moments about tipping point due to self weight of crane body

$$\begin{aligned}M_R &= 150 \text{ kip} \cdot 10 \text{ ft} \\&= 1500 \text{ kip} \cdot \text{ft}\end{aligned}$$

Overtipping moments about tipping point due to load and boom self weight

$$M_O = \text{Load} \cdot 60 \text{ ft} + 15 \text{ kip} \cdot 30 \text{ ft}$$

For equilibrium, including a factor of safety 1.5 against overturning

$$1.5 (\text{Load} \cdot 60 \text{ ft} + 15 \text{ kip} \cdot 30 \text{ ft}) = 1500 \text{ kip} \cdot \text{ft}$$

$$\text{Load} \cdot 60 \text{ ft} = 1500 \text{ kip} \cdot \text{ft} / 1.5 - 450 \text{ kip} \cdot \text{ft}$$

$$\text{Solve for Tipping load} = 9.166 \text{ kips} \approx \mathbf{9 \text{ kips}}$$

25. Problem

Correct Answer: (A) 5 kips/ft

Explanation:

Unit weight of concrete $\gamma = 150 \text{ pcf}$, Lateral pressure exerted by concrete $= \gamma \cdot (10 \text{ ft})$

Total lateral load exerted by 10ft high fresh concrete

$$L = 0.5 \cdot \text{Lateral Pressure} \cdot \text{height} = 0.5 \cdot \gamma \cdot (10 \text{ ft})^2$$

$$L = 0.5 \cdot 150 \text{ lb/ft}^3 \cdot 100 \text{ ft}^2 \quad \text{This gives, } L = 7500 \text{ lb/ft}$$

Following triangular distribution, the point of application of pressure is at $10/3 \text{ ft}$ from the base $= 3.33 \text{ ft}$

Taking moments about the base to be zero, the lateral force P at the point of bracing is

$$P \cdot 5 \text{ ft} = 7500 \text{ lb/ft} \cdot 3.33 \text{ ft}$$

$$P = 4995 \text{ lb/ft} \approx \mathbf{5 \text{ kip/ft}}$$

1.4 Wind loads

[Click here for Problems 1.4](#)

26. Problem

Correct Answer: (A)

Explanation:

Per ASCE 7-16 page 246,

For analysis under wind loads, diaphragms constructed of untopped steel decks, concrete-filled steel decks, and concrete slabs, each having a span-to-depth ratio of 2 or less, shall be permitted to be idealized as rigid.

The span to depth in the direction of wind for floor A is $50 \text{ ft}/20 \text{ ft} = 2.5$, exceeds 2. Thus it cannot be idealized as rigid although it is a concrete slab.

27. Problem

Correct Answer: (C) **15.24 lb/ft²**

Explanation:

Given Basic Wind Speed = 90 mph

Velocity pressure is given by ASCE 7-16 equation 26.10-1

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2$$

Since topographic effects can be neglected, $K_{zt} = 1$

From Table 26.9-1,

For ground elevation above 2000 ft, elevation factor $K_e = 0.93$

From Table 26.10-1,

For 80ft, Exposure B, Velocity pressure coefficient $K_z = 0.93$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for Buildings MWFRS

$$q_z = 0.00256 * 0.93 * 1 * 0.85 * 0.93 * 90^2$$

$$q_z = 0.00256 * 0.93 * 1 * 0.85 * 0.93 * 90^2$$

$$q_z = 15.24 \frac{\text{lb}}{\text{ft}^2}$$

28. Problem

Correct Answer: (C) Dead +Wind

Explanation:

Dead load is the only guaranteed permanent load that resists uplift when wind load is acting
Hence to evaluate uplift, the combination 'Dead + Wind' will be most critical and is the correct answer.

29. Problem

Correct Answer: (D) 0.9 kips

Explanation:

$V = 95 \text{ mph}$

Velocity pressure at any height is given by ASCE 7-16 equation 26.10-1

$$q_h = 0.00256 K_z K_{zt} K_d K_e V^2$$

Exposure category B can be assumed when no information is given.

From chapter 26, for 50ft, Exposure B, Velocity pressure coefficient $K_z = 0.81$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for rooftop equipment

$$q_h = 0.00256 * 0.81 * 1 * 0.85 * 1 * 95^2 = 15.9 \text{ psf}$$

Lateral pressure on equipment is given by ASCE 7-16 equation 29.4-2 page 322

$$F_h = q_h (GC_r) A_f$$

$$A_f = 6\text{ft} * 5\text{ft} = 30\text{ft}^2 \text{ (Projected area of vertical face of unit)}$$

Width of building = 40ft $h = 50\text{ft}$

$$0.1 * B * h = 200\text{ft}^2$$

Since projected area A_f is less than $(0.1B * h)$, $GC_r = 1.9$

$$F_h = 15.9\text{psf} * 1.9 * 30\text{ft}^2$$

$$F_h = 906 \text{ lbs} \approx \mathbf{0.9 \text{ kips}}$$

30. Problem

Correct Answer: (B) 3200 lbs

Explanation:

Per ASCE 7-16 section 28.3.4, minimum design wind loads should not be less than 16 psf for area of walls and 8 psf for area of roofs projected in vertical plane.

Projected roof area in vertical plane

$$= (12\text{ft} - 8\text{ft}) * 20\text{ ft} = 80\text{ft}^2$$

Projected wall area in vertical plane

$$= 8\text{ft} * 20\text{ft} = 160\text{ ft}^2$$

$$\text{Load for projected roof area} = 80\text{ft}^2 * 8\text{psf} = 640\text{ lbs}$$

$$\text{Load for projected wall area} = 160\text{ ft}^2 * 16\text{psf} = 2560\text{ lbs}$$

$$\text{Total load} = 640\text{ lbs} + 2560\text{ lbs} = \mathbf{3200\text{ lbs}}$$

31. Problem

Correct Answer: (C) Exposure C

Explanation:

Per ASCE 7-16 page 266, section 26.7, the surface with obstructions qualifies for Exposure category B, However, it only extends upto 2500 ft, beyond which the surface is Exposure C.

For buildings or other structures with a mean roof height greater than 30 ft (9.1 m),

Exposure B shall apply where Surface Roughness B prevails in the upwind direction for a distance greater than 2,600 ft (792 m) or 20 times the height of the building or structure, whichever is greater. Hence the structure does not qualify for category B and the correct answer is category C.

32. Problem

Correct Answer: (D) Partially open

Explanation: Ref ASCE 7-16 page 245 for openings definitions

Calculate total area of building envelope

$$\text{Area of walls} = 4 * 20\text{ft} * 8\text{ft} = 640\text{ ft}^2$$

$$\text{Area of roof} = 20\text{ft} * 20\text{ft} = 400\text{ ft}^2$$

Check condition for enclosed building

$$\text{Area of opening 01} = 2\text{ft} * 3\text{ft} = 6\text{ft}^2$$

$$\text{Gross area of wall} = 8\text{ft} * 20\text{ft} = 160\text{ft}^2$$

Since area of opening is greater than $0.01 \times \text{Area of wall}$, it is not enclosed.

Check condition for open building

Area of openings in front wall $01+02 = A_o = 2\text{ft} \times 3\text{ft} + 2\text{ft} \times 6\text{ft} = 18\text{ft}^2$

This is less than 80% of wall area. Thus building is not open

Check condition for partially enclosed.

Total area of openings excluding 01 and 02 $= A_{o1} = 03+04+05$
 $= 2\text{ft} \times 2\text{ft} + 2\text{ft} \times 2\text{ft} + 4\text{ft} \times 4\text{ft} = 24\text{ft}^2$

Check $A_o > 1.1 \times A_{o1}$, 18ft^2 is not greater than $1.1 \times 24\text{ft}^2$

Hence the shed does not qualify as partially enclosed.

Thus correct answer is partially open

33. Problem

Correct Answer: (A) 9.25 psf, 15.91 psf

Explanation:

From ASCE 7-16, chapter 26, Table 26.10-1 for 10ft, Exposure C, Velocity pressure coefficient $K_z = 0.85$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for MWFRS

$K_{zt} = 1$ $K_e = 1$

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2 \quad (\text{equation 26.10-1})$$

$$q_z = 0.00256 * 0.85 * 1 * 0.85 * 1 * 100^2 = 18.5 \text{ psf}$$

From page 276, Figure 27.3-1, for windward wall, for all values of L/B , $C_p = 0.8$

From page 271, Table 26.13-1, for enclosed buildings, value of internal coefficient (GC_{pi}) = +0.18 and -0.18

From page 269, section 26.11.1, gust effect factor $G = 0.85$

From page 274, equation 27.3-1,

Design pressure

$$p = q * G * C_p - q_i (GC_{pi}) \quad q_i = q_z = 18.5 \text{ psf}$$

Calculate design pressure for two internal pressure coefficients

$$p1 = 18.5 * 0.85 * 0.8 - 18.5 * (0.18) = 9.25 \text{ psf}$$

$$p2 = 18.5 * 0.85 * 0.8 - 18.5 * (-0.18) = 15.91 \text{ psf}$$

Correct answer is option A both positive (thrust pressure)

34. Problem

Correct Answer: (B) 77.5 mph

Explanation:

The basic wind speeds given in maps of ASCE 7-16 and IBC 2018 are ultimate design wind speeds to be used in LRFD design load combinations.

To convert it to ASD wind speeds and get ASD design wind pressures, the wind speed has to be scaled down using IBC 2018 equation 16-33, page 378

$$V_{asd} = V\sqrt{0.6}$$

$$V_{asd} = 100\text{mph}\sqrt{0.6}$$

$$V_{asd} = 77.46\text{mph} \approx \mathbf{77.5 \text{ mph}}$$

35. Problem

Correct Answer: (B) - 14.15psf, -2.83psf

Explanation:

From ASCE 7-16, chapter 26, Table 26.10-1, for 10ft, Exposure C, Velocity pressure coefficient $K_z = 0.85$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for MWFRS

$$K_{zt} = 1 \quad K_e = 1 \quad q_z = 0.00256 K_z K_{zt} K_d K_e V^2 \quad (\text{equation 26.10-1})$$

$$q_z = 0.00256 * 0.85 * 1 * 0.85 * 1 * 100^2 = 18.5 \text{ psf}$$

$$h/L = 10\text{ft}/20\text{ft} = 0.5$$

From page 276, Figure 27.3-1 for windward side of roof (wind normal to the ridge), for $h/L = 0.5$ and slope of roof = 10 degrees, $C_p = -0.9, -0.18$

From page 271, Table 26.13-1 for open buildings, value of internal coefficient (GC_{pi}) = 0

From page 269, section 26.11.1, gust effect factor $G = 0.85$

From page 274, equation 27.3-1,

Design pressure

$$p = q * G * C_p - q_i(GC_{pi})$$

Calculate design pressure for two external pressure coefficients

$$p_1 = 18.5 * 0.85 * (-0.9) - 18.5 * (0) = -14.15 \text{ psf}$$

$$p_2 = 18.5 * 0.85 * (-0.18) - 18.5 * (0) = -2.83 \text{ psf}$$

Correct answer is option A both negative (suction pressure)

1.5 Seismic loads

[Click here for Problems 1.5](#)

36. Problem

Correct Answer: (B) 2 kips

Explanation:

The plan shows two walls along the length of the diaphragm which will have high in-plane stiffness compared to the small central columns. Hence, the rigid diaphragm will transfer the loads equally to two walls on two sides with similar dimensions and stiffness.

Load shared by each wall = $(200 \text{ lb/ft} \times 20 \text{ ft}) / 2 = 2000 \text{ lb} = \mathbf{2 \text{ kip}}$

37. Problem

Correct Answer: (D) 25% of floor live load for slab on grade ground floor storage

Explanation:

Per ASCE 7-16 section 12.7.2, effective seismic weight, options A, B and C are all correct. If we look at option D, it says 25% of floor live load for areas used for storage (which is true per section 12.7.2). However, option D also says that the storage area is a ground floor slab on grade – meaning all the loads from that floor is directly transferred to the bearing soil via slab on grade independent of framing elements of the building. Hence it can be excluded from seismic weight calculation and the base shear can be reduced.

38. Problem

Correct Answer: (D) Seismic Design category D

Explanation:

Seismic design category is determined following procedures of ASCE 7-16 Chapter 11

Mapped short period spectral acceleration $S_s = 1g$

Mapped 1s period spectral acceleration $S_1 = 0.4g$

Per ASCE 7-16 page 204, table 20.3-1, the site with stiff soil is classified as class D.

From Page 84, table 11.4-1,

For class D and $S_s = 1$, $F_a = 1.1$

From Page 84, table 11.4-2,

For class D and $S_1 = 0.4$, $F_v = 1.9$

Calculate MCE_R parameters adjusted for site class from Section 11.4.4

$$S_{MS} = F_a * S_s$$

$$= 1.1 * 1g = 1.1g$$

$$S_{M1} = F_v * S_1$$

$$= 1.9 * 0.4g = 0.76g$$

Calculate Design spectral acceleration parameters per Section 11.4.5 page 84

$$S_{DS} = (2/3) * S_{MS} = 0.667 * 1.1g = 0.73g$$

$$S_{D1} = (2/3) * S_{M1} = 0.667 * 0.76g = 0.5g$$

From page 85 Table 11.6-1 For Risk category IV, for $S_{DS} > 0.5$, seismic design category is D

From page 85 Table 11.6-2 For Risk category IV, for $S_{D1} > 0.2$, seismic design category is D

Hence correct answer is seismic design category D

39. Problem

Correct Answer: (C) 0.55 seconds

Explanation:

Per ASCE 7-16, page 102 section 12.8.2.1, the approximate time period of framed structures can be calculated using Equation 12.8-7

Structural height of building, $h_n = 50ft$

From Table 12.8-2

$C_t = 0.016$ for concrete moment resisting frames

$$x = 0.9$$

$$T_a = C_t * h_n^x$$

$$T_a = 0.016 * 50^{0.9} = 0.54 \text{ seconds} \approx \mathbf{0.55 \text{ seconds}}$$

40. Problem

Correct Answer: (A) 118 kips

Explanation:

Base shear $V = 500$ kips

1st floor seismic weight $w_1 = 350$ kips, height from ground level $h_1 = 10ft$

2nd floor seismic weight $w_2 = 300$ kips, height from ground level $h_2 = 20ft$

3rd floor seismic weight $w_3 = 200$ kips, height from ground level $h_3 = 30$ ft

4th floor seismic weight $w_4 = 250$ kips, height from ground level $h_4 = 40$ ft

For time period less than 0.5 seconds, $k = 1$ from section 12.8.3

Per ASCE 7-16, page 102 section 12.8.3, the lateral force distribution is given by equation 12.8-11 and equation 12.8-12

$$F_x = C_{vx} * V$$

$$C_{vx} = \frac{w_x h_x^k}{\sum w_i h_i^k}$$

$$\sum w_i h_i^k = 350 * 10 + 300 * 20 + 200 * 30 + 250 * 40 = 25500 \text{ kip} * \text{ft}$$

For 3rd floor,

$$C_{v3} = \frac{w_3 h_3^k}{\sum w_i h_i^k} = \frac{200 * 30}{25500} = 0.235$$

$$F_3 = C_{v3} * V = 0.235 * 500 \text{ kips} = 117.5 \text{ kips}$$

Thus, force acting 3rd floor = 117.5 kips $\approx$ **118 kips**

41. Problem

Correct Answer: (C) Higher value of “R” indicates that the system is weaker and has lower resistance to lateral loads,

Explanation:

Per ASCE 7-16 section 12.2.3.1, option A is true

Option B is true

Option D is true per section 12.2.3.3

Option C is incorrect because higher value of R denotes robust system and higher resistance to lateral loads.

42. Problem

Correct Answer: (B) 1.26g

Explanation:

Given, vertical period of vibration of structure = 0.1s

From the given graph of design horizontal spectrum, design short period spectral acceleration, $S_{DS} = 1.1g$

Back calculate the value of S_{MS} from per ASCE 7-16 section 11.4.5, page 84

$$S_{DS} = (2/3) * S_{MS}$$

$$S_{MS} = (3/2) * S_{DS} = (3/2) * 1.1g = 1.65g$$

Back calculate the value of S_s from equation 11.4-1,

$$S_{MS} = F_a * S_s$$

$F_a = 1$ from table 11.4-1 for site class D and assumed $S_s > 1.5g$

Thus, $S_s = S_{MS} / F_a = 1.65g$ (Assumption $> 1.5g$ OK)

From ASCE 7-16, section 11.9.2, for vertical period of vibration greater than 0.05s and less than 0.15, equation 11.9-3 can be used to calculate MCE_R vertical response spectral acceleration

$$S_{aMv} = 0.8 * C_v * S_{MS}$$

From Table 11.9-1, value of coefficient $C_v = 1.3$ for $S_s = 1$ and $C_v = 1.5$ for $S_s = 2$

Interpolating for $S_s = 1.65$, we get $C_v = 1.43$

$$\text{Then, } S_{aMv} = 0.8 * C_v * S_{MS} = 0.8 * 1.43 * 1.65g = 1.89g$$

From section 11.9.3, the design vertical response spectral acceleration

$$S_{av} = (2/3) * S_{aMv} = 0.667 * 1.89g = \mathbf{1.26g}$$

43. Problem

Correct Answer: (C) 78 kips

Explanation:

From ASCE 7-16 page 5, Table 1.5-2, for risk category II,

Seismic Importance factor $I_e = 1$

From ASCE 7-16 page 91, Table 12.2-1, for ordinary reinforced concrete moment frame,

$R = 3$

Given, $S_{MS} = 0.5g$

$S_{DS} = (2/3) * S_{MS} = 0.333g$

From equation 12.8-2, page 101

$C_s = S_{DS} / (R / I_e) = 0.333 / (3/1) = 0.111$

Given, effective seismic weight $W = 700$ kips

From equation 12.8-1, Base shear is calculated as

$V = C_s * W = 0.111 * 700 \text{ kips} = 77.7 \text{ kips} \approx 78 \text{ kips}$

44. Problem

Correct Answer: (B) 30 kips

Explanation:

ASCE 7-16 chapter 13 gives provisions for calculation of seismic demands on non structural component

Weight $W_p = 50$ kips

Design short period spectral acceleration $S_{DS} = 0.5g$

Importance factor $I_p = 1$

From page 126 table 13.5-1

Response modification factor for unbraced parapet $R_p = 2.5$

Amplification factor for unbraced parapet $a_p = 2.5$

The seismic demand is given by page 123 Equation 13.3-1

$$F_p = \frac{0.4 * a_p * S_{DS} * W_p}{\frac{R_p}{I_p}} \left(1 + 2 \frac{z}{h} \right)$$

Since component is at the roof of the building, $z = h$

$$F_p = \frac{0.4 * 2.5 * 0.5 * 50 \text{ kips}}{\frac{2.5}{1}} \left(1 + 2 \frac{h}{h} \right) = 30 \text{ kips}$$

45. Problem

Correct Answer: (B) 150 kips

Explanation:

Given

Weight $W = 600$ kips

Mapped Short period spectral acceleration $S_s = 0.5g$

From ASCE 7-16, page 119, Section 12.14.8.1, for simplified procedure, for rocky sites, $F_a = 1$

$$S_{DS} = (2/3) * F_a * S_s$$

$$= 0.667 * 1 * 0.5g = 0.333g$$

Response modification factor for simplified procedure is given by Table 12.14-1, page 115

For ordinary plain masonry walls, $R = 1.5$

For two story building, from page 119 $F = 1.1$

Then, seismic base shear is given by equation 12.14-12

$$V = \frac{F * S_{DS}}{R} * W$$

$$V = \frac{1.1 * 0.333}{1.5} * 600 \text{ kips} = 146.5 \text{ kip} \approx \mathbf{150 \text{ kips}}$$

1.6 Moving loads

[Click here for Problems 1.6](#)

46. Problem

Correct Answer: (C) 80 psi

Explanation:

Ref AASHTO LRFD specs 2018 page 3-23, section 3.6.1.2.5 for tire contact area

$$A = 10\text{in} * 20\text{in} = 200\text{in}^2$$

Given, maximum weight of rear axle = 32 kip

This load is split in two sides

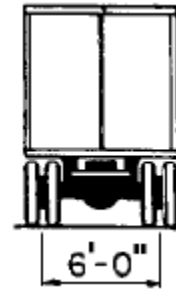
Total wheel load on each side

$$P = 32\text{ kips} / 2 = 16\text{ kips}$$

Pressure = Load/ area

$$= 16\text{ kips} / 200\text{in}^2$$

$$= 16000\text{ lb} / 200\text{in}^2 = \mathbf{80\text{ psi}}$$



47. Problem

Correct Answer: (D) Design truck or Design tandem plus 0.64 klf of uniformly distributed lane load

Explanation:

Per AASHTO LRFD section 3.6.1.2, the total design vehicular live load is the combination of either the design truck and design lane load, or the design tandem and design lane load.

Per section 3.6.1.2.4 the design lane load is a load of 0.64 klf uniformly distributed in the longitudinal direction. Hence, option D is the correct answer

48. Problem

Correct Answer: (B) 250 kip*ft

Explanation:

The wheel will act like a point load and will cause maximum bending moment at the beam when it is at the center of the span.

Ref Beam formulas from NCEES PE handbook page 238

$$P = 50 \text{ kips} \quad L = 20 \text{ ft}$$

$$M = P*L/4 = 50 \text{ kips} * 20 \text{ ft} / 4 = \mathbf{250 \text{ kip*ft}}$$

49. Problem

Correct Answer: (C) 31.4 kips

Explanation:

Reaction due to uniform lane load is constant regardless of the position of the concentrated load

$$R_1 = (0.64 \text{ kip/ft} * 20 \text{ ft}) / 2$$

$$= 12.8 \text{ kip} / 2$$

$$= 6.4 \text{ kips}$$

Reaction due to moving concentrated load will be maximum when the concentrated load is directly on top of the supports.

It will be equal to the concentrated load itself

$$R_2 = 25 \text{ kips}$$

$$\text{Total reaction} = R_1 + R_2 = 25 \text{ kips} + 6.4 \text{ kips} = 31.4 \text{ kips}$$

Thus maximum possible reaction on either supports = **31.4 kips**

50. Problem

Correct Answer: (D) 16 kips

Explanation:

At maximum radius of 40ft, the girder and column is controlled by flexure and limited by 1 ton weight. Per the question, all other components have sufficient residual capacity. So we need to calculate moment acting on the girder when 1 ton load is at maximum radius.

$$\text{Moment acting due to 1ton load at 40 ft} = 1 \text{ ton} * 40 \text{ ft} = 2 \text{ kips} * 40 \text{ ft} = 80 \text{ kip*ft}$$

We need to equate this moment to the moment caused by a load at 5ft

$$P \cdot 5\text{ft} = 80 \text{ kip}\cdot\text{ft}$$

$$P = 80 \text{ kip}\cdot\text{ft} / 5\text{ft}$$

$$= 16 \text{ kip}$$

Hence the crane can safely pick and land a weight of **16 kip** at the minimum radius 5ft.

51. Problem

Correct Answer: (C) 242 kip·ft

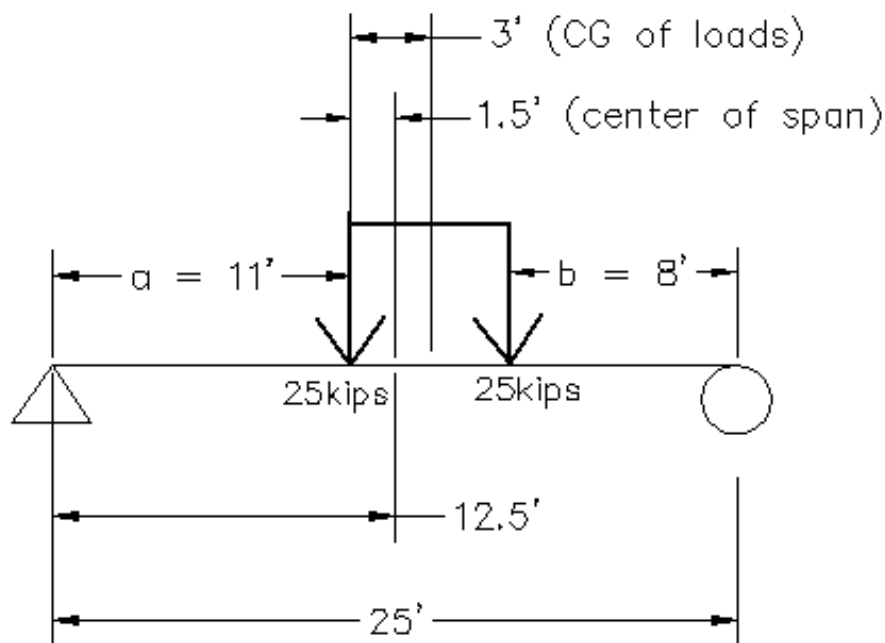
Explanation:

Per NCEES handbook page 232, Section 4.1.2, the absolute maximum moment produced in a beam by a set of n moving loads occurs when the resultant R of the load set and an adjacent load are equidistant from the centerline of the beam.

Per the question, two 25 kips loads are moving along the span, spaced 6 ft apart.

CG of loading group i.e. resultant of load is at $6\text{ft}/2 = 3\text{ft}$.

Thus, condition for maximum moment is when the center of the 25 ft long beam i.e. 12.5ft is between this CG and any one of the axle as shown in the sketch below



Then use the formula from NCEES handbook page 239 (Condition 10)

Here, $P = 25 \text{ kips}$ $a = 11\text{ft}$ $b = 8\text{ft}$ $l = 25\text{ft}$

$$R_1 = 25\text{kip}/25\text{ft} * (25\text{ft} - 11\text{ft} + 8\text{ft}) = 22 \text{ kip}$$

$$M_{\max} = R_1 * a = 22\text{kip} * 11\text{ft} = \mathbf{242 \text{ kip*ft}}$$

1.7 Snow, rain, and ice

[Click here for Problems 1.7](#)

52. Problem

Correct Answer: (D)

Explanation:

From ASCE 7-16 page 54, section 7.5.1, all other loading conditions are valid for partial except loading condition D. The provisions don't mention applying half the uniform loading on all spans.

53. Problem

Correct Answer: (B) **46.2 psf**

Explanation:

From ASCE 7-16, page 266, section 26.7.3, for urban and semi urban area, surface roughness and exposure is category B

From ASCE 7-16, Table 1.5-2, page 5, for risk category III, snow importance factor $I_s = 1.1$

From page 58, Table 7.3-1, for surface roughness B and partially exposed roof, exposure factor $C_e = 1.0$

From page 58, Table 7.3-2, for unheated roof, thermal factor $C_t = 1.2$

For anchorage Alaska, from Table 7.2-1 page 54,

Ground snow load, $p_g = 50\text{psf}$

From equation 7.3-1 page 51, flat roof snow load

$$p_f = 0.7 * C_e * C_t * I_s * p_g$$

$$p_f = 0.7 * 1 * 1.2 * 1.1 * 50\text{psf} = \mathbf{46.2 \text{ psf}}$$

54. Problem

Correct Answer: (B) 17 psf

Explanation:

From ASCE 7-16, Table 1.5-2, page 5, for risk category II,

snow importance factor $I_s = 1$

From page 58, Table 7.3-1, for surface roughness C and fully exposed roof,

exposure factor $C_e = 0.9$

From page 58, Table 7.3-2, for well heated roof,

thermal factor $C_t = 1$

Ground snow load, $p_g = 17\text{psf}$

From equation 7.3-1 page 51, flat roof snow load

$$p_f = 0.7 * C_e * C_t * I_s * p_g$$

$$p_f = 0.7 * 0.9 * 1 * 1 * 17\text{psf} = 10.7\text{ psf}$$

$$\text{Slope of roof} = \tan^{-1} (6\text{ft}/25\text{ft}) = 13.5 \text{ degrees}$$

From page 59, from 1st graph of figure 7.4-1, slope factor with $C_t = 1$, slope = 13.5 degrees and slippery surface

$$C_s = 0.9$$

From equation 7.4-1

$$P_s = C_s * p_f = 0.9 * 10.7 = 9.63\text{ psf}$$

Check criteria for minimum snow load from section 7.3.4, monoslope roof for ground snow load less than 20 psf

$$P_m = I_s * p_g = 1 * 17\text{psf} = 17\text{ psf}$$

Hence minimum load requirement governs, and correct answer is **17psf**

55. Problem

Correct Answer: (C)

Explanation:

From ASCE 7-16 page 62, section 7.10, all other conditions are valid except statement C.

Slope of roof = 20 degrees

$$W/50 = 20/50 = 0.4$$

Since $20 > 0.4$ rain surcharge does not apply and should be 0 psf.

56. Problem

Correct Answer: (D) 44.75 psf

Explanation:

From ASCE 7-16, section 7.7.1, page 59, calculate the density of the snow,

Ground snow load $p_g = 30$ psf

From equation 7.7-1, density of snow,

$$\gamma = 0.13 * p_g + 14 = 0.13 * 30 + 14 = 17.9 \text{ pcf}$$

Length of upper roof $l_u = 50$ ft

Importance factor $I_s = 1$

Get drift height from page 61, figure 7.6-1

For $l_u = 50$ ft and $p_g = 30$ psf

Since $I_s = 1$, we get $h_d = 2.5$ ft

Thus, maximum intensity of drift surcharge, from section 7.7.1

$$p_d = \gamma * h_d = 17.9 \text{ pcf} * 2.5 \text{ ft} = 44.75 \text{ psf}$$

57. Problem

Correct Answer: (D) 57 psf

Explanation:

From NCEES PE handbook, page 3

$$1 \text{ gallon} = 0.134 \text{ ft}^3$$

$$5000 \text{ gallons} = 5000 * 0.134 \text{ ft}^3 = 670 \text{ ft}^3$$

Depth of water due to blockage, i.e. static head = Volume / Area

$$d_s = 670 \text{ ft}^3 / (30 \text{ ft} * 30 \text{ ft})$$

$$d_s = 0.7444 \text{ ft} = 8.9 \text{ inches}$$

Given, Hydraulic head

$$d_h = 2 \text{ inches}$$

From ASCE 7-16 page 65, equation 8.3-1 or IBC 2018 page 391 equation 16-35

Rain load

$$R = 5.2 (d_s + d_h)$$

$$= 5.2 * (8.9 + 2)$$

$$= 56.68 \text{ psf} \approx \mathbf{57 \text{ psf}}$$

58. Problem

Correct Answer: (A) 24.6 lb/ft

Explanation:

From ASCE 7-16, page 5, for risk category III,

Ice importance factor $I_i = 1.15$

Given

Height of roof, $z = 100 \text{ ft}$

Nominal ice thickness, $t = 1.5 \text{ inches}$

Topography factor $K_{zt} = 1$

For L6X6X1/2 from figure 10.4-1, page 70

$$D_c = \sqrt{6^2 + 6^2} = 8.49 \text{ inches}$$

From page 70, equation 10.4-4, height factor

$$f_z = \left(\frac{z}{33}\right)^{0.1} = \left(\frac{100}{33}\right)^{0.1} = 1.117$$

Design ice thickness for freezing rain from page 71, equation 10.4-5

$$t_d = t * I_i * f_z * K_{zt}^{0.35} = 1.5 \text{ inch} * 1.15 * 1.117 * 1 = 1.93 \text{ inches}$$

Area of ice forming in structural shapes, prismatic members is given by page 70, equation 10.4-1

$$A_i = \pi * t_d (D_c + t_d) = \pi * 1.93 (8.49 + 1.93) = 63.18 \text{ sq. in}$$

From section 10.4.1

Density of ice = 56 pcf

Thus, load due to weight of ice = 56 pcf * (63.18/144) sq.ft = **24.6 lb/ft**



59. Problem

Correct Answer: (B)

Explanation:

All options are correct except option B. Both the weight of ice and wind on ice are considered together and are in the same load combination as given in ASCE 7-16 page 7, section 2.3.3. They are not considered separately.

1.8 Impact loads

[Click here for Problems 1.8](#)

60. Problem

Correct Answer: (C) 10 kips

Explanation:

Per AASHTO LRFD 2017 section 3.6.4, page 3-33, the braking force is taken as the greater of 25% if the weights of design truck/tandem or 5% of the sum of lane load and truck/tandem.

Here Reaction due to truck = 40 kips

25% of 40 kips = 10 kips

Sum of reactions due to truck and lane load

= 40 kips + 65 kips

= 105 kips

5% of 105 kips = 5.25 kips

Thus, 'truck only' governs and estimated braking force = 10 kips

61. Problem

Correct Answer: (D) 21.6 kip*ft

Explanation:

Per ASCE 7-16 section 4.5.3, page 16

Load to be considered for passenger vehicle barrier

$P = 6$ kips

This load is assumed to act at a height of 1ft 6in to 2ft 3in from floor level

For maximum moment, the load is applied at 2ft 3in above floor in the cantilevered barrier

Moment = $P \times 2\text{ft}3\text{in} = 6\text{kip} \times 2\text{ft}3\text{in} = 13.5$ kip*ft

Load factor for vehicle collision loads is the same as for live loads.

Thus, factored moment = 1.6×13.5 kip*ft = **21.6 kip*ft**

62. Problem

Correct Answer: (D) 1518 kip*ft

Explanation:

Per AASHTO LRFD 2017, page 3-31 section 3.6.2, dynamic load allowance i.e. impact factor will not applied to design lane load and will only be applied to truck load with a value 33%

Thus, moment due to truck load with impact factor

$$= 1.33 * 500 \text{ kip*ft}$$

$$= 665 \text{ kip*ft}$$

Thus total moment with impact/dynamic allowance

= truck load with impact + design lane load without impact

$$= 665 \text{ kip*ft} + 500 \text{ kip*ft} = 1265 \text{ kip*ft}$$

Also, for single lane load, from table 3.6.1.1.2-1 page 3-21, multiple presence factor = 1.2

$$\text{Thus total unfactored moment} = m * 1265 \text{ kip*ft} = 1.2 * 1265 \text{ kip*ft} = \mathbf{1518 \text{ kip*ft}}$$

63. Problem

Correct Answer: (B) 2 kips

Explanation:

Weight of cab = 1500 lbs

Weight of passengers = 2000 lbs

Total weight = 1500 lbs + 2000 lbs = 3500 lbs = 3.5 kip

Apply impact factor 100%

Service load with impact factor

$$= 3.5 \text{ kip} * (1 + 100\%)$$

$$= 3.5 \text{ kip} * (1 + 1)$$

$$= 7 \text{ kip}$$

Hence safe working load limit for each supporting cable

$$= 7 \text{ kip} / 4 = 1.75 \text{ kip}$$

Correct answer from options is **2 kips**

64. Problem

Correct Answer: (D)

Explanation:

Options A, B and C are correct per ASCE 7-16 page 16-18, section 4.6 and section 4.9.

Option D is incorrect because in addition to the recommended impact factors, for LRFD design, the loads need to include an additional 1.6 factor for live load and dead load combinations.

1.9 Earth pressure and surcharge loads

[Click here for Problems 1.9](#)

65. Problem

Correct Answer: (C) 120 kips

Explanation:

Calculate soil lateral loads per IBC 2018 section 1610 page 391.

Since water table is below the foundation mat, hydrostatic pressures don't apply.

The foundation wall is extended up to the first floor slab and can't freely move at the top.

Hence, the pressure that applies is earth pressure at rest

From page 392 table 1610.1 for well graded gravel-sand mixes,

At-rest pressure = 60psf/ ft depth.

For a 10feet deep wall, the maximum intensity at the bottom of the wall

= 60psf/ft depth * 10ft = 600psf

Soil pressure varies linearly from 0 at the top of wall to maximum intensity at the bottom.

Hence average lateral load along the vertical face of wall = $\frac{1}{2}$ *max intensity* height

= $\frac{1}{2}$ * 600 psf * 10 ft = 3000 lb/ft = 3kips/ft

Thus, total lateral load on a wall of 40 ft length = 3kip/ft * 40 ft = **120 kips**

66. Problem

Correct Answer: (B)

Explanation:

All options are incorrect except option B.

Sands and gravels generally do not swell and clayey soils are more susceptible. Per ASCE 7-16, page 12, section 3.2.1, The lateral pressure shall be increased if expansive soils are present at the site, as determined by a geotechnical investigation.

67. Problem

Correct Answer: (D) 2.4 kips/ft

Explanation:

Active earth pressure coefficient, $K_a = 0.33$

Dry density of soil, $\gamma = 110$ pcf

Submerged density of soil, $\gamma' = 65$ pcf

Density of water, $\gamma_w = 62.4$ pcf

Height of water table = 5ft

Pressure intensity due to dry soil upto 5ft depth

$$= K_a * \gamma * 5\text{ft}$$

$$= 0.33 * 110\text{pcf} * 5\text{ft} = 181.5 \text{ psf}$$

Pressure intensity due to submerged soil below 5ft

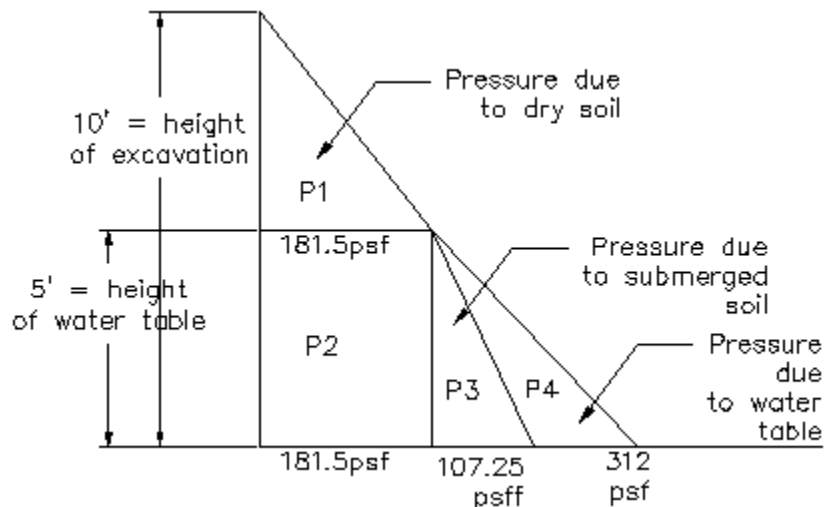
$$= K_a * \gamma' * (10\text{ft} - 5\text{ft})$$

$$= 0.33 * 65\text{pcf} * 5\text{ft} = 107.25\text{psf}$$

Pressure intensity due to water table below 5ft

$$= \gamma_w * (10\text{ft} - 5\text{ft})$$

$$= 62.4\text{pcf} * 5\text{ft} = 312\text{psf}$$



$$P1 = 0.5 * 5\text{ft} * 181.5 \text{ psf} = 453.75 \text{ plf}$$

$$P2 = 5\text{ft} * 181.5\text{psf} = 907.5 \text{ plf}$$

$$P3 = 0.5 * 5\text{ft} * 107.25\text{psf} = 268.125 \text{ plf}$$

$$P4 = 0.5 * 5\text{ft} * 312\text{psf} = 780\text{plf}$$

Total load is the sum of all pressure areas

$$P1 + P2 + P3 + P4 = (435.75 + 907.5 + 268.125 + 780)\text{plf} = 2391.375 \text{ lb/ft} \approx \mathbf{2.4 \text{ kips/ft}}$$

68. Problem

Correct Answer: (B) 2081.25 lb/ft

Explanation:

Given, wall face is vertical and wall friction can be neglected

Use Rankine's equation for active earth pressure coefficient

$$K_a = \tan^2(45 - \phi/2)$$

$$= \tan^2(45 - 30/2)$$

$$= 0.333$$

Dry density of soil, $\gamma = 115$ pcf

Surcharge, $q = 50$ psf

Height of wall, $H = 10$ ft

Total load due to surcharge and earth pressure

$$= K_a * q * H + \frac{1}{2} * K_a * \gamma * H^2 = 0.333 * 50 \text{ psf} * 10 \text{ ft} + \frac{1}{2} * 0.333 * 115 \text{ pcf} * 10 \text{ ft} * 10 \text{ ft} = \mathbf{2081.25 \text{ lb/ft}}$$

69. Problem

Correct Answer: (C) 1.2

Explanation:

Given,

Active earth pressure coefficient, $K_a = 0.33$

Passive earth pressure coefficient, $K_p = 3$

Density of soil, $\gamma = 115$ pcf

Density of concrete, $\gamma = 150$ pcf

Height of embankment for active pressure = 12 ft

Height of embankment for passive pressure = 3 ft

Calculate self weight of concrete that resists sliding

$$\text{Area of wall} = 0.5 * 12 \text{ ft} * (2 \text{ ft} + 3 \text{ ft}) = 30 \text{ ft}^2$$

$$\text{Self weight of wall} = \text{Area} * \text{density} = 30 \text{ ft}^2 * 150 \text{ pcf} = 4500 \text{ lb/ft}$$

Calculate active pressure

$$= \frac{1}{2} * K_a * \gamma * H^2$$

$$= 0.5 * 0.33 * 115 \text{ pcf} * 12 \text{ ft} * 12 \text{ ft}$$

$$= 2732.4 \text{ lb/ft}$$

Calculate resisting passive pressure

$$= \frac{1}{2} * K_p * \gamma * H_p^2$$

$$= 0.5 * 3 * 115 \text{ pcf} * 3 \text{ ft} * 3 \text{ ft}$$

$$= 1552.5 \text{ lb/ft}$$

Calculate resisting force due to friction between concrete and soil

= coefficient of friction * weight of concrete wall

$$= 0.4 * 4500 \text{ lb/ft}$$

$$= 1800 \text{ lb/ft}$$

$$\text{Total sliding force} = 2732.4 \text{ lb/ft}$$

$$\text{Total resisting force} = 1552.5 \text{ lb/ft} + 1800 \text{ lb/ft} = 3352.5 \text{ lb/ft}$$

$$\text{Factor of safety against sliding} = 3352.5 / 2732.4 = \mathbf{1.2}$$

70. Problem

Correct Answer: (B) 1848 lb/ft

Explanation:

The coefficient to be used here is earth pressure at rest

Earth pressure coefficient at rest, $K_o = 0.5$

Density of soil, $\gamma = 110 \text{ pcf}$

From table 3.11.6.4-1, page 3-142, the equivalent height of soil for vehicular loadings are

For 10 ft abutment = 3ft For 20ft abutment = 2ft (Abutments perpendicular to traffic)

Interpolating for 12ft, $h_{eq} = 2.8 \text{ ft}$

Then, from equation 3.11.6.4-1, surcharge loading pressure intensity

$$= K_o * \gamma * h_{eq} = 0.5 * 110 \text{ pcf} * 2.8 \text{ ft} = 154 \text{ psf}$$

Total load = area of rectangular pressure diagram of height 12ft

$$= 12 \text{ ft} * 154 \text{ psf} = \mathbf{1848 \text{ lb/ft}}$$

71. Problem

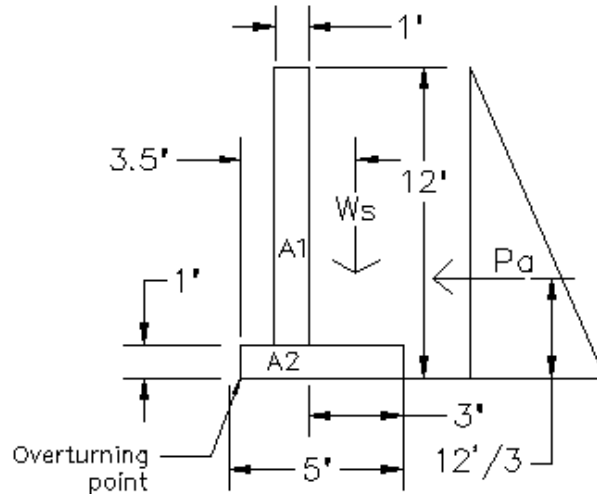
Correct Answer: (C) 1.6

Explanation:

Active Earth pressure coefficient, $K = 0.33$

Density of soil, $\gamma = 115 \text{ pcf}$

Density of concrete, $\gamma = 150 \text{ pcf}$



The forces resisting overturning are the self weight of the wall and the weight of soil retained by the toe of the wall.

Weight of soil retained by wall

$$W_s = (12\text{ft} - 1\text{ft}) * 3\text{ft} * 115\text{pcf}$$

$$= 3795 \text{ lb/ft}$$

Distance of centroid of retained soil weight to overturning point = $(5\text{ft} - 3\text{ft}/2) = 3.5\text{ft}$

Weight of wall is broken into A1 and A2

$$\text{Weight of stem A1} = (12\text{ft} - 1\text{ft}) * 1\text{ft} * 150\text{pcf} = 1650 \text{ lb/ft}$$

Distance of centroid of A1 to overturning point = $(5\text{ft} - 3\text{ft} - 1\text{ft}/2) = 1.5\text{ft}$

$$\text{Weight of base A2} = 5\text{ft} * 1\text{ft} * 150\text{pcf} = 750 \text{ lb/ft}$$

Distance of centroid of A2 to overturning point = $5\text{ft}/2 = 2.5\text{ft}$

$$\text{Resisting moment} = W_s * 3.5\text{ft} + A1 * 1.5\text{ft} + A2 * 2.5\text{ft}$$

$$= 3795 * 3.5 + 1650 * 1.5 + 750 * 2.5$$

$$= 17632.5 \text{ lb*ft/ft}$$

Overturning force, active earth pressure = $\frac{1}{2} * K_a * \gamma * H^2$

$$= 0.5 * 0.33 * 115 * 12\text{ft} * 12\text{ft} = 2732.4 \text{ lb/ft}$$

Overturning moment arm = $12\text{ft}/3 = 4\text{ft}$

$$\text{Overturning moment} = 2732.4\text{lb/ft} * 4\text{ft} = 10929.6 \text{ lb*ft/ft}$$

Factor of safety against overturning = Overturning moment/resisting moment

$$= (17632.5 \text{ lb*ft/ft}) / 10929.6 \text{ lb*ft/ft} = \mathbf{1.6}$$

72. Problem

Correct Answer: (B) 11.38 lb/ft²

Explanation:

Lateral pressure intensity due to infinitely long line load is given by AASHTO LRFD 2017 page 3-137 equation 3.11.6.2-3

Here, X = 5ft since load is acting 5ft away from the wall

Z = 4ft, depth at which intensity is required

Q = 150 lb/ft

Calculate R

$$R^2 = X^2 + Z^2$$

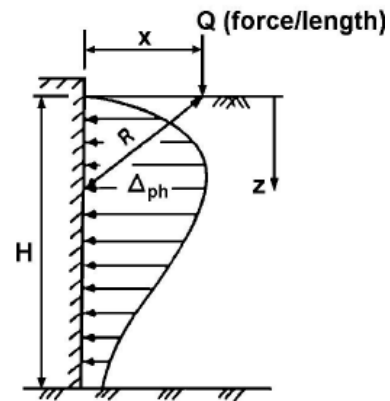
$$R^2 = 5^2 + 4^2$$

$$R = 6.4\text{ft}$$

At depth Z, the intensity of pressure is given by

$$\Delta_{ph} = \frac{4*Q}{\pi} * \frac{X^2*Z}{R^4}$$

$$\Delta_{ph} = \frac{4*150\text{lb/ft}}{\pi} * \frac{(5\text{ft})^2*4\text{ft}}{(6.4\text{ft})^4} = 11.38 \text{ lb/ft}^2$$



73. Problem

Correct Answer: (C)

Explanation:

All options are correct except option C.

Active earth pressure can be positive or negative for non granular soil with cohesion and develops tension cracks.

1.10 Tributary areas and load paths

[Click here for Problems 1.10](#)

74. Problem

Correct Answer: (C) 94.5 kips

Explanation:

Tributary area for interior column

$$= (15\text{ft}/2 + 15\text{ft}/2) * (12\text{ft}/2 + 18\text{ft}/2)$$

$$= 225\text{ft}^2$$

From each floor, load shared by interior column

$$\text{Dead load} = 60\text{psf} * 225\text{ft}^2 = 13500 \text{ lbs} = 13.5 \text{ kips}$$

$$\text{Superimposed dead load} = 30\text{psf} * 225\text{ft}^2 = 6750 \text{ lbs} = 6.75 \text{ kips}$$

$$\text{Live load} = 50\text{psf} * 225\text{ft}^2 = 11250 \text{ lbs} = 11.25 \text{ kips}$$

$$\text{Total load from 3 floors} = 3 * (13.5 + 6.75 + 11.25) \text{ kips} = \mathbf{94.5 \text{ kips}}$$

75. Problem

Correct Answer: (A) 500 lb/ft

Explanation:

Light gauge steel deck floor is idealized as a flexible diaphragm.

A flexible diaphragm distributes load according to tributary area and not stiffness

Tributary area for interior wall

$$= 20\text{ft}/2 * 12\text{ft}$$

$$= 120\text{ft}^2$$

$$\text{Total area of floor} = 20\text{ft} * 12\text{ft} = 240 \text{ ft}^2$$

Load shared by central wall

$$= (120\text{ft}^2 / 240\text{ft}^2) * (600 \text{ lb/ft} * 20\text{ft})$$

$$= 6000 \text{ lbs}$$

$$\text{Load per unit length of wall} = 6000 \text{ lbs} / 12\text{ft} = \mathbf{500 \text{ lb/ft}}$$

76. Problem

Correct Answer: (B) 2666.67 lbs

Explanation:

Reinforced concrete floor is idealized as rigid diaphragm.

A rigid diaphragm distributes load proportional to stiffness

Total load acting on diaphragm = (500 lb/ft*20ft) = 10000 lbs

Load shared by central wall

= $10000 * 4\text{ft} / (4\text{ft} + 3\text{ft} + 3\text{ft} + 2.5\text{ft} + 2.5\text{ft})$

= **2666.67 lbs**

77. Problem

Correct Answer: (C) 833.33 lbs

Explanation:

Diaphragms act as beams and the chord forces are the tension and compression forces that act in the outer boundaries of a diaphragm.

The diaphragm is analyzed as simply supported beam with Uniform loading of 200 lb/ft

Loading, $W = 200 \text{ lb/ft}$

Depth, $d = 12\text{ft}$

Span, $L = 20\text{ft}$

Moment = $W * L^2 / 8$

= $\{200 \text{ lb/ft} * (20\text{ft})^2\} / 8$

= 10,000 lb*ft

Chord force = M / d

= $\{10,000 \text{ lb*ft}\} / 12\text{ft}$

= **833.33 lbs**

78. Problem

Correct Answer: (B) Diagonals in Compression, vertical members in tension, upper chord in compression and bottom chord in tension,

Explanation:

All options are incorrect except option B.

This is a typical Howe bridge truss. A passing car exerts gravity loads on the truss. When there is no lateral load, the diagonals are in compression, the vertical members are in tension, the upper chord is in compression and the bottom chord is in tension.

79. Problem

Correct Answer: (C) 26 kips

Explanation:

All walls have equal in plane stiffness.

Let stiffness be 'k'

The center of rigidity is at (20ft/2, 12ft/2) from the bottom left corner.

Total lateral load

$$= 3 \text{ kip/ft} \times 12 \text{ ft} = 36 \text{ kips}$$

Total torsional moment

$$M = 360 \text{ kip}\cdot\text{ft}$$

Direct shear on North and south walls

$$V = 36 \text{ kips} \times k / (k + k)$$

$$V = 18 \text{ kips}$$

Calculate torsional shear.

Distance of walls from center of rigidity

$$r_1 = r_2 = 6 \text{ ft for north and south walls}$$

$$r_3 = r_4 = 10 \text{ ft for east and west wall}$$

Max shear is in north wall with combined direct and torsional shear

$$V_S = \frac{M \cdot k_i \cdot r_i}{\sum k_i \cdot r_i^2}$$

$$V_s = \frac{360 \text{ kip} \cdot \text{ft} \cdot k \cdot 6 \text{ ft}}{k \cdot 10 \text{ ft} \cdot 10 \text{ ft} + k \cdot 10 \text{ ft} \cdot 10 \text{ ft} + k \cdot 6 \text{ ft} \cdot 6 \text{ ft} + k \cdot 6 \text{ ft} \cdot 6 \text{ ft}}$$

$$V_s = \frac{360 \text{ kip} \cdot \text{ft} \cdot k \cdot 6 \text{ ft}}{k \cdot 272 \text{ ft}^2}$$

$$V_s = \frac{360 \text{ kip} \cdot \text{ft} \cdot k \cdot 6 \text{ ft}}{k \cdot 272 \text{ ft}^2}$$

$$V_s = 7.94 \text{ kips}$$

$$\text{Total shear} = V + V_s = 18 \text{ kips} + 7.94 \text{ kips} = 25.94 \text{ kips} \approx \mathbf{26 \text{ kips}}$$

1.11 Load combinations

[Click here for Problems 1.11](#)

80. Problem

Correct Answer: (C) 55 kips

Explanation:

$D = 25$ kips

$L = 5$ kips

$L_r = 10$ kips

$S = 8$ kips (does not govern over roof live load)

$W = 15$ kips

Roof live load is greater than live load from equipment, so combination 2 won't govern.

Check governing strength design combinations.

From ASCE 7-16, section 2.3, page 7

Combination 3

$$1.2D + 1.6L_r + 0.5W = 1.2*25 + 1.6*10 + 0.5*15 = 53.5 \text{ kips}$$

Combination 4

$$1.2D + 1W + L + 0.5L_r = 1.2*25 + 1*15 + 5 + 0.5*10 = 55 \text{ kips}$$

Hence governing factored load is **55 kips**

81. Problem

Correct Answer: (D) 137 kip*ft

Explanation:

$D = 40$ kip*ft

$F = 75$ kip*ft

$L_r = 5$ kip*ft (does not govern over rain load)

$S = 8.35$ kip*ft (does not govern over rain load)

$R = 10$ kip*ft

$E_h = 25$ kip*ft

$E_v = 5$ kip*ft

Check governing allowable stress design combinations.

From ASCE 7-16, section 2.4, page 8

$$(D+F)+R = 40+75+ 10 = 125$$

From ASCE 7-16, section 2.4.5, page 9

$$1.0(D+F)+0.7E_h+0.7E_v = 40+75+0.7*25+0.7*5 = 136 \text{ kip*ft}$$

$$1.0(D+F)+0.525E_h+0.525E_v+0.75S = 40+75+0.525*25+0.525*5+0.75*8.35 = 136.75 \text{ kip}$$

$\approx 137 \text{ kip*ft}$

82. Problem

Correct Answer: (C) Dead + Seismic or Wind

Explanation:

Code does not require us to combine seismic and wind loads at the same time. Live load helps to counter overturning so should not be used in calculation against uplift and overturning. Hence the correct combination is option C, dead + seismic or wind (whichever governs between seismic and wind)

83. Problem

Correct Answer: (C) Typically, 'Strength IV' combination governs over 'Strength I' if the dead load effects are at least two times the live load effects,

Explanation:

All other options are correct per AASHTO LRFD section 3.4 except option C. Per page 3-9, typically Strength IV governs if the dead load to live load ratio is above 7.

84. Problem

Correct Answer: (B) 2328 kip*ft

Explanation:

From AASHTO LRFD 2017 page 3-21, Table 3.6.1.1.2-1 multiple presence factor for 1 lane Bridge, $m = 1.2$

From page 3-31, Table 3.6.2.1-1, impact factor for truck loads $IM = 33\%$

From page 3-16, Table 3.4.1-2, for dead loads, max factor = 1.25

From page 3-15, Table 3.4.1-1, for live loads, max factor = 1.75

Dead load moment = 800 kip*ft

Live load moment including multiple presence factor and impact factor

$$= 1.2*(250 \text{ kip*ft}*1.33+300 \text{ kip*ft})=759 \text{ kip*ft (impact for truck only)}$$

$$\text{Thus strength I combination} = 1.25*800 \text{ kip*ft}+1.75*759 \text{ kip*ft} = \mathbf{2328 \text{ kip*ft}}$$

85. Problem

Correct Answer: (C) 24kip*ft/ft

Explanation:

Moment due to earth load, $EH = 10 \text{ kip*ft/ft}$

Moment due to live load surcharge, $LS = 5.5 \text{ kip*ft/ft}$

Moment due to wind load on structure, $WS = 1.5 \text{ kip*ft/ft}$

From AASHTO LRFD 2017 page 3-15,

Factor for $LS = 1.35$

Factor for $WS = 1$

From page 3-16, Factor for $EH = 1.5$

Thus, max factored moment $= 1.5EH + 1.35LS + 1WS$

$= 1.5 * 10 + 1.35 * 5.5 + 1 * 1.5 = 23.92 \text{ kip*ft/ft} \approx \mathbf{24 \text{ kip*ft/ft}}$

2. Analysis of Structures–Forces and Load Effects

2.1 Diagrams

[Click here for Problems 2.1](#)

86. Problem

Correct Answer: (D)

Explanation:

The section with the highest moment of inertia about Y-Y will have the highest flexural stiffness. From the given options, Option C will have the lowest moment of inertia about YY axis. Options A and B have higher moment of inertias than C but less than Option D, Hence D is the correct answer.

87. Problem

Correct Answer: (B)

Explanation:

For the same cross sectional area, the hollow tube will have the highest polar moment of inertia and hence will have the best torsional rigidity out of all the shapes. Thus it is best suited and the correct answer is option B.

88. Problem

Correct Answer: (D)

Explanation:

The given equation of bending moment is a cubic equation. So, the shear force equation should be parabolic and the loading diagram should be linear. Hence Option C is ruled out. Also, when we put $x = 0$ in the equation, $M = 13$. Hence the left support should be able to resist moments and is fixed. This rules out options A and B .

Option D is the answer.

89. Problem

Correct Answer: (B)

Explanation:

The shear force diagram is linear, so the load must be uniformly distributed.

This rules out option D.

Since the slope of shear force does not change abruptly, there is no concentrated loading in the span. This rules out option A

There is an abrupt decrease in bending moment at midspan so the acting moment must be in anticlockwise direction. Hence the correct answer is B.

90. Problem

Correct Answer: (C)

Explanation:

The bending moment at the hinge located at the third span must always be zero.

Hence this rules out options A and options B.

Option D does not show any changes in the slope of the bending moment diagram in the first and third spans when there is a concentrated load. Hence Option D is not correct.

Thus the correct answer is Option C.

91. Problem

Correct Answer: (C)

Explanation:

An external couple does not produce any abrupt change in the shear force diagram.

Hence Options B and D are ruled out.

The shear force diagram, when there is a uniformly distributed load should be linear. This rules out option A. Hence the correct answer is Option C.

92. Problem

Correct Answer: (D)

Explanation:

Since there is a hinge at the center of the beam, Option A and option B are ruled out because they don't show sufficient rotation about the hinge.

Option C shows a case when the uniformly distributed load would be acting in upwards direction. Hence the correct answer is Option D.

93. Problem

Correct Answer: (A)

Explanation:

When a lateral load acts in the direction shown, the left column will be in tension and the right column will be in compression. The center column will have very little axial force if any. Hence option B is ruled out. The beam will be in axial compression. As the axial load is transferred in the beam from the left bay to the right bay, a part of the axial load will be transferred to the center column as shear loads. Hence the axial compression in the right beam will be lesser than the left beam. Thus the correct answer is option A.

94. Problem

Correct Answer: (D)

Explanation:

An external moment at the center of the beam brings an abrupt change to the moment diagram. Option A and B have no changes in the slope of the moment diagram and are ruled out. Option C shows bending moment decreasing gradually at the point of application of moment. This is true if there was a point load, not a moment. Option D shows an abrupt decrease in the moment at the center of the beam and is the correct answer.

95. Problem

Correct Answer: (A)

Explanation:

Calculate the reactions from the load

Let H_1 and H_2 be horizontal reactions acting in opposite directions as shown.

Let V_1 and V_2 be vertical reactions acting as shown

Sum of forces in horizontal direction = 0

$$H_1 + 50 \text{ kip} - 50 \text{ kip} - H_2 = 0$$

$$H_1 = H_2$$

Taking sum of moments about left pinned support
(anti clockwise +ve)

$$V_2 * 10 \text{ ft} + 50 \text{ kip} * 10 \text{ ft} - 10 \text{ kip/ft} * 10 \text{ ft} * 5 \text{ ft} - 50 \text{ kip} * 5 \text{ ft} = 0$$

$$V_2 = 25 \text{ kip}$$

Taking sum of moments at the hinge = 0 and
considering forces from the right side of the hinge,

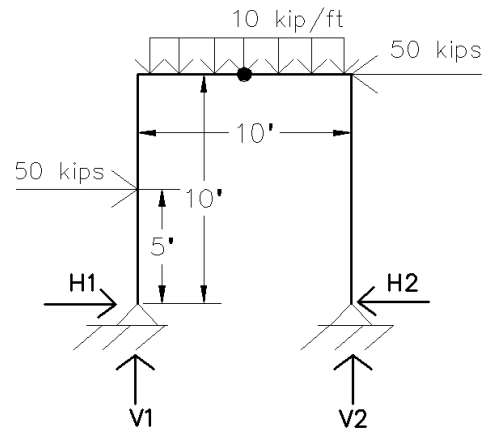
$$V_2 * 5 \text{ ft} - H_2 * 10 \text{ ft} - 10 \text{ kip/ft} * 5 \text{ ft} * 5 \text{ ft} / 2 = 0$$

$$25 \text{ kip} * 5 \text{ ft} + H_2 * 10 \text{ ft} - 50 \text{ kip} * 2.5 \text{ ft} = 0$$

This gives $H_2 = 0$

Hence both H_1 and $H_2 = 0$.

Since both horizontal reactions are zero, the only diagram that shows no shear at the reaction points is the correct answer, i.e. option A.



96. Problem

Correct Answer: (A)

Explanation:

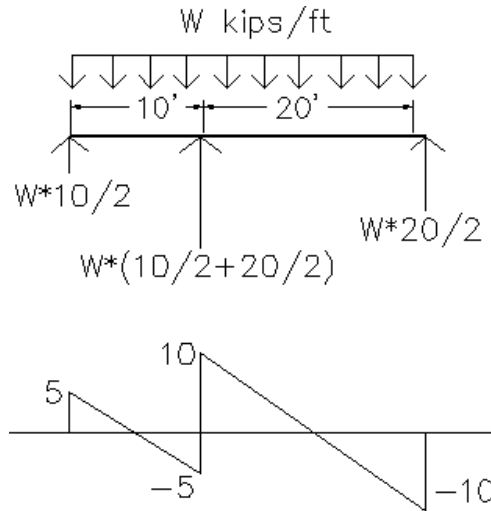
Since the diaphragm is flexible, the shear wall load distribution is per the tributary area.

If 'W' is the diaphragm load W kip/ft,

The left wall shear is $10 \text{ ft} / 2 * W = 5W$

The center wall shear is $(10 \text{ ft} / 2 + 20 \text{ ft} / 2)W = 15W$

The right wall shear is $20 \text{ ft} / 2 * W = 10W$



This can be thought of as a two span beam with 3 reactions as shown above. The correct shear force diagram is Option A for this beam.

97. Problem

Correct Answer: (B)

Explanation:

All options are correct except option B. The slope of the shear force diagram at any point gives the maximum intensity of loading at that point, not the moment.

98. Problem

Correct Answer: (B)

Explanation:

All options are correct except option B. Influence lines describe fundamental properties of the structure and the shape of the diagram is independent of the system of units used.

99. Problem

Correct Answer: (C)

Explanation:

The vertical member CG is a zero force member unless a vertical load acts directly on top of the member at node C. Here, the load is travelling along the bottom chord between nodes A,H,G,F and E. For all these cases, the force in member CG will be zero. Hence the correct answer is Option C that shows zero force throughout.

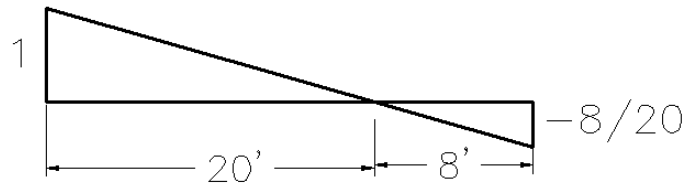
100. Problem

Correct Answer: (C)

Explanation:

The influence lines for reactions and internal forces for section between supports should be constructed as for beam without overhangs and extended along the overhangs,

For reaction at A, the maximum reaction will occur when the load is directly on top of point A and will be zero when the load is directly on top of point B. The value of ordinate at point A will be equal to 1. Thus the influence line diagram is



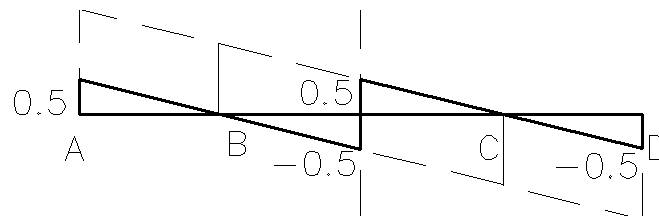
The correct answer is Option C.

101. Problem

Correct Answer: (C)

Explanation:

For shear at midspan of a simply supported beam, the maximum shear will occur when the load is directly on top of midspan at either side of the section and will be zero when the load is directly on top of supports. The overhang is then extrapolated on either side as shown below. The value of the sum of ordinates at midspan will be equal to 1. Thus the influence line diagram is:



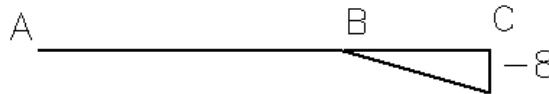
The correct answer is Option C.

102. Problem

Correct Answer: (D)

Explanation:

The moment at support B will be zero when the load is travelling between the two supports i.e. point A to point B. When the load is travelling along the overhanging portion, the maximum moment occurs when the load is at the edge of the overhang and zero when the load is directly on top of B. Thus it will have a linear distribution between B and C as shown below :



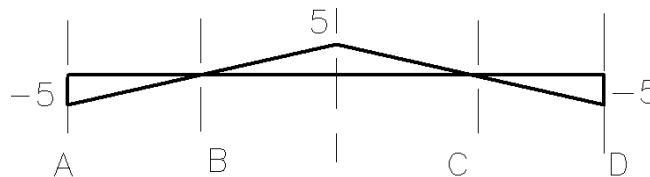
The correct answer is Option D.

103. Problem

Correct Answer: (B)

Explanation:

At midspan, the moment is negative (tension at top) when the load travels along the overhangs and positive when it travels between the supports. The maximum moment at the midspan occurs when the load is directly on top of it. The value of ordinate at that point is $(a*b/L)$, where 'a' is distance to left support from midspan = 10ft, 'b' is the distance to right support from midspan = 10 ft and 'L' is the span length between supports = 20ft. The overhang diagrams are then linearly extrapolated on both sides



The correct answer is Option B.

2.2 Axial

[Click here for Problems 2.2](#)

104. Problem

Correct Answer: (D) Cannot be determined

Explanation:

Ref to NCEES PE handbook page 34 for critical buckling load

The column is free on one side and pinned on the other side. For pinned=free condition, the effective length factor 'K' is very huge and indeterminate as the configuration itself is unstable. This makes the value of critical buckling load close to zero. Hence the best answer for this question is option D 'Cannot be determined'

105. Problem

Correct Answer: (B) 170.7 kips

Explanation:

Ref to NCEES PE handbook page 34 for critical buckling load and page 25 for section properties.

$K = 2$ for fixed - free condition

$L = 20\text{ft} = 240\text{ inches}$

Calculate moment of inertia of the section

$a = 4\text{ inches}$

$b = a - \text{wall thickness} = 4\text{ inches} - 1\text{ inches} = 3\text{ inches}$

$$I = \pi * (a^4 - b^4)/4$$

$$I = \pi * (4^4 - 3^4)/4 = 137.44\text{ in}^4$$

Calculate critical axial buckling load

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} = \frac{\pi^2 * 29000 * 137.44}{(2 * 240)^2} = \mathbf{170.7\text{ kips}}$$

Correct answer is option B.

106. Problem

Correct Answer: (C) 0.295 in

Explanation: Ref NCEES PE handbook page 29, section 1.6.3 for thermal deformations.

$$A = 20\text{in}^2$$

$$L = 30\text{ ft}$$

$$\alpha = 0.000007/^{\circ}\text{F}$$

$$P = 200\text{ kips}$$

$$\Delta T = 100 - 32 = 68^{\circ}\text{F}$$

Ref to NCEES PE handbook, page 28, section 1.6.2.6, for linearly elastic material,

$$E = \frac{\text{Stress}}{\text{Strain}}$$

Change in length due to load 'P'

$$\Delta L_1 = \frac{PL}{AE} = \frac{200 * 30\text{ft} * 12(\text{in}/\text{ft})}{20\text{in}^2 * 29000\text{ ksi}} = 0.124\text{ in}$$

Change in length due to load ' ΔT '

$$\Delta L_2 = \alpha * L * \Delta T = 0.000007/^{\circ}\text{F} * 30\text{ft} * 12(\text{in}/\text{ft}) * 68^{\circ}\text{F} = 0.171\text{in}$$

$$\text{Total change in length} = \Delta L_1 + \Delta L_2 = 0.124 + 0.171 = \mathbf{0.295\text{ in}}$$

Correct answer is option C.

107. Problem

Correct Answer: (B) 92.3 kips

Explanation: Ref NCEES PE handbook page 29, section 1.6.3, for thermal deformations.

$$A = 15\text{ in}^2$$

$$L = 30\text{ ft}$$

$$\alpha = 0.000007/^{\circ}\text{F}$$

$$\Delta T = 100 - 30 = 70^{\circ}\text{F}$$

Change in length due to change in temperature ' ΔT '

$$\Delta L = \alpha * L * \Delta T = 0.000007/^{\circ}\text{F} * 30\text{ft} * 12(\text{in}/\text{ft}) * 70^{\circ}\text{F} = 0.1764\text{in}$$

There is clearance of only 0.1 inch for expansion.

The restriction in expansion due to insufficient clearance induces thermal compressive stresses

$$\Delta L_r = 0.1764\text{ in} - 0.1\text{ in} = 0.0764\text{ in}$$

For linearly elastic material,

$$E = \frac{\text{Stress}}{\text{Strain}}$$

Change in length due to load 'P'

$$E = \frac{P/A}{\Delta L_r/L}$$

$$P = E \cdot A \cdot \Delta L_r / L = 29000 \text{ ksi} \cdot 15 \text{ in}^2 \cdot 0.0764 \text{ in} / \{30 \text{ ft} \cdot 12 (\text{in/ft})\} = \mathbf{92.3 \text{ kips}}$$

Correct answer is option B.

108. Problem

Correct Answer: (B) 3.75 kips

Explanation: For cables,

The moment at any point can be taken as zero.

Let vertical and horizontal reactions at point A be A_x and A_y

Let vertical and horizontal reactions at point D be D_x and D_y

Taking moment at point A = 0

$$D_y \cdot 18 \text{ ft} = 4 \text{ kips} \cdot (18 \text{ ft} - 4 \text{ ft}) + 2 \text{ kips} \cdot 5 \text{ ft}$$

$$D_y = 66/18 = 3.67 \text{ kips}$$

Sum of vertical forces = 0

$$D_y + A_y = 2 \text{ kips} + 4 \text{ kips}$$

$$A_y = 6 \text{ kips} - 3.67 \text{ kips} = 2.33 \text{ kips}$$

Taking moments about point C = 0

$$D_x \cdot 5 \text{ ft} = D_y \cdot 4 \text{ ft}$$

$$D_x = D_y \cdot (4 \text{ ft} / 5 \text{ ft}) = 2.936 \text{ kips}$$

Taking sum of horizontal forces = 0

$$A_x = D_x = 2.936 \text{ kips (in opposite direction)}$$

Now, total tension in section AB

$$T = \sqrt{A_x^2 + A_y^2} = \sqrt{2.936^2 + 2.33^2} = \mathbf{3.75 \text{ kips}} \quad \text{Correct answer is option B.}$$

109. Problem

Correct Answer: (C) Catenary curve

Explanation:

Chains, ropes and cables assume a catenary shape when they are hanging under their own weight. Under uniformly distributed external load, the shape will be parabolic, Hence the correct answer is option C

110. Problem

Correct Answer: (B) Mid span at lowest sag point

Explanation:

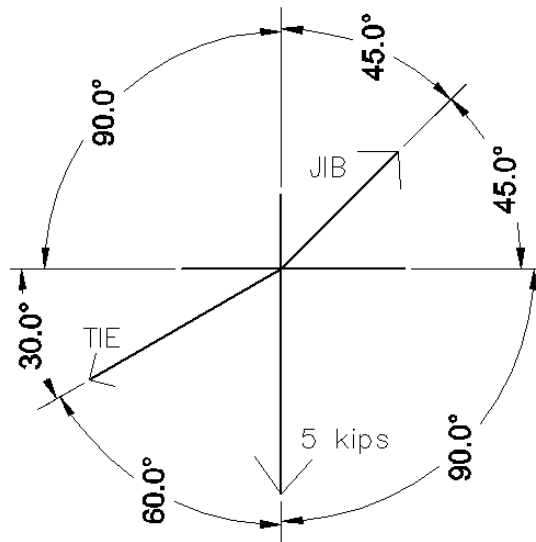
When a cable is subjected to uniformly distributed loads, it assumes a parabolic shape. The lowest point of the cable will have zero slope and zero vertical component of tension. Hence the tension is minimum at that point. Correct answer is option B.

111. Problem

Correct Answer: (C) 13.66 kips

Explanation:

The forces in the tie are tensile in nature and the forces in the jib are compressive. We can idealize the tip of the jib as a single point in equilibrium with three concurrent forces. Then from Lami's theorem, the force in any one member must be directly proportional to the sine of angle between other two forces



$$\frac{F_{jib}}{\sin(60)} = \frac{F_{tie}}{\sin(45+90)} = \frac{5 \text{ kips}}{\sin(30+90+45)}$$

$$\frac{F_{tie}}{\sin(135)} = \frac{5 \text{ kips}}{\sin(165)}$$

$$\frac{F_{tie}}{1} = \frac{5 \text{ kips} \cdot \sin(135)}{\sin(165)} = 13.66 \text{ kips}$$

Correct answer is option C.

112. Problem

Correct Answer: (B) 860 lbs

Explanation:

Let the point of attachment of the jib to the mast be 'A', the point of attachment of tie be point 'B' and the point of suspended load be point 'C'.

Let T be the tension in the Tie

Angle of inclination of tie to the horizontal

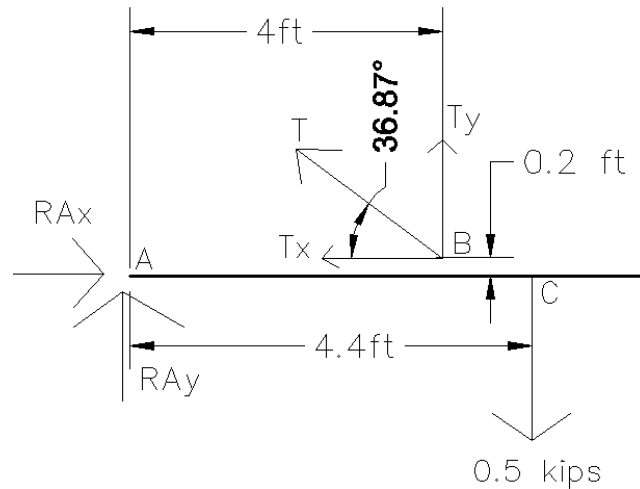
Horizontal distance between tie and upper support = 4ft

Vertical height between tie and upper support = 3.2ft - 0.2ft = 3ft

$$\Theta = \tan^{-1}(3\text{ft} / 4\text{ft}) = 36.87^\circ$$

Horizontal component of tension, $T_x = T\cos\Theta$

Vertical component of tension, $T_y = T\sin\Theta$



Taking sum of moments about reaction point 'A'

$$T_x * 0.2\text{ft} + T_y * 4\text{ft} = 0.5 \text{ kips} * 4.4\text{ft}$$

$$T * \cos(36.87^\circ) * 0.2 \text{ ft} + T * \sin(36.87^\circ) * 4 \text{ ft} = 0.5 \text{ kip} * 4.4\text{ft}$$

$$T = 0.859 \text{ kips} \approx \mathbf{860 \text{ lbs}}$$

Thus the correct answer is option B.

113. Problem

Correct Answer: (B) 450 kips

Explanation:

Here , total number of piles , $n = 5$

Total axial load = 2000 kips

Moments about X-X axis, $M_{xx} = 500 \text{ kip}\cdot\text{ft}$

Moments about Y-Y axis, $M_{yy} = 300 \text{ kip}\cdot\text{ft}$

The pile layout is symmetric and the centroid of pile group is at the center of middle pile

Let x and y be the distances of each pile from the centroid of the group.

$$\sum x^2 = (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (0) = 64 \text{ ft}^2$$

$$\sum y^2 = (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (0) = 64 \text{ ft}^2$$

Axial load distribution in each pile

$$= 2000 \text{ kips}/n$$

$$= 2000 \text{ kips}/5 = 400 \text{ kips}$$

Additional axial load due to M_{xx}

$$= M_{xx} * (x/\sum x^2) = 500 \text{ kip}\cdot\text{ft} * 4\text{ft}/64 \text{ ft}^2 = 31.25 \text{ kips}$$

Additional axial load due to M_{yy}

$$= M_{yy} * (y/\sum y^2) = 300 \text{ kip}\cdot\text{ft} * 4\text{ft}/64 \text{ ft}^2 = 18.75 \text{ kips}$$

$$\text{Max total axial load} = 400 \text{ kips} + 31.25 \text{ kips} + 18.75 \text{ kips} = \mathbf{450 \text{ kips}}$$

Correct answer is option B.

114. Problem

Correct Answer: (D) 4250 kip/in

Explanation:

Here, when horizontal load acts, the deformation in each braces will be equal. Thus the total stiffness of the system can be assumed as three springs in parallel and will be the sum of each individual stiffness

Total stiffness, $K = K_1 + K_2 + K_3$

For each brace,

Length, $L = 10\text{ft}$

Area, $A = 15\text{ in}^2$

Modulus of elasticity 'E' is variable depending on material property

For Wood,

$$K_1 = (E_{\text{wood}} * A) / L = 1500\text{ ksi} * 15\text{ in}^2 / \{10\text{ft} * 12(\text{in}/\text{ft})\} = 187.5\text{ kip/in}$$

For Steel,

$$K_2 = (E_{\text{steel}} * A) / L = 29000\text{ ksi} * 15\text{ in}^2 / \{10\text{ft} * 12(\text{in}/\text{ft})\} = 3625\text{ kip/in}$$

For Concrete,

$$K_3 = (E_{\text{concrete}} * A) / L = 3500\text{ ksi} * 15\text{ in}^2 / \{10\text{ft} * 12(\text{in}/\text{ft})\} = 437.5\text{ kip/in}$$

$$\text{Total stiffness } K = 187.5 + 3625 + 437.5 = \mathbf{4250\text{ kip/in}}$$

Correct answer is option D.

115. Problem

Correct Answer: (A) 378.5 kip/in

Explanation:

Here, when vertical load acts, the deformation in each braces will be proportional to the individual stiffness. Axial force acting in each layer will be equal to the total force and the deformation will be the sum of deformations in each layer. Thus the springs are in series and the total stiffness will be

$$1/K = 1/K_1 + 1/K_2 + 1/K_3$$

For first layer,

$$E_1 = 200\text{ ksi} \quad A_1 = 20\text{ in}^2 \quad L_1 = 4\text{ in}$$

$$K_1 = E_1 * A_1 / L_1 = 200\text{ ksi} * 20\text{ in}^2 / 4\text{ in} = 1000\text{ kip/in}$$

For second layer,

$$E_2 = 300\text{ ksi} \quad A_2 = 16\text{ in}^2 \quad L_2 = 5\text{ in}$$

$$K_2 = E_2 * A_2 / L_2 = 300\text{ ksi} * 16\text{ in}^2 / 5\text{ in} = 960\text{ kip/in}$$

For second layer,

$$E_3 = 250\text{ ksi} \quad A_3 = 20\text{ in}^2 \quad L_3 = 3\text{ in}$$

$$K_3 = E_3 * A_3 / L_3 = 250\text{ ksi} * 20\text{ in}^2 / 3\text{ in} = 1666.67\text{ kip/in}$$

Now,

$$1/K = 1/K_1 + 1/K_2 + 1/K_3 = 1/1000 + 1/960 + 1/1666.67 = 0.00264166\text{ in/kip}$$

$$\text{Thus, } K = 1/0.00264166\text{ in/kip} = \mathbf{378.5\text{ kip/in}}$$

Correct answer is option A.

116. Problem

Correct Answer: (D)

Explanation:

Ref to NCEES PE handbook page 34 for critical buckling load.

The column self weight is assumed insignificant to buckling load and is neglected.

The material of the column is assumed homogeneous, isotropic and elastic, obeying Hooke's law.

Deformation including axial shortening of columns due to compression is disregarded.

Column is assumed to be subjected to purely axial load only.

Hence all options are incorrect except option D.

117. Problem

Correct Answer: (A) **112 kips (compression)**

Explanation:

Calculate reaction at joint D

Sum of Moments at joint A = 0

$$R_d \cdot (4+8+3) = 80 \cdot 4 + 40 \cdot 6 + 60 \cdot (4+8)$$

$$R_d = 1280/15 = 85.33 \text{ kips}$$

Angle of inclination of member CD

$$\theta = \tan^{-1} (3.5/3) = 49.4^\circ$$

Analyze joint D

Vertical component of F_{cd}

$$F_{ycd} = F_{cd} \cdot \sin(49.4)$$

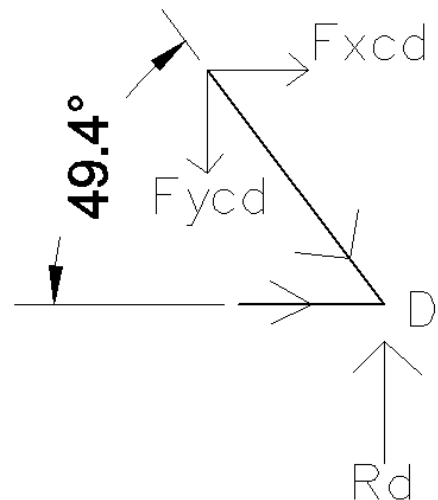
Taking sum of vertical forces at joint D = 0

$$F_{ycd} = R_d$$

$$F_{cd} \cdot \sin(49.4) = 85.33 \text{ kips}$$

$$F_{cd} \approx \mathbf{112 \text{ kips (compression)}}$$

Hence the correct answer is option A.



118. Problem

Correct Answer: (C) AB = 0 and CD = 60 kips

Explanation:

Analyze joint B,

There are no external loads and BA is the only vertical member.

Taking sum of vertical forces = 0 , we get

Force in BA = 0

Analyze joint D with an external load of 60 kips

Taking sum of vertical forces = 0

Force in CD = 60 kips (CD is the only vertical member)

Hence the correct answer is option C.

119. Problem

Correct Answer: (C) 70.7 kips

Explanation:

Angle of inclination of braces $\Theta = \tan^{-1} (10/10) = 45^\circ$

The frame is pin connected and braced. It can be analyzed as a truss.

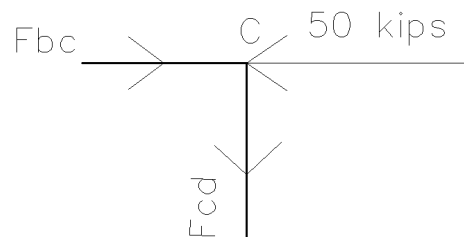
The threaded tie rods can only take tension. Hence at the given direction of loading, member AC is not engaged and can be ignored for analysis.

Analyze joint C,

BC is the only horizontal member.

Taking sum of horizontal forces = 0 , we get

Force in BC = 50 kips compressive



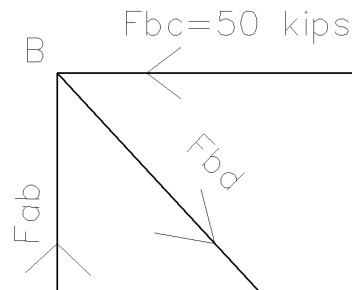
Analyze joint B of 60 kips

Taking sum of horizontal forces = 0

$F_{bd} \cos(45) = F_{bc} = 50$ kips

$F_{bd} = 50 \text{ kips} / 0.707 = 70.7$ kips

Hence the correct answer is option C.



120. Problem

Correct Answer: (A)

Explanation:

The given options of trusses B, C and D have mixed forces i.e. tension and compression in the diagonals.

Option A is a typical Howe truss in which the diagonals are only subjected to compressive forces. Hence Option A is the correct answer.

121. Problem

Correct Answer: (C)4

Explanation:

Member AG is a zero force member. This is because when we analyze joint A, the roller support does not produce any horizontal reaction and AG is the only horizontal member at that point. Thus force in AG = 0

When we analyze joint E, there is no external load. If there are ONLY two non-collinear members at a joint, with no external load, both are zero force members. Hence both ED and EF are zero force members.

When we analyze joint C, there is no external load. If there are ONLY three members at a joint, two of which are collinear, with no external load, then the non collinear member is zero force member. So CG is also a zero force member.

Thus AG, ED, EF and CG are zero force members. So the correct answer is Option C.

122. Problem

Correct Answer: (D) 6

Explanation:

When we analyze joint A and E, there is no external load. If there are ONLY two non-collinear members at a joint, with no external load, both are zero force members. Hence AB, AI, ED and EF are zero force members.

When we analyze joint C, there is no external load. If there are ONLY three members at a joint, two of which are collinear, with no external load, then the non collinear member is zero force member. So CH is also a zero force member.

Since AI is a zero force member, IH is also zero because force in AI = IH.

Hence AB, AI, ED, EF, CH and IH are zero force members and the correct answer is option D.

123. Problem

Correct Answer: (B) $BH = 0$ $DF = 0$

Explanation:

When we analyze joint H, there is no external load. BH is the only vertical member.

So when we take sum of forces in vertical direction = 0,

Force in BH = 0

When we analyze joint F, there is no external load. DF is the only vertical member.

So when we take sum of forces in vertical direction = 0,

Force in DF = 0

Hence both BH and DF are zero force members and the correct answer is option B. .

2.3 Shear

[Click here for Problems 2.3](#)

124. Problem

Correct Answer: (B) 42.5 kips

Explanation:

Per ACI 318-14 page 132, section 9.4.3.2, the critical section for shear in beams is at a distance 'd' from the face of support and beams shall be permitted to be designed for shear at that critical section

Effective depth, $d = 18 \text{ inches} = 1.5 \text{ ft}$

Load, $w = 5 \text{ kips/ft}$

Span, $L = 20 \text{ ft}$

Reaction at support $= wL/2$

$= 5 * 20 / 2 = 50 \text{ kips}$

Calculate shear at critical section i.e. at distance 'd'

Shear at 1.5 ft

$= \text{Reaction} - W * d$

$= 50 - 5 * 1.5 = 42.5 \text{ kips}$

Hence design shear is **42.5 kips** and the correct answer is option B. .

125. Problem

Correct Answer: (D)

Explanation:

Short beams and columns are more susceptible to failure in shear. Long beams and columns usually fail in flexure and buckling.

In short beam shear stress is more concentrated due to shorter span to depth ratio.

Sudden brittle failure of concrete occurs in over reinforced sections before the reinforcement yields.

The critical value of shear in a beam is not always at a distance 'd'. E.g. if there is a concentrated load acting very close to the support, this doesn't apply.

Hence all options are true except Option D and is the correct answer.

126. Problem

Correct Answer: (C) 92 kips

Explanation:

Per ACI 318-14 section 18.6.5 page 279, for special moment resisting frames, the beams shall be able to develop the shear due to factored gravity loads and maximum probable moment acting at the joint faces

Factored Dead load = $1.2 \times 2 \text{ kip/ft} = 2.4 \text{ kip/ft}$

Factored Live load = $1 \times 1 \text{ kip/ft} = 1 \text{ kip/ft}$

Total $w_u = 2.4 + 1 = 3.4 \text{ kip/ft}$

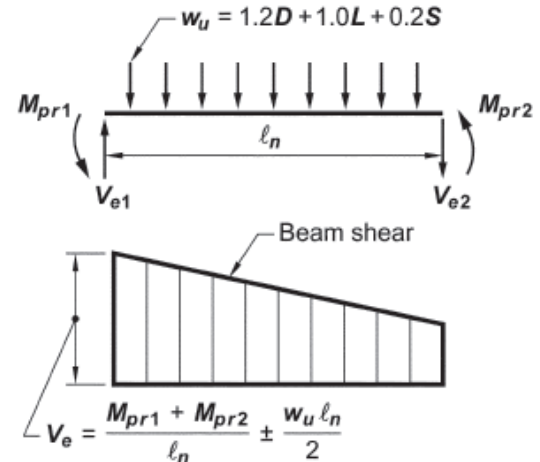
$M_{pr1} = 500 \text{ kip}\cdot\text{ft}$ $M_{pr2} = 600 \text{ kip}\cdot\text{ft}$

Clear span, $\ell_n = 18 \text{ ft}$

Maximum shear

$$\begin{aligned} &= (M_{pr1} + M_{pr2})/\ell_n + w_u \ell_n / 2 \\ &= (500 \text{ kip}\cdot\text{ft} + 600 \text{ kip}\cdot\text{ft})/18 \text{ ft} + (3.4 \text{ kip/ft} \times 18 \text{ ft})/2 \\ &= 91.7 \text{ kip} \approx \mathbf{92 \text{ kips}} \end{aligned}$$

Hence, the correct answer is option C.



127. Problem

Correct Answer: (B) 60 kips

Explanation:

Per ACI 318-14 section 18.4.3 page 271, for intermediate moment resisting frames, the columns design shear should be the least between the shear associated with developing nominal flexural strengths and the factored shear combination considering overstrength factor of 3 for EQ loads.

Factored Dead load shear = $1.2 \times 5 \text{ kips} = 6 \text{ kips}$

Factored Live load shear = $1 \times 10 \text{ kips} = 10 \text{ kips}$

Factored EQ load shear = $3 \times 20 \text{ kips} = 60 \text{ kips}$

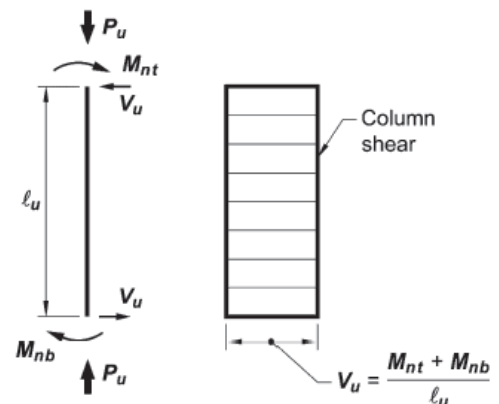
Total factored shear combination = $6 + 10 + 60 = 76 \text{ kips}$

$M_{nt} = 450 \text{ kip}\cdot\text{ft}$ $M_{nb} = 600 \text{ kip}\cdot\text{ft}$

Clear span, $\ell_u = 20 \text{ ft} - 24 \text{ in} = 18 \text{ ft}$

Flexural shear = $(M_{nt} + M_{nb})/\ell_u = (450 \text{ kip}\cdot\text{ft} + 600 \text{ kip}\cdot\text{ft})/18 \text{ ft}$

$= 58.33 \approx 60 \text{ kips}$ Least between the two shears is **60 kips**, thus Option B is correct.



128. Problem

Correct Answer: (A) -6 kips

Explanation:

Ref to NCEES PE handbook page 32.

Moment, $M = x^3 - 5x^2 - 3x$

The shear force is given by the first derivative of the bending moment.

$$V = \frac{d(Mx)}{dx}$$

$$V = \frac{d(x^3 - 5x^2 - 3x)}{dx}$$

$$V = \frac{d(x^3 - 5x^2 - 3x)}{dx}$$

$$V = 3x^2 - 10x - 3$$

When $x = 3\text{ft}$

$$V = 3 \cdot 3^2 - 10 \cdot 3 - 3 = -6 \text{ kips}$$

Thus, the correct answer is Option A.

129. Problem

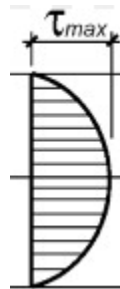
Correct Answer: (C)

Explanation:

Ref to NCEES PE handbook page 32 for shear flow distribution.

For a rectangular section, shear stress does not vary linearly between the top and bottom.

Shear stress is zero at the top and bottom and maximum at the neutral axis.



Hence the correct answer is Option C

130. Problem

Correct Answer: (B) 15.15 kips

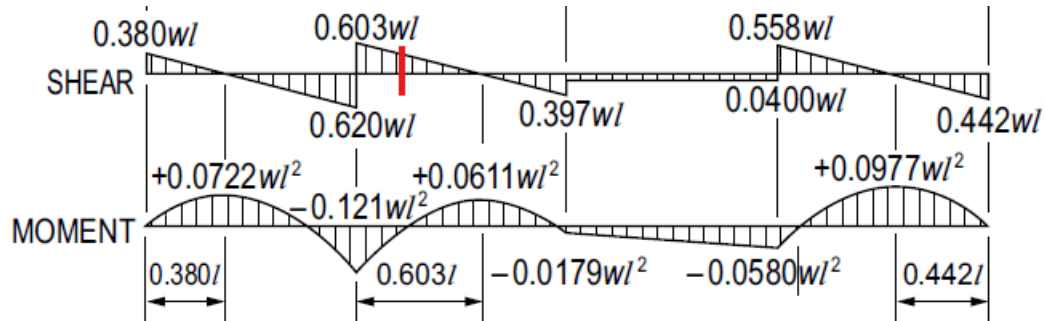
Explanation:

Ref to NCEES PE handbook page 254 for shear and bending moments in continuous beams.

Loading, $w = 5 \text{ kips/ft}$

Individual span length, $L = 10 \text{ ft}$

For a four span continuous system with third span not loaded, the shear force distribution is



At 13 ft from the exterior left support, the shear force varies from $0.603 \cdot w \cdot L$ to zero at the second span, $0.603 \cdot L$ from the second support.

$$0.603 \cdot w \cdot L = 0.603 \cdot 5 \text{ kips/ft} \cdot 10 \text{ ft} = 30.15 \text{ kips}$$

$$0.603 \cdot L = 0.603 \cdot 10 \text{ ft} = 6.03 \text{ ft}$$

Calculate the shear force at 13 ft i.e. at 3 ft from the second support,

$$V = 30.15 \text{ kips} / 6.03 \text{ ft} \cdot (6.03 \text{ ft} - 3 \text{ ft})$$

$V = \mathbf{15.15 \text{ kips}}$, Thus correct answer is option B.

131. Problem

Correct Answer: (C) +9 kips

Explanation:

Total load due to 5 kips/ft distributed load = $5 \text{ kips/ft} \cdot 6 \text{ ft} = 30 \text{ kips}$

Location from the hinge of centroid of the force = $4 \text{ ft} + 6 \text{ ft} / 2 = 7 \text{ ft}$

Let R_c be the reaction at point C

At the internal hinge, sum of moments = zero. Considering forces on the right of the hinge,

$$\sum M_B = 0 \quad R_c \cdot 10 \text{ ft} = 30 \text{ kips} \cdot 7 \text{ ft} \quad R_c = 21 \text{ kips}$$

$$\text{Shear at point B} = -R_c + 30 \text{ kips} = -21 \text{ kips} + 30 \text{ kips} = \mathbf{+9 \text{ kips}},$$

Correct answer is option C.

132. Problem

Correct Answer: (D) 2943 psi

Explanation:

Ref to NCEES PE handbook page 20,21 for centroid, moment of inertia and page 32 for shear stress distribution in beams

Calculate location of neutral axis

Area of flange, $A_f = 4 \text{ in} \times 0.5 \text{ in} = 2 \text{ in}^2$

Moment of inertia of flange, $I_f = \{4 \text{ in} \times (0.5 \text{ in})^3\}/12 = 0.042 \text{ in}^4$

Location of centroid of flange from base, $y_f = 2 \text{ in} + 0.25 \text{ in} = 2.25 \text{ in}$

Area of web, $A_w = 2 \text{ in} \times 0.5 \text{ in} = 1 \text{ in}^2$

Moment of inertia of web, $I_w = \{0.5 \text{ in} \times (2 \text{ in})^3\}/12 = 0.333 \text{ in}^4$

Location of centroid of web from base, $y_w = 2 \text{ in}/2 = 1 \text{ in}$

$y' = (2 \text{ in}^2 \times 2.25 \text{ in} + 1 \text{ in}^2 \times 1 \text{ in}) / (2 \text{ in}^2 + 1 \text{ in}^2) = 1.833 \text{ in}$

Calculate total moment of inertia about centroid

$$\begin{aligned} I_{xx} &= I_f + A_f(y_f - y')^2 + I_w + A_w(y' - y_w)^2 \\ &= 0.042 + 2(2.25 - 1.833)^2 + 0.333 + 1(1.833 - 1)^2 \\ &= 1.417 \text{ in}^4 \end{aligned}$$

Calculate first moment of area above the junction of web and flange, about the centroid,

$$Q = 4 \text{ in} \times 0.5 \text{ in} \times (2.25 \text{ in} - 1.833 \text{ in}) = 0.834 \text{ in}^3$$

Thickness of web, $b = 0.5 \text{ in}$

Shear force, $V = 2.5 \text{ kips} = 2500 \text{ lbs}$

Shear stress,

$$\begin{aligned} \tau &= (VQ)/(Ib) \\ &= (2500 \text{ lbs} \times 0.834 \text{ in}^3) / (1.417 \text{ in}^4 \times 0.5 \text{ in}) \\ &= \mathbf{2943 \text{ psi}} \end{aligned}$$

Thus correct option is option D

133. Problem

Correct Answer: (B) 16.5 kips

Explanation:

Ref to NCEES PE handbook page 33 for transverse shear stress flow in beams

Calculate location of neutral axis

Location of centroid of section from base,

$$y_w = (0.5\text{ in} + 0.5\text{ in} + 2\text{ in} + 0.5\text{ in} + 0.5\text{ in})/2 = 2\text{ in (due to symmetry)}$$

Calculate total moment of inertia about centroid

$$\begin{aligned} I_{xx} &= \{4\text{ in} * (4\text{ in})^3\}/12 - \{3.5\text{ in} * (2\text{ in})^3\}/12 \\ &= 19\text{ in}^4 \end{aligned}$$

Calculate the first moment of area above the junction of top flange plates

i.e. at depth 0.5 in from the top of section,

$$Q = 4\text{ in} * 0.5\text{ in} * (2\text{ in} - 0.25\text{ in}) = 3.5\text{ in}^3$$

Shear force, $V = 15\text{ kips} = 15000\text{ lbs}$

Transverse shear flow per unit length of beam along the span,

$$\begin{aligned} \tau &= (VQ)/I \\ &= (15000\text{ lbs} * 3.5\text{ in}^3)/(19\text{ in}^4) \\ &= 2763.15\text{ lbs/in} \end{aligned}$$

Spacing of bolts, $s = 12\text{ inches}$

There are 2 bolts, one on either side of the flange sharing this shear

Thus shear per bolt

$$\begin{aligned} &= (\tau * s)/2 \\ &= (2763.15\text{ lbs/in} * 12\text{ in})/2 \\ &= 16578.94\text{ lbs} \end{aligned}$$

$\approx 16.5\text{ kips}$

Hence, correct answer is option B

134. Problem

Correct Answer: (A) 21.23 ksi

Explanation:

The 1st (top) and the 2nd plate connection has only 1 shear plane.

The 2nd, 3rd and 4th plates have two shear planes.

Since both P1 and P2 are equal, shear stress will be maximum in the bolts under the top plate in which 6 bolts are under single shear.

No. of bolts, $n = 6$

$P1 = 100$ kips

Area of 1 inch bolts

$$= \pi/4 * (1 \text{ in})^2$$

$$= 0.785 \text{ in}^2$$

$$\text{Maximum nominal shear stress} = 100 \text{ kips} / (6 * 0.785 \text{ in}^2) = \mathbf{21.23 \text{ ksi}}$$

Correct answer is option A.

135. Problem

Correct Answer: (D) 5 kips/in

Explanation:

Here, the load is concentric to the centroid of the weld group.

So, there is no torsion or moments causing additional shear on the welds.

Total length of the welds (L)

$$= 6 \text{ in} + 6 \text{ in} + 4 \text{ in}$$

$$= 16 \text{ in}$$

Load = 80 kips

Shear on welds

$$= 80 \text{ kips} / 16 \text{ in}$$

$$= \mathbf{5 \text{ kips/in}}$$

Correct answer is Option D.

136. Problem

Correct Answer: (C)

Explanation:

When load passes through the shear center of a section, there will be no twisting. Hence there will be no torsion. There will be bending and shear in the span and also some shear deflection. Hence, the correct option is C.

137. Problem

Correct Answer: (B) 1604 lbs/ft

Explanation:

Calculate the maximum shear due to point loads.

Reaction at pinned end (Taking moments about roller = 0)

$$R_a * (4\text{ft} + 6\text{ft} + 8\text{ft}) = 15 \text{ kips} * 8\text{ft} + 10 \text{ kips} * (6\text{ft} + 8\text{ft})$$

$$R_a = 14.44 \text{ kips}$$

Reaction at roller end

Taking sum of vertical forces, $\sum V = 0$

$$R_b = 10 \text{ kips} + 15 \text{ kips} - R_a$$

$$R_b = 25 \text{ kips} - 14.44 \text{ kips} = 10.56 \text{ kips}$$

Thus maximum shear occurs at R_a and the value is 14.44 kips

Shear due to uniformly distributed load $w = wL/2$

$$\text{Thus, } w = 14.44 \text{ kips} * 2 / (8\text{ft} + 6\text{ft} + 4\text{ft})$$

$$= 1.604 \text{ kips/ft}$$

$$= \mathbf{1604 \text{ lbs/ft}}$$

Thus, the correct answer is option B.

138. Problem

Correct Answer: (D) 16.64 kips

Explanation:

Calculate the maximum reaction at pinned end per unit spacing of floor beam

(Taking moments about roller = 0)

$$R_a \cdot 20\text{ft} = 0.8 \text{ kips} \cdot 20 \text{ ft} + 0.16 \text{ kip/ft} \cdot 30\text{ft} \cdot 5\text{ft} - 0.6 \text{ kip} \cdot \text{ft} - 0.1 \text{ kip} \cdot 10\text{ft}$$

$$R_a = 1.92 \text{ kips}$$

Reaction at roller end (Sum of vertical forces = 0)

$$R_b = 0.8 + 0.16 \cdot 30 + 0.5 + 0.1 - R_a = 4.28 \text{ kips}$$

Thus maximum shear occurs at R_b and the value is

$$V_b = 0.1 + 0.16 \cdot 10 + 0.5 - R_b = -2.08 \text{ kips (calculated per unit spacing)}$$

For floor beams spaced at 8 ft c/c, total shear = $2.08 \cdot 8 = 16.64 \text{ kips}$

Thus, the correct answer is option D.

139. Problem

Correct Answer: (C) 9.67 kips

Explanation:

The maximum shear will occur when the heaviest axle travels past the maximum reaction point i.e. when the heavier rear axle is at the pinned support location and the front axle is at 8 ft from the pinned support.

Maximum shear

$$V = 8 \text{ kips} + 3 \text{ kips} \cdot (18\text{ft} - 8\text{ft}) / 18\text{ft}$$

$$= 9.67 \text{ kips} \quad \text{Thus, the correct answer is option C.}$$

140. Problem

Correct Answer: (B)

Explanation:

Increasing the depth of the beam section will not reduce the shear force.

Reducing the span b will reduce the vertical reaction at fixed support and will also reduce the shear at that location.

Shear is directly proportional to the magnitude of the load so reducing load will reduce the shear. Hence correct answer is option B.

2.4 Flexure

[Click here for Problems 2.4](#)

141. Problem

Correct Answer: (C) -11 kip*ft

Explanation: Ref NCEES PE handbook page 32.

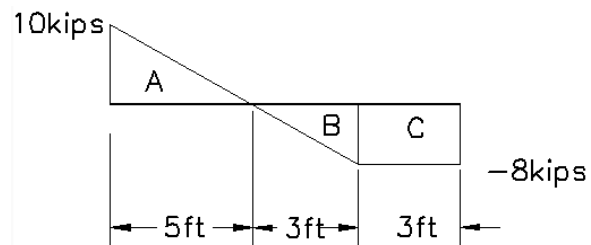
The difference in bending moment between any two points is given by the area of shear force diagram between the two points. Since the beam is simply supported, the moment at the left end is zero. So the value of the bending moment at 11 ft is given by the sum of areas of triangles A, B and rectangle C.

Total Area

$$= 0.5 \times 10 \text{ kips} \times 5 \text{ ft} - 0.5 \times 8 \text{ kips} \times 3 \text{ ft} - 8 \text{ kips} \times 3 \text{ ft}$$

$$= -11 \text{ kip*ft}$$

Correct Answer is Option C.



142. Problem

Correct Answer: (A) 21.7 kip*ft

Explanation:

The wind load is acting at 30 degree inclination. the effective surface area of the wind load is to be calculated.

For 40 ft length, effective length exposed to wind

$$= 40 \text{ ft} \times \sin(30) = 20 \text{ ft}$$

For 6 ft depth, effective depth exposed to wind

$$= 6 \text{ ft} \times \cos(30) = 5.19 \text{ ft}$$

$$\text{Total exposed length} = 20 \text{ ft} + 5.19 \text{ ft} = 25.19 \text{ ft}$$

Height of cap = 3 ft

$$\text{Total area} = 3 \text{ ft} \times 25.19 \text{ ft} = 75.57 \text{ ft}^2$$

$$\text{Wind load} = 50 \text{ psf} \times 75.57 \text{ ft}^2 = 3.78 \text{ kips}$$

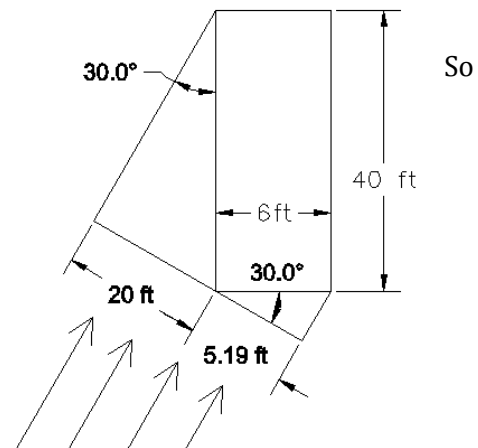
$$\text{Component of wind perpendicular to Y axis} = 3.78 \text{ kips} \times \sin(30) = 1.89 \text{ kips}$$

Height of pile = 10 ft

$$\text{Moment arm} = \text{height of pile} + \text{half the height of cap} = 10 \text{ ft} + 1.5 \text{ ft} = 11.5 \text{ ft}$$

$$\text{Moment about Y axis} = 1.89 \text{ kips} \times 11.5 \text{ ft} = \mathbf{21.7 \text{ kip*ft}}$$

Correct answer is option A..



143. Problem

Correct Answer: (B)

Explanation:

Ref to continuous beam formulas in NCEES PE handbook page 253-255

Maximum negative moment occurs in the 2nd support when the first, second and fourth spans are loaded and the third span is unloaded. The value of the moment is $-0.121wl^2$.

Thus correct answer is option B

144. Problem

Correct Answer: (D) 13 ksi

Explanation:

Ref to continuous beam formulas in NCEES PE handbook page 255.

Span length, $L = 10$ ft

Loading, $w = 5$ kips/ft

Maximum value of moment is $+0.0772wL^2$ for +ve moment and $-0.107wL^2$ for -ve moment.

Hence, governing value of bending moment is

$$M = -0.107 * 5 \text{ kips/ft} * 10\text{ft} * 10\text{ft} = 53.5 \text{ kip}\cdot\text{ft}$$

Elastic flexural stress in beam is given in NCEES PE handbook page 32

$$\sigma = M/s$$

Ref to AISC Steel manual for value of section modulus 'S' for W10X45

From page 1-29, For W10X45, $S_{xx} = 49.1 \text{ in}^3$

$$\sigma = \{53.5 \text{ kip}\cdot\text{ft} * 12 \text{ (in/ft)}\} / 49.1 \text{ in}^3$$

$$= 13.07 \text{ ksi} \approx \mathbf{13 \text{ ksi}}$$

Thus the correct answer is option D.

145. Problem

Correct Answer: (C) 90 kip *ft

Explanation:

Bending moment at the base of the column due to lateral load

$$M = 5 \text{ kips} \times 15 \text{ ft}$$

$$M = 75 \text{ kip} \cdot \text{ft}$$

Lateral stiffness of the column = 2.5 kips/in

Thus displacement at the top of the column due to lateral load of 5 kips

$$d = 5 \text{ kips} / (2.5 \text{ kips/in})$$

$$d = 2 \text{ in}$$

This deflection causes additional second order loading effects due to the axial load and increases the bending moment at the base.

Additional moment

= Axial load * deflection (i.e. eccentricity for the axial load)

$$= 85 \text{ kips} \times 2 \text{ in} \times (1/12 \text{ ft/in})$$

$$= 14.17 \text{ kip} \cdot \text{ft}$$

$$\text{Total moment} = 75 \text{ kip} \cdot \text{ft} + 14.17 \text{ kip} \cdot \text{ft} = 89.17 \text{ kip} \cdot \text{ft} \approx \mathbf{90 \text{ kip} \cdot \text{ft}}$$

Thus the correct answer is option C.

146. Problem

Correct Answer: (A) 23.2 ksi

Explanation:

The steel section is converted to

equivalent wood section using

modular ratio, n

$$n = (E_{\text{steel}}) / (E_{\text{wood}})$$

$$= 30000 / 2500 = 12$$

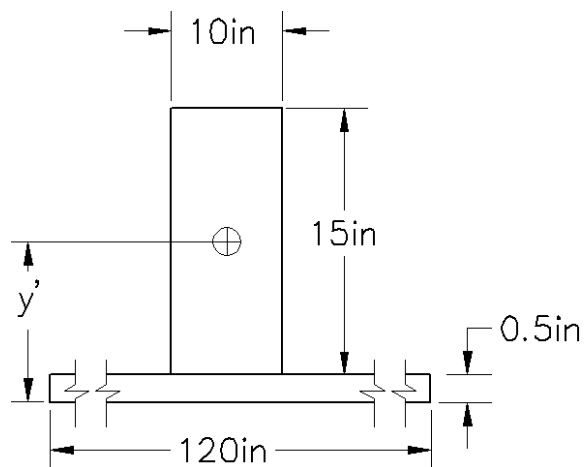
Thus, width of transformed steel plate

$$= 10 \text{ in} \times n$$

$$= 10 \text{ in} \times 12 = 120 \text{ in}$$

Calculate the centroid and MOI

of the transformed section,



Centroid from the base,

$$y' = \{120 \cdot 0.5 \cdot 0.25 + 15 \cdot 10 \cdot (7.5 + 0.5)\} / (120 \cdot 0.5 + 15 \cdot 10)$$

$$Y' = 5.786 \text{ in}$$

Calculate moment of inertia about centroid

$$I_a = 10 \cdot 15 \cdot 15 \cdot 15 / 12 = 2812.5 \text{ in}^4$$

$$I_b = 120 \cdot 0.5 \cdot 0.5 \cdot 0.5 / 12 = 1.25 \text{ in}^4$$

$$I = 2812.5 \text{ in}^4 + 10 \text{ in} \cdot 15 \text{ in} \cdot (8 \text{ in} - y')^2 + 1.25 \text{ in}^4 + 120 \text{ in} \cdot 0.5 \text{ in} \cdot (y' - 0.25 \text{ in})^2$$

$$I = 5388 \text{ in}^4$$

Section modulus for steel fiber

$$S = I / y'$$

$$= 5388 \text{ in}^4 / 5.786 \text{ in}$$

$$= 931.21 \text{ in}^3$$

Flexural stress in steel is given by

$$\sigma = n \cdot M / S$$

$$= \{12 \cdot 150 \text{ kip} \cdot \text{ft} \cdot 12 \text{ (in/ft)}\} / 931.21 \text{ in}^3$$

$$= 23.2 \text{ ksi}$$

Thus the correct answer is option A.

147. Problem

Correct Answer: (D) 3830 kip*ft

Explanation:

Since all story height are equal

$$\text{Story height} = 40 \text{ ft} / 4 = 10 \text{ ft}$$

$$\text{Lever arm for 55 kips load} = 40 \text{ ft}$$

$$\text{Lever arm for 37 kips load} = 30 \text{ ft}$$

$$\text{Lever arm for 21 kips load} = 20 \text{ ft}$$

$$\text{Lever arm for 10 kips load} = 10 \text{ ft}$$

Total overturning moment

$$= 55 \text{ kip} \cdot 40 \text{ ft} + 37 \text{ kip} \cdot 30 \text{ ft} + 21 \text{ kip} \cdot 20 \text{ ft} + 10 \text{ kip} \cdot 10 \text{ ft}$$

$$= 3830 \text{ kip} \cdot \text{ft}$$

Correct answer is option D

148. Problem

Correct Answer: (C) + 6.4 kip*ft , -8.4 kip*ft

Explanation:

Ref to ACI 318-14 page 70 for simplified method of analysis for continuous beams.

Uniform loading, $w = 0.4$ kips/ft

Calculate maximum positive moment

For longest end span

$$M_1 = w \cdot 15\text{ft} \cdot 15\text{ft} / 14 = 6.43 \text{ kip*ft}$$

For longest interior span,

$$M_2 = w \cdot 14\text{ft} \cdot 14\text{ft} / 16 = 4.9 \text{ kip*ft}$$

Calculate maximum negative moment

Based on the dimensions given, critical section is at exterior face of first interior support (from the right),

Average of end spans 14ft and 15ft = 14.5ft

$$M_3 = w \cdot 14.5\text{ft} \cdot 14.5\text{ft} / 10 = 8.41 \text{ kip*ft}$$

Hence max positive moment is **6.4 kip*ft** and max negative moment is **- 8.4 kip*ft**.

Correct answer is option C

149. Problem

Correct Answer: (B) 17 kip *ft

Explanation:

At the point of inflection, the bending moment is zero.

Due to symmetry, the shear acting on the each fixed support is

$$V = 5 \text{ kips} / 2$$

$$V = 2.5 \text{ kips}$$

The point of inflection is at 0.55H from the base

$$0.55 \cdot 15 \text{ ft} = 8.25 \text{ ft}$$

The shear at point of contraflexure = 2.5 kips

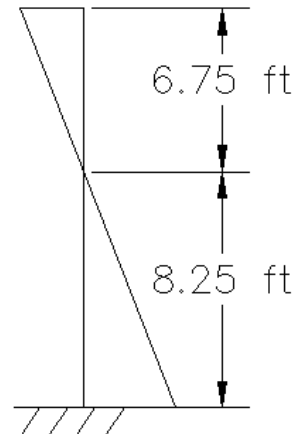
For the given loading, the beam moment is maximum at the column joint.

At the top of the column,

$$\text{Moment} = 2.5 \text{ kips} \cdot (15\text{ft} - 8.25 \text{ ft})$$

$$= 16.875 \text{ kip*ft} \approx \mathbf{17 \text{ kip *ft}}$$

Correct answer is option B.



150. Problem

Correct Answer: (D)

Explanation:

Point of inflection is a point on the elastic curve where the deflected shape changes directions and point of contraflexure is the point where the bending moment is zero and changes signs. Thus A, B and C are correct.

Option D is false and is the answer.

151. Problem

Correct Answer: (D) **2.125 in**

Explanation:

The plastic neutral axis divides the areas of tension and compression into two equal parts.

Let the plastic neutral axis be a distance x from the top.

Total area of the section

$$= 4\text{in} \cdot 0.5\text{in} + 2\text{in} \cdot 0.5\text{in}$$

$$= 3\text{ in}^2$$

If area is divided into half,

Area of upper half = $A/2$

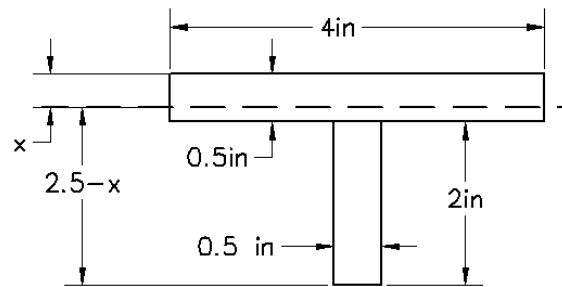
$$4\text{in} \cdot x = 3\text{ in}^2 / 2$$

$$x = 0.375\text{ in}$$

Location of plastic neutral axis from base = $2.5\text{ in} - x$

$$= 2.5\text{ in} - 0.375\text{ in} = \mathbf{2.125\text{ in}}$$

Thus correct answer is option D



152. Problem

Correct Answer: (D)

Explanation:

The shape factor of I beams is around 1.14 and of rectangular cross sections is 1.5.

Structures using plastic moment capacity are generally economical. PNA divides areas of stress blocks into two equal halves. However, for hybrid sections, the plastic section modulus is different from that of a homogenous section with the same shape and dimensions. So Option D is false and the answer.

153. Problem

Correct Answer: (B) 6.07 kip*ft

Explanation:

Ref NCEES PE handbook page 232

Let 'I' be the uniform moment of inertia,

Let 'E' be the uniform modulus of elasticity,

Ref to NCEES PE handbook page 232 section 4.1.3, rotational stiffness "k" for far end fixed condition is given by $k = 4EI/L$

If the far end is hinged instead of fixed, $k = 3EI/L$

Calculate the stiffness of each member accordingly.

$$K_{oa} = 4EI/9 = 0.4444 EI$$

$$K_{ob} = 3EI/11 = 0.2727 EI \text{ (far end hinged)}$$

$$K_{oc} = 3EI/13 = 0.2307 EI \text{ (far end hinged)}$$

$$K_{od} = 4EI/8 = 0.5 EI$$

$$K_{oe} = 4EI/5 = 0.8EI$$

Total joint stiffness

$$K = K_{oa} + K_{ob} + K_{oc} + K_{od} + K_{oe}$$

$$= 0.4444 EI + 0.2727 EI + 0.2307 EI + 0.5 EI + 0.8 EI$$

$$= 2.2478 EI$$

Balancing moment distributed to OB

$$= k_{ob}/K * 50 \text{ kip*ft}$$

$$= 0.2727EI/2.2478 EI * 50 \text{ kip*ft} = \mathbf{6.07 \text{ kip*ft}}$$
 Correct answer is option B.

154. Problem

Correct Answer: (B) 2.3 kip*ft

Explanation:

Ref NCEES PE handbook page 232 for distribution factors and page 241-244 for FEMs.

E is constant and not considered in the calculation since it cancels out in numerator and denominator for calculation of distribution factor.

Calculate joint stiffness and distribution factors for joint B (units and E not shown for calculation simplification).

Ref to NCEES PE handbook page 232 section 4.1.3, rotational stiffness “k” for far end fixed condition is given by $k = 4EI/L$

If the far end is hinged instead of fixed, $k = 3EI/L$

$$K_{bc} = 3 \cdot 800 / (12 \cdot 12) = 16.67$$

$$K_{ba} = 4 \cdot 550 / (16 \cdot 12) = 11.46$$

$$K = 16.67 + 11.46 = 28.13$$

$$DF_{bc} = 16.67 / 28.13 = 0.593$$

$$DF_{ba} = 11.46 / 28.13 = 0.407$$

	Joint A	Joint B		Joint C
	AB	BA	BC	CB
DF	0	0.407	0.593	0
FEM	$-(4 \cdot 8 \cdot 8^2) / 16^2$ = -8	$(4 \cdot 8 \cdot 8^2) / 16^2$ = 8	$-2 \cdot 12^2 / 8$ = -36	
Balancing moment		28		
Distribution		11.39	16.6	
Carryover	5.695			
Resultant	-2.305	19.39	-19.4	0

Hence, correct answer is option B, 2.3 kip*ft.

155. Problem

Correct Answer: (C) 38.24 kip*ft

Explanation:

Ref NCEES PE handbook page 232 for distribution factors and page 241-244 for FEMs.

E is constant and not considered in the calculation since it cancels out in numerator and denominator for calculation of distribution factor.

Calculate joint stiffness and distribution factors for joint B (units and E not shown for calculation simplification).

Ref to NCEES PE handbook page 232 section 4.1.3, rotational stiffness “k” for far end fixed condition is given by $k = 4EI/L$

If the far end is hinged instead of fixed, $k = 3EI/L$

$$K_{bc} = 4 \cdot 700 / (12 \cdot 12) = 19.444$$

$$K_{ba} = 3 \cdot 700 / (18 \cdot 12) = 9.722$$

$$K_{bd} = 3 \cdot 1100 / (9 \cdot 12) = 30.555$$

$$K = 19.444 + 9.722 + 30.555 = 59.721$$

$$DF_{bc} = 19.444 / 59.721 = 0.326$$

$$DF_{ba} = 9.722 / 59.721 = 0.163$$

$$DF_{bd} = 30.555 / 59.721 = 0.511$$

	Joint A	Joint B			Joint C	Joint D
	AB	BA	BC	BD	CB	DB
DF	0	0.163	0.326	0.511	0	0
FEM		$5 \cdot 18^2 / 8 = 202.5$	$-5 \cdot 12^2 / 12 = -60$	$-3 \cdot 6 \cdot 3 \cdot 15 / (2 \cdot 9^2) - 3 \cdot 3 \cdot 6 \cdot 12 / (2 \cdot 9^2) = -9$	$5 \cdot 12^2 / 12 = 60$	
Balancing moment		-133.5				
Distribution		-21.76	-43.521	-68.218		
Carryover	0				-21.76	
Resultant	0	180.74	-103.521	-77.218	38.24	0

Hence, correct answer is option C, 38.24 kip*ft.

156. Problem

Correct Answer: (B) 10.2 ksi

Explanation:

Maximum bending moment acting in the beam

Due to point load at center

$$= P \cdot L / 4$$

$$= 10 \text{ kips} \cdot 24 \text{ ft} / 4$$

$$= 60 \text{ kip} \cdot \text{ft}$$

Due to UDL

$$= w \cdot L^2 / 8$$

$$= 2 \text{ kips/ft} \cdot (24 \text{ ft})^2 / 8 = 144 \text{ kip} \cdot \text{ft}$$

$$\text{Total moment} = 60 + 144 = 204 \text{ kip} \cdot \text{ft}$$

Section modulus of rectangular section

$$S = b \cdot h^2 / 6$$

$$S = 10 \text{ in} \cdot 12 \text{ in} \cdot 12 \text{ in} / 6$$

$$S = 240 \text{ in}^3$$

Flexural stress

$$= M / S = 204 \text{ kip} \cdot \text{ft} \cdot 12 (\text{in/ft}) / 240 \text{ in}^3 = \mathbf{10.2 \text{ ksi}}, \text{ Correct answer is option B.}$$

157. Problem

Correct Answer: (C) 35.37 kip*ft

Explanation:

Ref NCEES PE handbook page 241-244 for FEMs.

Due to 10 kip point load at 7ft from joint C,

$$FEM_{CB} = \frac{P a^2 b}{L^2} = 10 \text{ kip} \cdot (3 \text{ ft} \cdot 3 \text{ ft}) \cdot 7 \text{ ft} / (10 \text{ ft} \cdot 10 \text{ ft}) = 6.3 \text{ kip} \cdot \text{ft}$$

Due to distributed load 5 kip/ft

$$FEM_{CD} = \frac{w L^2}{12} = -5 \text{ kip/ft} \cdot (10 \text{ ft} \cdot 10 \text{ ft}) / 12 = -41.67 \text{ kip} \cdot \text{ft}$$

$$\text{Thus unbalanced moment} = 41.67 \text{ kip} \cdot \text{ft} - 6.3 \text{ kip} \cdot \text{ft} = \mathbf{35.37 \text{ kip} \cdot \text{ft}}$$

Correct answer is option C.

158. Problem

Correct Answer: (A) 17.4 ft

Explanation:

Maximum bending moment is located where the shear force is zero.

Calculate reaction R_b

Sum of moments about A = 0

$$R_b * 30\text{ft} = 30 \text{ kips} * 18 \text{ ft} + 5 \text{ kips/ft} * (30\text{ft})^2 / 2$$

$$R_b = 93 \text{ kips}$$

Sum of vertical forces = 0

$$R_a = 30 \text{ kips} + 5 \text{ kip/ft} * 30\text{ft} - R_b$$

$$R_a = 87 \text{ kips}$$

Equation of shear force from left end is:

At any point x, (x = 0 to 18ft)

$$V_x = R_a - 5 \text{ kip/ft} * x$$

$$\text{At } x = 0, V_x = 87 \text{ kips}$$

$$\text{At } x = 18 \text{ ft}, V_x = 87 \text{ kips} - 5 \text{ kip/ft} * 18 = -3 \text{ kips}$$

Hence point of zero force lies between 0 ft and 18 ft

$$\text{If } V_x = 0$$

$$0 = R_a - 5 \text{ kip/ft} * x$$

$$x = 87 \text{ kip} / (5 \text{ kip/ft}) = \mathbf{17.4 \text{ ft}}$$

Correct answer is option A.

2.5 Combined stresses

[Click here for Problems 2.5](#)

159. Problem

Correct Answer: (B) 15.5 ksi compressive, 30.5 ksi tensile

Explanation:

Ref to NCEES PE handbook page 29 for principal stresses.

$\sigma_x = 25$ ksi (tensile stress)

$\sigma_y = -10$ ksi (compressive stress)

$\tau_{xy} = 15$ ksi (shear stress)

$$\sigma_a = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_a = \frac{25 - 10}{2} + \sqrt{\left(\frac{25 - (-10)}{2}\right)^2 + 15^2} = 30.5 \text{ ksi (tensile)}$$

$$\sigma_b = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_b = \frac{25 - 10}{2} - \sqrt{\left(\frac{25 - (-10)}{2}\right)^2 + 15^2} = -15.5 \text{ ksi (compressive)}$$

Correct answer is option B.

160. Problem

Correct Answer: (C) 8.2 ksi

Explanation:

Area of the section (A) = 8 in * 16 in = 128 in²

Section modulus about strong axis, $S_x = \{8 \text{ in} * (16 \text{ in})^2\} / 6 = 341.33 \text{ in}^3$

Section modulus about weak axis $S_y = \{16 \text{ in} * (8 \text{ in})^2\} / 6 = 170.67 \text{ in}^3$

P = 150 kip

$M_x = 100 \text{ kip}\cdot\text{ft}$

$M_y = 50 \text{ kip}\cdot\text{ft}$

Max normal stress = $P / A + M_x / S_x + M_y / S_y$

= $150 \text{ kip} / 128 \text{ in}^2 + 100 \text{ kip}\cdot\text{ft} * 12(\text{in}/\text{ft}) / 341.33 \text{ in}^3 + 50 \text{ kip}\cdot\text{ft} * 12(\text{in}/\text{ft}) / 170.67 \text{ in}^3$

= **8.2 ksi**

Correct answer is option C.

161. Problem

Correct Answer: (D) 15 ksi

Explanation:

Ref AISC steel manual page 1-28 and page 1-29 for section properties of W8X40.

$$\text{Area (A)} = 11.7 \text{ in}^2$$

$$\text{Section modulus (Sx)} = 35.5 \text{ in}^3$$

$$\text{Tensile load (P)} = 120 \text{ kips}$$

$$\text{Moment (Mx)} = 5 \text{ kips} \times 15 \times 12 \text{ in} = 900 \text{ kip}\cdot\text{in}$$

$$\text{Maximum compressive stress due to bending} = Mx/Sx$$

$$= 900 \text{ kip}\cdot\text{in} / 35.5 \text{ in}^3$$

$$= 25.35 \text{ ksi}$$

$$\text{Maximum tensile stress due to tensile load} = P/A$$

$$= 120 \text{ kips} / 11.7 \text{ in}^2$$

$$= 10.26 \text{ ksi}$$

$$\text{Net compressive stress} = 25.35 \text{ ksi} - 10.26 \text{ ksi} = 15.1 \text{ ksi} \approx \mathbf{15 \text{ ksi}}$$

Correct answer is option D.

162. Problem

Correct Answer: (D) 25.26 ksi

Explanation:

$$\text{Area of pipe (A)} = \pi * \{(7 \text{ in})^2 - (6 \text{ in})^2\} / 4 = 10.21 \text{ in}^2$$

$$\text{Moment of inertia (I)} = \pi * \{(7 \text{ in})^4 - (6 \text{ in})^4\} / 64 = 54.24 \text{ in}^4$$

$$\text{Section modulus (Sx)} = 54.24 \text{ in}^4 / (7/2) \text{ in} = 15.5 \text{ in}^3$$

$$\text{Tensile load (P)} = 5 \text{ kips}$$

$$\text{Moment (Mx)} = 4 \text{ kips} \times 8 \times 12 \text{ in} = 384 \text{ kip}\cdot\text{in}$$

$$\text{Combined stress} = 5 \text{ kips} / 10.21 \text{ in}^2 + 384 \text{ kip}\cdot\text{in} / 15.5 \text{ in}^3$$

$$= \mathbf{25.26 \text{ ksi}}$$

Correct answer is option D.

163. Problem

Correct Answer: (C)

Explanation:

When an object is in pure shear

Principal stresses, $\sigma_1 = \tau$ $\sigma_2 = -\tau$, that is $\sigma_2 = -\sigma_1$ and $\sigma_3 = 0$

Out of the three stress matrices, Option C resembles this stress state.

Thus, Option C is the correct answer.

164. Problem

Correct Answer: (D) 2175 psi compressive

Explanation:

Area, $A = 240 \text{ in}^2$

Section Modulus, $S = 800 \text{ in}^3$

Prestress, $P = 90 \text{ kips}$

eccentricity, $e = -4 \text{ in}$

Axial compression due to prestress

$$f_1 = P/A$$

$$= -90000 \text{ lbs} / 240 \text{ in}^2 = -375 \text{ psi}$$

Moment due to prestress eccentricity, $M_p = P \cdot e$

$$= 90000 \text{ lbs} \cdot 4 \text{ in} = 360000 \text{ lbs} \cdot \text{in}$$

Since Prestress is applied below the centroid, eccentricity will cause tension at the top,

$$f_2 = M_p / S$$

$$= 360000 \text{ lbs} \cdot \text{in} / 800 \text{ in}^3 = 450 \text{ psi}$$

Moment due to UDL, $M_{udl} = wL^2/8$

$$= 3 \text{ kip/ft} \cdot 20\text{ft} \cdot 20\text{ft} / 8$$

$$= 150 \text{ kip} \cdot \text{ft} = 1800000 \text{ lbs} \cdot \text{in}$$

This bending will cause compression at the top.

$$f_3 = M_{udl} / S$$

$$= -1800000 \text{ lbs} \cdot \text{in} / 800 \text{ in}^3 = -2250 \text{ psi}$$

Total stress at the top = $f_1 + f_2 + f_3$

$$= -375 \text{ psi} + 450 \text{ psi} - 2250 \text{ psi}$$

$$= -2175 \text{ psi} = \mathbf{2175 \text{ psi compressive}}$$

Correct answer is option D

165. Problem

Correct Answer: (D)

Explanation:

Based on the graph of failure envelope, the section fails if the sum of utilization is equal to or exceeds 100%.

For option D, sum of utilization = 60%+50% = 110%. Hence the section fails and option D is False. Thus, it is the correct answer.

166. Problem

Correct Answer: (C) 8 ksi

Explanation:

Area of section , $A = 6 \text{ in} \times 10 \text{ in} = 60 \text{ in}^2$

Eccentricity, $e = 10 \text{ in}/2 = 5 \text{ in}$

Load, $P = 120 \text{ kip}$

Section modulus about bending axis , $S = 6 \text{ in} \times 10 \text{ in} \times 10 \text{ in} / 6$

$S = 100 \text{ in}^3$

Moment, $M = P \times e$

$= 120 \text{ kip} \times 5 \text{ in}$

$= 600 \text{ kip} \cdot \text{in}$

Maximum compressive stress

$= P/A + M/S$

$= 120 \text{ kip} / 60 \text{ in}^2 + 600 \text{ kip} \cdot \text{in} / 100 \text{ in}^3$

= 8 ksi

Correct answer is option C

167. Problem

Correct Answer: (D) 83.5 kips

Explanation:

Ref to NCEES PE handbook page 257, section 4.2.1 for shear in eccentrically loaded bolts

No. of bolts, $n = 4$

Due to symmetry location of centroid of bolt group from bottom left bolt ,

$$x = 3\text{ in } y = 2\text{ in}$$

Eccentric moment acting on the group, $M = 50\text{ kips} \cdot (18\text{ in} + 6\text{ in}/2) = 1050\text{ kip}\cdot\text{in}$

Moment of inertia of bolt group (Area of all the bolts is assumed equal)

$$I_x = \sum x^2 = 4 \cdot 3\text{ in} \cdot 3\text{ in} = 36\text{ in}^2 \quad I_y = \sum y^2 = 4 \cdot 2\text{ in} \cdot 2\text{ in} = 16\text{ in}^2$$

$$I_p = I_x + I_y = 36 + 16 = 52\text{ in}^2$$

Vertical load on each bolt due to direct shear = $50\text{ kips} / 4 = 12.5\text{ kips}$

Vertical load on each bolt due to moment = $(M/I_p) \cdot x = 1050/52 \cdot 3 = 60.6\text{ kips}$

Total vertical shear = $12.5 + 60.6 = 73.1\text{ kips}$

Horizontal shear on each bolt due to moment = $(M/I_p) \cdot y = 1050/52 \cdot 2 = 40.4\text{ kips}$

$$\text{Total shear} = \sqrt{(73.1\text{ kips})^2 + (40.4\text{ kips})^2} = \mathbf{83.5\text{ kips}}$$

Correct answer is option D.

168. Problem

Correct Answer: (C) 3.75 kip/in

Explanation:

Here, there is no information about the eccentricity of the loads.

Thus eccentricity can be neglected.

Total length of all round welds = $12\text{ in} + 12\text{ in} + 12\text{ in} + 12\text{ in} = 48\text{ in}$

Tension (T) = 100 kips

Tension per unit length of welds

$$f_1 = T/L = 100\text{ kips} / 48\text{ in} = 2.08\text{ kip/in}$$

Horizontal load (H) = 150 kips

Shear per unit length of welds

$$f_2 = H/L = 150\text{ kips} / 48\text{ in} = 3.125\text{ kip/in}$$

Resultant stress acting on the welds

$$= \sqrt{(2.08\text{ kip/in})^2 + (3.125\text{ kip/in})^2} = \mathbf{3.75\text{ kip/in}} \quad \text{Correct answer is option C}$$

169. Problem

Correct Answer: (D) 14.54 kip/in

Explanation:

Maximum bending moment acting in the weld

$$M = 20 \text{ kip} \cdot 12 \text{ in}$$

$$M = 240 \text{ kip} \cdot \text{in}$$

The full penetration weld can be treated as a linear weld with height $h = 10 \text{ in}$ and unit width

$$\text{Moment of inertia, } I = (1) \cdot (10 \text{ in})^3 / 12 = 83.333 \text{ in}^3$$

$$\text{Section modulus, } S = 83.333 \text{ in}^3 / (10 \text{ in} / 2) = 16.67 \text{ in}^3$$

Maximum tension acting on the weld

$$T = M / S$$

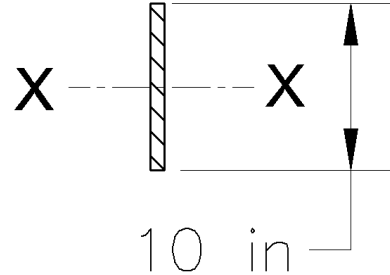
$$T = 240 \text{ kip} \cdot \text{in} / 16.67 \text{ in}^3 = 14.4 \text{ kip/in}$$

Maximum shear acting on the weld

$$H = 20 \text{ kip} / 10 \text{ in} = 2 \text{ kip/in}$$

Resultant stress

$$= \sqrt{(14.4 \text{ kip/in})^2 + (2 \text{ kip/in})^2} = \mathbf{14.54 \text{ kip/in}} \quad \text{Correct answer is option D}$$



170. Problem

Correct Answer: (B) 18 kip*ft

Explanation:

Maximum bending moment acting in the shaft,

$$M = 15 \text{ kip} \cdot \text{ft}$$

Maximum twisting moment acting in the shaft

$$T = 10 \text{ kip} \cdot \text{ft}$$

The equivalent twisting moment that produces the same maximum shear as both T and M is

$$T_{eq} = \sqrt{T^2 + M^2}$$

$$T_{eq} = \sqrt{(10 \text{ kip} \cdot \text{ft})^2 + (15 \text{ kip} \cdot \text{ft})^2} = \mathbf{18 \text{ kip} \cdot \text{ft}}$$

Correct answer is option B.

171. Problem

Correct Answer: (C) 18.76 ksi

Explanation:

Ref NCEES PE handbook page 28-29 for stresses, strains and thermal deformations,

Area, $A = 11 \text{ in}^2$

Length, $L = 20 \text{ ft}$

$\alpha = 0.000007 / ^\circ\text{F}$

$E = 29000 \text{ ksi}$

$\Delta T = 102 - 32 = 70^\circ\text{F}$

Change in length due to load ' ΔT '

$\Delta L = \alpha * L * \Delta T$

Thermal strain = $\Delta L / L = \alpha * \Delta T$

Thermal stress = Thermal strain * $E = \alpha * \Delta T * E$

$= 0.000007 * 70 * 29000 \text{ ksi}$

$= 14.21 \text{ ksi (compressive)}$

The axial load acts at the center of the span. Hence, the reaction is equal on both ends.

$R = 100 \text{ kips} / 2 = 50 \text{ kips}$

Axial stress due to compressive reaction on right end $= R / A$

$= 50 \text{ kips} / 11 \text{ in}^2 = 4.55 \text{ ksi}$

Total compressive stress = $14.21 \text{ ksi} + 4.55 \text{ ksi} = \mathbf{18.76 \text{ ksi}}$ Correct answer is option C .

172. Problem

Correct Answer: (D)

Explanation:

When a threaded rod is tightened, a torque is applied, however, it does not cause any bending in the rod. It causes pretension in the rod which is balanced by the clamping force via bearing under washers and nuts. Thus option D is the correct answer.

173. Problem

Correct Answer: (A) 0.27 ksi

Explanation:

Cross section area of wall, $A = 6 \text{ in} \times 10 \times 12 \text{ in} = 720 \text{ in}^2$

Section modulus, $S = 6 \text{ in} \times 120 \text{ in} \times 120 \text{ in} / 6 = 14400 \text{ in}^3$

Vertical load $P = 20 \text{ kip/ft} \times 10 \text{ ft} = 200 \text{ kips}$

Total moment $M, = 360 \text{ kip}\cdot\text{ft} + 50 \text{ kips} \times 6 \text{ ft}$

$= 660 \text{ kip}\cdot\text{ft}$

$= 7920 \text{ kip}\cdot\text{in}$

Tensile stress due to bending $= M/S$

$= 7920 \text{ kip}\cdot\text{in} / 14400 \text{ in}^3 = 0.55 \text{ ksi}$

Net tensile stress $= 0.55 \text{ ksi} - 200 \text{ kips} / 720 \text{ in}^2 = \mathbf{0.27 \text{ ksi}}$

Correct answer is option A.

174. Problem

Correct Answer: (B)

Explanation:

For a 2D body subjected to plane stress, there are three independent stress components - two normal and one shear.

For a 3D body, the total number of independent stress components are six - three normal and three shear.

Thus the correct answer is option B.

2.6 Deflection

[Click here for Problems 2.6](#)

175. Problem

Correct Answer: (D)

Explanation:

Shear forces are forces that act parallel to the cross section of the body i.e. tangentially to the surface of the body. Unlike point loads in a cantilever, this does not produce bending and the body is purely in shear. Hence, the correct answer is option D 'shear only'.

176. Problem

Correct Answer: (B) 0.15 in

Explanation:

Ref to AISC steel manual page 1-29 for properties of W10X45.

$I = 53.4 \text{ in}^4$ (about weaker axis)

Ref to NCEES PE handbook page 255 for maximum deflection.

Loading, $w = 2 \text{ kips/ft} = 0.1667 \text{ kips/in}$

Span, $l = 10 \text{ ft} = 120 \text{ in}$

Maximum deflection is given by

$$\Delta_{\max} = 0.0065 \frac{wl^4}{EI}$$

$$= 0.0065 * 0.1667 \text{ kips/in} * (120 \text{ in})^4 / (29000 \text{ ksi} * 53.4 \text{ in}^4)$$

$$= 0.145 \text{ in} \approx \mathbf{0.15 \text{ in}}$$

Correct answer is option B.

177. Problem

Correct Answer: (C) 1.3 kip/ft

Explanation:

Ref to NCEES PE handbook pages 233 and 237 for deflections due to uniform loadings

$$I = 60 \text{ in}^4$$

$$E = 29000 \text{ ksi}$$

$$\text{span, } l = 20 \text{ ft} = 240 \text{ in}$$

$$\text{Total loading due to uniformly varying load, } W = 0.5 * 20 \text{ ft} * 2 \text{ kips/ft} = 20 \text{ kips}$$

Maximum deflection is given by

$$\Delta_{\max} = \frac{Wl^3}{60EI}$$

$$= 20 \text{ kips} * (240 \text{ in})^3 / (60 * 29000 \text{ ksi} * 60 \text{ in}^4) = 2.65 \text{ in}$$

Deflection caused by a uniformly distributed loading of intensity w is given by

$$\Delta_{\text{max}} = 5w l^4 / (384EI)$$

$$w = \Delta_{\text{max}} * EI / l^4 * (384/5)$$

$$w = 2.65 \text{ in} * 29000 \text{ ksi} * 60 \text{ in}^4 / (240 \text{ in})^4 * (384/5)$$

$$w = 0.107 \text{ kip/in} = 1.28 \text{ kip/ft}$$

$$\approx \mathbf{1.3 \text{ kip/ft}}$$

Correct answer is option C.

178. Problem

Correct Answer: (D) 2.12 in

Explanation:

Ref to NCEES PE handbook page 242 for deflections due to uniform loadings.

$$I = 84 \text{ in}^4$$

$$E = 29000 \text{ ksi}$$

$$\text{span, } l = 15 \text{ ft} = 180 \text{ in}$$

$$\text{Uniform loading, } w = 12 \text{ kips/ft} = 1 \text{ kip/in}$$

$$\text{Point load, } P = 80 \text{ kips}$$

Maximum deflection by uniformly distributed load is given by

$$\Delta_1 = w l^4 / (384EI)$$

$$= 1 \text{ kips/in} * (180 \text{ in})^4 / (384 * 29000 \text{ ksi} * 84 \text{ in}^4) = 1.12 \text{ in}$$

Maximum deflection by point load is given by

$$\Delta_2 = P l^3 / (192EI)$$

$$= 80 \text{ kips} * (180 \text{ in})^3 / (192 * 29000 \text{ ksi} * 84 \text{ in}^4) = 1 \text{ in}$$

Both max deflections occur at mid span.

Hence max deflection

$$\Delta_{\text{max}} = \Delta_1 + \Delta_2$$

$$= 1.12 \text{ in} + 1 \text{ in} = \mathbf{2.12 \text{ in}}$$

Correct answer is option D.

179. Problem

Correct Answer: (C) 1.65 in

Explanation:

The deflection at any point is given by the moment of area of the M/EI diagram at that point.

The M/EI diagram is obtained by drawing a bending moment diagram for the beam and dividing the ordinates by EI.

For heavier section, $EI = 1000E$

For lighter section, $EI = 500E$

At support. Max moment = 5 kips*25ft = 1500 kip*in

$$1500/1000E = 1.5/E$$

At 10 ft from the tip, Moment = 5 kips* 120 in = 600 kip*in

For heavier section, $600/1000E = 0.6/E$

For lighter section, $600/500E = 1.2/E$

$$A1 = 0.5*(1.5-0.6)/E*180 = 81/E$$

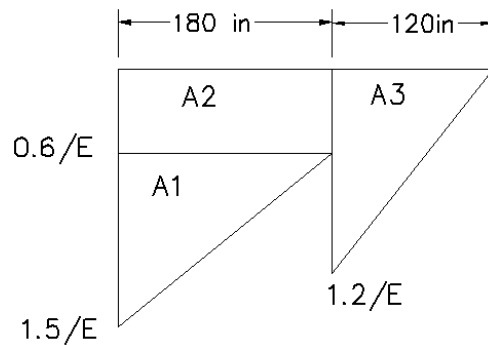
$$X1 = 180*2/3 + 120 = 240 \text{ in}$$

$$A2 = 0.6/E*180 = 108/E$$

$$X2 = 180*1/2 + 120 = 210 \text{ in}$$

$$A3 = 0.5*1.2/E*120 = 72/E$$

$$X3 = 2/3*120\text{in} = 80\text{in}$$



Moment of all areas about the tip

$$= A1*x1 + A2*x2 + A3*X3$$

$$= 81/E*240 + 108/E*210 + 72/E*80$$

$$= 47880/E$$

$$= 47880/29000$$

$$= \mathbf{1.65 \text{ in}}$$

Correct answer is option C.

180. Problem

Correct Answer: (D) 1.2 in

Explanation:

Ref IBC 2018 Section Table 1604.3, page 363

For combined dead and live loading, roof members not supporting ceiling, the deflection limits is

$$= L/120$$

$$= 12\text{ft}/120$$

$$= \mathbf{1.2\text{ in}}$$

Correct answer is option D.

181. Problem

Correct Answer: (A) 0.16 in

Explanation:

Ref TMS 402 page C-52, section 5.2.1.4.1.

Deflection limits for beams and lintels supporting unreinforced masonry is $L/600$

$$L = 8\text{ft}$$

Allowable deflection

$$= L/600$$

$$= 8\text{ft} \times 12/600$$

$$= \mathbf{0.16\text{ in}}$$

Correct answer is option A.

182. Problem

Correct Answer: (B) 2.27in

Explanation:

Ref ASCE 7-16 page 103, section 12.8.6 to calculate deflection associated with story drift

$$\delta_x = (C_d \cdot \delta_e) / I_e$$

Deflection amplification factor from page 91 ,

$$C_d = 5.5$$

For risk category IV, seismic importance factor From Table 1.2-2 page 5,

$$I = 1.5$$

$$\delta_5 = 5.5 \times 2.1 / 1.5 = 7.7 \text{ in}$$

$$\delta_4 = 5.5 \times 1.63 / 1.5 = 5.98 \text{ in}$$

$$\delta_3 = 5.5 \times 1.14 / 1.5 = 4.18 \text{ in}$$

$$\delta_2 = 5.5 \times 0.62 / 1.5 = 2.27 \text{ in}$$

$$\delta_1 = 0$$

Maximum difference in displacement between adjacent stories is the maximum story drift.

The maximum drift is observed for the 2nd floor

$$\Delta_1 = \delta_2 - \delta_1 = \mathbf{2.27 \text{ in}}$$

Correct answer is Option B.

183. Problem

Correct Answer: (A) 2 in

Explanation:

Per ASCE 7-16 page 109, table 12.12-1, allowable story drift for three story steel frame, risk category III

$$\Delta_a = 0.020 \times h_{sx}$$

$$\text{Story height, } h_{sx} = 11 \text{ ft} = 132 \text{ in}$$

$$\Delta_a = 0.020 \times 132 \text{ in}$$

$$\Delta_a = 2.64 \text{ in}$$

From section 12.12.1.1, page 109, for seismic design category

$$\text{Allowable drift} = \Delta_a / p$$

From section 12.3.4.2, page 98, $p = 1.3$ for extreme torsional irregularity

$$\text{Allowable drift} = 2.64 \text{ in} / 1.3 = 2.03 \text{ in} \approx \mathbf{2 \text{ in}}$$

Correct answer is option A.

184. Problem

Correct Answer: (B) 0.017 in

Explanation:

Ref NCEES PE handbook page 233 for truss deflection .

Forces on members due to 30 kips acting towards right on joint B and their lengths,

$$F_{ab} = -15 \text{ kips} \quad L_{ab} = 120 \text{ in}$$

$$F_{bd} = 25 \text{ kips} \quad L_{bd} = 180 \text{ in}$$

$$F_{ad} = 10 \text{ kips} \quad L_{ad} = 100 \text{ in}$$

$$F_{cd} = 0$$

$$F_{bc} = 0$$

Since the desired deflection is at joint B, unit load has to be applied at joint B.

Forces on members due to 1 kip acting towards right (can be directly scaled for this case)

$$f_{ab} = -15/30 = -0.5$$

$$f_{bd} = 25/30 = 0.833$$

$$f_{ad} = 10/30 = 0.333$$

$$f_{cd} = 0$$

$$f_{bc} = 0$$

Deflection at joint is given by

$$\Delta = \sum f \cdot (F \cdot L / AE)$$

Since $A = 10 \text{ in}^2$ and $E = 29000 \text{ ksi}$ is constant for all members,

$$\sum f \cdot F \cdot L = (-0.5) \cdot (-15) \cdot 120 + 0.833 \cdot 25 \cdot 180 + 0.333 \cdot 10 \cdot 100$$

$$= 4981.5$$

$$\Delta = 4981.5 / (10 \cdot 29000)$$

$$\Delta = \mathbf{0.017 \text{ in}}$$

Correct answer is option B.

185. Problem

Correct Answer: (D) 0.075in

Explanation:

Ref NCEES PE handbook page 244 for deflection on cantilever beam due to uniform load.

Dimensions of base plate = 5ftX5ft = 60inX60in

Dimensions of outrigger foot = 2ftX2ft = 24inX24in

Bearing pressure per unit width of the base plate due to outrigger load

$$w = 50 \text{ kips} / (60 \text{ in}) = 0.833 \text{ ksi}$$

The base plate can be idealized as a cantilevered beam from the tip of the plate to the tip of the outrigger foot.

Cantilevered length , $l = (60\text{in}-24\text{in})/2 = 18\text{in}$

Moment of inertia of plate section

$$= 60\text{in} \cdot (1\text{ in})^3 / 12 = 5\text{in}^4$$

$$\text{Deflection} = w \cdot l^4 / 8EI$$

$$= 0.833 \text{ (kip/in)} \cdot (18\text{ in})^4 / (8 \cdot 29000 \text{ ksi} \cdot 5\text{in}^4)$$

$$= \mathbf{0.075\text{in}}$$

Correct answer is option D.

186. Problem

Correct Answer: (C) 0.11 inch upwards

Explanation:

Ref NCEES PE handbook page 28-29 for stresses, strains and thermal deformations,

$$\text{Area, } A = \pi/4 \cdot (1.85\text{ in})^2 = 2.69\text{ in}^2$$

$$\text{Length, } L = 25\text{ ft}$$

$$\alpha = 0.000007 / ^\circ\text{F}$$

$$E = 29000\text{ ksi}$$

$$\Delta T = 0 - 100^\circ\text{F} = -100^\circ\text{F}$$

Change in length due to thermal strain

$$= \alpha \cdot \Delta T \cdot L = 0.000007 \cdot (-100) \cdot 25\text{ft} \cdot 12(\text{in/ft}) = -0.21\text{ in (decrease in length)}$$

Change in length due to load = PL/AE

$$= 25\text{ kip} \cdot 25\text{ft} \cdot 12(\text{in/ft}) / (2.69\text{ in}^2 \cdot 29000\text{ ksi}) = 0.096\text{ in (increase in length)}$$

$$\text{Total deflection at tip} = 0.096\text{in} - 0.21\text{ in} = -\mathbf{0.11\text{ in (upwards)}}$$

Correct answer is option C.

187. Problem

Correct Answer: (A) $184/3EI$

Explanation:

Ref NCEES PE handbook page 233 for frame deflection by unit load method,

$$\Delta = \sum \int m_i \cdot (M_i / EI) dx$$

m_i is the moment equation in member i due to unit load

M_i is the moment equation in member i due to load applied to frame

For this case, since unit load is acting in the frame, $m_i = M_i$

For member BC, moment equation

$$m_i = 1 \text{ kip} \cdot x = x$$

For member AB,

$$m_i = 1 \text{ kip} \cdot 4\text{ft} = 4 \text{ kip} \cdot \text{ft}$$

Moment of inertia of BC = I, limits = 0ft to 4ft

Moment of inertia of AB = 2I, limits = 0ft to 5ft

Then the integral becomes,

$$\Delta = \int_0^4 x \cdot x \, dx / (EI) + \int_0^5 4 \cdot 4 \, dx / (2EI)$$

$$\Delta = \{ (4^3/3 - 0) / (EI) \} + \{ 16 \cdot (5 - 0) / (2EI) \}$$

$$\Delta = 64/3EI + 40EI$$

$$\Delta = 184/3EI$$

Correct answer is option A.

188. Problem

Correct Answer: (D) One sixteenth

Explanation:

Ref NCEES PE handbook page 236 for beam deflections.

When depth of beam is doubled, the new moment of inertia is

$$I_{\text{new}} = b \cdot (2h)^3 / 12 = 8 \cdot (bh^3 / 12) = 8 \cdot I_{\text{original}}$$

Loadings, $w_{\text{new}} = (1/2) w_{\text{original}}$

Deflection is directly proportional to loading 'w' and inversely proportional to 'I'

$$\Delta_{\text{new}} / \Delta_{\text{original}} = w_{\text{new}} / w_{\text{original}} \cdot I_{\text{original}} / I_{\text{new}}$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = \{ (1/2) w_{\text{original}} \} / w_{\text{original}} \cdot I_{\text{original}} / (8 \cdot I_{\text{original}})$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (1/2) \cdot (1/8) = 1/16. \quad \text{Hence the correct answer is option D.}$$

189. Problem

Correct Answer: (B)

Explanation:

Modulus of elasticity of gray cast iron is less than mild steel and the deflections are inversely proportional to the modulus of elasticity. Hence, mild steel with higher 'E' will deflect less than gray cast iron beam for identical conditions. All other options are incorrect and the correct answer is option B.

190. Problem

Correct Answer: (A)

Explanation:

Ref NCEES PE handbook page 242 for beam deflections, $\Delta = PL^3/192EI$

When depth of beam is doubled, the new moment of inertia is

$$I_{\text{new}} = b(2h)^3/12 = 8(bh^3/12) = 8 * I_{\text{original}}$$

Length, $L_{\text{new}} = 2 L_{\text{original}}$

Deflection is directly proportional to cube of Length 'L' and inversely proportional to 'I'

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (L_{\text{new}} / L_{\text{original}})^3 * I_{\text{original}} / I_{\text{new}}$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (2 L_{\text{original}} / L_{\text{original}})^3 * I_{\text{original}} / (8 * I_{\text{original}})$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (8) * (1/8) = 1.$$

Thus deflections will be equal if depth is doubled and the correct answer is option A.

191. Problem

Correct Answer: (B) 2.33 in

Explanation:

Calculate deflections using the moment area method. Deflection at free end is given by the first moment of area of M/EI diagram between fixed end and free end, about the free end.

Moment diagram due to point load

Area of M/EI diagram

$$A_1 = -0.5 * 5 * 50 / EI$$

$$A_1 = -125 / EI$$

Centroid from B

$$x_1 = 25/3 \text{ ft}$$

Moment diagram due to
free end

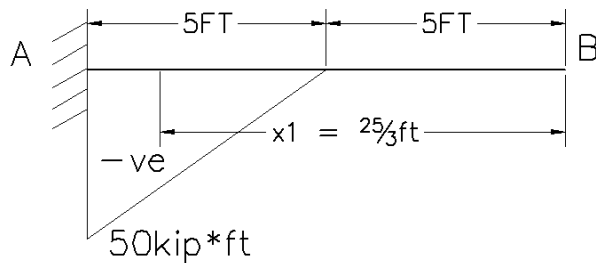
Area of M/EI diagram

$$A_2 = 10 * 60 / EI$$

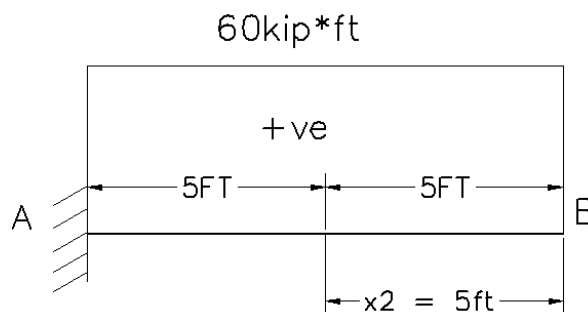
$$A_2 = 600 / EI$$

Centroid from B

$$x_2 = 5 \text{ ft}$$



moment at



Deflection = Sum of moment of areas due to A1 and A2 (all units in inches and kips)

$$= A1 \cdot x1 + A2 \cdot x2$$

$$= (-125/EI \cdot 25/3 + 600/EI \cdot 5) \cdot (12 \text{ in/ft})^3$$

$$= 1958.33/EI \cdot (12 \text{ in/ft})^3$$

$$= 1958.33/(29000 \cdot 50) \cdot (12 \text{ in/ft})^3$$

$$= \mathbf{2.33 \text{ in}}$$

Correct answer is option B.

192. Problem

Correct Answer: (C) $2PL/3AE$

Explanation:

Ref to NCEES PE handbook page 28 for axial deflection.

The three sections of the beam are subjected to same axial load P

Deflection in the first L/3 section

$$\Delta_1 = P \cdot (L/3) / (2A \cdot E)$$

$$\Delta_1 = PL/6AE$$

Deflection in the second L/3 section

$$\Delta_2 = P \cdot (L/3) / (A \cdot E)$$

$$\Delta_2 = PL/3AE$$

Deflection of the third L/3 section

$$\Delta_3 = P \cdot (L/3) / (2A \cdot E)$$

$$\Delta_3 = PL/6AE$$

$$\text{Total deflection} = \Delta_1 + \Delta_2 + \Delta_3$$

$$= PL/6AE + PL/3AE + PL/6AE$$

$$= \mathbf{2PL/3AE}$$

Correct answer is option C.

193. Problem

Correct Answer: (B) 0.03 in

Explanation:

Ref to NCEES PE handbook page 28 for axial deflection.

The three bars are parallel to each other and the stiffness of the system is the sum of stiffness of all three bars.

$$K_1 = AE/L$$

$$K_2 = 2AE/L$$

$$K_3 = AE/L$$

$$\text{Total stiffness} = k_1 + k_2 + k_3$$

$$= AE/L + 2AE/L + AE/L$$

$$= 4AE/L$$

$$\text{Total deflection} = \text{Load} / \text{stiffness}$$

$$= P / (4AE/L)$$

$$= PL / 4AE$$

$$= 200 \text{ kips} \cdot 180 \text{ in} / (4 \cdot 10 \text{ in}^2 \cdot 29000 \text{ ksi})$$

$$= \mathbf{0.031 \text{ in}}$$
 Correct answer is option B.

2.7 Special topics

[Click here for Problems 2.7](#)

194. Problem

Correct Answer: (C)

Explanation:

For pure torsion, the stress element only has shear stresses and no normal stresses.

For Mohr's circle, Principal stresses $\sigma_1 = \tau$ $\sigma_2 = -\tau$, and τ is the radius of the circle.

Hence, the correct answer is option C.

195. Problem

Correct Answer: (D) 625 kip*ft

Explanation:

Span of the beam, $L = 20\text{ft}$

Eccentricity, $e = 15\text{in}$

Load, $w = 25\text{ kips/ft}$

Total load $= w * L$

$= 25\text{ kips/ft} * 20\text{ft} = 500\text{kips}$

For beam restrained against torsion at one end only,

Torsional moment $= P * e$

$= 500\text{ kips} * 15\text{ in} * (1/12\text{ ft/in})$

$= 625\text{ kip*ft}$

Correct answer is option D.

196. Problem

Correct Answer: (B) 0.096

Explanation:

Ref to NCEES PE handbook page 31 for angle of twist and page 28 for shear modulus and page 25 for polar moment of inertia.

$G = E / 2(1 + \nu) = 29000\text{ ksi} / 2(1 + 0.3) = 11153.8\text{ ksi}$

Since the torque is applied at the center, torque is resisted equally by the two supports.

$T = 500\text{ kip*ft} / 2 = 250\text{ kip*ft} = 3000\text{ kip*in}$

Polar moment of inertia, $J = \pi * (8\text{ in})^4 / 32 = 402.12\text{ in}^4$

Angle of twist $= TL / GJ = (3000 * 144) / (11153.8 * 402.12) = 0.096$ Correct answer is Option B.

197. Problem

Correct Answer: (C) 0.5 ksi

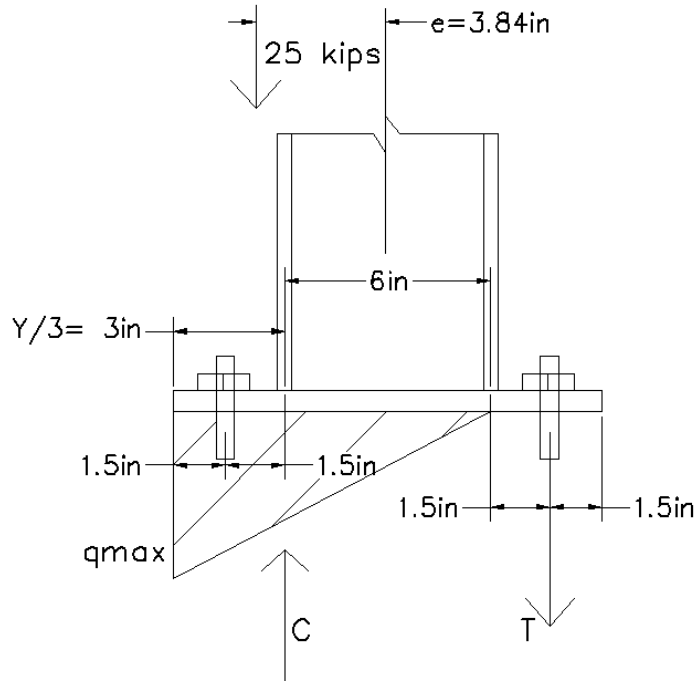
Explanation:

Calculate the eccentricity,

$$e = 8 \text{ kip}\cdot\text{ft}/25 \text{ kip}$$

$$e = 0.32\text{ft} = 3.84 \text{ in}$$

Eccentricity falls outside the column flange



The centroid of the compression area is at the center of the flange. Hence the length of bearing triangle, Y is given by $Y/3 = 3 \text{ in}$

$$Y = 9 \text{ in}$$

Width of base plate, $B = 12 \text{ in}$

To calculate compressive force ' C ', taking sum of moments about tension bolt = 0

$$C \cdot (6 \text{ in} + 1.5 \text{ in}) = 25 \text{ kips} \cdot (3.84 \text{ in} + 3 \text{ in} + 1.5 \text{ in})$$

$$C = 27.8 \text{ kips}$$

Max bearing pressure is obtained by equating area of bearing triangle to ' C '

$$0.5 \cdot q_{\max} \cdot Y \cdot B = C$$

$$0.5 \cdot q_{\max} \cdot 9 \text{ in} \cdot 12 \text{ in} = 27.8 \text{ kips}$$

$$q_{\max} = 0.51 \text{ ksi} \approx \mathbf{0.5 \text{ ksi}}$$
 Correct answer is option C.

198. Problem

Correct Answer: (C) **1.8 ksf**

Explanation:

$$L = 5 \text{ ft}$$

$$B = 5 \text{ ft}$$

Calculate the eccentricity,

$$e = 5 \text{ ft} / 2 - 2 \text{ ft} = 0.5 \text{ ft}$$

Eccentricity is less than $L/6$, so all areas of footing are in compression.

Calculate pressure due to eccentric axial load

$$\begin{aligned} q_1 &= P/A + P \cdot e / (B \cdot L^2 / 6) \\ &= 25 \text{ kips} / (5 \text{ ft} \cdot 5 \text{ ft}) + (25 \text{ kips} \cdot 0.5 \text{ ft}) / \{5 \text{ ft} \cdot 5 \text{ ft} \cdot 5 \text{ ft} / 6\} \\ &= 1.6 \text{ kips/ft}^2 \end{aligned}$$

Calculate pressure due to self weight of footing

$$q_2 = 0.15 \text{ kips/ft}^3 \cdot 15 \text{ in} \cdot (1/12 \text{ ft/in})$$

$$q_2 = 0.188 \text{ kips/ft}^2$$

Total bearing pressure = $q_1 + q_2$

$$\begin{aligned} &= 1.6 \text{ kips/ft}^2 + 0.188 \text{ kips/ft}^2 \\ &= 1.788 \text{ kips/ft}^2 \approx \mathbf{1.8 \text{ ksf}} \end{aligned}$$

Correct answer is option C.

199. Problem

Correct Answer: (A)

Explanation:

Inelastic buckling occurs when the material stress has exceeded its elastic limits and is always greater than the elastic buckling strength. Hence option A is false and is the correct answer. All other options are true.

200. Problem

Correct Answer: (C)

Explanation:

Lateral torsional buckling occurs when beams are subjected to transverse loads (or combination of transverse and axial loads) and have long unsupported spans. Hence, the correct answer is option C.

201. Problem

Correct Answer: (B) 419.5 kip*ft

Explanation:

The reactions and internal forces in the beam due to a settlement of support A is equivalent to removing support A and applying an equivalent point load that causes a deflection equal to the settlement at point A.

Calculate the load 'P'

Ref to NCEES PE handbook page 247, Case 26

$$L = 12 \text{ ft} = 144 \text{ in}$$

$$a = 144 \text{ in}$$

$$\Delta = 2 \text{ in}$$

$$\Delta = P a^2 (l+a) / (3EI)$$

$$P = \Delta * 3 EI / \{a^2 * (l+a)\}$$

$$P = 2 \text{ in} * 3 * 29000 \text{ ksi} * 1200 \text{ in}^4 / \{144 \text{ in} * 144 \text{ in} * (144 \text{ in} + 144 \text{ in})\}$$

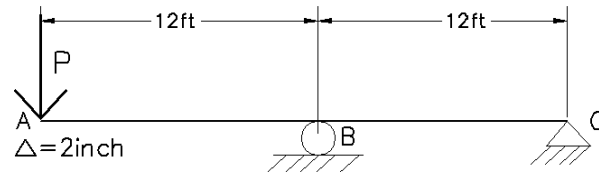
$$P = 34.96 \text{ kips}$$

$$\text{Moment at support B} = P * 12 \text{ ft}$$

$$= 34.96 \text{ kip} * 12 \text{ ft}$$

$$= \mathbf{419.5 \text{ kip*ft}}$$

Correct answer is Option B



202. Problem

Correct Answer: (D)

Explanation:

Static and dynamic analysis needs to be done in an analytical model without the failed column to ensure that all structural members sufficiently withstand the redistribution of loads. The bearing and uplift under the failed column does not need to be checked and is the correct answer.

203. Problem

Correct Answer: (A) $3M_p/10$

Explanation:

The collapse of the structure is due to the formation of a plastic hinge under load 'P'.

Let θ_1 and θ_2 be the rotations at support A and C

Total rotation of plastic hinge = $\theta_1 + \theta_2$

$$\Delta = \theta_1 * 10\text{ft}$$

$$\Delta = \theta_1 * 10\text{ft} = \theta_2 * 5\text{ft}$$

$$\theta_2 = \theta_1 * 10\text{ft} / 5\text{ft}$$

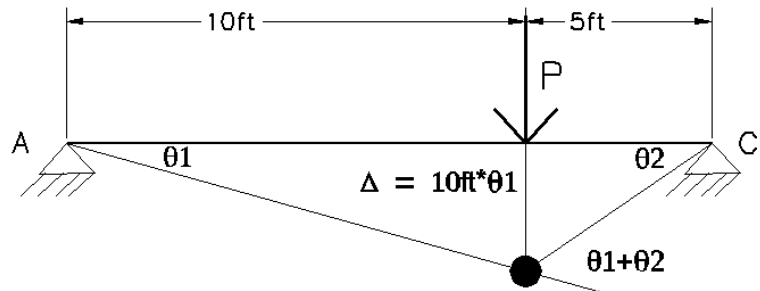
$$\theta_2 = 2 * \theta_1$$

External work done = internal work done

$$P * \Delta = M_p * (\theta_1 + \theta_2)$$

$$P = M_p * (\theta_1 + 2\theta_1) / (10 * \theta_1)$$

$$P = 3M_p/10 \quad \text{Correct answer is option A}$$



204. Problem

Correct Answer: (B) 30 kip

Explanation:

The collapse of the structure is due to the formation of a plastic hinge under load 'P' and fixed support A.

Let θ_1 and θ_2 be the rotations at support A and B

$$\Delta = \theta_1 * 10\text{ft}$$

$$\Delta = \theta_1 * 10\text{ft} = \theta_2 * 10\text{ft}$$

$$\theta_2 = \theta_1$$

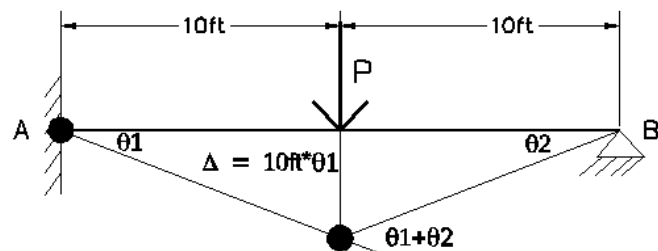
$$\text{Total rotation of plastic hinge} = \theta_1 + (\theta_1 + \theta_2) = 3\theta_1$$

External work done = internal work done

$$P * \Delta = M_p * 3\theta_1$$

$$P = M_p * 3\theta_1 / (10 \text{ ft} * \theta_1)$$

$$P = 3M_p/10 \text{ ft} = 3 * 100 \text{ kip} * \text{ft} / 10 \text{ ft} = 30 \text{ kip} \quad \text{Correct answer is option B}$$

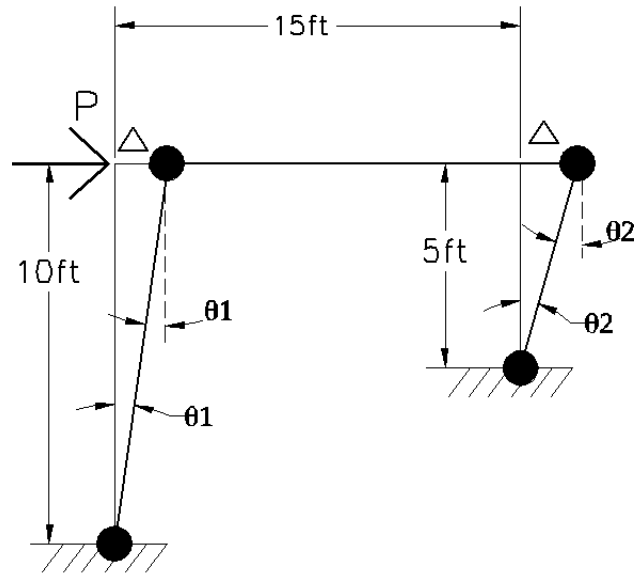


205. Problem

Correct Answer: (D) 90 kip

Explanation:

The collapse of the structure is due to the formation of four plastic hinges as shown below



Let θ_1 and θ_2 be the rotations at the taller and shorter columns

$$\Delta = \theta_1 \cdot 10\text{ft}$$

$$\Delta = \theta_2 \cdot 5\text{ft}$$

$$\theta_2 = \theta_1 \cdot 10\text{ft} / 5\text{ft} = 2\theta_1$$

$$\text{Total rotation of plastic hinge} = \theta_1 + \theta_1 + \theta_2 + \theta_2 = 6\theta_1$$

External work done = internal work done

$$P \cdot \Delta = M_p \cdot 6\theta_1$$

$$P = M_p \cdot 6\theta_1 / (10\text{ft} \cdot \theta_1)$$

$$P = 6M_p / 10\text{ft} = 6 \cdot 150\text{ kip}\cdot\text{ft} / 10\text{ft} = \mathbf{90\text{ kip}}$$
 Correct answer is option D

206. Problem

Correct Answer: (C) Infinite

Explanation:

A constant amplitude fatigue limit is the value of stress range under which the body can theoretically withstand an infinite number of loading cycles. Hence, the correct answer for this question is "Infinite".

207. Problem

Correct Answer: (A) 3.77 ksi

Explanation:

Ref AASHTO LRFD bridge design specifications page 6-50, section 6.6.1.2.5 for fatigue resistance.

AADT = 2000

From AASHTO LRFD Table 6.6.1.2.3-1 page 6-39. This detail belongs to Category C'.

Value of constant A = 44×10^8 ksi

From table 6.6.1.2.5-2 page 6-52,

Cycles per truck passage, for continuous girders , near interior support,

$n = 1.5$

From equation 6.6.1.2.5-3, page 6-50

$N = 365 \times 75 \times n \times \text{AADT}$

$N = 365 \times 75 \times 1.5 \times 2000$

$N = 82125000$

From equation 6.6.1.2.5-2, page 6-50

Fatigue resistance = $(A/N)^{1/3}$

$= (4400000000 / 82125000)^{1/3}$

= 3.77 ksi

Correct answer is option A

3. Temporary Structures and Other Topics

3.1 Special inspections

[Click here for Problems 3.1](#)

208. Problem

Correct Answer: (B)

Explanation:

The department of building official is the one who approves procedures, submittals and permits related to special inspection and cannot be the special inspector. A professional employed by the contractor and the subcontractor are not independent from the contractor and cannot work as a special inspector for that project. The correct answer is the engineer of record of the building who can work as both the design professional and a special inspector if they have the necessary qualifications and license/registration.

209. Problem

Correct Answer: (A)

Explanation:

Option A is incorrect because although the owner, engineer and building officials from the authority have the right to approve/disapprove the party serving as special inspectors, the contractor has no right to choose who serves as the inspector. All other options are true and not the answer.

210. Problem

Correct Answer: (C)

Explanation:

Ref to IBC 2018 page 418, section 1705.3.

Isolated concrete spread footings of buildings 3 story or less are exempt and Option A is incorrect.

Between Options B and C, Option C is the one that requires concrete strength, f'_c of $0.75 \times 3 \text{ ksi} = 2.25 \text{ ksi}$. Since the footing design is based on concrete less than 2500 psi, it is exempt from special inspections and is the correct answer.

211. Problem

Correct Answer: (A)

Explanation:

Ref to IBC 2018 page 417, section 1704.3.

The registered design professional in responsible charge shall prepare the statement of special inspection.

The 'engineer of record' hired by the owner is the design professional in responsible charge and is the correct answer i.e. Option A.

212. Problem

Correct Answer: (C) **13/16 inch.**

Explanation:

Ref to 'Specification for Structural Joints Using High-Strength Bolts-2014', page 16.2-23 Table 3.1 (part 16 of AISC steel manual 15th Ed.)

For 5/8 inch diameter bolts, oversized diameter of holes = **13/16 inch.**

Hence the correct answer is Option C.

213. Problem

Correct Answer: (C) **Option III**

Explanation:

Ref to AISC 360-16, page 16.1-495 Table C-N5.4-1 (part 16 of AISC steel manual)

Groove welds require inspections for joint preparation prior to welding. Fillet welds require little to no joint preparation. Option III is the only figure that shows groove welds. Hence, Option C is the correct answer.

214. Problem

Correct Answer: (B)

Explanation:

Ref to IBC 2018 page 419, section 1705.5.2

Site built wooden trusses with heights of 60 inches or greater and trusses with spans 60 feet or greater are subjected to the requirements of special inspections. Per IBC 1704.2.5.1, fabricators approved to work without special inspection requirements are exempt from this. Hence the correct answer is Option B, site built metal plate connected King truss with a span 65 feet.

215. Problem

Correct Answer: (A)

Explanation:

Ref to IBC 2018 page 420, Table 1705.3

Verifying rebar placement is under periodic inspection.

Adhesive anchors installed horizontally require continuous inspection.

Groove welds require continuous inspection.

The process of prestressing concrete members requires constant monitoring.

Hence the correct answer that requires intermittent i.e. periodic inspection is Option A.

216. Problem

Correct Answer: (C) 1.75 inch

Explanation:

Ref to ACI 318-14, page 457, section 26.4.2

Maximum aggregate size based on thickness of slab

$$= 1/3 * \text{slab thickness} = 1/3 * 6 \text{ inch} = 2 \text{ inch}$$

Maximum aggregate size based on clear spacing of rebars

$$= 3/4 * \text{Clear spacing between rebar} = 3/4 * (3 \text{ inch} - 5/8 \text{ inch})$$

$$= 1.78 \text{ inch} \approx \mathbf{1.75 \text{ inch}}$$
 Correct answer is Option C.

217. Problem

Correct Answer: (D)

Explanation:

All options require special inspections per IBC Chapter 17 except batching plant operations unless specifically required by the contract documents. Hence it is the correct answer.

3.2 Submittals

[Click here for Problems 3.2](#)

218. Problem

Correct Answer: (D)

Explanation:

The submittal document content and all associated procedures are project specific and governed by the contract document and specifications for the associated project. Hence the correct answer is Option D 'Contract documents and specifications'.

219. Problem

Correct Answer: (B)

Explanation:

Contractors do not approve submittal documents or revise any design assumptions. The engineer of record will be notified by the department of building directly when the design drawings are approved because it is usually the responsibility of the EOR to get the design approved. Hence the correct answer is option B, to compile and review all submittals before forwarding it to the building engineer of record, so that the EOR can approve it.

220. Problem

Correct Answer: (C)

Explanation:

Structural calculations for construction loads is not the responsibility of the general contractor. It is done by the structural engineer/ design professional responsible for the project. Hence Option C is not a part of the contractor's submittal package and is the correct answer.

221. Problem

Correct Answer: (B)

Explanation:

'Approved as noted' typically means that minor discrepancies found in the submittal package is acceptable/tolerable and the contractor can proceed with further construction work in the field. There is no requirement for resubmissions. Hence, the correct answer is Option B.

222. Problem

Correct Answer: (D)

Explanation:

Option A, B and C are all the intent of construction submittals. Option D is incorrect. Financial planning and budgeting is not based on contractor's submittals during the project. The owner should have detailed budget plans to manage project costs before the commissioning of the project.

3.3 Formwork, falsework, scaffolding, shoring and reshoring, bracing, and anchorage

[Click here for Problems 3.3](#)

223. Problem

Correct Answer: (C) 850 lbs/ft

Explanation:

Total lateral load due to uniform load = 12 ft * W

Point of action of uniform lateral load at mid height = 12 ft / 2 = 6 ft

Angle of inclination of the brace = $\tan^{-1}(5 \text{ ft} / 4 \text{ ft}) = 51.34 \text{ degree}$

Horizontal component of the allowable axial force on the brace
= 20 kips * cos (51.34 deg) = 12.5 kips

This component must be equal to the horizontal reaction at brace point.

Taking sum of moments at the base of wall due to lateral load and horizontal component of brace reaction = 0

$12 \text{ ft} * W * 6 \text{ ft} - 12.5 \text{ kips} * 5 \text{ ft} = 0$

$W = 12.5 \text{ kips} * 5 \text{ ft} / (12 \text{ ft} * 6 \text{ ft})$

$W = 0.868 \text{ kip/ft} \approx \mathbf{850 \text{ lbs/ft}}$

Correct answer is option C.

224. Problem

Correct Answer: (B) 6ft X6ft

Explanation:

Horizontal component of cable force, $H = 5 \text{ kips} * \cos (30 \text{ deg}) = 4.33 \text{ kips}$

Vertical component of cable force pulling upward, $V = 5 \text{ kips} * \sin (30 \text{ deg}) = 2.5 \text{ kips}$

Coefficient of friction, $\mu = 0.3$

Let 'W' be the weight of the block

Normal reaction * coefficient of friction = Factor of safety * Horizontal force

$(W - V) * \mu = 1.5 * H$

$W = 1.5 * 4.33 \text{ kips} / 0.3 + 2.5 \text{ kips} = 24.15 \text{ kips}$

6 ft height of block is constant, Try 6 ft X 5 ft X 5 ft block

Weight = $0.15 \text{ kips/ft}^3 * 6 \text{ ft} * 5 \text{ ft} * 5 \text{ ft} = 22.5 \text{ kips}$, Little less than required.

Try 6 ft X 6 ft X 6 ft block

Weight = $0.15 \text{ kips/ft}^3 \times 6 \text{ ft} \times 6 \text{ ft} \times 6 \text{ ft} = 32.4 \text{ kips}$, Sufficient

Check tipping of block

Overturning moment = $H \times 6 \text{ ft} = 4.33 \text{ kip} \times 6 \text{ ft} = 25.98 \text{ kip}\cdot\text{ft}$

Restoring moment = $(W-V) \times 6 \text{ ft} / 2 = (32.4 \text{ kips} - 2.5 \text{ kips}) \times 3 \text{ ft} = 89.7 \text{ kip}\cdot\text{ft}$

Thus, we have sufficient factor of safety against overturning.

Hence, 6ft X 6ft X 6ft block will satisfy the requirement and option B is correct.

225. Problem

Correct Answer: (C) 1/2 inch

Explanation:

Tension load in the anchor, $T = 2.5 \text{ kips} \times \sin(45 \text{ degree}) = 1.768 \text{ kips}$

Shear load in the anchor, $V = 2.5 \text{ kips} \times \cos(45 \text{ degree}) = 1.768 \text{ kips}$

Factor of safety = 2

Check with 3/8 inch anchor

Allowable forces = Ultimate forces / Factor of safety

Tension allowable, $T_a = 5.94 \text{ kips} / 2 = 2.97 \text{ kips}$

Shear allowable, $V_a = 4.14 \text{ kips} / 2 = 2.07 \text{ kips}$

Hence 3/8 inch anchor satisfies individual shear tension requirement.

Check interaction since both forces are acting together

$(T / T_a) + (V / V_a)$

$= 1.768 / 2.97 + 1.768 / 2.07 = 1.44$

This should be less than or equal to 1 for the anchor to be safe. Hence it doesn't work

Check 1/2 inch anchor

Tension allowable, $T_a = 9.64 \text{ kips} / 2 = 4.82 \text{ kips}$

Shear allowable, $V_a = 7.24 \text{ kips} / 2 = 3.62 \text{ kips}$

$(T / T_a) + (V / V_a)$

$= 1.768 / 4.82 + 1.768 / 3.62 = 0.855$, Safe

Hence, the correct answer is Option C, **1/2 inch**.

226. Problem

Correct Answer: (D) 6 kips

Explanation:

Weight of beam, $W = 200 \text{ lbs/ft} \times 20 \text{ ft}$

$W = 4 \text{ kips}$

This acts at the center of the beam

Distance of point of action of beam weight from wire rope attachment point

$= 13 \text{ ft} + 6 \text{ ft} - 20 \text{ ft} / 2$

$= 9 \text{ ft}$

Let V be the vertical reaction at the column

Taking sum of moment at wire rope support point $= 0$

$W \times 9 \text{ ft} - V \times 6 \text{ ft} = 0$

$V = W \times 9 \text{ ft} / 6 \text{ ft}$

$V = 4 \text{ kips} \times 1.5$

$V = 6 \text{ kips}$

Correct answer is option D.

227. Problem

Correct Answer: (B)

Explanation:

Reshoring is done after the initial setting and initial removal of formworks. This ensures the concrete slab can continue curing and gain full strength and redistribute any loads acting on it to the floors below and the foundation. Hence the correct answer is option B.

228. Problem

Correct Answer: (C)

Explanation:

Although all other factors influence the removal of formwork, the most important option is the compressive strength that the member has gained. The member may be cured at a very favorable temperature and may be cured with a compound which helps in rapid strength gain, but the ultimate factor that decides removal is whether it has gained sufficient strength to withstand the imposed loads. Hence option B is the correct answer.

229. Problem

Correct Answer: (B) 4.5 ft X 4.5 ft

Explanation:

Ref to NCEES handbook page 3 for conversion factors.

$$1 \text{ ft} = 0.3048 \text{ m}$$

$$15 \text{ ft} = 0.3048 * 15 = 4.57 \text{ m}$$

For 4.57m , the allowable load in a 550 prop from the table = 20 kN

$$1 \text{ N} = 0.225 \text{ lbf}$$

$$1 \text{ kN} = 1000 * 0.225 \text{ lbf} = 0.225 \text{ kip}$$

$$20 \text{ kN} = 20 * 0.225 \text{ kip} = 4.5 \text{ kip}$$

$$\text{Total load on shoring posts} = 100 \text{ psf} + 75 \text{ psf} + 15 \text{ psf} = 190 \text{ psf} = 0.19 \text{ ksf}$$

$$\text{Allowable area} = \text{Allowable prop load} / \text{Load acting}$$

$$= 4.5 \text{ kip} / 0.19 \text{ ksf}$$

$$= 23.684 \text{ ft}^2$$

$$\text{Spacing} = \sqrt{\text{Area}}$$

$$= \sqrt{23.684 \text{ ft}^2}$$

$$= 4.866 \text{ ft}$$

Hence maximum spacing of 4.5 ftX 4.5 ft is permitted and Option B is the answer.

230. Problem

Correct Answer: (B) 20 kips

Explanation:

$$\text{Maximum load transferred to 5th floor slab} = 50 \text{ kips} - 35 \text{ kips}$$

$$= 15 \text{ kips}$$

$$\text{Maximum load transferred to 4th floor slab} = 35 \text{ kips} - 15 \text{ kips}$$

$$= 20 \text{ kips}$$

$$\text{Maximum load transferred to 3rd floor slab} = 15 \text{ kips} - 5 \text{ kips}$$

$$= 10 \text{ kips}$$

$$\text{Maximum load transferred to 2nd floor slab} = 5 \text{ kips} - 0$$

$$= 5 \text{ kips}$$

Thus maximum load is transferred to 4th floor slab = **20 kips**, Option B is the answer.

231. Problem

Correct Answer: (D) 7.6 kips

Explanation:

Dead Load due to Slab weight = $150 \text{ pcf} \times 0.5 \text{ ft}$
= 75 psf

Total load from 6th floor at an area 5 ft X 5 ft
= $(75 \text{ psf} + 75 \text{ psf} + 15 \text{ psf}) \times 5 \text{ ft} \times 5 \text{ ft}$
= 4125 lbs
= 4.125 kips

Total load from 5th floor at an area 5 ft X 5 ft
= $(75 \text{ psf} + 50 \text{ psf} + 15 \text{ psf}) \times 5 \text{ ft} \times 5 \text{ ft}$
= 3500 lbs
= 3.5 kips

Total load on shoring post
= 4.125 kips + 3.5 kips
= 7.625 kips $\approx$ **7.6 kips**

Option D is the correct answer.

232. Problem

Correct Answer: (B) 4 kips

Explanation:

Spacing of vertical posts = 30 in c/c

Spacing of horizontal wales = 24 inches c/c

Since every alternate post is tied to the wales,

Tributary area for each tie point

= $(30 \text{ in} \times 2) \times 24 \text{ in}$
= 1440 in^2
= 10 ft^2

Lateral pressure = 400 psf

Load on each tie = $400 \text{ psf} \times 10 \text{ ft}^2$
= **4 kips**

Correct answer is Option B.

233. Problem

Correct Answer: (C) 3 kip*ft

Explanation:

Per OSHA, CFR Title 29, Part 1926, Subpart L, section 1926.451(a)(1)

The scaffold component shall be capable of supporting its own weight and at least 4 times the maximum intended load.

Maximum intended load on each beam = $20 \text{ psf} \times 5 \text{ ft} / 2 = 50 \text{ plf}$

Design load including self weight of the beam

$= 40 \text{ plf} + 4 \times 50 \text{ plf} = 0.24 \text{ kip/ft}$

Design moment for simply supported beam (Ref NCEES PE handbook page 236 Case 1)

$= w l^2 / 8 = 0.24 \text{ kip/ft} \times 10 \text{ ft} \times 10 \text{ ft} / 8$

$= \mathbf{3 \text{ kip*ft}}$, Correct answer is option C.

234. Problem

Correct Answer: (A) 50 lbs*ft

Explanation:

Per OSHA, CFR Title 29, Part 1926, Subpart L, section 1926.451(g)(4)(vii) and (ix),

Maximum load acting on mid rail for a guardrail with 100 pound top rail capacity i.e. single point adjustable suspension scaffold, = 75 lbs

Since the rail is welded between two posts, it can be assumed a fixed-fixed beam.

Design moment for simply supported beam (Ref NCEES PE handbook page 242 Case 16)

$= P l / 8 = 75 \text{ lbs} \times 5 \text{ ft} / 8$

$= 46.875 \text{ lbs*ft} \approx \mathbf{50 \text{ lbs*ft}}$ Correct answer is option A.

235. Problem

Correct Answer: (B) 8 ft X 5ft

Explanation:

Per OSHA, CFR Title 29, Part 1926, Subpart L, Appendix A, (2) (a)

From table for single pole wood pole scaffolds,

For medium duty (50 psf) scaffolds up to 60 feet high,

Max pole spacing using a 4 in X 4 in = **8 ft X 5ft**

Correct answer is option B.

3.4 Impact of construction on adjacent facilities

[Click here for Problems 3.4](#)

236. Problem

Correct Answer: (C) 9ft

Explanation:

Difference between depths of
foundation between two structures

$$= 15 \text{ ft} - 6 \text{ ft}$$

$$= 9 \text{ ft}$$

For a 45 degree line,

Horizontal Distance

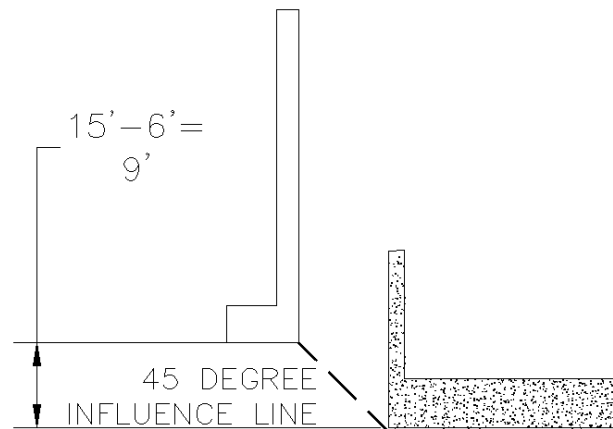
$$= 9 \text{ ft} * \tan (45 \text{ degree})$$

$$= 9 \text{ ft}$$

Thus, the minimum distance between
the two structures for existing

structure to be away from influence of new structure = 9ft

Correct answer is option C.



237. Problem

Correct Answer: (C)

Explanation:

Improving soil strength helps reduce the settlements.

Pile foundations transfer loads to a deeper depth away from the stress overlap zone
between the two buildings and is also a remedial measure.

Placing lateral load resisting systems away from the perimeter of the building towards the
center helps reduce the bearing pressures at overlap zones in the boundaries of adjacent
buildings and helps reduce settlement.

Option C, placing the foundation of the new building above the foundation of the existing
building does not help with the stress overlap and is the answer.

238. Problem

Correct Answer: (B) 12 in

Explanation:

Maximum inelastic response displacement of building 1

$$\delta_{M1} = 7 \text{ in}$$

Maximum inelastic response displacement of building 2

$$\delta_{M2} = 9 \text{ in}$$

Ref to ASCE 7-16, page 110, equation 12.12-2

Required structural separation

$$\delta_{MT} = \sqrt{\delta_{M1}^2 + \delta_{M2}^2}$$

$$\delta_{MT} = \sqrt{(7\text{in})^2 + (9\text{in})^2}$$

$$\delta_{MT} = 11.4 \text{ in} \approx \mathbf{12 \text{ in}}$$

Correct answer is option B.

239. Problem

Correct Answer: (C)

Explanation:

Per IBC 2018, page 609, section 3307, "Protection of adjoining property",

All options are correct except Option C.

IBC 2018 Chapter 33 only says that excavation should be done in a way to not endanger life or property. Vacating adjacent buildings is optional but not mandatory. Hence, the correct answer is option C.

3.5 Safety

[Click here for Problems 3.5](#)

240. Problem

Correct Answer: (C) 27.5 ft

Explanation:

Ref to NCEES PE handbook page 153, section 3.10.3

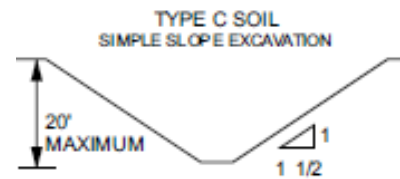
For type C soil,

Horizontal: Vertical = 1.5:1

Minimum undisturbed horizontal perimeter around the building = $25 \text{ ft} / 2 = 12.5 \text{ ft}$

Total distance from face of building to bottom of excavated slope = $12.5 \text{ ft} + 10 \text{ ft} \times 1.5 = \mathbf{27.5 \text{ ft}}$

Correct answer is option C



241. Problem

Correct Answer: (D) 200%

Explanation:

Ref to NCEES PE handbook page 70, section 2.6.2

For 90 dBA, time permitted = 8 hours, time spent = 4 hours

For 100 dbA, time permitted = 2 hours, time spent = 2 hours

For 95 dbA, time permitted = 4 hours, time spent = 2 hours

Noise dose, $D = 100 \times \sum (\text{time spent at specified sound pressure} / \text{time permitted})$

$$D = 100 \times (4/8 + 2/2 + 2/4)$$

$D = \mathbf{200\%}$

Correct answer is Option D.

242. Problem

Correct Answer: (D) **100 kips**

Explanation:

Per OSHA, CFR Title 29, Part 1910 ,Subpart F, 1910.66 Appendix D, (b) (20) (iii)

The minimum factor of safety against the manufacturer's rated load shall be 10.

Required rated load

= Static Load * 10

= 10 kips* 10

= **100 kips** Correct answer is Option D.

243. Problem

Correct Answer: (B)

Explanation:

Per OSHA, CFR Title 29, Part 1926 ,Subpart M, all options are correct except Option B.

Per 29 CFR 1926.502(d)(16)(iv)

Fall arrest system shall bring an employee to a complete stop and limit maximum deceleration distance an employee travels to 3.5 feet.

Hence, Option B is the answer.

244. Problem

Correct Answer: (B) **4**

Explanation:

Per OSHA, CFR Title 29, Part 1926 ,Subpart R, 1926.758(c)

Rigid frames shall have 50 percent of their bolts installed and tightened on both sides of the web adjacent to each flange before the hoisting equipment is released.

Number of bolts on each side of web = 8

Minimum number of bolts to be tightened = 50% * 8 = **4**

Correct answer is option B.

4. Design and Details of Structures–Materials and Material Properties

4.1 Soil classification and soil properties

[Click here for Problems 4.1](#)

245. Problem

Correct Answer: (B) I - Gap graded II- Well graded III- Uniform

Explanation:

Sample I shows a non uniform irregular curve which denotes that one or more ranges of particle sizes are missing. Hence it is gap graded.

Sample II shows a continuous curve spreading over a wide range of particle size which denotes a well graded sample. Sample III shows a narrow vertical curve which denotes uniformity in particle size and hence is uniformly graded.

Correct answer is option B.

246. Problem

Correct Answer: (C) CH

Explanation:

Ref to NCEES PE handbook page 110, section 3.7.2 for USCS classification.

% of particles passing through No. 200 (0.075 mm sieve)

$$= (1-0.31)*100\%$$

$$= 69\%$$

Hence the inorganic sample is silt and clay.

Liquid Limit = 51 i.e. greater than 50, so it is either CH or MH

From page 111,

$$PI = LL - PL$$

$$= 51 - 19 = 32$$

LL = 51 and PI = 32 plots above the A line from the plasticity chart on page 111.

Hence from table in page 110, it is classified as **CH**.

Correct answer is Option C.

247. Problem

Correct Answer: (D) A-7-6(22)

Explanation:

Ref to NCEES PE handbook page 118, section 3.7.3 for AASHTO classification.

% of particles passing through No. 200 (0.075 mm sieve)

$$F = (1 - 0.35) \times 100\%$$

$$F = 65\%$$

Hence it is between A-4 and A-7.

From page 111,

$$PI = LL - PL = 55 - 17 = 38$$

Liquid Limit = 51 i.e. greater than 41, and PI = 38 i.e. greater than 11

From page 119, AASHTO plasticity chart, it belongs to group A-7-6

Calculate group index from page 119

$$GI = (F - 35) [0.2 + 0.005(LL - 40)] + 0.01(F - 15)(PI - 10)$$

$$GI = (65 - 35) [0.2 + 0.005(55 - 40)] + 0.01(65 - 15)(38 - 10)$$

$$GI = 22.25, \text{ rounded off to } 22$$

Thus the correct answer is Option D, **A-7-6(22)**.

248. Problem

Correct Answer: (C) SW-SM

Explanation:

Ref to NCEES PE handbook page 115, for sieve sizes and page 110 for USCS classification.

No. 4 sieve = 4.75 mm, No. 200 sieve = 0.075 mm

% of particles passing through No. 200 = 11%, thus soil is Sand or Gravel

% of particles passing No. 4 sieve = 70%, thus soil is Sand and requires dual symbol as fines

(11%) are between 5% and 12%

$$C_u = D_{60}/D_{10} = 0.8/0.12 = 6.67 > 6 \text{ thus it is well graded 'SW'}$$

LL is less than 50

$$PI = 46 - 36 = 10$$

From plasticity chart in page 111, it is below A line

Thus it is silty 'M'

Hence the correct answer is option C, **SW-SM**.

249. Problem

Correct Answer: (B) Good

Explanation:

Ref to NCEES PE handbook page 120, section 3.7.4.1. for rock quality designation

Total Length of core run = 6 ft = 72in

Length of sound core pieces >4 in

$$= 19+8+11+6+6+12$$

$$= 62\text{in}$$

RQD = sum of length of sound core pieces/ total core run length

$$= 62 \text{ in} / 72 \text{ in}$$

$$= 0.86$$

$$= 86\% \text{ (Good from table in page 120)}$$

Thus the correct answer is Option B, **Good**.

250. Problem

Correct Answer: (A) 2920 psf

Explanation:

Ref to NCEES PE handbook page 2 for density of water,

$$\gamma_w = 62.4 \text{ pcf}$$

Effective stress at 45 ft = Total stress - Pore water pressure

$$= \gamma_1 * h_1 + \gamma_2 * h_2 + \gamma_3 * h_3 - \gamma_w * h_w$$

$$= 120 \text{ pcf} * (75 \text{ ft} - 60 \text{ ft}) + 130 \text{ pcf} * (60 \text{ ft} - 55 \text{ ft}) + 128 \text{ pcf} * (55 \text{ ft} - 45 \text{ ft}) \\ - 62.4 \text{ pcf} * (58 \text{ ft} - 45 \text{ ft})$$

$$= 2918.8 \text{ psf}$$

$$\approx \mathbf{2920 \text{ psf}}$$

Correct answer is option A.

251. Problem

Correct Answer: (D) 131.6 pcf

Explanation:

Ref to NCEES PE handbook page 285, section 5.5.1,

$$\text{Relative density, } D_R = \frac{e_{max} - e}{e_{max} - e_{min}}$$

e = in situ void ratio of soil

e_{max} and e_{min} are void ratios in loosest and densest conditions

From PE handbook page 126, chart showing correlation between SPT resistance and D_R ,

For overburden of 4 ksf and $N = 20$, relative density $\approx 60\%$

Then,

$$60\% = \frac{e_{max} - e}{e_{max} - e_{min}} = \frac{0.4 - e}{0.4 - 0.2}$$

Solve for ' e ',

$$e = 0.28$$

$\gamma_w = 62.4$ pcf, unit weight of water from PE handbook page 2,

$$G_s = 2.7$$

From PE handbook page 133

$$\gamma_d = \frac{G_s * \gamma_w}{1 + e}$$

$$\gamma_d = \frac{2.7 * 62.4 \text{ pcf}}{1 + 0.28}$$

$$= \mathbf{131.6 \text{ pcf}}$$

Correct answer is option D

252. Problem

Correct Answer: (C) 75%

Explanation:

Ref to NCEES PE handbook page 135

Water content w = weight of water/weight of solids

$$w = W_w/W_s$$

$$0.19 = W_w/W_s$$

$$W_w = 0.19W_s$$

Assume 1 m^3 volume of sample

Total weight of sample = 1900 kg

$$W_s + W_w = 1900 \text{ kg}$$

$$W_s + 0.19W_s = 1900 \text{ kg}$$

$$W_s = 1596.6 \text{ kg}$$

$$\text{Thus, } W_w = 0.19W_s$$

$$= 0.19 \times 1596.6 \text{ kg} = 303.4 \text{ kg}$$

Calculate volume of solids

From NCEES PE handbook page 2, mass of 1 cubic meter water = 1000 kg

$$\gamma_w = 1000 \text{ kg/m}^3$$

From page 134,

$$V_s = W_s / (G^* \gamma_w)$$

$$V_s = 1596.6 \text{ kg} / (2.7 \times 1000 \text{ kg/m}^3)$$

$$V_s = 0.591 \text{ m}^3$$

$$V_w = W_w / \gamma_w$$

$$V_w = 303.4 \text{ kg} / 1000 \text{ kg/m}^3$$

$$V_w = 0.3034 \text{ m}^3$$

Thus volume of air in 1 m^3 sample

$$V_a = 1 \text{ m}^3 - 0.591 \text{ m}^3 - 0.3034 \text{ m}^3$$

$$V_a = 0.106 \text{ m}^3$$

Total volume of voids = $V_a + V_w$

$$V_v = 0.106 \text{ m}^3 + 0.3034 \text{ m}^3 = 0.409 \text{ m}^3$$

Degree of saturation = V_w / V_v (Ref NCEES PE handbook page 135)

$$= 0.3034 \text{ m}^3 / 0.409 \text{ m}^3$$

$$= 0.742 \approx \mathbf{75\%}$$
 Correct answer is option C

253. Problem

Correct Answer: (B) 24750 m³

Explanation:

Ref to NCEES PE handbook page 134 for volume weight relationships

For filling, required dry unit weight

$$\gamma_d = 1800 \text{ kg/m}^3$$

Then, weight of solids for 1 m³ = 1800 kg

From borrow pit, 1800 kg of solids will be required per m³ fill

Weight of water in soil from borrow pit = water content in borrow pit * weight of solids

$$= 10\% * 1800 \text{ kg} = 180 \text{ kg}$$

Total weight of soil required from borrow pit per m³ fill =

(Weight of solids + weight of water)

$$(1800 + 180) \text{ kg} = 1980 \text{ kg}$$

Given, density of borrow pit = 1600 kg/m³

For Weight of soil 1980 kg,

Volume required = Total Weight required / in situ density of borrow pit

$$= 1980 \text{ kg} / (1600 \text{ kg/m}^3)$$

$$= 1.2375 \text{ m}^3$$

Thus for every m³ of fill, 1.2375 m³ of soil from borrow pit has to be excavated

For 20000 m³ of fill, 1.2375 * 20,000 m³ = **24750 m³** has to be excavated.

Option B is the correct answer.

254. Problem

Correct Answer: (D) 95%

Explanation:

Ref to NCEES PE handbook page 146 section 3.9 for standard proctor test.

Volume of mold = 1/30 ft³

By inspection of the given table, maximum dry density occurs at water content 10%

$$\gamma = \text{Weight} / \text{Volume}$$

$$= 4.35 \text{ lbf} / (1/30 \text{ ft}^3)$$

$$= 130.5 \text{ pcf}$$

Ref NCEES PE handbook page 147

$$\gamma_{Dmax} = \gamma / (1 + w)$$

$$\gamma_{Dmax} = 130.5 \text{ pcf} / (1 + 0.1)$$

$$\gamma_{Dmax} = 118.63 \text{ pcf}$$

Ref NCEES PE handbook page 148, section 3.9.2

Relative compaction = dry unit weight at field/ max dry unit weight from lab test *100%

$$= 112.7 \text{ pcf} / 118.63 \text{ pcf} * 100\%$$

$$= \mathbf{95\%}$$

Hence, the correct answer is Option D.

255. Problem

Correct Answer: (B) 99 days

Explanation:

Ref to NCEES PE handbook page 85 section 3.2.3 for time rate of settlement.

Coefficient of consolidation

$$C_v = 0.29 \text{ ft}^2 / \text{day}$$

$$\text{Depth of clay layer} = 155 \text{ ft} - 135 \text{ ft} = 20 \text{ ft}$$

Longest distance to drainage boundary

$$H_d = 20 \text{ ft} / 2 = 10 \text{ ft}$$

From table in page 86, for $U = 60\%$, Time factor

$$T_v = 0.287$$

From equation in page 85

$$T = T_v * (H_d)^2 / C_v$$

$$T = 0.287 * 10 \text{ ft} * 10 \text{ ft} / 0.29 \text{ ft}^2 / \text{day}$$

$$T = 98.9 \text{ days} \approx \mathbf{99 \text{ days}}$$

Correct answer is option B.

256. Problem

Correct Answer: (B)

Explanation:

Sandy soils are more susceptible to liquefaction. It is caused due to decrease of effective stresses due to increase in pore water pressure. As the flowability of soil increases, the lateral earth pressure exerted also increases. Options A, C and D are all true.

Clayey soils, although less susceptible, are not immune to liquefaction. Option B is false and the answer.

257. Problem

Correct Answer: (B) 4.33 ksf

Explanation:

Ref to NCEES PE handbook page 90 section 3.4.2.1 for bearing capacity of square footings.

Depth of water table (25ft) is beyond (1.5*breadth of footing + depth)

Hence water table correction factor is not needed.

$$B_f = 6\text{ft}$$

$$L_f = 6\text{ ft (Square footing)}$$

From table in page 91, bearing capacity factors for $\Phi = 20$ degrees

$$N_c = 14.8$$

$$N_q = 6.4$$

$$N_\gamma = 5.4$$

From table in page 90, shore correction factors

$$S_c = 1 + (B_f/L_f) * (N_q/N_c)$$

$$S_c = 1 + 1 * (6.4/14.8)$$

$$S_c = 1.432$$

$$S_\gamma = 1 - 0.4 * (B_f/L_f)$$

$$S_\gamma = 1 - 0.4 * 1$$

$$S_\gamma = 0.6$$

$$S_q = 1 + (B_f/L_f) * \tan\Phi$$

$$S_q = 1 + 1 * \tan 20$$

$$S_q = 1.364$$

From equation in page 90, section 3.4.2.1

$$Q_{ult} = c * N_c * S_c + q * N_q * S_q + 0.5 * \gamma * B_f * N_\gamma * S_\gamma$$

$$Q_{ult} = 350 \text{ psf} * 14.8 * 1.432 + (115 \text{ pcf} * 5 \text{ ft}) * 6.4 * 1.364 + 0.5 * 115 \text{ pcf} * 6 \text{ ft} * 5.4 * 0.6$$

$$Q_{ult} = 13555.08 \text{ psf}$$

Calculate Q_{net}

$$Q_{net} = Q_{ult} - \gamma * D_f$$

$$Q_{net} = 13555.08 \text{ psf} - 115 \text{ pcf} * 5 \text{ ft} = 12.98 \text{ ksf}$$

Calculate allowable bearing pressure

$$Q_{all} = Q_{net} / \text{Factor of safety}$$

$$= 12.98 \text{ ksf} / 3 = \mathbf{4.33 \text{ ksf}}$$

Correct answer is option B.

258. Problem

Correct Answer: (A) 0.9 cubic feet/ day / feet

Explanation:

Ref to NCEES PE handbook page 190 section 3.16.4 for flow nets

Permeability, $k = 0.0001$ ft/min

Number of flow channels, $N_f = 5$

Number of equipotential drops, $N_d = 12$

Head difference, $h = 15$ ft

From equation in page 191

Flow rate $Q = k \cdot h \cdot (N_f / N_d)$

$Q = 0.0001$ ft/ min $\cdot 15$ ft $\cdot (5/12)$

$Q = 0.000625$ cubic feet/min per feet width

Convert to per day

1 day = $24 \cdot 60$ min = 1440 min

Q per day = $0.000625 \cdot 1440$ cubic feet/ day/feet

= 0.9 cubic feet/ day / feet

Correct answer is option A.

259. Problem

Correct Answer: (D) 8 ft³/ min

Explanation:

Ref to NCEES PE handbook page 187 section 3.16 for Darcy's law

Permeability, $k = 0.004$ ft/min

Hydraulic gradient, $i = \Delta h / \Delta S$

$i = (65 \text{ ft} - 45 \text{ ft}) / 90 \text{ ft}$

$I = 0.2222$

Area of flow = $30 \text{ ft} \cdot 300 \text{ ft}$

$Q = k \cdot i \cdot A$

$Q = 0.004 \text{ ft/min} \cdot 0.2222 \cdot (30 \text{ ft} \cdot 300 \text{ ft})$

$Q = 8 \text{ ft}^3 / \text{min}$

Correct answer is option D.

4.2 Concrete

[Click here for Problems 4.2](#)

260. Problem

Correct Answer: (D)

Explanation:

Ref to ACI 318-14, page 318 Table 19.3.1.1

For sea water exposure, the exposure class is S1

From page 323, Table 19.3.2.1, for Class S1, the cementitious material specified is Type II.

Generally, chemical accelerators are not good to improve sulphate resistance.

Superplasticizers help lower the water cement ratio and help reduce the permeability of concrete. They help make concrete more durable and more resistant to sulphate attacks.

Hence Option D “Type II cement, sand, aggregates, water, super plasticizer” is the correct answer.

261. Problem

Correct Answer: (C) 4032 psi

Explanation:

Ref to ACI 318-14, page 478 Section 26.12.1.1,

Strength test shall be average of at least three 4X8 in cylinders

Area of samples

$$A = \{\pi * (4 \text{ in})^2\} / 4$$

$$A = 12.566 \text{ in}^2$$

$$\text{Sample 1 Strength} = 50 \text{ kips} / 12.566 \text{ in}^2 = 3.979 \text{ ksi} = 3979 \text{ psi}$$

$$\text{Sample 2 Strength} = 45 \text{ kips} / 12.566 \text{ in}^2 = 3.581 \text{ ksi} = 3581 \text{ psi}$$

$$\text{Sample 3 Strength} = 57 \text{ kips} / 12.566 \text{ in}^2 = 4.536 \text{ ksi} = 4536 \text{ psi}$$

$$\text{Average strength} = (3979 \text{ psi} + 3581 \text{ psi} + 4536 \text{ psi}) / 3$$

$$\text{Average strength} = \mathbf{4032 \text{ psi}}$$

Correct answer is option C.

262. Problem

Correct Answer: (D) 4

Explanation:

Ref to ACI 318-14, page 480 Section 26.12.3.1(b),

For compliance, the arithmetic average of any three consecutive strength tests equals or exceeds f_c' and no strength test falls below f_c' by more than 500 psi if f_c' is 5000 psi or less.

Option A is not true because Day 1 and Day 2 results although less than 3500 psi is not less than $f_c' - 500 \text{ psi} = 3500 \text{ psi} - 500 \text{ psi} = 3000 \text{ psi}$

Option B and C is not true because the mix fails because of results from day 5, the results are less than 3000 psi.

Hence Option D is true and the answer.

263. Problem

Correct Answer: (C) Target air content 6 % and maximum w/c ratio 0.45

Explanation:

Ref to ACI 318-14, page 318 Table 19.3.1.1,

For frequent exposure to water, the exposure class is F2.

From page 323, table 19.3.2.1,

Maximum water cement ratio = 0.45 for class F2

From page 324, table 19.3.3.1, target air content = 6 % for class F2 and 3/4 in aggregates.

Hence the correct answer is Option C "Target air content 6 % and maximum w/c ratio 0.45"

264. Problem

Correct Answer: (A) 0.35 ksi

Explanation:

Ref to ACI 318-14, page 316 Section 19.2.3.1,

Modulus of rupture

$$f_r = 7.5 \cdot \lambda \cdot \sqrt{f_c'}$$

For lightweight concrete from page 316, table 19.2.4.2,

$$\lambda = 0.75$$

$$f_r = 7.5 \cdot 0.75 \cdot \sqrt{4000}$$

$$f_r = 355.75 \text{ psi} = \mathbf{0.35 \text{ ksi}}$$

Correct answer is option A.

265. Problem

Correct Answer: (B) 2.5 in

Explanation:

Ref to ACI 318-14, page 336 Table 20.6.1.3.3,

Tensile strength in bending is given by the modulus of rupture

For members except walls, exposed to weather with bars between number 6 through number 11, specified concrete cover = 1.5 inches

From page 334, R20.6.1.1, specified cover should be measured to the outer edge of stirrups.

Diameter of number 4 bars = $4/8$ in = 0.5 in

Diameter of number 8 bars = $8/8$ in = 1 in

Half the diameter of number 8 bars = 0.5 in

Clear cover = 1.5 in

$c = 0.5 \text{ in} + 0.5 \text{ in} + 1.5 \text{ in} = \mathbf{2.5 \text{ in}}$

Correct answer is option B.

266. Problem

Correct Answer: (C) 60 bags

Explanation:

The weight ratio is 1 cement : 2 Fine aggregate : 3 Coarse aggregate

Calculate volume of each component per bag of cement i.e. per 94 lbf weight.

Cement Volume

$$= 1 * 94 \text{ lbf} / 195 \text{ lbf} / \text{ft}^3$$

$$= 0.482 \text{ ft}^3$$

Fine aggregate volume

$$= 2 * 94 \text{ lbf} / 160 \text{ lbf} / \text{ft}^3$$

$$= 1.175 \text{ ft}^3$$

Coarse aggregate volume

$$= 3 * 94 \text{ lbf} / 164 \text{ lbf} / \text{ft}^3$$

$$= 1.72 \text{ ft}^3$$

Water volume (Ref to NCEES PE handbook page 3 for volume of water per gallon)

$$= 6 \text{ gal} * 0.134 \text{ ft}^3 / \text{gal.}$$

$$= 0.804 \text{ ft}^3$$

Total volume yielded by 1 bag of cement

$$= (0.482 + 1.175 + 1.75 + 0.804) \text{ ft}^3$$

$$= 4.181 \text{ ft}^3$$

Number of bags required for 250 ft^3

$$= 250 \text{ ft}^3 / 4.181 \text{ ft}^3$$

$$= 59.79$$

Hence **60 bags** are required and Option C is the correct answer

267. Problem

Correct Answer: (C) 4

Explanation:

Effective depth, d = total depth - effective cover

$$d = 18 \text{ in} - 2 \text{ in} = 16 \text{ in}$$

$$B = 12 \text{ in}$$

Ref to ACI 318-14, page 137 section 9.6.1.2,

Minimum area of rebar should be greater of

$$\text{A) } (3 \sqrt{f'_c} / f_y) * b * d$$

$$= (3 \sqrt{3500} / 60000) * 12 \text{ in} * 16 \text{ in}$$

$$= 0.57 \text{ in}^2$$

$$\text{B) } 200 / f_y * b * d$$

$$= 200 / 60000 * 12 \text{ in} * 16 \text{ in}$$

$$= 0.64 \text{ in}^2$$

Thus minimum area of steel = 0.64 in^2

From NCEES PE handbook, page 260 section 4.3.1

Area of #4 bars = 0.2 in^2

No of bars required = $0.64 / 0.2 = 3.2$,

Hence **4 bars** is required and the correct answer is Option C.

268. Problem

Correct Answer: (D) 0.9

Explanation:

From NCEES PE handbook page 260, area of #6 bars = 0.44 in²

$$A_s = 5 \times 0.44 \text{ in}^2 = 2.2 \text{ in}^2$$

Calculate depth of stress block 'a' from equation in page 264 section 4.3.2.2

$$a = A_s \cdot f_y / (0.85 \cdot f'_c \cdot b)$$

$$a = 2.2 \cdot 60000 / (0.85 \cdot 4000 \cdot 12)$$

$$a = 3.24 \text{ in}$$

Ref to NCEES PE handbook, page 263, section 4.3.2.1

Ratio of depth of stress block "a" to neutral axis depth "c"

$$\beta_1 = 0.85 \text{ for } 4000 \text{ psi concrete}$$

$$a/c = 0.85$$

$$c = a / 0.85 = 3.81 \text{ in}$$

$$\text{Strain in concrete} = 0.003$$

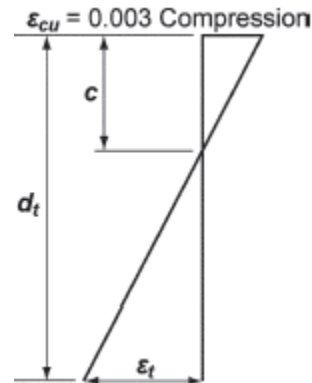
$$\text{Strain in rebar} = 0.003 \cdot (d - c) / c$$

$$= 0.003 \cdot (20 - 3.81) / 3.81$$

$$= 0.0127$$

From ACI 318-14, page 343, table 21.2.2, section is tension controlled and $\Phi = 0.9$

Correct answer is option D.



269. Problem

Correct Answer: (C) 4494 in⁴

Explanation:

From NCEES PE handbook page 260, area of #6 bars = 0.44 in²

$$A_s = 4 \times 0.44 \text{ in}^2 = 1.76 \text{ in}^2$$

$$\text{Modular ratio, } n = 29000 / 3600 = 8.06$$

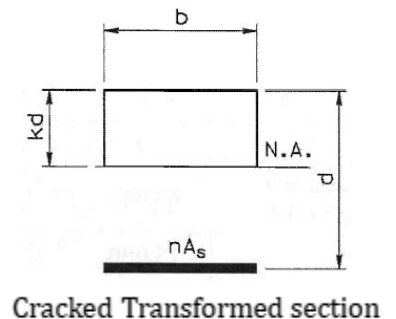
Calculate depth of N.A. from equation in page 265, section 4.3.2.3

$$p = A_s / bd = 1.76 \text{ in}^2 / (12 \text{ in} \times 22 \text{ in}) = 0.0067$$

$$k = \sqrt{[(p \cdot n)^2 + 2p \cdot n]} - p \cdot n$$

$$k = \sqrt{[(0.0067 \cdot 8.06)^2 + 2 \cdot 0.0067 \cdot 8.06]} - 0.0067 \cdot 8.06$$

$$k = 0.279$$



Depth of neutral axis

$$c = kd$$

$$c = 0.279 * 22 \text{ in}$$

$$c = 6.138 \text{ in}$$

Moment of inertia of the cracked transformed section about the N.A. is given by

Sum of transformed rebar area $n * A_s$ and depth of concrete block $c * b$ about the base

MOI of rectangular block about the base

$$= 1/3 * b * c^3$$

$$= 1/3 * 12 \text{ in} * (6.138 \text{ in})^3$$

$$= 924.99 \text{ in}^4$$

MOI of transformed rebar area about NA

$$= n * A_s * (d - c)^2$$

$$= 8.06 * 1.76 \text{ in}^2 * (22 \text{ in} - 6.138 \text{ in})^2$$

$$= 3569.14 \text{ in}^4$$

$$\text{Total MOI} = 924.99 \text{ in}^4 + 3569 \text{ in}^4 = 4494.13 \text{ in}^4$$

Hence the correct answer is Option C **4494 in⁴**.

270. Problem

Correct Answer: (D) 47

Explanation:

Ref to ACI 318-14 page 72, Table 6.6.3.1.1 (a)

Moment of inertia for Columns = 0.7 I_g

Moment of inertia of beams = 0.35 I_g

For beams,

$$I_{\text{beam}} = 0.35 * 1/12 * 12 \text{ in} * (24 \text{ in})^3$$

$$I_{\text{beam}} = 0.35 * 13824 \text{ in}^4 = 4838.4 \text{ in}^4$$

For columns,

$$I_{\text{column}} = 0.7 * 1/12 * 14 \text{ in} * (14 \text{ in})^3$$

$$I_{\text{column}} = 0.7 * 3201.33 \text{ in}^4 = 2240.9 \text{ in}^4$$

For the same grade of concrete, the modulus of elasticity is the same.

From ACI 318-14 page 65, Fig R6.2.5,

For top joint

$$\Psi_a = (I_{\text{column}}/15 \text{ ft}) / (2 * I_{\text{beam}}/12.5 \text{ ft})$$

$$\Psi_a = [2240.9/(15*12)] / [2*4838.4/(12.5*12t)]$$

$$\Psi_a = 0.193$$

For bottom joint

$$\Psi_b = (2*I_{column}/15ft) / (2*I_{beam}/12.5ft)$$

$$\Psi_b = [2*2240.9/(15*12)] / [2*4838.4/(12.5*12t)]$$

$$\Psi_b = 0.386$$

For sway frames, from ACI 318-14 page 65, Fig R6.2.5, $k = 1.1$

Calculate radius of gyration for rectangular columns from page 64, section 6.2.5.1 (b)

$$r = 0.3*14 \text{ in} = 4.2 \text{ in}$$

$$\text{Slenderness} = kL/r$$

$$= 1.1*15*12\text{in}/4.2\text{in} = 47.14$$

Correct answer is Option D, **47**.

271. Problem

Correct Answer: (B) 18in

Explanation:

Diameter of #8 bar = 8/8 inch = 1 inch

Ref to ACI 318-14 page 427, section 25.4.9.2

Taking $\Psi_r = 1$

For normal weight concrete $\lambda = 1$

L_{dc} is the greater of

$$\begin{aligned} \text{a) } & f_y \Psi_r / (50 \lambda \sqrt{f'_c}) * d_b \\ & = 60000 * 1 / (50 * 1 * \sqrt{6000}) * 1 \text{ in} = 15.5 \text{ in} \end{aligned}$$

$$\begin{aligned} \text{b) } & 0.0003 * f_y \Psi_r * d_b \\ & = 0.0003 * 60000 * 1 \text{ in} = \mathbf{18 \text{ in}} \end{aligned}$$

Correct answer is option B.

272. Problem

Correct Answer: (B) 2925 psi

Explanation:

28 day design strength of concrete = 6500 psi

Ref to ACI 318-14 page 409, Table 24.5.4.1

For sustained loading, allowable compressive stress

$$= 0.45 * f'_c$$

$$= 0.45 * 6500 \text{ psi}$$

$$= \mathbf{2925 \text{ psi}}$$

Correct answer is option B.

273. Problem

Correct Answer: (C) 1.75 in

Explanation:

Ref to ACI 318 -14, page 401 section 24.2.4.1.1

Calculate p' that is the ratio of compression rebar to $b*d$

$$b = 15 \text{ in}$$

$$d = 24 \text{ in} - 2 \text{ in} = 22 \text{ in}$$

Area of compression rebar

From NCEES PE handbook page 260, area of #5 bar = 0.31 in^2

$$A_s = 3 * 0.31 \text{ in}^2 = 0.93 \text{ in}^2$$

$$p' = 0.93 \text{ in}^2 / (15 \text{ in} * 22 \text{ in})$$

$$p' = 0.0028$$

From ACI 318-14 page 402, table 24.2.4.1.3

$$\xi = 2.0 \text{ for more than 60 months}$$

From page 401, equation 24.2.4.1.1

$$\lambda = \xi / (1 + 50 * p')$$

$$\lambda = 2 / (1 + 50 * 0.0028)$$

$$\lambda = 1.754$$

Long term deflection = λ * short term deflection

$$= 1.754 * 1 \text{ in} = 1.754 \text{ in}$$

Hence the correct answer is Option C, **1.75 in**.

274. Problem

Correct Answer: (A) 5.1

Explanation:

Ref to ACI 318-14 page 327, section 20.2.2.2.

Modulus of elasticity of non prestressed bars and wires

$$E_s = 29000 \text{ ksi}$$

Ref to ACI 318-14 page 315, section 19.2.2.1,

$$E_c = w^{1.5} * 33 * \sqrt{f'_c}$$

$$E_c = 155^{1.5} * 33 * \sqrt{8000} \text{ psi}$$

$$E_c = (155^{1.5} * 33 * \sqrt{8000}) / 1000 \text{ ksi}$$

$$E_c = 5695.8 \text{ ksi}$$

$$\text{Modular ratio} = E_s / E_c$$

$$n = 29000 / 5695.8 = 5.1$$

Correct answer is option A.

275. Problem

Correct Answer: (B) 4.46 ksi

Explanation:

Ref to PCI Handbook 7th edition, page 5-84, section 5.7.3 for loss due to elastic shortening

Ref to ACI 318-14 page 315, section 19.2.2.1, for modulus of elasticity of concrete

$$E_c = 57000 * \sqrt{f'_c}$$

$$E_{ci} = 57000 * \sqrt{5000}$$

$$E_{ci} = 4030.5 \text{ ksi}$$

From PCI handbook page 5-84, equation 5-65

For concentric strands, $e = 0$

$$f_{cir} = K_{cir} * (P_i / A_g)$$

$$f_{cir} = 0.9 * (350 \text{ kips} / 500 \text{ in}^2) = 0.63 \text{ ksi}$$

$E_{ps} = 28500 \text{ ksi}$ from page 5-84

From equation 5-64

$$ES = K_{es} * E_{ps} * f_{cir} / E_{ci}$$

$$ES = 1 * 28500 \text{ ksi} * 0.63 \text{ ksi} / 4030.5 \text{ ksi} = 4.46 \text{ ksi}$$

Correct answer is option B.

4.3 Steel

[Click here for Problems 4.3](#)

276. Problem

Correct Answer: (B)

Explanation:

A is the proportionality limit for which the linear relationship is valid.

B is the elastic limit and C is the yield point. For steel B and C are very close to each other.

D is the ultimate strength and E is the fracture point.

Hence Option B is the correct answer.

277. Problem

Correct Answer: (B)

Explanation:

All options are true except option B. Strain hardening increases the hardness and reduces the ductility, however, strain hardening increases the yield strength of the material. Hence option B is false and the answer.

278. Problem

Correct Answer: (D)

Explanation:

Option A is false because increase in temperature decreases the yield and ultimate strength of steel.

Option B is false because ductility of steel reduces with reducing temperature and the steel becomes more brittle.

Option C is false because the modulus of elasticity of steel increases with decreasing temperature.

Option D is true because decreasing temperatures result in increasing Modulus of Elasticity and consequently increase the stiffness of the members. Hence Option D is the answer.

279. Problem

Correct Answer: (A) 56 ksi and 33 ksi

Explanation:

Ref to AISC steel manual 15th edition page 16.1-129, AISC 360-16 Table J3.2

Nominal tensile strength = $0.75 F_u$

= $0.75 \times 150 \text{ ksi} = 112.5 \text{ ksi}$

Nominal shear strength = $0.45 F_u$

= $0.45 \times 150 \text{ ksi} = 67.5 \text{ ksi}$

From page 16.1-131, section J3.6 Strength reduction factor for allowable stress design = 2

Allowable tensile strength = $112.5 \text{ ksi} / 2 = 56.25 \text{ ksi}$

Allowable shear strength = $67.5 \text{ ksi} / 2 = 33.75 \text{ ksi}$

Thus correct answer is option A, **56 ksi and 33 ksi**

280. Problem

Correct Answer: (B) 12 ksi

Explanation:

Ref to AISC steel manual 15th edition page 16.1-129, AISC 360-16 Table J3.2

For A307 bolts,

Nominal tensile strength, $F_{nt} = 45 \text{ ksi}$

Nominal shear strength, $F_{nv} = 27 \text{ ksi}$

From page 16.1-133, section J3.7 equation J3-3b,

Modified nominal tensile strength to include effects of shear

= $1.3 F_{nt} - \Omega F_{nt} / F_{nv} \phi_{rv}$

= $1.3 \times 45 \text{ ksi} - 2 \times 45 \text{ ksi} / 27 \text{ ksi} \times 10 \text{ ksi}$

= 25.167 ksi

Allowable tensile stress

= $25.167 \text{ ksi} / \Omega$

= $25.167 \text{ ksi} / 2$

= 12.58 ksi

Correct answer is option B, **12 ksi.**

281. Problem

Correct Answer: (B) 8.35 kip/in

Explanation:

Ref to AISC steel manual 15th edition page 8-8 for available strength of fillet welds,

For 3/8 in weld,

Size of weld in 16th of an inch

Size = $(3 \times 2) / (8 \times 2)$ in = 6/16 in, Thus D = 6

F_{exx} = 70 ksi

From equation 8-1,

Nominal strength per unit length

$R_n = 0.6 \times F_{exx} \times (1/\sqrt{2}) \times (D/16)$

$R_n = 0.6 \times 70 \times (1/\sqrt{2}) \times (6/16)$

$R_n = 11.137$ kip/in

Allowable Strength = $R_n / \Omega = 11.137 / 2 = 5.57$ kip/in

From page 8-9, effect of loading angle on strength, when angle = 90 degrees for transversely loaded welds, increase in strength of 50% is applicable

Thus, Allowable strength = $5.57 \text{ kip/in} \times 1.5 = \mathbf{8.35 \text{ kip/in}}$

Correct answer is option B.

282. Problem

Correct Answer: (D) III only

Explanation:

Per AISC Steel manual 15th edition, page 16.1-7, AISC 306-16 section A3, 1c and 1d,

Rolled heavy shapes and built up members with flanges and plate thickness exceeding 2 inches need Charpy V notch impact tests per ASTM A673. Hence the correct answer is option D, III only, which has a flange of 3 in.

283. Problem

Correct Answer: (B) I - A572 II- A500 III - A36

Explanation:

Per AISC Steel manual 15th edition, page 2-48, Table 2-4

For L sections (III) , Preferred specification is A36

For rectangular HSS section (II). Preferred specification is A500 Gr C

Section I is plate girder with plate thickness 1.5 inch and 2 inch

From page 2-50, Table 2-5 , preferred specification for plate is A572 or A36

Thus the correct answer is option B.

284. Problem

Correct Answer: (A) 32.5 ksi

Explanation:

Per AISC Steel manual 15th edition, AISC 360-16, page 16.1-338

Elastic buckling stress is given by equation C-F9-1

$$b/t = 36 \text{ in} / 1 \text{ in} = 36$$

$$E = 29000 \text{ ksi}$$

$$\text{Poisson's ratio , } \nu = 0.3$$

From page 16.1-339, plate buckling coefficient $k = 1.61$

$$F_{cr} = \pi^2 * E * k / [12 * (1 - \nu^2) * (b/t)^2]$$

$$F_{cr} = \pi^2 * 29000 \text{ ksi} * 1.61 / [12 * (1 - 0.3^2) * (36)^2]$$

$$F_{cr} = 32.5 \text{ ksi} \quad \text{Correct answer is option A.}$$

285. Problem

Correct Answer: (C) 1.83

Explanation:

Per AISC Steel manual 15th edition, AISC 360-16, page 16.1-46 section F1(c)

$$C_b = 12.5 * M_{max} / (2.5 M_{max} + 3M_a + 4 M_b + 3M_c)$$

$$M_{max} = 25 \text{ kip*ft} \quad M_a = 20 \text{ kip*ft} \quad M_b = 0 \quad M_c = 16 \text{ kip*ft}$$

$$C_b = 12.5 * 25 / (2.5 * 25 + 3 * 20 + 4 * 0 + 3 * 16)$$

$$C_b = 1.83$$

Correct answer is option C.

286. Problem

Correct Answer: (C) 9750 kip*in

Explanation:

Per AISC Steel manual 15th edition, AISC 360-16, page 16.1-67 section F13 for reduction in moment strength due to holes

$$F_u = 65 \text{ ksi}$$

$$F_y = 50 \text{ ksi}$$

Gross area of flange,

$$A_{fg} = 7 \text{ in} \times 0.8 \text{ in} = 5.6 \text{ in}^2$$

Net area of flange for bolt holes on both sides of the web,

$$A_{fn} = 5.6 \text{ in}^2 - 2 \times 0.875 \text{ in} \times 0.8 \text{ in}$$

$$A_{fn} = 4.2 \text{ in}^2$$

$$F_y/F_u = 50/65 = 0.77 < 0.8.$$

$$\text{Hence } Y_t = 1$$

$$F_u \times A_{fn} = 65 \text{ ksi} \times 4.2 \text{ in}^2$$

$$F_u \times A_{fn} = 273 \text{ kip}$$

$$Y_t \times F_y \times A_{fg} = 1 \times 50 \text{ ksi} \times 5.6 \text{ in}^2$$

$$Y_t \times F_y \times A_{fg} = 280 \text{ kip}$$

$F_u \times A_{fn} < Y_t \times F_y \times A_{fg}$, Hence limit state of tensile rupture applies

From equation F13-1

$$M_n = F_u \times (A_{fn}/A_{fg}) \times S_x$$

$$= 65 \text{ ksi} \times (4.2 \text{ in}^2 / 5.6 \text{ in}^2) \times 200 \text{ in}^3$$

$$= \mathbf{9750 \text{ kip*in}}$$

Correct answer is Option C.

287. Problem

Correct Answer: (D)

Explanation:

All options are true except Option D.

Dimensional tolerances for cold formed steel are tighter than that for hot rolled sections.

This is because cold working allows for more precise control over the dimensions. Hence Option D is false and the answer.

288. Problem

Correct Answer: (B) 5.34

Explanation:

Ref to AISC Steel manual 15th edition, AISC 360-16, page 16.1-71 section G2.1(2) for web plate shear buckling coefficient k_v

Clear distance between the flanges

$$h = 15 \text{ in} - 2 \text{ in} - 2 \text{ in}$$

$$h = 11 \text{ in}$$

$$a = 40 \text{ in}$$

$$a/h = 40/11$$

$$a/h = 3.63 > 3$$

Thus, $k_v = 5.34$

Correct answer is option B.

289. Problem

Correct Answer: (A) 11.6 ksi

Explanation:

Total depth of section = 15 in + 1 in (plate)

$$H = 16 \text{ in}$$

Depth of extreme fiber of cover plate from the centroid

$$y = H - 9.8 \text{ in}$$

$$y = 6.2 \text{ in}$$

Section modulus = I/y

$$= 2238 \text{ in}^4 / 6.2 \text{ in}$$

$$= 360.97 \text{ in}^3$$

Flexural stress in cover plate = Moment / section modulus

$$= 350 \text{ kip}\cdot\text{ft} \cdot (12 \text{ in/ft}) / 360.97 \text{ in}^3$$

$$= 11.6 \text{ ksi}$$

Correct answer is Option A.

290. Problem

Correct Answer: (C) 40 kips

Explanation:

Ref to AISC steel manual page 16.1-134, Section J3.8 for slip critical connections.

From page 16.1-135, for class A surface, $\mu = 0.3$

Pretension, $T_b = 102$ kips

Filler factor, $h_f = 1.0$

Here, factor $D_u = 1$ because T_b during installation is directly given.

Number of slip planes, $n_s = 2$ for double shear.

From equation J3-4

$$R_n = \mu * D_u * h_f * T_b * n_s$$

$$R_n = 0.3 * 1 * 1 * 102 \text{ kips} * 2 = 61.2 \text{ kips}$$

$\Omega = 1.5$ for standard size holes

$$\text{Allowable shear} = R_n / \Omega$$

$$= 61.2 \text{ kips} / 1.5 = 40.8 \text{ kips}$$

Correct answer is option C, **40 kips**.

4.4 Timber

[Click here for Problems 4.4](#)

291. Problem

Correct Answer: (C)

Explanation:

Knots are structural defects in wood. While it has minor impacts on density, stiffness and compressive strength, the tensile strength perpendicular to the grains is significantly weakened and risks of splitting increases. Hence the correct answer is Option C.

292. Problem

Correct Answer: (D)

Explanation:

All options are false except Option D. Wood with higher density has a higher percentage of cell wall material which contributes more to overall shrinkage and swelling. Hence Option D is correct and the answer.

293. Problem

Correct Answer: (C) 600 psi

Explanation:

Ref to NDS 2018 Supplement page 33 Table 4A

For Aspen No. 1, $F_c = 600$ psi

(All adjustment values below are for F_c)

From page 32, Size factor for 10 in width and No 1, $C_f = 1$

From page 32, Wet service factor, $C_m = 0.8$

Also, check $F_c * C_f = 600 \text{ psi} * 1 < 750 \text{ psi}$

Thus, $C_m = 1$

From NDS 2018 page 11, Table 2.3.2, For live load, $C_d = 1$

From NDS 2018 page 11, Table 2.3.3, For 90°F, $C_t = 1$

From NDS 2018 page 29, Table 4.3.1,

$F'_c = F_c * C_d * C_m * C_t * C_f * C_i * C_p$

$F'_c = 600 \text{ psi} * 1 * 1 * 1 * 1 * 1 * 1 = 600 \text{ psi}$ Correct answer is option C

294. Problem

Correct Answer: (D) 1258 psi

Explanation:

Moisture Less than or equal to 19% is dry condition for lumber.

Ref to NDS 2018 Supplement page 33 Table 4A

For Alaska Yellow cedar No. 2, $F_b = 800$ psi (All adjustment values below are for F_b)

All adjustment values below are for F_b

From NDS 2018 page 11, Table 2.3.2, For wind load, $C_d = 1.6$

From NDS 2018 page 11, Table 2.3.3, For 110°F Dry, $C_t = 0.8$

From NDS supplement page 32, $C_m = 1$ for moisture less than 19% (dry)

From page 32, Size factor for 8 in width and No. 2 with thickness 4 in , $C_f = 1.3$

From page 32, Flat use factor for 8 in width and thickness 4 in , $C_{fu} = 1.05$

In absence of specific usage and spacing of beam, Repetitive member factor $C_r = 1$

From NDS 2018 page 29, Table 4.3.1,

$$F'_b = F_b * C_d * C_m * C_t * C_L * C_f * C_{fu} * C_i * C_r$$

$$F'_b = 800 \text{ psi} * 1.6 * 1 * 0.8 * 0.9 * 1.3 * 1.05 * 1 * 1 = 1257.98 \text{ psi}$$

Correct answer is option D, **1258 psi**.

295. Problem

Correct Answer: (B) 675 psi

Explanation:

Moisture exceeding 19% is wet condition for lumber.

Ref to NDS 2018 Supplement page 40 Table 4B

For Southern pine No. 2, 6 in wide, $F_t = 600$ psi (All adjustment values below are for F_t)

From NDS 2018 page 11, Table 2.3.2, For construction load, $C_d = 1.25$

From NDS 2018 page 11, Table 2.3.3, For 95°F, $C_t = 1$

From NDS supplement page 39, $C_m = 1$ (For F_t)

From NDS supplement page 39 for 6 in wide lumber, no size factor adjustment required,
 $C_f = 1$

From NDS 2018 page 29, Table 4.3.1,

$$F'_t = F_t * C_d * C_m * C_t * C_f * C_i$$

$$F'_t = 600 \text{ psi} * 1.25 * 1 * 1 * 1 * 0.9 = 675 \text{ psi}$$

Correct answer is option B.

296. Problem

Correct Answer: (A) 794 ksi

Explanation:

Ref to NDS 2018 Supplement page 67 Table 5B

For 5 DF L1,

$E_{min} = 1060000 \text{ psi}$ (All adjustment values below are for E min

From NDS 2018 page 11, Table 2.3.3, For 120°F, $C_t = 0.9$

From NDS supplement, page 66, $C_m = 0.833$

From NDS 2018 page 37 Table 5.3.1,

$$E'_{min} = E_{min} * C_m * C_t$$

$$E'_{min} = 1060000 \text{ psi} * 0.833 * 0.9$$

$$E'_{min} = 794682 \text{ psi}$$

Hence the correct answer is option A, **794 ksi**.

297. Problem

Correct Answer: (B) 117 lbs

Explanation:

Ref to NDS 2018 page 111 Table 12N

For DFR with side member thickness 1.5 in and 16d common nails,

Nail diameter, $D = 0.162 \text{ in}$

Reference $Z = 141 \text{ lbs}$

Calculate penetration of nail

$$L/3 = 3.5 \text{ in} / 3$$

$$L/3 = 1.17 \text{ in}$$

$$\text{Penetration} = L * \cos 30^\circ - L/3$$

$$= 3.5 * \cos 30^\circ - 1.17 \text{ in}$$

$$= 1.86 \text{ in}$$

Check $10 * D = 10 * 0.162 = 1.62 \text{ in}$. Hence penetration is greater than $10D$,

No adjustment required for penetration

From NDS 2018 page 67, for dry conditions and 100°F, $C_m = C_t = 1$

From page 11, Table 2.3.2, for live loads, $C_d = 1$

From page 91, section 12.5.4, $C_{tn} = 0.83$

From page 89, section 12.5.1, $C_\Delta = 1$ for $D < 0.25 \text{ in}$

For this connection geometry, C_g and C_{di} are both equal to 1

From page 66, Table 11.3.1,

$$Z' = Z * C_d * C_m * C_t * C_g * C_{\Delta} * C_{eg} * C_{di} * C_{tn}$$

$$Z' = 141 \text{ lbs} * 1 * 1 * 1 * 1 * 1 * 1 * 0.83$$

$$Z' = 117.03 \text{ lbs}$$

Correct answer is Option B, **117 lbs**.

298. Problem

Correct Answer: (A) **31 in⁴**

Explanation:

Ref to NDS 2018 Supplement page 34 Table 4A.

For select structural DFL, $E = 1900000 \text{ psi}$

For dry conditions, $C_m = 1$

From NDS 2018 page 11, Table 2.3.3, for $<100^\circ\text{F}$, $C_t = 1$

From NDS 2018 page 29, Table 4.3.1

$$E' = E * C_m * C_t * C_i$$

$$E' = 1900000 \text{ psi} * 1 * 1$$

$$E' = 1900000 \text{ psi}$$

Per IBC 2018 page 363 Table 1604.3, for live loads in floor members,

$$\Delta = L/360$$

$$\Delta = (12\text{ft} * 12 \text{ in/ft}) / 360 = 0.4 \text{ in}$$

$$w = 50 \text{ plf}$$

$$w = 50/12 \text{ lbs/in} = 4.167 \text{ lbs/in}$$

From NCEES handbook page 236 Case I,

$$\Delta = 5 * w * L^4 / (384 EI)$$

$$I = 5 * w * L^4 / (384 E \Delta)$$

$$I = [5 * 4.167 \text{ lb/in} * (144\text{in})^4] / (384 * 1900000 \text{ psi} * 0.4 \text{ in})$$

$$I = 30.7 \text{ in}^4$$

Correct answer is Option A, **31 in⁴**.

4.5 Masonry

[Click here for Problems 4.5](#)

299. Problem

Correct Answer: (D) 65 psi

Explanation:

Ref to TMS 402/602 -16 page C-110 Table 8.2.4.2

Allowable flexural tensile stresses for clay and concrete masonry

For hollow, fully grouted unit and Type M mortar,

Allowable stress = **65 psi**

Correct answer is Option D.

300. Problem

Correct Answer: (C) 195 psi

Explanation:

Ref to TMS 402/602-16, page C-45 Section 5.1.3.1,

Vertical loads are dispersed at 2V:1H on both sides of the bearing plate.

At mid height i.e. 3ft, the load dispersion on each side of the bearing plate

$= 3\text{ft}/2 = 1.5\text{ ft} = 18\text{ in}$

Thus total effective length of dispersion

$= 1.5\text{ ft} + 5\text{in bearing plate} + 1.5\text{ ft}$

$= 41\text{ in}$

Since mortar is not provided at the webs, thickness of masonry resisting the loads are only the two faces of the CMU unit and not webs.

Effective thickness $= 2 \times 1.25\text{ in} = 2.5\text{ in}$

Axial stress due to load $= P/A$

$= 20\text{ kips} / (41\text{ in} \times 2.5\text{ in})$

$= 0.195\text{ ksi}$

$= 195\text{ psi}$

Correct answer is option C.

301. Problem

Correct Answer: (B)

Explanation:

Option A is correct per TMS 402/602-16, page S-30 to S-33, section 1.8C and 1.8D.

Options C and D are correct per TMS 402/602-16, page S-36, Table SC-1 and SC-2,

If the mortar is stronger than the individual masonry units, the masonry units will crack before the mortar which is not the desired mode of failure of the overall masonry system.

Hence it is always advisable for the mortar to be weaker than the units. Option B is false and the answer.

302. Problem

Correct Answer: (B) **10 kips**

Explanation:

Shear depth, $d_v = 32$ in

Spacing, $s = 10$ in

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #4 bar, $A_v = 0.2$ in²

From TMS 402/602-16, page C-114, section 8.3.3.1,

Allowable stress in grade 60 rebar

$f_s = 32000$ psi

From TMS 402/602-16, page C-118, equation 8-27,

Shear stress carried by rebar

$$f_{vs} = 0.5 * [(A_v * f_s * d_v) / (A_{nv} * s)]$$

Rearranging,

$$f_{vs} * A_{nv} = 0.5 * [(A_v * f_s * d_v) / s]$$

$$\text{Shear force} = 0.5 * (0.2 \text{ in}^2 * 32000 \text{ psi} * 32 \text{ in}) / 10 \text{ in}$$

$$\text{Shear force} = 10240 \text{ lbs} \approx \mathbf{10 \text{ kips}}$$

Correct answer is option B.

303. Problem

Correct Answer: (C) 8 in

Explanation:

Ref to TMS 402/602-16, page C-213 section A 6.2.1,

For one story load bearing walls, minimum thickness shall be 6 in.

Ref to TMS 402/602-16, page C-212 Table A .5.1,

For fully grouted load bearing walls,

Maximum $h/t = 20$

$12 \text{ ft}/t = 20$

$t = 12 * 12 \text{ in} / 20$

$t = 144 \text{ in} / 20 = 7.2 \text{ in}$

Hence, the correct answer is option C, **8 in**.

304. Problem

Correct Answer: (C) 43in

Explanation:

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Diameter of No. 6 bar, $d_b = 0.75 \text{ in}$

Ref to TMS 402/602-16, page C-65 section 6.1.5.1.1 for development length

$K =$ smallest between cover, clear bar spacing between splices and $6*d_b$

$K = 3 \text{ in}$

$\gamma = 1.3$ for No. 6 bar

$f_y = 60 \text{ ksi}$

From equation 6-1

$$L_d = (0.13 * d_b^2 * f_y * \gamma) / (K * \sqrt{f'_m})$$

$$L_d = [0.13 * (0.75 \text{ in})^2 * 60000 * 1.3] / (3 \text{ in} * \sqrt{2000})$$

$$L_d = 42.5 \text{ in}$$

Correct answer is Option C, **43in**.

305. Problem

Correct Answer: (C) 17 ksi

Explanation:

Ref to TMS 402/602-16 page C-32, Table 4.2.2

$$E_m = 900 \cdot f'_m = 900 \cdot 2000 \text{ psi} = 1800000 \text{ psi}$$

$$E_s = 29000000 \text{ psi}$$

$$\text{Modular ratio, } n = E_s / E_m = 29000000 \text{ psi} / 1800000 \text{ psi} = 16.11$$

Ref to NCEES PE handbook page 260, section 4.3.1

$$\text{Area of \#9 bar, } A_s = 1 \text{ in}^2$$

$$d = 24 \text{ in} - 4 \text{ in} = 20 \text{ in}$$

$$b_w = 7.625 \text{ in}$$

$$p = A_s / (b_w \cdot d)$$

$$p = 1 \text{ in}^2 / (7.625 \text{ in} \cdot 20 \text{ in}) = 0.00656$$

Calculate depth of N.A. from equation in NCEES PE handbook page 265, section 4.3.2.3

$$k = \sqrt{[(p \cdot n)^2 + 2p \cdot n]} - p \cdot n$$

$$k = \sqrt{[(0.00656 \cdot 16.11)^2 + 2 \cdot 0.00656 \cdot 16.11]} - 0.00656 \cdot 16.11$$

$$k = 0.366$$

$$j = 1 - k/3$$

$$j = 1 - 0.366/3 = 0.878$$

$$f_s = M / (A_s \cdot j \cdot d)$$

$$f_s = 25 \text{ kip} \cdot \text{ft} \cdot (12 \text{ in/ft}) / (1 \text{ in}^2 \cdot 0.878 \cdot 20 \text{ in})$$

$$f_s = 17.08 \text{ ksi}$$

Hence, the correct answer is Option C, **17 ksi**.

306. Problem

Correct Answer: (D) Four #3

Explanation:

Ref to TMS 402/602-16 page C-56, Section 5.3.1.3,

Vertical reinforcement shall not be less than $0.0025 A_n$ and minimum 4 bars should be provided.

Minimum area required = $0.0025 A_n$

= $0.0025 \times 9.625 \text{ in} \times 9.625 \text{ in}$

= 0.23 in^2

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #3 bars = 0.11 in^2

Area of #4 bars = 0.2 in^2

Although three #3 and two #4, satisfy the total area required, a minimum of four bars is needed.

Hence correct answer is Option D, **Four #3**.

307. Problem

Correct Answer: (C) 2.75 hour

Explanation:

Ref to IBC 2018 page 189, Table 722.3.2

For concrete masonry constructed with calcareous gravel and equivalent thickness 5 inches,

Fire resistance rating = **2.75 hour**

Correct answer is Option C.

4.6 Material test methods and specification conformance

[Click here for Problems 4.6.](#)

308. Problem

Correct Answer: (C) 29.5 in

Explanation:

Per ACI 318-14, page 469, Table 26.6.2.1 (a)

For beams with 'd' > 8, the tolerance on 'd' is ± 0.5 in

Hence minimum tolerable dimension is 30 in - 0.5 in = **29.5 in**

Correct answer is Option C.

309. Problem

Correct Answer: (A)

Explanation:

Samples of the concrete are made at the point where the mix is delivered i.e. the point of placement. Hence, Option B is correct

Per ACI 318-14, page 480, Section 23.12.3.1 (a), Options C and Option D are correct.

Per ACI 318-14, page 478 section 26.12.1.1 (a), an average of three 4X8 cylinders are required. Hence Option A is not true and is the answer.

310. Problem

Correct Answer: (D)

Explanation:

Marking on rebar S denotes conformance to A615 carbon steel specs and W denotes conformance to A706 low alloy steel specs. Hence both Options A and B are not true. A706 rebars have lesser carbon content than A615 carbon steel. A706 has higher ductility and weldability. Option D is the correct answer.

311. Problem

Correct Answer: (A) Toughness test

Explanation:

The given procedure records energy required to cause failure of the specimen due to impact from the pendulum. This is a measure of energy absorbed due to the impact and is the procedure followed in standard Charpy V- notch testing. Hence, the correct answer is Option A, **Toughness test**.

312. Problem

Correct Answer: (D) 11

Explanation:

Hammer efficiency measured

$$= \text{Energy transferred} / \text{Potential energy} * 100\%$$

$$= (375 \text{ J} / 467 \text{ J}) * 100\%$$

$$= 80.3\%$$

Ref to NCEES PE handbook page 124, section 3.8.1.1.

$$N_{60} = (\text{Efficiency} / 60) * N$$

$$N_{60} = (80.3 / 60) * 8$$

$$N_{60} = 10.7.$$

Hence, the correct answer is Option D, **11**.

313. Problem

Correct Answer: (C) I, II and III

Explanation:

The number of blows from an SPT test can be correlated to different soil properties like shear modulus, unconfined compressive strength, angle of internal friction, relative density and consistency, etc. Hence the correct answer is Option C.

314. Problem

Correct Answer: (B) 1750 psi

Explanation:

Ref to Unit strength method from TMS 402/602, Masonry specification page S-18, Table-2.

If individual masonry unit of 2000 psi strength is used with Type N mortar, the specified strength of masonry assembly is given as

Correct answer is Option B, **1750 psi**.

315. Problem

Correct Answer: (A) I and II

Explanation:

Per AISC steel manual 15th edition, RCSC specification page 16.2-8, ASTM specifications do not permit the galvanization of A490 bolts. Hence, coating test is not required. Both tensile strength test and proof load test is required. Hence, the correct answer is Option A, **I and II**.

5. Design and Details of Structures–Component Design and Detailing

5.1 Horizontal members

[Click here for Problems 5.1](#)

316. Problem

Correct Answer: (C) **W18X106**

Explanation:

Per IBC 2018 page 363, Table 1604.3, For floor beams subjected to live loads, the deflection limit is $L/360$

$$\Delta = 25 \text{ ft} \times 12 \text{ (in/ft)} / 360$$

$$\Delta = 0.833 \text{ in}$$

From AISC steel manual 15th Ed, page 3-210, Table 3-23 case 7,

$$\Delta = P \cdot L^3 / (48 \cdot E \cdot I)$$

$$I = P \cdot L^3 / (48 \cdot E \cdot \Delta)$$

$$I = 80 \text{ kip} \cdot (25 \cdot 12 \text{ in})^3 / (48 \cdot 29000 \text{ ksi} \cdot 0.833 \text{ in})$$

$$I = 1862.8 \text{ in}^4$$

From Steel manual page 3-29, Table 3-3

The lightest section W18 that satisfies the requirement is

W18X106 with $I_x = 1910 \text{ in}^4$.

Hence Option C is the answer

317. Problem

Correct Answer: (B) **W18X50**

Explanation:

Here max unbraced length due to bracing at mid point = $25 \text{ ft} / 2$

$$L_b = 12.5 \text{ ft}$$

Max moment due to uniformly distributed = $w \cdot L^2 / 8$ (Ref. NCEES PE handbook page 236)

$$M = 3 \text{ kip/ft} \cdot (25 \text{ ft})^2 / 8 = 234.37 \text{ kip} \cdot \text{ft}$$

From AISC Steel Manual 15th Ed, page 3-26, Table 3-2, M_p / Ω for W18x40 = 196 kip*ft which is less than required, so it doesn't satisfy the requirement.

From AISC Steel Manual 15th Ed, page 3-18, Table 1, for this case, $C_b = 1.3$

From page 3-25, Table 3-2, for W18X50,

$M_p/\Omega = 252 \text{ kip}\cdot\text{ft}$, $L_p = 5.83 \text{ ft}$, $L_r = 16.9 \text{ ft}$, $BF/\Omega = 8.76 \text{ kips}$

From AISC steel manual page 3-10, when $L_p < L_b < L_r$

From equation 3-4b

$M_n/\Omega = C_b * [M_p/\Omega - BF/\Omega * (L_b - L_p)]$

$= 1.3 * (252 \text{ kip}\cdot\text{ft} - 8.76 \text{ kip} * (12.5 \text{ ft} - 5.83 \text{ ft}))$

$= 251.6 \text{ kip}\cdot\text{ft}$ (Check this should be less than M_p/Ω , OK here)

Hence the lightest W18 section that satisfies flexural strength requirement is **W18X50**.

Correct answer is Option B.

318. Problem

Correct Answer: (B) W18X50

Explanation:

For compact sections, bending about minor axis, lateral torsional buckling and flange local buckling don't apply.

From AISC steel manual, Part ASIC 360-16, page 16.1-56, section 2(a), W18 sections are compact, hence only yielding controls.

Maximum moment by uniformly distributed load $= w * L^2 / 8$

$M = 0.5 \text{ kip/ft} * (25 \text{ ft})^2 / 8 = 39.06 \text{ kip}\cdot\text{ft}$

From steel manual page 3-32, Table 3-4

Lightest W18 section that satisfies moment M_n/Ω is

W18X50 with $M_n/\Omega = 41.4 \text{ kip}\cdot\text{ft}$

Correct answer is Option B

319. Problem

Correct Answer: (C) HSS 20X20X3/4

Explanation:

Per IBC 2018 page 363, Table 1604.3, For floor beams subjected to live loads, the deflection limit is $L/360$ and beam subjected to dead plus live load, the deflection limit is $L/240$

$L = 30 \text{ ft} = 360 \text{ in}$

Case 1, Live only

$w = 5 \text{ kip/ft} = 0.417 \text{ kip/in}$

$\Delta_1 = 360 \text{ in} / 360$

$$\Delta 1 = 1 \text{ in}$$

From AISC steel manual 15th Ed, page 3-208, Table 3-23 case 1,

$$\Delta 1 = 5 * w * L^4 / (384 * E * I)$$

$$I = 5 * w * L^4 / (384 * E * \Delta 1)$$

$$I = 5 * 0.417 \text{ kip/in} * (360 \text{ in})^4 / (384 * 29000 \text{ ksi} * 1 \text{ in})$$

$$I = 3144.76 \text{ in}^4$$

Case 2, Dead + Live

$$w = 5 \text{ kip/ft} + 1 \text{ kip/ft} = 0.5 \text{ kip/in}$$

$$\Delta 2 = 360 \text{ in} / 240$$

$$\Delta 2 = 1.5 \text{ in}$$

$$\Delta 2 = 5 * w * L^4 / (384 * E * I)$$

$$I = 5 * w * L^4 / (384 * E * \Delta 2)$$

$$I = 5 * 0.5 \text{ kip/in} * (360 \text{ in})^4 / (384 * 29000 \text{ ksi} * 1.5 \text{ in})$$

$$I = 2513.8 \text{ in}^4$$

Hence governing I is for case 1 = 3144.76 in⁴

Ref to AISC steel manual page 1-96, Table 1-12

The lightest section that satisfies the requirement is

HSS 20 X 20 X 3/4 with I = 3230 in⁴

Hence, the correct answer is Option C.

320. Problem

Correct Answer: (C) HSS 18.000 X 0.375

Explanation:

Max moment due to uniformly distributed = $w * L^2 / 8$

$$M = 3 \text{ kip/ft} * (25 \text{ ft})^2 / 8 = 234.37 \text{ kip*ft}$$

From AISC steel manual, page 3-141, Table 3-14,

For **HSS 18.000 X 0.375**

Allowable moment = 242 kip*ft

This is the lightest section that will satisfy the requirement.

Hence the correct answer is Option C.

321. Problem

Correct Answer: (A) Compact

Explanation:

Given $F_y = 50 \text{ ksi}$

Modulus of elasticity for steel, $E = 29000 \text{ ksi}$

Ref to AISC Steel manual, Part AISC 360-16, page 16.1-18, Table B4.1b Case 11,

For compact sections, compact limit

$$\lambda_p = 0.38 \sqrt{E/F_y}$$

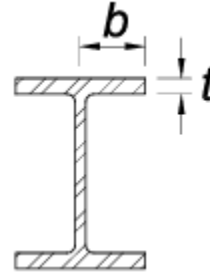
$$= 0.38 \sqrt{(29000 \text{ ksi}/50 \text{ ksi})}$$

$$= 9.15$$

$$b/t$$

$$= (8 \text{ in} / 2) / 0.5 \text{ in}$$

$$= 8 < \lambda_p$$



Hence the flange is **compact** and the correct answer is Option A.

322. Problem

Correct Answer: (D) 5.5 in

Explanation:

Reactions due to uniformly distributed factored load on simply supported end is

$$R_u = w \cdot L / 2$$

$$R_u = 10 \text{ kip/ft} \cdot 25 \text{ ft} / 2 = 125 \text{ kip}$$

$$\Phi R_n = R_u = 125 \text{ kip}$$

From AISC steel manual 15th ed. page 9-48, Table 9-4,

$$\Phi R_1 = 74 \text{ kips}$$

$$\Phi R_2 = 23.5 \text{ kip/in}$$

$$\Phi R_3 = 90.9 \text{ kips}$$

$$\Phi R_4 = 6.25 \text{ kip/in}$$

From AISC Steel manual page 9-23, eq. 9-46a

For web local yielding, since load is throughout the span,

$$\Phi R_n = \Phi R_1 + l_b \cdot \Phi R_2$$

Rearranging,

$$l_b = (\Phi R_n - \Phi R_1) / \Phi R_2$$

$$l_b = (125 \text{ kips} - 74 \text{ kips}) / 23.5 \text{ kip/in}$$

$$l_b = 2.17 \text{ in}$$

From AISC Steel manual page 9-24, eq. 9-48a

For web local crippling, assuming $l_b/d \leq 0.2$

$$\Phi R_n = \Phi R_3 + l_b \cdot \Phi R_4$$

Rearranging,

$$l_b = (\Phi R_n - \Phi R_3) / \Phi R_4$$

$$l_b = (125 \text{ kips} - 90.9 \text{ kips}) / 6.25 \text{ kip/in}$$

$$l_b = 5.45 \text{ in}$$

Hence the second case governs.

Correct answer is Option D, **5.5 in**.

323. Problem

Correct Answer: (D) 150%

Explanation:

Component of force acting along major axis

$$W_x = 1 \text{ kip/ft} \cdot \cos(30^\circ) = 0.866 \text{ kip/ft}$$

$$M_x = W_x \cdot L^2 / 8$$

$$M_x = 0.866 \text{ kip/ft} \cdot (12 \text{ ft})^2 / 8 = 15.6 \text{ kip}\cdot\text{ft}$$

Component of force acting along minor axis

$$W_y = 1 \text{ kip/ft} \cdot \sin(30^\circ) = 0.5 \text{ kip/ft}$$

$$M_y = W_y \cdot L^2 / 8$$

$$M_y = 0.5 \text{ kip/ft} \cdot (12 \text{ ft})^2 / 8 = 9 \text{ kip}\cdot\text{ft}$$

Total Utilization

$$= M_x / M_{xa} + M_y / M_{ya}$$

$$= 15.6 / 30 + 9 / 10 = 1.42 = 142\%$$

Hence the closest answer is Option D, **150%**.

324. Problem

Correct Answer: (B) 910 kip·ft

Explanation:

Ref to AISC steel manual 15th ed., page 1-26 and 1-27, Table 1-1

For W12X58, Overall depth $d = 12.2 \text{ in}$

Section modulus of W12X58, $Z_x = 86.4 \text{ in}^3$

Total depth of built up section including top and bottom 10 in X 1 in thick plates

$$D = d + 1 \text{ in} + 1 \text{ in} = 12.2 \text{ in} + 1 \text{ in} + 1 \text{ in} = 14.2 \text{ in}$$

Distance to centroid of symmetric section from plate centroid (y) = $14.2 \text{ in} / 2 - 1 \text{ in} / 2 = 6.6 \text{ in}$

Total plastic section modulus, $Z_{xx} = Z_x \text{ of rolled section} + 2 * \text{Area of plates} * y$

$$Z_{xx} = 86.4 \text{ in}^3 + 2 * 10 \text{ in} * 1 \text{ in} * 6.6 \text{ in} = 218.4 \text{ in}^3$$

Nominal plastic moment of resistance = $Z_{xx} * F_y$

$$= 218.4 \text{ in}^3 * 50 \text{ ksi} = 10920 \text{ kip} * \text{in} = \mathbf{910 \text{ kip} * \text{ft}}$$

Correct answer is Option B.

325. Problem

Correct Answer: (C) 234 kips

Explanation:

Ref to AISC steel manual 15th ed., page 1-38, Table 1-5

For C15X40

$d = 15 \text{ in}$

$t_w = 0.52 \text{ in}$

$h = T = 12.125 \text{ in}$

$E = 29000 \text{ ksi}$

$F_y = 50 \text{ ksi}$

From AISC steel manual, Page 16.1-71, (b)(2)

$k_v = 5.34$ for webs without stiffeners

$1.1 \sqrt{(k_v \cdot E / F_y)}$

$= 1.1 \sqrt{(5.34 \cdot 29000 / 50)}$

$= 61.2$

$h/t_w = 12.125 \text{ in} / 0.52 \text{ in} = 23.3$

Thus $h/t_w < 1.1 \sqrt{(k_v \cdot E / F_y)}$

From Page 16.1-71, (b)(1) $C_v = 1$

From Page 16.1-70, equation G2-1

$V_n = 0.6 \cdot F_y \cdot A_w \cdot C_v$

$V_n = 0.6 \cdot 50 \text{ ksi} \cdot 15 \text{ in} \cdot 0.52 \text{ in} \cdot 1$

$V_n = 0.6 \cdot 50 \text{ ksi} \cdot 15 \text{ in} \cdot 0.52 \text{ in} \cdot 1$

$V_n = 234 \text{ kips}$. Hence the closest answer is Option C, **234 kips**.

326. Problem

Correct Answer: (C) S8X23

Explanation:

Ref to AISC steel manual 15th ed., page 3-77, Table 3-7

The only section that satisfies the shear requirement is **S8X23**.

With $V_n / \Omega = 50.8 \text{ kips}$,

Options A and B have allowable shear strength less than 46 kips.

Hence the most economical and correct answer is Option C.

327. Problem

Correct Answer: (C) 14.9 kip*ft

Explanation:

Ref to AISC steel manual 15th ed., part AISC 360-16, page 16.1-65, section F11

$$F_y = 50 \text{ ksi}$$

$$E = 29000 \text{ ksi}$$

$$L_b = 3\text{ft} = 36 \text{ in}$$

$$D = 5\text{in}$$

$$t = 1\text{in}$$

Section modulus

$$S = b*d^2/6 = 1 \text{ in} * 5\text{in} * 5\text{in} / 6 = 4.167 \text{ in}^3$$

$$Z = b*d^2/4 = 1 \text{ in} * 5\text{in} * 5\text{in} / 4 = 6.25 \text{ in}^3$$

Elastic and plastic yielding moments

$$M_y = F_y * S = 50 \text{ ksi} * 4.167 \text{ in}^3 = 208.3 \text{ kip*in}$$

$$M_p = F_y * Z = 50 \text{ ksi} * 6.25 \text{ in}^3 = 312.5 \text{ kip*in}$$

Check lateral torsional buckling

$$L_b*d/t^2 = 36 \text{ in} * 5 \text{ in} / (1 \text{ in})^2 = 180$$

$$0.08E/F_y = 0.08*29000 \text{ ksi}/50 \text{ ksi} = 46.4$$

$$1.9*E/F_y = 1.9*29000 \text{ ksi}/50 \text{ ksi} = 1102$$

$$\text{Thus, } 0.08E/F_y < L_b*d/t^2 < 1.9*E/F_y$$

From page 16.1-66, equation F11-2

$$M_n = C_b * [1.52 - 0.274*(L_b*d/t^2) * (F_y/E)] * M_y$$

$$M_n = 1 * [1.52 - 0.274*180*50 \text{ ksi}/29000 \text{ ksi}] * 208.3 \text{ kip*in}$$

$$M_n = 298.9 \text{ kip*in} (< M_p \text{ OK})$$

Allowable flexural strength

$$= M_n/\Omega_b$$

$$= 298.9 \text{ kip*in} / 1.67$$

$$= 178.98 \text{ kip*in}$$

$$= \mathbf{14.9 \text{ kip*ft}}$$

Hence the correct answer is option C.

328. Problem

Correct Answer: (A) 48 kips

Explanation:

$$F_y = 50 \text{ ksi}$$

$$l_b = 1 \text{ in}$$

Ref to AISC steel manual 15th ed., part AISC 360-16, page 16.1-142, equation J10-1

For Flange local bending,

$$R_n = 6.25 * t_f^2 * F_y$$

$$R_n = 6.25 * (0.51 \text{ in})^2 * 50 \text{ ksi}$$

$$R_n = 81.2 \text{ kips}$$

$$R_n / \Omega = 81.2 \text{ kips} / 1.67 = 48.6 \text{ kips}$$

From page 16.1-143, equation J10-2, for loads sufficiently away from supports,

For Web local yielding,

$$R_n = F_y * t_w * (5k + l_b)$$

$$R_n = 50 \text{ ksi} * 0.3 \text{ in} * (5 * 0.81 \text{ in} + 1 \text{ in})$$

$$R_n = 75.8 \text{ kips}$$

$$R_n / \Omega = 75.8 \text{ kips} / 1.5 = 50.5 \text{ kips}$$

Hence the 1st case governs allowable strength and the closest answer is Option A.

329. Problem

Correct Answer: (B) 7.5 ft

Explanation:

$$F_y = 50 \text{ ksi}$$

$$E = 29000 \text{ ksi}$$

Calculate radius of gyration about weaker axis

$$r_y = \sqrt{I_y / A} = \sqrt{311.7 \text{ in}^4 / 68 \text{ in}^2} = 2.141 \text{ in}$$

Ref to AISC steel manual 15th ed., part AISC 360-16, page 16.1-48, equation F2-5

Limiting unbraced length for yielding

$$L_p = 1.76 * r_y * \sqrt{E / F_y}$$

$$L_p = 1.76 * 2.141 \text{ in} * \sqrt{29000 \text{ ksi} / 50 \text{ ksi}}$$

$$L_p = 90.74 \text{ in} \approx \mathbf{7.5 \text{ ft}}$$

Hence the correct answer is Option B.

330. Problem

Correct Answer: (B) 70 kips

Explanation:

$$P_r = 20 \text{ kips}$$

Required Strength Available Strength

$$M_r = 220 \text{ kip}\cdot\text{in} \qquad M_c = 500 \text{ kip}\cdot\text{in}$$

$$V_r = 4 \text{ kips} \qquad V_c = 20 \text{ kips}$$

$$T_r = 45 \text{ kip}\cdot\text{in} \qquad T_c = 200 \text{ kip}\cdot\text{in}$$

Ref to AISC steel manual 15th ed. Part ASIC360-16, page 16.1-83, section H3.2

$$T_r/T_c = 45 \text{ kip}\cdot\text{in}/200 \text{ kip}\cdot\text{in} = 22.5\%$$

Hence the interaction is given by equation H3-6

$$(P_r/P_c + M_r/M_c) + (V_r/V_c + T_r/T_c)^2 \leq 1$$

Check with $P_c = 50 \text{ kips}$

$$(20/50 + 220/500) + (4/20 + 45/200)^2 = 1.02 > 1, \text{ not good}$$

Check with $P_c = 70 \text{ kips}$

$$(20/70 + 220/500) + (4/20 + 45/200)^2 = 0.906 \text{ O.K.}$$

Hence correct answer is **70 kips**, i.e. Option B.

331. Problem

Correct Answer: (D) 36.75 kip\cdot ft

Explanation:

$$\text{Effective depth, } d = 24 \text{ in} - 4 \text{ in} = 20 \text{ in}$$

Ref to TMS 402/602-16 page C-114, section 8.3.3.1

Allowable stress in 60 ksi rebar, $f_s = 32000 \text{ psi}$

From TMS 402/602-16, page C-64 Table CC-6.1.2,

$$\text{Area of \#9 bar, } A_s = 1 \text{ in}^2$$

Max tension force based on max rebar stress = $A_s \cdot f_s$

$$T = 1 \text{ in}^2 \cdot 32000 \text{ psi} = 32000 \text{ lbs}$$

Ref to TMS 402/602-16 page C-115, section 8.3.4.2.2

Allowable stress in masonry due to flexural compression = $0.45f'_m$

$$f_b = 0.45 \cdot 2000 \text{ psi}$$

$$f_b = 900 \text{ psi}$$

Max compression force based on max masonry stress (for triangular distribution of stress)

$$C = 0.5 * f_b * (0.366 * d) * \text{width of section}$$

$$C = 0.5 * 900 \text{ psi} * 0.366 * 20 \text{ in} * 7.625 \text{ in}$$

$$C = 25116.75 \text{ lbs}$$

$C_{max} < T_{max}$, so stress in masonry controls the moment strength.

$$\text{Couple arm } j * d = 0.878 * d = 0.878 * 20 \text{ in} = 17.56 \text{ in}$$

$$\text{Allowable moment} = C * jd$$

$$= 25116.75 \text{ lbs} * 17.56 \text{ in}$$

$$= 441050.13 \text{ lbs} * \text{in}$$

$$= \mathbf{36.75 \text{ kip} * \text{ft}}$$

Correct answer is Option D.

332. Problem

Correct Answer: (B) $b = 7.625 \text{ in}$, $dv = 24 \text{ in}$

Explanation:

Ref to TMS 402/602-16 page C-118, section 8.3.5.1.2 (b)

Max allowable shear stress in masonry for $M/Vdv > 1$

$\gamma_g = 1$ for fully grouted section

$$F_v = 2[\sqrt{f'_m}] * \gamma_g$$

$$F_v = 2[\sqrt{1500}] * 1$$

$$F_v = 77.46 \text{ psi}$$

Shear stress acting on net shear area

$$F_v = V / (b * dv)$$

$$77.46 \text{ psi} = 13000 \text{ lbs} / (7.625 \text{ in} * dv)$$

$$dv = 13000 \text{ lbs} / (77.46 \text{ psi} * 7.625 \text{ in})$$

$$dv = 22.01 \text{ in}$$

Hence **$b = 7.625 \text{ in}$, $dv = 24 \text{ in}$** is adequate and Option B is the answer.

333. Problem

Correct Answer: (D) 27.5 psi

Explanation:

Shear depth, $d_v = 24$ in

Total shear stress $F_v = V / (b * d_v)$

$F_v = 13000 \text{ lbs} / (7.625 \text{ in} * 24 \text{ in}) = 71.04 \text{ psi}$

Calculate F_{vm} , shear stress carried by masonry

From TMS 402/602-16, page C-118, eq 8-26,

$F_{vm} = 0.5 * [\{ 4.0 - 1.75 * (M / V d_v) \} * \sqrt{f'_m}] + 0.25 * P / A_n$ ($P = 0$ here)

$F_{vm} = 0.5 * [\{ 4.0 - 1.75 * 1 \} * \sqrt{1500}] = 43.57 \text{ psi}$

From eq. 8-22, for F_{vs} i.e. shear carried by reinforcement,

$F_v = (F_{vm} + F_{vs}) * \gamma_g$ ($\gamma_g = 1.0$ for fully grouted section)

$F_{vs} = 71.04 \text{ psi} - 43.57 \text{ psi} \approx \mathbf{27.5 \text{ psi}}$

Correct answer is Option D.

334. Problem

Correct Answer: (B) 9in

Explanation:

Shear depth, $d_v = 24$ in

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #4 bar, $A_v = 0.2 \text{ in}^2$

From TMS 402/602-16, page C-114, section 8.3.3.1,

Allowable stress in grade 60 rebar

$F_s = 32000 \text{ psi}$

From TMS 402/602-16, page C-118, equation 8-27,

Shear stress carried by rebar with spacing 's'

$F_{vs} = 0.5 * [(A_v * F_s * d_v) / (A_{nv} * s)]$

$s = 0.5 * [(A_v * F_s * d_v) / (A_{nv} * F_{vs})]$

$s = 0.5 * [(0.2 \text{ in}^2 * 32000 \text{ psi} * 24 \text{ in}) / (7.625 \text{ in} * 24 \text{ in} * 27.5 \text{ psi})] = 15.26 \text{ in}$

From TMS 402/602-16, page C-119, section 8.3.5.2.1, spacing is limited to $d/2$

$d/2 = 20 \text{ in} / 2 = 10 \text{ in}.$

Hence the correct answer is Option B, **9in**.

335. Problem

Correct Answer: (B) 10.5 ft

Explanation:

From TMS 402/602-16, page C-52, section 5.2.1.1.1,

Span of beams not built integrally with support shall be taken as clear span plus depth of beam and need not exceed the c/c distance between supports

Height of beam , $h = 2.5$ ft

Clear span, $l = 8$ ft

Effective span = $l+h$

= $8\text{ft} + 2.5$ ft

= **10.5 ft**

Correct answer is Option B.

336. Problem

Correct Answer: (B) I and II

Explanation:

From TMS 402/602-16, page C-16, definition of deep beam

For simple span, $l_{eff}/d < 2$

For continuous span $l_{eff}/d < 3$

Option I simple span, $l_{eff}/d = 10 \times 12 / 65 = 1.84 < 2$

Option II continuous span , $l_{eff}/d = 15 \times 12 / 65 = 2.76 < 3$

For option III, effective length is not provided so its type cannot be determined.

Correct answer is Option B, **I and II**.

337. Problem

Correct Answer: (C) 8.083 ft

Explanation:

Per AASHTO LRFD bridge design specification 2018, page 9-9, section 9.7.2.3

For empirical design method, the effective length of the slab is the distance between flange tips, plus the flange overhang, taken as the distance from the extreme flange tip to the face of the web.

Flange width, $b = 20\text{in}$

Center to center distance, $L = 9\text{ft}$

Clear distance between flanges

$$= L - 20\text{in}/2 + 20\text{in}/2 = 9\text{ft} - 20\text{in} = 7.333\text{ ft}$$

$$\text{Flange overhang} = 20\text{in}/2 - 2\text{in}/2 = 9\text{in}$$

$$\text{Effective length} = 7.333\text{ ft} + 9\text{in} = \mathbf{8.083\text{ ft}}$$

Correct answer is option C.

338. Problem

Correct Answer: (C) 0.43 ksi (compressive)

Explanation:

$$\text{Area of total 10 strands, } A_s = 10 * 0.153\text{ in}^2 = 1.53\text{in}^2$$

$$\text{Total prestressing force, } P = 160\text{ ksi} * 1.53\text{in}^2 = 244.8\text{ kips}$$

$$\text{Area of section, } A = 40\text{ in} * 24\text{ in}$$

$$A = 960\text{ in}^2$$

$$\text{Section modulus, } S = b*d^2/6$$

$$S = 24\text{ in} * 40\text{ in} * 40\text{ in} / 6 = 6400\text{ in}^3$$

Eccentricity of prestressing strands from center of section

$$e = 40\text{ in}/2 - 4\text{in} - 2\text{ in}/2 = 15\text{ in}$$

$$\text{Bending stress due to prestress} = P*e/S$$

$$= 244.8\text{ kips} * 15\text{in} / 6400\text{ in}^3 = 0.574\text{ ksi (tension at top)}$$

$$\text{Compressive stress} = P/A$$

$$= 244.8\text{ kips} / 960\text{ in}^2 = 0.255\text{ ksi (compression at top)}$$

$$\text{Due to self weight, stress} = 0.75\text{ ksi (compression at top)}$$

$$\text{Total top fiber stress} = 0.75\text{ ksi} + 0.255\text{ ksi} - 0.574\text{ ksi} = \mathbf{0.43\text{ ksi (compressive)}}$$

Correct answer is Option C.

339. Problem

Correct Answer: (B) 1

Explanation:

24F-V1 SP/SP denotes that the timber is of Southern Pine species.

From NDS-2018 -Supplement page 62,

$x = 20$ for southern pine

$L = 15\text{ft}$ for simply supported beam

$b = 4\text{ in}$

$d = 16\text{ in}$

$$C_v = (21/L)^{\frac{1}{x}} (12/d)^{\frac{1}{x}} (5.125/b)^{\frac{1}{x}}$$

$$C_v = (21/15)^{\frac{1}{20}} (12/16)^{\frac{1}{20}} (5.125/4)^{\frac{1}{20}}$$

$C_v = 1.01$, must be less than or equal to 1

Hence Option B, **1** is the correct answer.

340. Problem

Correct Answer: (B) 2X4

Explanation:

From NDS-2018, page 11, Table 2.3.2

For Live, load, $C_d = 1$

For normal temperature and moisture conditions, $C_m = C_t = 1$

From NDS-2018, page 29, Table 4.3.1

$$F'_b = F_b * C_d * C_m * C_t * C_L * C_f * C_{fu} * C_i * C_r$$

From NDS-2018 supplement page 34, Table 4A

For DFL, select structural , $F_b = 1500\text{ psi}$

Choose 2X4.

From NDS -2018 Supplement, page 32, $C_f = 1.5$ for 2X4

$C_{fu} = 1$, because it is not said that beam is loaded on flat face

Since no information on spacing is given, $C_r = 1$

$$F'_b = 1500\text{ psi} * 1 * 1 * 1 * 1 * 1.5 * 1 * 1 * 1 = 2250\text{ psi}$$

$$\text{Allowable moment} = F'_b * (b*d^2/6) = 2250\text{ psi} * (2\text{in} * 4\text{ in} * 4\text{ in} / 6)$$

$$= 12000\text{ lb-in} = 1000\text{ lb-ft} . \text{ Hence it is adequate}$$

Correct answer is Option B.

341. Problem

Correct Answer: (C) 9

Explanation:

Unbraced length, $l_u = 6\text{ft}$

From NDS-2018 supplement page 14, table 1B,

For nominal size 4X8, standard dressed size = 3.5X7.25

$$l_u / d = 6 * 12 \text{ in} / 7.25 \text{ in} = 9.9$$

From NDS-2018, page 16, Table 3.3.3, for uniformly distributed load

When $l_u / d > 7$

Effective length, $l_e = 1.63 * l_u + 3 * d$

$$l_e = 1.63 * 6 * 12 \text{ in} + 3 * 7.25 \text{ in}$$

$$l_e = 139.11 \text{ in}$$

From NDS 2018 page 17, eq 3.3-5

$$R_b = \sqrt{(l_e * d / b^2)}$$

$$R_b = \sqrt{[139.11 \text{ in} * 7.25 \text{ in} / (3.5 \text{ in})^2]}$$

$$R_b = 9.07$$

Correct answer is Option C, **9**.

342. Problem

Correct Answer: (C) 10.8 ksi

Explanation:

From NDS-2018 supplement page 34, Table 4A

For DFL, select structural, $E_{min} = 690000 \text{ psi}$

For normal temperature, from NDS 2018, page 11, Table 2.3.3, $C_t = 1$

For wet condition, from NDS-2018 supplement, page 32, $C_m = 0.9$ for E_{min}

From NDS-2018 page 29, Table 4.3.1

$$E_{min}' = E_{min} * C_m * C_t * C_i * C_T$$

$$E_{min}' = 690000 \text{ psi} * 0.9 * 1 * 1 * 1 = 621000 \text{ psi}$$

From NDS -2018, page 17, section 3.3.3.8

$$F_{bE} = 1.2 * E_{min}' / R_b^2$$

$$F_{bE} = 1.2 * 621000 \text{ psi} / 8.3^2 = 10817.24 \text{ psi} = \mathbf{10.8 \text{ ksi}}$$

Correct answer is Option C.

343. Problem

Correct Answer: (A) 10.2 in³

Explanation:

From NDS-2018, page 11, Table 2.3.2, for snow load, $C_d = 1.15$

For dry condition and normal temperature, $C_m = 1$ and $C_t = 1$

Since it is loaded along the thickness and not along the flat (width) face,

Flat use factor, $C_{fu} = 1$

From NDS-2018, page 30, section 4.3.9

For rafter spaced less than 24 in, $C_r = 1.15$

From NDS-2018 supplement page 35, Table 4A

For Eastern white pine, select structural, $F_b = 1250$ psi

From NDS-2018, page 29, Table 4.3.1

$$F'_b = F_b * C_d * C_m * C_t * C_L * C_f * C_{fu} * C_i * C_r$$

$$F'_b = 1250 \text{ psi} * 1.15 * 1 * 1 * 1 * 1 * 1 * 1 * 1 * 1.15 = 1653.125 \text{ psi}$$

$$\text{Moment acting on the beam} = w * L^2 / 8$$

$$= 50 \text{ plf} * 15 \text{ ft} * 15 \text{ ft} / 8 = 1406.25 \text{ lb-ft} = 16875 \text{ lb-in}$$

$$\text{Required section modulus} = M / F'_b$$

$$= 16875 \text{ lb-in} / 1653.125 \text{ psi} = \mathbf{10.2 \text{ in}^3}$$

Correct answer is option A.

344. Problem

Correct Answer: (A) 360 plf

Explanation:

From AWC- SDPWS 2015, page 21, Table 4.2A,

For Structural I grade sheathing with thickness 3/8 inch and 8d nails @ 4in spacing, for seismic load, nominal shear strength assuming 2 in width of nailed face at boundaries,

$$V_s = 720 \text{ plf}$$

No adjustments are required for Southern Pine blocking.

From AWC- SDPWS 2015, page 16, section 4.2.3, ASD reduction factor = 2.0

$$\text{Hence allowable diaphragm shear} = 720 \text{ plf} / 2 = \mathbf{360 \text{ plf}}$$

Correct answer is Option A.

345. Problem

Correct Answer: (D) Flexible

Explanation:

Story drift of wall A = 0.45 in

Story drift of wall B = 0.6 in

Average story drift

$$\Delta = (0.45 \text{ in} + 0.6 \text{ in}) / 2 = 0.525 \text{ in}$$

Diaphragm in plane deflection

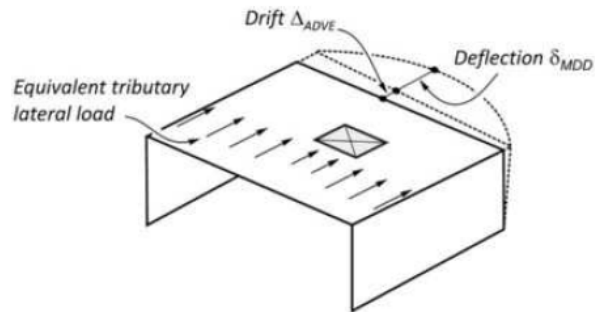
$$\delta = 1.1 \text{ in}$$

From ASCE 7-16, page 96, section 12.3.1.3, for flexible diaphragm

$$\delta / \Delta > 2$$

$$1.1 \text{ in} / 0.525 \text{ in} = 2.09 > 2.$$

Hence, the diaphragm is flexible and flexible diaphragms are not associated with torsional regularity. The correct answer is Option D.



346. Problem

Correct Answer: (D) 10in

Explanation:

Ref to ACI 318-14, page 84, Table 7.3.1.1

For one end continuous i.e. exterior span,

$$t = L / 24 = 17 \text{ ft} / 24 = 8.5 \text{ in}$$

For both ends continuous i.e. interior span,

$$t = L / 28 = 21 \text{ ft} / 28 = 9 \text{ in (governs)}$$

From page 84, section 7.3.1.1.2, for unit weight 110 pcf,

$$\text{Modification factor} = 1.65 - 0.005 * w_c$$

$$= 1.65 - 0.005 * 110 = 1.1$$

$$\text{Required thickness} = 9 \text{ in} * 1.1 = 9.9 \text{ in}$$

Hence the correct answer is Option D, **10in**.

347. Problem

Correct Answer: (B) #5

Explanation:

From NCEES PE handbook page 260, section 4.3.1,

$$\text{Area of \#4 bars} = 0.2 \text{ in}^2$$

$$\text{Area of \#5 bars} = 0.31 \text{ in}^2$$

Try with five #4 bars

$$\text{Total } A_s = 5 \times 0.2 \text{ in}^2 = 1 \text{ in}^2$$

From NCEES PE handbook page 264, section 4.3.2.2

$$a = (A_s \cdot f_y) / (0.85 \cdot f'_c \cdot b)$$

$$a = (1 \text{ in}^2 \cdot 60 \text{ ksi}) / (0.85 \cdot 4 \text{ ksi} \cdot 9 \text{ in})$$

$$a = 1.961 \text{ in}$$

$$M_n = A_s \cdot f_y \cdot (d - a/2)$$

$$M_n = 1 \text{ in}^2 \cdot 60 \text{ ksi} \cdot (18 \text{ in} - 1.961 \text{ in} / 2)$$

$$M_n = 1021.17 \text{ kip} \cdot \text{in} = 85.1 \text{ kip} \cdot \text{ft} \text{ (not enough)}$$

Try five #5 bars

$$A_s = 5 \times 0.31 \text{ in}^2 = 1.55 \text{ in}^2$$

$$a = (1.55 \text{ in}^2 \cdot 60 \text{ ksi}) / (0.85 \cdot 4 \text{ ksi} \cdot 9 \text{ in})$$

$$a = 3.039 \text{ in}$$

$$M_n = 1.55 \text{ in}^2 \cdot 60 \text{ ksi} \cdot (18 \text{ in} - 3.039 \text{ in} / 2)$$

$$M_n = 1532.68 \text{ kip} \cdot \text{in} = 127.7 \text{ kip} \cdot \text{ft}$$

From ACI 318-14, page 343, Table 21.2.2 for tension controlled section,

$$\Phi = 0.9$$

$$\text{Design strength} = \Phi \cdot M_n$$

$$= 0.9 \cdot 127.7 \text{ kip} \cdot \text{ft}$$

$$= 114.9 \text{ kip} \cdot \text{ft}$$

Hence #5 bars satisfies the requirement and the correct answer is Option B.

348. Problem

Correct Answer: (C) 101 in

Explanation:

From ACI-318-14, page 68, Section 6.3.2.1 and Table 6.3.2.1,

Effective overhang is the least of

- a) $8 \times \text{slab thickness} = 8 \times 6 \text{ in} = 48 \text{ in}$
- b) $\text{Clear distance to adjacent web} / 2 = (9 \text{ ft} - 20 \text{ in}) / 2 = 44 \text{ in}$
- c) $\text{Clear span of beam} / 12 = 27 \text{ ft} / 8 = 40.5 \text{ in}$

Effective flange width for the center beam = beam width + 2*effective overhang
 $= 20 \text{ in} + 2 \times 40.5 \text{ in} = \mathbf{101 \text{ in}}$

Correct answer is option C.

349. Problem

Correct Answer: (C) 190 kip*ft

Explanation:

From NCEES PE handbook page 260, section 4.3.1,

Area of #6 bars = 0.44 in^2

Total $A_s = 5 \times 0.44 \text{ in}^2 = 2.2 \text{ in}^2$

Assume that the depth of the compression block is less than 6in, i.e. lies inside the flange thickness.

From NCEES PE handbook page 264, section 4.3.2.2

$$a = (A_s \cdot f_y) / (0.85 \cdot f'_c \cdot b_f)$$

$$a = (2.2 \text{ in}^2 \cdot 60 \text{ ksi}) / (0.85 \cdot 4 \text{ ksi} \cdot 40 \text{ in})$$

$a = 0.971 \text{ in} < 6 \text{ in}$, assumption is valid.

$$M_n = A_s \cdot f_y \cdot (d - a/2)$$

$$M_n = 2.2 \text{ in}^2 \cdot 60 \text{ ksi} \cdot (18 \text{ in} - 0.971 \text{ in} / 2)$$

$$M_n = 2311.9 \text{ kip} \cdot \text{in}$$

$$M_n = 192.7 \text{ kip} \cdot \text{ft}$$

Closest answer is Option C , **190 kip*ft.**

350. Problem

Correct Answer: (A) 6in

Explanation:

From NCEES PE handbook page 260, section 4.3.1,

Area of #3 bars = 0.11 in^2

For 2 legged #3 stirrups, $A_v = 2 * 0.11 \text{ in}^2 = 0.22 \text{ in}^2$

From ACI 318-14, page 341 Table 21.2.1, strength reduction factor for shear

$\Phi = 0.75$

Required Shear strength, $V_u = 51 \text{ kips}$

From ACI 318-14, page 316, Table 19.2.4.2, for normal weight concrete, $\lambda = 1$

From ACI 318-14, page 354, section 22.5.5.1,

Shear capacity provided by concrete

$$V_c = 2\lambda(\sqrt{f'_c}) * b_w * d$$

$$V_c = 2 * 1 * (\sqrt{4000}) * 12 \text{ in} * 18 \text{ in}$$

$$V_c = 27322.1 \text{ lbs} = 27.3 \text{ kips}$$

$V_u > \Phi V_c$, so stirrups are required

From ACI 318-14, page 359, section 22.5.10

$$V_s = V_u / \Phi - V_c$$

$$V_s = 51 \text{ kips} / 0.75 - 27.3 \text{ kips}$$

$$V_s = 40.7 \text{ kips}$$

From ACI 318-14, page 359, equation 22.5.10.5.3

$$V_s = A_v * f_y * d / s$$

$$s = A_v * f_y * d / V_s$$

$$s = 0.22 \text{ in}^2 * 60 \text{ ksi} * 18 \text{ in} / 40.7 \text{ kips}$$

$$s = 5.8 \text{ in}$$

Hence, the correct answer is Option A, **6in**.

351. Problem

Correct Answer: (B) 9 in

Explanation:

Here, diameter of stirrups to be used is not given, however, ACI 318-14 limits the tie spacing regardless of tie diameter.

From ACI 318-14, page 341 Table 21.2.1, strength reduction factor for shear

$$\Phi = 0.75$$

Effective depth of section

$$d = 24 \text{ in} - \text{cover}$$

$$d = 24 \text{ in} - 2.5 \text{ in} = 21.5 \text{ in}$$

$$b_w = 12 \text{ in}$$

From ACI 318-14, page 316, Table 19.2.4.2, for normal weight concrete, $\lambda = 1$

From ACI 318-14, page 354, section 22.5.5.1,

Shear capacity provided by concrete

$$V_c = 2\lambda(\sqrt{f'_c}) * b_w * d$$

$$V_c = 2 * 1 * (\sqrt{4000}) * 12 \text{ in} * 21.5 \text{ in}$$

$$V_c = 32634.7 \text{ lbs} = 32.63 \text{ kips}$$

From ACI 318-14, page 359, section 22.5.10

$$V_s = V_u / \Phi - V_c$$

$$V_s = 70 \text{ kips} / 0.75 - 32.63 \text{ kips}$$

$$V_s = 60.7 \text{ kips}$$

Ref to ACI 318-14 page 146, Table 9.7.6.2.2 for max permissible spacing of stirrups

$$4(\sqrt{f'_c}) * b_w * d$$

$$= 4(\sqrt{4000}) * 12 \text{ in} * 21.5 \text{ in}$$

$$= 65269.4 \text{ lbs} = 65.27 \text{ kips}$$

$$\text{Here } V_s < 4(\sqrt{f'_c}) * b_w * d$$

Thus the maximum permissible spacing is the least of $d/2$ or 24 in

$$s = d/2$$

$$s = 21.5 \text{ in} / 2$$

$$s = 10.75 \text{ in}$$

Hence spacing cannot be greater than 10.75 in. Here, the correct answer is Option B, **9 in**.

352. Problem

Correct Answer: (C) 83 kips

Explanation:

Effective depth of slab, $d = 8 \text{ in} - 1.5 \text{ in} = 6.5 \text{ in}$

From ACI 318-14, page 341 Table 21.2.1,
strength reduction factor for shear, $\Phi = 0.75$

From ACI 318-14, page 362, Section 22.6.4.1,

The critical perimeter for edge column is

$$b_o = 2*(c_1 + d/2) + (c_2 + d)$$

$$b_o = 2*(12 \text{ in} + 6.5 \text{ in}/2) + (18 \text{ in} + 6.5 \text{ in})$$

$$b_o = 55 \text{ in}$$

Calculate two way shear strength from ACI 318-14, page 364, section 22.6.5

From page 364, Table 22.6.5.2

$$\beta = 18 \text{ in}/12 \text{ in} = 1.5$$

From page 365, section 22.6.5.3, $\alpha_s = 30$ for edge column

Assume normal weight concrete and $\lambda = 1$

v_c is the least of

$$a) \quad 4\lambda(\sqrt{f'_c}) = 4*1*\sqrt{6000} = 309.8 \text{ psi}$$

$$b) \quad (2+4/\beta)*\lambda*\sqrt{f'_c} = (2+4/1.5)*1*\sqrt{6000} = 361.48 \text{ psi}$$

$$c) \quad (2+\alpha_s*d/b_o)*\lambda*\sqrt{f'_c} = (2+30*6.5 \text{ in}/55 \text{ in})*1*\sqrt{6000} = 429.5 \text{ psi}$$

$$v_c = 309.8 \text{ psi}$$

Shear strength

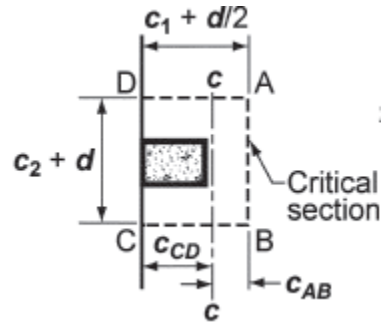
$$V = \Phi*v_c*b_o*d$$

$$V = 0.75*309.8 \text{ psi}*55 \text{ in}*6.5 \text{ in}$$

$$V = 83065.13 \text{ lbs}$$

$$V \approx \mathbf{83 \text{ kips}}$$

Correct answer is option C.



353. Problem

Correct Answer: (C) 160 kips

Explanation:

From ACI 318-14, page 341 Table 21.2.1,

strength reduction factor for shear, $\Phi = 0.75$

From ACI 318-14, page 362, Section 22.6.4.1,

The critical perimeter for interior columns with drop panels is located at distance $d/2$ from the edge of the drop panel.

Effective depth, $d = 6 \text{ in} - 1.25 \text{ in} = 4.75 \text{ in}$

$b_o = 4*(16 \text{ in} + 12 \text{ in} + 12 \text{ in} + d/2 + d/2)$

$b_o = 179 \text{ in}$

Design two way shear strength

$= \Phi * v_c * b_o * d$

$= 0.75 * 250 \text{ psi} * 179 \text{ in} * 4.75 \text{ in}$

$= 159421.87 \text{ lbs}$

$\approx \mathbf{160 \text{ kips}}$

Correct answer is Option C.

354. Problem

Correct Answer: (C) 15 in

Explanation:

From ACI 318-14, page 88, section 7.7.2.3

Max permitted spacing for deformed bars shall be lesser of $3h$ and 18 in .

$3*h = 3*5 \text{ in} = 15 \text{ in}$

Hence the maximum permitted rebar spacing for 5 inch thick slab is **15 in**.

Correct answer is Option C.

355. Problem

Correct Answer: (B) 0.086 in²/ft

Explanation:

$$M_u = 0.7 \text{ kip}\cdot\text{ft}/\text{ft} = 8.4 \text{ kip}\cdot\text{in}/\text{ft}$$

$$\text{Effective depth, } d = 4 \text{ in} - 1 \text{ in} = 3 \text{ in}$$

$$b = 12 \text{ in} \text{ (Design is done per foot width)}$$

From ACI 318-14, page 86, Table 7.6.1.1

Minimum area of rebar required for slab with Gr 60 rebar

$$= 0.0018 \cdot \text{gross area}$$

$$= (0.0018 \cdot 4 \text{ in} \cdot 12 \text{ in}) \text{ per ft}$$

$$= 0.086 \text{ in}^2/\text{ft}$$

Calculate area of rebar required for applied moment.

Assume depth of compression block, $a = 0.2 \cdot d$

$$a = 0.2 \cdot 3 \text{ in} = 0.6 \text{ in}$$

From ACI 318-14, page 343, Table 21.2.2 for tension controlled section, $\Phi = 0.9$

From NCEES PE handbook page 264, section 4.3.2.2

$$M_n = M_u / \Phi = A_s \cdot f_y \cdot (d - a/2)$$

$$A_s = M_u / [\Phi \cdot f_y \cdot (d - a/2)]$$

$$A_s = 8.4 \text{ kip}\cdot\text{in} / [0.9 \cdot 60 \text{ ksi} \cdot (3 \text{ in} - 0.6 \text{ in}/2)] \text{ per ft}$$

$$A_s = 0.058 \text{ in}^2/\text{ft}$$

Check validity of 'a' (if calculated value of a is less than assumed value, O.K.)

$$a = (A_s \cdot f_y) / (0.85 \cdot f'_c \cdot b)$$

$$a = (0.058 \text{ in}^2 \cdot 60 \text{ ksi}) / (0.85 \cdot 4 \text{ ksi} \cdot 12 \text{ in})$$

$$a = 0.085 \text{ in} < \text{initial assumption, i.e. required area of steel will be lesser.}$$

Hence moment applied does not govern the rebar area and minimum area required per

ACI 318-14 Table 7.6.1.1 as calculated initially, governs.

Correct answer is Option B, **0.086 in²/ft.**

5.2 Vertical members

[Click here for Problems 5.2](#)

356. Problem

Correct Answer: (C) 440 kips

Explanation:

From AISC steel manual 15th ed., page 16.1-570, table C-A-7.1,

Effective length factor for fixed-fixed column, $k = 0.65$

Effective length

$$k * L = 0.65 * 19.25 \text{ ft} * (12 \text{ in/ft})$$

$$k * L = 150.15 \text{ in}$$

From steel manual page 1-97, table 1-12,

Area of HSS 8X8X1/2

$$A_g = 13.5 \text{ in}^2$$

Radius of gyration, $r = 3.04$

$$K * l / r = 150.15 \text{ in} / 3.04 \text{ in}$$

$$K * l / r = 49.4$$

From steel manual page 4-230, table 4-14

For $F_y = 70 \text{ ksi}$ and $L/r = 49$

$$F_{cr} / \Omega = 32.8 \text{ ksi}$$

Hence, allowable axial load

$$= (F_{cr} / \Omega) * A_g$$

$$= 32.8 \text{ ksi} * 13.5 \text{ in}^2$$

$$= 442.8 \text{ kips}$$

Closest answer is Option C, **440 kips**.

357. Problem

Correct Answer: (A) HSS 3 X 3 X 3/16

Explanation:

From AISC steel manual 15th ed., page 16.1-570, table C-A-7.1,

Effective length factor for pinned- fixed column, $k = 0.8$

Effective length

$$k * L = 0.8 * 13.5 \text{ ft}$$

$$k * L = 10.8 \text{ ft}$$

From steel manual page 4-82, table 4-4, for $Kl = 11 \text{ ft}$,

HSS 3 X 3 X 3/16 can withstand compression of 21.2 kips

Hence it is the lightest section that satisfies the requirement.

Correct answer is Option A.

358. Problem

Correct Answer: (C) 250 kips

Explanation:

$$Kl_x = 28 \text{ ft} = L_{cx}$$

$$Kl_y = 14 \text{ ft}$$

From AISC steel manual 15th ed., page 1-29, table 1-1

For W8X40, $r_x = 3.53 \text{ in}$, $r_y = 2.04 \text{ in}$

Since $Kl_x > Kl_y$ from steel manual page 4-5, equation 4-1

Equivalent effective length,

$$L_{cy.eq} = L_{cx} / (r_x / r_y)$$

$$= 28 \text{ ft} / (3.53 \text{ in} / 2.04 \text{ in})$$

$$= 16.18 \text{ ft} > Kl_y, \text{ so use this effective length.}$$

From steel manual page 4-24, table 4-1 a

For W8X40 with $L_c = 17 \text{ ft}$, $\Phi P_n = 253 \text{ kips}$

Hence the correct answer is Option C, **250 kips**.

359. Problem

Correct Answer: (C) 82.2%

Explanation:

$$Kl_x = Kl_y = 12 \text{ ft}$$

From AISC steel manual page 4-24, table 4-1 a

For W8X40 with $L_c = 12 \text{ ft}$,

$$P_n/\Omega = P_c = 243 \text{ kips}$$

$$P_r/P_c = 100 \text{ kips} / 243 \text{ kips} = 0.412 > 0.2$$

For flexure, lateral torsional buckling is prevented, so yield strength is verified.

From AISC Steel manual page 3-27, table 3-2

For W-8X40

$$M_{px}/\Omega = 99.3 \text{ kip}\cdot\text{ft} = M_{cx}$$

From AISC Steel manual page 3-32, table 3-4

For W-8X40

$$M_{ny}/\Omega = 46.2 \text{ kip}\cdot\text{ft} = M_{cy}$$

From steel manual page 16.1-77, equation H1-1a

$$P_r/P_c + (8/9) * (M_{rx}/M_{cx} + M_{ry}/M_{cy})$$

$$= 100 \text{ kips} / 243 \text{ kips} + (8/9) * (25 \text{ kip}\cdot\text{ft} / 99.3 \text{ kip}\cdot\text{ft} + 10 \text{ kip}\cdot\text{ft} / 46.2 \text{ kip}\cdot\text{ft})$$

$$= \mathbf{82.8\%}$$

Hence the correct answer is Option C.

360. Problem

Correct Answer: (D) 40 in

Explanation:

From AISC steel manual page 15th ed. Page 16.1-17, Table B4.1a, Case II

$$\text{Limiting } \lambda_r = 0.64 * \sqrt{(k_c * E / F_y)}$$

$$h = 12 \text{ in} - 2 * 1.5 \text{ in} = 9 \text{ in}$$

$$t = 1.5 \text{ in}, t_w = 0.75 \text{ in}$$

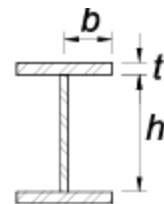
$$k_c = 4 / [\sqrt{(h/t_w)}] = 4 / [\sqrt{(9 \text{ in} / 0.75 \text{ in})}] = 1.155$$

$$\text{Limiting value of } k_c = 0.76$$

$$b/t = 0.64 * \sqrt{(k_c * E / F_y)} = 0.64 * \sqrt{(0.76 * 29000 \text{ ksi} / 50 \text{ ksi})}$$

$$b = 13.437 * t = 13.437 * 1.5 \text{ in} = 20.155 \text{ in}$$

Total permissible width = $2 * b = 40.3 \text{ in}$. Correct answer is Option D, **40 in**.



361. Problem

Correct Answer: (B) 528 in²

Explanation:

From AISC steel manual page 15th ed. Page 16.1-140 and 16.1-141, section J8

The bearing frustum A₂ is the maximum area of the portion of the supporting surface that is geometrically similar to and concentric with the loaded area.

Loaded Plate area = 18 in X 16 in = 288 in² = A₁

Pedestal area = 24 in X 24 in

Maximum dimension of pedestal = 24 in

Maximum dimension of plate = 18 in

24 in - 18 in = 6 in

Hence the area of bearing frustum geometrically similar to the plate

= (18 in + 6 in) X (16 in + 6 in)

= **528 in²**

Check A₂ < 4 A₁ 528 in² < 4*288 in² O.K.

Hence correct answer is Option B, **528 in²**.

362. Problem

Correct Answer: (B) 1.125 in

Explanation:

From AISC steel manual page 15th ed. Page 14-5, eq 14-2, eq 14-3 and eq 14-4

$m = (N - 0.95 \cdot d) / 2 = (15 \text{ in} - 0.95 \cdot 11 \text{ in}) / 2 = 2.275 \text{ in}$

$n = (B - 0.8 \cdot b_f) / 2 = (15 \text{ in} - 0.8 \cdot 12 \text{ in}) / 2 = 2.7 \text{ in}$

$n' = [\sqrt{(d \cdot b_f)}] / 4 = [\sqrt{(11 \cdot 12)}] / 4 = 2.872 \text{ in}$

Since concrete strength is not given, take $\lambda = 1$

Bearing length $l = \text{maximum between } m, n, \text{ and } \lambda \cdot n'$

$l = 2.872 \text{ in}$

From Page 14-6, equation 14-7a

$t_{\min} = l \cdot \sqrt{[2 \cdot P_u / (0.9 \cdot F_y \cdot B \cdot N)]}$

$t_{\min} = 2.872 \text{ in} \cdot \sqrt{[2 \cdot 650 \text{ kips} / (0.9 \cdot 50 \text{ ksi} \cdot 15 \text{ in} \cdot 15 \text{ in})]}$

$t_{\min} = 1.03 \text{ in}$

Hence 1 in plate doesn't work, provide **1.125 in** plate.

Correct answer is option B.

363. Problem

Correct Answer: (D) 10.5 ksi

Explanation:

From AISC steel manual page 15th ed. Page 16.1-140 and 16.1-141, section J8

$$\Omega_c = 2.31$$

Area of bearing plate

$$A_1 = 15\text{ in} \times 15\text{ in} = 225\text{ in}^2$$

Area of frustum

$$A_2 = 18\text{ in} \times 18\text{ in} = 324\text{ in}^2$$

$$\text{Check } A_2 < 4 \cdot A_1 \quad 324\text{ in}^2 < 4 \cdot 225\text{ in}^2 \quad \text{O.K.}$$

From eq. J8-2

$$P_p = 0.85 f'_c A_1 \sqrt{A_2/A_1}$$

$$1000\text{ kips} = 0.85 \cdot f'_c \cdot 225\text{ in}^2 \cdot \sqrt{324\text{ in}^2 / 225\text{ in}^2}$$

$$f'_c = 1000\text{ kips} / 229.5\text{ in}^2$$

$$f'_c = 4.357\text{ ksi}$$

$$\text{Using ASD factor } \Omega_c \cdot f'_c = 2.31 \cdot 4.357\text{ ksi} = 10.1\text{ ksi}$$

Hence the correct answer is Option D, **10.5 ksi**.

364. Problem

Correct Answer: (B) I and II

Explanation:

From AISC steel manual page 15th ed. Page 16.1-38, section E5

Effects of eccentricity can be ignored if

L_c/r as determined in the section does not exceed 200

Members are loaded at the ends in compression through the same one leg.

Condition III is not a requirement.

Hence, the correct answer is Option B, **I and II**.

365. Problem

Correct Answer: (B) 20 ksi

Explanation:

From AISC steel manual page 15th ed. Page 1-118, Table 1-16

For 2C12X30 spaced at 3/8in , $r_y = 1.15\text{in}$

$$KL/r_y = 10\text{ ft} * 12\text{ (in/ft)} / 1.15\text{ in}$$

$$KL/r_y = 104.34$$

From steel manual page 4-231, Table 4-14,

For 36 ksi steel and $L/r = 105$

$$F_{cr}/\Omega = 12.1\text{ ksi}$$

$$F_{cr} = 12.1\text{ ksi} * \Omega$$

$$F_{cr} = 12.1\text{ ksi} * 1.67$$

$$F_{cr} = 20.2\text{ ksi}$$

Hence the correct answer is Option B, **20 ksi**.

366. Problem

Correct Answer: (D) 1250 kips

Explanation:

$$KL = 15\text{ ft} = 180\text{ in} = L_c$$

$$F_y = 60\text{ ksi}$$

From AISC steel manual page 15th ed. Page 1-20 and 1-21, Table 1-1

For W21X111

$$A = 32.6\text{ in}^2$$

$$r_x = 9.05\text{ in} \quad r_y = 2.9\text{ in}$$

From steel manual page 16.1-36, eq. E3-4

$$F_e = \pi^2 * E / (L_c/r_y)^2$$

$$F_e = \pi^2 * 29000\text{ ksi} / (180\text{ in}/2.9\text{ in})^2 = 74.29\text{ ksi}$$

From page 16.1-35, $F_y/F_e = 60\text{ ksi} / 74.29\text{ ksi} = 0.808 < 2.25$. Hence case (a) applies

$$F_{cr} = 0.658^{\frac{F_y}{F_e}} * F_y$$

$$F_{cr} = 0.658^{0.808} * 60\text{ ksi} = 42.78\text{ ksi} \quad \Phi_c = 0.9 \text{ from page 16.1-33 section E1}$$

$$\text{Design compressive strength} = \Phi_c * F_{cr} * A = 0.9 * 42.78\text{ ksi} * 32.6\text{ in}^2 = 1255.1\text{ kips}$$

Correct answer is Option D, **1250 kips**.

367. Problem

Correct Answer: (C) 1

Explanation:

All beams and columns seem to have hinges at the connection points with each other.

The center column has a hinge at the base and the frame can be assumed to be pin connected.

Also, X- bracing prevents any sidesway. Hence the effective length factor 'K' = 1.

Correct answer is Option C.

368. Problem

Correct Answer: (C) 1.3

Explanation:

Length of column, $L_c = 12$ ft

Length of beam, $L_b = 15$ ft

From AISC Steel Manual 15th ed. page 1-28 and page 1-29, table 1-1

For W10X60, $I_b = 341$ in⁴

From page 1-26 and 1-27, table 1-1

For W12X40, $I_c = 307$ in⁴

For rigid frames, from page 16.1-571 to 16.1-573

For base of column, $G_b = 1$ for rigidly connected to base from page 16.1-573

For top of column, from page 16.1-572

$$G_a = \frac{\sum(E_c \cdot I_c / L_c)}{\sum(E_b \cdot I_b / L_b)}$$

Since E is equal, it cancels out

$$G_a = (307/12) / (341/15)$$

$$G_a = 1.125$$

From Fig C-A-7.2 for side sway uninhibited,

$$K = 1.32$$

Hence the correct answer is Option C, **1.3**.

369. Problem

Correct Answer: (B) 38

Explanation:

From AISC Steel Manual 15th ed. page 1-26 and page 1-27, table 1-1

For W12X120

$$r_x = 5.51 \text{ in}$$

$$r_y = 3.13 \text{ in}$$

From steel manual page 16.1-570, table C-A-7.1

For pin fixed condition $K = 0.8$

For pin pin condition $K = 1.0$

For weaker axis, $L_y = 10\text{ft}$ with $K = 1$ (pin-pin for top half) governing

$$K \cdot L_y / r_y$$

$$= 1 \cdot (10 \cdot 12\text{in}) / 3.13\text{in} = 38.33$$

For stronger axis, $L_x = 20 \text{ ft}$ with $K = 0.8$ for fixed-pinned

$$K \cdot L_x / r_x$$

$$= 0.8 \cdot (20 \cdot 12\text{in}) / 5.51\text{in} = 34.8$$

Hence weaker axis governs and the correct answer is Option B, **38**.

370. Problem

Correct Answer: (C) 233 kips

Explanation:

From AISC Steel Manual 15th ed. page 4-17, table 4-1a

For W14X82 with 15ft effective length

Allowable compression, $P_c = 489 \text{ kips}$

Lateral torsional buckling is not a concern so get allowable yielding moments.

From steel manual page 3-25, table 3-2 for W14X82, $M_{cx} = M_{px} / \Omega = 347 \text{ kip}\cdot\text{ft}$

From steel manual page 3-31, table 3-4 for W14X82, $M_{cy} = M_{ny} / \Omega = 112 \text{ kip}\cdot\text{ft}$

From steel manual page 16.1-77, eq H1-1a for interaction,

$$P_r / P_c + (8/9) \cdot (M_{rx} / M_{cx} + M_{ry} / M_{cy}) = 1$$

$$P_r / 489 \text{ kips} + (8/9) \cdot (80 \text{ kip}\cdot\text{ft} / 347 \text{ kip}\cdot\text{ft} + 40 \text{ kip}\cdot\text{ft} / 112 \text{ kip}\cdot\text{ft}) = 1$$

$$P_r / 489 \text{ kip} = 0.478$$

$$P_r = 233.7 \text{ kips}$$

Correct answer is option C, **233 kips**.

371. Problem

Correct Answer: (B)

Explanation:

Per TMS 402/602-16 page C-56 and C-57, section 5.3.1

Minimum vertical reinforcement should be $0.0025A_n$

For 16 in X 16 in column $A_{smin} = 0.0025 * 16 \text{ in} * 16 \text{ in} = 0.64 \text{ in}^2$.

Option A is not true.

Minimum side dimension should be 8 in, so Option C is not true.

Per TMS 402/602-16 page C-57, section 5.3.1.4 (b)

Tie spacing should not exceed $48 * \text{tie diameter}$

For #4 tie, spacing = $48 * 4 / 8 \text{ in} = 24 \text{ in}$. So Option D is not true

From section 5.3.1.1 (a), $h < 99 * r$

$h < 99 * 4 \text{ in}$

$h < 33 \text{ ft}$

For cantilevered column, from figure CC-5.3-1, $h = 2 * \text{height}$

For 17 ft cantilevered column, $h = 17 \text{ ft} * 2 = 34 \text{ ft}$ i.e. h exceeds 33ft and is not permitted

Hence Option B is true and the answer.

372. Problem

Correct Answer: (B) 385 kips

Explanation:

Per TMS 402/602-16 page C-64, Table CC-6.1.2,

Area of No. 8 bar, $A_s = 0.79 \text{ in}^2$

$A_{st} = 8 * 0.79 \text{ in}^2 = 6.32 \text{ in}^2$

Per TMS 402/602-16 page C-114, section 8.3.3 (b)

$F_s = 32000 \text{ psi}$ for grade 60 rebar.

Calculate radius of gyration

For 23.625in X 23.625in section, $I = b * h^3 / 12$

$I = 23.625 \text{ in} * (23.625 \text{ in})^3 / 12 = 25960.08 \text{ in}^4$

$A_n = 23.625 \text{ in} * 23.625 \text{ in} = 558.14 \text{ in}^2$

$r = \sqrt{I / A_n}$

$r = \sqrt{(25960.08 \text{ in}^4 / 558.14 \text{ in}^2)}$

$r = 6.82 \text{ in}$

$H / r = 20 \text{ ft} * 12 (\text{in/ft}) / 6.82 \text{ in} = 35.2 < 99$

From page C-115, section 8.3.4.2.1 (a), eq 8-18,

$$P_a = (0.25 \cdot f'_m \cdot A_n + 0.65 \cdot A_{st} \cdot F_s) \cdot [1 - (H/140r)^2]$$

$$P_a = (0.25 \cdot 2000 \text{ psi} \cdot 558.14 \text{ in}^2 + 0.65 \cdot 6.32 \text{ in}^2 \cdot 32000 \text{ psi}) \cdot [1 - (35.2/140)^2]$$

$$P_a = 384574.05 \text{ lbs}$$

$$P_a = 384.6 \text{ kips}$$

Hence the correct answer is Option B, **385 kips**.

373. Problem

Correct Answer: (A) 0

Explanation:

First, assume that there is no tension in the wall and calculate the moment of inertia of the uncracked section without considering rebar.

Wall is fully grouted so the entire cross section is engaged.

$$L = 5 \text{ ft} = 60 \text{ in}$$

$$t = 7.625 \text{ in}$$

$$H = 8 \text{ ft} = 96 \text{ in}$$

$$\text{Area, } A = L \cdot t$$

$$A = 60 \text{ in} \cdot 7.625 \text{ in}$$

$$A = 457.5 \text{ in}^2$$

Section modulus

$$S = t \cdot L^2 / 6$$

$$S = 7.625 \text{ in} \cdot (60 \text{ in})^2 / 6 = 4575 \text{ in}^3$$

Total bending stress due to V

$$F_b = V \cdot H / S$$

$$F_b = 5 \text{ kips} \cdot 96 \text{ in} / 4575 \text{ in}^3 = 0.105 \text{ ksi}$$

Total compressive stress due to P

$$F_c = P \cdot L / A$$

$$F_c = (12 \text{ kip/ft} \cdot 5 \text{ ft}) / 457.5 \text{ in}^2$$

$$F_c = 0.131 \text{ ksi}$$

Minimum stress acting on the section

$$= F_c - F_b = 0.131 \text{ ksi} - 0.105 \text{ ksi} = 0.026 \text{ ksi (compressive, i.e. initial assumption valid)}$$

Hence there is no tension in the section and the correct answer is Option A, **0**.

374. Problem

Correct Answer: (A) 22 kips /ft

Explanation:

Per TMS 402/602-16 page C-57, section 5.3.1.4 (b)

Spacing of lateral ties should not exceed the least dimension of the member.

Here #3 ties are spaced at 12 in c/c which is greater than 7.625 in

Thus rebars are not effectively braced per requirements

Effective area of rebars, $A_{st} = 0$

Calculate available axial length per foot length

From page C-115, section 8.3.4.2.1 (b), eq 8-19

$$P_a = (0.25 \cdot f'_m \cdot A_n + 0.65 \cdot A_{st} \cdot F_s) \cdot (70r/H)^2$$

$$P_a = 0.25 \cdot 2000 \text{ psi} \cdot [12 \text{ (in/ft)} \cdot 7.625 \text{ in}] \cdot (70/100)^2$$

$$P_a = 22417.5 \text{ lbs /ft}$$

$$P_a \approx \mathbf{22 \text{ kips /ft}}$$

Correct answer is Option A.

375. Problem

Correct Answer: (A) 610 plf

Explanation:

From AWC- SDPWS 2015, page 35, Table 4.3A,

For Structural I panel, 3/8 in thick, 8d nails with spacing 2 in,

Nominal seismic shear strength = 1220 plf

No adjustment for southern pine framing is required.

From AWC- SDPWS 2015, page 25, section 4.3.3, ASD reduction factor = 2.0

$$\text{Allowable shear} = 1220 \text{ plf} / 2 = \mathbf{610 \text{ plf}}$$

Correct answer is Option A.

376. Problem

Correct Answer: (B) 5 kips

Explanation:

From NDS-2018 supplement page 14, Table 1B, standard dressed size for 2X6 = 1.5X5.5

From NDS-2018 supplement page 34, Table 4A, for select structural DFL,

Compression parallel to grain $F_c = 1700$ psi

From NDS-2018, page 29, Table 4.3.1

$$F_c' = F_c * C_d * C_m * C_t * C_f * C_i * C_p$$

$$F_c' = 1700 \text{ psi} * 1 * 1 * 1 * 1.15 * 1 * 0.33$$

$$F_c' = 645.15 \text{ psi}$$

Allowable compression, $P = F_c' * A$

$$P = 645.15 \text{ psi} * 1.5 \text{ in} * 5.5 \text{ in}$$

$$P = 5322.5 \text{ lbs}$$

Correct answer is Option B, **5 kips**.

377. Problem

Correct Answer: (A) 339 psi

Explanation:

From NDS-2018 supplement page 14, Table 1B, standard dressed size for 2X8 = 1.5X7.25

From NDS-2018 supplement page 34, Table 4A, for DFL No. 1 & Btr,

$E_{min} = 660000$ psi

For normal temperature and dry conditions, $C_m = C_t = 1$

From NDS-2018, page 29, Table 4.3.1

$$E_{min}' = E_{min} * C_m * C_t * C_i * C_T$$

$$E_{min}' = 660000 \text{ psi} * 1 * 1 * 1 * 1 = 660000 \text{ psi}$$

Planks are pin connected, effective length factor = 1

$$L_x/d_1 = 10 \text{ ft} / 7.25 \text{ in} = 120 \text{ in} / 7.25 \text{ in} = 16.55$$

$$L_y/d_2 = 5 \text{ ft} / 1.5 \text{ in} = 60 \text{ in} / 1.5 \text{ in} = 40 \text{ (} L_y \text{ governs)}$$

From NDS-2018, page 21, section 3.7.1, critical buckling stress is given by

$$F_{cE} = 0.822 * E_{min}' / (l_e/d)^2$$

$$F_{cE} = 0.822 * 660000 \text{ psi} / (40)^2 = 339.07 \text{ psi}$$

Hence the correct answer is Option A, **339 psi**.

378. Problem

Correct Answer: (C) Wall C

Explanation:

Wall A is a blocked shear wall as it has framing on all sides. Wall B, C and D are unblocked walls because framing is not present on all sides.

From AWC- SDPWS 2015, page 28, Table 4.3.4,

Max aspect ratio for blocked WSP walls = 3.5:1 ,

Wall A aspect ratio = $12/4 = 3$. O.K.

Max aspect ratio for unblocked WSP walls = 2:1

Wall B, C and D aspect ratios are all less than or equal to 2. O.K.

From AWC- SDPWS 2015, page 25, Section 4.3.3.2, the height of unblocked shear wall is limited to 16ft.

Hence **Wall C**, 18 ft tall ,does not meet the requirement and Option C is the answer.

379. Problem

Correct Answer: (C) 8

Explanation:

From ACI 318-14, page 156, section 10.6.1.1, maximum area of longitudinal bars permitted is $0.08 \times \text{Gross area}$

$$= 0.08 \times 10 \text{ in} \times 10 \text{ in}$$

$$= 8 \text{ in}^2$$

From ACI 318-14, page 507, Appendix A

$$\text{Area of \#9 bar} = 1 \text{ in}^2$$

Max number of bars permitted

$$= 8 \text{ in}^2 / 1 \text{ in}^2$$

$$= \mathbf{8}$$

Correct answer is Option C.

380. Problem

Correct Answer: (B) 1322 kips

Explanation:

Gross area, $A_g = 16 \text{ in} \times 16 \text{ in} = 256 \text{ in}^2$

From ACI 318-14, page 507, Appendix A

Area of #8 bar = 0.79 in^2

$A_{st} = 8 \times 0.79 \text{ in}^2 = 6.32 \text{ in}^2$

From ACI 318-14, page 350, equation 22.4.2.2,

$P_o = 0.85 \cdot f'_c \cdot (A_g - A_{st}) + f_y \cdot A_{st}$

$P_o = 0.85 \cdot 6 \text{ ksi} \cdot (256 \text{ in}^2 - 6.32 \text{ in}^2) + 60 \text{ ksi} \cdot 6.32 \text{ in}^2$

$P_o = 1652.6 \text{ kips}$

From ACI 318-14, page 350, Table 22.4.2.1,

For tied columns, $P_{n,\max} = 0.8 \cdot P_o$

$P_{n,\max} = 0.8 \cdot 1652.6 \text{ kips}$

$P_{n,\max} \approx \mathbf{1322 \text{ kips}}$ Correct answer is Option B.

381. Problem

Correct Answer: (A) 855 kips

Explanation:

Gross area, $A_g = 18 \text{ in} \times 18 \text{ in} = 324 \text{ in}^2$

$A_{st} = 8 \times 0.6 \text{ in}^2 = 4.8 \text{ in}^2$

From ACI 318-14, page 350, equation 22.4.2.2,

$P_o = 0.85 \cdot f'_c \cdot (A_g - A_{st}) + f_y \cdot A_{st}$

$P_o = 0.85 \cdot 5 \text{ ksi} \cdot (324 \text{ in}^2 - 4.8 \text{ in}^2) + 60 \text{ ksi} \cdot 4.8 \text{ in}^2 = 1644.6 \text{ kips}$

From ACI 318-14, page 350, Table 22.4.2.1,

$P_{n,\max} = 0.8 \cdot P_o = 0.8 \cdot 1644.6 \text{ kips} = 1315.7 \text{ kips}$

$\Phi \cdot P_{n,\max} = 0.65 \cdot 1315.7 \text{ kips} = 855.2 \text{ kips}$

Per ACI 318-14, page 350, section R22.4.2.1, the 0.8 factor applied to 'Po' is to account for eccentricity to depth ratios of 0.1 for tied columns.

Eccentricity considered = $0.1 \cdot 18 \text{ in} = 1.8 \text{ in}$

Eccentricity due to applied moment = $M / (\Phi \cdot P_{n,\max})$

$= 100 \text{ kip} \cdot \text{ft} / 855.2 \text{ kips} = 0.117 \text{ ft} = 1.4 \text{ in}$, this is less than 1.8 in

Thus axial load capacity is not compromised and capacity $\approx \mathbf{855 \text{ kips}}$

Correct answer is Option A.

382. Problem

Correct Answer: (D) 10.5 in^2

Explanation:

Gross area, $A_g = 16 \text{ in} \times 16 \text{ in} = 256 \text{ in}^2$

Factored axial load, $P_u = 798 \text{ kips}$

Nominal axial strength, $P_n = P_u / \Phi$

$P_n = 798 \text{ kips} / 0.65 = 1227.7 \text{ kips}$

$K_n = P_n / (f'_c * A_g)$

$K_n = 1227.7 \text{ kips} / (4 \text{ ksi} * 256 \text{ in}^2)$

$K_n = 1227.7 \text{ kips} / (4 \text{ ksi} * 256 \text{ in}^2)$

$K_n = 1.199$

Factored moment, $M_u = 66.5 \text{ kip}\cdot\text{ft}$

Nominal moment, $M_n = M_u / \Phi$

$M_n = 66.5 \text{ kip}\cdot\text{ft} / 0.65 = 102.308 \text{ kip}\cdot\text{ft}$

$M_n = 1227.69 \text{ kip}\cdot\text{in}$

$R_n = P_n * e / (f'_c * A_g * h)$

$R_n = M_n / (f'_c * A_g * h)$

$R_n = 1227.69 \text{ kip}\cdot\text{in} / (4 \text{ ksi} * 256 \text{ in}^2 * 16 \text{ in})$

$R_n = 0.0749$

Plot the values of R_n and K_n in the interaction chart.

We get, ratio of Area of steel, $p = 0.04$

Required area of steel = $p * A_g$

$= 0.04 * 256 \text{ in}^2$

$= 10.24 \text{ in}^2$

Closest answer is Option D, **10.5 in²**.

383. Problem

Correct Answer: (B) 12 in

Explanation:

Ref to ACI 318-14, page 507, Appendix A for bar size diameters

Per ACI 318-14, page 438, section 25.7.2.1 (b)

Center-to-center spacing shall not exceed the least of $16 * d_b$ of longitudinal bar, $48 * d_b$ of tie bar, and smallest dimension of member

1) For #7 longitudinal bar, $16 * d_b = 16 * 7/8 \text{ in} = 14 \text{ in}$

2) For #3 Tie bar, $48 * 3/8 \text{ in} = 18 \text{ in}$

3) Minimum dimension of the member = 12in
Hence **12 in** governs and the correct answer is Option B.

384. Problem

Correct Answer: (B) 38.3 kips

Explanation:

Here, information on ties is not provided. Hence, only the strength of concrete is calculated.

Effective cover for columns is assumed as 3 in.

$$d = 20 \text{ in} - 3 \text{ in}$$

$$d = 17 \text{ in}$$

$$b_w = 20 \text{ in}$$

$$\lambda = 1 \text{ (Assume normal weight concrete)}$$

$$A_g = 20 \text{ in} * 20 \text{ in} = 400 \text{ in}^2$$

$$f'_c = 4000 \text{ psi}$$

From ACI 318-14, page 354, equation 22.5.6.1

$$V_c = 2 \left[1 + \frac{N_u}{(2000 * A_g)} \right] * \lambda * (\sqrt{f'_c}) * b_w * d$$

$$V_c = 2 \left[1 + \frac{150000 \text{ lbs}}{(2000 * 400 \text{ in}^2)} \right] * 1 * (\sqrt{4000}) * 20 \text{ in} * 17 \text{ in}$$

$$V_c = 51070.78 \text{ lbs}$$

From ACI 318-14, page 341, Table 21.2.1

For shear, $\Phi = 0.75$

$$\text{Design strength} = \Phi * V_c = 0.75 * 51070.78 \text{ lbs} \approx \mathbf{38.3 \text{ kips}}$$

Correct answer is Option B.

385. Problem

Correct Answer: (A) 3.5 in

Explanation:

When it is assumed that entire shear is taken by reinforcement only, $V_c = 0$

$$d = 18 \text{ in} - 3 \text{ in} = 15 \text{ in}$$

From ACI 318-14, page 341, Table 21.2.1

For shear, $\Phi = 0.75$

Ref to ACI 318-14, page 507, Appendix A for bar size areas

For #4 Ties (two legs of the tie are engaged),

$$A_v = 2 \times 0.2 \text{ in}^2 = 0.4 \text{ in}^2$$

From ACI 318-14, page 359, equation 22.5.10.5.3

$$V_s = (A_v \times f_y \times d) / s$$

$$\text{Required strength } V_s = 75 \text{ kips} / \Phi = 75 \text{ kips} / 0.75 = 100 \text{ kips}$$

$$s = (A_v \times f_y \times d) / 100 \text{ kips}$$

$$s = (0.4 \text{ in}^2 \times 60 \text{ ksi} \times 15 \text{ in}) / 100 \text{ kips}$$

$s = 3.6 \text{ in}$, spacing should be less than this.

Hence the correct answer is Option A, **3.5 in**.

386. Problem

Correct Answer: (B) 310 kips

Explanation:

Thickness of wall, $h = 12 \text{ in} = 1 \text{ ft}$

Unsupported length, $l_c = 15 \text{ ft}$

$F'_c = 5 \text{ ksi}$

Gross area of section, $A_g = 10 \text{ ft} \times 12 \text{ in} = 1440 \text{ in}^2$

From ACI 318-14, page 341, Table 21.2.1

For compression controlled sections, $\Phi = 0.65$

From ACI 318-14, page 167, Table 11.5.3.2, Effective length factor, $k = 2$

From ACI 318-14, page 166, section 11.5.3.1, equation 11.5.3.1

$$P_n = 0.55 \times F'_c \times A_g \times [1 - (k \times l_c / 32h)^2]$$

$$P_n = 0.55 \times 5 \text{ ksi} \times 1440 \text{ in}^2 \times [1 - \{2 \times 15 \text{ ft} / (32 \times 1 \text{ ft})\}^2] = 479.5 \text{ kips}$$

$$\Phi \times P_n = 0.65 \times 479.5 \text{ kips} = 311.7 \text{ kips}$$

Hence the correct answer is Option B, **310 kips**.

387. Problem

Correct Answer: (D) 1.5

Explanation:

Critical buckling load, $P_c = 1400 \text{ kips}$

Factored compression, $P_u = 350 \text{ kips}$

Moments acting on the column

$$M_1 = 360 \text{ kip}\cdot\text{ft} \quad M_2 = 360 \text{ kip}\cdot\text{ft}$$

The column is bending in single curvature.

From ACI 318-14, page 76, section 6.6.4.5.3 (a)

M1/M2 is negative for single curvature

$$M1/M2 = -360 \text{ kip}\cdot\text{ft}/360 \text{ kip}\cdot\text{ft} = -1$$

$$C_m = 0.6 - 0.4(M1/M2) = 0.6 - 0.4(-1) = 1.0$$

From ACI 318-14, page 75, equation 6.6.4.5.2

$$\delta = C_m / [1 - P_u/(0.75 P_c)]$$

$$\delta = 1 / [1 - 350 \text{ kips}/(0.75 \cdot 1400 \text{ kips})]$$

$$\delta = 1.5$$

Correct answer is Option D

388. Problem

Correct Answer: (B) 25 in

Explanation:

Ref to ACI 318-14, page 507, Appendix A for bar size diameters and area

For #8 longitudinal bar, $d_b = 1 \text{ in}$

$$\text{Effective area of (two legs of) \#3 tie} = 2 \cdot 0.11 \text{ in}^2 = 0.22 \text{ in}^2$$

From ACI 318-14 page 431, section 25.5.5.1 (a)

For Gr 60 rebar,

Compression lap splice should be greater of 12 in or

$$0.0005 \cdot f_y \cdot d_b = 0.0005 \cdot 60000 \cdot 1 \text{ in} = 30 \text{ in (governs)}$$

From ACI 318-14, page 159, section 10.7.5.2.1 (a) reduction is possible if

Effective Area of tie $> 0.0015 \cdot \text{column dimension} \cdot \text{tie spacing}$

$$0.22 \text{ in}^2 > 0.0015 \cdot 24 \text{ in} \cdot 5 \text{ in}$$

$$0.22 \text{ in}^2 > 0.18 \text{ in}^2 \text{ i.e. reduction is valid}$$

$$\text{Splice length} = 0.83 \cdot 30 \text{ in} \approx \mathbf{25 \text{ in}} \quad \text{Correct answer is Option B.}$$

389. Problem

Correct Answer: (A) 65 kips

Explanation:

Here, the column is cast over hardened concrete and no information about intentional surface roughening is given.

From ACI 318-14, page 382, Table 22.9.4.2,

Coefficient of friction matches case (c)

$$\mu = 0.6 \cdot \lambda = 0.6 \cdot 1 = 0.6 \text{ (Assume normal weight concrete)}$$

$$\text{Area of dowels, } A_{vf} = 4 \cdot 0.6 \text{ in}^2 = 2.4 \text{ in}^2$$

From ACI 318-14, page 381, equation 22.9.4.2

$$V_n = \mu * A_v f * f_y$$

$$V_n = 0.6 * 2.4 \text{ in}^2 * 60 \text{ ksi} = 86.4 \text{ kips}$$

From ACI 318-14 page 383, Table 22.9.4.4, V_n cannot exceed lesser of

$$(d) 0.2 * f'_c * A_c = 0.2 * 3 \text{ ksi} * 20 \text{ in} * 20 \text{ in} = 240 \text{ kips}$$

$$\text{Or } (e) 800 * A_c = 800 * 20 * 20 = 3200000 \text{ lbs} = 320 \text{ kips}$$

$$\text{Hence design strength} = \Phi * V_n = 0.75 * 86.4 \text{ kips} = 64.8 \text{ kips} \approx \mathbf{65 \text{ kips}}$$

Correct answer is Option A.

390. Problem

Correct Answer: (B) #4 at 18 inches O.C. both faces

Explanation:

$$V_u = 50 \text{ kips} \quad \Phi * V_c = 110 \text{ kips}$$

$$V_u \leq \Phi * V_c$$

thickness, $t = 8 \text{ in}$

From ACI 318-14, page 169, Table 11.6.1

Minimum transverse rebar ratio 'pt' for 60 ksi rebar = 0.002

For 1 ft length,

$$\text{Area} = 0.002 * \text{thickness} * 1 \text{ ft} / \text{ft} = 0.002 * 8 \text{ in} * 12 \text{ in} / \text{ft} = 0.192 \text{ in}^2 / \text{ft}$$

From ACI 318-14, page 507, Appendix A,

$$\text{Area of \#3} = 0.11 \text{ in}^2 \quad \text{Area of \#4} = 0.2 \text{ in}^2$$

$$\text{For \#3 in both faces @ 18 in, Area} = 0.11 \text{ in}^2 * 2 / 18 \text{ in} = 0.122 \text{ in}^2 / \text{ft}, \text{ not good}$$

$$\text{For \#4 in both faces @ 18 in, Area} = 0.2 \text{ in}^2 * 2 / 18 \text{ in} = 0.222 \text{ in}^2 / \text{ft}, \text{ O.K.}$$

Hence Option B is the correct answer.

5.3 Systems

[Click here for Problems 5.3](#)

391. Problem

Correct Answer: (B) 50 kips

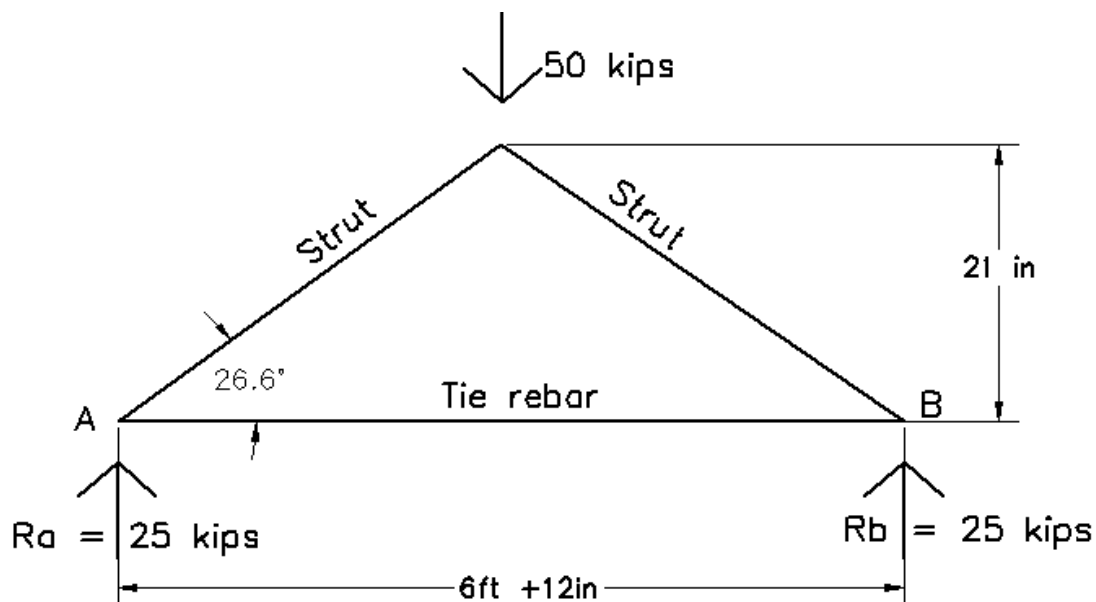
Explanation:

$$d = 21 \text{ in}$$

$$\text{Span of beam} = 6\text{ft} + 12\text{in}/2 + 12\text{in}/2 = 84 \text{ in}$$

$$\theta = \tan^{-1} [21 \text{ in} / (0.5 \cdot 84 \text{ in})] = 26.6^\circ$$

The given problem can be idealized as a truss with two struts and a tie as follows:



Reaction at each end = $0.5 \cdot 50 \text{ kips} = 25 \text{ kips}$ due to symmetry

Force in strut, C

Taking sum of vertical forces = 0 at end A

$$R_a - C \sin (26.6^\circ) = 0$$

$$C = R_a / \sin (26.6^\circ)$$

Taking sum of horizontal forces = 0 at end A, Force in horizontal rebar tie 'T'

$$C \cdot \cos (26.6^\circ) - T = 0$$

$$T = C \cdot \cos (26.6^\circ) = [R_a / \sin (26.6^\circ)] \cdot \cos (26.6^\circ) = R_a / \tan (26.6^\circ)$$

$$T = 25 \text{ kips} / 0.5 = 50 \text{ kips}$$

Correct answer is Option B.

392. Problem

Correct Answer: (D) Enhance lateral stiffness and resistance while the stiffer wall takes the majority of the lateral load.

Explanation:

Moment frames are combined with shear walls to enhance the overall strength and ductility of the shear wall only systems. The walls are typically much stiffer than the framing elements and carry the majority of the load. Hence Option D is the correct answer.

393. Problem

Correct Answer: (D) Combined moment frame and shear wall

Explanation:

Cantilevered columns provide the least redundancy .

Flat slabs with drop panels primarily resist gravity loads and are not for lateral application.

Reinforced masonry walls have average seismic performance. Dual systems i.e. combined wall and moment frame systems have the highest performance and redundancy. Hence Option D is the answer.

394. Problem

Correct Answer: (B) Vertical bracing

Explanation:

In braced frames, the diaphragms assist in distributing the loads efficiently from the floors and roof to the bracing elements. However, the primary element that contributes to lateral story stiffness, strength and stability in a braced frame are the vertical bracings. Moment connections are typically not designed in braced frames and could only serve as redundant elements. Foundation anchors help transfer the superstructure load to the substructure and is not the answer. Hence, Option B is the correct answer.

395. Problem

Correct Answer: (C) Install a tuned mass damper

Explanation:

Increasing the overall mass of the structure by beefing up superstructure and substructure elements is not an economical solution to the problem and may have consequences with seismic design. Increasing the height is incorrect and impractical. Reducing the stiffness of the bracing may alter the frequency but will also have severe consequences to lateral and gravity load resisting capacity. Tuned mass dampers are advanced devices that help mitigate vibrations caused due to wind and seismic loads. Hence Option C is the correct answer.

396. Problem

Correct Answer: (B) 4ft

Explanation:

Ref to AISC steel manual 15th ed., part ASIC 360-16, page 16.1-94, section I3.1a

Effective width on either side of beam centerline is lesser of

- a) $1/8$ th of c/c between supports = $(1/8) * 16\text{ft} = 2\text{ft}$
- b) $1/2$ distance to centerline of adjacent beam = $(1/2) * 6\text{ft} = 3\text{ft}$
- c) Distance to slab edge = 3ft

Hence (a) controls and total effective width

= $2\text{ft} + 2\text{ft} = 4\text{ft}$

Correct answer is option B.

397. Problem

Correct Answer: (A) 372 kips

Explanation:

For W12X72, from AISC Steel manual 15th Ed, page 1-26 Table 1-1,

Area, $A_s = 21.1 \text{ in}^2$

From AISC Steel manual 15th edition, page 16.1-91 section 1b

Strength reduction factor for axial compression of composites, $\Omega = 2$

$F_y = 50 \text{ ksi}$

$f'_c = 4 \text{ ksi}$

Area of concrete, $A_c = 16 \text{ in} \times 16 \text{ in} - A_s = 16 \text{ in} \times 16 \text{ in} - 21.1 \text{ in}^2$

$A_c = 234.9 \text{ in}^2$

From page 16.1-91 equation I2-4

$P_{no} = F_y A_s + F_{ysr} A_{sr} + 0.85 f'_c A_c$ (Area of rebar, $A_{sr} = 0$)

$P_{no} = 50 \text{ ksi} \times 21.1 \text{ in}^2 + 0.85 \times 4 \text{ ksi} \times 234.9 \text{ in}^2$

$P_{no} = 1853.7 \text{ kips}$

Given, elastic critical buckling strength, $P_e = 850 \text{ kips}$

$P_{no}/P_e = 1853.7 \text{ kips} / 850 \text{ kips} = 2.181$

From page 16.1-91 equation I2-2

When $P_{no} / P_e \leq 2.25$

$$P_n = P_{no} \left[0.658^{\frac{P_{no}}{P_e}} \right]$$

$$P_n = 1853.7 \text{ kips} \left[0.658^{2.181} \right]$$

$P_n = 744.03 \text{ kips}$

Allowable strength

$P_n / \Omega = 744 \text{ kips} / 2 = \mathbf{372 \text{ kips}}$

Correct answer is Option A.

398. Problem

Correct Answer: (A) 263 kips

Explanation:

For HSS14.000X0.5, from AISC Steel manual 15th Ed, page 1-101 Table 1-13,

Area, $A_s = 19.8 \text{ in}^2$

$D/t = 30.1$

Given, $F_y = 46 \text{ ksi}$

$E_s = 29000 \text{ ksi}$

$f'_c = 4 \text{ ksi}$

From AISC Steel manual 15th edition, page 16.1-89, Table I1.1a

Limiting D/t for compact section $= 0.15 * E / F_y$

$= 0.15 * 29000 \text{ ksi} / 46 \text{ ksi} = 94.5$, i.e. section is compact

From AISC Steel manual 15th edition, page 16.1-91 section 1b

Strength reduction factor for axial compression of composites, $\Omega = 2$

Area of concrete, $A_c = (\pi/4) * D^2 - A_s = (\pi/4) * 14 \text{ in} * 14 \text{ in} - 19.8 \text{ in}^2$

$A_c = 134.1 \text{ in}^2$

From page 16.1-93 equation I2-9(a) and (b), for compact sections,

$P_{no} = P_p$

$P_p = F_y * A_s + C_2 * f'_c * (A_c + A_{sr} * E_s / E_c)$ (Area of rebar, $A_{sr} = 0$)

$P_p = 46 \text{ ksi} * 19.8 \text{ in}^2 + 0.95 * 4 \text{ ksi} * (134.1 \text{ in}^2)$

$P_p = 1420.4 \text{ kips} = P_{no}$

Given, elastic critical buckling strength, $P_e = 600 \text{ kips}$

$P_{no} / P_e = 1420.4 \text{ kips} / 600 \text{ kips} = 2.367$

From page 16.1-91 equation I2-3

When $P_{no} / P_e > 2.25$

$P_n = 0.877 * P_e$

$P_n = 0.877 * 600 \text{ kips} = 526.2 \text{ kips}$

Allowable strength

$P_n / \Omega = 526.2 \text{ kips} / 2 \approx \mathbf{263 \text{ kips}}$

Correct answer is Option A.

399. Problem

Correct Answer: (C) 98

Explanation:

From AISC Steel manual 15th edition, page 1-22, Table 1-1,

Area of W18X50, $A_s = 14.7 \text{ in}^2$

Ref to AISC steel manual 15th edition, page 16.1-96, section 2d.1

For full composite action, the nominal shear force between the steel beam and the concrete slab transferred by steel anchors V' , between the point of maximum positive moment and the point of zero moment shall be determined as the lowest value in accordance with the limit states of

- a) concrete crushing, (equation I3-1a, page 16.1-97)

$$V' = 0.85 \cdot f'_c \cdot A_c$$

$$V' = 0.85 \cdot 4 \text{ ksi} \cdot (6 \text{ ft} \cdot 5 \text{ in}) = 1224 \text{ kips}$$

- b) tensile yielding of the steel section (equation I3-1b, page 16.1-97)

$$V' = F_y \cdot A_s$$

$$V' = 50 \text{ ksi} \cdot 14.7 \text{ in}^2 = 735 \text{ kips (governs)}$$

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length

Number of studs required for half the span = $V' / Q_n = 735 \text{ kips} / 15 \text{ kips} = 49$

Number of studs required for full span = $49 \cdot 2 = 98$

Correct answer is option C

400. Problem

Correct Answer: (C) 9.57 kips

Explanation:

From AISC Steel manual 15th edition, page 16.1-105, section 2a

When steel deck ribs are perpendicular to the beam and distance between edge of stud shank and deck web is > 2 inches, $R_p = 0.75$ (strong position)

From AISC steel manual, page 3-203, Table 3-21,

For $F_u = 65 \text{ ksi}$, Normal weight 4ksi concrete and deck perpendicular with $R_p = 0.75$

For 1 stud per rib, 0.5 inch diameter stud, $Q_n = 9.57 \text{ kips}$

Correct answer is Option C.

401. Problem

Correct Answer: (A) 5.19 in

Explanation:

From AISC Steel manual 15th edition, page 1-22, Table 1-1,

Area of W18X50, $A_1 = 14.7 \text{ in}^2$

Depth, $d = 18 \text{ in}$

Location of centroid of the section from the top of composite section

$$y_1 = 4.5 \text{ in} + 2.5 \text{ in} + 18 \text{ in} / 2 = 16 \text{ in}$$

Modular ratio, $n = 8$

Effective width of concrete, $b_{\text{eff}} = 8 \text{ ft} = 96 \text{ in}$

Effective width of transformed concrete, $b = 96 \text{ in} / n = 96 \text{ in} / 8 = 12 \text{ in}$

Area of transformed concrete, $A_2 = 4.5 \text{ in} * b = 4.5 \text{ in} * 12 \text{ in} = 54 \text{ in}^2$

Distance of centroid of transformed concrete slab from the top of composite section,

$$y_2 = 4.5 \text{ in} / 2 = 2.25 \text{ in}$$

Taking first moment of area from the top, distance of centroid of composite section from top of section, $y = (A_1 * y_1 + A_2 * y_2) / (A_1 + A_2)$

$$y = (14.7 \text{ in}^2 * 16 \text{ in} + 54 \text{ in}^2 * 2.25 \text{ in}) / (14.7 \text{ in}^2 + 54 \text{ in}^2) = \mathbf{5.19 \text{ in}}$$

Correct answer is Option A.

402. Problem

Correct Answer: (B) 446 kip*ft

Explanation:

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length, number of studs, $n = 64 / 2 = 32$

$$\Sigma Q_n = 16.28 \text{ kip} * 32 = 520.96 \text{ kips}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-7,

$$a = \Sigma Q_n / (0.85 * f'_c * b) = 520.96 \text{ kips} / [0.85 * 4.25 \text{ ksi} * 6 \text{ ft} * (12 \text{ in} / \text{ft})] = 2 \text{ in}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-6,

$y_2 = Y_{\text{con}} - a/2$, where Y_{con} is the distance between top of steel beam to top of concrete

$$y_2 = (3 \text{ in} + 2.5 \text{ in}) - 2 \text{ in} / 2 = 4.5 \text{ in}$$

From AISC Steel manual 15th edition, page 3-174 and 3-175, Table 3-19,

For W18X50, $y_2 = 4.5 \text{ in}$ and $\Sigma Q_n = 521 \text{ kips}$, $M_{\text{allowable}} = \mathbf{446 \text{ kip*ft}}$

Correct answer is Option B.

403. Problem

Correct Answer: (C) 1970 in⁴

Explanation:

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length, number of studs, $n = 40/2 = 20$

$$\Sigma Q_n = 16.8 \text{ kip} * 20 = 336 \text{ kips}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-7,

$$a = \Sigma Q_n / (0.85 * f'_c * b) = 336 \text{ kips} / [0.85 * 5.5 \text{ ksi} * 6 \text{ ft} * (12 \text{ in/ft})] = 1 \text{ in}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-6,

$y_2 = Y_{con} - a/2$, where Y_{con} is the distance between top of steel beam to top of concrete

$$y_2 = (4 \text{ in} + 2.5 \text{ in}) - 1 \text{ in}/2 = 6 \text{ in}$$

From AISC Steel manual 15th edition, page 3-197, Table 3-20,

For W18X55, $y_2 = 6 \text{ in}$ and $\Sigma Q_n = 336 \text{ kips}$, $I_{LB} = \mathbf{1970 \text{ in}^4}$

Correct answer is Option C.

404. Problem

Correct Answer: (C) 63.8 kips

Explanation:

Area of stud

$$A_{sa} = \pi * \Phi^2 / 4 = \pi * (1 \text{ in})^2 / 4 = 0.785 \text{ in}^2$$

$$F'_c = 6000 \text{ psi} = 6 \text{ ksi}$$

$$E = 4400 \text{ ksi}$$

From AISC steel manual 15th ed., page 16.1-105, eqn. I8-1

$$Q_n = 0.5 * A_{sa} * \sqrt{(f'_c * E)}$$

$$Q_n = 0.5 * 0.785 \text{ in}^2 * \sqrt{(6 \text{ ksi} * 4400 \text{ ksi})}$$

$$Q_n = 0.5 * 0.785 \text{ in}^2 * \sqrt{(6 \text{ ksi} * 4400 \text{ ksi})}$$

$$Q_n = \mathbf{63.8 \text{ kips}}$$

Correct answer is Option C.

405. Problem

Correct Answer: (B) 450 kips

Explanation:

From AISC steel manual 15th ed., page 3-14, Fig. 3-3, the distance between top of steel and location of effective concrete flange force is Y_2

$$Y_2 = 3.5 \text{ in}$$

$$\text{Total service level moment} = 150 \text{ kip}\cdot\text{ft} + 290 \text{ kip}\cdot\text{ft} = 440 \text{ kip}\cdot\text{ft}$$

From AISC steel manual 15th ed., page 3-174, Table 3-19

For $Y_2 = 3.5 \text{ in}$ and W18X55 with allowable moment 441 kip·ft,

$$\Sigma Q_n = 454 \text{ kips}$$

Closest answer is Option B, **450 kips**.

406. Problem

Correct Answer: (D) 8

Explanation:

Length of channel, $l_a = 6 \text{ in}$

$$f'_c = 4 \text{ ksi}$$

$$E = 3600 \text{ ksi}$$

From AISC steel manual 15th ed., page 1-38, Table 1-5,

For C 3X6,

$$t_w = 0.356 \text{ in}$$

$$t_f = 0.273 \text{ in}$$

From AISC steel manual 15th ed., page 16.1-106, eqn. I8-2,

$$Q_n = 0.3(t_f + 0.5t_w)l_a \sqrt{f'_c E}$$

$$Q_n = 0.3(0.273 \text{ in} + 0.5 \cdot 0.356 \text{ in}) \cdot 6 \text{ in} \cdot \sqrt{(4 \text{ ksi} \cdot 3600 \text{ ksi})}$$

$$Q_n = 97.4 \text{ kips}$$

Total force to be transferred = 380 kips.

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length,

$$\text{Total number of channels required over the full span} = 2 \cdot 380 \text{ kips} / Q_n$$

$$= 2 \cdot 380 \text{ kips} / 97.4 \text{ kips} = 7.8$$

Hence total number of channels required = **8**, correct answer is Option D

407. Problem

Correct Answer: (C) 515 kips

Explanation:

From AISC steel manual 15th ed., page 3-14, Fig. 3-3, the distance between top of steel and location of effective concrete flange force is Y_2

$Y_2 = 4$ in

Total factored moment = $240 \text{ kip}\cdot\text{ft} + 400 \text{ kip}\cdot\text{ft} = 640 \text{ kip}\cdot\text{ft}$

From AISC steel manual 15th ed., page 3-174 and 3-175, Table 3-19

For $Y_2 = 4$ in and W18X50 with LRFD moment capacity $650 \text{ kip}\cdot\text{ft}$,

$\Sigma Q_n = 521 \text{ kips}$

Closest answer is Option C, **515 kips**.

408. Problem

Correct Answer: (B) 5 kips

Explanation:

Span of the truss, $L = 20$ ft

Depth of the truss, $d = 5$ ft

Factored Load, $P = 5$ kips

Max factored moment acting on the span with point load at the center

$$= P \cdot L / 4$$

$$= 5 \text{ kips} \cdot 20 \text{ ft} / 4$$

$$= 25 \text{ kip}\cdot\text{ft}$$

Max design tension at the bottom chord

$$T = P \cdot d$$

$$T = 25 \text{ kip}\cdot\text{ft} / 5 \text{ ft}$$

$$T = \mathbf{5 \text{ kips}}$$

Correct answer is Option B.

409. Problem

Correct Answer: (C) 2 L 2X2X5/16

Explanation:

Span of the truss, $L = 20$ ft

Net service load from Case A

$= 5 \text{ kip} + 15 \text{ kip} = 20 \text{ kip}$ Tension

Net service load from Case B

$= 5 \text{ kip} - 18 \text{ kip} = 13 \text{ kip}$ compression

Unbraced length $= 8$ ft

Ref to AISC steel manual 15th ed. Page 4-149, Table 4-8

For unbraced length 8ft, Allowable Compression for 2 L 2X2X5/16 $= 13.5$ kips, O.K.

Ref to AISC steel manual 15th ed. Page 5-51, Table 5-8

For 2 L 2X2X5/16, minimum tensile strength $= 50$ kips, O.K.

Hence the correct answer is Option C.

410. Problem

Correct Answer: (C) 1000 kip*ft

Explanation:

For strong -column-weak-beam approach,

Per ACI 318-14, page 281, Equation 18.7.3.2

$$\Sigma M_{nc} \geq (6/5)\Sigma M_{nb}$$

$$\text{i.e. } \Sigma M_{nb} \leq (5/6)\Sigma M_{nc}$$

$$\Sigma M_{nb} \leq (5/6) * 1200 \text{ kip*ft}$$

$$\Sigma M_{nb} \leq \mathbf{1000 \text{ kip*ft}}$$

Hence the correct answer is Option C.

411. Problem

Correct Answer: (A)

Explanation:

The nominal joint shear strength is not limited to $20\lambda\sqrt{f_c'}$ times joint area and varies per framing conditions (ACI 318-14, page 287, Table 18.8.4.1).

There are no clauses that confirm Option C and D and both are incorrect.

From ACI 318-14, page 286, section 18.8.3.1, joint transverse reinforcement should satisfy the requirements set for columns in the preceding sections except for the case in section 18.8.3.2 which does not apply here. Hence correct answer is Option A.

412. Problem

Correct Answer: (C) Four #6 bars

Explanation:

Per ACI 318-14 provisions for special moment frames, page 277, Section 18.6.3.2

Positive moment strength at joint face shall be at least one-half the negative moment strength provided at that face of the joint.

From ACI 318-14, page 507, appendix A

Area of #8 bars = 0.79 in^2

Total area of negative reinforcement = $4 * 0.79 \text{ in}^2 = 3.16 \text{ in}^2$

Required area of positive reinforcement = $0.5 * 3.16 \text{ in}^2 = 1.58 \text{ in}^2$

Area of six #4 bars = $6 * 0.2 \text{ in}^2 = 1.2 \text{ in}^2$, Not enough

Area of four #5 bars = $4 * 0.31 \text{ in}^2 = 1.24 \text{ in}^2$, Not enough

Area of four #6 bars = $4 * 0.44 \text{ in}^2 = 1.76 \text{ in}^2$, O.K.

Hence correct answer is Option C, **Four #6 bars**.

413. Problem

Correct Answer: (B) W12X40

Explanation:

Linear load due to weight of 13 ft tall wall = $80 \text{ lbf/ft}^2 * 13 \text{ ft} = 1.04 \text{ kip/ft}$

Linear load due to floor dead load = $75 \text{ lbf/ft}^2 * (8 \text{ ft} / 2) = 0.3 \text{ kip/ft}$

Linear load due to floor live load = $100 \text{ lbf/ft}^2 * (8 \text{ ft} / 2) = 0.4 \text{ kip/ft}$

Total load = $1.04 \text{ kip/ft} + 0.3 \text{ kip/ft} + 0.4 \text{ kip/ft} = 1.74 \text{ kip/ft}$

Moment due to uniform loading on a simply supported beam = $w * L^2 / 8$

$M = 1.74 \text{ kip/ft} * (25 \text{ ft})^2 / 8 = 135.93 \text{ kip*ft}$

Maximum shear = $w * L / 2$

$V = 1.74 \text{ kip/ft} * 25 \text{ ft} / 2 = 21.75 \text{ kip}$

Since compression flanges are fully supported, lateral torsional buckling won't control.

From AISC steel manual 15th edition, page 3-26, Table 3-2,

All beams satisfy shear requirements.

Allowable plastic moment capacity for W12X30 = 108 kip*ft, not enough.

Allowable plastic moment capacity for **W12X40** = 142 kip*ft, O.K.

Hence, the correct answer is Option B.

414. Problem

Correct Answer: (B) 7.1 ksi

Explanation:

For an unshored construction, the entire load is taken by the beam. Immediately after pouring, the concrete won't have hardened and there is no composite action. All the load is taken by the steel beam. The given section modulus in the question is of no use.

Also, before the concrete has hardened, there won't be any live load application so only dead load is considered.

From AISC steel manual 15th Ed. page 1-22 and 1-23, Table 1-1,

For W18X71, $S_x = 127 \text{ in}^3$

Factored bending stress = Factored Moment / Section modulus

$= 75 \text{ kip*ft} * 12 (\text{in/ft}) / 127 \text{ in}^3 = \mathbf{7.1 \text{ ksi}}$

Hence, the correct answer is Option B.

415. Problem

Correct Answer: (C) 11 in

Explanation:

The foundation wall is adequately supported at the top and bottom. Hence earth pressure to be used is “earth pressure at rest”.

From IBC 2018, page 392, Table 1610.1,

For well graded clean gravel mixes GW, At-rest pressure = 60 psf.

From IBC 2018, page 436, Table 1807.1.6.2,

For unbalanced backfill height = 6ft , Height of wall = 8 ft and 60 psf lateral soil load,

Permissible minimum thickness for plain concrete (PC) = 9.5 in

From IBC 2018, page 435, Section 1807.1.6.1,

The foundation wall thickness should not be less than the thickness of the wall supported.

Minimum thickness = Thickness of masonry wall supported = 11 in.

Hence the correct answer is Option C.

5.4 Connections

[Click here for Problems 5.4](#)

416. Problem

Correct Answer: (B)

Explanation:

Fully restrained connections have negligible relative rotation between the members and are simpler to analyze, compared to partially restrained connections which add additional relative rotational indeterminacy to the analysis. Option A is true.

Option C and D are true.

Option B is not true. Partially restrained connections, although transmit moments partially, do not allow unrestrained relative rotation. They have partial fixity.

Hence, Option B is the correct answer.

417. Problem

Correct Answer: (C) **Min 3/16 in, Max 7/16 in**

Explanation:

From AISC steel manual 15th ed. , page 16.1-119, Table J2.4, minimum size of fillet welds for joining plates 0.5 in thick is 3/16 in.

From AISC steel manual 15th ed. , page 16.1-119, along edges of material 1/4 in. or more in thickness; weld size should not be greater than the thickness of the material minus 1/16 in

$$= 1/2 \text{ in} - 1/16 \text{ in}$$

$$= 7/16 \text{ in}$$

Hence the correct answer is Option C , **Min 3/16 in, Max 7/16 in**.

418. Problem

Correct Answer: (A) 15/32 in

Explanation:

From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with 1/2 in. diameter bolts = 9/16 in

From AISC steel manual 15th ed. , page 16.1-132, Table J3.4, minimum edge distance from center of hole to the edge of connected part for 1/2 in. diameter bolts = 3/4 in

Clear distance between the edge of plate and the hole

$$= 3/4 \text{ in} - (9/16 \text{ in})/2$$

$$= \mathbf{15/32 \text{ in}}$$

Correct answer is Option A.

419. Problem

Correct Answer: (A) 33 kips

Explanation:

From AISC steel manual 15th ed. , page 16.1-116, Section J1.8 (d)

The high-strength bolts shall have an available strength of not less than 33% of the required strength of the connection.

$$33/100 * 100 \text{ kips} = \mathbf{33 \text{ kips}}$$

Correct answer is Option A.

420. Problem

Correct Answer: (C) 45 in

Explanation:

Size of weld, $w = 0.25 \text{ in}$

Length of weld, $l = 70 \text{ in} > 100 * w$

From AISC steel manual 15th ed. , page 16.1-120, eqn. J2-1

$$\beta = 1.2 - 0.002 * (l/w)$$

$$\beta = 1.2 - 0.002 * (70 \text{ in} / 0.25 \text{ in})$$

$$\beta = 0.64$$

Check, $300 * w = 300 * 0.25 \text{ in} = 75 \text{ in}$, i.e. length does not exceed $300 * w$

Hence effective length = $\beta * 70 \text{ in} = 0.64 * 70 \text{ in} = 44.8 \text{ in} \approx \mathbf{45 \text{ in}}$

Correct answer is Option C.

421. Problem

Correct Answer: (A) 13.2 kips

Explanation:

Ref to NCEES PE handbook page 257, section 4.2.1 for shear in eccentrically loaded bolts

No. of bolts, $n = 4$

Due to symmetry location of centroid of bolt group from bottom left bolt ,

$$x = 3\text{ in } y = 2\text{ in}$$

$$\text{Eccentric moment acting on the group, } M = P * (18\text{ in} + 6\text{ in}/2) = 21 * P$$

Moment of inertia of bolt group (Area of all the bolts is assumed equal)

$$I_x = \sum x^2 = 4 * 3\text{ in} * 3\text{ in} = 36\text{ in}^2 \quad I_y = \sum y^2 = 4 * 2\text{ in} * 2\text{ in} = 16\text{ in}^2$$

$$I_p = I_x + I_y = 36 + 16 = 52\text{ in}^2$$

$$\text{Vertical load on each bolt due to direct shear} = P / 4 = 0.25 P$$

$$\text{Vertical load on each bolt due to moment} = (M/I_p) * x = 21 P / 52 * 3 = 1.212 P$$

$$\text{Total vertical shear} = 0.25 P + 1.212 P = 1.462 P$$

$$\text{Horizontal shear on each bolt due to moment} = (M/I_p) * y = 21 * P / 52 * 2 = 0.8077 P$$

$$\text{Total shear on each bolt} = \sqrt{(1.462 P)^2 + (0.8077 P)^2} = 1.67 P$$

Allowable shear in bolts = 22 kips

$$1.67 * P = 22\text{ kips}$$

$$P = 22\text{ kips} / 1.67 = \mathbf{13.2\text{ kips}}$$
 Correct answer is Option A.

422. Problem

Correct Answer: (C) 105 kips

Explanation:

Vertical Spacing of bolts, $s = 3\text{ in}$

Number of bolts in a vertical row, $n = 6$

$$\text{Eccentricity, } e = 7.25\text{ in} + (5.5\text{ in} / 2) = 10\text{ in}$$

From AISC steel manual 15th ed. Page 7-43, Table 7-8,

$$C = 5.23$$

Thus allowable load in the connection

$$P = C * \text{allowable bolt shear strength}$$

$$P = 5.23 * 20\text{ kips} = 104.6\text{ kips} \approx \mathbf{105\text{ kips}}$$

Correct answer is Option C.

423. Problem

Correct Answer: (C) 5 X 5 X ½

Explanation:

Since assumed failure is gross section failure, net area, bolt bearing etc. does not have to be checked.

From AISC steel manual 15th Ed. page 5-15 and page 5-16, Table 5-2

Tension capacity of L 5 X 3 X ½ = 80.8 kips

Tension capacity of L 5 X 3½ X ½ = 86.2 kips

Tension capacity of L 5 X 5 X ½ = 103 kips, Meets requirement

Hence Correct answer is Option C.

424. Problem

Correct Answer: (C) 26 kips

Explanation:

From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with ¾ in. diameter bolts = 13/16 in

For edge bolt,

Clear distance, $l_c = 2 \text{ in} - (13/16 \text{ in})/2 = 1.594 \text{ in}$

For interior bolts,

Clear distance, $l_c = 2.5 \text{ in} - (13/16) \text{ in} = 1.688 \text{ in}$

Hence edge bolt governs

From AISC steel manual 15th ed. , page 16.1-136, eqn. J3-6a

Nominal Bearing strength = $2.4 \cdot d \cdot t \cdot F_u$

$= 2.4 \cdot 0.75 \text{ in} \cdot 0.5 \text{ in} \cdot 58 \text{ ksi} = 52.2 \text{ kips}$

From AISC steel manual 15th ed. , page 16.1-136, eqn. J3-6c

Nominal Tearout strength = $1.2 \cdot l_c \cdot t \cdot F_u$

$= 1.2 \cdot 1.594 \text{ in} \cdot 0.5 \text{ in} \cdot 58 \text{ ksi} = 55.5 \text{ kips}$

Allowable bearing/tearout strength = Nominal strength / Ω

$= 52.2 \text{ kips} / 2 \approx \mathbf{26 \text{ kips}}$

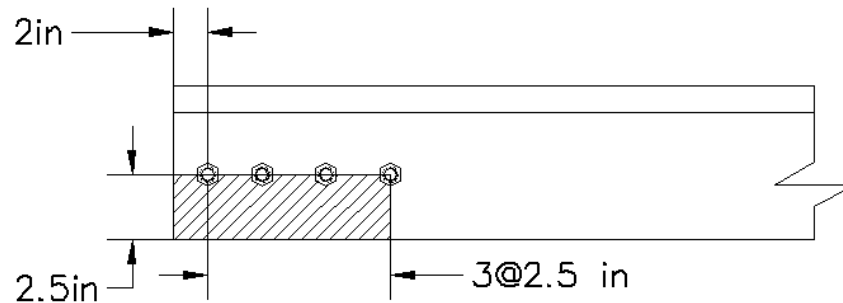
Hence the correct answer is Option C.

425. Problem

Correct Answer: (B) 81 kips

Explanation:

The shaded area represents area that splits from the member during block shear failure



From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with 3/4 in. diameter bolts = 13/16 in.

For net area calculation per page 16.1-20, section B4.3b, width of a bolt hole shall be taken as 1/16 in. greater than the nominal dimension of the hole,

$$d = 13/16 \text{ in} + 1/16 \text{ in} = 0.875 \text{ in}$$

Net tension area, A_{nt}

$$= (2.5 \text{ in} - 0.875 \text{ in} / 2) * t$$

$$= 2.06 \text{ in} * 0.5 \text{ in} = 1.03 \text{ in}^2$$

Net shear area, A_{nv}

$$= (3 * 2.5 \text{ in} + 2 \text{ in} - 3.5 * 0.875 \text{ in}) * 0.5 \text{ in} = 3.21 \text{ in}^2$$

Gross shear area, A_{gv}

$$= (3 * 2.5 \text{ in} + 2 \text{ in}) * 0.5 \text{ in} = 4.75 \text{ in}^2$$

From AISC steel manual 15th ed. Page 16.1-138, equation J4-5,

$$R_n = 0.6 * F_u * A_{nv} + U_{bs} * F_u * A_{nt} \quad (U_{bs} = 1)$$

$$R_n = 0.6 * 58 \text{ ksi} * 3.21 \text{ in}^2 + 1 * 58 \text{ ksi} * 1.03 \text{ in}^2$$

$$= 171.45 \text{ kips}$$

Check $0.6 * F_y * A_{gv} + U_{bs} * F_u * A_{nt}$

$$= 0.6 * 36 \text{ ksi} * 4.75 \text{ in}^2 + 1 * 58 \text{ ksi} * 1.03 \text{ in}^2$$

$$= 162.34 \text{ kips (governs)}$$

Allowable block shear strength = $162.34 \text{ kips} / \Omega$

$$= 162.34 \text{ kips} / 2 \approx \mathbf{81 \text{ kips}}$$

Hence the correct answer is Option B.

426. Problem

Correct Answer: (B) 5/8 in

Explanation:

Ref to AISC steel manual 15th ed. Page 9-4, Fig 9-1 for width of Whitmore section.

Distance between center of end bolts

$$L = 3 * 2.5 \text{ in} = 7.5 \text{ in}$$

Effective width of gusset

$$L_w = 2 * L * \tan(30^\circ)$$

$$L_w = 2 * 7.5 \text{ in} * \tan(30^\circ) = 8.66 \text{ in}$$

Gross area of plate = 8.66 * thickness

$$A_g = 8.66 \text{ in} * t$$

From AISC steel manual 15th ed.

Page 16.1-137, eqn. J4-1 ($\Omega = 1.67$)

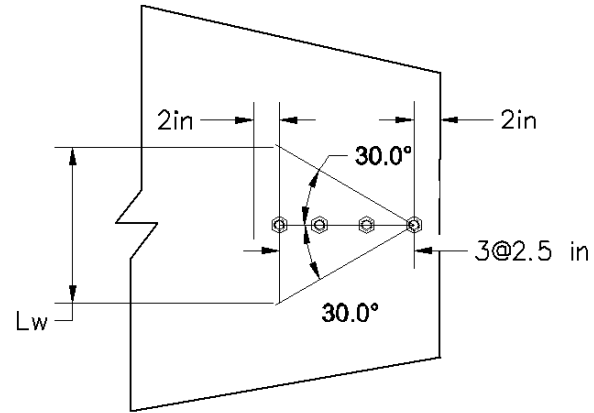
For yielding, Allowable strength = Service load = 100 kips

$$\text{Allowable strength} = F_y * A_g / \Omega$$

$$100 \text{ kips} = 36 \text{ ksi} * 8.66 \text{ in} * t / 1.67$$

$$t = 100 \text{ kips} * 1.67 / (36 \text{ ksi} * 8.66 \text{ in}) = 0.54 \text{ in}$$

Hence minimum thickness from the option is **5/8 in**. Correct answer is Option B.



427. Problem

Correct Answer: (B) 2.98 in²

Explanation:

Per AISC steel manual 15th ed. page 16.1-20, section B4.3b, width of a bolt hole shall be taken as 1/16 in. greater than the nominal hole dimension, = 13/16 in + 1/16 in = 0.875 in

From AISC steel manual 15th ed, page 1-48 and 1-49, Table 1-7, for L 5 X 3 X 1/2

Height of centroid, x = 0.746 in Area of section, $A_g = 3.75 \text{ in}^2$

$$\text{Net area after subtracting bolt hole, } A_n = 3.75 \text{ in}^2 - 0.875 \text{ in} * 0.5 \text{ in} = 3.3125 \text{ in}^2$$

$$\text{Length of connection, } l = 3 * 2.5 \text{ in} = 7.5 \text{ in}$$

From AISC steel manual 15th ed, page 16.1-30, Table D3.1, Case 2

$$U = 1 - x/l = 1 - 0.746 \text{ in} / 7.5 \text{ in} = 0.9$$

From page 16.1-29, Eqn. D3-1, $A_e = A_n * U$

$$A_e = 0.9 * 3.3125 \text{ in}^2 = \mathbf{2.98 \text{ in}^2}$$
. Correct answer is Option B.

428. Problem

Correct Answer: (C) 0.93

Explanation:

The shear lag factor for W section with plates at top and bottom is calculated by splitting the W section into two equivalent half T sections

From AISC steel manual 15th ed, page 1-28 , Table 1-1, for W8X40

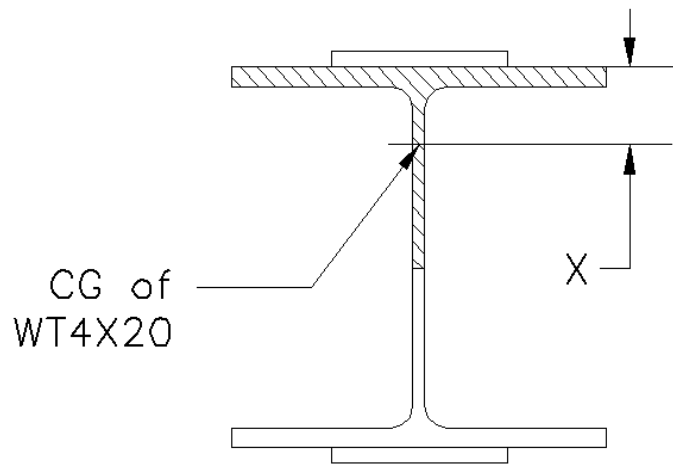
Depth = 8.25 in and weight = 40 lbs/ft

The half T section depth should be 4.125in and weight should be 20 lb/ft

WT 4 X 20 matches the criteria

From AISC steel manual 15th ed, page 1-70 and 1-71 , Table 1-8, for WT 4X20

Distance from centroid of the T section to top of the flange = 0.735 in



Thus connection eccentricity, $X = 0.735$ in

Length of connection, $L = 10$ in

From AISC steel manual 15th ed, page 16.1-30, Table D3.1, Case 2

$$U = 1 - x/l$$

$$= 1 - 0.735 \text{ in} / 10 \text{ in}$$

$$= \mathbf{0.93}$$

Correct answer is Option C.

429. Problem

Correct Answer: (B) 7.87 kips

Explanation:

Thickness of filler plate, $t = 3/8$ in

From AISC steel manual 15th ed, page 16.1-139, Section J5.2 (a)

Modification factor for shear strength of bolts

$$= 1 - 0.4 * (t - 0.25)$$

$$= 1 - 0.4 * (3/8 - 0.25)$$

$$= 0.95 > 0.85 \text{ O.K.}$$

From AISC steel manual 15th ed page 7-22, Table 7-1

For $5/8$ inch diameter A307 bolts in double shear,

Allowable shear = 8.29 kips

For the connection with filler, allowable bolt shear with modification factor

$$= 0.95 * 8.29 \text{ kips} = \mathbf{7.87 \text{ kips}}$$

Correct answer is Option B.

430. Problem

Correct Answer: (B) 3/8 in

Explanation:

The connection is in double shear. So the splice plates share half of the loads each.

$$P_u = 100 \text{ kips} / 2 = 50 \text{ kips}$$

Width of plate, $b = 6$ in thickness, $t = ?$

$$F_y = 36 \text{ ksi} \quad F_u = 58 \text{ ksi}$$

From AISC steel manual 15th ed. Page 16.1-137, Section J4.1

For tensile yielding,

$$P_u = \Phi_y * F_y * A_g$$

$$50 \text{ kips} = 0.9 * 36 \text{ ksi} * 6 \text{ in} * t$$

$$t = 0.257 \text{ in} > 1/4 \text{ in, try with } t = 3/8 \text{ inches}$$

Per AISC steel manual 15th ed. page 16.1-20, section B4.3b,

$$\text{hole dimension for net area} = 13/16 \text{ in} + 1/16 \text{ in} = 0.875 \text{ in}$$

$$\text{Net area using thickness} = (6 \text{ in} - 2 * 0.875 \text{ in}) * 3/8 \text{ in} = 1.59 \text{ in}^2$$

$$\text{Design rupture strength} = \Phi_r * F_u * A_n = 0.75 * 58 \text{ ksi} * 1.59 \text{ in}^2 = 69 \text{ kips, O.K.}$$

Hence the correct answer is Option B, **3/8 in.**

431. Problem

Correct Answer: (C) 359 kips

Explanation:

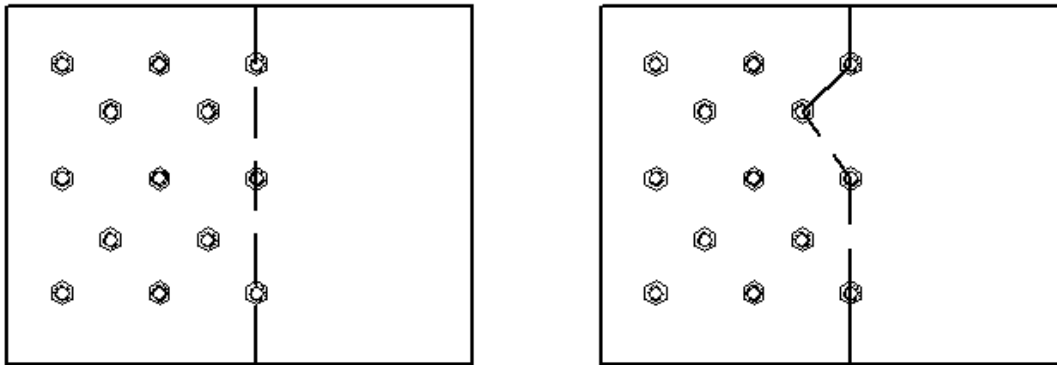
For net area, per AISC steel manual 15th ed. page 16.1-20, section B4.3b, width of a bolt hole shall be taken as 1/16 in. greater than the nominal hole dimension ,

$$= 13/16\text{in} + 1/16\text{in} = 0.875\text{ in}$$

Longitudinal center to center spacing, $s = 3\text{in}$

Transverse spacing, $g = 2.5\text{ in}$

The plates can fail in tensile rupture in two ways as shown below



Per AISC steel manual 15th ed. page 16.1-20, section B4.3b

$$\text{Additional length for each diagonal stagger} = s^2 / 4g = 3\text{ in}^2 / (4 * 2.5\text{ in}) = 0.9\text{ in}$$

$$\text{For two diagonal staggers, additional length due to staggered path} = 0.9\text{ in} * 2 = 1.8\text{ in}$$

$$\text{Reduction in length of staggered path due to additional hole} = 0.875\text{ in} < 1.8\text{ in}$$

Hence staggered path with 4 holes does not govern, straight path with 3 holes governs

For straight path,

$$\text{Net area, } A_n = (15\text{in} - 3 * 0.875\text{ in}) * 0.5\text{ in}$$

$$A_n = 6.188\text{ in}^2$$

$$\text{Nominal tensile rupture strength} = A_n * F_u$$

$$= 6.188\text{ in}^2 * 58\text{ ksi}$$

$$\approx \mathbf{359\text{ kips}}$$

Correct answer is Option C.

432. Problem

Correct Answer: (D) 55 ksi

Explanation:

Total number of bolts, $n = 9$

Shear Load per bolt, $P = 250 \text{ kips}/n$

$= 250 \text{ kips}/9 = 27.78 \text{ kips}$

From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with $3/4$ in. diameter bolts $= 13/16$ in.

Thickness of plate, $t = 5/8$ in

Projected area of bolt hole for bearing, $A = 13/16 \text{ in} * 5/8 \text{ in}$

$A = 0.508 \text{ in}^2$

Bearing stress $= P/A$

$= 27.78 \text{ kips} / 0.508 \text{ in}^2 \approx \mathbf{55 \text{ ksi}}$

Correct answer is Option D.

433. Problem

Correct Answer: (A) 137 kips

Explanation:

The governing failure mode is single shear of bolts and bolt bearing/tearout.

From AISC steel manual, page 7-27, Table 7-4

For 3 inch spacing, standard holes with $F_u = 58 \text{ ksi}$, for $7/8$ inch diameter bolts

Allowable bolt bearing $= 60.9 \text{ kip/in}$

For $1/4$ inch plate, Allowable bolt bearing/tearout $= 60.9 \text{ kip/in} * 1/4 \text{ in} = 15.23 \text{ kips}$

From AISC steel manual, page 7-22, Table 7-1, For Group A $7/8$ inch diameter bolts

For single shear, threads not included in shear plane,

$V_{\text{allowable}} = 20.4 \text{ kips}$

Thus bearing/tearout governs

Total number of bolts $= 9$

Allowable strength of connection $= 9 * 15.23 \text{ kips} \approx \mathbf{137 \text{ kips}}$

Correct answer is Option A.

434. Problem

Correct Answer: (A) 21.1 kips

Explanation:

From AISC steel manual, page 10-91, the single plate shear connection meets all requirements for Table 10-10a to be used.

From Page 10-94, Table 10-10 a, for 3/8 inch plate, 3/4 inch bolt and $n = 2$,

For Group A bolts (A325) with threads not in shear plane,

Allowable strength = **21.1 kips**

Correct answer is Option A.

435. Problem

Correct Answer: (B)

Explanation:

Slip critical connections have lesser shear strength than bearing type connections for the same bolts. Slip critical connections do not increase the bolt hole tolerance or eliminate the need for bearing/tearout checks. They are designed to prevent bolt slip under service loads, ensuring stiffness and minimizing movements, especially in fatigue-sensitive or dynamically loaded structures. Hence Option B is the correct answer.

436. Problem

Correct Answer: (D)

Explanation:

The grade of steel does not affect the rotational stiffness, only the strength of the joint.

The beam web welds primarily resist shear.

The diameter of the bolts used in the flange plates only increases the capacity of the connection. The geometry and thickness of the flange plate and angles primarily affect the rotational stiffness. Correct answer is Option D.

437. Problem

Correct Answer: (C)

Explanation:

High strength bolts and larger welds contribute to strength and not overall ductility.

Reducing the number of bolts does not necessarily increase ductility and may complicate the load distribution. Designing the connection so that yielding occurs in the connected parts (e.g., plates or angles) before bolt or weld failure ensures adequate energy dissipation ductile behavior during cyclic loadings.

Correct answer is Option C.

438. Problem

Correct Answer: (A) 134.4 kips

Explanation:

From AISC steel manual 15th ed, page 1-47, Table 1-7, for L 5X5X1/2

Distance from centroid of the base of leg, $x = 1.42$ in

Gross area of section, $A_g = 4.79$ in²

Length of connection in the direction of tension, $L = 4$ in

From AISC steel manual 15th ed, page 16.1-30, Table D3.1, Case 2

$$U = 1 - x/L$$

$$U = 1 - 1.42 \text{ in} / 4 \text{ in} = 0.645$$

For welded connections, no hole reductions for net area

From AISC steel manual 15th ed page 16.1-29, Eqn. D3-1, $A_e = A_n * U$

$$A_e = 0.645 * 4.79 \text{ in}^2 = 3.09 \text{ in}^2.$$

From AISC steel manual 15th ed page 16.1-28, section D2,

For tensile yielding,

$$\Phi * F_y * A_g = 0.9 * 36 \text{ ksi} * 4.79 \text{ in}^2 = 155.2 \text{ kips}$$

For tensile rupture,

$$\Phi * F_u * A_e = 0.75 * 58 \text{ ksi} * 3.09 \text{ in}^2 = \mathbf{134.4 \text{ kips}}$$

Correct answer is Option A.

439. Problem

Correct Answer: (A) 130.5 kips

Explanation:

Diameter of pin, $d = 2.875$ in

Hole diameter, $d_h = 3$ in

Thickness of plate, $t = 1$ in

From AISC steel manual 15th ed, page 16.1-29 and 16.1-31, Section D5.1

$$b_e = 2t + 0.63 = 2(1 \text{ in}) + 0.63 = 2.63 \text{ in}$$

' b_e ' must be less than $3 \text{ in} - d_h/2 = 3 \text{ in} - 1.5 \text{ in} = 1.5 \text{ in}$, governs

$$b_e = 1.5 \text{ in}$$

$$a = 4.5 \text{ in} - d_h/2 = 4.5 \text{ in} - 3 \text{ in}/2 = 3 \text{ in}$$

$$A_s = 2t(a + d/2)$$

$$A_s = 2(1 \text{ in})(3 \text{ in} + 2.875 \text{ in}/2) = 8.875 \text{ in}^2$$

For tensile rupture in net area,

$$\Phi F_u (2t b_e) = 0.75(58 \text{ ksi})(2(1 \text{ in})(1.5 \text{ in})) = \mathbf{130.5 \text{ kips}} \text{ (governs)}$$

For shear rupture in net area,

$$\Phi 0.6 F_u A_s = 0.75(0.6)(58 \text{ ksi})(8.875 \text{ in}^2) = 231.6 \text{ kips}$$

From AISC steel manual 15th ed, page 16.1-28, Equation D2-1

$$\text{Strength in gross tension yielding} = \Phi F_y A_g = 0.9(36 \text{ ksi})(6 \text{ in} \times 1 \text{ in}) = 194.4 \text{ kips}$$

Correct answer is Option A.

440. Problem

Correct Answer: (C) 145.8 kips

Explanation:

Diameter of pin, $d = 3$ in

Thickness of plate, $t = 1.5$ in

From AISC steel manual 15th ed, page 16.1-140, Equation J7-1

$R_n = 1.8 F_y$ * projected area in bearing

$$R_n = 1.8(36 \text{ ksi})(3 \text{ in} \times 1.5 \text{ in})$$

$$R_n = 291.6 \text{ kips}$$

Allowable bearing strength = R_n / Ω

$$= 291.6 \text{ kips} / 2 = \mathbf{145.8 \text{ kips}}$$
 Correct answer is Option C.

441. Problem

Correct Answer: (B) 1.2 kips

Explanation:

The maximum force that can be applied to the angle is governed by the flexural yielding of the upright leg at the angle bend due to the applied load 'P'.

Thickness of leg, $t = 0.5$ in

Width, $b = 4$ in

Plastic section modulus,

$$Z = b * t^2 / 4 = 4 \text{ in} * (0.5 \text{ in})^2 / 4 = 0.25 \text{ in}^3$$

Elastic section modulus

$$S = b * t^2 / 6 = 4 \text{ in} * (0.5 \text{ in})^2 / 6 = 0.167 \text{ in}^3$$

$$1.6 * S > Z,$$

Hence allowable flexural strength

$$= F_y * Z / \Omega \quad (\Omega = 1.67 \text{ from AISC steel manual 15th ed page 16.1-46, section F1})$$

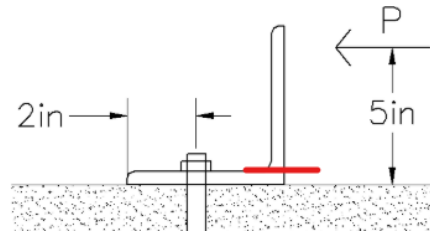
$$= 36 \text{ ksi} * 0.25 \text{ in}^3 / 1.67 = 5.39 \text{ kip*in}$$

$$\text{Moment arm due to } P = (5 \text{ in} - 0.5 \text{ in}) = 4.5 \text{ in}$$

$$\text{Allowable moment} = P * \text{moment arm}$$

$$P = 5.39 \text{ kip*in} / 4.5 \text{ in} = \mathbf{1.2 \text{ kips}}$$

Correct answer is Option B.



442. Problem

Correct Answer: (D) 12.5 kips

Explanation:

The tipping point is the tip of the horizontal leg of the angle.

$$\text{Overturning moment} = 5 \text{ kips} * 5 \text{ in}$$

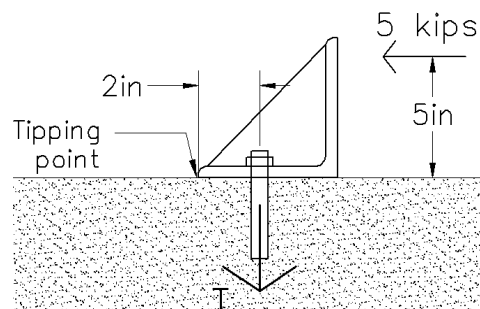
$$\text{Resisting moment} = T * 2 \text{ in}$$

$$T * 2 \text{ in} = 5 \text{ kips} * 5 \text{ in}$$

$$T = 5 \text{ kips} * 5 \text{ in} / 2 \text{ in}$$

$$T = \mathbf{12.5 \text{ kips}}$$

Correct answer is Option D.



443. Problem

Correct Answer: (C) 6

Explanation:

The connection satisfies all requirements per AISC steel manual 15th ed. Page 10-9, 'Design table discussions'. So Table 10-1 can be directly used.

Total reaction to be transferred = Dead load + Live Load
 = 20 kip + 50 kips = 70 kips

From AISC steel manual Page 10-16, Table 10-1

For 6 rows of bolts, Group A (A325) SC class A, 3/4 in diameter bolts, STD holes,

Available strength for 5/16 inch thick angle = 75.9 kips, satisfies requirement.

Hence the correct answer is Option C, **6**.

444. Problem

Correct Answer: (D) 41.2 kips

Explanation:

Length of welds (single line), $L = 6$ in

Total vertical load in connection = P

Total moment in the connection = $P \cdot 3$ in

From AISC steel manual 15th ed. Page 8-14, Fig 8-6,

For single line vertical weld, $I = L^3/12$

For two lines, $I = 2 \cdot L^3/12 = 2 \cdot (6 \text{ in})^3/12 = 36 \text{ in}^3$

Section modulus, $S = I / (L/2) = 36 \text{ in}^3 / 3 \text{ in} = 12 \text{ in}^2$

Shear per unit length of weld, $V = P / (2 \cdot L) = P/12 = 0.0833 P$

Max tension in the weld, $T = \text{Moment} / S = P \cdot 3/12 = P/4 = 0.25 P$

Resultant per unit length = $\sqrt{V^2 + T^2} = \sqrt{[(0.0833 P)^2 + (0.25 P)^2]} = 0.2635 P$

Angle between weld line and resultant, $\Theta = \tan^{-1} (T/V) = \tan^{-1} (0.25 P/0.0833 P) = 71.6^\circ$

From AISC steel manual 15th ed. Page 8-9, effect of load angle = $(1 + 0.5 \sin^{1.5} \Theta)$

Size of 1/2 inch weld in 16th of an inch, $1/2 \text{ in} = 8/16 \text{ in}$. Hence $D = 8$

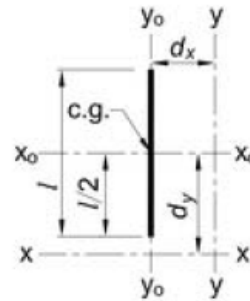
From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n/\Omega = 0.928 \cdot D \cdot l$

Resultant = Allowable stress per unit length of weld including load angle effect

Resultant = $0.928 \cdot D \cdot (1 + 0.5 \sin^{1.5} \Theta)$

$0.2635 P = 0.928 \cdot 8 \cdot (1 + 0.5 \sin^{1.5} 71.6^\circ)$

$P = 41.2 \text{ kips}$ Correct answer is Option D.



445. Problem

Correct Answer: (D) 70.9 kips

Explanation:

Length of welds (single line), $L = 6\text{ in}$

Total vertical load in connection = P

Total moment in the connection = $P \cdot 3\text{ in}$

From AISC steel manual 15th ed. Page 8-14, Fig 8-6,

For single line horizontal weld, $I = L \cdot (dy)^2$

For two lines, $I = 2 \cdot L \cdot (dy)^2 = 2 \cdot 6\text{ in} \cdot (6\text{ in}/2)^2 = 108\text{ in}^3$

Section modulus, $S = I / (L/2) = 108\text{ in}^3 / 3\text{ in} = 36\text{ in}^2$

Shear per unit length of weld, $V = P / (2 \cdot L) = P/12 = 0.0833 P$

Max tension in the weld, $T = \text{Moment} / S = P \cdot 3 / 36 = P/12 = 0.0833 P$

Resultant per unit length = $\sqrt{(V^2 + T^2)} = \sqrt{[(0.0833 P)^2 + (0.0833 P)^2]} = 0.1178 P$

Here, both T and V are acting transverse to the weld axis.

From AISC steel manual 15th ed. Page 8-9, "Effect of Load angle" for transversely loaded welds, 50% increase can be taken

Size of 3/8 inch weld in 16th of an inch, $3/8\text{ in} = 6/16\text{ in}$. Hence $D = 6$

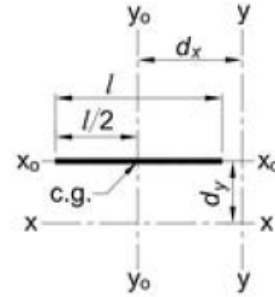
From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n/\Omega = 0.928 \cdot D \cdot l$

Resultant = Allowable stress per unit length of weld including load angle effect

Resultant = $0.928 \cdot D \cdot 1.5$

$0.1178 P = 0.928 \cdot 6 \cdot 1.5$

$P = 70.9\text{ kips}$ Correct answer is Option D.



446. Problem

Correct Answer: (B) 3.25 in

Explanation:

Total load = Dead + Live = 10 kips + 25 kips = 35 kips

Size of 3/8 inch weld in 16th of an inch, $3/8\text{ in} = 6/16\text{ in}$. Hence $D = 6$

From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n/\Omega = 0.928 \cdot D \cdot l$

$35\text{ kips} = 0.928 \cdot 6 \cdot l$

$l = 6.28\text{ in}$

$L_w = l/2 = 6.28\text{ in} / 2 = 3.14\text{ in}$

Hence the correct answer is option B, **3.25 in**.

447. Problem

Correct Answer: (B) 2 in

Explanation:

Total load = Dead + Live = 20 kips + 30 kips = 50 kips

Size of 3/8 inch weld in 16th of an inch, $3/8 \text{ in} = 6/16 \text{ in}$. Hence $D = 6$

From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n/\Omega = 0.928 * D * l$

From AISC steel manual 15th ed. Page 16.1-124, eqn. J2-6a and eqn. J2-6b,

For transverse and longitudinal welds, the nominal resistance can be taken as greater of

(i) $R_n = R_{nwl} + R_{nwt}$

i.e. $R_n/\Omega = (R_{nwl}/\Omega) + (R_{nwt}/\Omega)$

(i) $R_n = 0.85 R_{nwl} + 1.5 * R_{nwt}$

i.e. $R_n/\Omega = 0.85 (R_{nwl}/\Omega) + 1.5 * (R_{nwt}/\Omega)$

Using strength reduction factor, Ω

Try, $L_w = 1.5 \text{ in}$

Total longitudinal length = $2 * 1.5 \text{ in} = 3 \text{ in}$

Total transverse length = 4 in

Allowable strength (i) = $0.928 * 6 * 3 + 0.928 * 6 * 4 = 38.97 \text{ kips}$

Allowable strength (ii) = $0.85 * 0.928 * 6 * 3 + 1.5 * 0.928 * 6 * 4 = 47.6 \text{ kips}$ (not enough)

Try, $L_w = 2 \text{ in}$

Total longitudinal length = $2 * 2 \text{ in} = 4 \text{ in}$

Total transverse length = 4 in

Allowable strength (i) = $0.928 * 6 * 4 + 0.928 * 6 * 4 = 44.5 \text{ kips}$

Allowable strength (ii) = $0.85 * 0.928 * 6 * 4 + 1.5 * 0.928 * 6 * 4 = 52.3 \text{ kips}$ (enough)

Hence the correct answer is Option B, **2 in**.

448. Problem

Correct Answer: (C) 62.4 kips

Explanation:

From AISC steel manual 15th edition, page 8-68, Table 8-4,

$$e_x = a'l,$$

$$\text{i.e. } a = 3 \text{ in}/6 \text{ in} = 0.5$$

$$K'l = 6 \text{ in}$$

$$k = 6 \text{ in}/6 \text{ in} = 1$$

For $a = 0.5$ and $k = 1$, $C = 2.6$

Size of $1/2$ inch weld in 16th of an inch ,

$$1/2 \text{ in} = 8/16 \text{ in} . \text{ Hence } D = 8$$

From AISC steel manual 15th edition, page 8-67, Table 8-3,

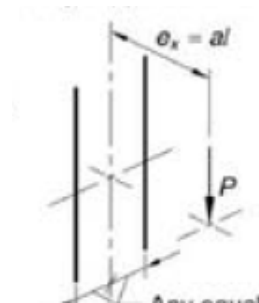
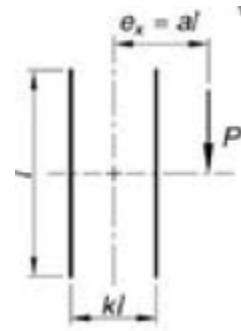
$C_1 = 1$ for $F_{exx} = 70 \text{ ksi}$

$$\text{Allowable Load} = C * C_1 * D * l / \Omega$$

$$= 2.6 * 1 * 8 * 6 / 2$$

$$= \mathbf{62.4 \text{ kips}}$$

Correct answer is Option C.



449. Problem

Correct Answer: (D) 91.2 kips

Explanation:

From AISC steel manual 15th edition, page 8-92, Table 8-8,

$$e_x = a'l,$$

$$\text{i.e. } a = 4 \text{ in}/8 \text{ in} = 0.5$$

$$K'l = 4 \text{ in}$$

$$k = 4 \text{ in}/8 \text{ in} = 0.5$$

For $a = 0.5$ and $k = 0.5$, $C = 2.85$

Size of $1/2$ inch weld in 16th of an inch ,

$$1/2 \text{ in} = 8/16 \text{ in} . \text{ Hence } D = 8$$

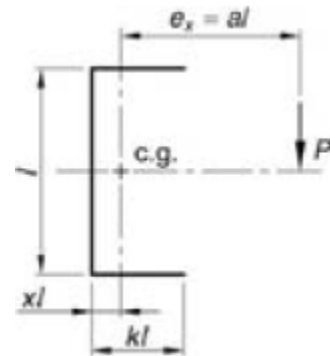
From AISC steel manual 15th edition, page 8-67, Table 8-3,

$C_1 = 1$ for $F_{exx} = 70 \text{ ksi}$

$$\text{Allowable Load} = C * C_1 * D * l / \Omega$$

$$= 2.85 * 1 * 8 * 8 / 2 = \mathbf{91.2 \text{ kips}}$$

Correct answer is Option D.



450. Problem

Correct Answer: (C) 3/4 in.

Explanation:

Eccentricity, $e = 9$ in,

No. of bolts in one row, $n = 4$

From AISC steel manual 15th edition, page 7-43, Table 7-8,

For $s = 3$ in, $e = 9$ in, $n = 4$, we get, $C = 3.01$

Allowable Strength of individual bolt = r_n / Ω

Available strength of the connection, $P_{allowable} = C * r_n / \Omega = 20$ kips

$r_n / \Omega = 20$ kips / C

$r_n / \Omega = 20$ kips / $3.01 = 6.64$ kips

From AISC steel manual 15th edition, page 7-25, Table 7-3,

Capacity of 5/8 in. diameter bolt for STD holes in single shear = 5.42 kips

Capacity of 3/4 in. diameter bolt for STD holes in single shear = 7.91 kips

Hence the correct answer is Option C, **3/4 in.**

451. Problem

Correct Answer: (D) 35.3 kips

Explanation:

From AISC steel manual 15th ed. Page 16.1-133 and 16.1-134, section J3.7,

When the required stress, in either shear or tension, is less than or equal to 30% of the corresponding available stress, the effects of combined stress need not be investigated.

From AISC steel manual 15th ed. Page 7-22, Table 7-1

Allowable shear strength of 1 inch diameter Group A bolts in single shear and threads excluded from shear plane,

$V_r = 26.7$ kips

Acting shear, $V = 5$ kips

$V/V_r = 5$ kips / 26.7 kips = 18.7 % < 30%

Hence no reduction in tension strength required.

From AISC steel manual 15th ed. Page 7-23, Table 7-2

Allowable tensile strength of Group A bolts, 1 inch diameter = **35.3 kips**

Correct answer is Option D.

452. Problem

Correct Answer: (C) 70.6 kip*ft

Explanation:

From AISC steel manual 15th ed. Page 7-23, Table 7-2

Allowable tensile strength of Group A bolts, 1 inch diameter = 35.3 kips

Moment arm = 12 in = 1 ft

Resisting moment capacity = (2 * bolt tension capacity) * moment arm

= 2*35.3 kips* 1 ft

= **70.6 kip*ft**

Correct answer is Option C.

453. Problem

Correct Answer: (B)

Explanation:

All options are true about the lack of fusion in welds except Option B.

Lack of fusion significantly affects the fatigue life during stress reversals but also significantly affects the static strength. Hence Option B is partially incorrect and is the answer.

454. Problem

Correct Answer: (C) 4.83 kips

Explanation:

No. of bolts, nb = 1

From AISC steel manual 15th edition, page 16.1-134, section J3.8, Du = 1.13

Tension in the bolt, Ta = 5 kips

From AISC steel manual 15th edition, page 7-24, Table 7-3, For 3/4 in. diameter bolts

Allowable shear resistance, Vr = 6.33 kips (STD hole)

Min.Bolt pretension , Tb = 28 kips

From AISC steel manual 15th edition, page 16.1-135, eqn. J3-5b

$K_{sc} = 1 - 1.5 \cdot T_a / (D_u \cdot T_b \cdot n_b)$

$K_{sc} = 1 - 1.5 \cdot 5 \text{ kips} / (1.13 \cdot 28 \text{ kips} \cdot 1) = 0.763$

Available shear strength due to interaction = $K_{sc} \cdot V_r = 0.763 \cdot 6.33 \text{ kips} = \mathbf{4.83 \text{ kips}}$

Correct answer is Option C.

455. Problem

Correct Answer: (B) 39 kips

Explanation:

$F_{exx} = 70 \text{ ksi (SMAW)}$

Total length of welds (two sides of plate), $L = 2 \times 6 \text{ in} = 12 \text{ in}$

From AISC steel manual 15th edition, page 1-97, table 1-12

For HSS 8X8X1/4, design wall thickness, $t = 0.233 \text{ in}$

From AISC steel manual 15th edition, page 16.1-118, table J2.2

$R = 2 \times t = 2 \times 0.233 \text{ in} = 0.466 \text{ in}$

For flare bevel done by SMAW, effective throat thickness

$= 5/16 \times R = 5/16 \times 0.466 \text{ in} = 0.146 \text{ in}$

Throat area = effective throat thickness $\times L = 0.146 \text{ in} \times 12 \text{ in} = 1.752 \text{ in}^2$

From AISC steel manual 15th edition, page 16.1-123, table J2.5

For flare bevel in tension normal to weld axis,

Allowable strength = $0.6 \times F_{exx} \times \text{throat area} / \Omega$

Allowable strength = $0.6 \times 70 \text{ ksi} \times 1.752 \text{ in}^2 / 1.88 \approx 39 \text{ kips}$

Correct answer is Option B.

456. Problem

Correct Answer: (A) 1

Explanation:

The slab is not restrained on all four sides and not a part of seismic force resisting system.

From ACI 318-14 page 507, Appendix A, area of #4 tie = 0.2 in^2

Effective area of tie in transverse shear, $A = 2 \times 0.2 \text{ in}^2 = 0.4 \text{ in}^2$

From ACI 318-14, page 204, section 15.4.2, tie spacing shall be at least the greater of

a) $A = 0.75 \times (\sqrt{f'_c}) \times b \times s / f_y$

$s = A \times f_y / [0.75 \times (\sqrt{f'_c}) \times b] = 0.4 \text{ in}^2 \times 60000 / [0.75 \times (\sqrt{4000}) \times 18 \text{ in}] = 28.1 \text{ in}$

b) $A = 50 \times b \times s / f_y$

$s = A \times f_y / [50 \times b] = 0.4 \text{ in}^2 \times 60000 / [50 \times 18 \text{ in}] = 26.6 \text{ in}$

Since the tie spacing is greater than the joint depth available i.e. 8 in, only **1** tie is required.

Correct answer is Option A.

457. Problem

Correct Answer: (B) 6in

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #8 bar, $d_b = 1\text{ in}$

From ACI 318-14 page 142, section 9.7.3.3, from point of inflection, bars should extend at least by a length that is greater of 'd' or $12*d_b$

$$d = 21\text{ in} \quad 12*d_b = 12\text{ in}$$

Point of inflection is 2.5ft i.e 30 in. from the face of support.

Bars should be extended to $30\text{ in} - 21\text{ in} = 9\text{ in.}$ away from support

Hence this clause does not govern extension of bars into the column

From ACI 318-14 page 143, section 9.7.3.8.2, at least one-fourth of the maximum positive moment reinforcement shall extend along the beam bottom into the support at least **6 in.**

Hence the correct answer is Option B, **6in.**

458. Problem

Correct Answer: (D) 48 in.

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #8 bar, $d_b = 1\text{ in}$

From ACI 318-14, page 417, Table 25.4.2.4

$\Psi_t = 1.0$ for bottom bars with less than 12 in. of concrete below.

From ACI 318-14, page 415, Table 25.4.2.2, development length for #8 bar

$$L_d = (f_y * \Psi_t * \Psi_e) / (20 * \lambda * \sqrt{f'_c}) * d_b$$

$$L_d = (60000 * 1 * 1) / (20 * 1 * \sqrt{4000}) * 1\text{ in} = 47.4\text{ in}$$

From ACI 318-14 page 142, section 9.7.3.3, from point of inflection, bars should extend at least by a length that is greater of 'd' or $12*d_b$

$$d = 21\text{ in} \quad 12*d_b = 12\text{ in}$$

Thus, point of inflection i.e 2.5ft (30 in) is sufficiently away from the face of support.

From ACI 318-14 page 143, section 9.7.3.8.2, at least one-fourth of the maximum positive moment reinforcement shall extend along the beam bottom into the support. If the beam is part of the primary lateral-load-resisting system, bars shall be anchored to develop ' f_y ' at the face of the support.

Hence required extension = l_d . Correct answer is Option D, **48 in.**

459. Problem

Correct Answer: (D) 45 in

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #6 bar, $d_b = 0.75$ in

From ACI 318-14, page 417, Table 25.4.2.4

$\Psi_t = 1.3$ for top bars with more than 12 in. of concrete below.

From ACI 318-14, page 415, Table 25.4.2.2, development length for #6 bar

$$L_d = (f_y * \Psi_t * \Psi_e) / (25 * \lambda * \sqrt{f'_c}) * d_b$$

$$L_d = (60000 * 1.3 * 1) / (25 * 1 * \sqrt{4000}) * 0.75 \text{ in} = 37 \text{ in}$$

From ACI 318-14, page 433, Section 25.6.1.5

Development length for individual bars within a bundle, in tension or compression, shall be that of the individual bar, increased 20 percent for a three-bar bundle.

$$L_d = 1.2 * 37 \text{ in} = 44.4 \text{ in}$$

Correct answer is Option D, **45 in.**

460. Problem

Correct Answer: (B) 10 in

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #6 bar, $d_b = 0.75$ in

From ACI 318-14, page 418, Table 25.4.3.2

Since the bar is hooked into a column that continues to the floors below, cover under the 90 degree extension is > 2 in, $\Psi_c = 0.7$

Since information about stirrups is not provided, $\Psi_r = 1.0$

From ACI 318-14, page 417, Section 25.4.3.1, L_{dh} should be greater of

a) $L_{dh} = (f_y * \Psi_e * \Psi_c * \Psi_r) / (50 * \lambda * \sqrt{f'_c}) * d_b$

$$L_{dh} = (60000 * 1 * 0.7 * 1) / (50 * 1 * \sqrt{4000}) * 0.75 \text{ in} = 9.96 \text{ in}$$

b) $8 * d_b = 8 * 0.75 \text{ in} = 6 \text{ in}$

c) 6 in

Correct answer is Option B, **10 in.**

461. Problem

Correct Answer: (C) 12.4 kips

Explanation:

Diameter of bolt = 0.5 in

Effective embedment depth, $h_{ef} = 4 \text{ in} - 0.5 \text{ in} = 3.5 \text{ in}$

Check $1.5 * h_{ef} = 1.5 * 3.5 \text{ in} = 5.25 \text{ in}$,

Edge distance = 12 in

From ACI 318-14, page 239, section 17.4.2.5 and page 235, section 17.4.2.1, eqn. 17.4.2.1c

Edge distance $> 1.5 h_{ef}$, $\psi_{ed} = 1.0$ and $A_{nc} = A_{co} = 9 * (h_{ef})^2$

From ACI 318-14, page 239, section 17.4.2.6, $\psi_c = 1.25$ for cast-in anchors and no cracking

From ACI 318-14, page 239, section 17.4.2.7, $\Psi_{cp} = 1$ for cast in place anchors

From ACI 318-14, page 236, section. 17.4.2.2, $K_c = 24$ for cast in place anchor

$$N_b = k_c * \lambda_a * (\sqrt{f'_c}) * (h_{ef})^{1.5}$$

$$N_b = 24 * 1 * (\sqrt{4000}) * (3.5)^{1.5} = 9939.01 \text{ lbs} = 9.94 \text{ kips}$$

From ACI 318-14, page 235, eqn. 17.4.2.1a,

$$N_{cb} = (A_{nc}/A_{co}) * \psi_{ed} * \psi_c * \Psi_{cp} * N_b$$

$$N_{cb} = 1 * 1 * 1.25 * 1 * 9.94 \text{ kips} = \mathbf{12.4 \text{ kips}}, \quad \text{Correct answer is Option C.}$$

462. Problem

Correct Answer: (A) 2.5 kips

Explanation:

Diameter of anchor bolt, $d_a = 0.5 \text{ in}$

From ACI 318-14, page 242, section. 17.4.3.5,

Distance to outer tip of J bolt from inner surface of shaft, $e_h = 2.5 \text{ in} - 0.5 \text{ in} = 2 \text{ in}$

Check $3 * d_a \leq e_h \leq 4.5 * d_a$, $1.5 \text{ in} \leq 2 \text{ in} \leq 2.25 \text{ in}$ O.K.

$$N_p = 0.9 * f'_c * e_h * d_a$$

$$N_p = 0.9 * 4 \text{ ksi} * 2 \text{ in} * 0.5 \text{ in} = 3.6 \text{ kips}$$

From ACI 318-14, page 242, section. 17.4.3.6, $\Psi_c = 1.0$, for cracking at service loads

From ACI 318-14, page 241, section. 17.4.3.1,

$$N_{pn} = \Psi_c * N_p = 1 * 3.6 \text{ kips} = 3.6 \text{ kips}$$

From ACI 318-14, page 233, section. 17.3.3, $\Phi = 0.7$ for pullout strength

$$\text{Design strength} = \Phi * N_{pn} = 0.7 * 3.6 \text{ kips} = \mathbf{2.5 \text{ kips}}, \quad \text{Correct answer is Option A.}$$

463. Problem

Correct Answer: (C) 12 kips

Explanation:

Diameter of bolt = 0.5 in

Effective embedment depth, $h_{ef} = 3.5$ in

Check $1.5 * h_{ef} = 1.5 * 3.5$ in = 5.25 in,

Edge distance = 12 in

From ACI 318-14, page 239, section 17.4.2.5, and page 235, section 17.4.2.1, eqn. 17.4.2.1c

Edge distance $> 1.5 h_{ef}$, $\psi_{ed} = 1.0$ and $A_{nc} = A_{co} = 9 * (h_{ef})^2$

From ACI 318-14, page 239, section 17.4.2.6, $\psi_c = 1$ for cast-in anchors and cracking

From ACI 318-14, page 239, section 17.4.2.7, $\Psi_{cp} = 1$ for cast in place anchors

From ACI 318-14, page 236, section. 17.4.2.2, $K_c = 24$ for cast in place anchor

$N_b = k_c * \lambda_a * (\sqrt{f'_c}) * (h_{ef})^{1.5}$

$N_b = 24 * 1 * (\sqrt{3000}) * (3.5)^{1.5} = 8607.4$ lbs = 8.6 kips

From ACI 318-14, page 235, eqn. 17.4.2.1a,

$N_{cb} = (A_{nc}/A_{co}) * \psi_{ed} * \psi_c * \Psi_{cp} * N_b$

$N_{cb} = 1 * 1 * 1 * 1 * 8.6$ kips = 8.6 kips

From ACI 318-14, page 257, section. 17.5.3.1,

For $h_{ef} > 2.5$ in, $K_{cp} = 2$

$V_{cp} = K_{cp} * N_{cp}$

For cast-in anchors, $N_{cp} = N_{cb}$

$V_{cp} = 2 * 8.6$ kip

$V_{cp} = 17.2$ kip

From ACI 318-14, page 233, section. 17.3.3,

$\Phi = 0.7$ for pryout strength in shear loads and supplementary reinforcement not present.

Design strength = $\Phi * N_{pn} = 0.7 * 17.2$ kips $\approx$ **12 kips**,

Correct answer is Option C.

464. Problem

Correct Answer: (C) 300 plf

Explanation:

From NDS 2018 page 11, Table 2.3.2, for dead loads, $C_d = 0.9$

From NDS 2018 supplement page 14, Table 1B

Dressed size of 4X8 = 3.5 in X 7.25 in

From NDS 2018, page 99, Table 12E

For Douglas Fir-Larch with side member 3.5 in thick, and 1 inch diameter bolt, loaded perpendicular to grain, $Z_{\perp} = 1330$ lbs

Since all adjustment factors are 1 except load factor,

$$Z_{\perp}' = C_d * Z_{\perp} = 0.9 * 1330 \text{ lbs} = 1197 \text{ lbs}$$

Allowable dead load = $Z_{\perp}' / \text{spacing}$

$$= 1197 \text{ lbs} / 4 \text{ ft} \approx \mathbf{300 \text{ plf}}$$

Correct answer is Option C.

465. Problem

Correct Answer: (C) 1826 lbs

Explanation:

From NDS 2018 supplement page 14, Table 1B

Dressed size of 6X8 = 5.5 in X 7.5 in (main member)

Dressed size of 4X6 = 3.5 in X 5.5 in (side member)

From NDS 2018, page 94, Table 12A,

For DFL with $t_m = 5.5$ in and $t_s = 3.5$ in

$$Z_{m\perp} = 1390 \text{ lbs} \quad Z_{||} = 2660 \text{ lbs} \quad (\text{no adjustment required since all values} = 1)$$

From NDS 2018, page 176, Appendix J, equation J-3

$$Z' = (Z_{||} * Z_{m\perp}) / (Z_{||} * \sin^2\theta + Z_{m\perp} * \cos^2\theta)$$

$$Z' = (2660 * 1390) / (2660 * \sin^2 45^\circ + 1390 * \cos^2 45^\circ)$$

$$Z' \approx \mathbf{1826 \text{ lbs}}$$

Correct answer is Option C.

466. Problem

Correct Answer: (B) 250 lbs

Explanation:

From NDS 2018 page 181, Table L2, for standard hex lag screws with

Diameter, $D = 3/8$ in Length, $L = 7$ in

Minimum thread length, $T = 4$ in

Length of thread penetration in member,

$p = T - E = 3.78$ in

From NDS 2018, page 87, Table 12.3.3A

Specific gravity for DFL, $G = 0.5$

From NDS 2018, page 77, Table 12.2A

For $G = 0.5$, using $3/8$ inch diameter lag screws,

Withdrawal value, $W = 305$ lbs/in

From NDS 2018, page 106, Table 12J

For DFL side member thickness 2.5 in. and $D = 3/8$ in

$Z_{||} = 200$ lbs

$Z_{\perp} = 140$ lbs (governs)

From NDS 2018, page 89, eqn. 12.4-1

$Z_{\alpha} = W \cdot p \cdot Z / (W \cdot p \cdot \cos^2 \alpha + Z \cdot \sin^2 \alpha)$

$Z_{\alpha} = 305 \cdot 3.78 \cdot 140 / (305 \cdot 3.78 \cdot \cos^2 45^\circ + 140 \cdot \sin^2 45^\circ)$

$Z_{\alpha} = 249.7$ lbs $\approx$ **250 lbs**

Correct answer is Option B.

467. Problem

Correct Answer: (C) 20

Explanation:

From NDS 2018 page 182, Table L4

Diameter of 16d common nails = 0.162 in Length = 3.5 in

From NDS 2018 page 112, Table 12P

For 12 gauge thick steel member into DFL member with 16d common nails,

$Z = 149$ lbs (All adjustment factors = 1)

Required penetration = $10 \cdot D = 10 \cdot 0.162$ in = 1.62 in (enough for 4X8 member)

Required number of nails = 2800 lbs / $Z = 2800$ lbs / 149 lbs = 18.79

Correct answer is Option C, **20**.

5.5 Shallow foundations

[Click here for Problems 5.5](#)

468. Problem

Correct Answer: (D) 14.24 ksf

Explanation:

From NCEES PE handbook page 91, Bearing capacity factors,

For $\Phi = 25^\circ$, $N_c = 20.7$ $N_q = 10.7$ $N_\gamma = 10.9$

From NCEES PE handbook page 90, section 3.4.2

$q = \gamma * D_f = 115 \text{ pcf} * 5 \text{ ft} = 575 \text{ psf}$ (Assuming surcharge $Q_{appl} = 0$)

$q_{ult} = c * N_c + q * N_q + 0.5 * \gamma * B_f * N_\gamma$

$q_{ult} = 300 \text{ psf} * 20.7 + 575 \text{ psf} * 10.7 + 0.5 * 115 \text{ pcf} * 3 \text{ ft} * 10.9$

$q_{ult} = 14242.75 \text{ psf} = \mathbf{14.24 \text{ ksf}}$

Correct answer is Option D.

469. Problem

Correct Answer: (A) 9130 psf

Explanation:

Depth of footing $D_f = 5 \text{ ft}$

Depth of water table, $D_w = 2.5 \text{ ft} = 5 \text{ ft} / 2 = D_f / 2$

From NCEES PE handbook page 91, Bearing capacity factors,

For $\Phi = 30^\circ$, $N_q = 18.4$ $N_\gamma = 22.4$

When $D_w = D_f / 2$, correction factor

$C_{wq} = 0.5$ $C_{w\gamma} = (1 + 0.5) / 2 = 0.75$

Corrected bearing capacity factors

$N_q = 18.4 * C_{wq} = 18.4 * 0.75 = 13.8$

$N_\gamma = 22.4 * C_{w\gamma} = 22.4 * 0.5 = 11.2$

From NCEES PE handbook page 90, section 3.4.2

$q_{ult} = c * N_c + q * N_q + 0.5 * \gamma * B_f * N_\gamma$

$q_{ult} = 0 + 110 \text{ pcf} * 5 \text{ ft} * 13.8 + 0.5 * 110 \text{ pcf} * 2.5 \text{ ft} * 11.2 = \mathbf{9130 \text{ psf}}$

Correct answer is Option A.

470. Problem

Correct Answer: (A)

Explanation:

Option B, C and D are all true about controlling differential settlement in buildings.

Option A is false. Trees and vegetation absorb moisture from soil and cause shrinkage and uneven settlement in the surrounding soil. Roots of trees around foundations can also exert pressure that leads to foundation movement and settlement. Hence, Option A is the answer.

471. Problem

Correct Answer: (C) 3ft

Explanation:

Weight of soil , $W_s = 110 \text{ pcf} \times 5 \text{ ft} \times 25 \text{ ft} \times 50 \text{ ft} = 687500 \text{ lbs} = 687.5 \text{ kips}$

Dead load, $D = 1500 \text{ kips}$

Weight of Mat = W_m

Uplift = 1800 kips

Factor of safety, $FS = 1.5$

Resisting weight / Uplift = FS

$W_s + D + W_m = \text{Uplift} \times FS$

$687.5 \text{ kips} + 1500 \text{ kips} + W_m = 1800 \text{ kips} \times 1.5$

$W_m = 512.5 \text{ kips}$

Required depth of mat = $W_m / (\text{Area} \times \text{unit weight})$

$= 512.5 \text{ kips} / (25 \text{ ft} \times 50 \text{ ft} \times 0.145 \text{ kcf})$

$= 2.83 \text{ ft}$

Correct answer is Option C, **3ft**

472. Problem

Correct Answer: (C) 4.44 ksf

Explanation:

Length of footing, $L = 5\text{ ft}$

Breadth of footing, $B = 5\text{ ft}$

Axial load, $P = 50\text{ kips}$

Moment, $M = 50\text{ kip}\cdot\text{ft}$

Eccentricity, $e = M/P$

$e = 50\text{ kip}\cdot\text{ft} / 50\text{ kip} = 1\text{ ft} = 12\text{ in}$

Check $B/6 = 5\text{ ft} / 6 = 60\text{ in} / 6 = 10\text{ in}$

Thus $e > B/6$

From NCEES PE handbook page 93,

$Q_{\max} = 4P/[3L(B-2e)]$

$Q_{\max} = 4 \cdot 50\text{ kips} / [3 \cdot 5\text{ ft} \cdot (5\text{ ft} - 2 \cdot 1\text{ ft})]$

$Q_{\max} = \mathbf{4.44\text{ ksf}}$

Correct answer is Option C.

473. Problem

Correct Answer: (B)

Explanation:

Footings need to be placed below the frost line to prevent damage from freezing and thawing cycles.

Insulating the soil against frost effects is not the correct measure. The changes in footing size won't help against cycles of freezing and thawing. Pile foundation is a very expensive counter measure and is not the most appropriate solution.

Hence correct answer is Option B.

474. Problem

Correct Answer: (C) 0.55 in

Explanation:

Lateral load = 200 kips

Total moment arm

$$= 2\text{ft} + 1.5\text{ft} + 7\text{ft} = 10.5\text{ft}$$

The moment will be resisted by a couple acting at the center of the two footings.

Center to center distance between the footings,

$$= 12\text{ft} - 1\text{ft}/2 - 1\text{ft}/2 = 11\text{ft}$$

Maximum thrust acting in the footing

$$= 200\text{ kips} \times 10.5\text{ft} / 11\text{ft}$$

$$= 190.9\text{ kips}$$

Maximum bearing pressure = $190.9\text{ kips} / (4\text{ft} \times 4\text{ft})$

$$= 190.9\text{ kips} / (48\text{ in} \times 48\text{ in}) = 0.0829\text{ kip/in}^2 = 82.9\text{ lb/in}^2$$

Modulus of subgrade reaction, $k = 150\text{ lb/in}^3$

$k = \text{Bearing pressure} / \text{settlement}$

$$\text{Settlement} = \text{Bearing pressure} / k = [82.9\text{ lbs/in}^2] / [150\text{ lb/in}^3] = \mathbf{0.55\text{ in}}$$

Correct answer is option C

475. Problem

Correct Answer: (C) 3.25 ft

Explanation:

Total Axial load, $P = 100\text{ kips} + 150\text{ kips} = 250\text{ kips}$

Total Moment, $M = 50\text{ kip}\cdot\text{ft} + 80\text{ kip}\cdot\text{ft} = 130\text{ kip}\cdot\text{ft}$

Eccentricity, $e = M/P = 130\text{ kip}\cdot\text{ft} / 250\text{ kips} = 0.52\text{ ft}$

For entire footing area to be effective in bearing the loads,

$$e \leq B/6$$

$$B \geq 6 \cdot e$$

$$B \geq 6 \cdot 0.52\text{ ft}$$

$$B \geq 3.12\text{ ft}$$

Correct answer is option C, **3.25 ft**.

476. Problem

Correct Answer: (D) 2.03 ksf

Explanation:

Weight of footing = $1.8 \text{ klf} \times 25 \text{ ft} = 45 \text{ kips}$

Total Vertical load, $P = 50 \text{ kips} - 15 \text{ kips} + 45 \text{ kips} = 80 \text{ kips}$

Total overturning Moment about the left edge of footing,

$M_o = 15 \text{ kip} \times (25 \text{ ft} - 2.5 \text{ ft}) = 337.5 \text{ kip}\cdot\text{ft}$

Total resisting moment about the left edge of the footing

$M_r = 50 \text{ kip} \times 2.5 \text{ ft} + 45 \text{ kips} \times 25 \text{ ft} / 2 = 687.5 \text{ kip}\cdot\text{ft}$

Location of resultant from left edge

$X = (M_r - M_o) / P = (687.5 \text{ kip}\cdot\text{ft} - 337.5 \text{ kip}\cdot\text{ft}) / 80 \text{ kips} = 4.375 \text{ ft}$

Eccentricity from center of footing, $e = B/2 - x = 25 \text{ ft} / 2 - 4.375 \text{ ft} = 8.125 \text{ ft}$

Thus $e > B/6$

From NCEES PE handbook page 93,

$Q_{\max} = 4P / [3L \times (B - 2e)]$

$Q_{\max} = 4 \times 80 \text{ kips} / [3 \times 6 \text{ ft} \times (25 \text{ ft} - 2 \times 8.125 \text{ ft})]$

$Q_{\max} = \mathbf{2.03 \text{ ksf}}$

Correct answer is Option D.

477. Problem

Correct Answer: (B) 22 ft

Explanation:

For uniform bearing pressure, the resultant of the total load must lie at the center of the combined footing.

Total resultant Vertical load, $P = 50 \text{ kips} + 40 \text{ kips} + 100 \text{ kips} + 80 \text{ kips} = 270 \text{ kips}$

Location of the resultant from left edge = $L/2$

Taking first moment about the left edge,

$P \times L/2 = \text{Right column load} \times (15 \text{ ft} + 1 \text{ ft}) + \text{Left column load} \times 1 \text{ ft}$

$270 \text{ kip} \times L/2 = (100 \text{ kip} + 80 \text{ kip}) \times 16 \text{ ft} + (50 \text{ kip} + 40 \text{ kip}) \times 1 \text{ ft}$

$L = 2 \times 2970 \text{ kip}\cdot\text{ft} / 270 \text{ kip}\cdot\text{ft} = \mathbf{22 \text{ ft}}$

Hence length of the footing must be 22ft for uniform bearing pressure.

Correct answer is Option B.

478. Problem

Correct Answer: (B)

Explanation:

The assumptions stated by Options A, C and D are incorrect or applicable only in typical design scenarios.

The soil can be modelled as a series of independent linear springs with a subgrade modulus, which is also known as The Winkler model.

The best answer is Option B.

479. Problem

Correct Answer: (C) 2.9 ksf

Explanation:

The objective of a strap footing is to use a rigid strap to connect two footings so as to achieve equal soil pressure under both footings. The strap beam does not provide any bearing resistance.

Calculate weight of strap and footings.

Self Weight, $W = (4\text{ft} \times 6\text{ft} \times 2.5\text{ft} + 5\text{ft} \times 10\text{ft} \times 2.5\text{ft} + 2\text{ft} \times 2\text{ft} \times 10\text{ft}) \times 0.15 \text{ kcf}$

$W = 33.75 \text{ kips}$

$P_a = 58 \text{ kips}$

$P_b = 121 \text{ kips}$

Total vertical load, $P = W + P_a + P_b$

$= 33.75 \text{ kips} + 58 \text{ kips} + 121 \text{ kips}$

$= 212.75 \text{ kips}$

Total bearing area, $A = \text{area of footing A} + \text{area of footing B}$

$= 4 \text{ ft} \times 6\text{ft} + 5 \text{ ft} \times 10 \text{ ft}$

$= 74 \text{ ft}^2$

Bearing pressure under footings $= P/A$

$= 212.75 \text{ kips} / 74 \text{ ft}^2$

$= \mathbf{2.9 \text{ ksf}}$

Correct answer is Option C.

480. Problem

Correct Answer: (B) 4ft

Explanation:

From IBC 2018 page 434, Table 1806.2,

Presumptive load bearing value for GW and GP for vertical foundation pressure
= 3000 psf = 3 ksf

Pressure due to column load +footing weight +soil weight = Presumptive value

$$P/B^2 + 0.15 \text{ kcf} \cdot 2\text{ft} + 0.115 \text{ kcf} \cdot (3\text{ft} - 2\text{ft}) = 3 \text{ ksf}$$

$$40 \text{ kips} / B^2 = 2.585 \text{ ksf}$$

$$B^2 = 40 \text{ kips} / 2.585 \text{ ksf}$$

$$B = 3.93 \text{ ft}$$

Hence , the correct answer is Option B, **4ft**.

481. Problem

Correct Answer: (C) 21 kip*ft

Explanation:

$$\text{Bearing pressure, } q = 50 \text{ kips} / (6\text{ft} \cdot 6\text{ft}) = 1.39 \text{ ksf}$$

From ACI 318-14 page 191, Table 13.2.7.1, Location of critical section for Mu is the face of the column

Lever arm is the distance between edge of the footing to face of the column

$$L = 6\text{ft}/2 - 18\text{in}/2$$

$$L = 2.25 \text{ ft}$$

Moment acting at the face of the column due to bearing pressure, q

$$= q \cdot L \cdot L/2$$

$$= 1.39 \text{ ksf} \cdot 2.25 \text{ ft} \cdot 2.25 \text{ ft}/2$$

$$= 3.5 \text{ kip*ft/ft}$$

For total width of 6ft,

$$\text{Total flexural strength required} = 3.5 \text{ kip*ft/ft} \cdot 6\text{ft} = \mathbf{21 \text{ kip*ft}}$$

Correct answer is Option C.

482. Problem

Correct Answer: (C) 25 in

Explanation:

Factored shear, $V_u = 150$ kips

Total width of the section, $b_w = 6 \text{ ft} = 72 \text{ in}$

$f'_c = 4000$ psi

From ACI 318-14, page 341, Table 21.2.1, $\Phi = 0.75$ for shear

From ACI 318-14, page 354, equation 22.5.5.1

$$V_c = 2\lambda(\sqrt{f'_c}) * b_w * d$$

$$V_u = \Phi * V_c$$

$$V_u = 0.75 * 2 * (\sqrt{4000}) * 72 \text{ in} * d$$

$$d = 150 * 1000 \text{ lbs} / [0.75 * 2 * (\sqrt{4000}) * 72 \text{ in}] = 21.96 \text{ in}$$

Depth of footing = d + effective cover = $21.96 \text{ in} + 3 \text{ in} = 24.96 \text{ in}$

Hence, the correct answer is Option C, **25 in**.

483. Problem

Correct Answer: (C) 622 kips

Explanation:

From ACI 318-14, page 362, section 22.6.4.1,

Critical perimeter, $b_o = 4 * (21 \text{ in} / 2 + 18 \text{ in} + 21 / 2 \text{ in}) = 4 * 39 \text{ in} = 156 \text{ in}$

Column at the center of footing is considered 'interior'

From ACI 318-14, page 365, section 22.6.5.3, $\alpha = 40$

From ACI 318-14, page 364, Table 22.6.5.2, $\beta = 18 \text{ in} / 18 \text{ in} = 1$

v_c is the least of

$$a) \quad 4\lambda(\sqrt{f'_c}) = 4 * 1 * \sqrt{4000} = 252.98 \text{ psi}$$

$$b) \quad (2 + 4/\beta) * \lambda * \sqrt{f'_c} = (2 + 4/1) * 1 * \sqrt{4000} = 379.47 \text{ psi}$$

$$c) \quad (2 + \alpha * d/b_o) * \lambda * \sqrt{f'_c} = (2 + 40 * 21 \text{ in} / 156 \text{ in}) * 1 * \sqrt{4000} = 467.04 \text{ psi}$$

$$v_c = 252.98 \text{ psi}$$

$$V_n = v_c * b_o * d = 252.98 \text{ psi} * 156 \text{ in} * 21 \text{ in} = 828762.5 \text{ lbs} = 828.8 \text{ kips}$$

From ACI 318-14, page 341, Table 21.2.1, $\Phi = 0.75$ for shear

$$\text{Design strength} = \Phi * V_n = 0.75 * 828.8 \text{ kips} \approx \mathbf{622 \text{ kips}}$$

Correct answer is Option C.

484. Problem

Correct Answer: (C) #6 @ 12 in. c/c

Explanation:

Depth, $D = 18$ in

Breadth, $B = 1$ ft = 12 in (rebar per ft of width calculated)

From ACI 318-14, page 104, Table 8.6.1.1

For $f_y = 60000$ psi

$A_{smin} = 0.0018 \cdot A_g$

$A_{smin} = 0.0018 \cdot D \cdot B$

$A_{smin} = 0.0018 \cdot 18 \text{ in} \cdot 12 \text{ in} = 0.389 \text{ in}^2$

From ACI 318-14, page 507, Appendix A

Area of #4 bar = 0.2 in^2

Area of #5 bar = 0.31 in^2

Area of #6 bar = 0.44 in^2 (satisfies requirement)

Hence the correct answer is Option C, **#6 @ 12 in. c/c**.

485. Problem

Correct Answer: (B) 716 kips

Explanation:

The column is located at the edge of the footing.

Supporting surface is not wider on all sides than the loaded area and the bearing frustum cannot be developed. Hence, only the area of column will be used for bearing calculation.

From ACI 318-14, page 379, Table 22.8.3.2 case (c)

$B_n = 0.85 \cdot f'_c \cdot A_1$

$B_n = 0.85 \cdot 4 \text{ ksi} \cdot (18 \text{ in} \cdot 18 \text{ in})$

$B_n = 1101.6 \text{ kips}$

From ACI 318-14, page 341, Table 21.2.1, $\Phi = 0.65$ for bearing

Design bearing capacity

$= \Phi \cdot B_n$

$= 0.65 \cdot 1101.6 \text{ kips} \approx \mathbf{716 \text{ kips}}$

Correct answer is Option B.

486. Problem

Correct Answer: (C) Four #8

Explanation:

The column only transfers a compression load of 1500 kips and the design bearing capacity is 2500 kips i.e. greater than the load.

Hence dowel does not have to transfer force and minimum ACI requirement applies.

Area of column, $A_g = 24\text{ in} \times 24\text{ in}$

From ACI 318-14, page 211, section 16.3.4.1, minimum reinforcement

$$A_s = 0.005 \times A_g$$

$$A_s = 0.005 \times 24\text{ in} \times 24\text{ in} = 2.88\text{ in}^2$$

From ACI 318-14, page 507, Appendix A

$$\text{Area of \#6 bar} = 0.44\text{ in}^2$$

$$\text{Area of \#7 bar} = 0.6\text{ in}^2$$

$$\text{Area of \#8 bar} = 0.79\text{ in}^2$$

Four #6 and #7 don't satisfy the minimum requirement.

$$\text{Area of four \#8} = 4 \times 0.79\text{ in}^2 = 3.16\text{ in}^2 \text{ (satisfies requirement)}$$

Hence the correct answer is Option C, **Four #8**.

487. Problem

Correct Answer: (D) 20 in

Explanation:

From ACI 318-14, page 507, Appendix A, Diameter of #8 bar, $d_b = 1\text{ in}$

Hooks are not considered effective in compression and are used to stabilize the dowels during construction. The embedment into the footing must be deep enough to develop the dowels in compression. From ACI 318-14 page 427, section 25.4.9.2

Taking $\Psi_r = 1$, L_{dc} is the greater of

$$\begin{aligned} \text{a) } & f_y \Psi_r / (50 \lambda \sqrt{f'_c}) \times d_b \\ & = 60000 \times 1 / (50 \times 1 \times \sqrt{4000}) \times 1\text{ in} = 18.97\text{ in} \end{aligned}$$

$$\begin{aligned} \text{b) } & 0.0003 \times f_y \Psi_r \times d_b \\ & = 0.0003 \times 60000 \times 1\text{ in} = 18\text{ in} \end{aligned}$$

Correct answer is option D, **20 in**.

5.6 Deep foundations

[Click here for Problems 5.6.](#)

488. Problem

Correct Answer: (C)

Explanation:

Option A is false because end bearing piles are not always shorter than friction piles. The length depends on geotechnical conditions.

The bearing capacity of skin friction piles are affected by the pile material so option B is false.

Both end bearing and friction piles can be used in cohesive or cohesionless soil so Option D is also false.

Negative skin friction affects both end bearing and friction piles. Hence Option C is true and is the answer.

489. Problem

Correct Answer: (B)

Explanation:

Group effects and efficiency is required depending on the center to center spacing of piles.

Option A, C and D all represent generalized scenarios with may or may not warrant the analysis for group effects.

From IBC 2018 page 445, section 1810.2.5, analysis shall include group effects if the spacing of piles is less than three times least horizontal dimension the piles. Hence , it is required if the diameter is more than 1/3rd of the spacing. Option B is the correct answer.

490. Problem

Correct Answer: (A)

Explanation:

Negative skin friction is the downward drag caused in the piles due to downward movement of surrounding soil due to settlement/consolidation.

Option B, C and D are all incorrect.

Option A is the answer.

491. Problem

Correct Answer: (B) 2.5 ft

Explanation:

Thrust, C = 75 kips

Uplift, U = 45 kips

Check required diameter for uplift

Uplift = Allowable skin friction resistance

$$45 \text{ kips} = L * \pi * D * 300 \text{ psf}$$

$$D = 45 \text{ kips} / (20 \text{ ft} * \pi * 0.3 \text{ ksf})$$

$$D = 2.387 \text{ ft}$$

Check this diameter for thrust capacity

Thrust $\leq$ Allowable skin friction resistance + Allowable end bearing resistance

$$75 \text{ kips} \leq L * \pi * D * 0.3 \text{ ksf} + (\pi * D^2 / 4) * 12 \text{ ksf}$$

$$75 \text{ kips} \leq 98.7 \text{ kips O.K.}$$

Hence the correct answer is Option B, **2.5 ft**.

492. Problem

Correct Answer: (A) 10 kips

Explanation:

Scour depth = 8 ft

Hence effective length of pile in the first layer = 10 ft - 8 ft = 2 ft

First layer

$$q_1 = 200 \text{ plf} \quad L_1 = 2 \text{ ft}$$

Second layer

$$q_2 = 300 \text{ plf} \quad L_2 = 12 \text{ ft}$$

Third layer

$$q_3 = 400 \text{ plf} \quad L_3 = 15 \text{ ft}$$

$$\text{Total frictional resistance} = q_1 * L_1 + q_2 * L_2 + q_3 * L_3$$

$$= 200 \text{ plf} * 2 \text{ ft} + 300 \text{ plf} * 12 \text{ ft} + 400 \text{ plf} * 15 \text{ ft}$$

$$= 10000 \text{ lbs}$$

$$= \mathbf{10 \text{ kips}}$$

Correct answer is option A.

493. Problem

Correct Answer: (B) 19 ton

Explanation:

Weight of hammer, $W = 700 \text{ lbs}$

Height of drop, $h = 12 \text{ ft}$

Driving energy, $E = W \cdot h$

$E = 700 \text{ lbs} \cdot 12 \text{ ft} = 8400 \text{ ft-lbs}$

Penetration of pile per hammer blow, $S = 1 \text{ inch} / 3 = 0.333 \text{ in}$

From NCEES PE handbook, page 59, section 2.3.4.2

$P = 2 \cdot E / (S + k)$ [$k = 0.1$ for single acting air hammer]

$P = 2 \cdot 8400 \text{ lbs} / (0.333 + 0.1) = 38799.08 \text{ lbs}$

From NCEES PE handbook page 3,

$2000 \text{ lbs} = 1 \text{ ton}$

$38799.08 \text{ lbs} \approx \mathbf{19 \text{ ton}}$

Correct answer is option B.

494. Problem

Correct Answer: (D) 448 kips

Explanation:

Area of rebar, $A_s = 2 \cdot 4 \text{ in}^2 = 8 \text{ in}^2$

Area of concrete, $A_c = (\pi \cdot D^2 / 4) - A_s$

$A_c = (\pi \cdot 12 \text{ in} \cdot 12 \text{ in} / 4) - 8 \text{ in}^2$

$A_c = 105.1 \text{ in}^2$

From IBC 2018, page 447, Table 1810.3.2.6

Allowable stress in concrete cast in rock = $0.33 f'_c$

$f_{ca} = 0.33 \cdot 6 \text{ ksi} = 1.98 \text{ ksi}$

Allowable stress in compression reinforcement = $0.4 \cdot f_y$

$f_{sa} = 0.4 \cdot 75 \text{ ksi} = 30 \text{ ksi}$

Allowable load in pile = $f_{ca} \cdot A_c + f_{sa} \cdot A_s$

$= 1.98 \text{ ksi} \cdot 105.1 \text{ in}^2 + 30 \text{ ksi} \cdot 8 \text{ in}^2$

$= 448.1 \text{ kips}$

Hence the correct answer is Option D, **448 kips**.

5.7 Retaining walls

[Click here for Problems 5.7](#)

495. Problem

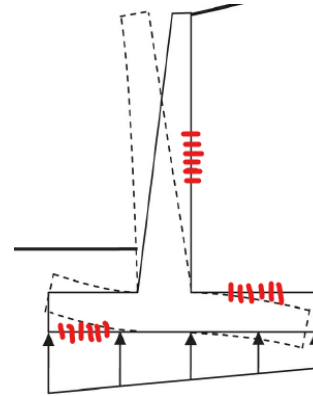
Correct Answer: (D)

Explanation:

The cantilevered retaining wall will be subjected to lateral earth pressure from left to right that will cause tension in the areas marked by red in the figure.

Those areas need tensile reinforcement.

The correct answer is Option D.



496. Problem

Correct Answer: (D) 2091 lb*ft/ft

Explanation:

Active earth pressure coefficient, $K_a = 0.33$

Density of soil, $\gamma = 110$ pcf

Height of stem = 7ft- 1ft = 6ft

Total lateral load acting on the stem follows triangular distribution

$$\text{Total load, } P = (1/2) * K_a * \gamma * H^2$$

$$P = (1/2) * 0.33 * 110 \text{ pcf} * (6 \text{ ft})^2$$

$$P = 653.4 \text{ lbs/ft}$$

Moment arm = distance from stem base to centroid of triangle = $6\text{ft}/3 = 2\text{ft}$

Max moment in the stem = $P * \text{moment arm}$

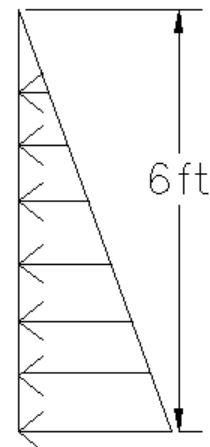
$$= 653.4 \text{ lbs/ft} * 2\text{ft}$$

$$= 1306.8 \text{ lb*ft/ft}$$

From ACI 318-14, page 60, section 5.3.8, load factor for earth pressure = 1.6

$$\text{Factored moment} = 1.6 * 1306.8 \text{ lb ft/ft} \approx \mathbf{2091 \text{ lb*ft/ft}}$$

Correct answer is option D.



497. Problem

Correct Answer: (D) 22152 lb*ft/ft

Explanation:

The critical section of heel for flexure is at the face of the stem as marked in the figure with red.

The bearing pressure under the heel is negligible.

So the loads acting on the heel are self weight of wall, surcharge and weight of soil above the stem.

From ACI 318-14, page 60, section 5.3.8, load factor for earth pressure = 1.6

From ACI 318-14, page 58, Table 5.3.1, load factor for dead load = 1.2

Self weight of wall = $150 \text{ pcf} \times 15 \text{ in} = 150 \text{ pcf} \times 1.25 \text{ ft} = 187.5 \text{ psf}$

Weight of soil above the heel = $120 \text{ pcf} \times (11.5 \text{ ft} - 15 \text{ in}) = 120 \text{ pcf} \times 10.25 \text{ ft} = 1230 \text{ psf}$

Surcharge = 360 psf

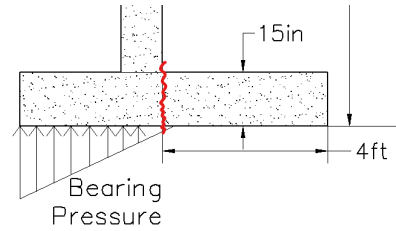
Moment arm for heel = 4ft ,

Sum of moments from edge of the heel to critical section with appropriate load factors,

$= 1.2 \times 187.5 \text{ psf} \times 4\text{ft} \times 4\text{ft}/2 + 1.6 \times (360 \text{ psf} + 1230 \text{ psf}) \times 4\text{ft} \times 4\text{ft}/2$

$= 22152 \text{ lb*ft/ft}$

Correct answer is option D.



498. Problem

Correct Answer: (B) 9 ft

Explanation:

The closest inclined crack to the support of the cantilevered wall will extend upward from the face of the base reaching the compression zone approximately 'd' from the face of the base as marked in the figure with red.

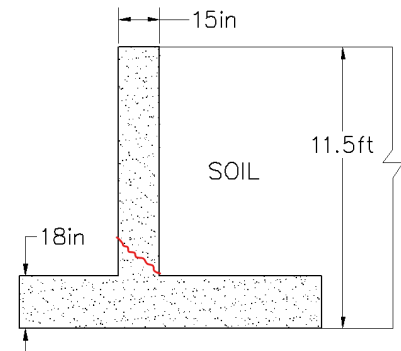
The critical section of stem for shear is at a distance 'd' from the face of the base.

$d = 15 \text{ in} - \text{effective cover} = 15 \text{ in} - 3 \text{ in} = 12 \text{ in}$

Distance from the top of the wall

$D = 11.5 \text{ ft} - 18 \text{ in} - 12 \text{ in} = 9 \text{ ft}$

Correct answer is option B.



499. Problem

Correct Answer: (B) 0.583 in²/ft

Explanation:

$$d = 18 \text{ in} - 3.5 \text{ in} = 14.5 \text{ in}$$

$$M_u = 13583 \text{ lb}\cdot\text{ft}/\text{ft} = 13583 \cdot (12/1000) \text{ kip}\cdot\text{in}/\text{ft}$$

$$M_u = 163 \text{ kip}\cdot\text{in}/\text{ft}$$

Calculate area of steel for 12 in width

From NCEES PE handbook page 264, section 4.3.2.2

$$a = A_s \cdot f_y / (0.85 \cdot f'_c \cdot b)$$

$$a = A_s \cdot 60 / (0.85 \cdot 4.5 \cdot 12)$$

$$a = 1.307 \cdot A_s$$

Assume tension controlled section, From ACI 318-14, page 343, Table 21.2.3

$$\Phi = 0.9$$

From NCEES PE handbook page 264, section 4.3.2.2

$$M_n = A_s \cdot f_y \cdot (d - a/2)$$

$$M_n = A_s \cdot 60 \cdot (14.5 - 1.307 \cdot A_s / 2)$$

$$M_n = 870 \cdot A_s - 39.21 \cdot A_s^2$$

$$M_u = \Phi \cdot M_n$$

$$163 = 0.9 \cdot (870 \cdot A_s - 39.21 \cdot A_s^2)$$

$$35.29 \cdot A_s^2 - 783 \cdot A_s + 163 = 0$$

Solving, we get $A_s = 0.21 \text{ in}^2$

For flexure at cantilevered heel, the minimum flexural reinforcement limit is determined from ACI 318-14, page 137, section 9.6.1.2, because of lack of redundancy (similar to statically determinate beam).

Minimum reinforcement should be greater of

$$\text{a) } 3(\sqrt{f'_c}) \cdot b_w \cdot d / f_y = 3(\sqrt{4500}) \cdot 12 \text{ in} \cdot 14.5 \text{ in} / 60000 = 0.583 \text{ in}^2$$

$$\text{b) } 200 \cdot b_w \cdot d / f_y = 200 \cdot 12 \text{ in} \cdot 14.5 / 60000 = 0.58 \text{ in}^2$$

Hence minimum requirement controls and required reinforcement = **0.583 in²/ft**

Correct answer is option B.

500. Problem

Correct Answer: (C) 32 in

Explanation:

The bars to be spliced are tension reinforcements.

From ACI 318-14 page 507, Appendix A, diameter of #5 bar, $d_b = 0.625$ in

From ACI 318-14, page 417, Table 25.4.2.4

$\Psi_e = 1.0$ for uncoated bars

$\Psi_t = 1.0$ for bottom bars with less than 12 in. of concrete below.

From ACI 318-14, page 415, Table 25.4.2.2, development length for #5 bar

$$L_d = (f_y * \Psi_t * \Psi_e) / (25 * \lambda * \sqrt{f'_c}) * d_b$$

$$L_d = (60000 * 1 * 1) / (25 * 1 * \sqrt{4000}) * 0.625 \text{ in} = 23.7 \text{ in}$$

From ACI 318-14, page 429, Table 25.5.2.1, due to limited information in the question about Area of steel required and provided, Class B is assumed.

$$\text{Splice length} = 1.3 * L_d = 1.3 * 23.7 \text{ in} = 30.81 \text{ in}$$

Hence the correct answer is Option C, **32 in**.

501. Problem

Correct Answer: (A) 25 kip*ft

Explanation:

Maximum intensity of earth pressure at the base of the stem

$$= K_a * \gamma * H = 0.33 * 120 \text{ pcf} * 10 \text{ ft} = 396 \text{ psf}$$

$$\text{Max load acting on the bottom 1 ft strip} = 396 \text{ psf} * 1 \text{ ft} = 0.396 \text{ kip/ft}$$

From ACI 318-14, page 60, section 5.3.8,

load factor for earth pressure = 1.6,

$$\text{factored load} = 1.6 * 0.396 \text{ kip/ft} = 0.634 \text{ kip/ft}$$

The stem is considered a continuous beam system with a uniform loading, $w_u = 0.634$ kip/ft and clear span, $l_n = 20$ ft

From ACI 318-14, page 70, Table 6.5.2, for interior segment of wall with more than two

spans, max negative moment = $w_u * l_n^2 / 10$

$$= (0.634 \text{ kip/ft}) * (20 \text{ ft})^2 / 10 = 25.36 \text{ kip*ft}$$

Correct answer is option A, **25 kip*ft**.